The Republic of Ambrosia v. Miguel Ortega

Digital Forensic Report

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Specialist Field:
Digital Forensics

On Behalf of:
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Subject Matter:
Digital Forensics Report

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1 Introduction

This report intends to provide corroborate evidence into the motivation, opportunity, and intent leading to the data exfiltration conducted by Miguel Ortega against the Republic of Ambrosia. The facts presented within this report are those within the preparer's own area of expertise and knowledge and do not extend to matters and knowledge outside such expertise.

1.1 Summary of Case and Tasking

In the period between January 31 and February 3, 2021, Miguel Ortega was working for the Republic of Ambrosia executing government policies under the direction of Eleanor Roach. During this time, he leaked sensitive information regarding the Republic of Ambrosia’s intention to poison and undercut Avalon’s banana market.

As a result of his actions, the Republic of Ambrosia faced retaliation and complains from people both inside and outside the island. The nation is at risk of international sanctions, threats, break of relations. Additionally, Jeannette Hassan suffered from malicious software infecting her computer.

The forensic experts were therefore hired to reconstruct to the best of their ability, the documentary evidence showing the activities of Miguel Ortega as well as find corroborative evidence of how the information has been leaked.

1.2 Statement of Compliance

I understand my duty as an expert witness to provide independent assistance by way of objective unbiased opinion in relation to matters within my expertise. I will inform all parties in the event that my opinion changes on any material issues.

2 Incident Overview

2.1 Parties Involved

1. **Richard T. Ater** - President of Ambrosia
2. **Thomas K. Chappel** - Vice President
3. **Eleanor M. Roach** - In email contact with Miguel Ortega
4. **Julia K. Dudley** - Secretary of State
5. **Martin K. Snyder** - Internal Police
6. **Eugene T. Barnes** - Secretary of the Treasury
7. **Richard Gauthier** - Dianne Gauthier's husband
8. **Jeanette Hassan** - Victim
9. **Louis Montoya** - Human Resources
10. **Dianne Gauthier** - Richard Ater's sister, Richard Gauthier's wife
11. **Molly Schmitt** - Richard Gauthier's mistress
12. **Jeff Schmitt** - Molly Schmitt's husband
13. **Farmer John** -Farmer in contact with Julia
14. **Emma Garner** - Eugene T. Barnes' financial advisor
15. **Miguel Ortega** - Suspect
16. **Y&S Farms** - The only farm excluded from the subsidy payments
17. **Major local banana producers** - Received subsidy payments from the government

2.2 Timeline

2021-01-31 Sun	
21:31:57 UTC	1. Roach proposes dumping program plan Chappel_N_0007
22:38:11 UTC	1. securityupdate.zip, containing securityupdate.exe was downloaded Jeanette's to the device Jeanette_N_0003
22:40:02 UTC	1. securityupdate.exe was executed for the first

	time Jeanette_N_0003
23:12:25 UTC	1. Roach and Barnes support the plan. Dudley opposes the plan. Snyder thinks it's worth trying to get a trade deal with Avalon. Chappel_N_0006
23:17:36 UTC	1. Ater approves the dumping program plan Chappel_N_0006
23:35 UTC	1. An email from HelpDesk102@ambrosia.it was sent to Jeanette Hassan containing a securityupdate.zip, which we confirmed to be a malicious file. Services_N_0003

2021-02-01 Mon	
00:10:53 UTC	1. Eugene has Emma short Avalon's banana producers and related shippers, then panics and orders her to liquidate every traceable asset immediately, saying he's coming to her office and it's time to retire. Eugene_N_0003
00:21:01 UTC	1. Eleanor tells Marius to prep contraband for a banana shipment, then cancels it after a snitch tips off Avalon. 2. When she orders all shipments halted that night (including one already en route), Marius agrees but warns his crew will want an explanation for the lost payday. Eleanor_N_0003
00:27 UTC	1. Eleanor sent an email to Miguel stating that they are moving ahead with the subsidy proposal. Services_N_0004
00:44 UTC	1. Miguel responds with disagreement, indicating apprehension for going ahead with the subsidy proposal. Services_N_0004

00:48 UTC	1. In response to the disagreements voiced by Miguel, Eleanor threatens him and tells him to enforce the policy. Services_N_0004
00:52 UTC	1. Miguel apologizes for disagreeing and says that he has faith in Richard's leadership. Services_N_0004
14:59:12 UTC	1. curl.exe installs update2.exe Jeanette_N_0008
15:00:06 UTC	1. update2.exe was installed to Jeanette's device Jeanette_N_0003
15:00:18 UTC	1. update2.exe was executed for the first time Jeanette_N_0003
15:30:16 UTC	1. Start of port scan from 192.168.2.42 to target 192.168.2.41. 3.4.8.2 PfSense_N_0001 Filter Log
15:30:17 UTC	1. End of port scan from 192.168.2.42 to target 192.168.2.41. Total of 794 packets sent in 1 second. 3.4.8.2 PfSense_N_0001 Filter Log
19:04:16 UTC	1. The government arranged for subsidy payments, with the exception of Y&S Farms Chappel_N_0005
20:17:33 UTC	1. Avalon learns about the dumping program Chappel_N_0005
20:39:21 UTC	1. Ater angrily denounces leakers as traitors and cowards, vows to find out who they are, and orders a team of forensics experts hired to identify who betrayed him. Chappel_N_0004
21:02:08 UTC	1. Ater orders a team of forensics investigators hired at any price to find the real leaker, not a scapegoat. He plans to lure them to the island with promises of big money, then make them

	work for free once they arrive. Chappel_N_0004

2021-02-02 Tue	
18:03:25 UTC	1. First shipments of discounted bananas sold 2. Y&S farms complained about unfair subsidies; they do not care what the government does, as long as they get a cut of the action Chappel_N_0005
18:37:58 UTC	1. Initial reports from two farms show blackened bananas wilting 2. Both farms are part of Y&S holdings, so it's very likely that it could be Fulsarium from the mainland Chappel_N_0012
19:33:58 UTC	1. Eugene informs the cabinet that Avalon announced they are screening everything coming onto the island, and they are the only producer in the region unaffected by the blight. 2. Eugene proposes poisoning Avalon bananas with blights. Julia opposes the proposal Gauthier_N_0005

2021-02-03 Wed	
00:45:44 UTC	1. Ater approves the operation to poison Avalon bananas Gauthier_N_0005
16:46:59 UTC	1. Ambrosia lost contact with the team sent to Avalon. They were ambushed by forces waiting at the island. 2. Cabinet has to assume that they have been captured. All cabinet members are instructed to meet at Ater's compound. Gauthier_N_0005
16:58:22 UTC	1. Start of blocked packets from a potential

	listener on 192.168.2.42:4444 to 192.168.2.41:45209. PfSense_N_0001
16:59:15 UTC	1. End of blocked packets from a potential listener on 192.168.2.42:4444 to 192.168.2.41:45209. PfSense_N_0001

3 Forensic Examination

All examinations, measurements, tests and experiments related to the Digital image of the accused's computer were performed by the expert and we have summarised our findings on which the expert relies.

3.1 Tools

The forensic tools employed in the performance of this investigation were as follows:

1. **File-Sharing Platform -** Box

   Box is a secure cloud-based content management and file-sharing platform used to store, organize, access, and collaborate on digital files. The platform allows users to manage documents and folders, share content with others, control permissions, and collaborate from different devices. Box also provides tools for workflow automation, integrations, security, and AI-powered content management.

2. **Operating System -** Windows 11 25H2

   Windows 11 is a modern operating system developed by Microsoft for personal computers. It provides a redesigned user interface, improved performance and security features, and built-in tools for productivity, communication, and file management. Windows 11 also supports a wide range of applications, hardware devices, and development environments, making it suitable for both everyday computing and professional use.

3. **Operating System -** Ubuntu 25.04

   Ubuntu is an open-source Linux distribution developed by Canonical, built on the Debian architecture and widely used for desktop computing, cloud infrastructure, and server environments. Known for its user-friendly interface, strong security model, and vast package repository, it provides a stable open-source operating system with both standard short-term releases and long-term support (LTS) editions favored for enterprise deployment.

4. **Artifact Extraction and Analysis -** Autopsy 4.23.1

   Autopsy is a open-source digital forensics platform and graphical interface used by law enforcement, cybersecurity professionals, and forensic investigators to examine hard drives, mobile devices, and media images. Built on top of The Sleuth Kit, it automates forensic analysis by parsing file systems, carving deleted files, extracting browser history and artifacts, and organizing evidence into a searchable interface to support incident response and digital investigations.

5. **Artifact Extraction and Analysis -** Sleuth Kit 4.12.1

   The Sleuth Kit (TSK) is an open-source library and collection of command-line digital forensics tools designed to analyze volume and file system data on forensic disk images. Created by Brian Carrier, it enables investigators to examine raw disk sectors, list active and deleted files, extract artifact metadata, and inspect partition layouts across file systems like NTFS, FAT, ext, and HFS+ without mounting the target media or risking evidence alteration.

6. **Data Visualization** - Google Sheets

   Google Sheets is a cloud-based spreadsheet application developed by Google that allows users to create, edit, organize, analyze, and share data online. It supports features such as formulas, charts, data sorting, filtering, and real-time collaboration, allowing multiple users to work on the same spreadsheet simultaneously. Google Sheets files can be stored and accessed through Google Drive from various devices with an internet connection.

7. **Custom Scripts** - Python 3.13

   Python is a high-level, interpreted programming language celebrated for its clean, readable syntax and design philosophy that prioritizes code simplicity. Supporting multiple programming paradigms—including procedural, object-oriented, and functional programming—it features an extensive ecosystem of libraries and frameworks that make it a versatile tool widely used across web development, data science, machine learning, automation, and software engineering.

8. **Online File Converter** – https://convert.guru/converter

   This website was used to examine the contents of the .frm and .ibd files.

9. **Static Analysis Tool** - PEstudio 9.61

   PeStudio is a free and pro Windows-based static analysis utility created by Marc Ochsenmeier used to inspect and triage Portable Executable (PE) files like .exe and .dll without running them. It extracts metadata, flags suspicious indicators, and checks file hashes against VirusTotal.

10. **Prefetch File Analysis Tool** - WinPrefetchView v1.31

    WinPrefetchView is a small utility that reads the Prefetch files stored in your system and displays the information stored in them. By looking in these files, you can learn which files every application is using, and which files are loaded on Windows boot.

11. **EDB File Viewer** - Shoviv EDB Viewer 2021

    Shoviv EDB Viewer is a standalone, free utility for Windows designed to open, scan, and preview Microsoft Exchange Database (.edb) files without requiring a live Exchange Server environment or Microsoft Outlook setup. It allows system administrators and forensically-oriented users to access both healthy and corrupted private/public EDB files of any size, preserving the original folder hierarchy while inspecting emails, attachments, contacts, calendars, and notes.

12. **Virtual Machine** - VMware

    VMware is a pioneering enterprise software company specializing in cloud computing, server virtualization, and platform virtualization. Its technology allows organizations to abstract physical hardware, running multiple isolated virtual machines (VMs) and operating systems on a single physical server to maximize resource efficiency, scalability, and security.

To the experts' knowledge, there is no software in use above that possesses any material issues whose severity would invalidate the findings thereof.

3.2 Chain of Custody

Ater Image (I)

Consultant Name: Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn
Client: Thomas K. Chappel
Auditee: Desktop image used by Ater
Date/Time of Custody: 2021-02-04 Thu 15:11:38 UTC
Location of Seizure: Ambrosia

Description of Evidence

Item #	Quantity	Description of Item (Model, Serial #, Condition, Marks, Scratches)
1	1	Unique description: ater
		Evidence Number: 001-002
		Source data size: 2047 MB

Chain of Custody				
Item #	**Date/Time**	**Released by** (Signature & ID#)	**Received by** (Signature & ID#)	**Comments/Location**
001-002	2021-02-04 Thu 15:11:38 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Impact on integrity of data will be assessed through metadata tag analysis Impact of integrity on deleted artifacts to be assessed during recovery process Imaging of the device was done through AccessData® FTK® Imager 4.2.0.13
001-002	2021-02-04 Thu 15:11:38 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Case Information: Acquired using: ADI4.2.0.13 Case Number: 001 Evidence Number: 001-002 Unique description: ater Examiner: Notes: desktop
001-002	2021-02-04 Thu 15:11:38 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Information for I:\ater: Physical Evidentiary Item (Source) Information: [Device Info] Source Type: Physical [Drive Geometry] Cylinders: 4,161

				Heads: 16 Sectors per Track: 63 Bytes per Sector: 512 Sector Count: 4,194,288 [Physical Drive Information] Drive Interface Type: ide [Image] Image Type: VMWare Virtual Disk Source data size: 2047 MB Sector count: 4194288 [Computed Hashes] MD5 checksum: d9af0774d042d5ad61d391e17e538adb SHA1 checksum: 8ec49b4a600073b07a0667f4ed635c1c646b872e
001-002	2021-02-04 Thu 15:11:38 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Information: Acquisition started: Thu Feb 4 15:11:38 2021 Acquisition finished: Thu Feb 4 15:12:31 2021 Segment list: I:\ater.001 I:\ater.002
001-002	2021-02-04 Thu 15:12:31 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Verification Results: Verification started: Thu Feb 4 15:12:31 2021 Verification finished: Thu Feb 4 15:12:38 2021 MD5 checksum: d9af0774d042d5ad61d391e17e538adb : verified

				SHA1 checksum: 8ec49b4a600073b07a0667f4ed635c1c646b872e : verified

Chappel Image (H)

Consultant Name: Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn
Client: Thomas K. Chappel
Auditee: Phone image used by Chappel
Date/Time of Custody: 2021-02-04 Thu 15:22:55 UTC
Location of Seizure: Ambrosia

Description of Evidence		
Item #	**Quantity**	**Description of Item** (Model, Serial #, Condition, Marks, Scratches)
001-002	1	Unique description: chappel
		Evidence Number: 001-002
		Source data size: 8191 MB

Chain of Custody				
Item #	**Date/Time**	**Released by** (Signature & ID#)	**Received by** (Signature & ID#)	**Comments/Location**
001-002	2021-02-04 Thu 15:22:55 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Impact on integrity of data will be assessed through metadata tag analysis Impact of integrity on deleted artifacts to be assessed during recovery process Imaging of the device was done through AccessData® FTK® Imager 4.2.0.13
001-002	2021-02-04 Thu 15:22:55 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Case Information: Acquired using: ADI4.2.0.13 Case Number: 001 Evidence Number: 001-002 Unique description: chappel

				Examiner: Notes: phone
001-002	2021-02-04 Thu 15:22:55 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Information for H:\chappel: Physical Evidentiary Item (Source) Information: [Device Info] Source Type: Physical [Drive Geometry] Cylinders: 16,644 Heads: 16 Sectors per Track: 63 Bytes per Sector: 512 Sector Count: 16,777,152 [Physical Drive Information] Drive Interface Type: ide [Image] Image Type: VMWare Virtual Disk Source data size: 8191 MB Sector count: 16777152 [Computed Hashes] MD5 checksum: 97d5e433cefb4abe3e93ca642b7e527a SHA1 checksum: 92879469c10e40e26bac5be784c0b49fb5a8ad40
001-002	2021-02-04 Thu 15:22:55 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Information: Acquisition started: Thu Feb 4 15:22:55 2021

				Acquisition finished: Thu Feb 4 15:26:25 2021 Segment list: H:\chappel.001 H:\chappel.002 H:\chappel.003 H:\chappel.004 H:\chappel.005 H:\chappel.006
001-002	2021-02-04 Thu 15:26:50 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Verification Results: Verification started: Thu Feb 4 15:26:25 2021 Verification finished: Thu Feb 4 15:26:50 2021 MD5 checksum: 97d5e433cefb4abe3e93ca642b7e527a : verified SHA1 checksum: 92879469c10e40e26bac5be784c0b49fb5a8ad40

Gauthier Image (J)

Consultant Name: Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn
Client: Thomas K. Chappel
Auditee: Desktop image used by Gauthier
Date/Time of Custody: 2021-02-04 Thu 15:39:08 UTC
Location of Seizure: Ambrosia

Description of Evidence		
Item #	**Quantity**	**Description of Item** (Model, Serial #, Condition, Marks, Scratches)
001-004	1	Unique description: gauthier
		Evidence Number: 001-004
		Source data size: 61439 MB

Chain of Custody				
Item #	**Date/Time**	**Released by** (Signature & ID#)	**Received by** (Signature & ID#)	**Comments/Location**

001-004	2021-02-04 Thu 15:39:08 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Impact on integrity of data will be assessed through metadata tag analysis Impact of integrity on deleted artifacts to be assessed during recovery process Imaging of the device was done through AccessData® FTK® Imager 4.2.0.13
001-004	2021-02-04 Thu 15:39:08 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Case Information: Acquired using: ADI4.2.0.13 Case Number: 001 Evidence Number: 001-004 Unique description: gauthier Examiner: Notes: desktop
001-004	2021-02-04 Thu 15:39:08 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Information for J:\gauthier: Physical Evidentiary Item (Source) Information: [Device Info] Source Type: Physical [Drive Geometry] Cylinders: 124,828 Heads: 16 Sectors per Track: 63 Bytes per Sector: 512 Sector Count: 125,827,200 [Physical Drive Information] Drive Interface Type: ide [Image] Image Type: VMWare Virtual

				Disk Source data size: 61439 MB Sector count: 125827200 [Computed Hashes] MD5 checksum: a504f14804a8adff6eb7c70849001898 SHA1 checksum: 6dee8a39c871014854151f1b729501cfdcb2cb92
001-004	2021-02-04 Thu 15:39:08 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Information: Acquisition started: Thu Feb 4 15:39:08 2021 Acquisition finished: Thu Feb 4 16:26:27 2021 Segment list: J:\gauthier.001 J:\gauthier.002 J:\gauthier.003 J:\gauthier.004 J:\gauthier.005 J:\gauthier.006 J:\gauthier.007 J:\gauthier.008 J:\gauthier.009 J:\gauthier.010 J:\gauthier.011 J:\gauthier.012 J:\gauthier.013 J:\gauthier.014 J:\gauthier.015 J:\gauthier.016 J:\gauthier.017 J:\gauthier.018 J:\gauthier.019 J:\gauthier.020 J:\gauthier.021 J:\gauthier.022 J:\gauthier.023 J:\gauthier.024 J:\gauthier.025 J:\gauthier.026 J:\gauthier.027 J:\gauthier.028 J:\gauthier.029

				J:\gauthier.030 J:\gauthier.031 J:\gauthier.032 J:\gauthier.033 J:\gauthier.034 J:\gauthier.035 J:\gauthier.036 J:\gauthier.037 J:\gauthier.038 J:\gauthier.039 J:\gauthier.040 J:\gauthier.041
001-004	2021-02-04 Thu 16:26:30 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Verification Results: Verification started: Thu Feb 4 16:26:30 2021 Verification finished: Thu Feb 4 17:06:02 2021 MD5 checksum: a504f14804a8adff6eb7c70849001898 : verified SHA1 checksum: 6dee8a39c871014854151f1b729501cfdcb2cb92 : verified

Jeanette Image (G)

Consultant Name: Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn
Client: Thomas K. Chappel
Auditee: Desktop image used by Jeanette
Date/Time of Custody: 2021-02-04 Thu 14:24:10 UTC
Location of Seizure: Ambrosia

Description of Evidence		
Item #	**Quantity**	**Description of Item** (Model, Serial #, Condition, Marks, Scratches)
001-001	1	Unique description: jean_pc
		Evidence Number: 001-001
		Source data size: 51199 MB

Chain of Custody

Item #	**Date/Time**	**Released by** (Signature & ID#)	**Received by** (Signature & ID#)	**Comments/Location**
001-001	2021-02-04 Thu 14:24:10 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Impact on integrity of data will be assessed through metadata tag analysis Impact of integrity on deleted artifacts to be assessed during recovery process Imaging of the device was done through AccessData® FTK® Imager 4.2.0.13
001-001	2021-02-04 Thu 14:24:10 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Case Information: Acquired using: ADI4.2.0.13 Case Number: 001 Evidence Number: 001-001 Unique description: jean_pc Examiner: Notes: desktop
001-001	2021-02-04 Thu 14:24:10 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Information for G:\jeanhassan: Physical Evidentiary Item (Source) Information: [Device Info] Source Type: Physical [Drive Geometry] Cylinders: 104,023 Heads: 16 Sectors per Track: 63 Bytes per Sector: 512 Sector Count: 104,856,000 [Physical Drive Information] Drive Interface Type: ide

				[Image] Image Type: VMWare Virtual Disk Source data size: 51199 MB Sector count: 104856000 [Computed Hashes] MD5 checksum: 09fa59081a6d0fa181dced5354edccb0 SHA1 checksum: 9ad041bcad931a90cbdef4cd82780cafb8de4082
001-001	2021-02-04 Thu 14:24:10 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Information: Acquisition started: Thu Feb 4 14:24:10 2021 Acquisition finished: Thu Feb 4 14:45:53 2021 Segment list: G:\jeanhassan.001 G:\jeanhassan.002 G:\jeanhassan.003 G:\jeanhassan.004 G:\jeanhassan.005 G:\jeanhassan.006 G:\jeanhassan.007 G:\jeanhassan.008 G:\jeanhassan.009 G:\jeanhassan.010 G:\jeanhassan.011 G:\jeanhassan.012 G:\jeanhassan.013 G:\jeanhassan.014 G:\jeanhassan.015 G:\jeanhassan.016 G:\jeanhassan.017 G:\jeanhassan.018 G:\jeanhassan.019 G:\jeanhassan.020 G:\jeanhassan.021 G:\jeanhassan.022 G:\jeanhassan.023 G:\jeanhassan.024 G:\jeanhassan.025 G:\jeanhassan.026

				G:\jeanhassan.027 G:\jeanhassan.028 G:\jeanhassan.029 G:\jeanhassan.030 G:\jeanhassan.031 G:\jeanhassan.032 G:\jeanhassan.033 G:\jeanhassan.034 G:\jeanhassan.035
001-001	2021-02-04 Thu 14:45:54 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Verification Results: Verification started: Thu Feb 4 14:45:54 2021 Verification finished: Thu Feb 4 15:07:18 2021 MD5 checksum: 09fa59081a6d0fa181dced5354edccb0 : verified SHA1 checksum: 9ad041bcad931a90cbdef4cd82780cafb8de4082 : verified

Mattermost Image (L)

Consultant Name: Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn
Client: Thomas K. Chappel
Auditee: Server/desktop image used by Mattermost
Date/Time of Custody: 2021-02-08 Mon 23:13:14 UTC
Location of Seizure: Ambrosia

Description of Evidence		
Item #	**Quantity**	**Description of Item** (Model, Serial #, Condition, Marks, Scratches)
001-007	1	Unique description: mattermostsrv
		Evidence Number: 001-007
		Source data size: 20479 MB

Chain of Custody				
Item #	**Date/Time**	**Released by** (Signature & ID#)	**Received by** (Signature & ID#)	**Comments/Location**

001-007	2021-02-08 Mon 23:13:14 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Impact on integrity of data will be assessed through metadata tag analysis Impact of integrity on deleted artifacts to be assessed during recovery process Imaging of the device was done through AccessData® FTK® Imager 4.2.0.13
001-007	2021-02-08 Mon 23:13:14 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Case Information: Acquired using: ADI4.2.0.13 Case Number: 001 Evidence Number: 001-007 Unique description: mattermostsrv Examiner: Notes: server/desktop
001-007	2021-02-08 Mon 23:13:14 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Information for L:\mmsrv: Physical Evidentiary Item (Source) Information: [Device Info] Source Type: Physical [Drive Geometry] Cylinders: 41,610 Heads: 16 Sectors per Track: 63 Bytes per Sector: 512 Sector Count: 41,942,880 [Physical Drive Information] Drive Interface Type: ide [Image] Image Type: VMWare Virtual

				Disk Source data size: 20479 MB Sector count: 41942880 [Computed Hashes] MD5 checksum: edd9ed8606087f054534adfff83aa4e7 SHA1 checksum: f636552bee232ce41d69a2f3f18732230663fd15
001-007	2021-02-08 Mon 23:13:14 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Information: Acquisition started: Mon Feb 8 23:13:14 2021 Acquisition finished: Mon Feb 8 23:24:52 2021 Segment list: L:\mmsrv.001 L:\mmsrv.002 L:\mmsrv.003 L:\mmsrv.004 L:\mmsrv.005 L:\mmsrv.006 L:\mmsrv.007 L:\mmsrv.008 L:\mmsrv.009 L:\mmsrv.010 L:\mmsrv.011 L:\mmsrv.012 L:\mmsrv.013 L:\mmsrv.014
001-007	2021-02-08 Mon 23:24:52 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Verification Results: Verification started: Mon Feb 8 23:24:52 2021 Verification finished: Mon Feb 8 23:33:31 2021 MD5 checksum: edd9ed8606087f054534adfff83aa4e7 : verified SHA1 checksum: f636552bee232ce41d69a2f3f1873

				2230663fd15 : verified

Julia Image (M)

Consultant Name: Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn
Client: Thomas K. Chappel
Auditee: Desktop image used by Julia
Date/Time of Custody: 2021-02-15 Mon 11:07:55 UTC
Location of Seizure: Ambrosia

Description of Evidence		
Item #	**Quantity**	**Description of Item** (Model, Serial #, Condition, Marks, Scratches)
001-008	1	Unique description: julia_pc
		Evidence Number: 001-008
		Source data size: 61439 MB

Chain of Custody				
Item #	**Date/Time**	**Released by** (Signature & ID#)	**Received by** (Signature & ID#)	**Comments/Location**
001-008	2021-02-15 Mon 11:07:55 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Impact on integrity of data will be assessed through metadata tag analysis Impact of integrity on deleted artifacts to be assessed during recovery process Imaging of the device was done through AccessData® FTK® Imager 4.2.0.13
001-008	2021-02-15 Mon 11:07:55 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Case Information: Acquired using: ADI4.2.0.13 Case Number: 001 Evidence Number: 001-008 Unique description: julia_pc Examiner:

				Notes: desktop
001-008	2021-02-15 Mon 11:07:55 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Information for M:\julia_pc: Physical Evidentiary Item (Source) Information: [Device Info] Source Type: Physical [Drive Geometry] Cylinders: 124,828 Heads: 16 Sectors per Track: 63 Bytes per Sector: 512 Sector Count: 125,827,200 [Physical Drive Information] Drive Interface Type: ide [Image] Image Type: VMWare Virtual Disk Source data size: 61439 MB Sector count: 125827200 [Computed Hashes] MD5 checksum: 5c5c9949114d7f455c148a12c6346856 SHA1 checksum: 2f403b33b78fe95c5f4fc466a519fd7708ab85fd
001-008	2021-02-15 Mon 11:07:55 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Information: Acquisition started: Mon Feb 15 11:07:55 2021 Acquisition finished: Mon Feb 15 11:34:24 2021

				Segment list: M:\julia_pc.001 M:\julia_pc.002 M:\julia_pc.003 M:\julia_pc.004 M:\julia_pc.005 M:\julia_pc.006 M:\julia_pc.007 M:\julia_pc.008 M:\julia_pc.009 M:\julia_pc.010 M:\julia_pc.011 M:\julia_pc.012 M:\julia_pc.013 M:\julia_pc.014 M:\julia_pc.015 M:\julia_pc.016 M:\julia_pc.017 M:\julia_pc.018 M:\julia_pc.019 M:\julia_pc.020 M:\julia_pc.021 M:\julia_pc.022 M:\julia_pc.023 M:\julia_pc.024 M:\julia_pc.025 M:\julia_pc.026 M:\julia_pc.027 M:\julia_pc.028 M:\julia_pc.029 M:\julia_pc.030 M:\julia_pc.031 M:\julia_pc.032 M:\julia_pc.033 M:\julia_pc.034 M:\julia_pc.035 M:\julia_pc.036 M:\julia_pc.037 M:\julia_pc.038 M:\julia_pc.039 M:\julia_pc.040 M:\julia_pc.041
001-008	2021-02-15 Mon 11:34:25 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Verification Results: Verification started: Mon Feb 15 11:34:25 2021 Verification finished: Mon Feb 15 12:00:20 2021 MD5 checksum: 5c5c9949114d7f455c148a12c6346856 : verified

				SHA1 checksum: 2f403b33b78fe95c5f4fc466a519fd7708ab85fd : verified

Eleanor Image (N)

Consultant Name: Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn
Client: Thomas K. Chappel
Auditee: Desktop image used by Eleanor
Date/Time of Custody: 2021-02-22 Mon 15:54:38 UTC
Location of Seizure: Ambrosia

Description of Evidence		
Item #	**Quantity**	**Description of Item** (Model, Serial #, Condition, Marks, Scratches)
001-010	1	Unique description: eleanor
		Evidence Number: 001-010
		Source data size: 61439 MB

Chain of Custody				
Item #	**Date/Time**	**Released by** (Signature & ID#)	**Received by** (Signature & ID#)	**Comments/Location**
001-010	2021-02-22 Mon 15:54:38 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Impact on integrity of data will be assessed through metadata tag analysis Impact of integrity on deleted artifacts to be assessed during recovery process Imaging of the device was done through AccessData® FTK® Imager 4.2.0.13
001-010	2021-02-22 Mon 15:54:38 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Case Information: Acquired using: ADI4.2.0.13 Case Number: 001 Evidence Number: 001-010 Unique description: eleanor

				Examiner: Notes: desktop
001-010	2021-02-22 Mon 15:54:38 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Information for N:\eleanor: Physical Evidentiary Item (Source) Information: [Device Info] Source Type: Physical [Drive Geometry] Cylinders: 124,828 Heads: 16 Sectors per Track: 63 Bytes per Sector: 512 Sector Count: 125,827,200 [Physical Drive Information] Drive Interface Type: ide [Image] Image Type: VMWare Virtual Disk Source data size: 61439 MB Sector count: 125827200 [Computed Hashes] MD5 checksum: 3e187fae63acd9d39dbe1b6485ac6ed0 SHA1 checksum: e3c5a1d1e3e84291e96c5bb757c32fe2526a26f3
001-010	2021-02-22 Mon 15:54:38 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Information: Acquisition started: Mon Feb 22 15:54:38 2021

				Acquisition finished: Mon Feb 22 16:21:07 2021 Segment list: N:\eleanor.001 N:\eleanor.002 N:\eleanor.003 N:\eleanor.004 N:\eleanor.005 N:\eleanor.006 N:\eleanor.007 N:\eleanor.008 N:\eleanor.009 N:\eleanor.010 N:\eleanor.011 N:\eleanor.012 N:\eleanor.013 N:\eleanor.014 N:\eleanor.015 N:\eleanor.016 N:\eleanor.017 N:\eleanor.018 N:\eleanor.019 N:\eleanor.020 N:\eleanor.021 N:\eleanor.022 N:\eleanor.023 N:\eleanor.024 N:\eleanor.025 N:\eleanor.026 N:\eleanor.027 N:\eleanor.028 N:\eleanor.029 N:\eleanor.030 N:\eleanor.031 N:\eleanor.032 N:\eleanor.033 N:\eleanor.034 N:\eleanor.035 N:\eleanor.036 N:\eleanor.037 N:\eleanor.038 N:\eleanor.039 N:\eleanor.040 N:\eleanor.041

PFsense Image (O)

Consultant Name: Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn
Client: Thomas K. Chappel
Auditee: Router image used by PFsense
Date/Time of Custody: 2021-02-22 Mon 14:20:26 UTC

Location of Seizure: Ambrosia

Description of Evidence		
Item #	**Quantity**	**Description of Item** (Model, Serial #, Condition, Marks, Scratches)
001-008	1	Unique description: pfsense
		Evidence Number: 001-008
		Source data size: 16384 MB

Chain of Custody				
Item #	**Date/Time**	**Released by** (Signature & ID#)	**Received by** (Signature & ID#)	**Comments/Location**
001-008	2021-02-22 Mon 14:20:26 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Impact on integrity of data will be assessed through metadata tag analysis Impact of integrity on deleted artifacts to be assessed during recovery process Imaging of the device was done through AccessData® FTK® Imager 4.2.0.13
001-008	2021-02-22 Mon 14:20:26 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Case Information: Acquired using: ADI4.2.0.13 Case Number: 001 Evidence Number: 001-008 Unique description: pfsense Examiner: Notes: router
001-008	2021-02-22 Mon 14:20:26 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Information for O:\pfsense: Physical Evidentiary Item (Source) Information: [Device Info] Source Type: Physical

				[Drive Geometry] Cylinders: 16,383 Heads: 16 Sectors per Track: 63 Bytes per Sector: 512 Sector Count: 33,554,432 [Physical Drive Information] Drive Interface Type: ide [Image] Image Type: VMWare Virtual Disk Source data size: 16384 MB Sector count: 33554432 [Computed Hashes] MD5 checksum: b5f74cf9fa7ea6e758ae9bc3da99a317 SHA1 checksum: dc00b8ad6b552ae5108763f0918a9dc107d91b6e
001-008	2021-02-22 Mon 14:20:26 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Information: Acquisition started: Mon Feb 22 14:20:26 2021 Acquisition finished: Mon Feb 22 14:27:32 2021 Segment list: O:\pfsense.001 O:\pfsense.002 O:\pfsense.003 O:\pfsense.004 O:\pfsense.005 O:\pfsense.006 O:\pfsense.007 O:\pfsense.008 O:\pfsense.009 O:\pfsense.010

				O:\pfsense.011
001-008	2021-02-22 Mon 14:27:32 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Verification Results: Verification started: Mon Feb 22 14:27:32 2021 Verification finished: Mon Feb 22 14:28:30 2021 MD5 checksum: b5f74cf9fa7ea6e758ae9bc3da99a317 : verified SHA1 checksum: dc00b8ad6b552ae5108763f0918a9dc107d91b6e : verified

Eugene Image (Q)

Consultant Name: Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn
Client: Thomas K. Chappel
Auditee: Desktop image used by Eugene
Date/Time of Custody: 2021-03-08 Mon 16:50:49 UTC
Location of Seizure: Ambrosia

Description of Evidence		
Item #	**Quantity**	**Description of Item** (Model, Serial #, Condition, Marks, Scratches)
001-010	1	Unique description: eugene
		Evidence Number: 001-010
		Source data size: 61439 MB

Chain of Custody				
Item #	**Date/Time**	**Released by** (Signature & ID#)	**Received by** (Signature & ID#)	**Comments/Location**
001-010	2021-03-08 Mon 16:50:49 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Impact on integrity of data will be assessed through metadata tag analysis Impact of integrity on deleted artifacts to be assessed during recovery process

				Imaging of the device was done through AccessData® FTK® Imager 4.2.0.13
001-010	2021-03-08 Mon 16:50:49 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Case Information: Acquired using: ADI4.2.0.13 Case Number: 001 Evidence Number: 001-010 Unique description: eugene Examiner: Notes: desktop
001-010	2021-03-08 Mon 16:50:49 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Information for Q:\eugenepc: Physical Evidentiary Item (Source) Information: [Device Info] Source Type: Physical [Drive Geometry] Cylinders: 124,828 Heads: 16 Sectors per Track: 63 Bytes per Sector: 512 Sector Count: 125,827,200 [Physical Drive Information] Drive Interface Type: ide [Image] Image Type: VMWare Virtual Disk Source data size: 61439 MB Sector count: 125827200 [Computed Hashes]

				MD5 checksum: 54f8043c8d73b577846db6713c02210f SHA1 checksum: b57df8fc537112de075373922c7759a0755804fe
001-010	2021-03-08 Mon 16:50:49 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Information: Acquisition started: Mon Mar 8 16:50:49 2021 Acquisition finished: Mon Mar 8 17:17:51 2021 Segment list: Q:\eugenepc.001 Q:\eugenepc.002 Q:\eugenepc.003 Q:\eugenepc.004 Q:\eugenepc.005 Q:\eugenepc.006 Q:\eugenepc.007 Q:\eugenepc.008 Q:\eugenepc.009 Q:\eugenepc.010 Q:\eugenepc.011 Q:\eugenepc.012 Q:\eugenepc.013 Q:\eugenepc.014 Q:\eugenepc.015 Q:\eugenepc.016 Q:\eugenepc.017 Q:\eugenepc.018 Q:\eugenepc.019 Q:\eugenepc.020 Q:\eugenepc.021 Q:\eugenepc.022 Q:\eugenepc.023 Q:\eugenepc.024 Q:\eugenepc.025 Q:\eugenepc.026 Q:\eugenepc.027 Q:\eugenepc.028 Q:\eugenepc.029 Q:\eugenepc.030 Q:\eugenepc.031 Q:\eugenepc.032 Q:\eugenepc.033 Q:\eugenepc.034 Q:\eugenepc.035 Q:\eugenepc.036 Q:\eugenepc.037

				Q:\eugenepc.038 Q:\eugenepc.039 Q:\eugenepc.040 Q:\eugenepc.041
001-010	2021-03-08 Mon 17:17:52 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Verification Results: Verification started: Mon Mar 8 17:17:52 2021 Verification finished: Mon Mar 8 17:44:11 2021 MD5 checksum: 54f8043c8d73b577846db6713c02210f : verified SHA1 checksum: b57df8fc537112de075373922c7759a0755804fe : verified

Martin Image (P)

Consultant Name: Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn
Client: Thomas K. Chappel
Auditee: Desktop image used by Martin
Date/Time of Custody: 2021-03-08 Mon 15:02:38 UTC
Location of Seizure: Ambrosia

Description of Evidence		
Item #	**Quantity**	**Description of Item** (Model, Serial #, Condition, Marks, Scratches)
001-009	1	Unique description: martin_pc
		Evidence Number: 001-009
		Source data size: 61439 MB

Chain of Custody				
Item #	**Date/Time**	**Released by** (Signature & ID#)	**Received by** (Signature & ID#)	**Comments/Location**
001-009	2021-03-08 Mon 15:02:38 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Impact on integrity of data will be assessed through metadata tag analysis Impact of integrity on deleted

				artifacts to be assessed during recovery process Imaging of the device was done through AccessData® FTK® Imager 4.2.0.13
001-009	2021-03-08 Mon 15:02:38 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Case Information: Acquired using: ADI4.2.0.13 Case Number: 001 Evidence Number: 001-009 Unique description: martin_pc Examiner: Notes: desktop
001-009	2021-03-08 Mon 15:02:38 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Information for P:\martinpc: Physical Evidentiary Item (Source) Information: [Device Info] Source Type: Physical [Drive Geometry] Cylinders: 124,828 Heads: 16 Sectors per Track: 63 Bytes per Sector: 512 Sector Count: 125,827,200 [Physical Drive Information] Drive Interface Type: ide [Image] Image Type: VMWare Virtual Disk Source data size: 61439 MB Sector count: 125827200

				[Computed Hashes] MD5 checksum: 1a9b2879361d869adae4192b5c9d7525 SHA1 checksum: 57e960e20e043ed75a847ba88429bd76a4a44c15
001-009	2021-03-08 Mon 15:02:38 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Information: Acquisition started: Mon Mar 8 15:02:38 2021 Acquisition finished: Mon Mar 8 15:29:09 2021 Segment list: P:\martinpc.001 P:\martinpc.002 P:\martinpc.003 P:\martinpc.004 P:\martinpc.005 P:\martinpc.006 P:\martinpc.007 P:\martinpc.008 P:\martinpc.009 P:\martinpc.010 P:\martinpc.011 P:\martinpc.012 P:\martinpc.013 P:\martinpc.014 P:\martinpc.015 P:\martinpc.016 P:\martinpc.017 P:\martinpc.018 P:\martinpc.019 P:\martinpc.020 P:\martinpc.021 P:\martinpc.022 P:\martinpc.023 P:\martinpc.024 P:\martinpc.025 P:\martinpc.026 P:\martinpc.027 P:\martinpc.028 P:\martinpc.029 P:\martinpc.030 P:\martinpc.031 P:\martinpc.032 P:\martinpc.033 P:\martinpc.034

				P:\martinpc.035 P:\martinpc.036 P:\martinpc.037 P:\martinpc.038 P:\martinpc.039 P:\martinpc.040 P:\martinpc.041
001-009	2021-03-08 Mon 15:29:09 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Verification Results: Verification started: Mon Mar 8 15:29:09 2021 Verification finished: Mon Mar 8 15:54:54 2021 MD5 checksum: 1a9b2879361d869adae4192b5c9d7525 : verified SHA1 checksum: 57e960e20e043ed75a847ba88429bd76a4a44c15 : verified

Services Image (K)

Consultant Name: Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn
Client: Thomas K. Chappel
Auditee: Domain controller image used by services
Date/Time of Custody: 2021-02-08 Mon 21:36:30 UTC
Location of Seizure: Ambrosia

Description of Evidence		
Item #	**Quantity**	**Description of Item** (Model, Serial #, Condition, Marks, Scratches)
001-006	1	Unique description: services
		Evidence Number: 001-006
		Source data size: 61440 MB

Chain of Custody				
Item #	**Date/Time**	**Released by** (Signature & ID#)	**Received by** (Signature & ID#)	**Comments/Location**
001-006	2021-02-08 Mon 21:36:30	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava,	Impact on integrity of data will be assessed through metadata tag

	UTC		Supavit Pitakanrongphorn	analysis Impact of integrity on deleted artifacts to be assessed during recovery process Imaging of the device was done through AccessData® FTK® Imager 4.2.0.13
001-006	2021-02-08 Mon 21:36:30 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Case Information: Acquired using: ADI4.2.0.13 Case Number: 001 Evidence Number: 001-006 Unique description: services Examiner: Notes: domainController
001-006	2021-02-08 Mon 21:36:30 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Information for K:\srvSVC: Physical Evidentiary Item (Source) Information: [Device Info] Source Type: Physical [Drive Geometry] Cylinders: 16,383 Heads: 16 Sectors per Track: 63 Bytes per Sector: 512 Sector Count: 125,829,120 [Physical Drive Information] Drive Interface Type: ide [Image] Image Type: VMWare Virtual Disk

				Source data size: 61440 MB Sector count: 125829120 [Computed Hashes] MD5 checksum: fa3de17e72581a6c4f7756dc76f13957 SHA1 checksum: 8472ee3690417db629f40f02e8a9cddf548f51a8
001-006	2021-02-08 Mon 21:36:30 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Information: Acquisition started: Mon Feb 8 21:36:30 2021 Acquisition finished: Mon Feb 8 22:03:18 2021 Segment list: K:\srvSVC.001 K:\srvSVC.002 K:\srvSVC.003 K:\srvSVC.004 K:\srvSVC.005 K:\srvSVC.006 K:\srvSVC.007 K:\srvSVC.008 K:\srvSVC.009 K:\srvSVC.010 K:\srvSVC.011 K:\srvSVC.012 K:\srvSVC.013 K:\srvSVC.014 K:\srvSVC.015 K:\srvSVC.016 K:\srvSVC.017 K:\srvSVC.018 K:\srvSVC.019 K:\srvSVC.020 K:\srvSVC.021 K:\srvSVC.022 K:\srvSVC.023 K:\srvSVC.024 K:\srvSVC.025 K:\srvSVC.026 K:\srvSVC.027 K:\srvSVC.028 K:\srvSVC.029 K:\srvSVC.030 K:\srvSVC.031

				K:\srvSVC.032 K:\srvSVC.033 K:\srvSVC.034 K:\srvSVC.035 K:\srvSVC.036 K:\srvSVC.037 K:\srvSVC.038 K:\srvSVC.039 K:\srvSVC.040 K:\srvSVC.041
001-006	2021-02-08 Mon 22:03:18 UTC	Thomas K. Chappel	Abhi Nagalla, Pallavi Srivastava, Supavit Pitakanrongphorn	Image Verification Results: Verification started: Mon Feb 8 22:03:18 2021 Verification finished: Mon Feb 8 22:29:14 2021 MD5 checksum: fa3de17e72581a6c4f7756dc76f13957 : verified SHA1 checksum: 8472ee3690417db629f40f02e8a9cddf548f51a8 : verified

3.3 Confidence Level

The Admiralty Code (also known as the NATO System or NATO Intelligence Rating Scale) is a standardized evaluation system used in intelligence analysis to rate the reliability of a source and the credibility of reported information. The code combines a Letter (A–F) representing the source reliability and a Number (1–6) representing the information credibility to form a two-character rating (e.g., A1, B2, F6).

Source Reliability (A–F):

Evaluates the historical trustworthiness, technical competency, or capability of where the data originated

1. **A: Completely Reliable** – A source that consistently and over time has provided verifiable data, information, and intelligence. In other words, this is your top-shelf stuff that has a history of reliability. You ran malware through your internal sandbox and checked the hash against two or more vendors to verify.

2. **B: Usually Reliable** – Reputable threat intelligence feeds or intelligence shared in closed communities by researchers you've come to trust over time because they are mostly correct. So probably true.

3. **C: Fairly Reliable** – Open-source intelligence from sources, like that security researcher on X/Twitter who is often correct.

4. **D: Not Usually Reliable** – Unverified dark web chatter from a sketchy user on a forum who’s made past claims that sometimes turned out to be true.

5. **E: Unreliable** – Potentially compromised/deceptive sources with a history of making false claims. Think bots, part of a mis- or disinformation campaign.

6. **F: Reliability Cannot be Judged** – New or unknown sources, such as a new handle/user or tool you're not sure about or that you have not verified yet.

Information Credibility (1–6):

Evaluates the logical likelihood and degree of corroboration of the specific data point or claim, independent of the source.

1. **1: Credible and verified** – Multiple independent and trusted sources confirm that a ransomware strain is active and has been observed by other independent sources.

2. **2: Probably True** – Not yet confirmed, but consistent with other known threat intelligence that lines up with other known intel about the threat actor group you are investigating.

3. **3: Possibly True** – Not confirmed, somewhat consistent with other intelligence and fits the Modus Operandi of that hacktivist group, but it needs more digging and requires further verification.

4. **4: Doubtful** – A new zero-day that hasn't been confirmed yet. It may be plausibly true the current threat landscape and known activities, so this requires more investigation.

5. **5: Improbable based on current understanding** – For example, claims that don't match known tactics, techniques, and procedures (TTPs) of the threat actor.

6. **6: Truth Cannot be Judged** – A post about a massive breach with zero details to back it up, you are unable to verify the claim due to a lack of context or conflicting information. Also, we use this when we haven't assessed it yet.

3.4 Evidence Notes

In line with our objective, we focused on obtaining evidence to support our investigation of a potential leaker in the Republic of Ambrosia, leading to the implication of Miguel Ortega as a malicious actor during this investigation.

3.4.1 Ater Image (I)

3.4.1.1 Ater_N_0001 Image Processing

Steps	**Analysis**
1. Using Box, we downloaded the image to our device Ater_A_0001 2. Using Windows, we extracted the image Ater_A_0002 3. Using Ubuntu, we assembled image segments. Then, we computed the checksum of the resulting disk image Ater_A_0003 4. Using Autopsy, we created a new case in	1. The disk image was successfully downloaded, extracted, assembled, and imported to Autopsy. 2. We verified that the evidence we worked with was not modified since the time we received it from the chain of custody. 3. Confidence level: A1. The tools used are completely reliable in their own respective functions. The images are credible and verified through the chain of custody.

Autopsy and imported the assembled disk image Ater_A_0004	

3.4.1.2 Ater_N_0002 Image Information

Steps	Analysis
1. We checked documented information to obtain information about the disk Ater_A_0003 2. Using autopsy, we navigated to File Views > By Extension > Documents > Plain Text > f2064505.txt, f2096878.txt and export the files Ater_A_0005 Ater_E_0001 f2096878.txt Ater_E_0003 f2064505.txt	1. Owner: Ater 2. Types: Desktop 3. Operating System: IBM OS/2 Warp 4 4. Confidence level: A1. Autopsy is completely reliable as a forensic analysis tool. Multiple files within the image and the chain of custody also verified the information derived from this analysis.

3.4.1.3 Ater_N_0003 Email - Cheese It

Steps	Analysis
1. Using Autopsy, we navigated to: Data Artifacts > E-Mail Messages > Default ([Default]) > Default > f0147879.txt and export the file Ater_A_0006 Ater_E_0002 f0147879.txt	1. Time: 2021-02-03 Wed 10:36:37 UTC 2. To: chappel@ambrosia.it 3. From: ater@ambrosia.it 4. Subject: Cheese It 5. Ater is going to Disney 6. Confidence level: A2. Autopsy is completely reliable as a forensic analysis tool. The email has a certain level of expectation of truth and officiality, but no other sources are found to confirm the analysis.

3.4.1.4 Ater_N_0004 Strings extraction

Steps	Analysis
1. Using Ubuntu, we use strings command to extract any readable bytes from the image and store them in ater_strings.txt Ater_A_0007 Ater_E_0004 ater_strings.txt	1. All readable bytes were extracted successfully 2. Confidence level: A1. Ubuntu is completely reliable. The information is from a verified image.

3.4.1.5 Ater_N_0005 IP addresses

Steps	Analysis
1. Using ater_strings.txt, we found information regarding the device IP address Ater_A_0008 Ater_E_0004 ater_strings.txt	1. Address = 192.168.1.75 2. Subnet mask = 255.255.255.0 3. Gateway = 192.168.1.1 4. Hostname = ater 5. Domain name = ambrosia.it 6. Name server = 192.168.1.100 7. Confidence level: A2. Ubuntu is completely reliable. The information is not confirmed by other sources, but is probably true.

3.4.1.6 Ater_N_0006 Email Credentials

Steps	Analysis
1. Using ater_strings.txt, we found information regarding email credentials Ater_A_0009 Ater_E_0004 ater_strings.txt	1. Email: ater@ambrosia.it 2. Password: MeenGreen75 8. Confidence level: A2. Ubuntu is completely reliable. The information is not confirmed by other sources, but is probably true.

3.4.2 Chappel Image (H)

3.4.2.1 Chappel_N_0001 Image Processing

Steps	Analysis
1. Using Box, we downloaded the image to our device Chappel_A_0001 2. Using Windows, we extracted the image Chappel_A_0002 3. Using Ubuntu, we assembled image segments. Then, we computed the checksum of the resulting disk image Chappel_A_0003 4. Using Autopsy, we created a new case in	1. The disk image was successfully downloaded, extracted, assembled, and imported to Autopsy. 2. We verified that the evidence we worked with was not modified since the time we received it from the chain of custody. 3. Confidence level: A1. The tools used are completely reliable in their own respective functions. The images are credible and verified through the chain of custody.

Autopsy and imported the assembled disk image Chappel_A_0004	

3.4.2.2 Chappel_N_0002 Image Information

Steps	Analysis
1. We checked documented information to obtain information about the disk Chappel_A_0003 2. Using Autopsy, we navigated to Data Artifacts > Installed Programs Chappel_A_0005 3. Using Autopsy, we navigated to Data Artifacts > Web Accounts Chappel_A_0006 4. Using Autopsy, we navigated to Data Sources > chappel_1 Host > vol2 > cm-x86-14.1-r4 > data > misc > wifi > ipconfog.txt and exported the file Chappel_A_0011 Chappel_E_0045 ipconfig.txt	1. Owner: Chappel 2. Types: Phone 3. Operating System: Android 4. IP address: 192.168.1.35 5. Gateway: 192.168.1.1, 8.8.8.8 6. IP address: 192.168.1.36 7. Gateway: 192.168.1.1, 192.168.1.100 8. Confidence level: A1. Autopsy is completely reliable as a forensic analysis tool. All analyses are verified through multiple sources.

3.4.2.3 Chappel_N_0003 Email Database

Steps	Analysis
1. Using Autopsy, we navigated to File Views > File Types > By Extensions > Databases > EmailProfider.db > Message and export it as a csv file Chappel_A_0009 Chappel_E_0002 EmailProvider_db.csv 2. Using Autopsy, we navigated to File Views > File Types > By Extensions > Documents > HTML > [1-42].html and exported the files Chappel_A_0010 Chappel_E_0004 1.html Chappel_E_0005 2.html Chappel_E_0006 3.html Chappel_E_0007 4.html	1. While the Email Database contains only snippets of the message, the Email HTML files contain the entire body of the message. Thus, we were able to observe the metadata from the database, and the message body from the HTML files and consolidate them into a more complete analysis 2. There are multiple email threads as well as actors involved in this communication channel to be investigated further 3. Confidence level: A1. The image and Autopsy are completely reliable, and multiple sources in the image suggest that the evidence found is credible.

Chappel_E_0008 5.html Chappel_E_0009 6.html Chappel_E_0010 7.html Chappel_E_0011 8.html Chappel_E_0012 9.html Chappel_E_0013 10.html Chappel_E_0014 11.html Chappel_E_0015 12.html Chappel_E_0016 13.html Chappel_E_0017 14.html Chappel_E_0018 15.html Chappel_E_0019 16.html Chappel_E_0020 17.html Chappel_E_0021 18.html Chappel_E_0022 19.html Chappel_E_0023 20.html Chappel_E_0024 21.html Chappel_E_0025 22.html Chappel_E_0026 23.html Chappel_E_0027 24.html Chappel_E_0028 25.html Chappel_E_0029 26.html Chappel_E_0030 27.html Chappel_E_0031 28.html Chappel_E_0032 29.html Chappel_E_0033 30.html Chappel_E_0034 31.html Chappel_E_0035 32.html Chappel_E_0036 33.html Chappel_E_0037 35.html Chappel_E_0038 36.html Chappel_E_0039 37.html Chappel_E_0040 38.html Chappel_E_0041 39.html Chappel_E_0042 40.html Chappel_E_0043 41.html Chappel_E_0044 42.html 3. Using the email database and email HTML files, we compiled the evidence into a single spreadsheet using Google Sheets for a more readable format Chappel_E_0003 EmailProvider_db M…	

3.4.2.4 Chappel_N_0004 Email Possible Leaker

Steps	**Analysis**

1. Using the email Google Sheets we compiled from existing evidence, we investigated the Posible Leaker thread Chappel_E_0003 EmailProvider_db M…	1. Mon 1 Feb 2021 20:31:59 Chappel informed Ater that Avalon already knows about the plans for dumping on the banana market Chappel is suspecting a possible leaker. Yet, the leaker is unknown 2. Mon 1 Feb 2021 20:39:21 Ater instructed Chappel to hire a team of forensics experts 3. Mon 1 Feb 2021 21:36:44 Ater instructed Chappel to continue with the plan and to deny all accusations. 4. Tue 2 Feb 2021 19:32:18 Chappel told Ater that this may be a good idea, and it might be the chance for Ater to sneak in the vacation time to go to Disney 5. Confidence level: A1. The image is completely reliable through the chain of custody, and Autopsy is completely reliable as a digital forensics tool. The evidence is credible and verified through multiple sources within the image, as well as the secure nature of an email.

3.4.2.5 Chappel_N_0005 Email Dumping Program Update

Steps	**Analysis**
1. Using the email Google Sheets we compiled from existing evidence, we investigated the Dumping Program Update thread Chappel_E_0003 EmailProvider_db M…	1. Mon 1 Feb 2021 19:04:16 Roach informed the cabinet that they have already arranged for subsidy payments to be made discreetly to all major local banana producers in exchange for them lowering their prices, with the exception of Y&S Farms, as they are the only local producer with a foreign interest 2. Mon 1 Feb 2021 20:17:33 Dudley expresses opposition to the dumping program. She says that she is already fielding complaints from Avalon 3. Mon 1 Feb 2021 21:45:51

	Chappel informed the cabinet to stick to the plan and want to be informed of any new developments immediately 4. Tue 2 Feb 2021 18:03:25 Roach informed the cabinet that the first shipments of discounted bananas have been sold. Y&S farms complained about unfair subsidies; they do not care what the government does, as long as they get a cut of the action 5. Tue 2 Feb 2021 18:31:50 Barnes informed Roach that the primary target is Avalon banana production. Due to Fusarium TR4, it is unlikely for Avalon to resume production quickly. This is a chance to capitalize on the broader market. The primary goal should be to purchase so much control over Avalon agriculture. 6. Confidence level: A1. The image is completely reliable through the chain of custody, and Autopsy is completely reliable as a digital forensics tool. The evidence is credible and verified through multiple sources within the image, as well as the secure nature of an email.

3.4.2.6 Chappel_N_0006 Email Proposal: Banana Dumping Program

Steps	**Analysis**
1. Using the email Google Sheets we compiled from existing evidence, we investigated the Proposal: Banana Dumping Program thread Chappel_E_0003 EmailProvider_db M…	1. Sun 31 Jan 2021 23:12:25 Chappel informed Ater that the secretaries are discussing about a plan to subsidize banana exports. Roach and Barnes support the plan. Dudley opposes the plan. Snyder thinks it's worth trying to get a trade deal with Avalon. 2. Sun 31 Jan 2021 23:17:36 Ater approves the plan 3. Confidence level: A1. The image is completely reliable through the chain of custody, and Autopsy is completely reliable as a digital forensics tool. The evidence is credible and verified through multiple sources within the image, as well as the secure nature

	of an email.

3.4.2.7 Chappel_N_0007 Email Proposal: Banana Market Share Prioritization

Steps	**Analysis**
1. Using the email Google Sheets we compiled from existing evidence, we investigated the Proposal: Banana Market Share Prioritization thread Chappel_E_0003 EmailProvider_db M…	1. Sun 31 Jan 2021 21:31:57 Roach informed the cabinet that the mainland is losing banana production due to fungus spores. Thus, she proposes to subsidize banana farmers, undercut Avalon banana farms, then overtake Avalon banana farms before banana prices shoot up 2. Confidence level: A1. The image is completely reliable through the chain of custody, and Autopsy is completely reliable as a digital forensics tool. The evidence is credible and verified through multiple sources within the image, as well as the secure nature of an email.

3.4.2.8 Chappel_N_0008 Email [Mattermost] administrator invited you to join Cabinet Team

Steps	**Analysis**
1. Using the email Google Sheets we compiled from existing evidence, we investigated the [Mattermost] administrator invited you to join Cabinet Team thread Chappel_E_0003 EmailProvider_db M…	1. Sun 31 Jan 2021 20:07:27 Chappel was invited to the Mattermost server 2. Confidence level: A1. The image is completely reliable through the chain of custody, and Autopsy is completely reliable as a digital forensics tool. The evidence is credible and verified through multiple sources within the image, as well as the secure nature of an email.

3.4.2.9 Chappel_N_0009 Email New Job Posting

Steps	**Analysis**
1. Using the email Google Sheets we compiled from existing evidence, we investigated the New Job Posting thread Chappel_E_0003 EmailProvider_db M…	1. Mon 1 Feb 2021 21:12:50 Chappel informed Montoya that he needs HR to start recruiting a team of forensic investigators 2. Confidence level: A1. The image is completely reliable through the chain of custody, and Autopsy is completely reliable as a digital forensics tool. The evidence is

	credible and verified through multiple sources within the image, as well as the secure nature of an email.

3.4.2.10 Chappel_N_0010 Email Proposal Approved

Steps	**Analysis**
1. Using the email Google Sheets we compiled from existing evidence, we investigated the Proposal Approved thread Chappel_E_0003 EmailProvider_db M…	1. Sun 31 Jan 2021 23:20:07 Chappel informed the cabinet that Ater has approved the plan 2. Confidence level: A1. The image is completely reliable through the chain of custody, and Autopsy is completely reliable as a digital forensics tool. The evidence is credible and verified through multiple sources within the image, as well as the secure nature of an email.

3.4.2.11 Chappel_N_0011 Email POLICY REMINDER - COMPLIANCE IS MANDATORY

Steps	**Analysis**
1. Using the email Google Sheets we compiled from existing evidence, we investigated the POLICY REMINDER - COMPLIANCE IS MANDATORY thread Chappel_E_0003 EmailProvider_db M…	1. Sun 31 Jan 2021 20:09:52 Chappel informed the cabinet that all work-related written communication must be conducted on Mattermost or via email. 2. Confidence level: A1. The image is completely reliable through the chain of custody, and Autopsy is completely reliable as a digital forensics tool. The evidence is credible and verified through multiple sources within the image, as well as the secure nature of an email.

3.4.2.12 Chappel_N_0012 Email UPDATE

Steps	**Analysis**
1. Using the email Google Sheets we compiled from existing evidence, we investigated the UPDATE thread Chappel_E_0003 EmailProvider_db M…	1. Tue 2 Feb 2021 18:37:58 Roach informed the cabinet that initial reports from two farms on different parts of the island show blackened bananas wilting 2. Tue 2 Feb 2021 18:43:55 Roach informed the cabinet that both farms are part of Y&S holdings, so it's very likely

	that it could be Fulsarium from the mainland 3. Confidence level: A1. The image is completely reliable through the chain of custody, and Autopsy is completely reliable as a digital forensics tool. The evidence is credible and verified through multiple sources within the image, as well as the secure nature of an email.

3.4.2.13 Chappel_N_0013 Email Containment

Steps	**Analysis**
1. Using the email Google Sheets we compiled from existing evidence, we investigated the Containment thread Chappel_E_0003 EmailProvider_db M…	1. Tue 2 Feb 2021 19:03:15 Chappel instructed Snyder to make sure the area is cordoned off, and anyone who asks is detained. 2. Tue 2 Feb 2021 19:04:42 Snyder confirms that internal police will handle it 3. Confidence level: A1. The image is completely reliable through the chain of custody, and Autopsy is completely reliable as a digital forensics tool. The evidence is credible and verified through multiple sources within the image, as well as the secure nature of an email.

3.4.3 Gauthier Image (J)

3.4.3.1 Gauthier_N_0001 Image Processing

Steps	**Analysis**
1. Using Box, we downloaded the image to our device Gauthier_A_0001 2. Using Windows, we extracted the image Gauthier_A_0002 3. Using Ubuntu, we assembled image segments. Then, we computed the checksum of the resulting disk image Gauthier_A_0003	1. The disk image was successfully downloaded, extracted, assembled, and imported to Autopsy. 2. We verified that the evidence we worked with was not modified since the time we received it from the chain of custody. 3. Confidence level: A1. The tools used are completely reliable in their own respective functions. The images are credible and verified through the chain of custody.

4. Using Autopsy, we created a new case in Autopsy and imported the assembled disk image Gauthier_A_0004	

3.4.3.2 Gauthier_N_0002 Image Information

Steps	**Analysis**
1. We checked documented information to obtain information about the disk Gauthier_A_0003 2. Using Autopsy, we navigated to Data Artifacts > Operating System Information Gauthier_A_0007 3. Using Autopsy, we navigated to Data Sources > gauthier_1 Host > gauthier > vol3 > Windows > System32 > config > SYSTEM and exported the file. We investigated for IP address information Gauthier_A_0010 Gauthier_E_0010 SYSTEM	1. Owner: richard 2. Types: Desktop 3. Operating System: Windows 10 Education 4. IP address: 192.168.1.31 5. Subnet mask: 255.255.255.0 6. Default gateway: 192.168.1.1 7. Name server: 192.168.1.100 8. DHCP server: 255.255.255.255 9. Confidence level: A1. Autopsy is completely reliable as a forensic analysis tool. All analyses are verified through multiple sources.

3.4.3.3 Gauthier_N_0003 Trillian diannegauthier

Steps	**Analysis**
1. Using Autopsy, we navigated to Data Sources > gauthier_1 Host > gauthier > vol3 > Users > richard.AMBROSIA > AppData > Roaming > Trillian > users > richardbgauthier > logs > ASTRA > Query > diannegauthier.log Gauthier_A_0008 2. Using Autopsy, we exported the file for investigation Gauthier_E_0008 diannegauthier.log	1. Mon Feb 01 00:42:14 2021 UTC Gauthier is telling who seems to be his wife that he will be going home late 2. Confidence level: A2. Autopsy is completely reliable as a forensic analysis tool. The information is not verified, but the information is probably true.

3.4.3.4 Gauthier_N_0004 Trillian mollieschmitt

Steps	**Analysis**

1. Using Autopsy, we navigated to Data Sources > gauthier_1 Host > gauthier > vol3 > Users > richard.AMBROSIA > AppData > Roaming > Trillian > users > richardbgauthier > logs > ASTRA > Query > mollieschmitt.log Gauthier_A_0009 3. Using Autopsy, we exported the file for investigation 2. Gauthier_E_0009 mollieschmitt.log	1. Mon Feb 01 00:35:33 2021 UTC 2. Gauthier is cheating on his wife, Dianne, with Molly. Molly wants Gauthier to leave Diane for her, but Gauthier refuses to break up his family. 3. Confidence level: A2. Autopsy is completely reliable as a forensic analysis tool. The information is not verified, but the information is probably true.

3.4.3.5 Gauthier_N_0005 Mattermost

Steps	**Analysis**
1. Using Autopsy, we navigated to Data Sources > gauthier_1 Host > gauthier > vol3 > Users > richard.AMBROSIA > AppData > Local > Microsoft > Edge > User Data > Default > Cache > data_1 Gauthier_A_0011 Gauthier_E_0011 data_1 2. Using we wrote python script to extract relevant information and store in a csv file Gauthier_E_0012 mattermost.py Gauthier_E_0013 mattermost.csv	1. Channel: Town Square 2. Team: Cabinet 3. 2021-02-02 Tue 19:33:58 UTC Eugene informs the cabinet that Avalon announced they are screening everything coming onto the island, and they are the only producer in the region unaffected by the blight. Eugene proposes poisoning Avalon bananas with blights. Julia opposes the proposal. 4. 2021-02-03 Wed 00:45:44 UTC Ater approves the operation to poison Avalon bananas 5. 2021-02-03 Wed 16:46:59 UTC Ambrosia lost contact with the team sent to Avalon. They were ambushed by forces waiting at the island. Cabinet has to assume that they have been captured. All cabinet members are instructed to meet at Ater's compound. 6. Confidence level: A2. Autopsy and python are completely reliable tools in handling data. The information are not confirmed from other independent source, but are probably true.

3.4.4 Jeanette Image (G)

3.4.4.1 Jeanette_N_0001 Image Processing

Steps	Analysis
1. Using Box, we downloaded the image to our device Jeanette_A_0001 2. Using Windows, we extracted the image Jeanette_A_0002 3. Using Ubuntu, we assembled image segments. Then, we computed the checksum of the resulting disk image Jeanette_A_0003 4. Using Autopsy, we created a new case in Autopsy and imported the assembled disk image Jeanette_A_0004	1. The disk image was successfully downloaded, extracted, assembled, and imported to Autopsy. 2. We verified that the evidence we worked with was not modified since the time we received it from the chain of custody. 3. Confidence level: A1. The tools used are completely reliable in their own respective functions. The images are credible and verified through the chain of custody.

3.4.3.2 Jeanette_N_0002 Image Information

Steps	Analysis
1. We checked documented information to obtain information about the disk Jeanette_A_0003 2. Using Autopsy, we navigated to Data Artifacts > Operating System Information Jeanette_A_0007 3. Using Autopsy, we navigated to Data Sources > jeanhassan Host > jeanhassan > vol3 > Windows > System32 > config > SYSTEM and exported the file. We investigated for IP address information Jeanette_A_0008 Jeanette_E_0006 SYSTEM	1. Owner: jean 2. Operating System: Windows 10 Education 3. IP address: 192.168.1.20 4. Subnet mask: 255.255.255.0 5. Default gateway: 192.168.1.1 6. Name server: 192.168.1.100 7. DHCP server: 255.255.255.255 8. Confidence level: A1. Autopsy is completely reliable as a forensic analysis tool. All analyses are verified through multiple sources.

3.4.3.3 Jeanette_N_0003 Download Folder

Steps	Analysis

1. Using Autopsy, we navigated to Data Sources > jeanhassan_1 Host > jeanhassan > vol 3 > Users > jean.AMBROSIA > Downloads > securityupdate and exported securityupdate.zip, securityupdate.exe, and update2.exe Jeanette_A_0009 Jeanette_E_0007 securityupdate.exe Jeanette_E_0008 update2.exe Jeanette_E_0011 securityupdate.zip 2. The securityupdate.exe and update2.exe triggered Windows Security Defender, labeling them as a trojans Jeanette_A_0010 Jeanette_E_0009 Trojan_Win64_Meter…	1. 2021-01-31 Sun 22:38:11 UTC securityupdate.zip was downloaded to the device, containing securityupdate.exe Downloaded from: https://mail.ambrosia.it/owa/service.svc/s/GetFileAttachment?id=AAMkAGMwODgwNjYzLTIzN2MtNDRiMS1iNWU0LTM0ZmVhNjc4ZDAzNwBGAAAAAAAeQW%2Bc3qmgQpVwu0yqM6yBBwDgDE2cB0zLQ6xpTL0h87EYAAAAAAEMAADgDE2cB0zLQ6xpTL0h87EYAAAEahyZAAABEgAQAPy4pQs9G5hOj1edib3H8RY%3D&X-OWA-CANARY=xkN-EYYy5kKbfTFTYv0t_8A4f6k4xtgIBZ293ce3JpSt503lT8wMupZl5rYvp6ubtEY_gP1VFd0. 2. 2021-01-31 Sun 22:39:09 UTC securityupdate.exe was extracted to the device 3. There were 3 running instances of the program: 2021-01-31 Sun 22:40:02 UTC 2021-02-03 Wed 17:20:38 UTC 2021-02-03 Wed 17:24:11 UTC 4. 2021-02-01 Mon 15:00:06 UTC update2.exe was created in the device 5. There were 5 running instances of the program: 2021-02-01 Mon 15:00:18 UTC 2021-02-01 Mon 15:01:08 UTC 2021-02-01 Mon 15:01:39 UTC 2021-02-03 Wed 17:26:48 UTC 2021-02-03 Wed 17:29:32 UTC 6. Malwares were delivered through email and downloaded onto the device. 7. Confidence level: B2. Autopsy is completely reliable, while Windows Defender has some error rate in detecting malicious programs. The information are not confirmed by different independent source, but is probably true.

3.4.3.4 Jeanette_N_0004 securityupdate.exe Static Analysis

Steps	Analysis
1. Using PEstudio, we conducted static analysis on securityupdate.exe Jeanette_A_0011	1. The executable receives a virustotal score of 62/70, 0/70 being safe 2. Imports: KERNEL32.dll: VirtualAlloc, ExitProcess 3. Confidence level: B2. PEstudio is a third-party software; the reputation is unsure, but is usually reliable. The information are not confirmed by different independent source, but is probably true.

3.4.3.5 Jeanette_N_0005 securityupdate.pf prefetch analysis

Steps	Analysis
1. Using Autopsy, we navigated to Data Artifacts > Run Programs, and exported SECURITYUPDATE.EXE-1C0E1048.pf Jeanette_A_0012 Jeanette_E_0012 SECURITYUPDATE… 2. Using WinPrefetchViewer, we investigated the prefetch file Jeanette_A_0013 3. We researched on all of relevant references Jeanette_E_0013 Definition of CMD.E… Jeanette_E_0014 ws2_32.dll - What is … Jeanette_E_0015 rpcrt4.dll - What is rp… Jeanette_E_0016 mswsock.dll Window… Jeanette_E_0017 sechost.dll Windows …	1. 2021-01-31 Sun 22:40:12 UTC Securityupdate.exe was first executed 2. References: AHCACHE.SYS, APPHELP.DLL, CMD.EXE, KERNEL32.DLL, KERNELBASE.DLL, LOCALE.NLS, MSWSOCK.DLL, NTDLL.DLL, RPCRT4.DLL, SECHOST.DLL, SECURITYUPDATE.EXE, SYSMAIN.SDB, WS2_32.DLL 3. The executable has network and process-creation behavior. The executable is likely a remote network connection establishment program. 4. Confidence level: B2. WinPrefetchViewer is a third-party software; the reputation is unsure, but is usually reliable. The information are not confirmed by different independent source, but is probably true.

3.4.3.6 Jeanette_N_0006 update2.exe Static Analysis

Steps	Analysis
1. Using PEstudio, we conducted static analysis on update2.exe Jeanette_A_0014	1. The executable receives a virustotal score of 63/70, 0/70 being safe 2. Imports: KERNEL32.dll: VirtualAlloc,

	ExitProcess 3. A string “tcp://192.168.2.42:5555” was also found in update2.exe, which may indicate the attacker address 4. Confidence level: B2. PEstudio is a third-party software; the reputation is unsure, but is usually reliable. The information are not confirmed by different independent source, but is probably true.

3.4.3.7 Jeanette_N_0007 update2.exe Prefetch Analysis

Steps	**Analysis**
1. Using Autopsy, we navigated to Data Artifacts > Run Programs, and exported UPDATE2.EXE-7783C67A.pf Jeanette_A_0015 Jeanette_E_0018 UPDATE2.EXE-7783… 2. Using WinPrefetchViewer, we investigated the prefetch file Jeanette_A_0016	1. 2021-02-01 Mon 15:00:28 UTC update2.exe was first executed 2. The executable contains substantially more references that securityupdate.exe, indicating that the executable may have more capabilities. 3. The executable is likely the attack payload and contain the main part of the attack vector. 4. Confidence level: B2. WinPrefetchViewer is a third-party software; the reputation is unsure, but is usually reliable. The information are not confirmed by different independent source, but is probably true.

3.4.3.8 Jeanette_N_0008 Running Programs

Steps	**Analysis**
1. Using Autopsy, we navigated to Data Artifacts > Run Programs to investigate what processes are running during the time of malware execution Jeanette_E_0017	1. 2021-01-31 Sun 22:40:02 UTC SECURITYUPDATE.EXE 2. 2021-01-31 Sun 22:40:03 UTC CMD.EXE 3. 2021-02-01 Mon 14:33:10 UTC WHOAMI.EXE 4. 2021-02-01 Mon 14:59:12 UTC CURL.EXE

	5. 2021-02-01 Mon 15:00:18 UTC UPDATE2.EXE 6. The programs indicate that securityupdate.exe may be attempting to identify the host before downloading and executing update.exe 7. Confidence level: A2. Autopsy are completely reliable tools in handling data. The information are not confirmed from other independent source, but are probably true.

5.2.4.9 Jeanette_N_0009 curl.exe prefetch

Steps	**Analysis**
1. Using Autopsy, we navigated to Data Artifacts > Run Programs, and exported CURL.EXE-9AA757BE.pf Jeanette_A_0017 Jeanette_E_0019 CURL.EXE-9AA757… 2. Using WinPrefetchViewer, we investigated the prefetch file Jeanette_A_0018	1. curl.exe has references to update2.exe, indicating that update2.exe may have been installed through curl. Thus, it is likely that securityupdate.exe executes curl to install update2.exe, which is the attack method. 2. Confidence level: B2. WinPrefetchViewer is a third-party software; the reputation is unsure, but is usually reliable. The information are not confirmed by different independent source, but is probably true.

3.4.5 Mattermost Image (L)

3.4.5.1 MatterMost_N_0001 Image Processing

Steps	**Analysis**
1. Using Box, we downloaded the image to our device 5.2.5.1 Mattermost_A_0001 Image Download 2. Using Windows, we extracted the image. 5.2.5.2 Mattermost_A_0002 Image Extraction 3. Using Ubuntu, we assembled image segments. Then, we computed the checksum of the resulting disk image. 5.2.5.3 Mattermost_A_0003 Image Verification 4. Using Autopsy, we created a new case in	1. The disk image was successfully downloaded, extracted, assembled, and imported to Autopsy. 2. We verified that the evidence we worked with was not modified since the time we received it from the chain of custody. 3. Confidence level: A1. The tools used are completely reliable in their own respective functions. The images are credible and verified through the chain of custody.

Autopsy and imported the assembled disk image. 5.2.5.4 Mattermost_A_0004 Autopsy Case	

3.4.5.2 MatterMost_N_0002 MySQL Version

Steps	Analysis
1. Using Autopsy, we navigated to Data Sources > mmsrv.raw_1.host> mmsrv.raw > var > lib > mysql > mysql_upgrade_info 5.2.5.5 MatterMost_A_0005 mysql_upgrade_info	1. Examination of this file enabled us to determine the storage format of the MatterMost server logs. The MySQL version being used was 5.7.33. This was necessary to discover to understand the format the logs related to mattermost were being stored in. 2. Confidence level A2: Autopsy and MySQL are completely reliable. The information are not confirmed by other independent sources, but are probably true.

3.4.5.3 MatterMost_N_0003 Channels

Steps	**Analysis**
1. Using Autopsy, we navigated to Data Sources > mmsrv.raw_1.host> mmsrv.raw > var > lib > mysql > mattermost. In the mattermost folder the Posts.ibd, Posts.frm,Channels.frm, Channels.ibd, 587,539-Users.ibd, and Users.frm.were extracted and placed onto a folder called Forensic Evidence in the forensic cloud environment. 5.2.5.6 MatterMost_A_0006 Posts.idb, Posts.frm, Public Channels.ibd, and Public Channels.frm 2. To examine these files a fresh installation of MySQL 5.7 was needed. To acquire this version with limited dependencies, a Ubuntu 18.04 virtual machine was created and docker was installed using sudo docker pull mysql:5.7.33. Additionally, the necessary files were downloaded from the cloud environment. 5.2.5.7 MatterMost_A_0007 MySQL 5.7.33 Docker Container Installation 5.2.5.8 MatterMost_A_0008 Downloaded Files	1. In the MatterMost Server there were 3 channels that were discovered – Town-Square, Off Topic, and Bossmanchat. 2. Confidence level A2: Autopsy and MySQL are completely reliable. The information are not confirmed by other independent sources, but are probably true.

3. The I used the command sudo docker run --name mysql57 \ -e MYSQL_ROOT_PASSWORD=password \ -p 3306:3306 \ -d mysql:5.7.33 to run and enter the mysql container.

   5.2.5.9 MatterMost_A_0009 Run Docker Container

4. Then the recovery database was created using the commands CREATE DATABASE recovery; USE recovery.

   5.2.5.10 MatterMost_A_0010 Create Database

5. Then we recreated the Channels, Posts, and Users table structure using the commands below. This command was recovered using this online tool https://convert.guru/converter :

   5.2.5.11 MatterMost_A_0011 Create Channels Table Structure

   5.2.5.12 MatterMost_A_0012 Create Posts Table Structure

   5.2.5.13 MatterMost_A_0013 Create Users Table Structure

6. Then empty tablespaces were discarded to add the recovered .ibd files using the command.

   5.2.5.14 MatterMost_A_0014 Discard Empty Tablespace

7. Then we copied the ibd files into the Docker database with the commands:

```
sudo docker cp ~/Downloads/Channels.ibd \
  mysql57:/var/lib/mysql/recovery/Channels.ibd

sudo docker cp ~/Downloads/Posts.ibd \
  mysql57:/var/lib/mysql/recovery/Posts.ibd

sudo docker cp \
~/Downloads/'587,539-Users.ibd' \
mysql57:/var/lib/mysql/recovery_test/Users.ibd
```

8. Then we modified ownership to allow access within the Docker container using the command:

```
 sudo docker exec -u root mysql57 \
  chown mysql:mysql \
  /var/lib/mysql/recovery/PublicChannels.ibd \
```

/var/lib/mysql/recovery/Posts.ibd sudo docker exec -u root mysql57 \ chown mysql:mysql /var/lib/mysql/recovery_test/Users.ibd 9. Then we reopened MySQL using the command sudo docker exec -it mysql57 mysql -uroot -p. 10. Then we entered USE recovery; to get into the recovery database. Then the respective tablespaces were imported with the commands: ALTER TABLE Channels IMPORT TABLESPACE; ALTER TABLE Posts IMPORT TABLESPACE; ALTER TABLE Users IMPORT TABLESPACE; 11. To view the recovered messages the select statement below was used within MySQL. SELECT FROM_UNIXTIME(CreateAt / 1000) AS CreateAt, UserId, ChannelId, Message FROM Posts ORDER BY CreateAt; 5.2.5.15 MatterMost_A_0015 Posts Table 12. To view the recovered channel names the select statement below was used within MySQL. SELECT * FROM Channels LIMIT 10; 5.2.5.16 MatterMost_A_0016 Channels Table 13. To view the recovered users and user IDs we used the SQL query below: SELECT Id, Username, FirstName, LastName, Email FROM Users; 5.2.5.17 MatterMost_A_0017 Users Table	

3.4.5.4 MatterMost_N_0004 Posts Table to csv.

Steps	**Analysis**
1. First we ran this command to extract the necessary data. 5.2.5.18 MatterMost_A_0018 Command to view Recovered Posts Messages as a readable	1. Examining the Posts Table, we discovered that there was a poll created by Chappel titled "What should we do per the proposal?" at 2021-01-31 22:59:59 UTC. The options were: "Yes, subsidize the exports" and "No, focus on

CSV 2. Then we copied the created file and put it inside the home directory of the virtual machine. sudo docker cp \ mysql57:/var/lib/mysql-files/posts_all_columns.csv \ /home/vamp/posts_all_columns.csv 3. Then we changed the ownership to our user. sudo chown vamp:vamp /home/vamp/posts_all_columns.csv 4. Then the file was uploaded to the Forensic Cloud Environment. MatterMost_E_0002	the trade agreement” 2. Between 2021-01-31 23:19:37 UTC – 2021-02-02 19:47:36 UTC several members of the Town Square channel discussed potentially intentionally sabotaging Avalon. With user Julia highlighting the potential serious consequences of such actions at 2021-02-02 19:37:45 UTC. 3. A second poll was created by chappel at 2021-02-02 19:48:17 UTC titled “are we going ahead” with the options “Yes” or “No”. However, the members are unable to come up with a satisfying conclusion as noted by user Chappel at 2021-02-02 19:52:52 who says “well, that was no help at all.” 4. At 2021-02-02 19:53:05 user Chappel says “I'll talk to Dick and let you know what we'll be doing shortly.” 5. Between 2021-02-03 0:35:37 – 2021-02-03 0:45:09 Dick and Chappel discuss what to do in Channel BossmanChat. User Ater says “do it, pull the trigger” 6. At 2021-02-03 0:45:44 Chappel returns to the Town Square channel and says “I've spoken with Dick, he's authorizing the operation” and tells Martin to “... make sure it's done before sunrise” at 2021-02-03 0:45:52. To which Martin agrees at 2021-02-03 0:46:33. 7. At 2021-02-03 16:47:39 user Martin reports that “we have to assume they've been captured [by avalon]” 8. User Chappel says “All cabinet members are to meet IMMEDATELY at dick's compound” at 2021-02-03 16:49:04. Then instructs members of the Town Square Channel to not speak to anyone else (2021-02-03 16:49:08) and that they will be escorting other guests (2021-02-03 16:49:19). 9. Confidence level A2: Autopsy and MySQL are completely reliable. The information are not confirmed by other independent sources, but are probably true.

3.4.5.5 MatterMost_N_0005 PDFs

Steps	**Analysis**

1. We navigated to File Views > By Extension > Documents > PDF 5.2.5.19 MatterMost_A_0019 Classified.pdf 5.2.5.20 MatterMost_A_0020 Unclassified.pdf 5.2.5.21 MatterMost_A_0021 Top secret.pdf 5.2.5.22 MatterMost_A_0022 Secret.pdf 5.2.5.23 MatterMost_A_0023 Confidential.pdf 5.2.5.24 MatterMost_A_0024 Cover Page 5.2.25 MatterMost_A_0025 Files	1. These were not deemed to have high value as they did not contain any additional information. 2. Confidence level A2: Autopsy and MySQL are completely reliable. The information are not confirmed by other independent sources, but are probably true.

3.4.6 Julia Image (M)

3.4.6.1 Julia_N_0001 Image Processing

Steps	**Analysis**
1. Using Box, we downloaded the image to our device Julia_A_0001 2. Using Windows, we extracted the image Julia_A_0002 3. Using Ubuntu, we assembled image segments. Then, we computed the checksum of the resulting disk image Julia_A_0003 4. Using Autopsy, we created a new case in Autopsy and imported the assembled disk image Julia_A_0004	1. The disk image was successfully downloaded, extracted, assembled, and imported to Autopsy. 2. We verified that the evidence we worked with was not modified since the time we received it from the chain of custody. 3. Confidence level: A1. The tools used are completely reliable in their own respective functions. The images are credible and verified through the chain of custody.

3.4.6.2 Julia_N_0002 Device Information

Steps	**Analysis**
1. We checked documented information to obtain information about the disk	1. Owner: julia

Julia_A_0003 2. Using Autopsy, we navigated to Data Artifacts > Operating System Information Julia_A_0005 3. Using Autopsy, we navigated to Data Sources > julia_pc_1 Host > julia_pc > vol3 > Windows > System32 > config > SYSTEM and exported the file. We investigated for IP address information Julia_A_0006 Julia_E_0002 SYSTEM	2. Types: Desktop 3. Operating System: Windows 10 Education 4. IP address: 192.168.1.10 5. Subnet mask: 255.255.255.0 6. Default gateway: 192.168.1.1 7. Name server: 192.168.1.100 8. DHCP server: 255.255.255.255 9. Confidence level: A1. Autopsy is completely reliable as a forensic analysis tool. All analyses are verified through multiple sources.

3.4.6.3 Julia_N_0003 Trillian Log

Steps	**Analysis**
1. Using Autopsy, we navigated to Data Sources > julia_pc_1 Host > julia_pc > vol3 > Users > julia.AMBROSIA > AppData > Roaming > Trillian > users > juliadudley > logs > ASTRA > Query > farm3rjohn.log 2. Using Autopsy, we exported the file for investigation Julia_A_0007 Julia_E_0001 farm3rjohn.log	1. Mon Feb 01 19:18:20 2021 UTC Farm3rjohn (John) confronts Julia, saying he's been generous to her and is angry that her friends are "dumping" on his business. Julia says she's on his side and voted against it but has little control, since "what Dick says goes," and John replies that she needs to do more than try or find a new place to live, because he can't keep funding her housing if his business is hurting. 2. Confidence level: A2. Autopsy is completely reliable as a forensic analysis tool. The information is not verified, but the information is probably true.

3.4.6.4 Julia_N_0004 To-Do List

Steps	**Analysis**
1. Using Autopsy, we navigated to Data Sources > julia_pc_1 Host > julia_pc > vol3 > Users > julia.AMBROSIA > Documents > state 2. Using Autopsy, we exported the file for investigation Julia_A_0008 Chappel_E_0003 EmailProvider_db M…	1. A to-do list called to-do.txt was created, with the text "fix complaint forms". This was accompanied by a file called complaints.jpg, which is a complaint form template. This aligns with her statement of 'receiving many complaints' (as mentioned in the emails on the Chappel drive). 2. Confidence level: A1. Autopsy is completely reliable as a forensic analysis tool. All

	analyses are verified through multiple sources.

3.4.7 Eleanor Image (N)

3.4.7.1 Eleanor_N_0001 Image Processing

Steps	Analysis
1. Using Box, we downloaded the image to our device Eleanor_A_0001 Image Download 2. Using Windows, we extracted the image. Eleanor_A_0002 Image Extract 3. Using Ubuntu, we assembled image segments. Then, we computed the checksum of the resulting disk image. Eleanor_A_0003 Image Verification 4. Using Autopsy, we created a new case in Autopsy and imported the assembled disk image. Eleanor_A_0004 Autopsy Case	1. The disk image was successfully downloaded, extracted, assembled, and imported to Autopsy. 2. We verified that the evidence we worked with was not modified since the time we received it from the chain of custody. 3. Confidence level: A1. The tools used are completely reliable in their own respective functions. The images are credible and verified through the chain of custody.

3.4.7.2 Eleanor_N_0002 Device Information

Steps	Analysis
1. We checked documented information to obtain information about the disk Eleanor_A_0003 2. Using Autopsy, we navigated to Data Artifacts > Operating System Information Eleanor_A_0006 3. Using Autopsy, we navigated to Data Sources > eleanor_pc_1 Host > eleanor > vol3 > Windows > System32 > config > SYSTEM and exported the file. We investigated for IP address information Eleanor_A_0010 4. Using Autopsy, we discovered an account of Eleanor's laptop with the username 'eugene'.	1. Owner: eleanor 2. Types: Desktop 3. Operating System: Windows 10 Education 4. IP address: 192.168.1.19 5. Subnet mask: 255.255.255.0 6. Default gateway: 192.168.1.1 7. Name server: 192.168.1.100 8. DHCP server: 255.255.255.255 9. Confidence level: A1. Autopsy is completely reliable as a forensic analysis tool. All analyses are verified through multiple sources.

Eleanor_A_0007	

3.4.7.3 Eleanor_N_0003 Trillian Log

Steps	Analysis
1. Using Autopsy, we navigated to Data Sources > img_eleanor > vol3 > Users > eleanor.AMBROSIA > AppData > Roaming > Trillian > users > eleanoroach > logs > ASTRA > Query > mariusalvarez.log 2. Using Autopsy, we exported the file for investigation Eleanor_A_0011 Eleanor_E_0001 log	1. Mon Feb 01 00:21:01 2021 UTC Eleanor tells Marius to prepare a shipment of contraband to be hidden among a large banana shipment, then abruptly cancels the plan after learning someone snitched to Avalon. Marius insists the leak didn't come from their side, and later, when Eleanor warns that "big shit" is coming and orders him to stop all shipments that night (even though one is already on its way out), he agrees but warns that his crew will want an explanation for losing their payday. 2. Confidence level: A2. Autopsy is completely reliable as a forensic analysis tool. The information is not verified, but the information is probably true.

3.4.8 Pfsense Image (O)

3.4.8.1 PfSense_N_0001 Image Processing

Steps	Analysis
1. Using Box, we downloaded the image to our device Pfsense_A_0001 2. Using Windows, we extracted the image. Pfsense_A_0002 3. Using Ubuntu, we assembled image segments. Then, we computed the checksum of the resulting disk image. Pfsense_A_0003 4. Using Autopsy, we created a new case in Autopsy and imported the assembled disk image.	1. The disk image was successfully downloaded, extracted, assembled, and imported to Autopsy. 2. We verified that the evidence we worked with was not modified since the time we received it from the chain of custody. 3. Confidence level: A1. The tools used are completely reliable in their own respective functions. The images are credible and verified through the chain of custody.

3.4.8.2 PfSense_N_0002 Filter Log

Steps	Analysis
1. Using Autopsy, we navigated to Data Sources> pfsense.raw_1 Host > vol 2 > var > log > filter.log 2. Right-click filter.log 3. Select Extract Files(s) and save to a safe location. 4. Navigate to destination location and open filter.log using VsCode or other text editor. 5. Manually examine files for discrepancies. PFSense_E_0001 log.log	1. Manual examination revealed port scanning like activity and potential data exfiltration between two internal hosts. 2. Between Feb 1 15:30:16 and Feb 1 15:30:17 there were 794 packets sent in 1 second with different destination ports from the source address 192.168.2.42 to destination address 192.168.2.41. 3. On February 3, from 16:58:22 to 16:59:15, pfSense blocked 11 TCP packets from a potential listener at 192.168.2.42:4444 to 192.168.2.41:45209. 4. Confidence level: A2. Autopsy is completely reliable as a forensic analysis tool. The information is not verified, but the information is probably true.

3.4.8.3 PfSense_N_0003 192.168.2.41

Steps	Analysis
1. Using Autopsy, we navigated to Data Sources> pfsense.raw_1 Host > vol 2 > > img_pfsense.raw > vol_vol2 > $CarvedFiles > 1 > f0000184.txt 2. Right-click f0000184.txt 3. Select Extract Files(s) and save to a safe location. 4. Navigate to destination location and open filter.log using VsCode or other text editor. 5. Manually examine files for discrepancies. 5.2.8.7 PFSense_A_0007 f0000184.txt 6. Using Autopsy, we navigated to Data Sources> pfsense.raw_1 Host > img_pfsense.raw > vol_vol2 > usr > local > etc > suricata > suricata_21256_em1 > suricata.yaml. Then repeat steps 2 – 5. 5.2.8.8 PFSense_A_0008 suricata.yaml	1. Examination of the f0000184.txt revealed that 192.168.1.20 was using 192.168.2.41 as the NATed External address to communicate with 192.168.2.42:4444. 2. Examination of suricata.yaml revealed that 192.168.2.41 was on the home net. 3. Confidence level: A2. Autopsy is completely reliable as a forensic analysis tool. The information is not verified, but the information is probably true

3.4.9 Eugene Image (Q)

3.4.9.1 Eugene_N_0001 Image Processing

Steps	Analysis
1. Using Box, we downloaded the image to our device Eugene_A_0001 2. Using Windows, we extracted the image. Eugene_A_0002 3. Using Ubuntu, we assembled image segments. Then, we computed the checksum of the resulting disk image. Eugene_A_0003 4. Using Autopsy, we created a new case in Autopsy and imported the assembled disk image. Eugene_A_0004	1. The disk image was successfully downloaded, extracted, assembled, and imported to Autopsy. 2. We verified that the evidence we worked with was not modified since the time we received it from the chain of custody. 3. Confidence level: A1. The tools used are completely reliable in their own respective functions. The images are credible and verified through the chain of custody.

3.4.9.2 Eugene_N_0002 Device Information

Steps	Analysis
1. We checked documented information to obtain information about the disk Eugene_A_0003 2. Using Autopsy, we navigated to Data Artifacts > Operating System Information Eugene_A_0007 3. Using Autopsy, we navigated to Data Sources > eleanor_pc_1 Host > eleanor > vol3 > Windows > System32 > config > SYSTEM and exported the file. We investigated for IP address information Eugene_A_0005	1. Owner: eugene 2. Types: Desktop 3. Operating System: Windows 10 Education 4. IP address: 192.168.1.12 5. Subnet mask: 255.255.255.0 6. Default gateway: 192.168.1.1 7. Name server: 192.168.1.100 8. DHCP server: 255.255.255.255 9. Confidence level: A1. Autopsy is completely reliable as a forensic analysis tool. All analyses are verified through multiple sources.

3.4.9.3 Eugene_N_0003 Trillian Logs

Steps	Analysis

1. Using Autopsy, we navigated to Data Sources > img_eugene > vol3 > Users > eugene.AMBROSIA > AppData > Roaming > Trillian > users > eugenebarnes > logs > ASTRA > Query > emmagarner.log 2. Using Autopsy, we exported the file for investigation Eugene_A_0006 Eugene_E_0001 log 3. Using Autopsy, we navigated to Data Sources > pfsense.raw_1 Host > pfsense.raw > vol2 > var > log > system.log PFSense_A_0006 PFsense_E_0002 system.log	1. Mon Feb 01 00:10:53 2021 UTC Eugene tells Emma to put at least 25% of his portfolio into shorting Avalon's publicly traded banana producers and any shipping companies that depend on their exports. Later he panics and orders her to immediately liquidate every traceable asset regardless of penalties, saying he's coming to her office to decide where to move the cash and that it's time for him to retire. 2. As of 2021-01-31 Sun 12:20:29 UTC, the WAN interface is assigned with IP address 192.168.2.41. 3. Confidence level: A2. Autopsy is completely reliable as a forensic analysis tool. The information is not verified, but the information is probably true.

3.4.10 Martin Image (P)

3.4.10.1 Martin_N_0001 Image Processing

Steps	**Analysis**
1. Using Box, we downloaded the image to our device Martin_A_0001 2. Using Windows, we extracted the image Martin_A_0002 3. Using Ubuntu, we assembled image segments. Then, we computed the checksum of the resulting disk image Martin_A_0003 4. Using Autopsy, we created a new case in Autopsy and imported the assembled disk image Martin_A_0004	1. The disk image was successfully downloaded, extracted, assembled, and imported to Autopsy. 2. We verified that the evidence we worked with was not modified since the time we received it from the chain of custody. 3. Confidence level: A1. The tools used are completely reliable in their own respective functions. The images are credible and verified through the chain of custody.

3.4.10.2 Martin_N_0002 Image Information

Steps	**Analysis**
1. We checked documented information to obtain	1. Owner: martin

information about the disk Martin_A_0003 2. Using Autopsy, we navigated to Data Artifacts > Operating System Information Martin_A_0006 3. Using Autopsy, we navigated to Data Sources > gauthier_1 Host > gauthier > vol3 > Windows > System32 > config > SYSTEM and exported the file. We investigated for IP address information Martin_A_0007 Martin_E_0001 SYSTEM	2. Types: Desktop 3. Operating System: Windows 10 Pro 4. IP address: 192.168.1.21 5. Subnet mask: 255.255.255.0 6. Default gateway: 192.168.1.1 7. Name server: 192.168.1.100 8. DHCP server: 255.255.255.255 9. Confidence level: A1. Autopsy is completely reliable as a forensic analysis tool. All analyses are verified through multiple sources.

3.4.11 Services Image (K)

3.4.11.1 Services_N_0001 Image Processing

Steps	**Analysis**
1. Using Box, we downloaded the image to our device Services_A_0001 2. Using Windows, we extracted the image Services_A_0002 3. Using Ubuntu, we assembled image segments. Then, we computed the checksum of the resulting disk image Services_A_0003 4. Using Autopsy, we created a new case in Autopsy and imported the assembled disk image Services_A_0004	1. The disk image was successfully downloaded, extracted, assembled, and imported to Autopsy. 2. We verified that the evidence we worked with was not modified since the time we received it from the chain of custody. 3. Confidence level: A1. The tools used are completely reliable in their own respective functions. The images are credible and verified through the chain of custody.

3.4.11.2 Services_N_0002 Email MetaData – IP

Steps	**Analysis**
1. We navigated to Data Sources > srvSVC.raw_1 Host > srvSVC.raw > vol 3> Program Files > Microsoft > Exchange Server	1. The ip addresses correspond to the sender 2. 192.168.1.10: julia@ambrosia.it

> V15 > TransportRoles > Logs > MessageTracking. We downloaded all the *.LOG files present in this folder.

Services_A_0006

- Services_E_0001 135260-MSGTRKM…
- Services_E_0002 135327-MSGTRK20…
- Services_E_0003 135329-MSGTRK20…
- Services_E_0004 135331-MSGTRK20…
- Services_E_0005 135333-MSGTRK20…
- Services_E_0006 135335-MSGTRK20…
- Services_E_0007 135337-MSGTRK20…
- Services_E_0008 135339-MSGTRK20…
- Services_E_0009 135341-MSGTRK20…
- Services_E_0010 135343-MSGTRK20…
- Services_E_0011 135345-MSGTRK20…
- Services_E_0012 135347-MSGTRK20…
- Services_E_0013 135349-MSGTRK20…
- Services_E_0014 135351-MSGTRKM…
- Services_E_0015 135353-MSGTRKM…
- Services_E_0016 135355-MSGTRKM…
- Services_E_0017 135357-MSGTRKM…
- Services_E_0018 135359-MSGTRKM…
- Services_E_0019 135361-MSGTRKM…
- Services_E_0020 135363-MSGTRKM…
- Services_E_0021 135365-MSGTRKM…
- Services_E_0022 135367-MSGTRKM…
- Services_E_0023 135369-MSGTRKM…
- Services_E_0024 135371-MSGTRKM…
- Services_E_0025 135373-MSGTRKM…
- Services_E_0026 135375-MSGTRKM…
- Services_E_0027 135377-MSGTRKM…
- Services_E_0028 135379-MSGTRKM…
- Services_E_0029 135381-MSGTRKM…
- Services_E_0030 135383-MSGTRKM…
- Services_E_0031 135385-MSGTRKM…
- Services_E_0032 135387-MSGTRKM…
- Services_E_0033 135389-MSGTRKM…
- Services_E_0034 135391-MSGTRKM…

2. We write python script to retrieve email metadata and match the IP addresses with email address and users

- Services_E_0035 email_ip.py
- Services_E_0036 email_ip_summary.csv

3. 192.168.1.12: eugene@ambrosia.it
4. 192.168.1.19: eleanor@ambrosia.it
5. 192.168.1.20: jean@ambrosia.it
6. 192.168.1.21: martin@ambrosia.it
7. 192.168.1.31: richard@ambrosia.it
8. 192.168.1.101: chappel@ambrosia.it
9. 192.168.2.42: helpdesk102@ambrosia.it, xxmiguelortegaxx3456@yahoo.com, xxmiguelortegaxx@yahoo.com
10. Confidence level: A1. Autopsy and python are completely reliable. The information is confirmed from other images.

3.4.11.3 Services_N_0003 Mailbox Database 0038886738.edb – Security Update

Steps	Analysis
1. We navigated to Data Sources > srvSVC.raw_1 Host > srvSVC.raw > vol 3> Program Files > Microsoft > Exchange Server > V15 > Mailbox > Mailbox Database 0038886738 > Mailbox Database 0038886738.edb. We downloaded this file and performed a manual examination of this file using a EDB Viewer tool. 5.2.11.7 Services_A_0007 Email HelpDesk102@ambrosia.it 5.2.11.8 Services_A_0008 192.168.2.42 5.2.11.9 Services_A_0009 Jeannette Forward Email to Jeff Schmitt Services_E_0037 Mailbox Database 00…	2. An email from HelpDesk102@ambrosia.it was sent to Jeanette at 01/31/2021 at 23:35 PM UTC containing a securityupdate.zip, which we previously confirmed to be a malicious file. 3. This email was sent from the ip address 192.168.2.42 which was previously identified as a potentially malicious ip as it attempted a port scan and was a listener in a data exfiltration attempt. 4. Jeannette forwarded this email to Jeff Schmitt (jschmitt@ambrosia.it) at 02/01/2021 at 1:41:30 AM UTC 5. Confidence level: B2. Autopsy is completely reliable, but Shoviv EDB Viewer reputation is not established. The information are not confirmed by other independent source, but are probably true.

3.4.11.4 Services_N_0004 Mailbox Database 0038886738.edb – Messages between Miguel and Eleanor

Steps	Analysis
1. We navigated to Data Sources > srvSVC.raw_1 Host > srvSVC.raw > vol 3> Program Files > Microsoft > Exchange Server > V15 > Mailbox > Mailbox Database 0038886738 > Mailbox Database 0038886738.edb. We downloaded this file and performed a manual examination of this file using a EDB Viewer tool. 2. In the EDB viewer we navigated to Eleanor M. Roach > Inbox and Sent Items to get the files below. 5.2.11.10 Services_A_00010 Subject: Subsidy Proposal between Eleanor and Miguel Services_E_0037 Mailbox Database 00…	1. On 02/01/2021 at 00:27 AM UTC Eleanor sent an email to Miguel stating that they are moving ahead with the subsidy proposal. 2. 02/01/2021 at 00:44 AM UTC Miguel responds with disagreement, indicating apprehension for going ahead with the subsidy proposal. 3. In response on 02/01/2021 at 00:48 AM UTC Eleanor threatens him and tells him to enforce the policy. 4. Then Miguel apologizes at 00:52 AM UTC 5. Confidence level: B2. Autopsy is completely reliable, but Shoviv EDB Viewer reputation is not established. The information are not confirmed by other independent source, but are probably true.

3.4.11.5 Services_N_0005 IP Address 192.168.2.42

Steps	Analysis
1. Using Autopsy, we navigated to Data Sources > srvSVC.raw_1 Host > srvSVC.raw > vol 3> Program Files > Microsoft > Exchange Server > V15 > TransportRoles > data > Queue > mail.que. We extracted the file. 2. Within mail.que, we found an email correlating IP address 192.168.2.42 to helpdesk102@ambrosia.it, which is responsible for sending the malware. We also found an email correlating IP address 192.168.2.42 to Miguel Ortega Services_A_0011 Services_E_0038 mail.que	1. Miguel Ortega is responsible for sending the malware to Jeanette, as well as performing remote shell code, data exfiltration, and leaking of the information. 2. Confidence level: A2. Autopsy is completely reliable. The information are not confirmed by other independent source, but are probably true.

3.4.11.6 Services_N_0006 Jeff Schmitt x Richard Gauthier - Pay Up

Steps	Analysis
1. Using Shoviv EDB Viewer, we investigated mailbox database edb file and found Pay Up email thread. Services_A_0012 Services_E_0037 Mailbox Database 00…	1. 2021-02-03 Wed 01:08 UTC Jeff Schmitt → Richard Gauthier Schmitt demands $10,000 from Gauthier for cheating on Dianne Gauthier with Mollie Schmitt 2. 2021-02-03 Wed 01:14 UTC Jeff Schmitt → Richard Gauthier Schmitt reveals that Gauthier is married to Ater's little sister 3. 2021-02-03 Wed 01:23 UTC Richard Gauthier → Jeff Schmitt Gauthier Paid Schmitt 4. 2021-02-03 Wed 01:26 UTC Jeff Schmitt → Richard Gauthier Schmitt instructs Gauthier that he must keep paying him as long as Ater runs the island. 5. Confidence level: B2. Autopsy is completely reliable, but Shoviv EDB Viewer reputation is

	not established. The information are not confirmed by other independent source, but are probably true.

4 Summary of Conclusions Reached

In the opinion of the forensic experts hired, Michael Ortega is responsible for leaking information regarding Ambrosia's secret plans regarding a banana dumping and intentional sabotage of Avalon's banana market.

This is likely to have occurred through a malicious securityupdate.exe sent to Jeanette through a phishing email establishing a connection to an attacker controlled ip address.

Date: October 22, 2026

Signature: Supavit Pitaknarongphorn

Signature: Pallavi Srivastava

Signature: Abhinaya Nagalla

5 Appendix

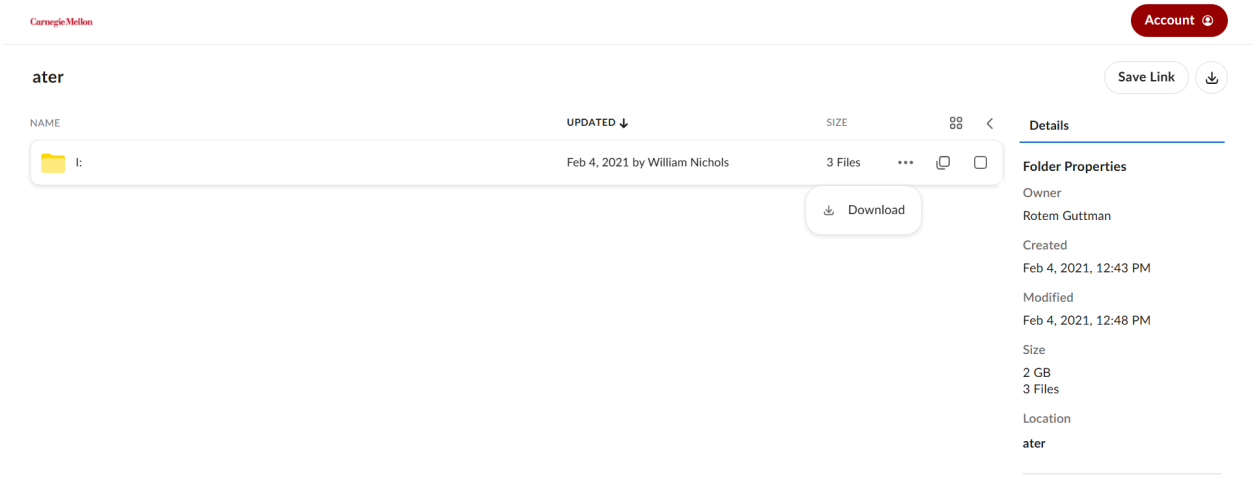
5.1 Report Supplement

Evidence Archive : Archive of all relevant evidence.

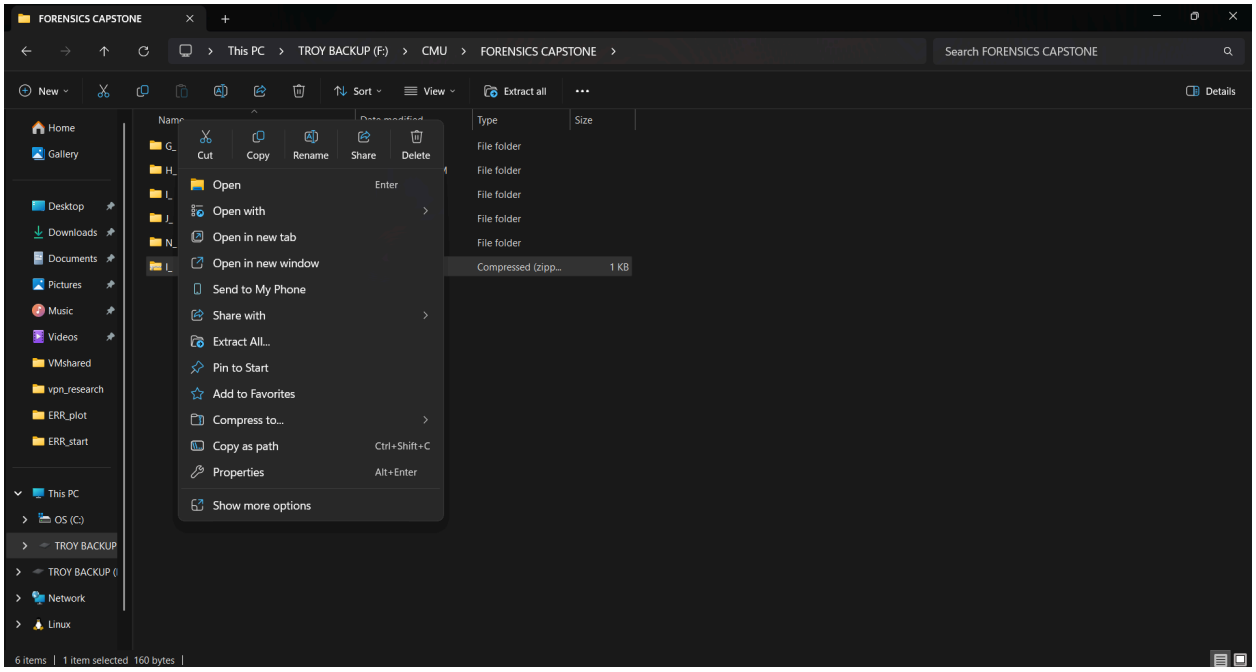
5.2 Examination Screenshot

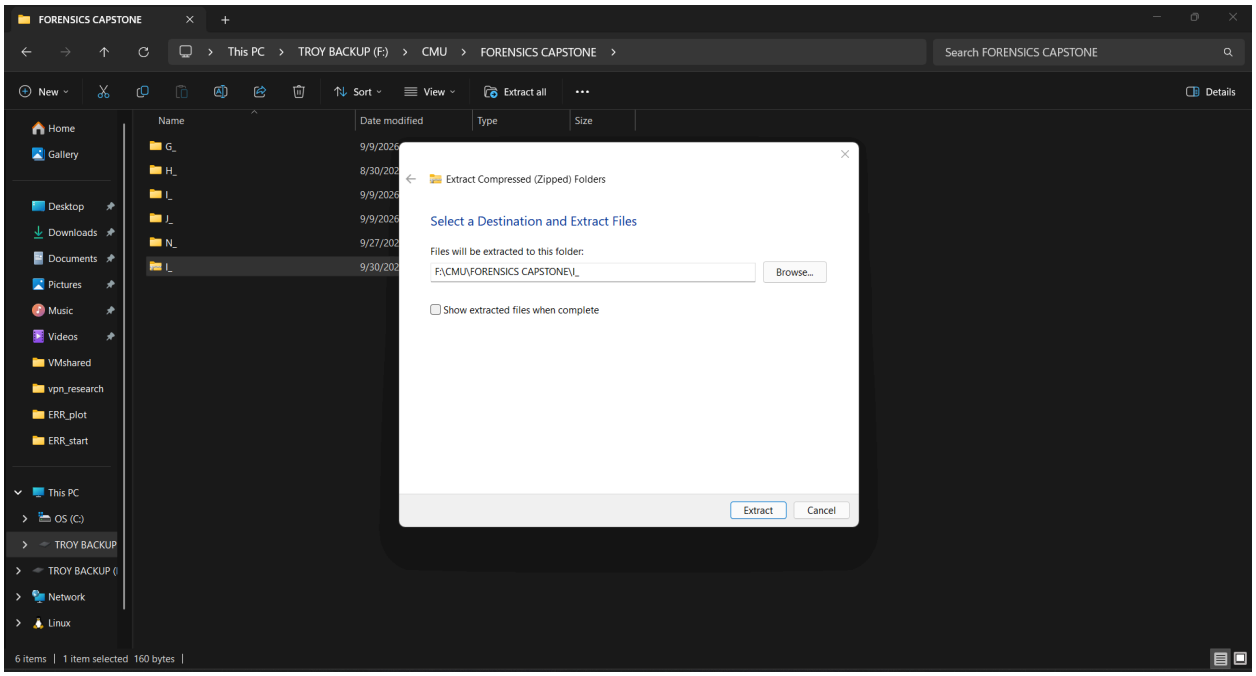
5.2.1 Ater Image (I)

5.2.1.1 Ater_A_0001 Image Download

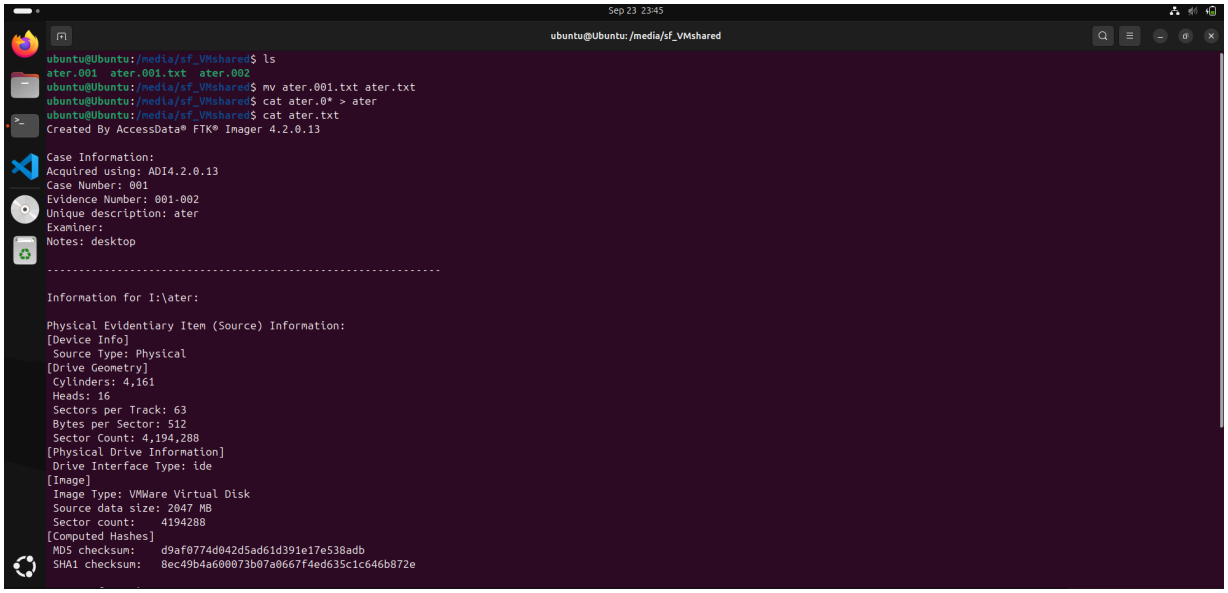


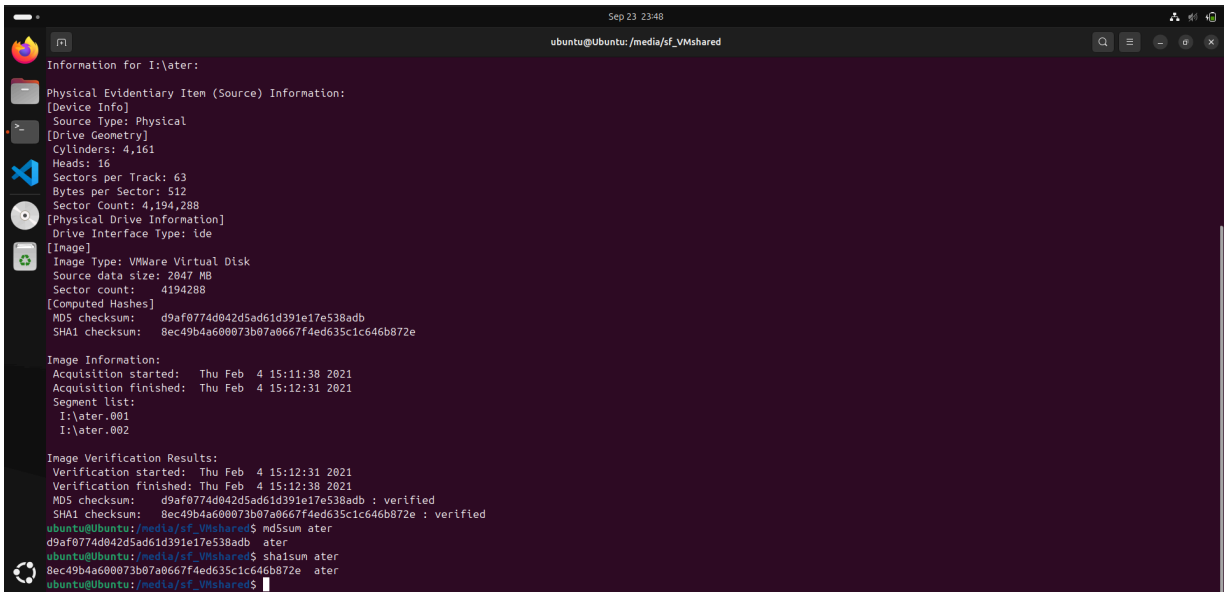
5.2.1.2 Ater_A_0002 Image Extraction



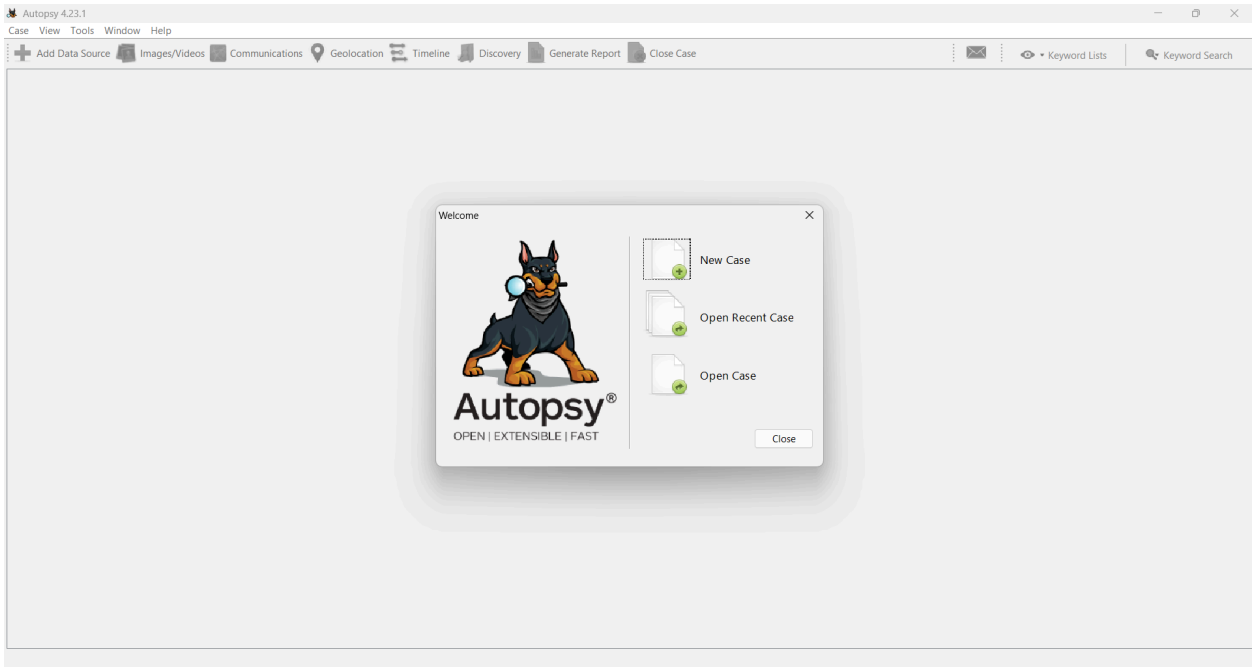


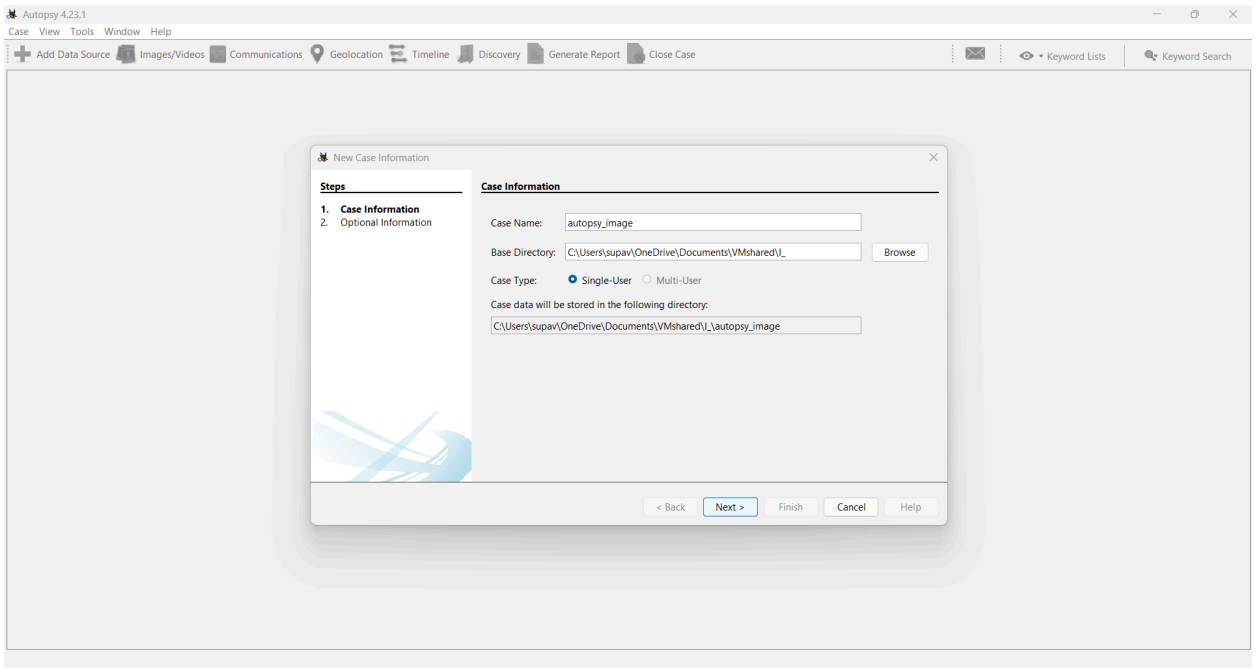
5.2.1.3 Ater_A_0003 Image Verification

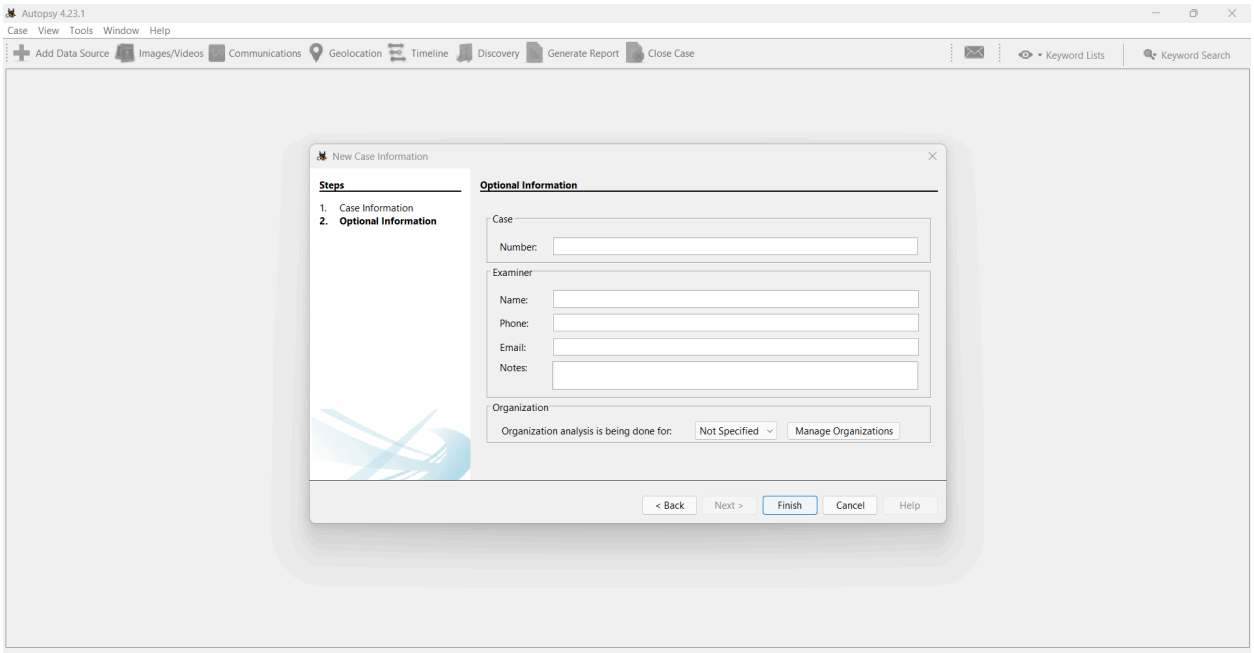


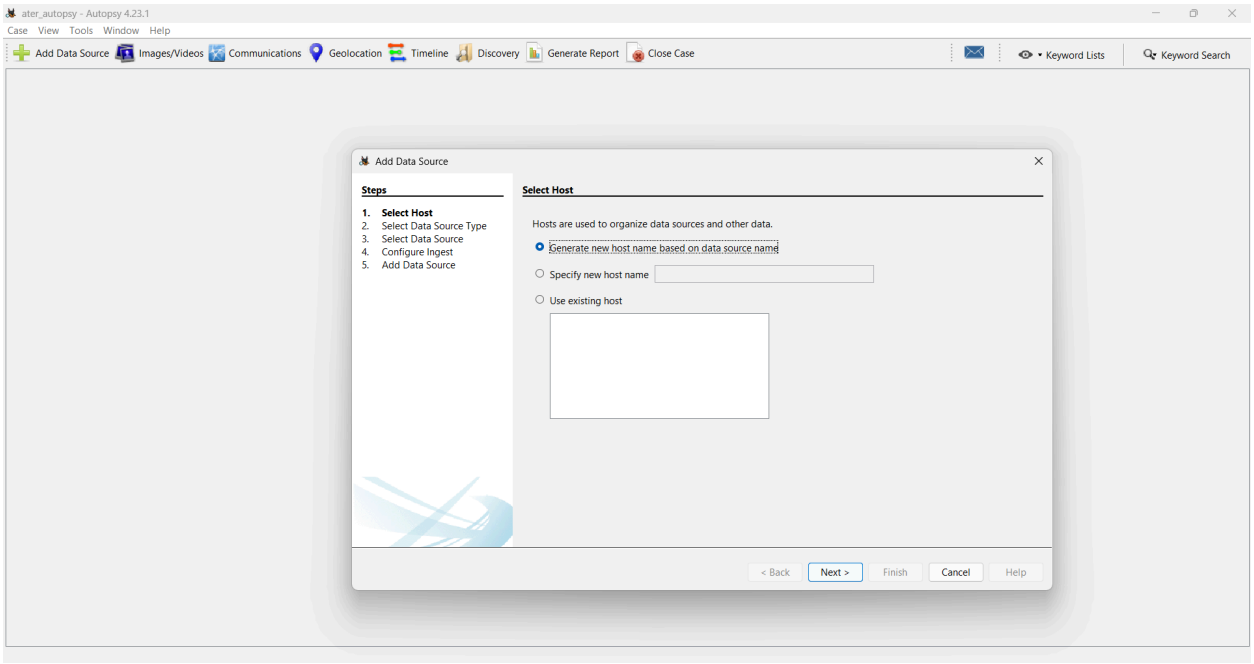


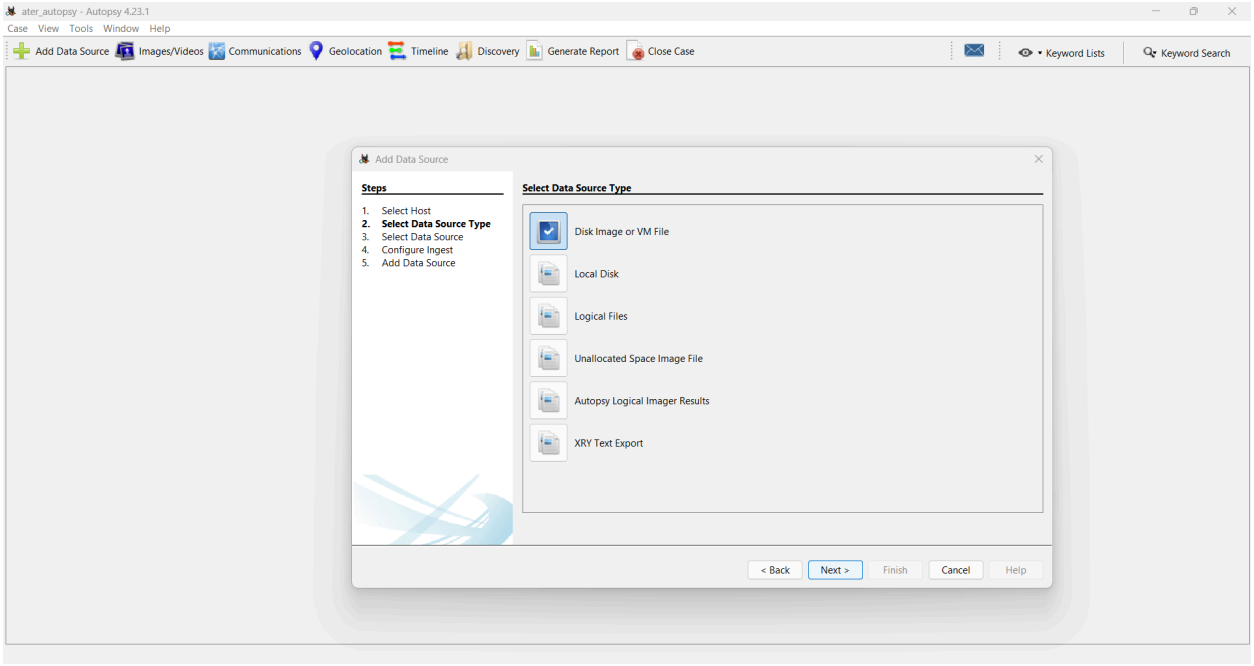
5.2.1.4 Ater_A_0004 Autopsy Case

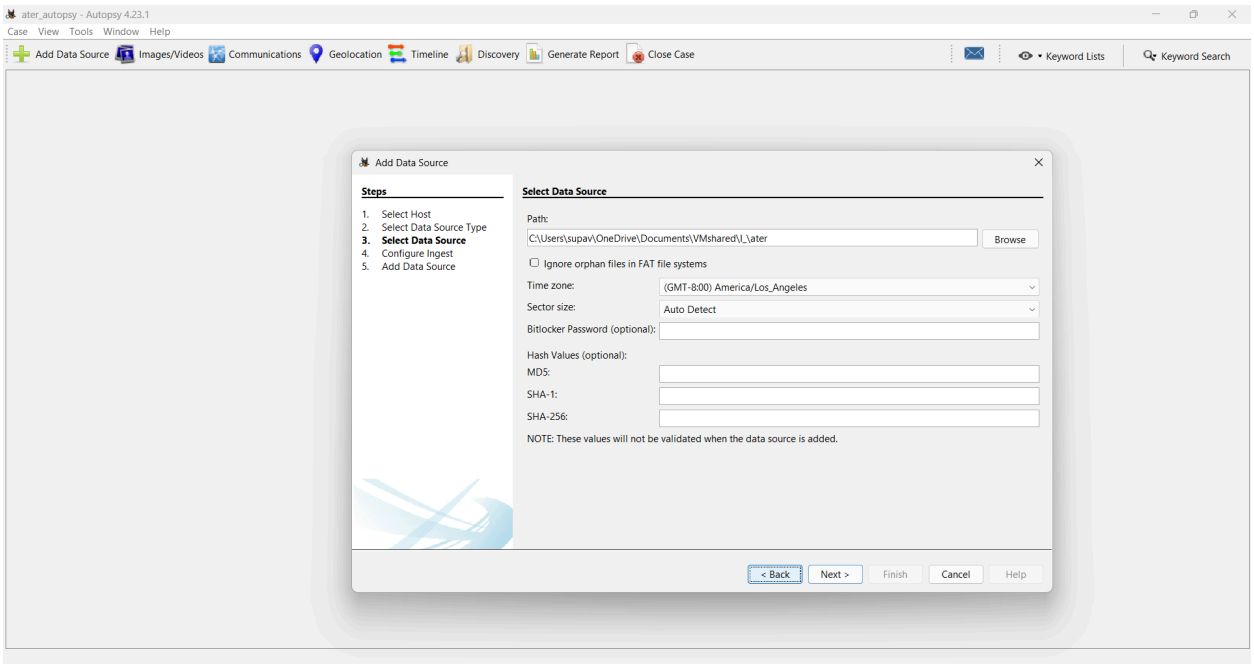


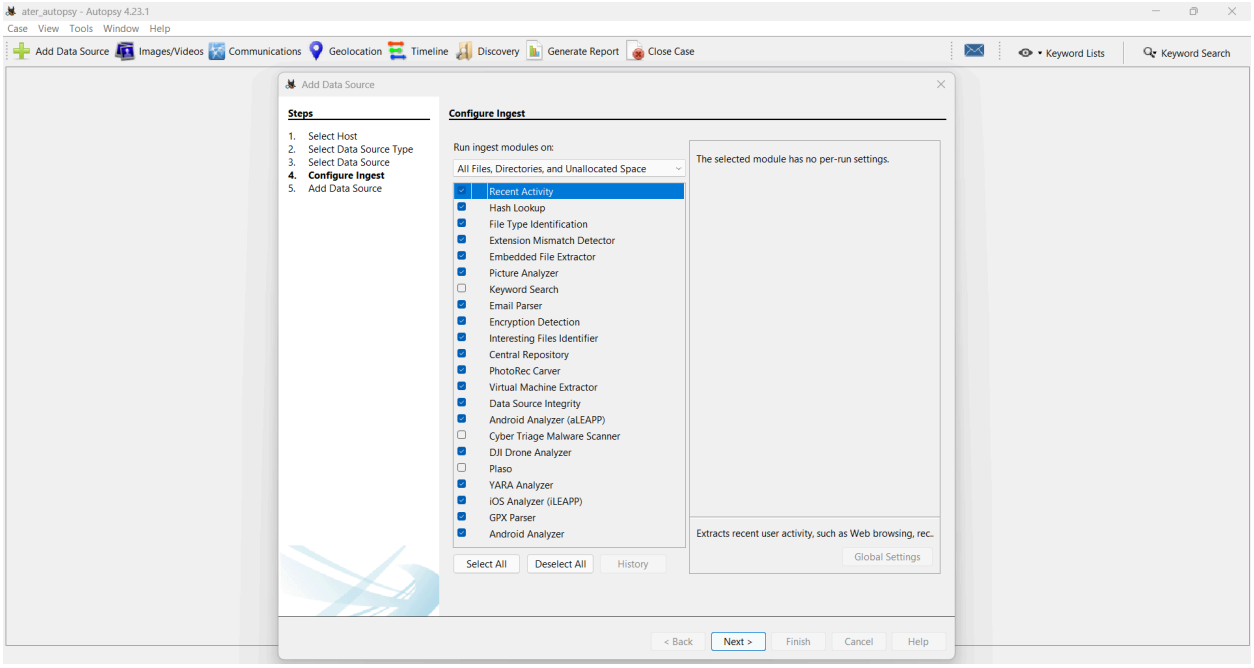


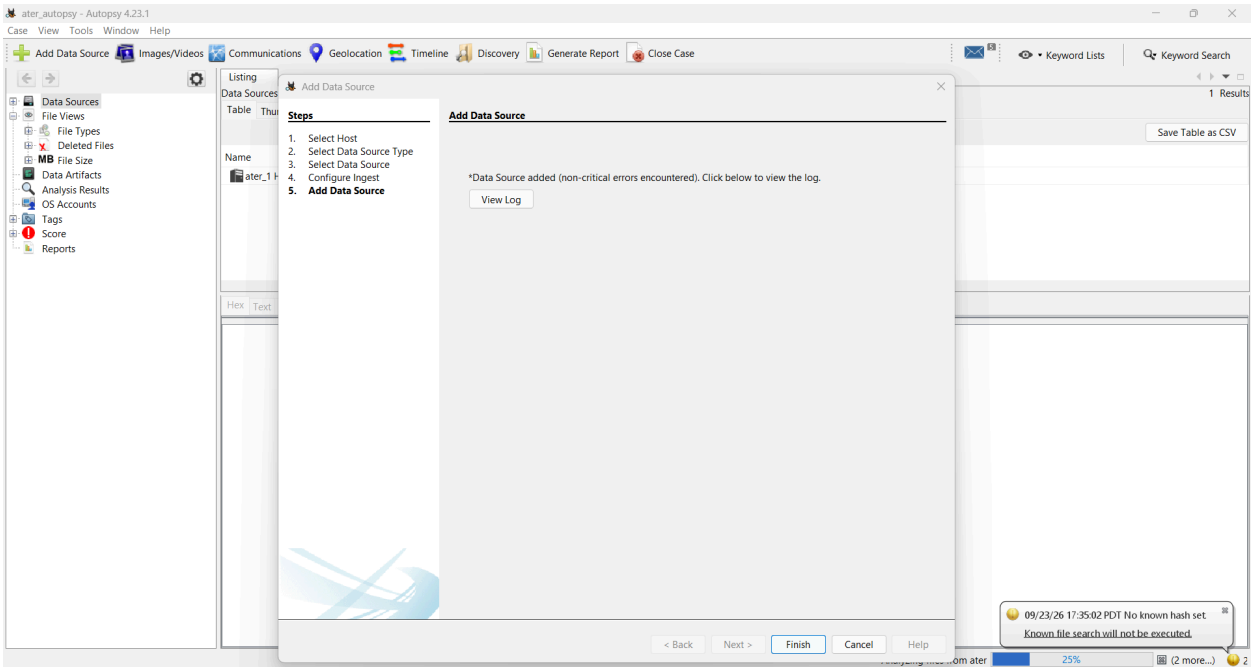

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Optional Information
Case
Number:
Examiner
Name:
Phone:
Email:
Notes:
Organization
Organization analysis is being done for:
Not Specified
Manage Organizations
< Back
Next >
Finish
Cancel
Help

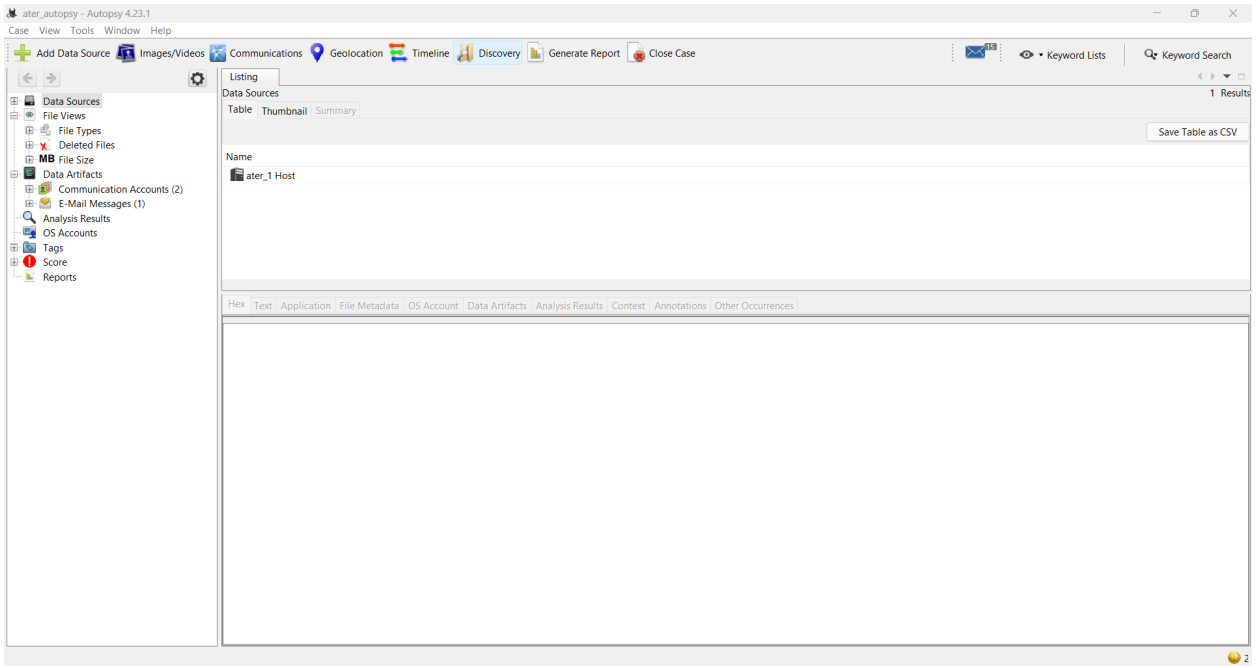

ater_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Host
Hosts are used to organize data sources and other data.
Generate new host name based on data source name
Specify new host name
Use existing host
< Back
Next >
Finish
Cancel
Help


ater_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source Type
Disk Image or VM File
Local Disk
Logical Files
Unallocated Space Image File
Autopsy Logical Imager Results
XRY Text Export
< Back
Next >
Finish
Cancel
Help

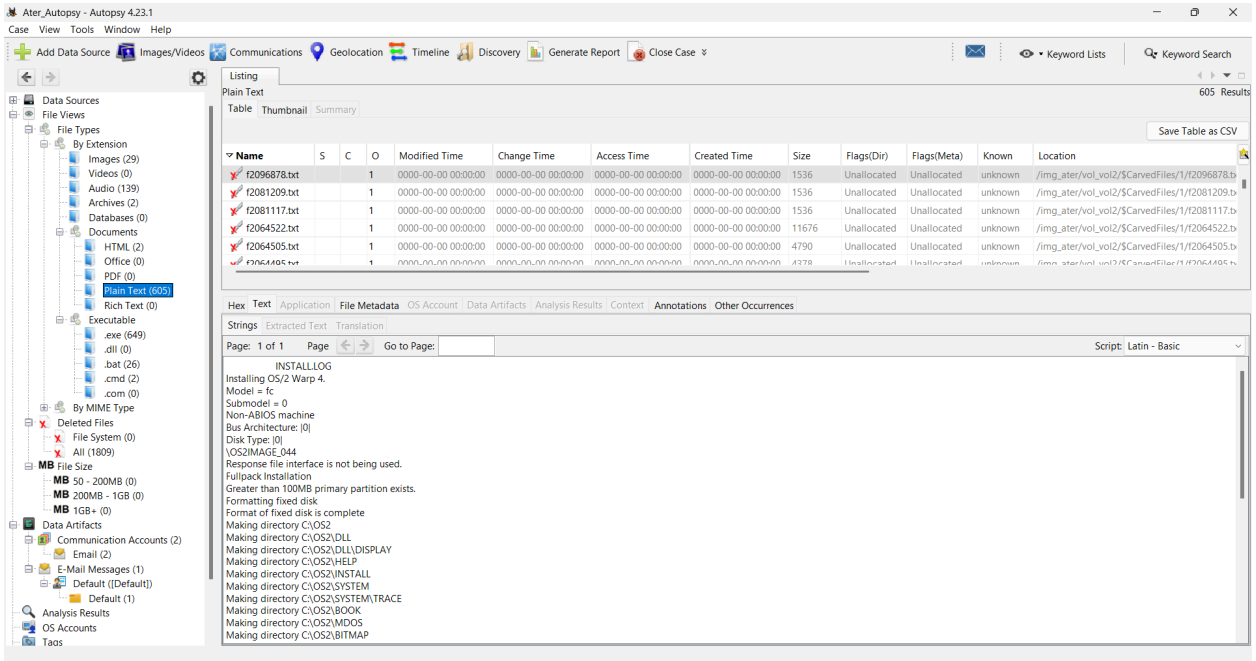

ater_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source
Path:
C:\Users\supav\OneDrive\Documents\VMshared\I_\ater
Browse
Ignore orphan files in FAT file systems
Time zone:
(GMT-8:00) America/Los_Angeles
Sector size:
Auto Detect
Bitlocker Password (optional):
Hash Values (optional):
MD5:
SHA-1:
SHA-256:
NOTE: These values will not be validated when the data source is added.
< Back
Next >
Finish
Cancel
Help


ater_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
Data Source Integrity
Android Analyzer (aLEAPP)
Cyber Triage Malware Scanner
DJI Drone Analyzer
Plaso
YARA Analyzer
iOS Analyzer (iLEAPP)
GPX Parser
Android Analyzer
The selected module has no per-run settings.
Extracts recent user activity, such as Web browsing, rec..
Global Settings
Select All
Deselect All
History
< Back
Next >
Finish
Cancel
Help

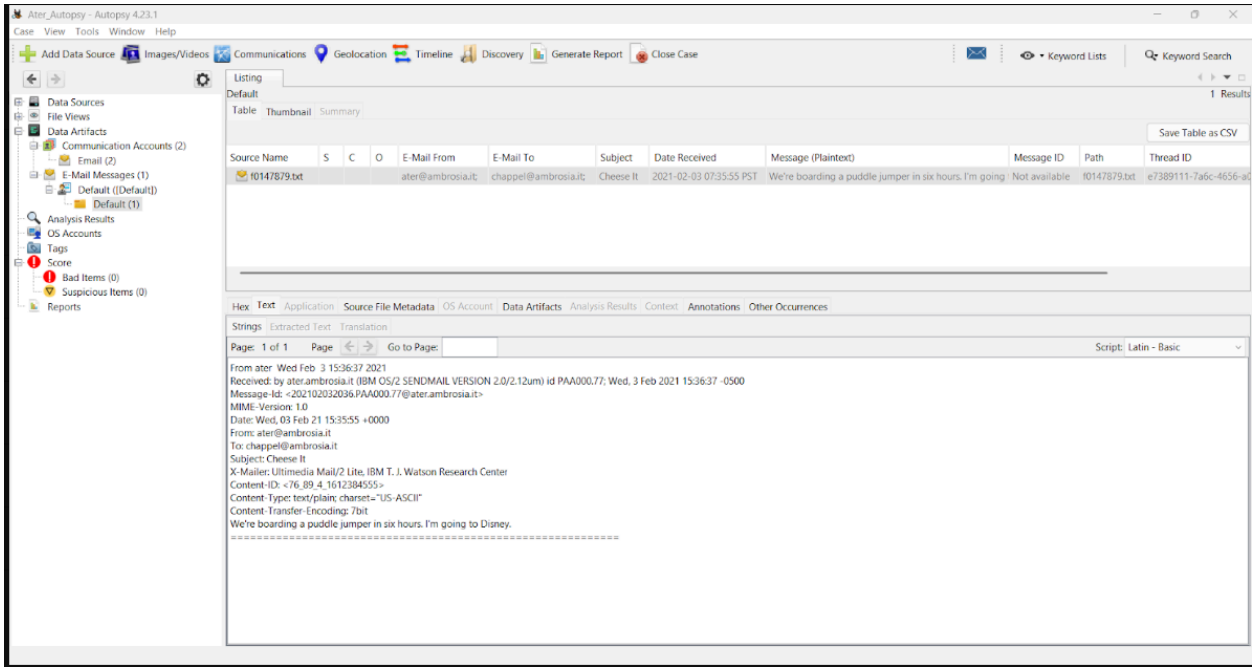

ater_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Analysis Results
OS Accounts
Tags
Score
Reports
Listing
Data Sources
Table
Name
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
*Data Source added (non-critical errors encountered). Click below to view the log.
View Log
1 Results
Save Table as CSV
Hex
Text
< Back
Next >
Finish
Cancel
Help
09/23/26 17:35:02 PDT No known hash set
Known file search will not be executed
25%
(2 more...)



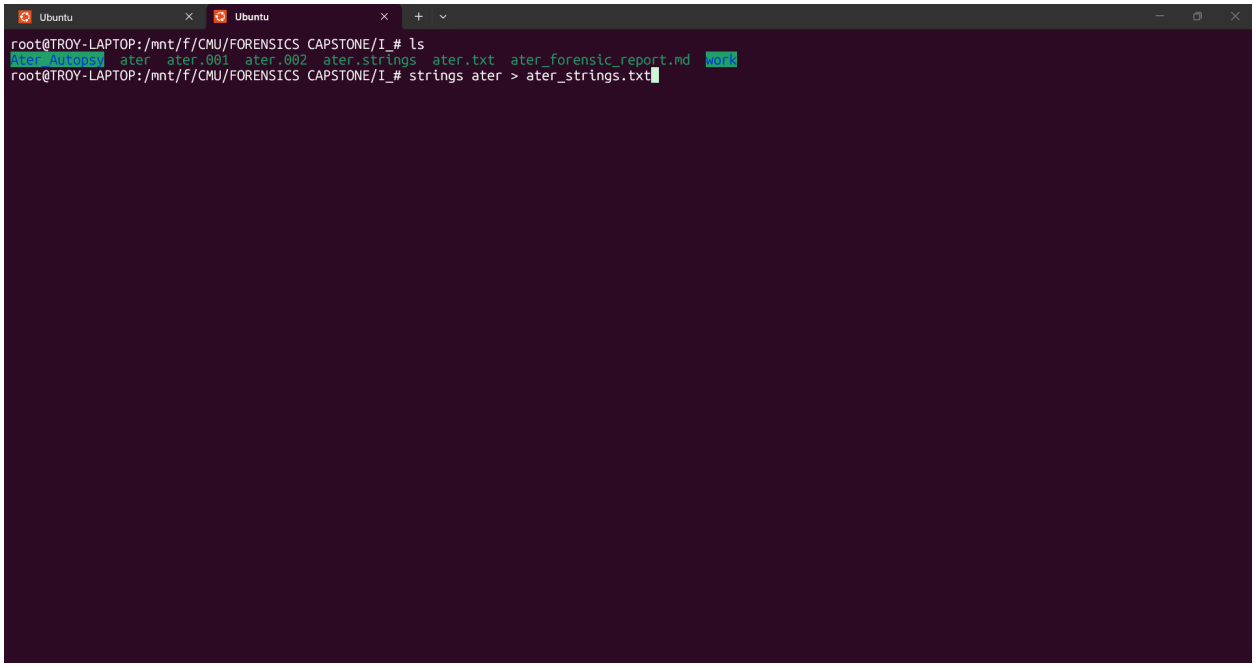
5.2.1.5 Ater_A_0005 Operating System



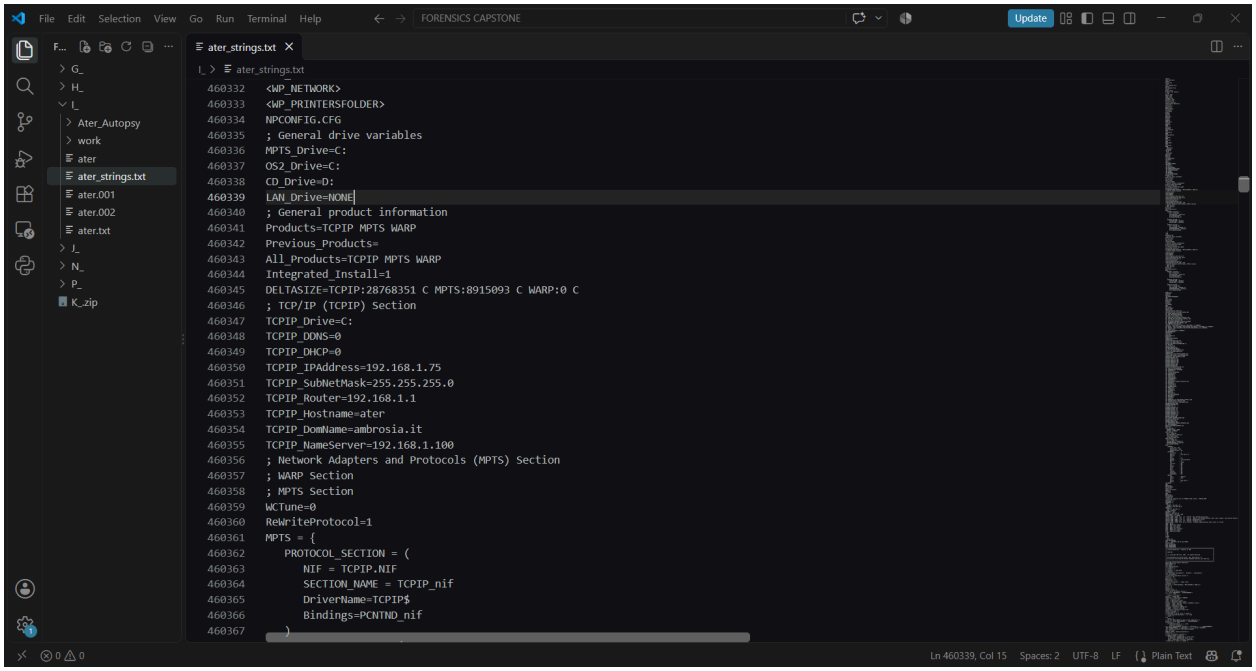
5.2.1.6 Ater_A_0006 Email - Cheese It



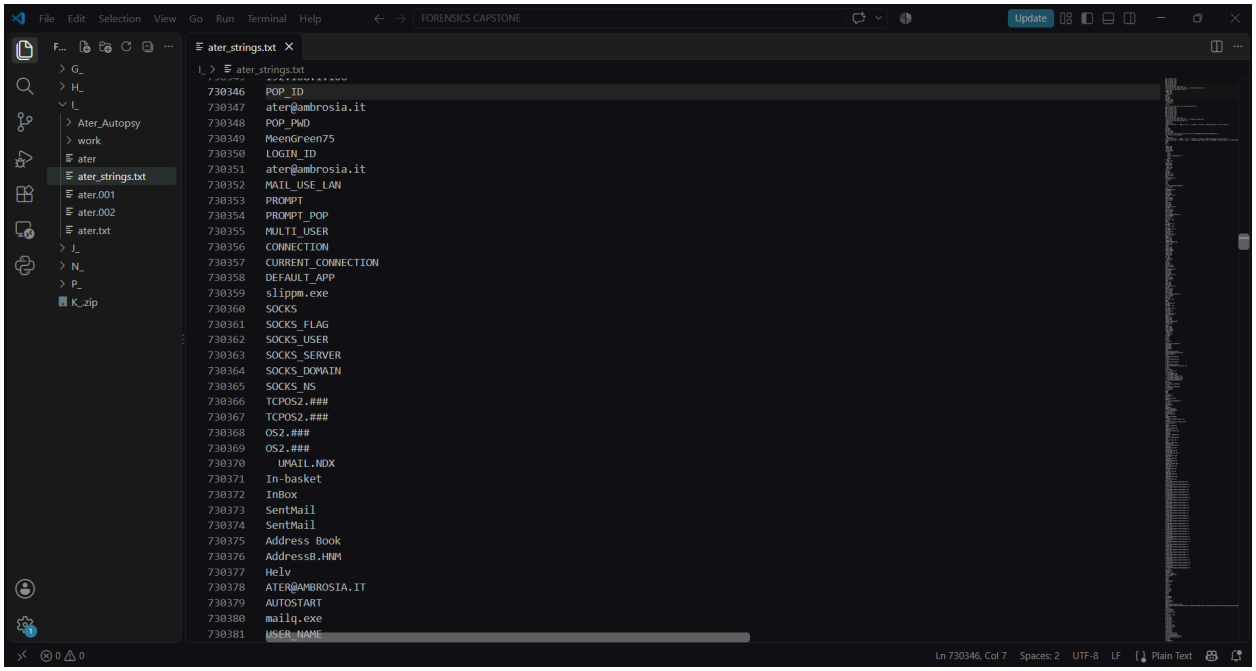
5.2.1.7 Ater_A_0007 Strings



5.2.1.8 Ater_A_0008 TCPIP Information

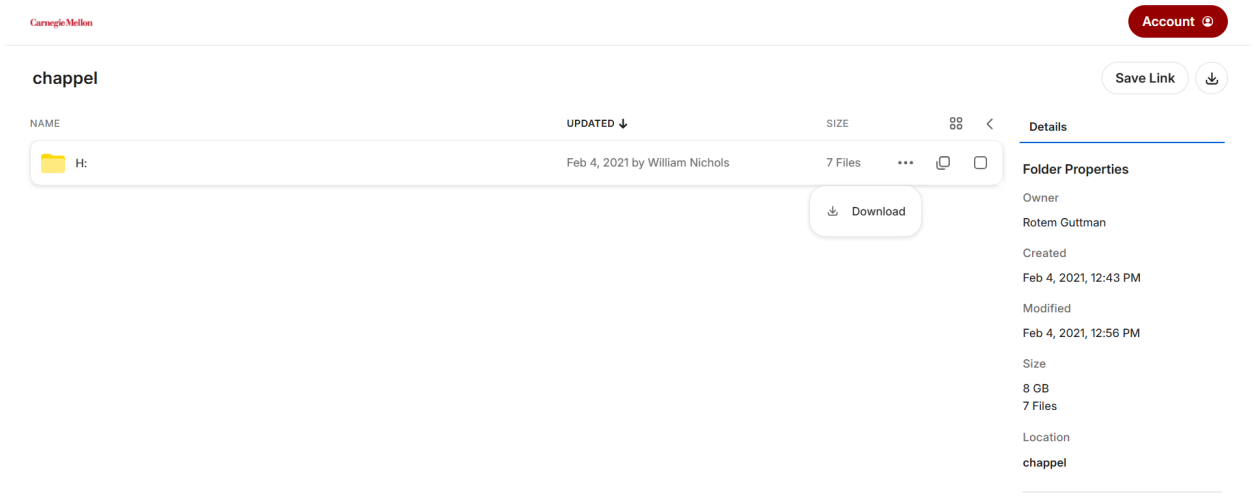


5.2.1.9 Ater_A_0009 Email Credentials

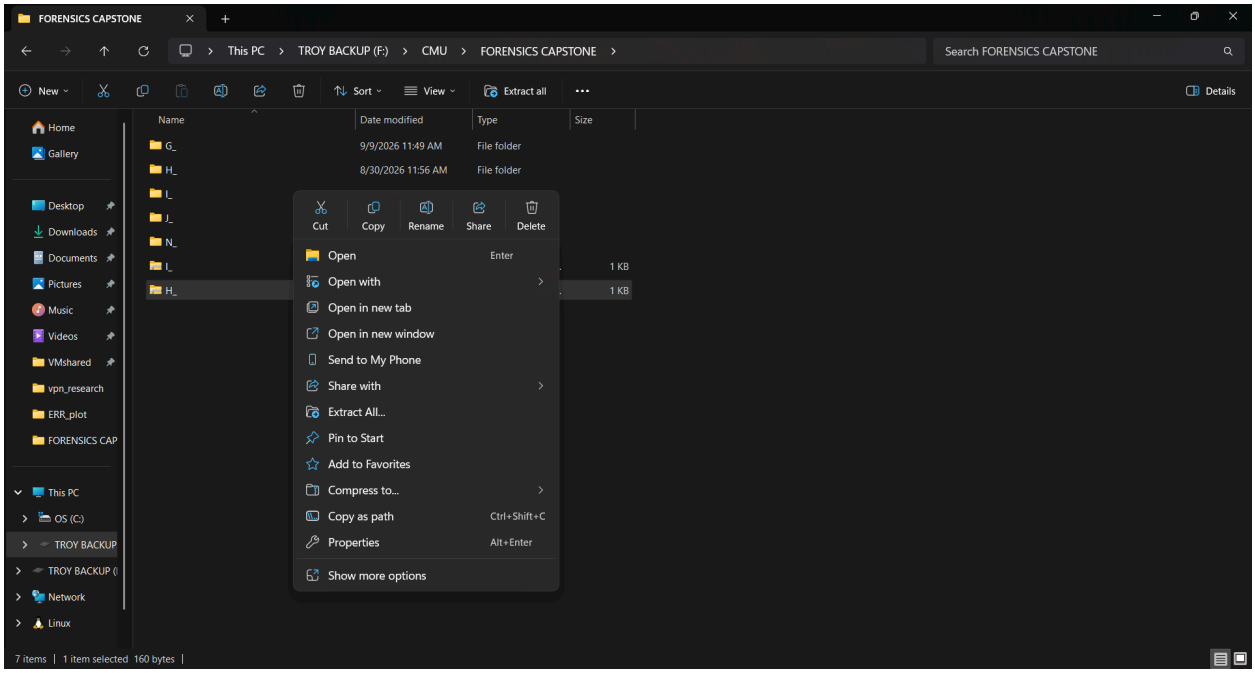


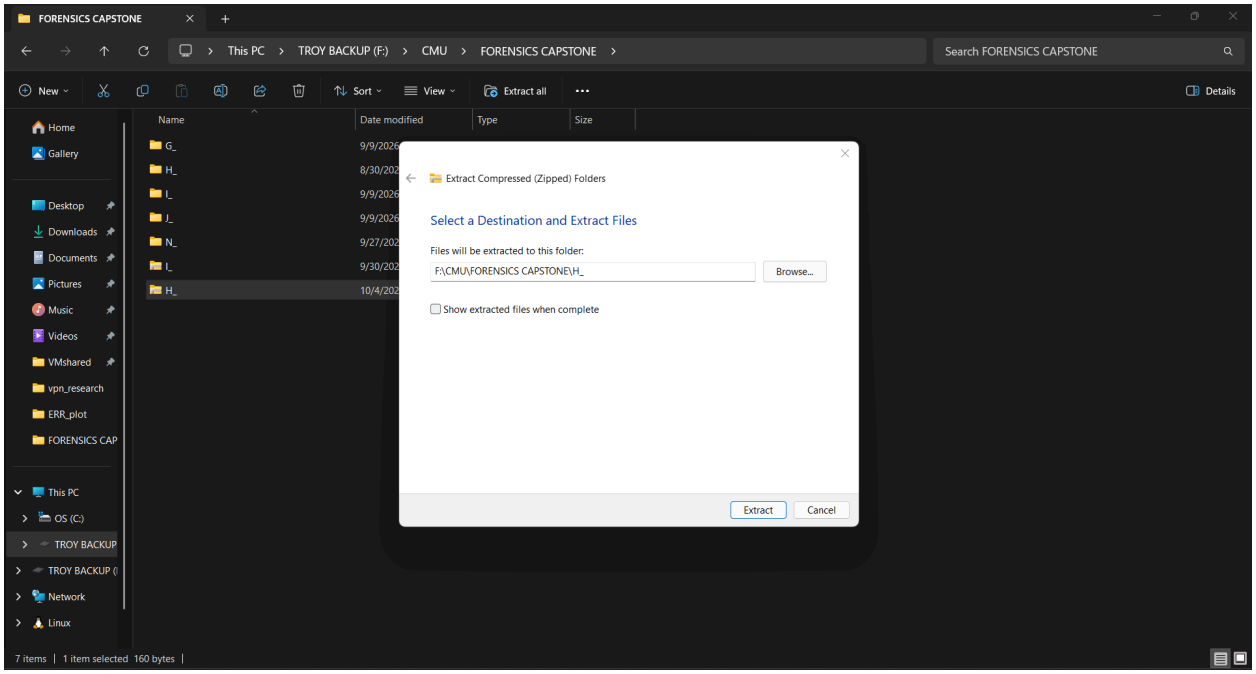
5.2.2 Chappel Image (H)

5.2.2.1 Chappel_A_0001 Image Download



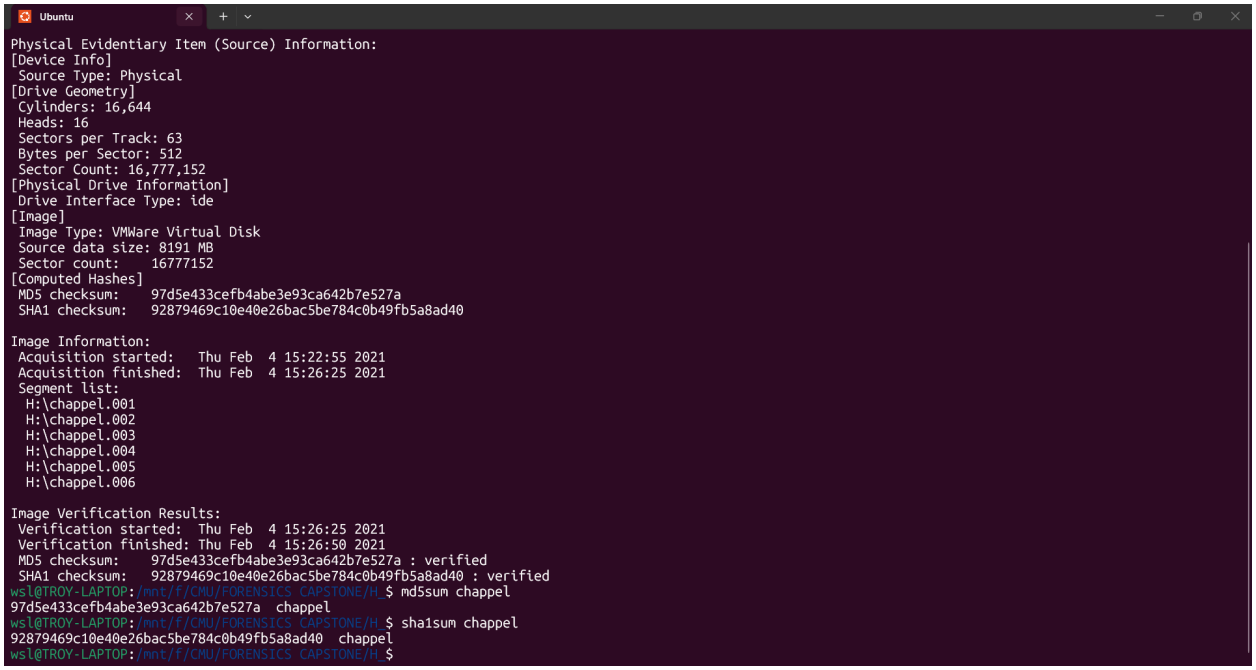
5.2.2.2 Chappel_A_0002 Image Extraction



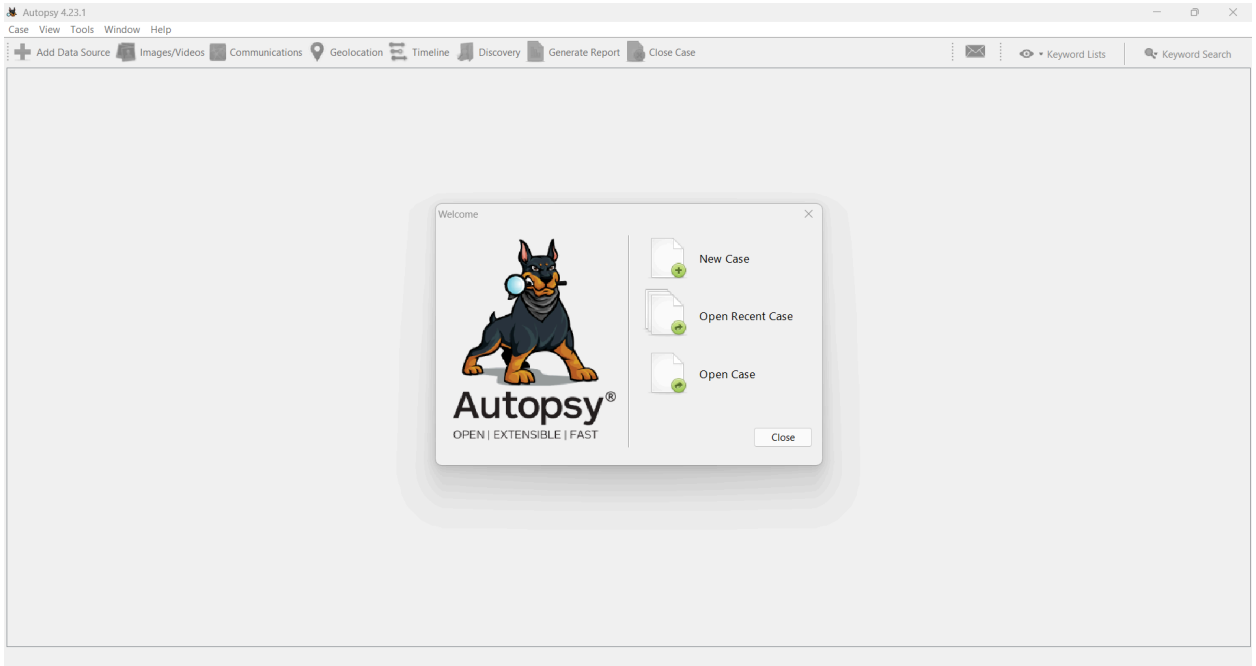


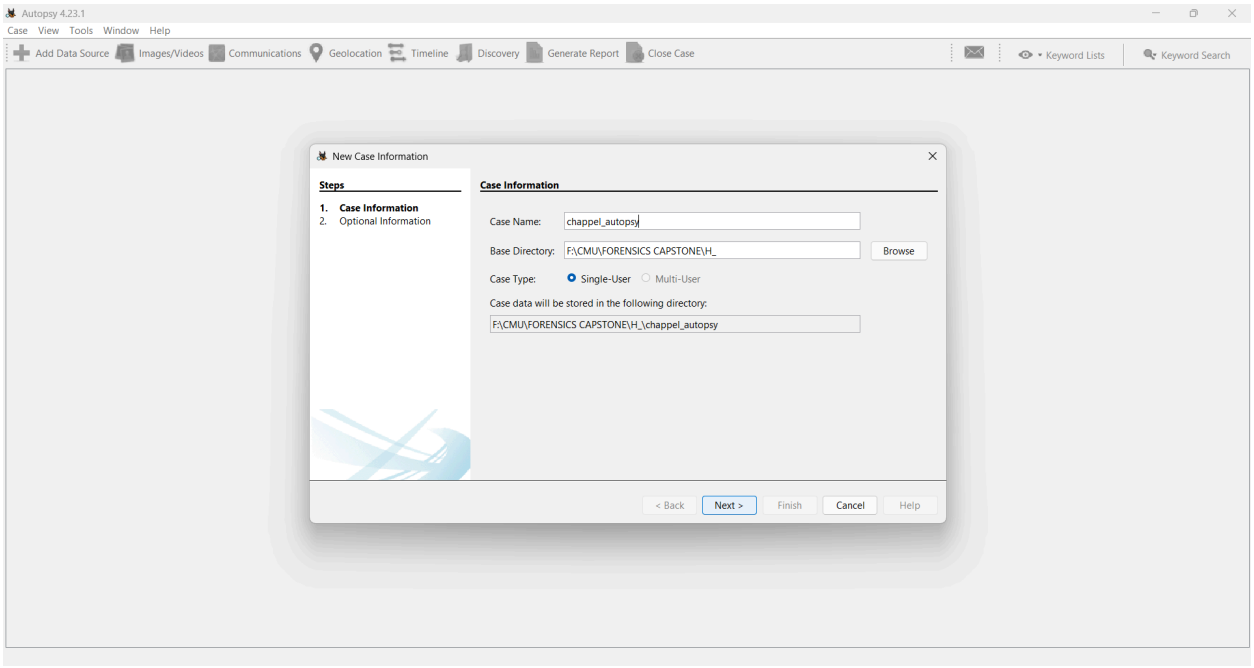
5.2.2.3 Chappel_A_0003 Image Verification

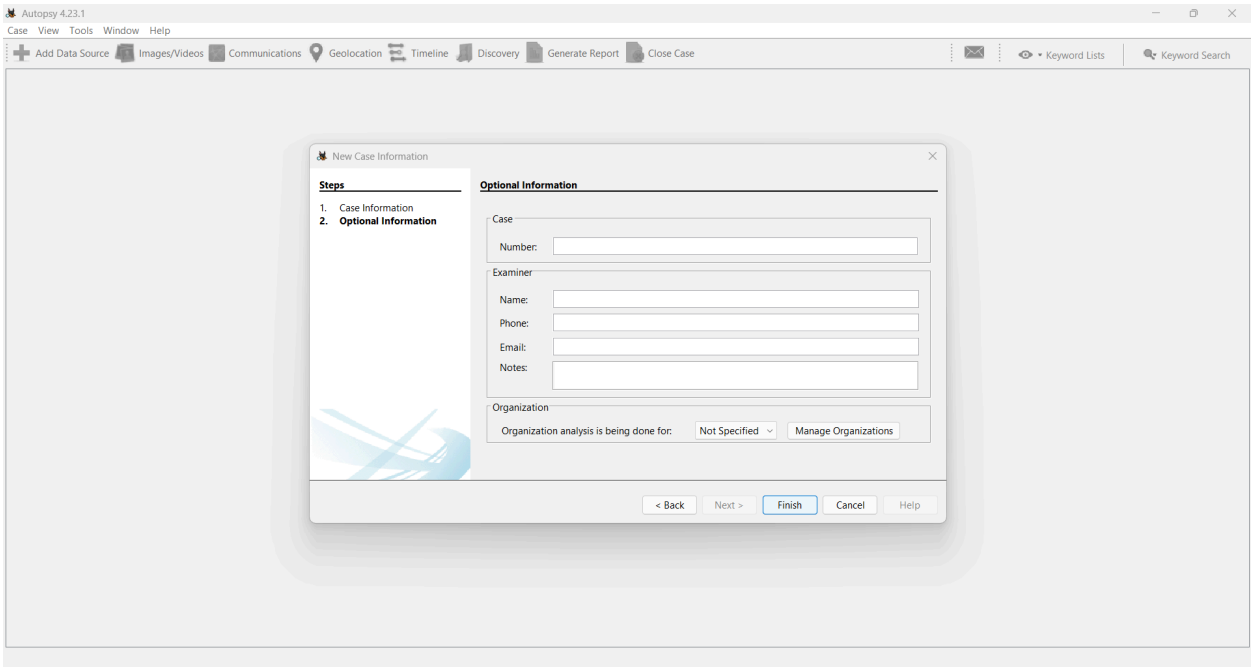
```
Ubuntu
wsl@TROY-LAPTOP:/mnt/f/CMU/FORENSICS CAPSTONE/H_$ ls
chappel.001  chappel.001.txt  chappel.002  chappel.003  chappel.004  chappel.005  chappel.006
wsl@TROY-LAPTOP:/mnt/f/CMU/FORENSICS CAPSTONE/H_$ mv chappel.001.txt chappel.txt
wsl@TROY-LAPTOP:/mnt/f/CMU/FORENSICS CAPSTONE/H_$ cat chappel.0* > chappel
wsl@TROY-LAPTOP:/mnt/f/CMU/FORENSICS CAPSTONE/H_$ cat chappel.txt
 Created By AccessData® FTK® Imager 4.2.0.13

Case Information:
Acquired using: ADI4.2.0.13
Case Number: 001
Evidence Number: 001-002
Unique description: chappel
Examiner:
Notes: phone

---------------------------------------------------------------

Information for H:\chappel:

Physical Evidentiary Item (Source) Information:
[Device Info]
 Source Type: Physical
[Drive Geometry]
 Cylinders: 16,644
 Heads: 16
 Sectors per Track: 63
 Bytes per Sector: 512
 Sector Count: 16,777,152
[Physical Drive Information]
 Drive Interface Type: ide
[Image]
 Image Type: VMWare Virtual Disk
 Source data size: 8191 MB
 Sector count:    16777152
[Computed Hashes]
 MD5 checksum:    97d5e433cefb4abe3e93ca642b7e527a
 SHA1 checksum:   92879469c10e40e26bac5be784c0b49fb5a8ad40

Image Information:
 Acquisition started:   Thu Feb  4 15:22:55 2021
```

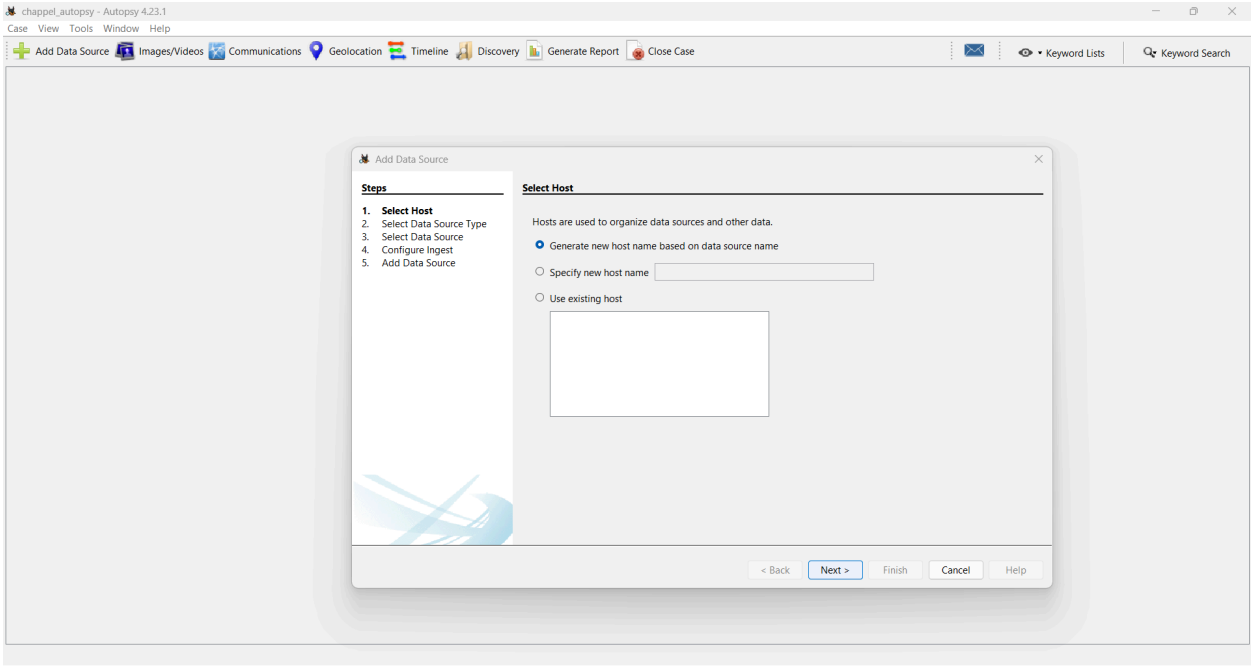


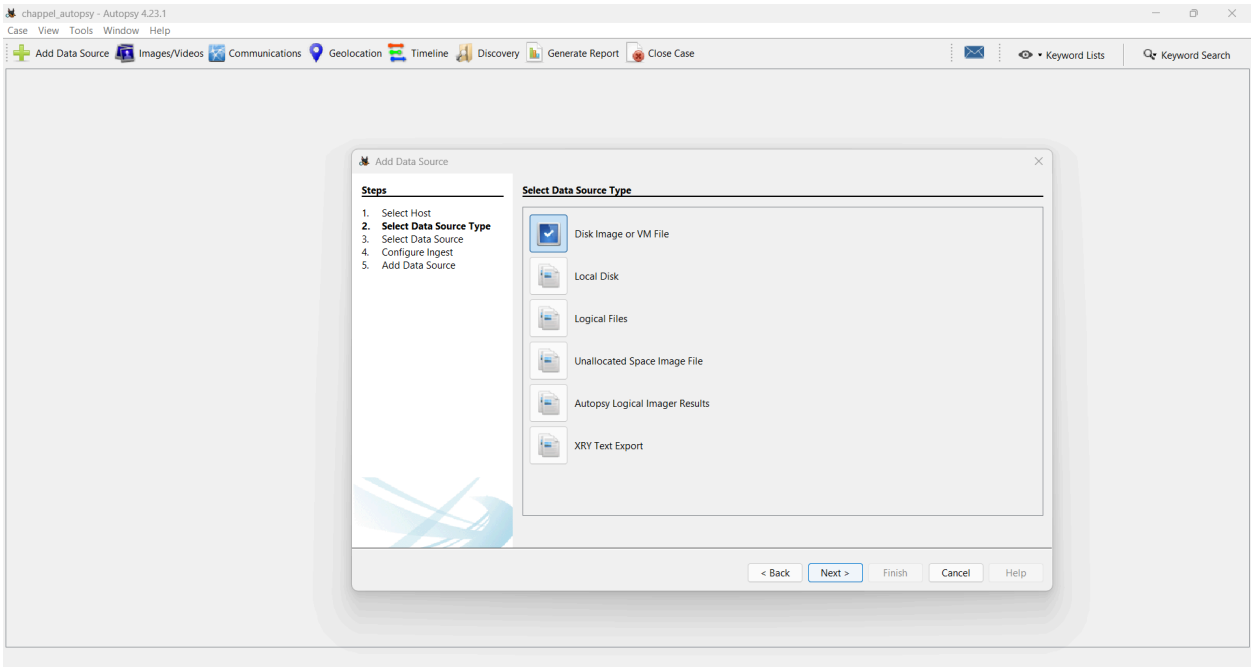
5.2.2.4 Chappel_A_0004 Autopsy Case

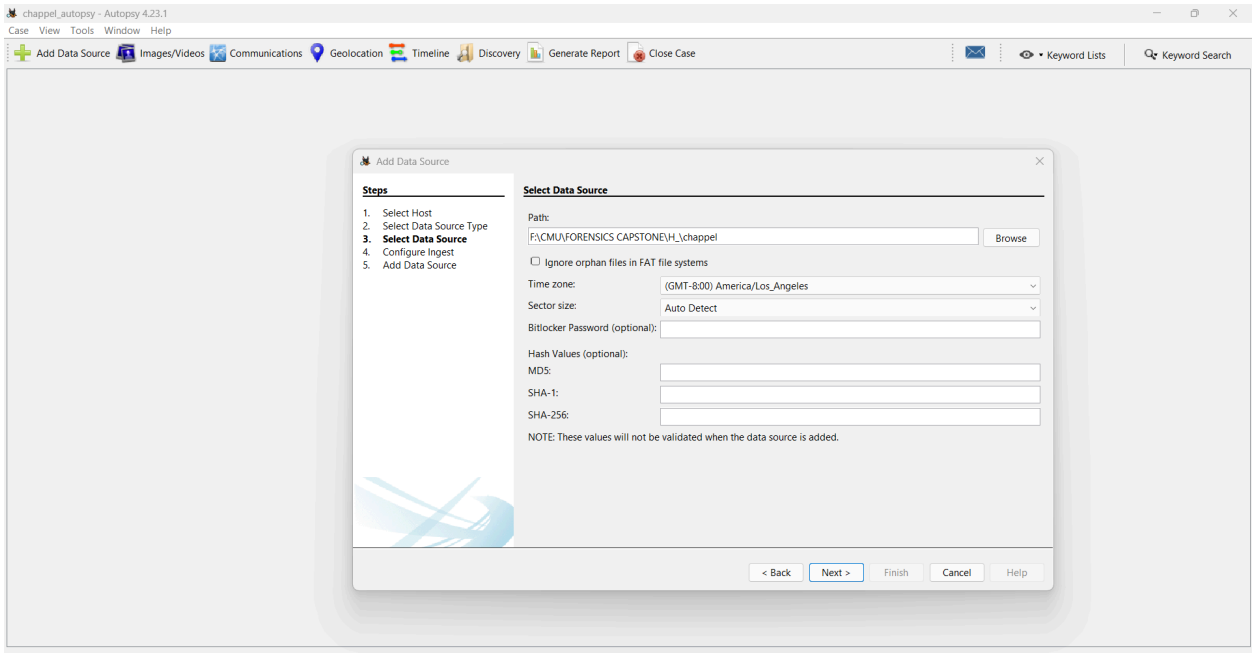


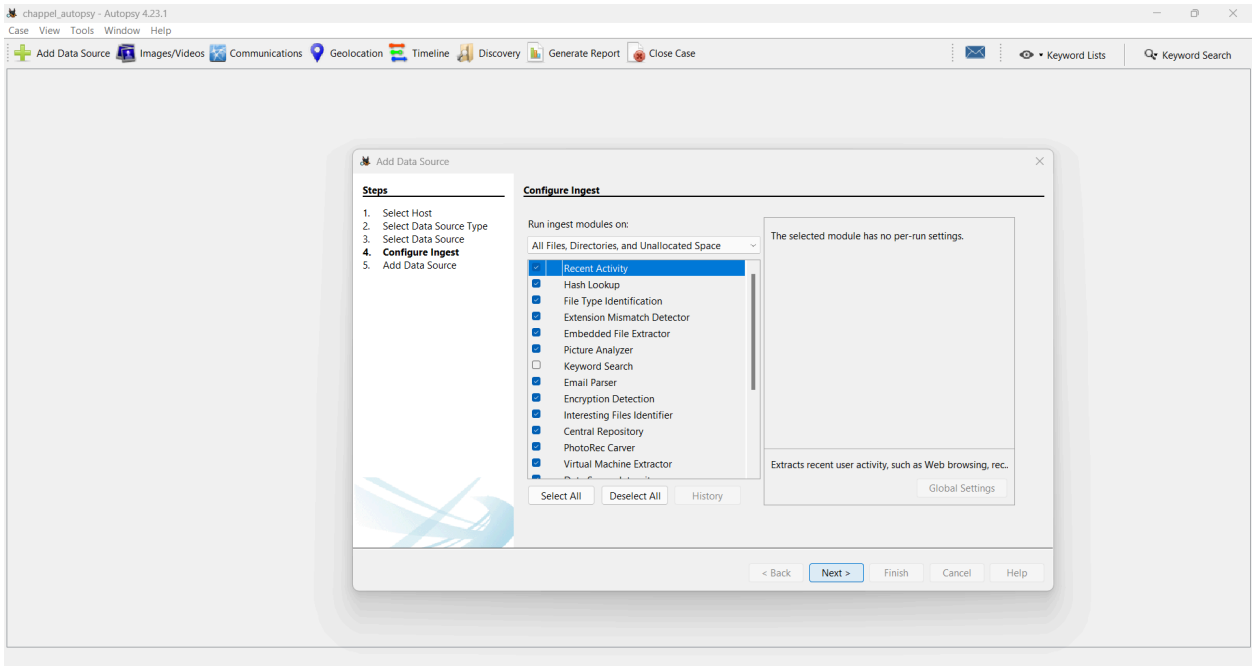

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Case Information
Case Name: chappel_autopsy
Base Directory: F:\CMU\FORENSICS CAPSTONE\H_
Browse
Case Type: Single-User Multi-User
Case data will be stored in the following directory:
F:\CMU\FORENSICS CAPSTONE\H_\chappel_autopsy
< Back Next > Finish Cancel Help

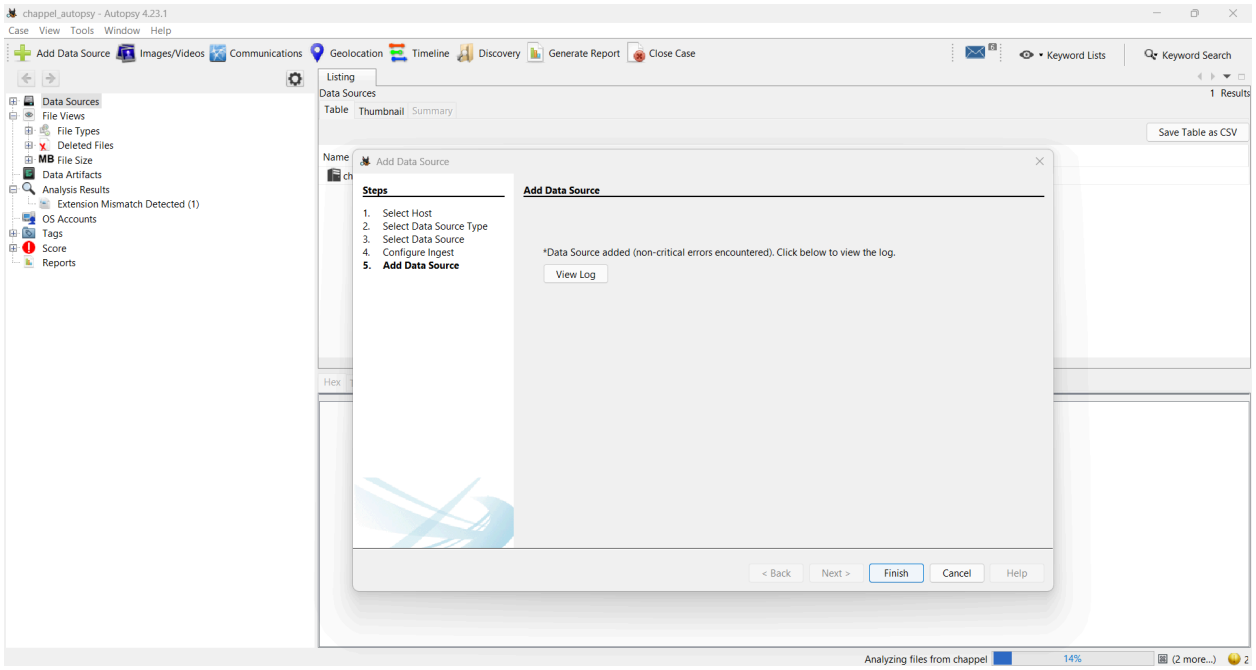

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Optional Information
Case
Number:
Examiner
Name:
Phone:
Email:
Notes:
Organization
Organization analysis is being done for: Not Specified Manage Organizations
< Back Next > Finish Cancel Help

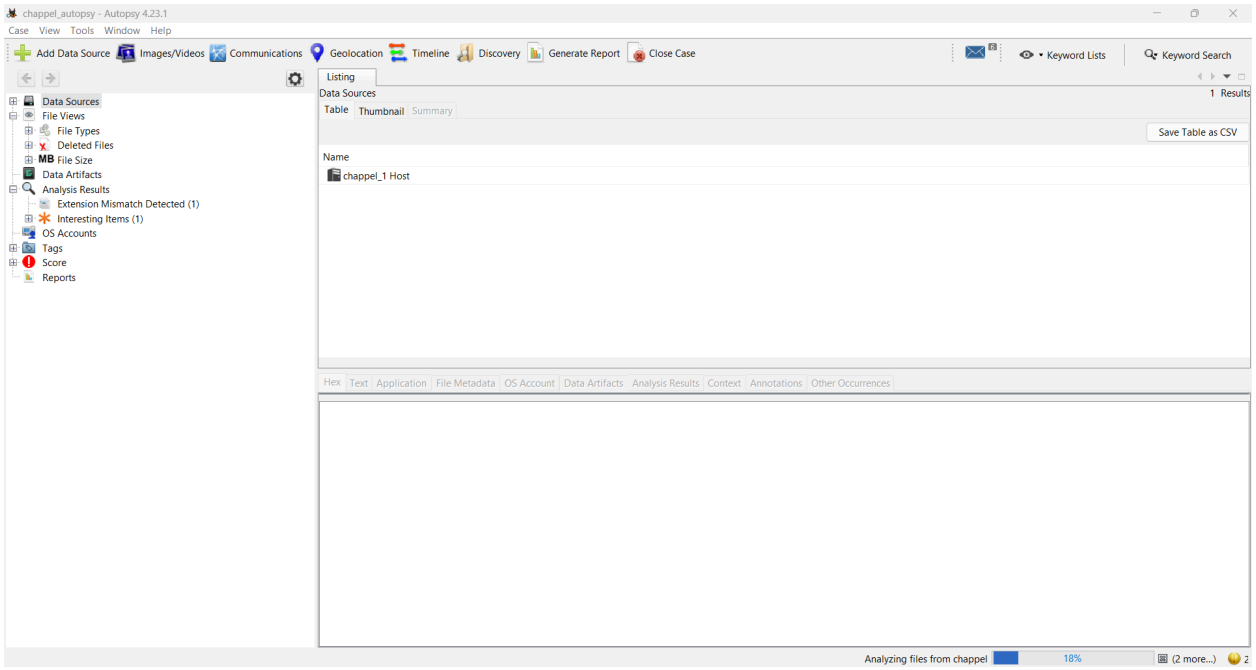

chappel_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Host
Hosts are used to organize data sources and other data.
Generate new host name based on data source name
Specify new host name
Use existing host
< Back
Next >
Finish
Cancel
Help


chappel_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source Type
Disk Image or VM File
Local Disk
Logical Files
Unallocated Space Image File
Autopsy Logical Imager Results
XRY Text Export
< Back
Next >
Finish
Cancel
Help

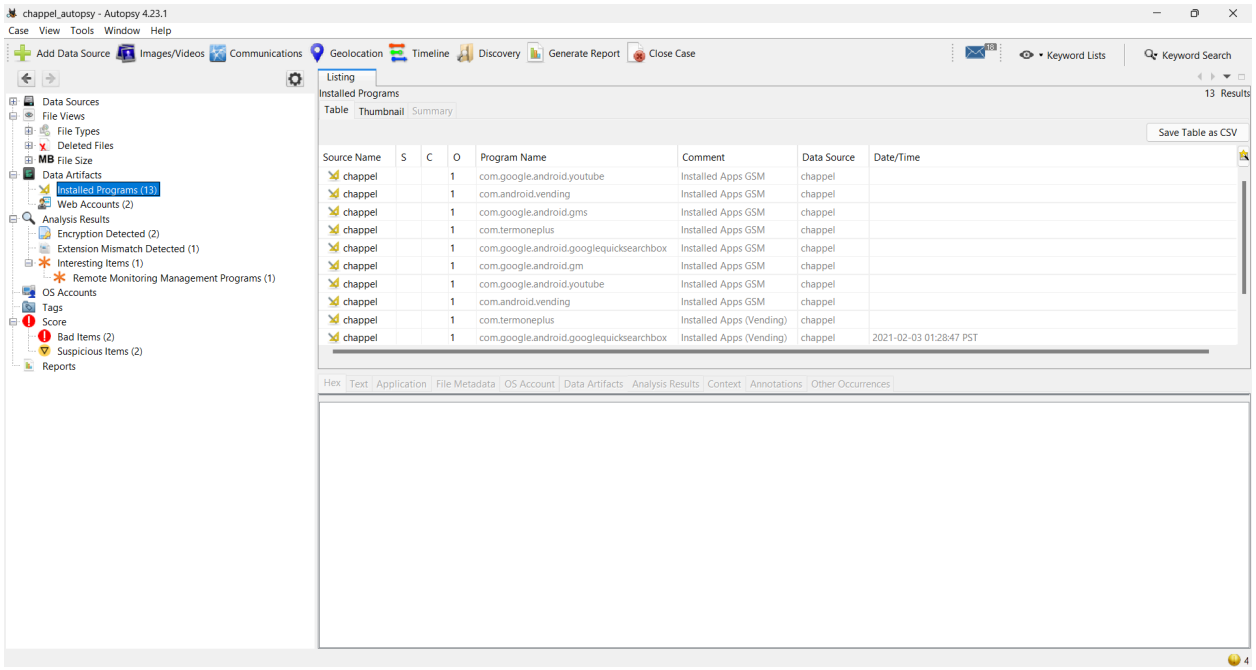

chappel_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source
Path:
F:\CMU\FORENSICS CAPSTONE\H_\chappel
Browse
Ignore orphan files in FAT file systems
Time zone:
(GMT-8:00) America/Los_Angeles
Sector size:
Auto Detect
Bitlocker Password (optional):
Hash Values (optional):
MD5:
SHA-1:
SHA-256:
NOTE: These values will not be validated when the data source is added.
< Back
Next >
Finish
Cancel
Help


chappel_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
Select All
Deselect All
History
The selected module has no per-run settings.
Extracts recent user activity, such as Web browsing, rec..
Global Settings
< Back
Next >
Finish
Cancel
Help

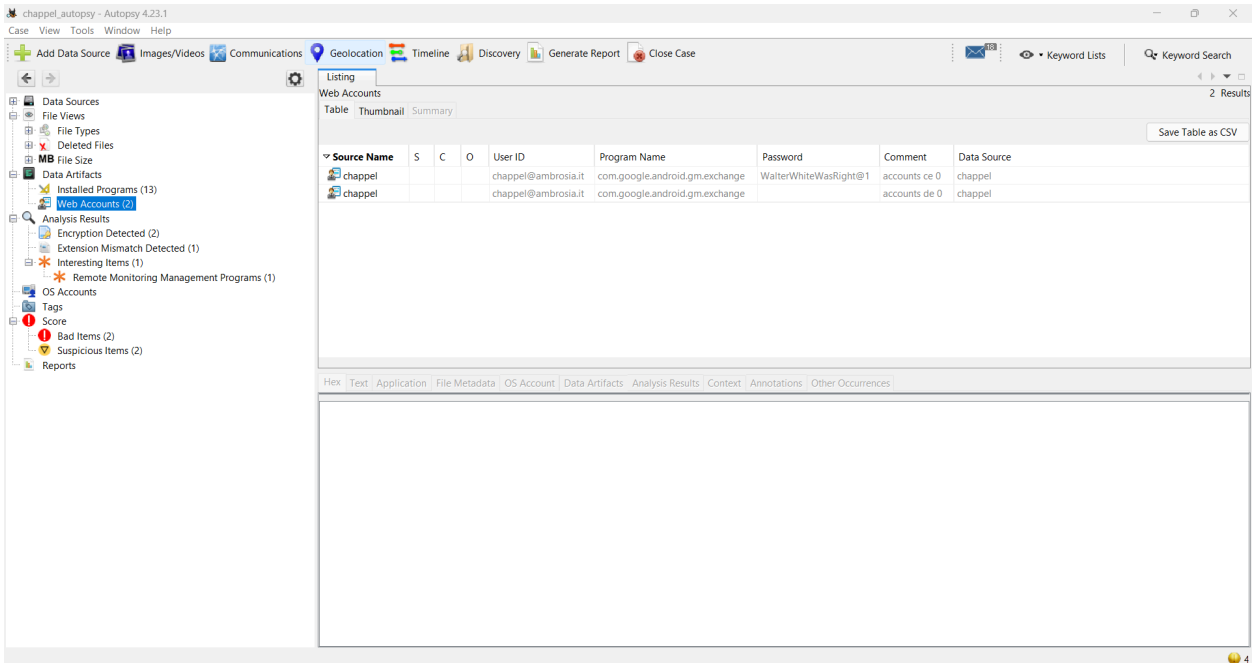

chappel_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Analysis Results
Extension Mismatch Detected (1)
OS Accounts
Tags
Score
Reports
Listing
Data Sources
1 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
*Data Source added (non-critical errors encountered). Click below to view the log.
View Log
< Back
Next >
Finish
Cancel
Help
Hex
Analyzing files from chappel
14%
(2 more...)


chappel_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Analysis Results
Extension Mismatch Detected (1)
Interesting Items (1)
OS Accounts
Tags
Score
Reports
Listing
Data Sources
1 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
chappel_1 Host
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Analyzing files from chappel
18%
(2 more...)

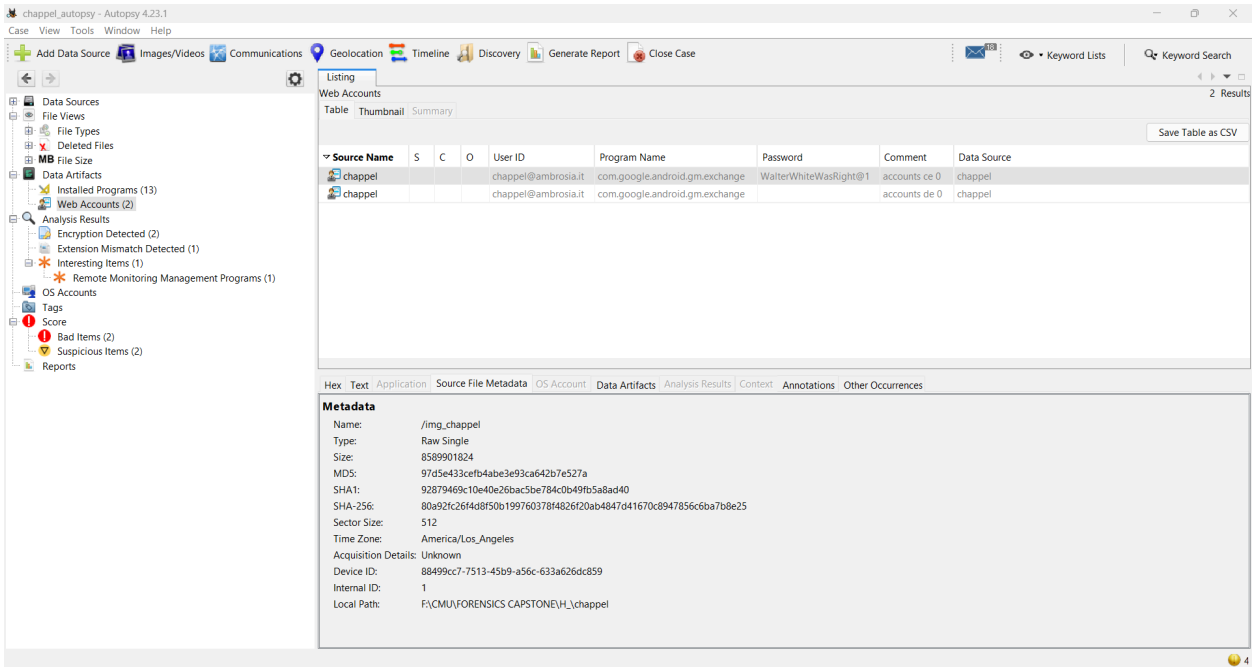
5.2.2.5 Chappel_A_0005 Operating System



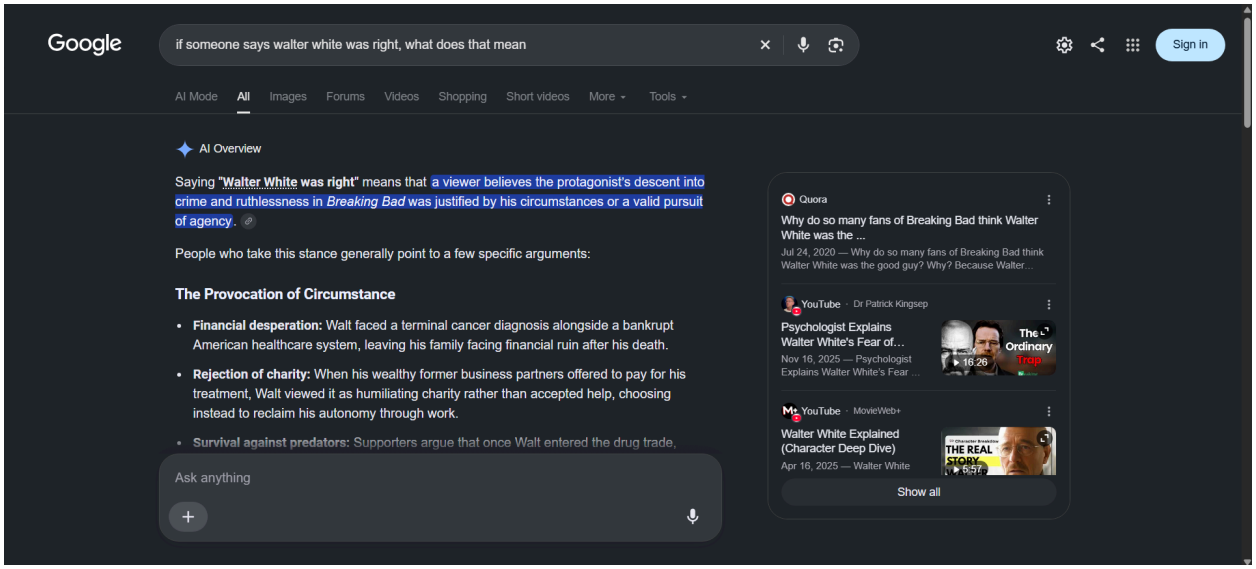
5.2.2.6 Chappel_A_0006 Web Accounts



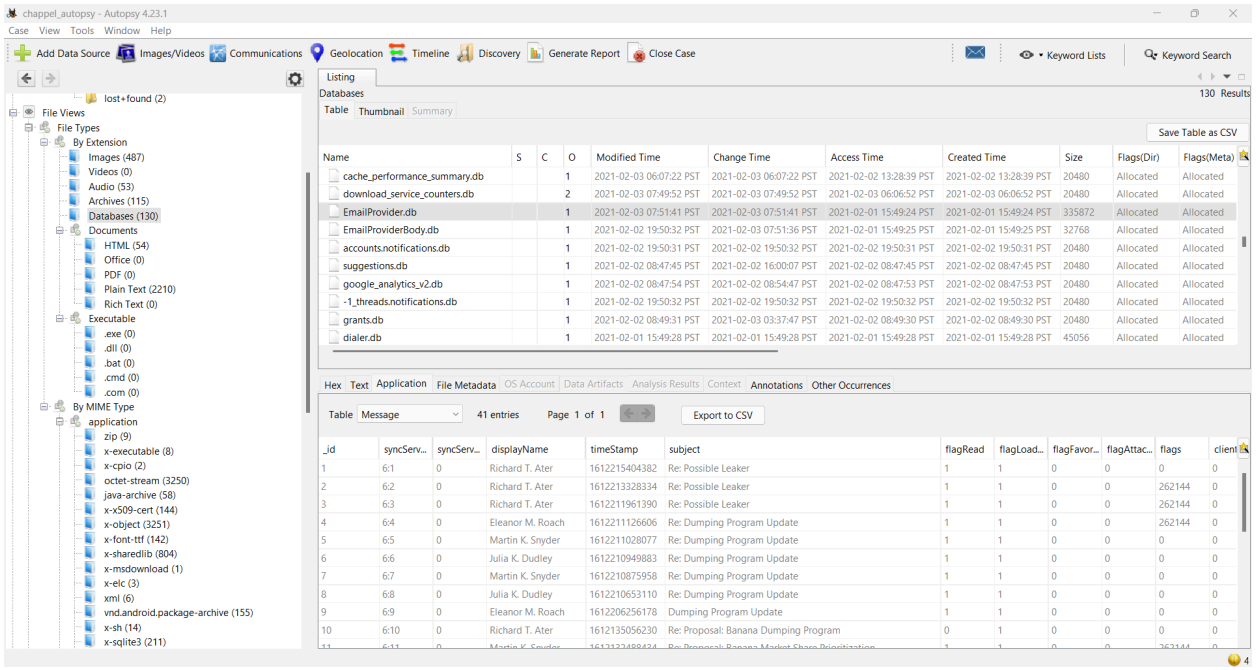
5.2.2.7 Chappel_A_0007 Walter White Password



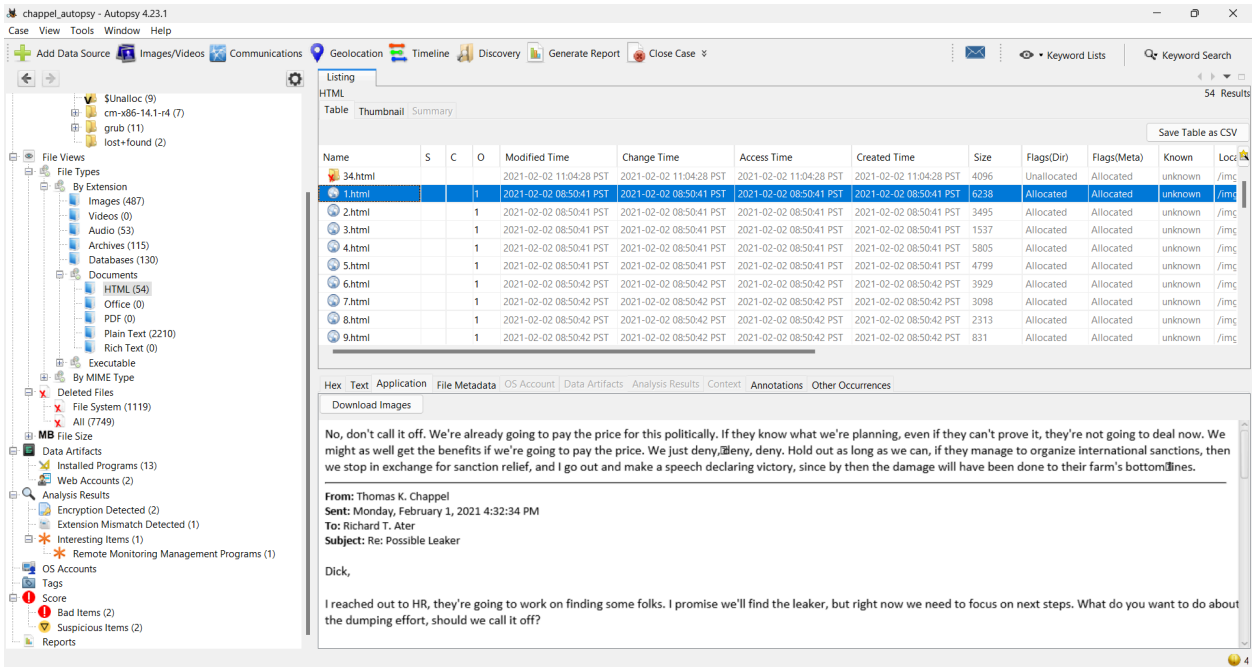
5.2.2.8 Chappel_A_0008 Walter White Google



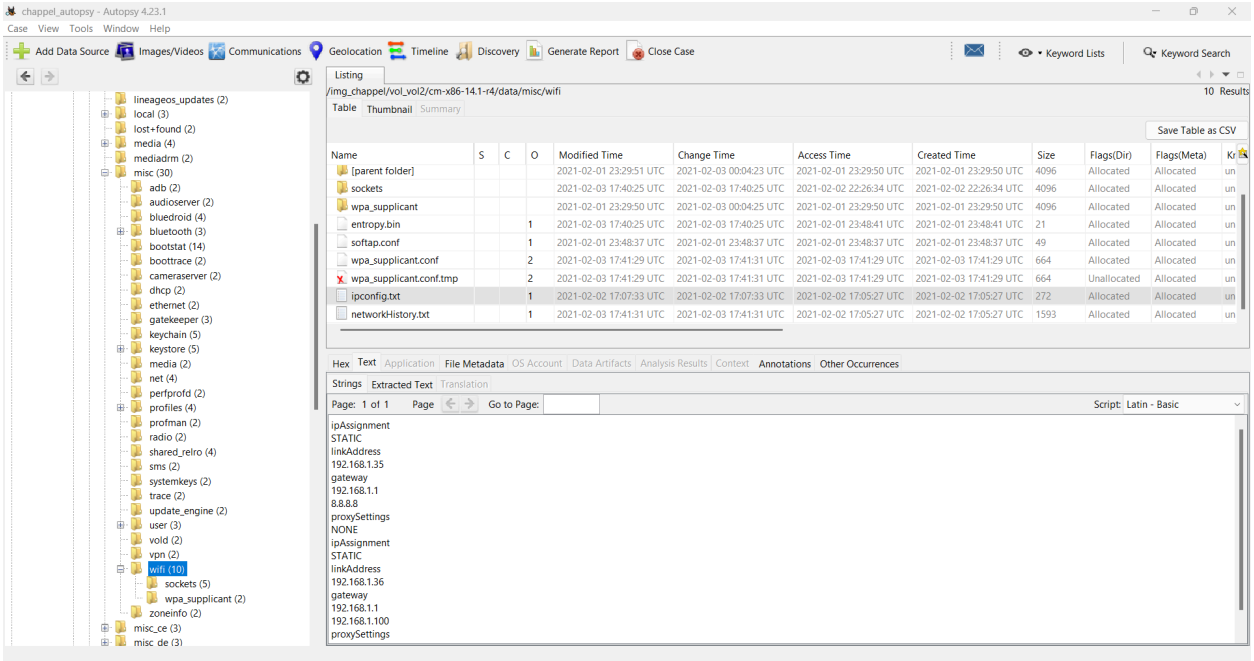
5.2.2.9 Chappel_A_0009 EmailProvider.db



5.2.2.10 Chappel_A_0010 Email HTML

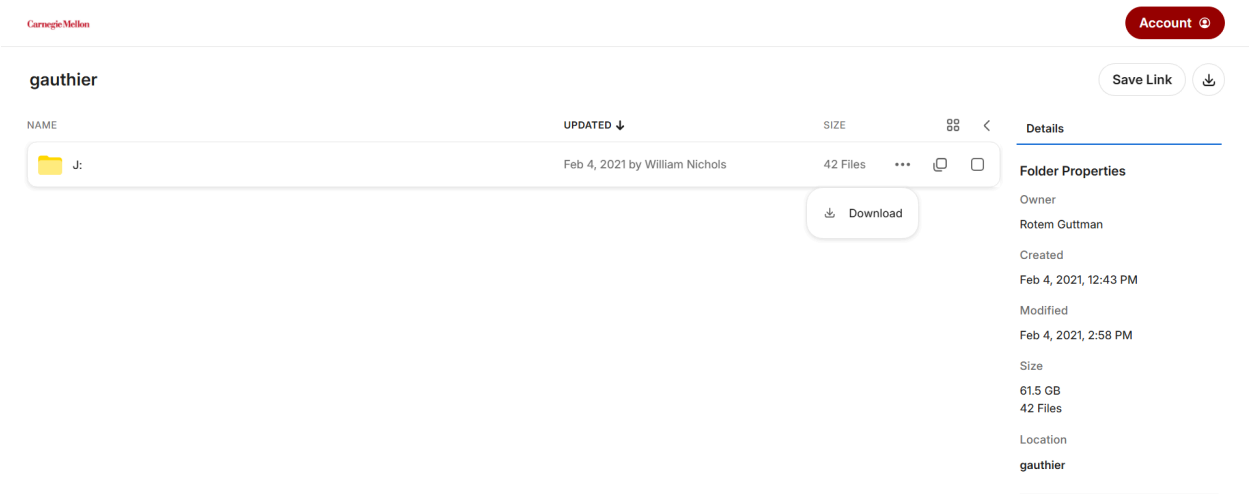


5.2.2.11 Chappel_A_0011 IPconfig.txt

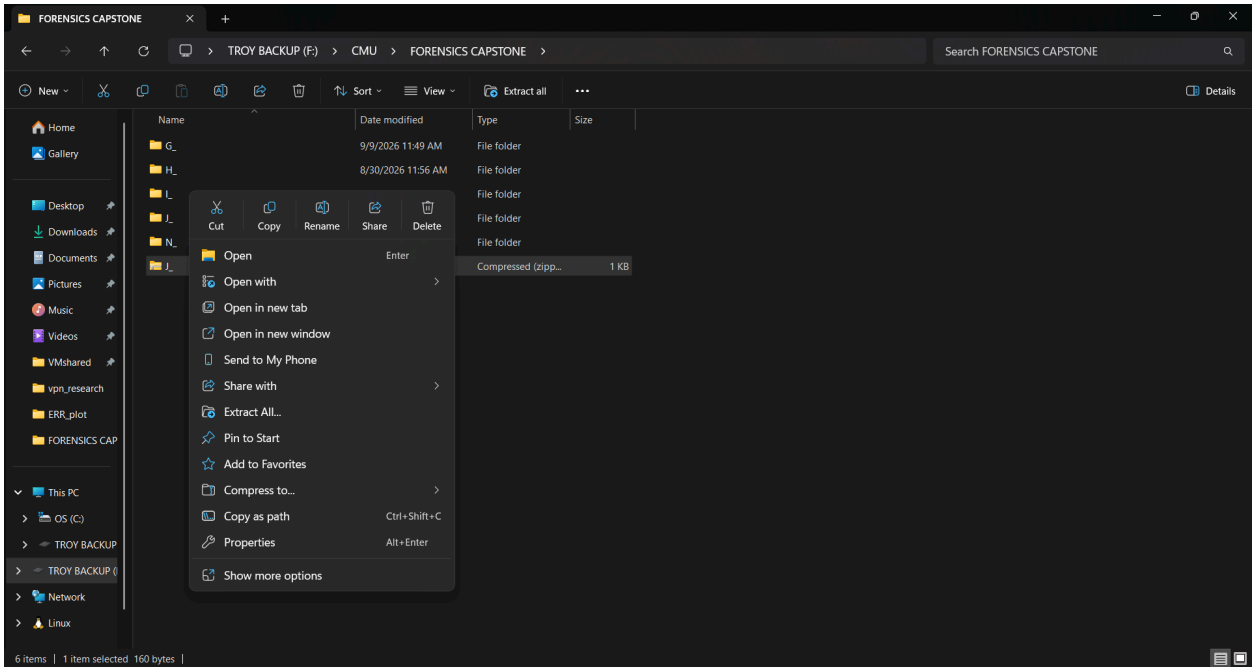


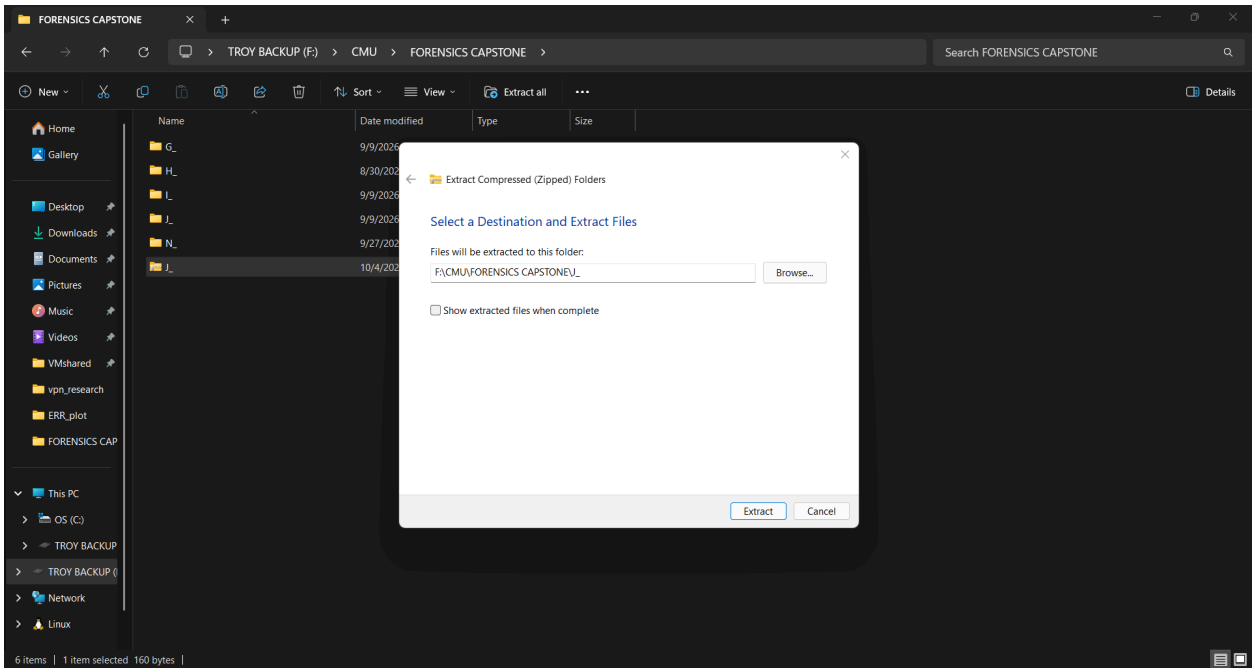
5.2.3 Gauthier Image (J)

5.2.3.1 Gauthier_A_0001 Image Download

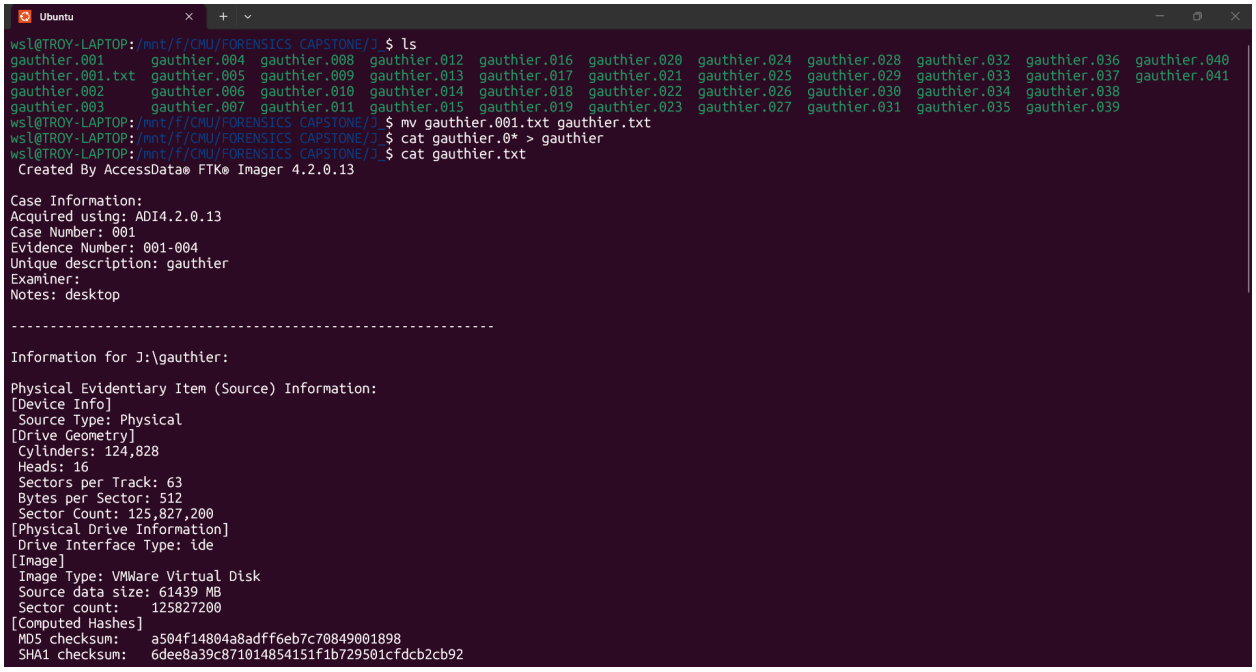


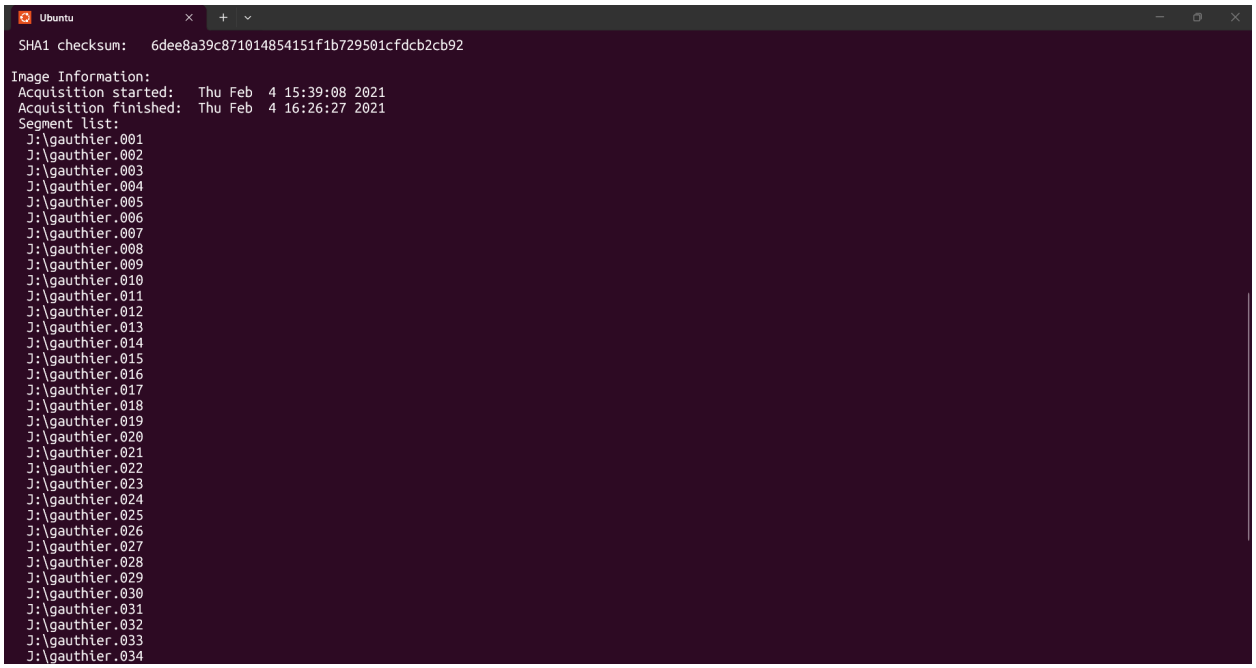
5.2.3.2 Gauthier_A_0002 Image Extraction

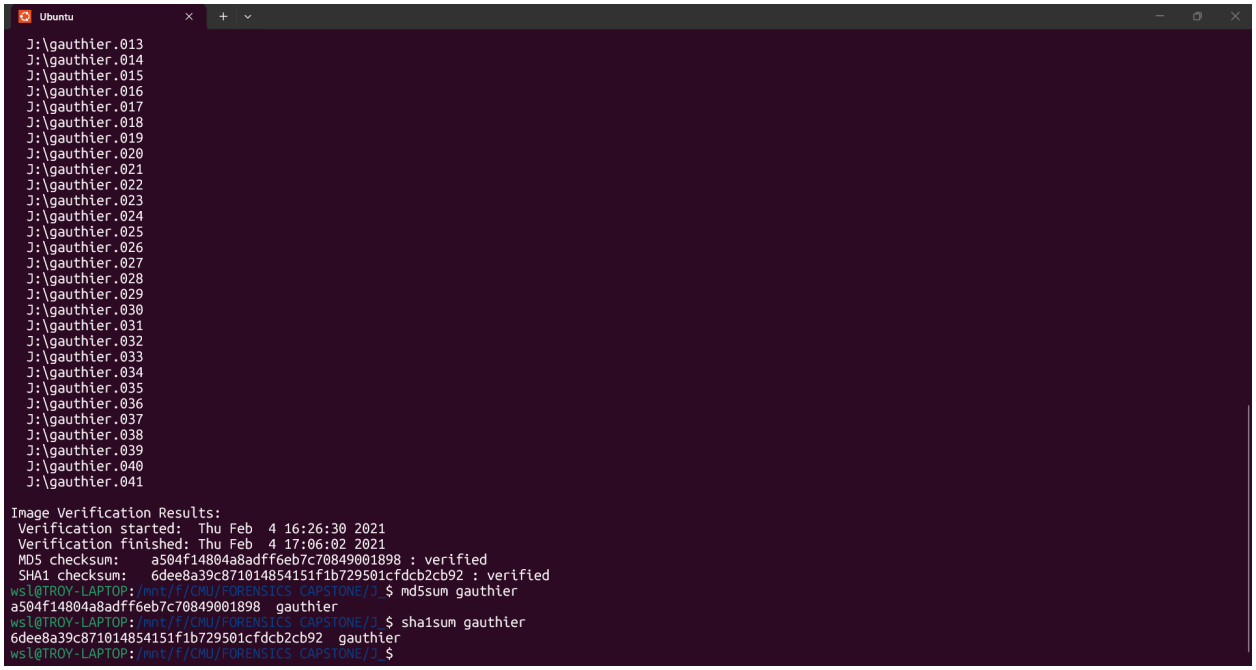




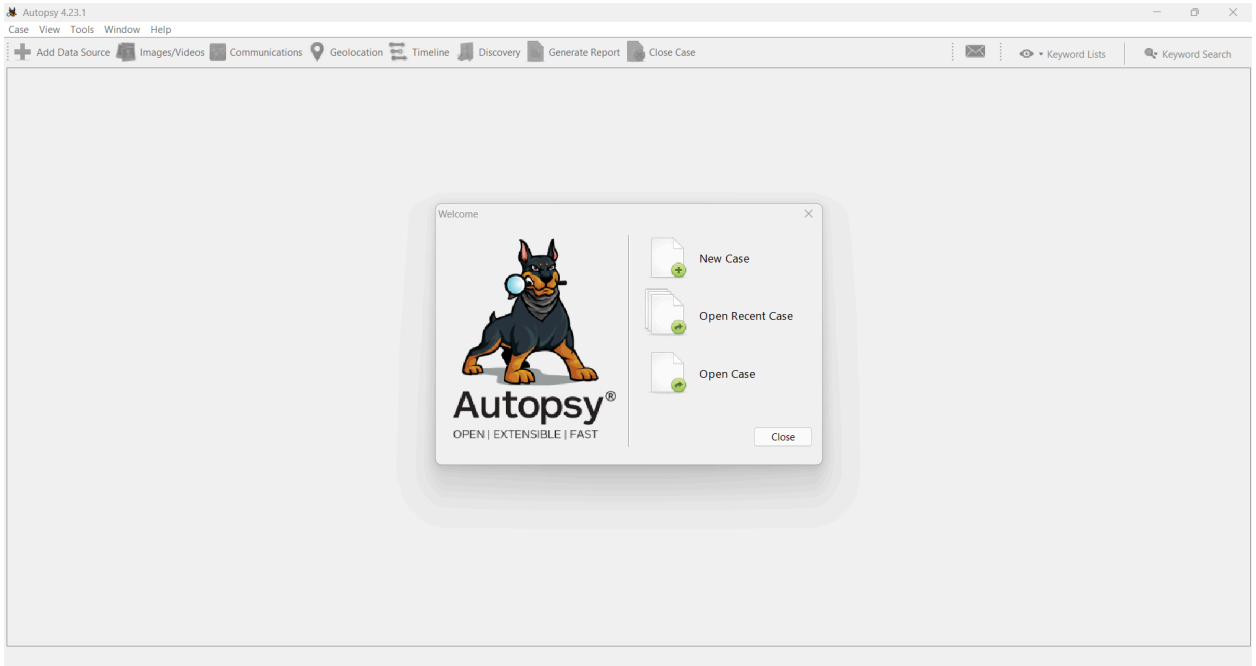
5.2.3.3 Gauthier_A_0003 Image Verification

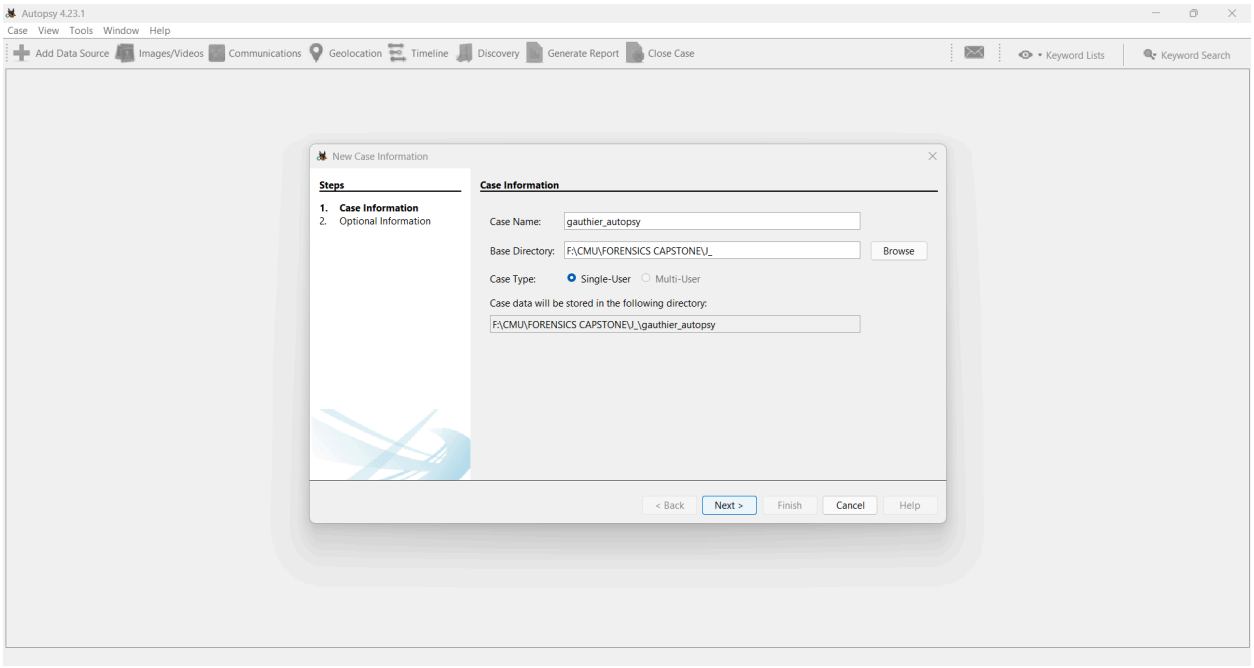


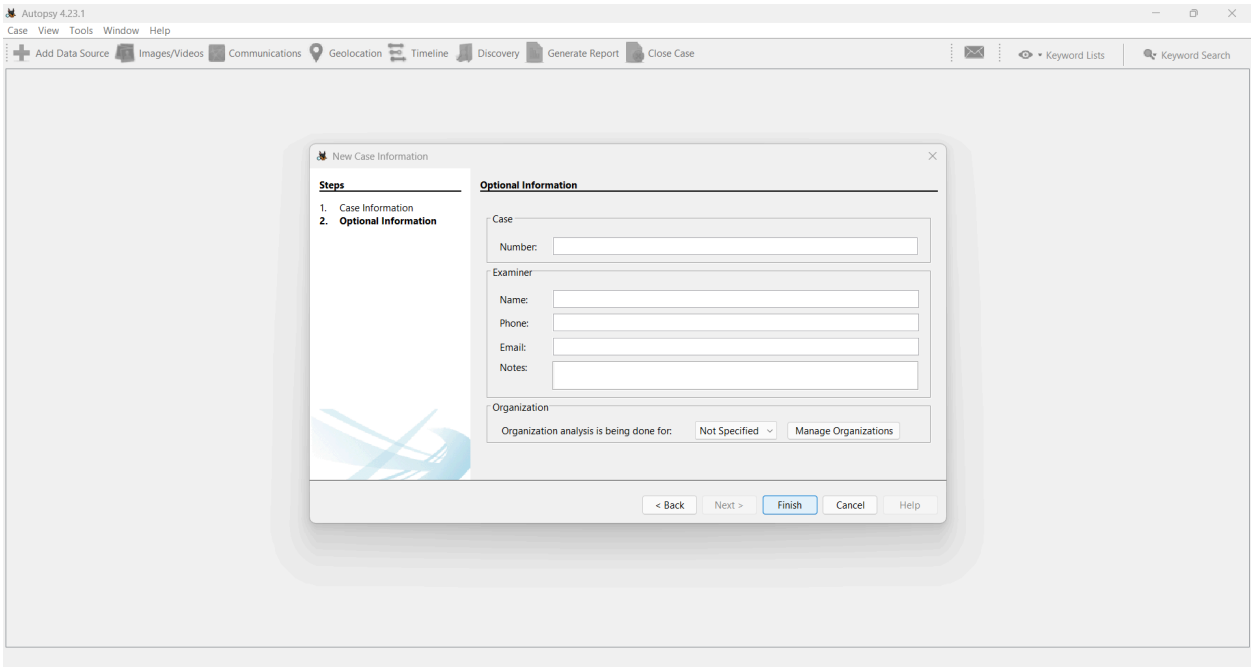


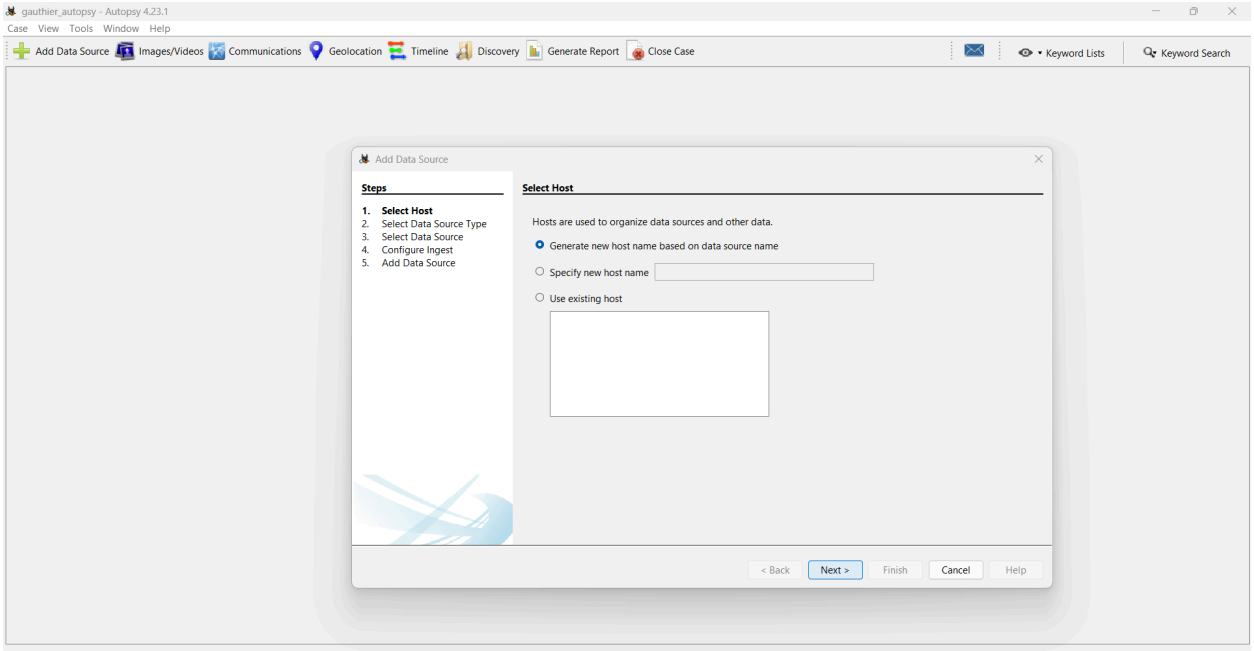


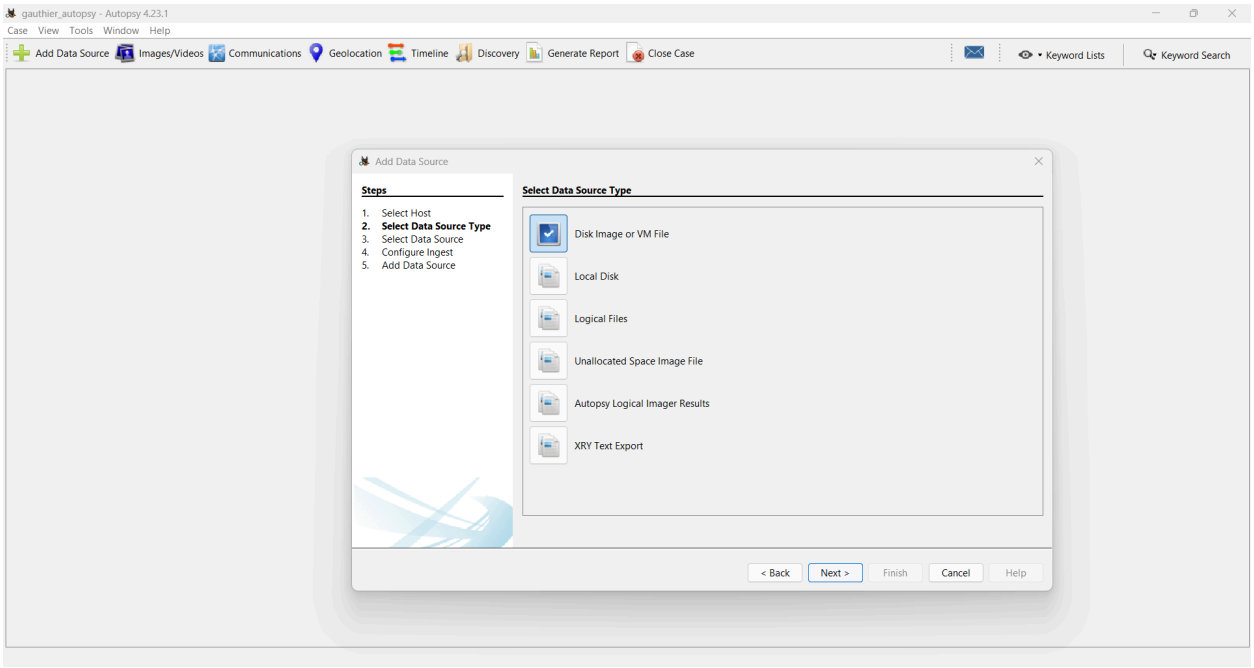
5.2.3.4 Gauthier_A_0004 Autopsy Case

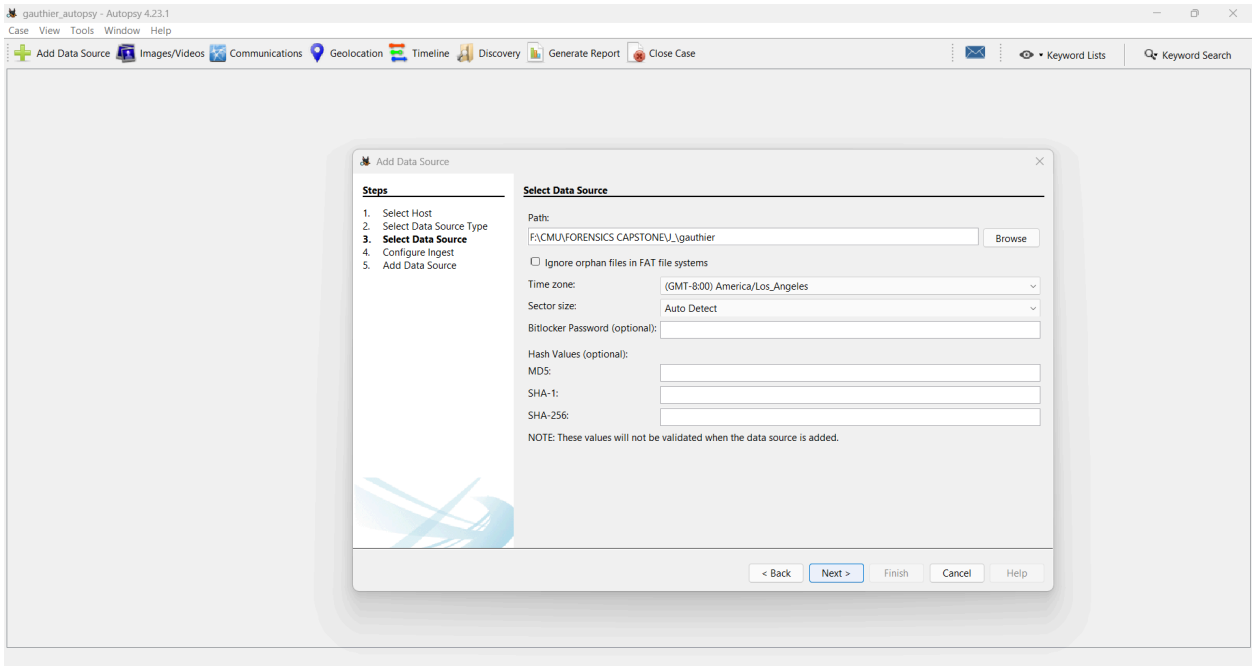


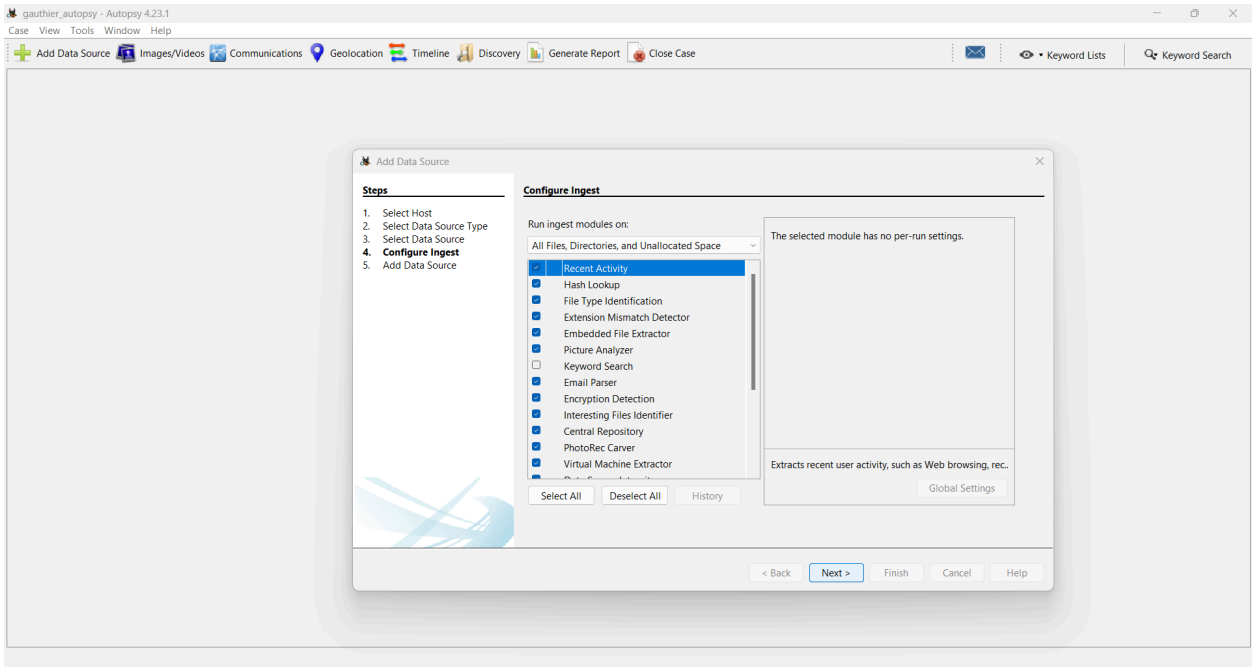

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Case Information
Case Name: gauthier_autopsy
Base Directory: F:\CMU\FORENSICS CAPSTONE\J_
Browse
Case Type: Single-User Multi-User
Case data will be stored in the following directory:
F:\CMU\FORENSICS CAPSTONE\J_\gauthier_autopsy
< Back Next > Finish Cancel Help

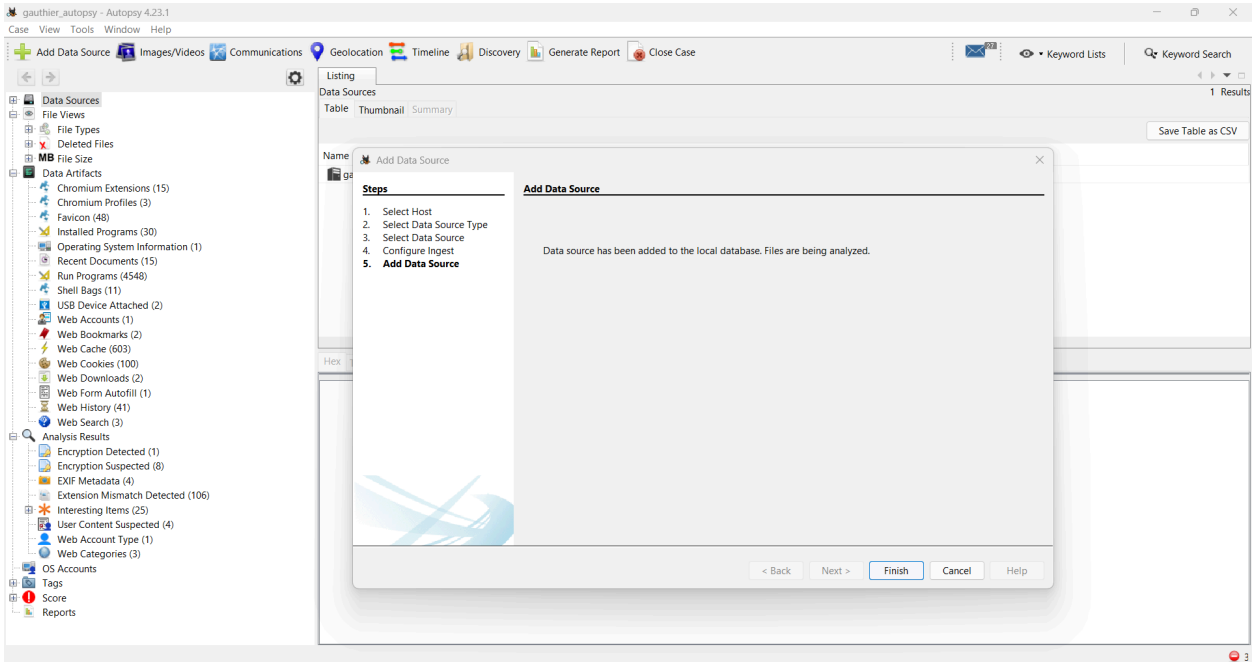

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Optional Information
Case
Number:
Examiner
Name:
Phone:
Email:
Notes:
Organization
Organization analysis is being done for: Not Specified Manage Organizations
< Back Next > Finish Cancel Help

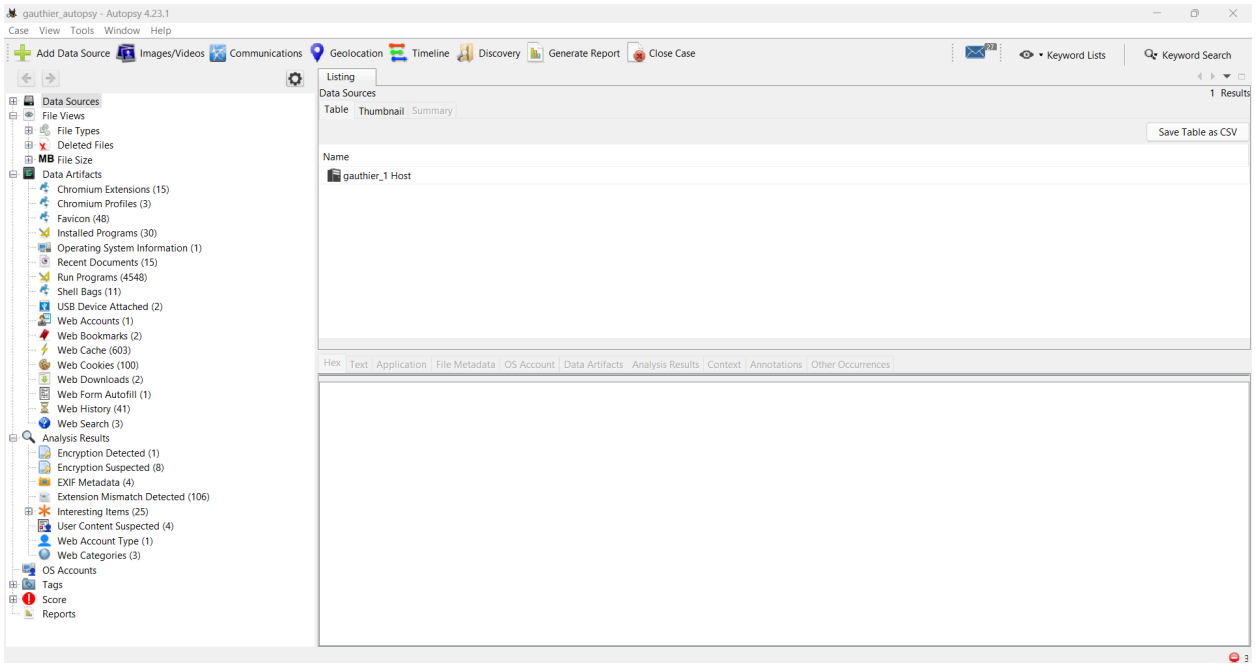

gauthier_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Host
Hosts are used to organize data sources and other data.
Generate new host name based on data source name
Specify new host name
Use existing host
< Back
Next >
Finish
Cancel
Help


gauthier_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source Type
Disk Image or VM File
Local Disk
Logical Files
Unallocated Space Image File
Autopsy Logical Imager Results
XRY Text Export
< Back
Next >
Finish
Cancel
Help

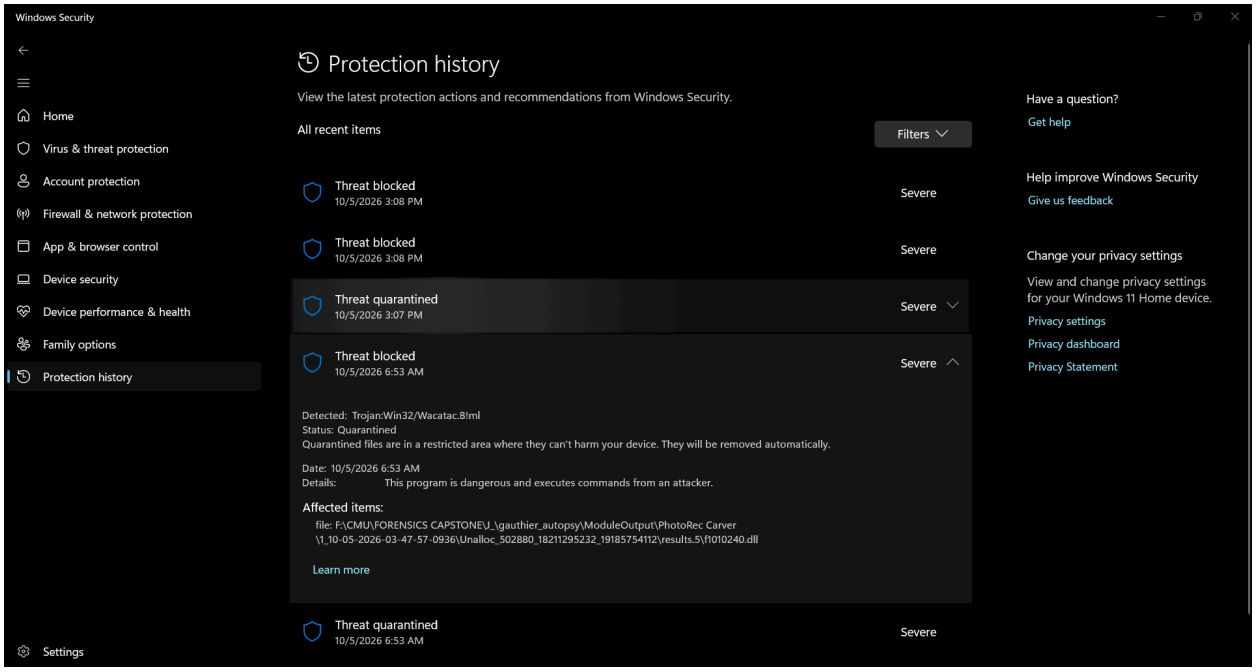

gauthier_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source
Path:
F:\CMU\FORENSICS CAPSTONE\J_\gauthier
Browse
Ignore orphan files in FAT file systems
Time zone:
(GMT-8:00) America/Los_Angeles
Sector size:
Auto Detect
Bitlocker Password (optional):
Hash Values (optional):
MD5:
SHA-1:
SHA-256:
NOTE: These values will not be validated when the data source is added.
< Back
Next >
Finish
Cancel
Help

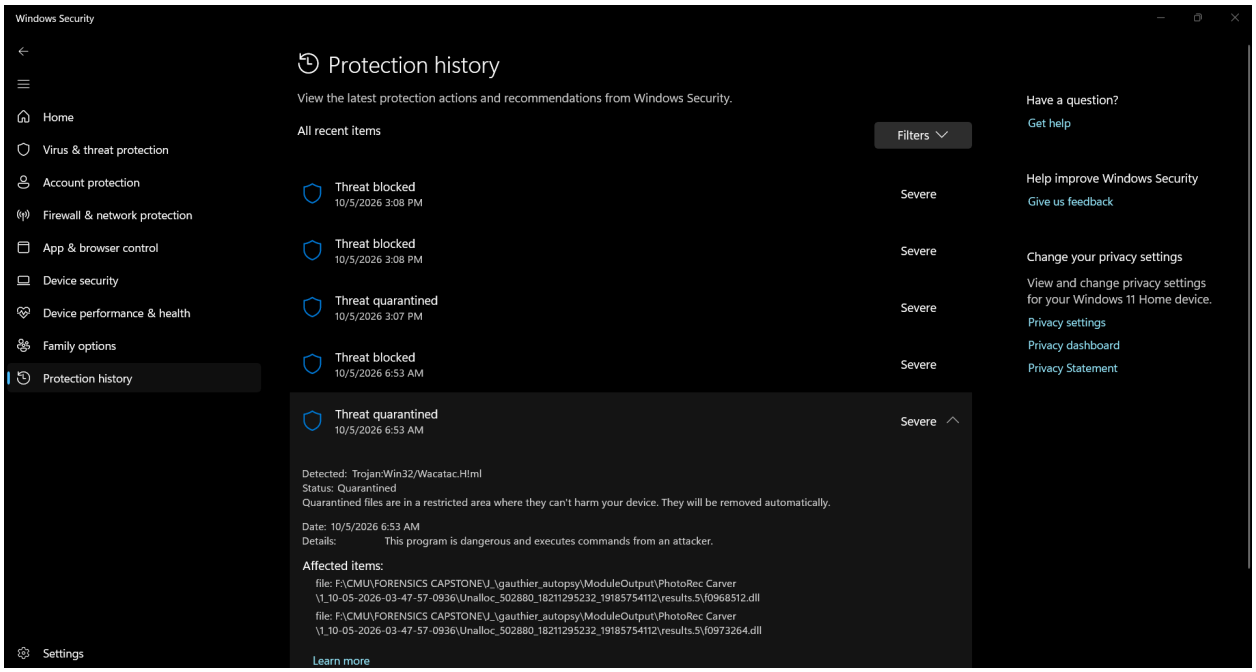

gauthier_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
Select All
Deselect All
History
The selected module has no per-run settings.
Extracts recent user activity, such as Web browsing, rec..
Global Settings
< Back
Next >
Finish
Cancel
Help

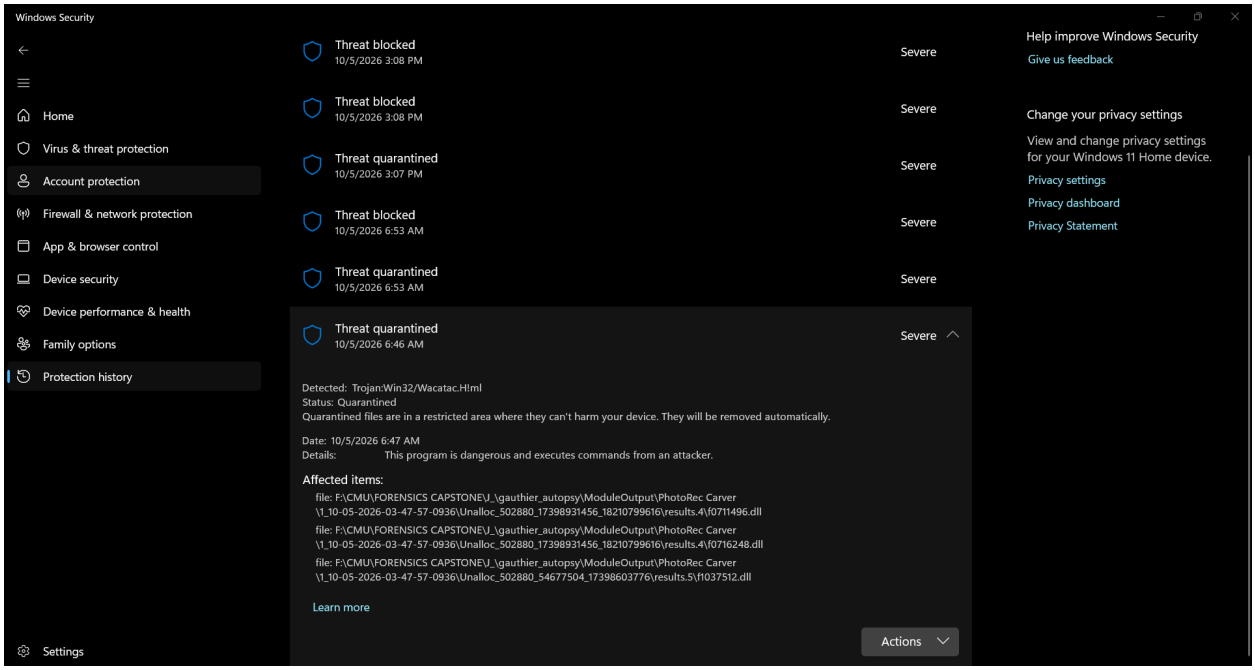

gauthier_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Listing
Data Sources
1 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
Data source has been added to the local database. Files are being analyzed.
< Back
Next >
Finish
Cancel
Help
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Chromium Extensions (15)
Chromium Profiles (3)
Favicon (48)
Installed Programs (30)
Operating System Information (1)
Recent Documents (15)
Run Programs (4548)
Shell Bags (11)
USB Device Attached (2)
Web Accounts (1)
Web Bookmarks (2)
Web Cache (603)
Web Cookies (100)
Web Downloads (2)
Web Form Autofill (1)
Web History (41)
Web Search (3)
Analysis Results
Encryption Detected (1)
Encryption Suspected (8)
EXIF Metadata (4)
Extension Mismatch Detected (106)
Interesting Items (25)
User Content Suspected (4)
Web Account Type (1)
Web Categories (3)
OS Accounts
Tags
Score
Reports


gauthier_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Listing
Data Sources
1 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
gauthier_1 Host
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Chromium Extensions (15)
Chromium Profiles (3)
Favicon (48)
Installed Programs (30)
Operating System Information (1)
Recent Documents (15)
Run Programs (4548)
Shell Bags (11)
USB Device Attached (2)
Web Accounts (1)
Web Bookmarks (2)
Web Cache (603)
Web Cookies (100)
Web Downloads (2)
Web Form Autofill (1)
Web History (41)
Web Search (3)
Analysis Results
Encryption Detected (1)
Encryption Suspected (8)
EXIF Metadata (4)
Extension Mismatch Detected (106)
Interesting Items (25)
User Content Suspected (4)
Web Account Type (1)
Web Categories (3)
OS Accounts
Tags
Score
Reports

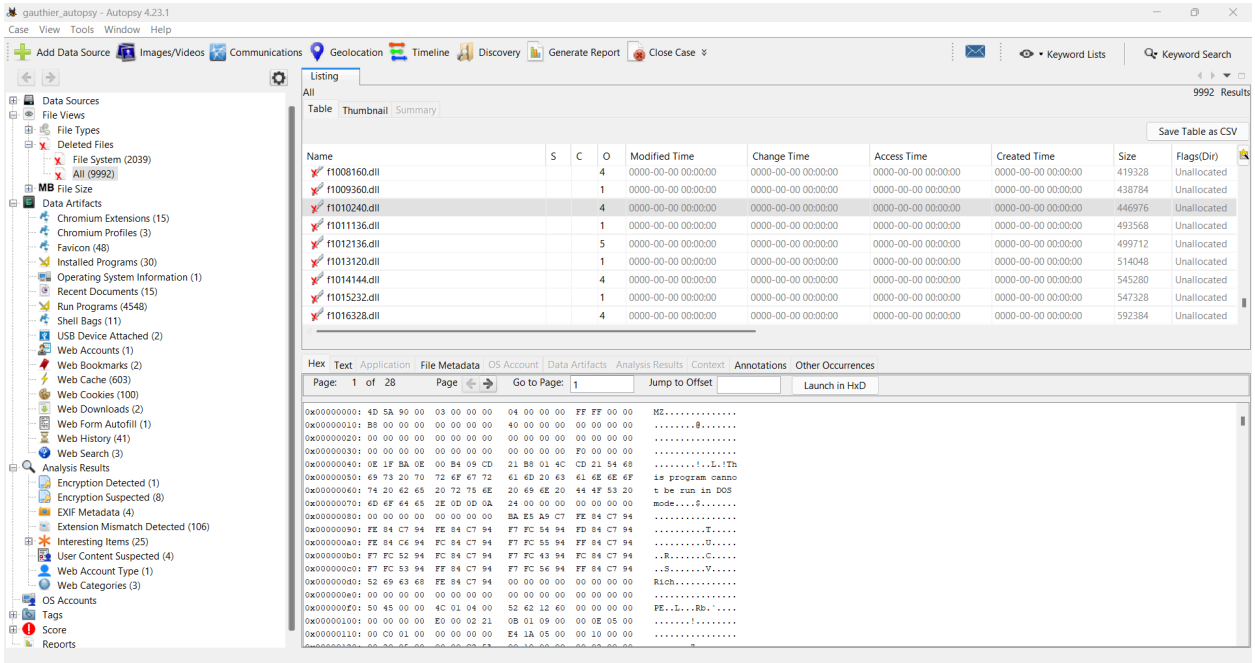
5.2.3.5 Gauthier_A_0005 Windows Security



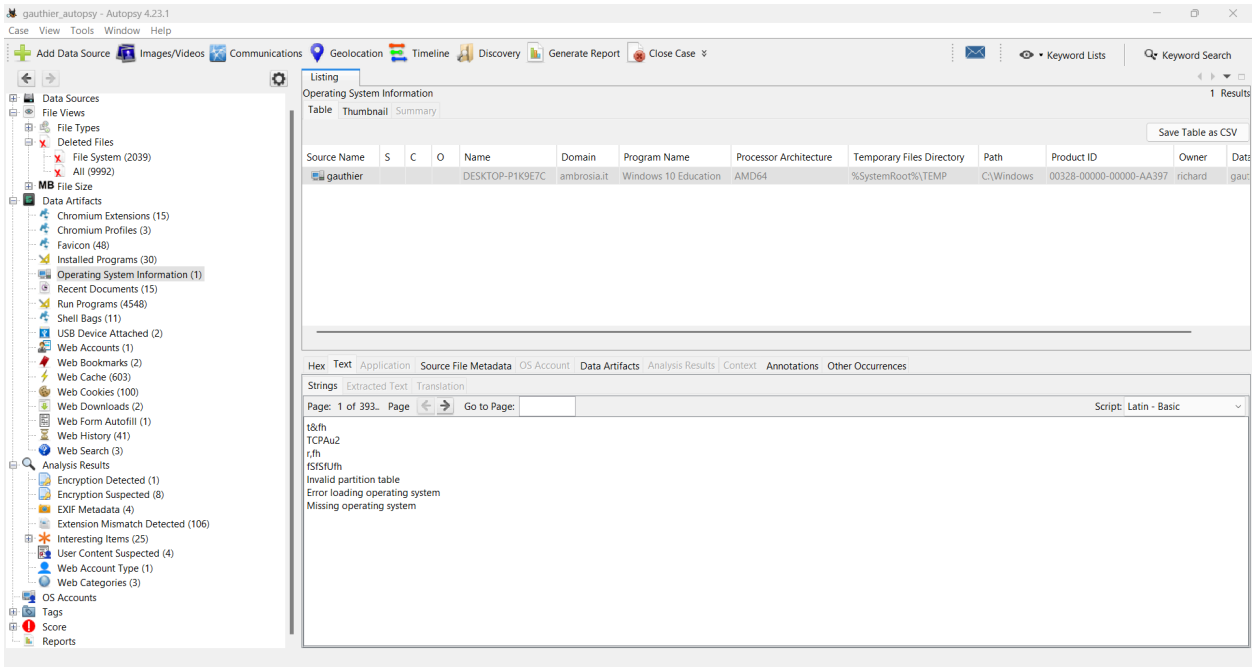




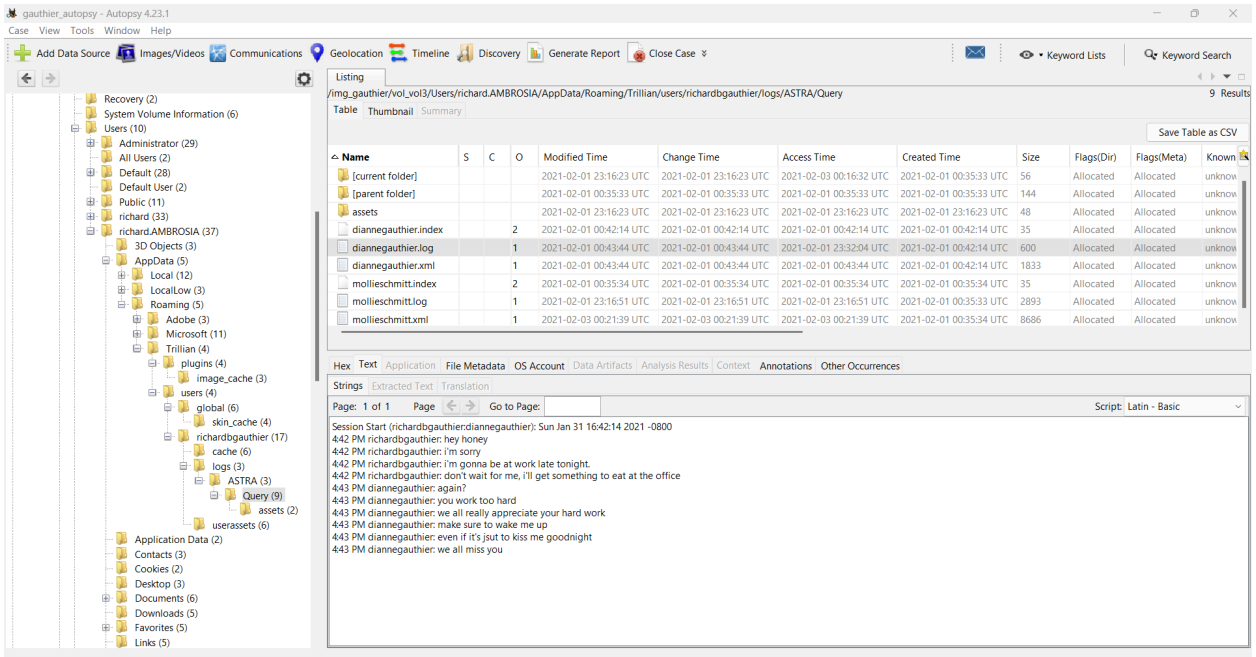
5.2.3.6 Gauthier_A_0006 Windows Security Flagged Files



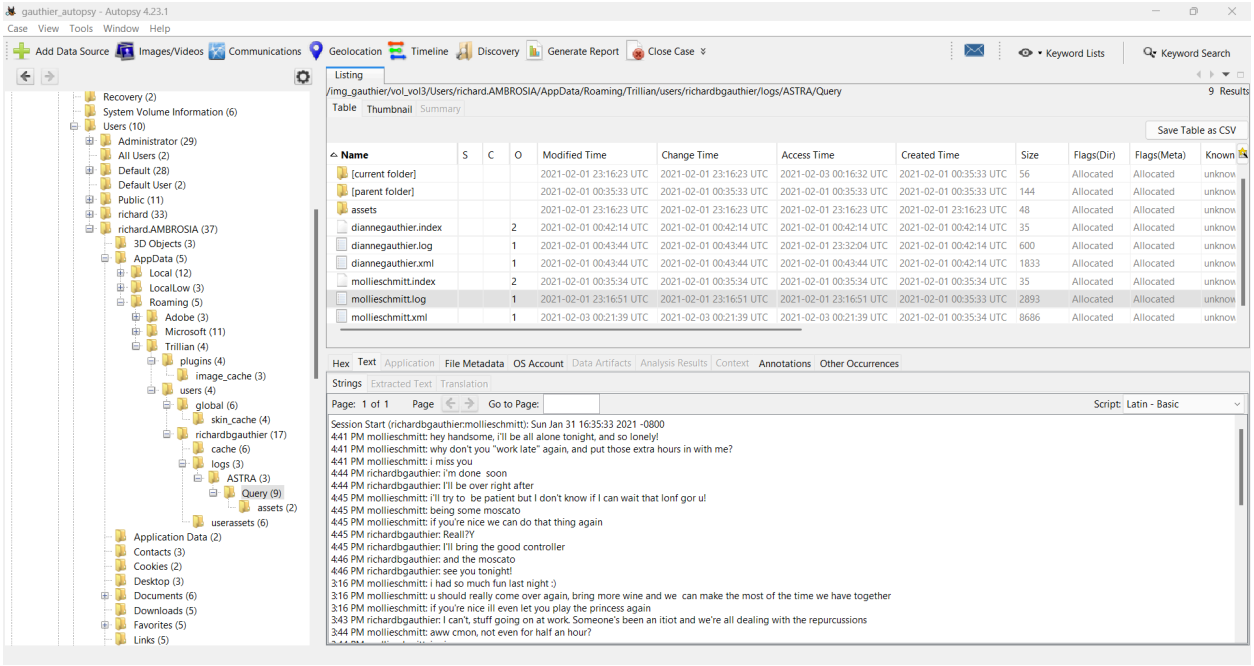
5.2.3.7 Gauthier_A_0007 Operating System Information



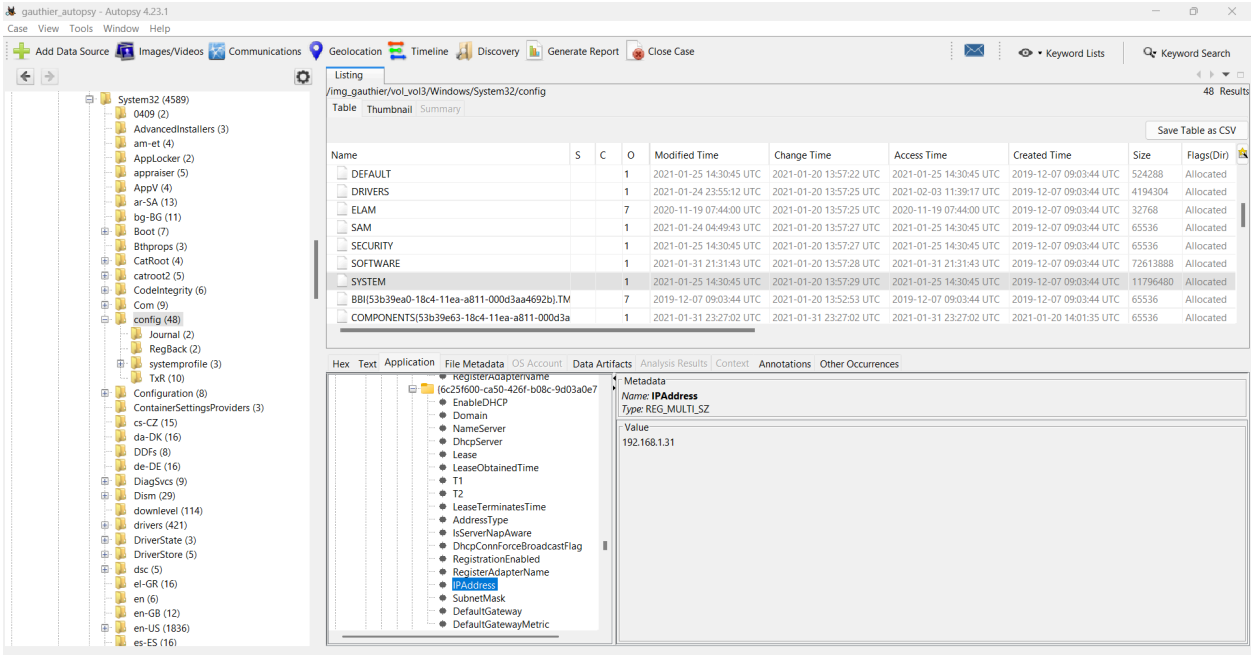
5.2.3.8 Gauthier_A_0008 Trillian diannegauthier log

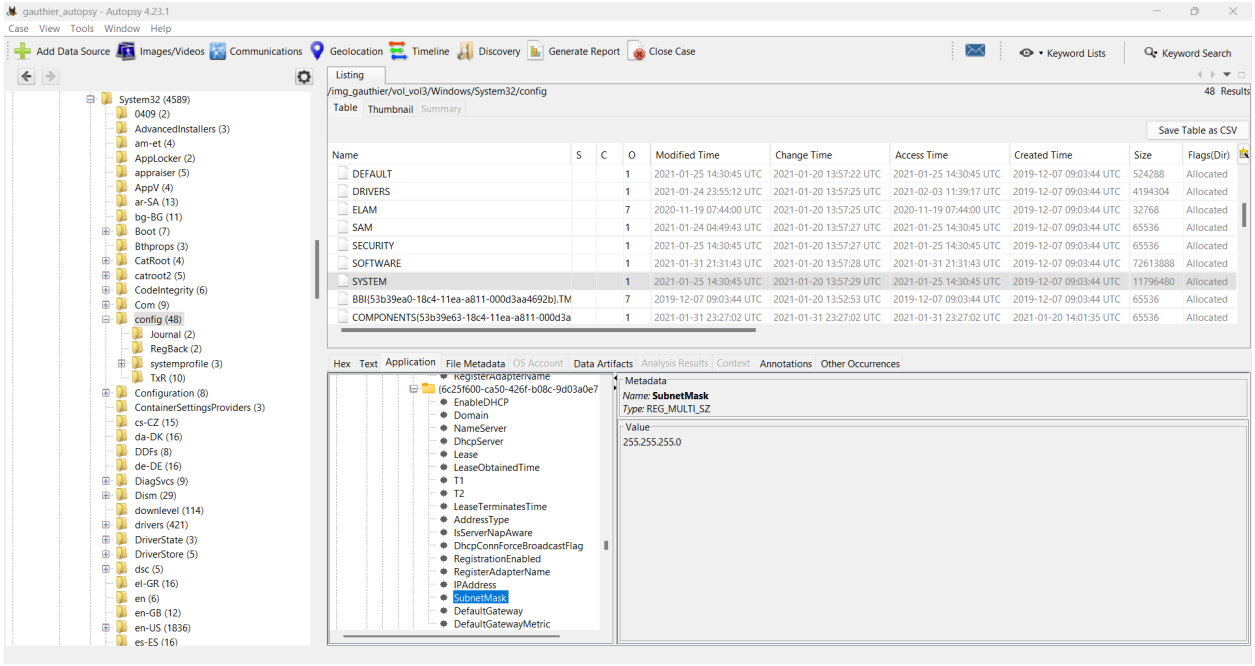


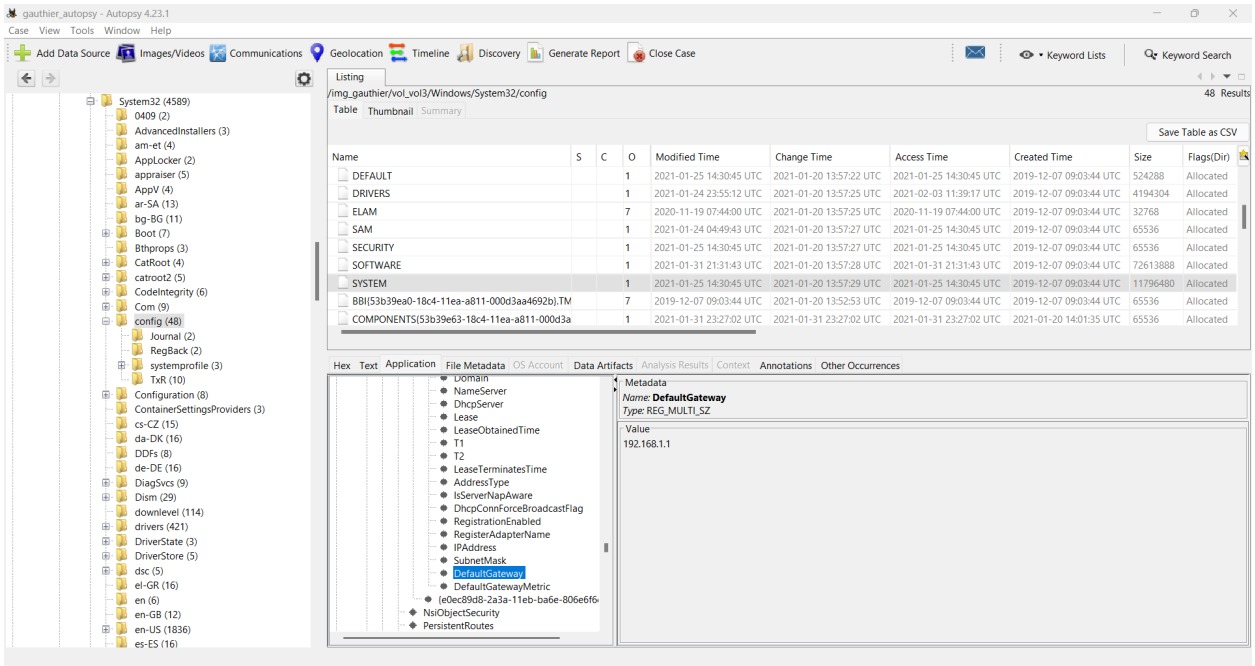
5.2.3.9 Gauthier_A_0009 Trillian mollieschmitt log

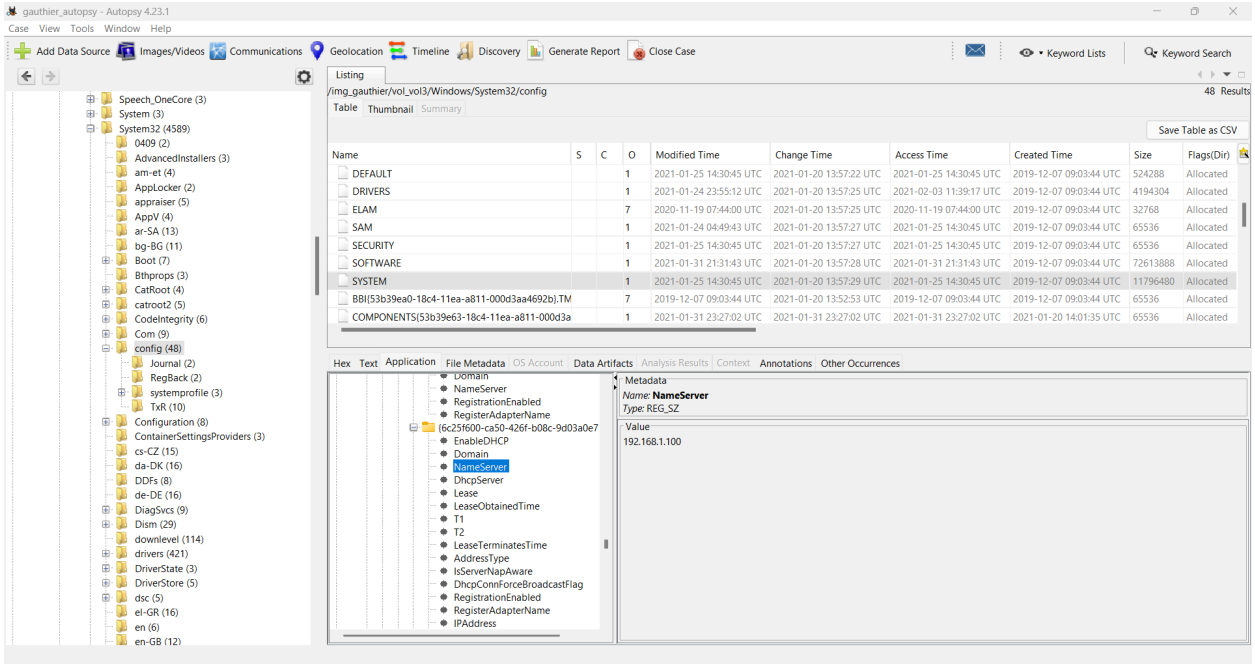


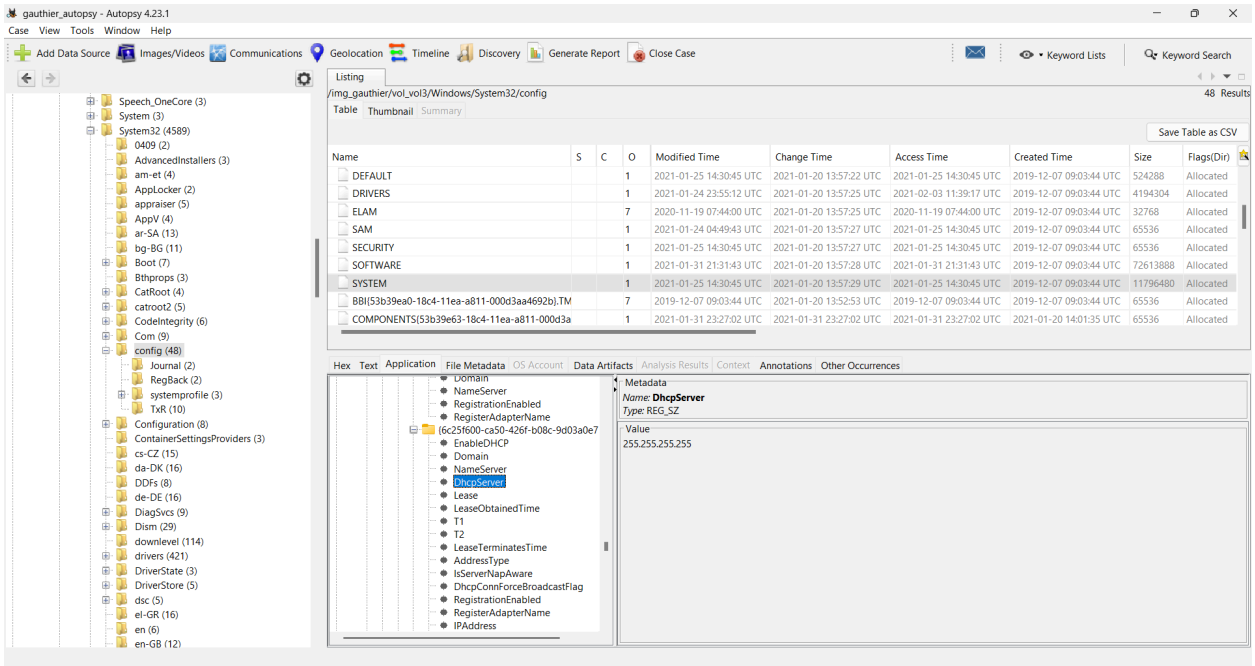
5.2.3.10 Gauthier_A_0010 IP address



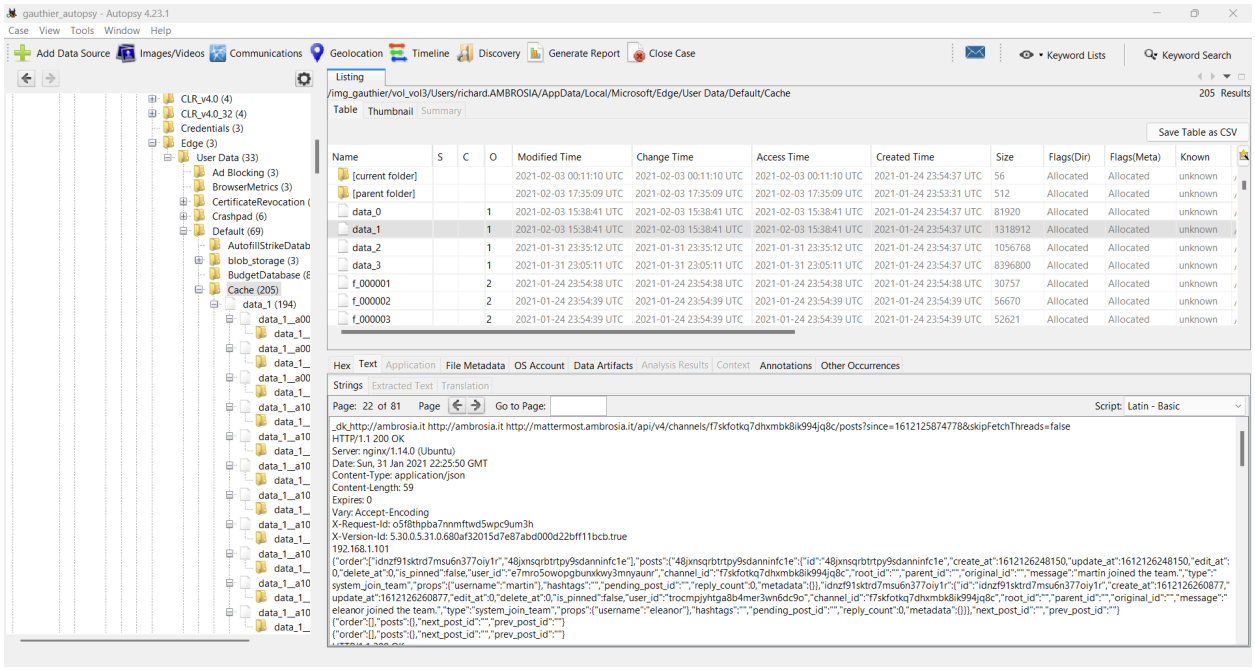

gauthier_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Listing
/img_gauthier/vol_vol3/Windows/System32/config
48 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(Dir)
DEFAULT
DRIVERS
ELAM
SAM
SECURITY
SOFTWARE
SYSTEM
BBI{53b39ea0-18c4-11ea-a811-000d3aa4692b}.TM
COMPONENTS{53b39e63-18c4-11ea-a811-000d3a
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
{6c25f600-ca50-426f-b08c-9d03a0e7
EnableDHCP
Domain
NameServer
DhcpServer
Lease
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
SubnetMask
DefaultGateway
DefaultGatewayMetric
Metadata
Name: SubnetMask
Type: REG_MULTI_SZ
Value
255.255.255.0
System32 (4589)
0409 (2)
AdvancedInstallers (3)
am-et (4)
AppLocker (2)
appraiser (5)
AppV (4)
ar-SA (13)
bg-BG (11)
Boot (7)
Bthprops (3)
CatRoot (4)
catroot2 (5)
CodeIntegrity (6)
Com (9)
config (48)
Journal (2)
RegBack (2)
systemprofile (3)
TxR (10)
Configuration (8)
ContainerSettingsProviders (3)
cs-CZ (15)
da-DK (16)
DDFs (8)
de-DE (16)
DiagSvcs (9)
Dism (29)
downlevel (114)
drivers (421)
DriverState (3)
DriverStore (5)
dsc (5)
el-GR (16)
en (6)
en-GB (12)
en-US (1836)
es-ES (16)


gauthier_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Listing
/img_gauthier/vol_vol3/Windows/System32/config
48 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(Dir)
DEFAULT
DRIVERS
ELAM
SAM
SECURITY
SOFTWARE
SYSTEM
BBI{53b39ea0-18c4-11ea-a811-000d3aa4692b}.TM
COMPONENTS{53b39e63-18c4-11ea-a811-000d3a
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Domain
NameServer
DhcpServer
Lease
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
SubnetMask
DefaultGateway
DefaultGatewayMetric
{e0ec89d8-2a3a-11eb-ba6e-806e6f6
NsiObjectSecurity
PersistentRoutes
Metadata
Name: DefaultGateway
Type: REG_MULTI_SZ
Value
192.168.1.1
System32 (4589)
0409 (2)
AdvancedInstallers (3)
am-et (4)
AppLocker (2)
appraiser (5)
AppV (4)
ar-SA (13)
bg-BG (11)
Boot (7)
Bthprops (3)
CatRoot (4)
catroot2 (5)
CodeIntegrity (6)
Com (9)
config (48)
Journal (2)
RegBack (2)
systemprofile (3)
TxR (10)
Configuration (8)
ContainerSettingsProviders (3)
cs-CZ (15)
da-DK (16)
DDFs (8)
de-DE (16)
DiagSvcs (9)
Dism (29)
downlevel (114)
drivers (421)
DriverState (3)
DriverStore (5)
dsc (5)
el-GR (16)
en (6)
en-GB (12)
en-US (1836)
es-ES (16)


gauthier_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Listing
/img_gauthier/vol_vol3/Windows/System32/config
48 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(Dir)
DEFAULT
DRIVERS
ELAM
SAM
SECURITY
SOFTWARE
SYSTEM
BBI{53b39ea0-18c4-11ea-a811-000d3aa4692b}.TM
COMPONENTS{53b39e63-18c4-11ea-a811-000d3a
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Domain
NameServer
RegistrationEnabled
RegisterAdapterName
{6c25f600-ca50-426f-b08c-9d03a0e7
EnableDHCP
Domain
NameServer
DhcpServer
Lease
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
Metadata
Name: NameServer
Type: REG_SZ
Value
192.168.1.100
Speech_OneCore (3)
System (3)
System32 (4589)
0409 (2)
AdvancedInstallers (3)
am-et (4)
AppLocker (2)
appraiser (5)
AppV (4)
ar-SA (13)
bg-BG (11)
Boot (7)
Bthprops (3)
CatRoot (4)
catroot2 (5)
CodeIntegrity (6)
Com (9)
config (48)
Journal (2)
RegBack (2)
systemprofile (3)
TxR (10)
Configuration (8)
ContainerSettingsProviders (3)
cs-CZ (15)
da-DK (16)
DDFs (8)
de-DE (16)
DiagSvcs (9)
Dism (29)
downlevel (114)
drivers (421)
DriverState (3)
DriverStore (5)
dsc (5)
el-GR (16)
en (6)
en-GB (12)

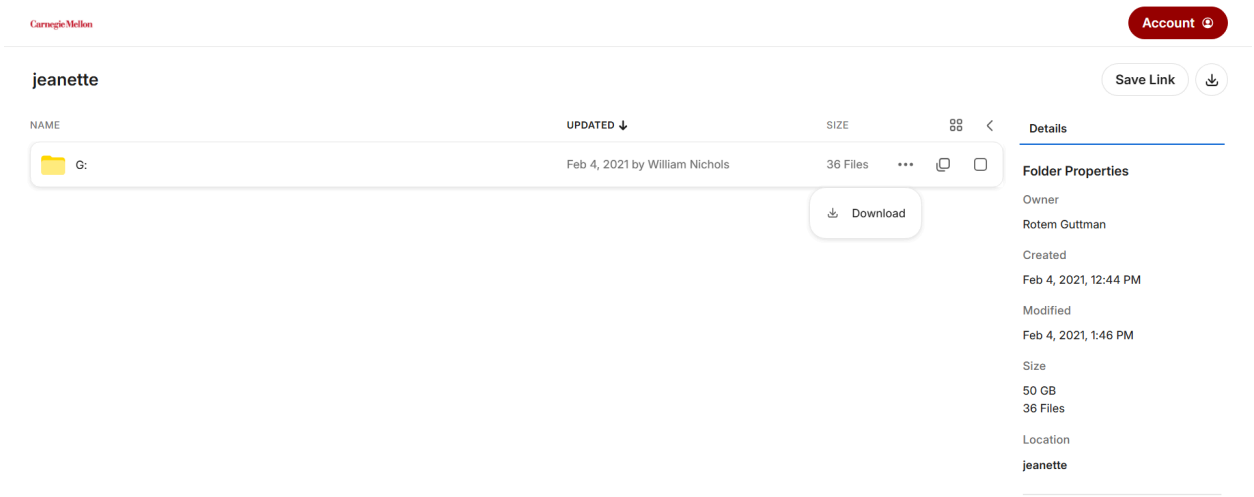

gauthier_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Listing
/img_gauthier/vol_vol3/Windows/System32/config
48 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(Dir)
DEFAULT
DRIVERS
ELAM
SAM
SECURITY
SOFTWARE
SYSTEM
BBI{53b39ea0-18c4-11ea-a811-000d3aa4692b}.TM
COMPONENTS{53b39e63-18c4-11ea-a811-000d3a
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Domain
NameServer
RegistrationEnabled
RegisterAdapterName
{6c25f600-ca50-426f-b08c-9d03a0e7
EnableDHCP
Domain
NameServer
DhcpServer
Lease
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
Metadata
Name: DhcpServer
Type: REG_SZ
Value
255.255.255.255
Speech_OneCore (3)
System (3)
System32 (4589)
0409 (2)
AdvancedInstallers (3)
am-et (4)
AppLocker (2)
appraiser (5)
AppV (4)
ar-SA (13)
bg-BG (11)
Boot (7)
Bthprops (3)
CatRoot (4)
catroot2 (5)
CodeIntegrity (6)
Com (9)
config (48)
Journal (2)
RegBack (2)
systemprofile (3)
TxR (10)
Configuration (8)
ContainerSettingsProviders (3)
cs-CZ (15)
da-DK (16)
DDFs (8)
de-DE (16)
DiagSvcs (9)
Dism (29)
downlevel (114)
drivers (421)
DriverState (3)
DriverStore (5)
dsc (5)
el-GR (16)
en (6)
en-GB (12)

5.2.3.11 Gauthier_A_0011 Mattermost

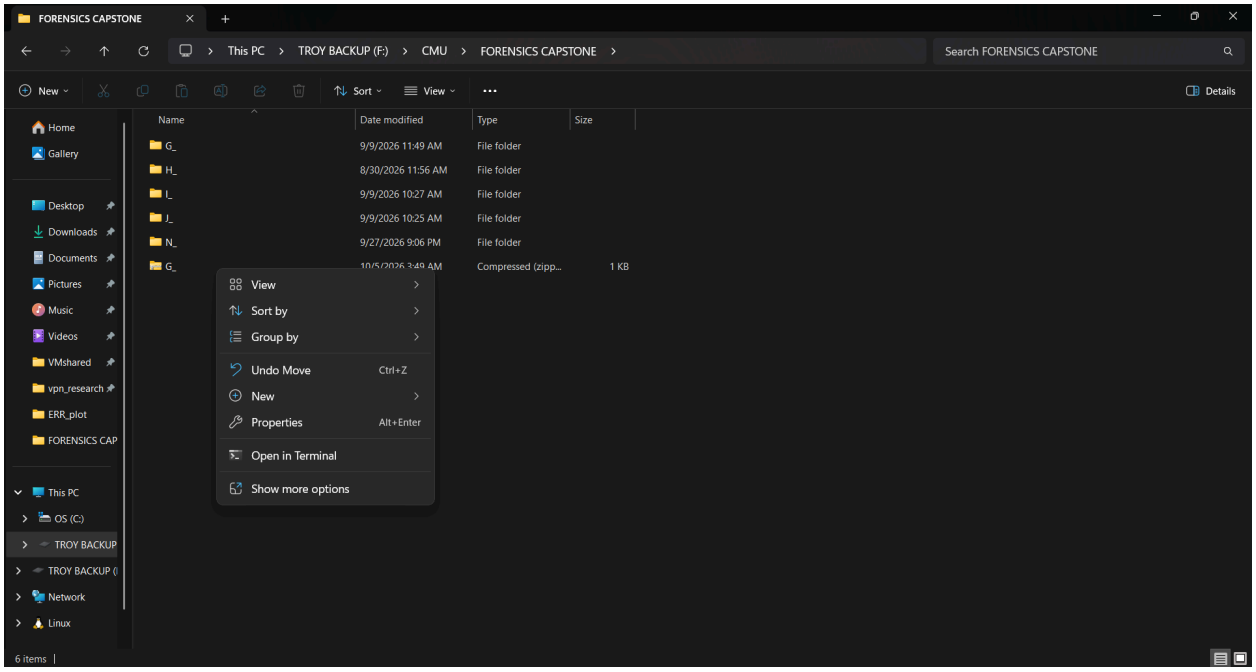


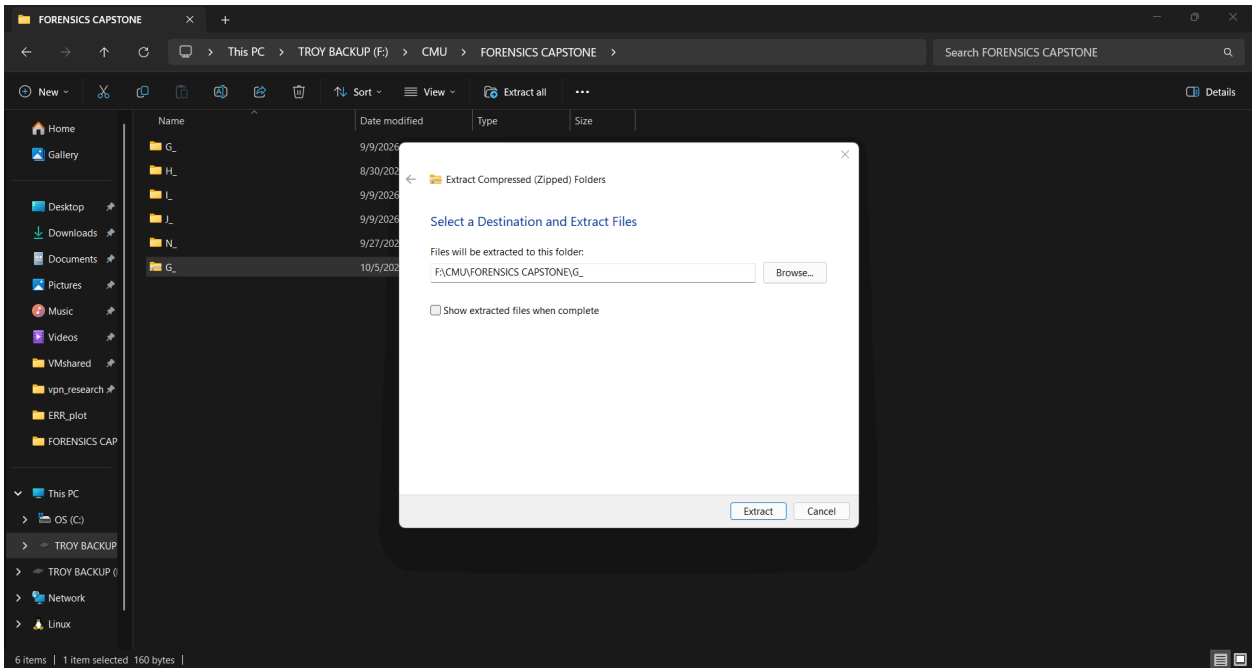
5.2.4 Jeanette Image (G)

5.2.4.1 Jeanette_A_0001 Image Download

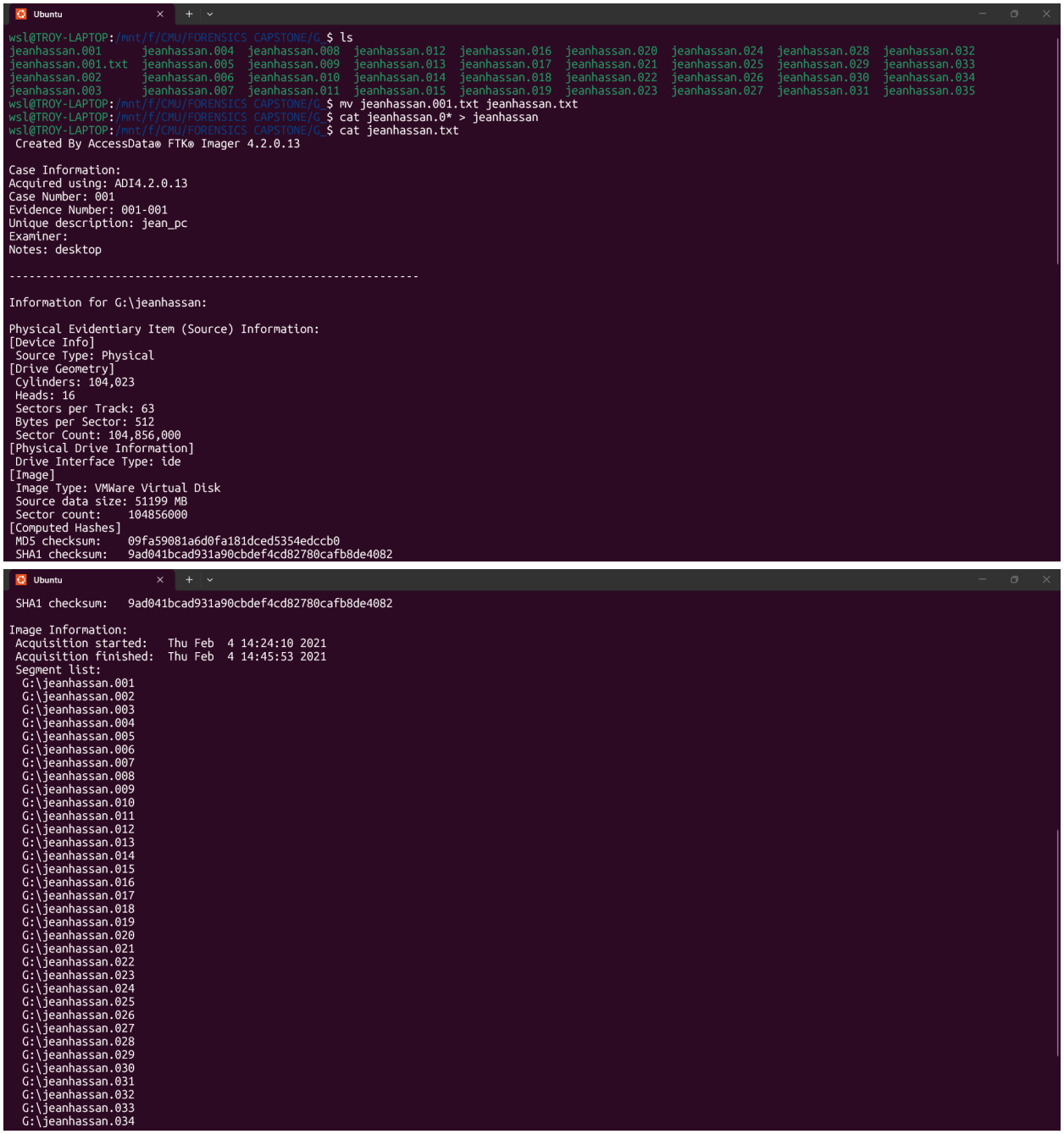


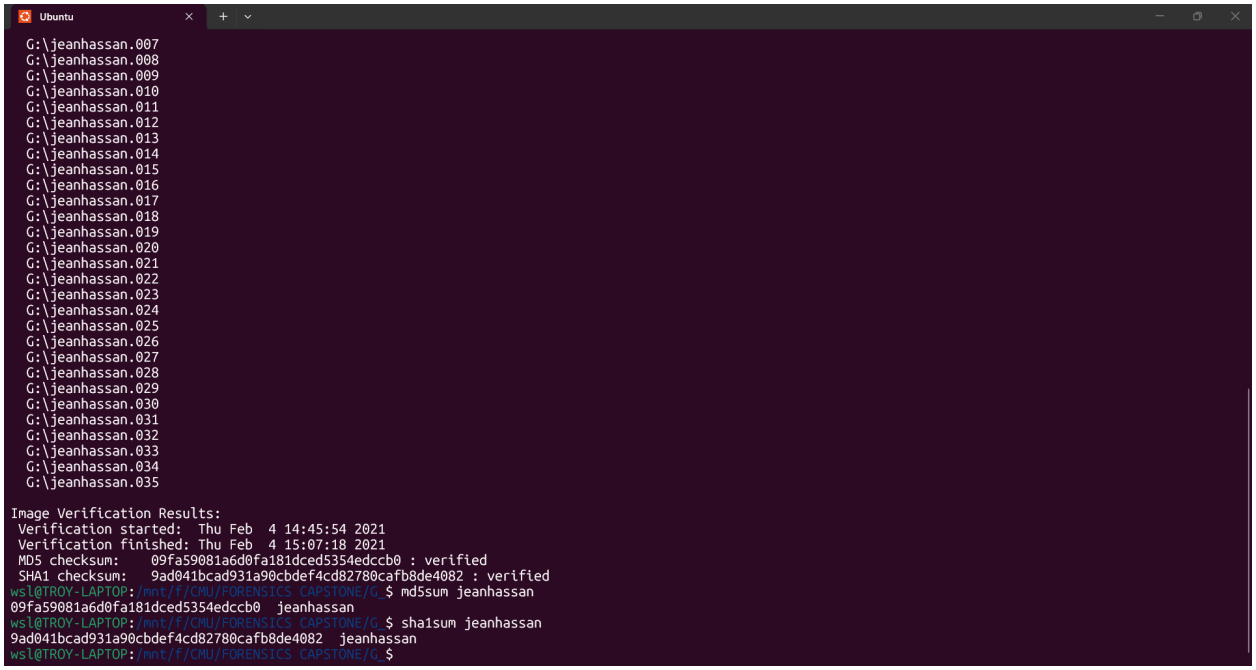
5.2.4.2 Jeanette_A_0002 Image Extraction



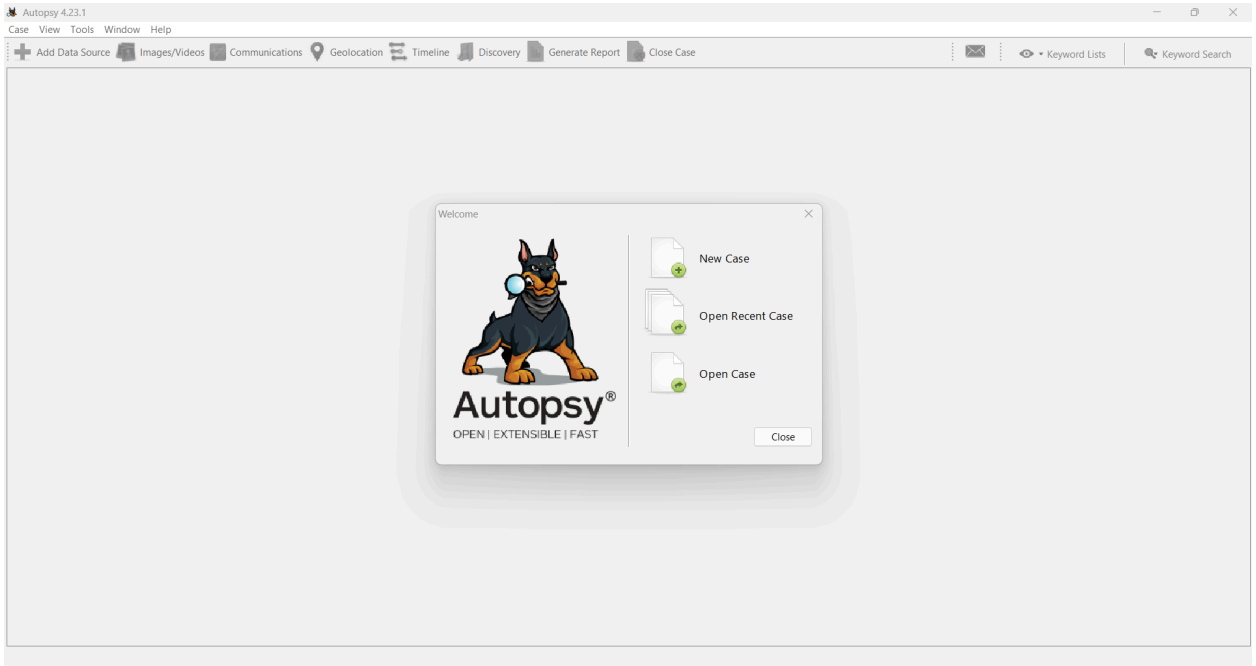


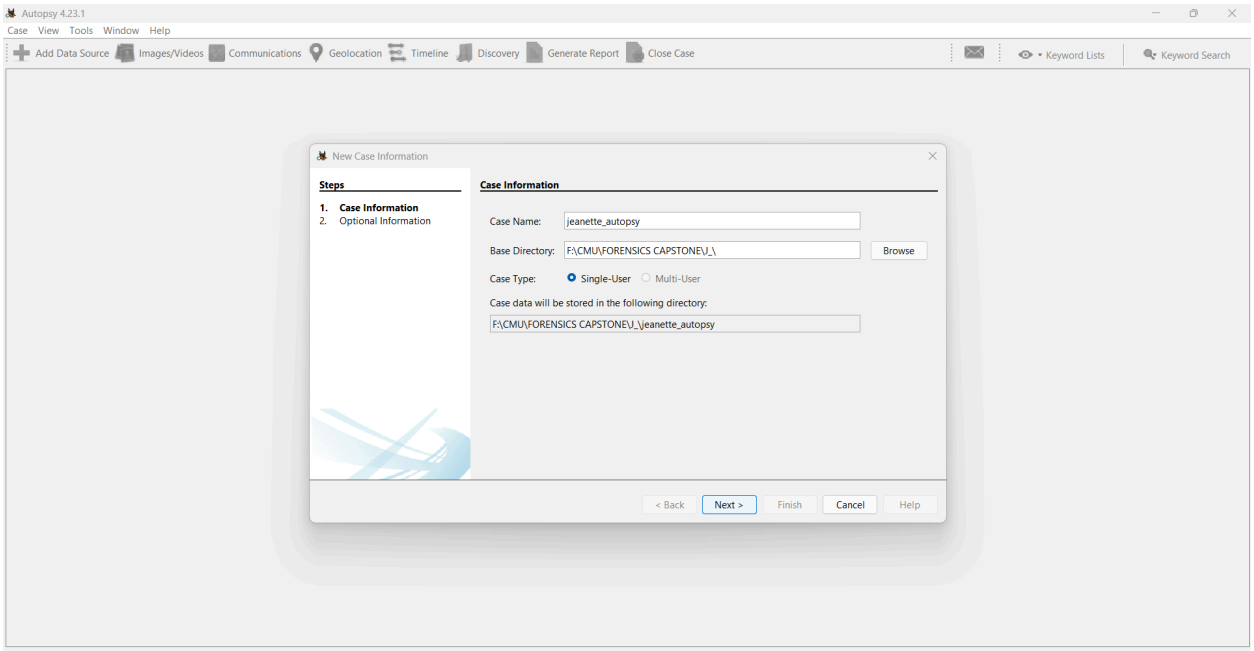
5.2.4.3 Jeanette_A_0003 Image Verification

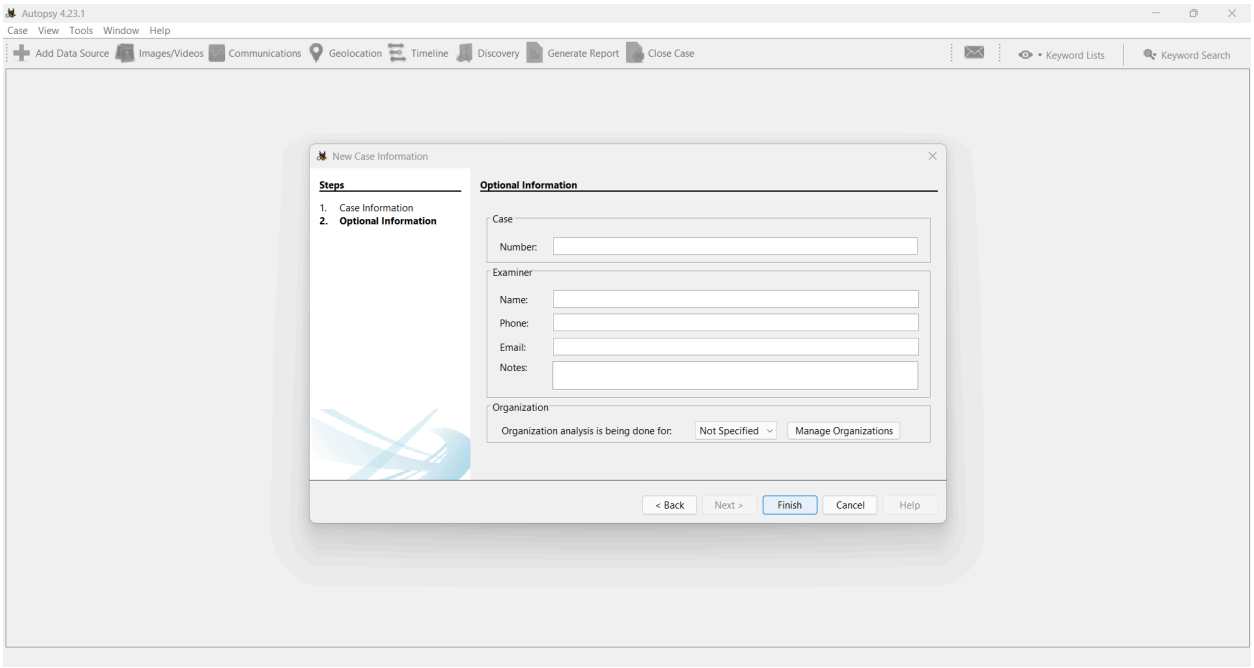


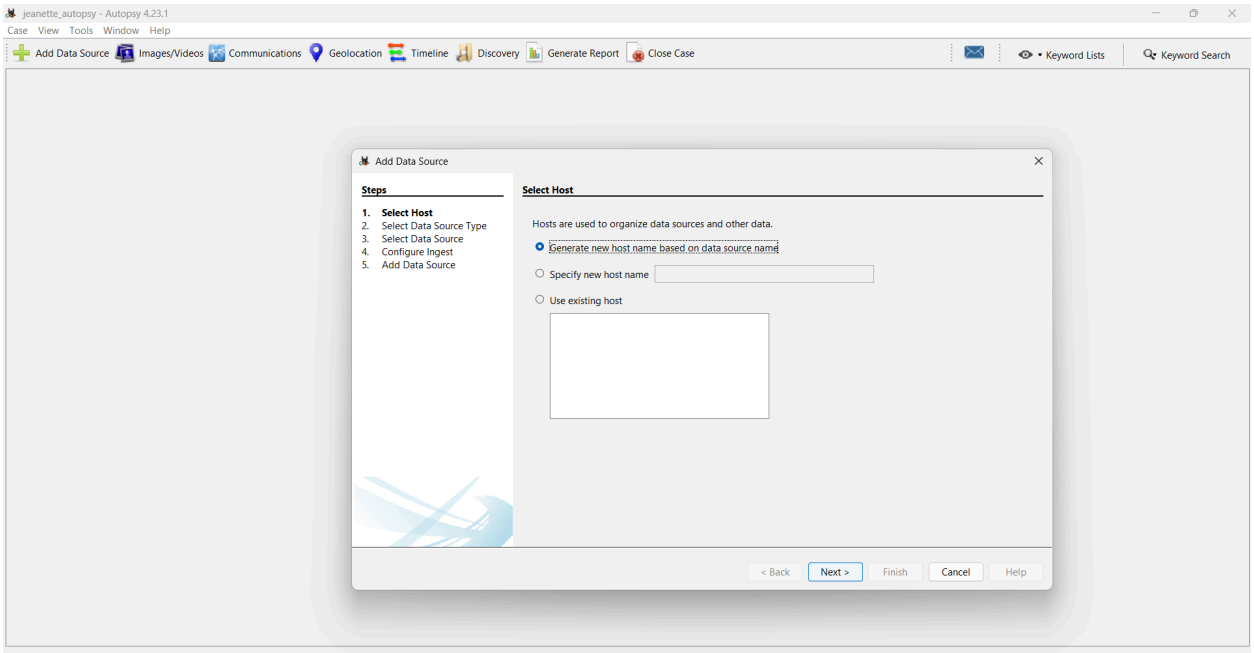


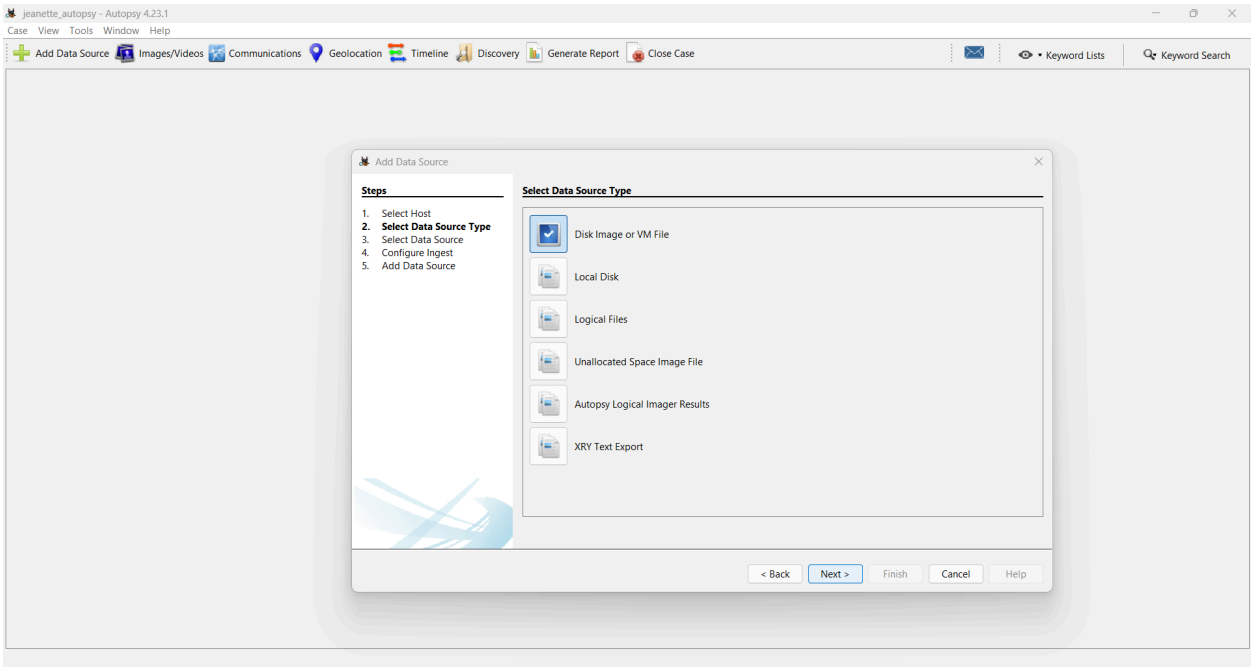
5.2.4.4 Jeanette_A_0004 Autopsy Case

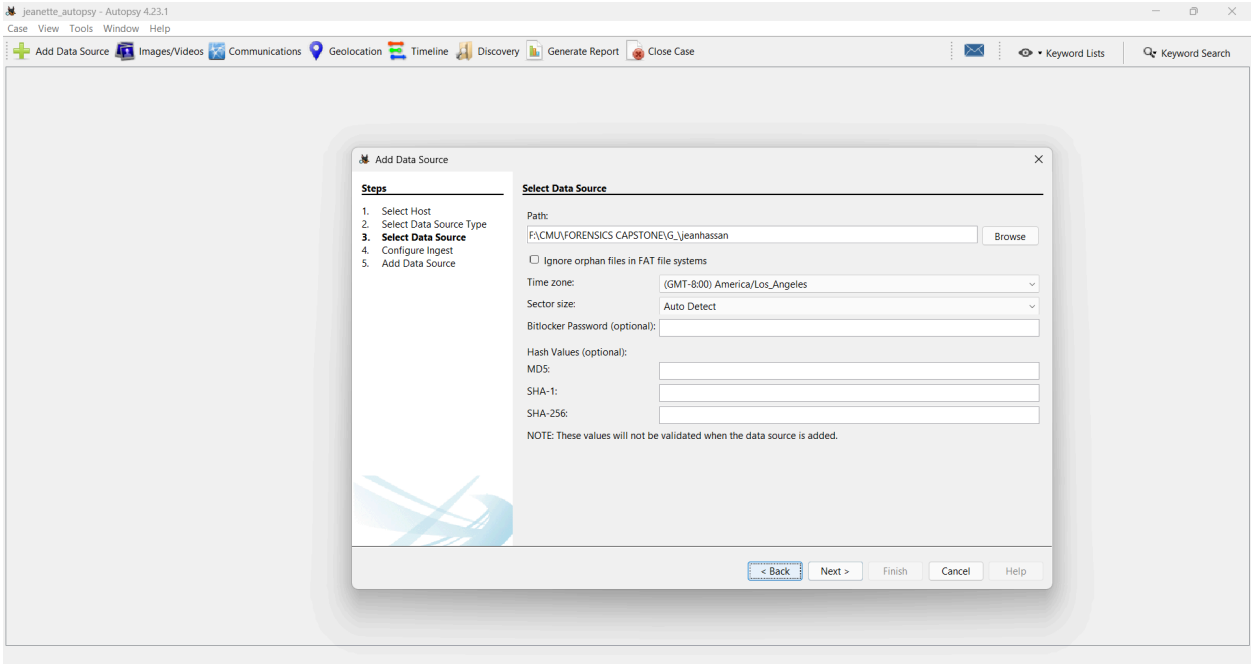


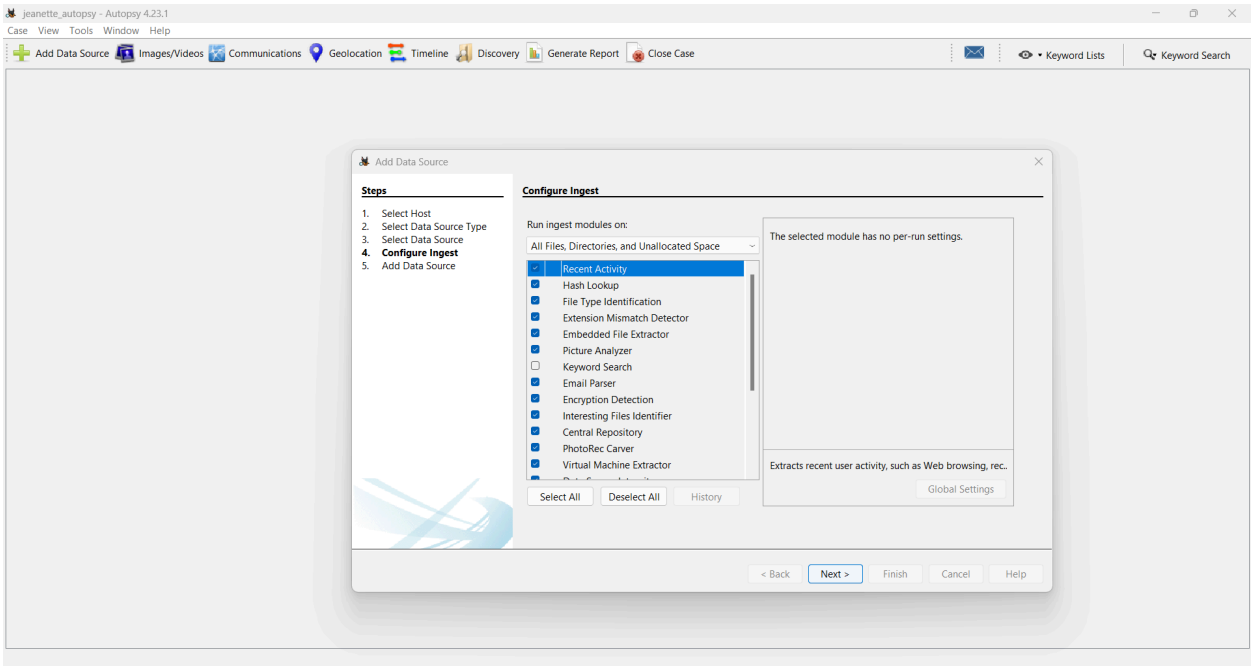

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Case Information
Case Name: jeanette_autopsy
Base Directory: F:\CMU\FORENSICS CAPSTONE\J_\
Browse
Case Type: Single-User Multi-User
Case data will be stored in the following directory:
F:\CMU\FORENSICS CAPSTONE\J_\jeanette_autopsy
< Back Next > Finish Cancel Help

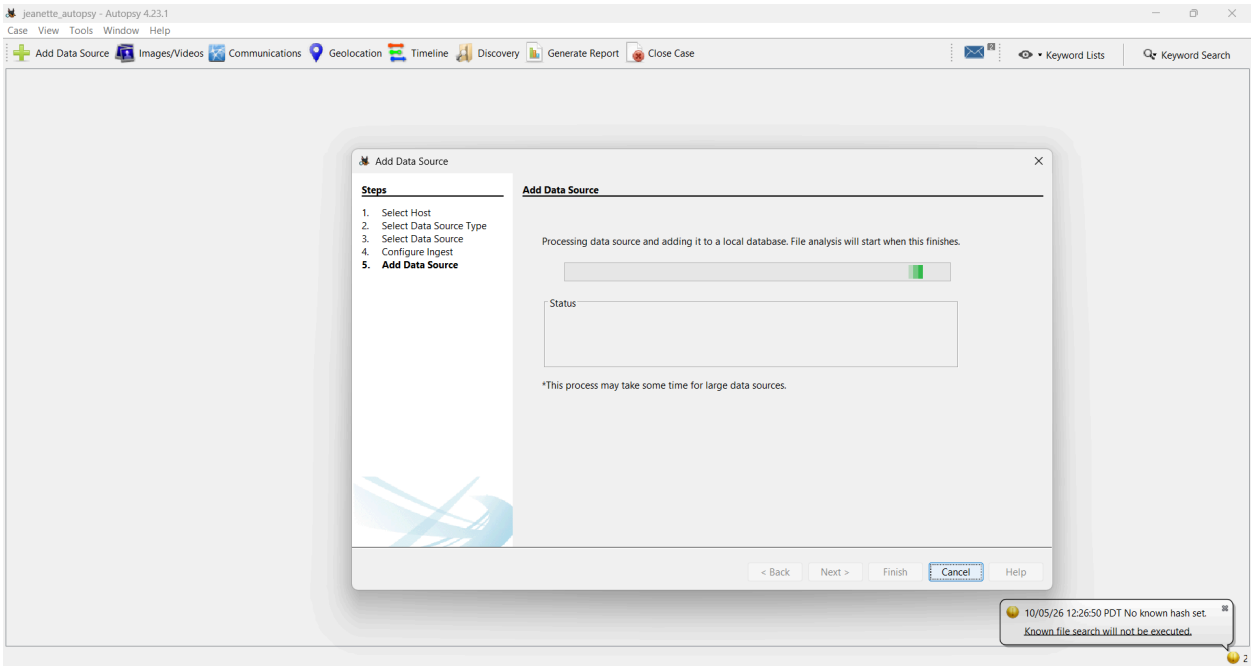

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Optional Information
Case
Number:
Examiner
Name:
Phone:
Email:
Notes:
Organization
Organization analysis is being done for: Not Specified Manage Organizations
< Back Next > Finish Cancel Help

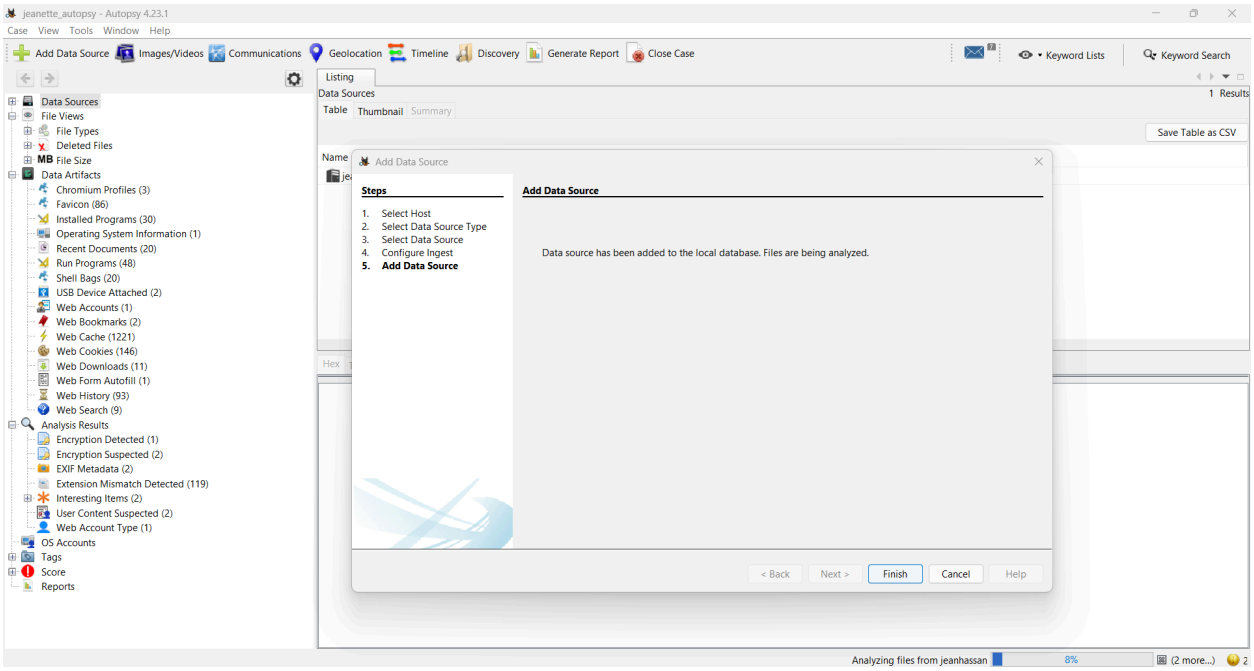

jeanette_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Host
Hosts are used to organize data sources and other data.
Generate new host name based on data source name
Specify new host name
Use existing host
< Back
Next >
Finish
Cancel
Help

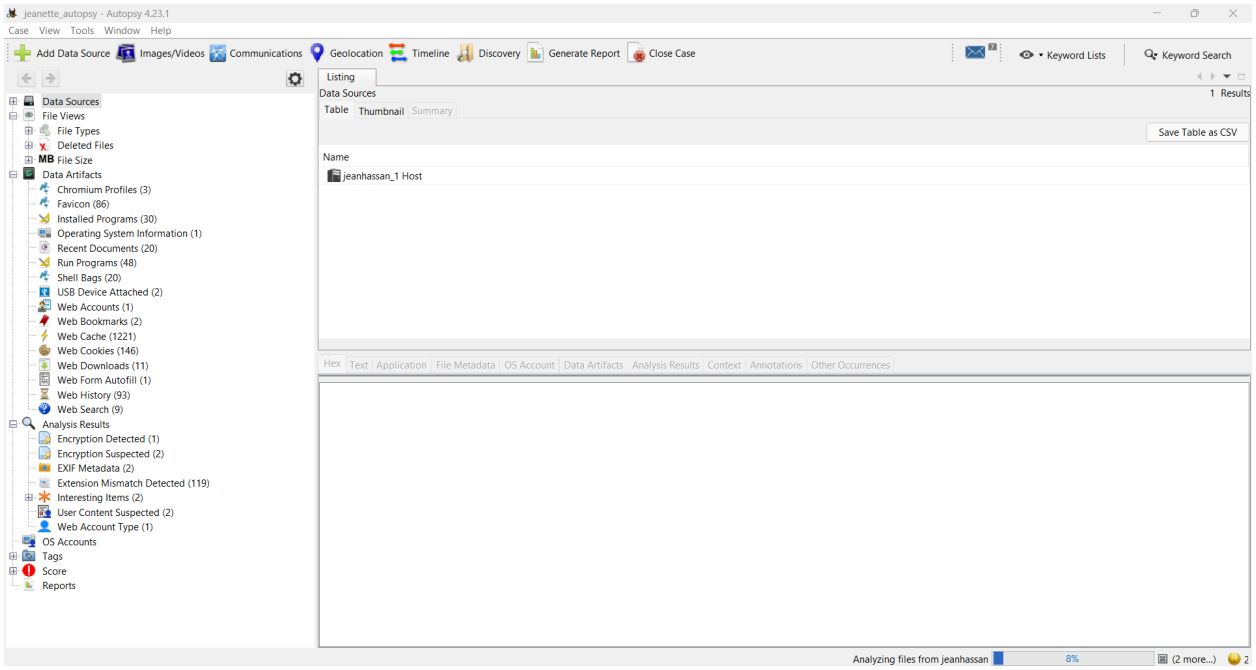

jeanette_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source Type
Disk Image or VM File
Local Disk
Logical Files
Unallocated Space Image File
Autopsy Logical Imager Results
XRY Text Export
< Back
Next >
Finish
Cancel
Help


jeanette_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source
Path:
F:\CMU\FORENSICS CAPSTONE\G_\jeanhassan
Browse
Ignore orphan files in FAT file systems
Time zone:
(GMT-8:00) America/Los_Angeles
Sector size:
Auto Detect
Bitlocker Password (optional):
Hash Values (optional):
MD5:
SHA-1:
SHA-256:
NOTE: These values will not be validated when the data source is added.
< Back
Next >
Finish
Cancel
Help

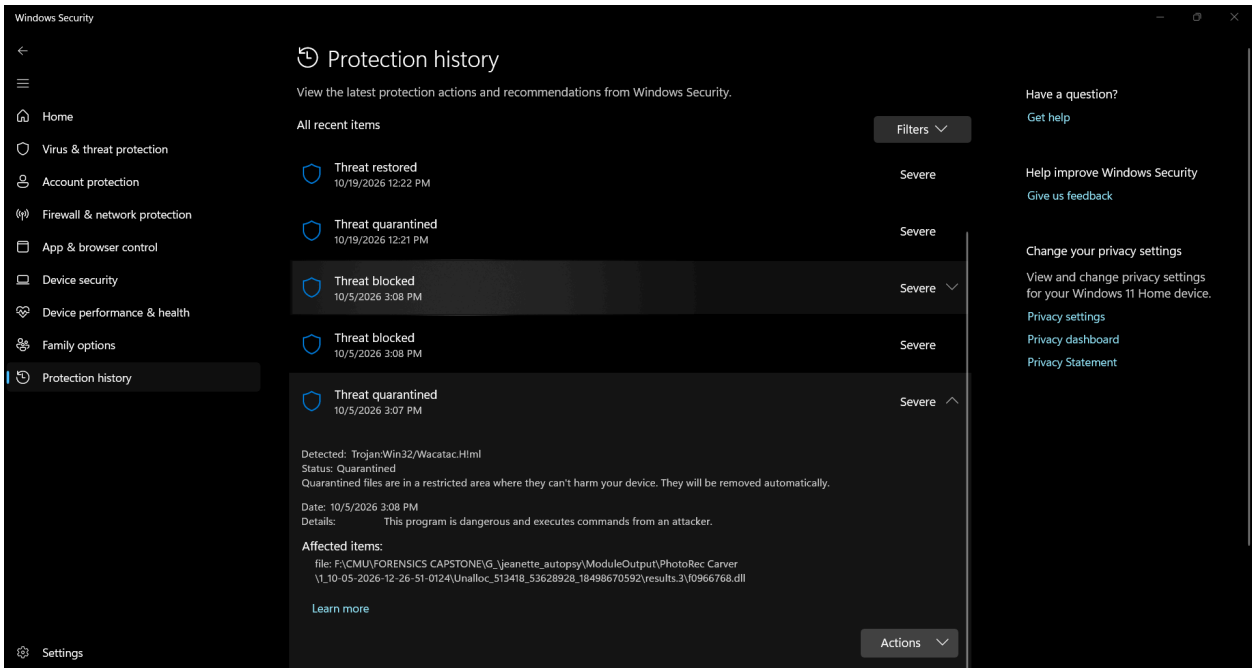

jeanette_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
Select All
Deselect All
History
The selected module has no per-run settings.
Extracts recent user activity, such as Web browsing, rec..
Global Settings
< Back
Next >
Finish
Cancel
Help

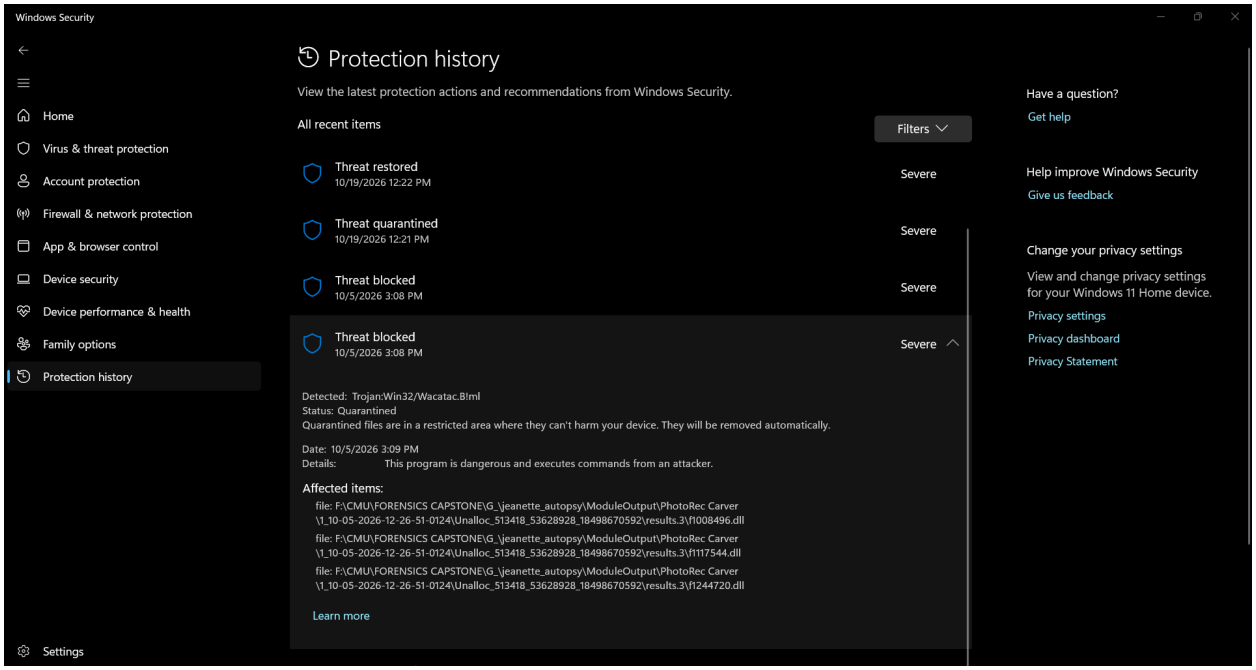

jeanette_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
Processing data source and adding it to a local database. File analysis will start when this finishes.
Status
*This process may take some time for large data sources.
< Back
Next >
Finish
Cancel
Help
10/05/26 12:26:50 PDT No known hash set.
Known file search will not be executed.

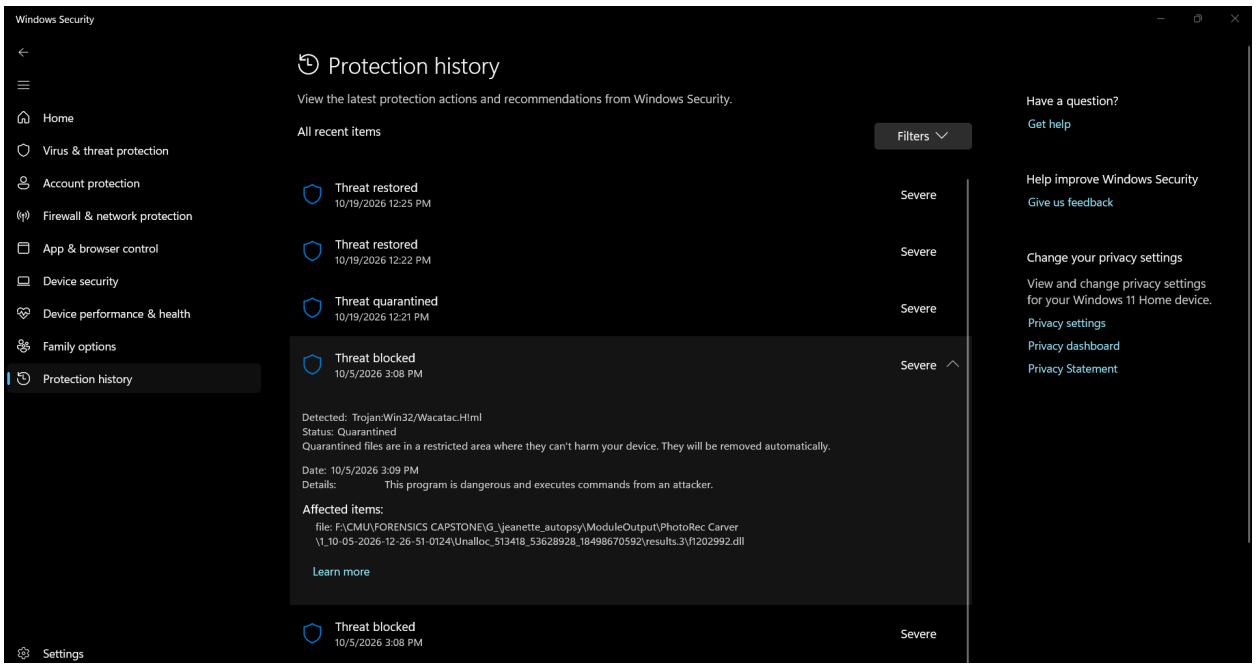

jeanette_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Chromium Profiles (3)
Favicon (86)
Installed Programs (30)
Operating System Information (1)
Recent Documents (20)
Run Programs (48)
Shell Bags (20)
USB Device Attached (2)
Web Accounts (1)
Web Bookmarks (2)
Web Cache (1221)
Web Cookies (146)
Web Downloads (11)
Web Form Autofill (1)
Web History (93)
Web Search (9)
Analysis Results
Encryption Detected (1)
Encryption Suspected (2)
EXIF Metadata (2)
Extension Mismatch Detected (119)
Interesting Items (2)
User Content Suspected (2)
Web Account Type (1)
OS Accounts
Tags
Score
Reports
Listing
Data Sources
1 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
Data source has been added to the local database. Files are being analyzed.
Hex
< Back
Next >
Finish
Cancel
Help
Analyzing files from jeanhassan
8%
(2 more...)



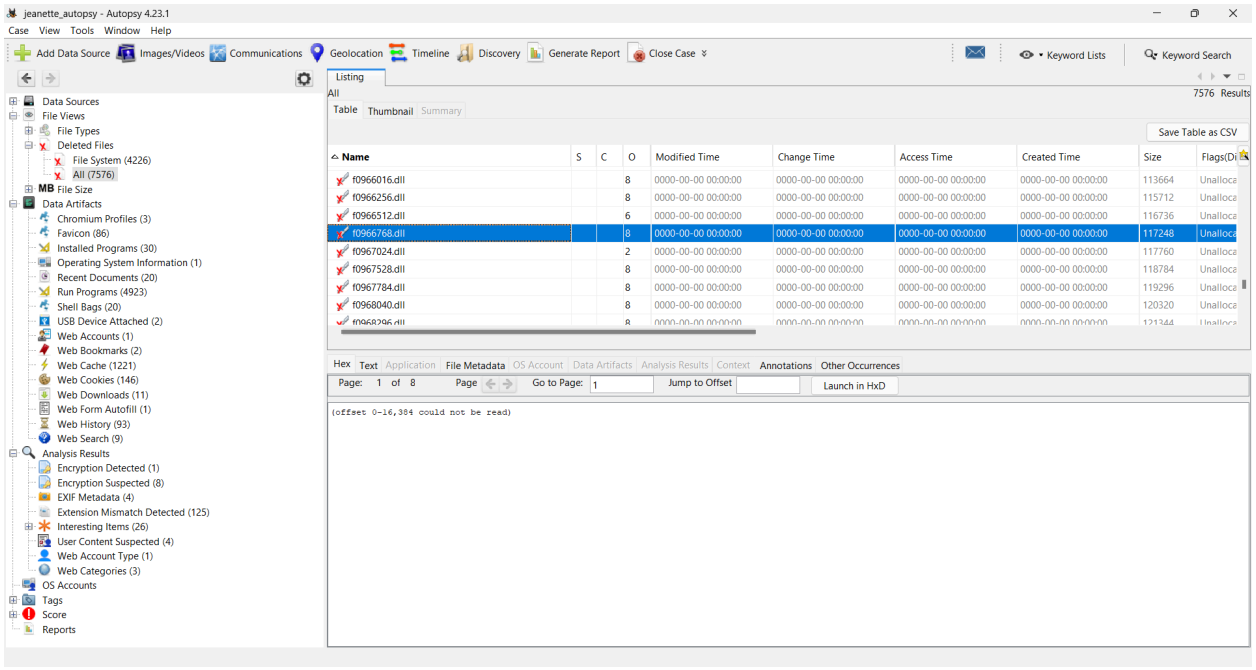
5.2.4.5 Jeanette_A_0005 Windows Security



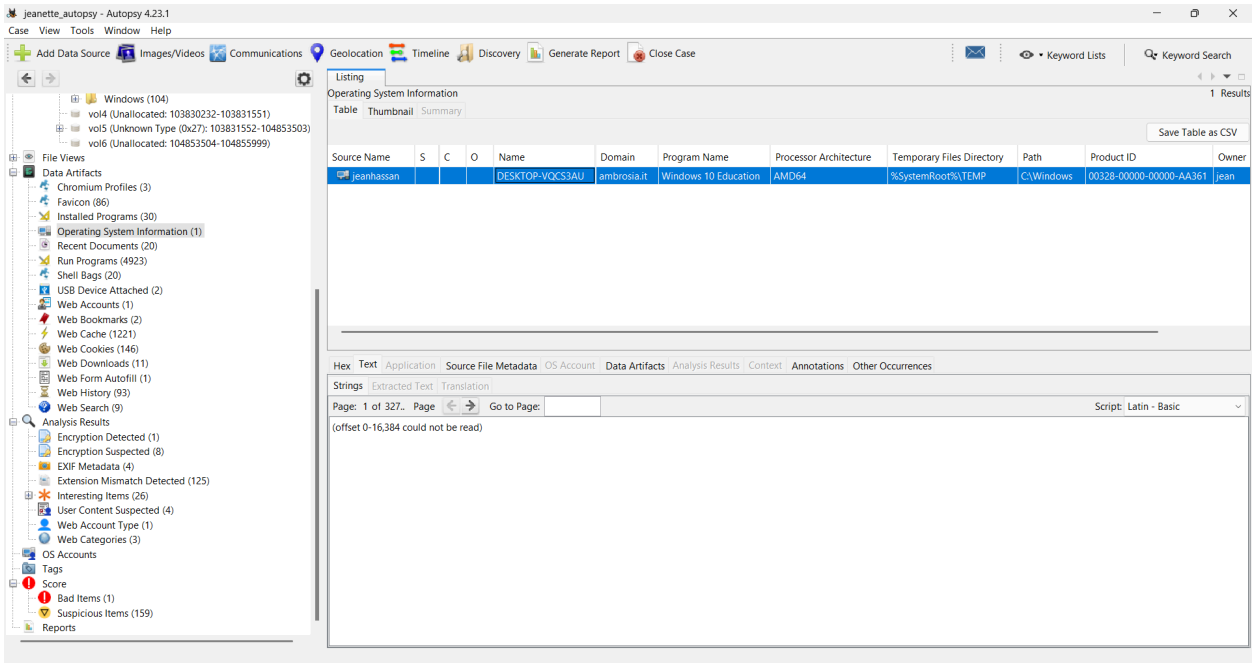

Windows Security
Protection history
View the latest protection actions and recommendations from Windows Security.
All recent items
Filters
Home
Virus & threat protection
Account protection
Firewall & network protection
App & browser control
Device security
Device performance & health
Family options
Protection history
Settings
Threat restored
10/19/2026 12:22 PM
Severe
Threat quarantined
10/19/2026 12:21 PM
Severe
Threat blocked
10/5/2026 3:08 PM
Severe
Threat blocked
10/5/2026 3:08 PM
Severe
Detected: Trojan:Win32/Wacatac.B!ml
Status: Quarantined
Quarantined files are in a restricted area where they can't harm your device. They will be removed automatically.
Date: 10/5/2026 3:09 PM
Details: This program is dangerous and executes commands from an attacker.
Affected items:
file: F:\CMU\FORENSICS CAPSTONE\G_\jeanette_autopsy\ModuleOutput\PhotoRec Carver \1_10-05-2026-12-26-51-0124\Unalloc_513418_53628928_18498670592\results.3\f1008496.dll
file: F:\CMU\FORENSICS CAPSTONE\G_\jeanette_autopsy\ModuleOutput\PhotoRec Carver \1_10-05-2026-12-26-51-0124\Unalloc_513418_53628928_18498670592\results.3\f1117544.dll
file: F:\CMU\FORENSICS CAPSTONE\G_\jeanette_autopsy\ModuleOutput\PhotoRec Carver \1_10-05-2026-12-26-51-0124\Unalloc_513418_53628928_18498670592\results.3\f1244720.dll
Learn more
Have a question?
Get help
Help improve Windows Security
Give us feedback
Change your privacy settings
View and change privacy settings for your Windows 11 Home device.
Privacy settings
Privacy dashboard
Privacy Statement


Windows Security
Protection history
View the latest protection actions and recommendations from Windows Security.
All recent items
Filters
Home
Virus & threat protection
Account protection
Firewall & network protection
App & browser control
Device security
Device performance & health
Family options
Protection history
Settings
Threat restored
10/19/2026 12:25 PM
Severe
Threat restored
10/19/2026 12:22 PM
Severe
Threat quarantined
10/19/2026 12:21 PM
Severe
Threat blocked
10/5/2026 3:08 PM
Severe
Detected: Trojan:Win32/Wacatac.H!ml
Status: Quarantined
Quarantined files are in a restricted area where they can't harm your device. They will be removed automatically.
Date: 10/5/2026 3:09 PM
Details: This program is dangerous and executes commands from an attacker.
Affected items:
file: F:\CMU\FORENSICS CAPSTONE\G_\jeanette_autopsy\ModuleOutput\PhotoRec Carver \1_10-05-2026-12-26-51-0124\Unalloc_513418_53628928_18498670592\results.3\f1202992.dll
Learn more
Threat blocked
10/5/2026 3:08 PM
Severe
Have a question?
Get help
Help improve Windows Security
Give us feedback
Change your privacy settings
View and change privacy settings for your Windows 11 Home device.
Privacy settings
Privacy dashboard
Privacy Statement

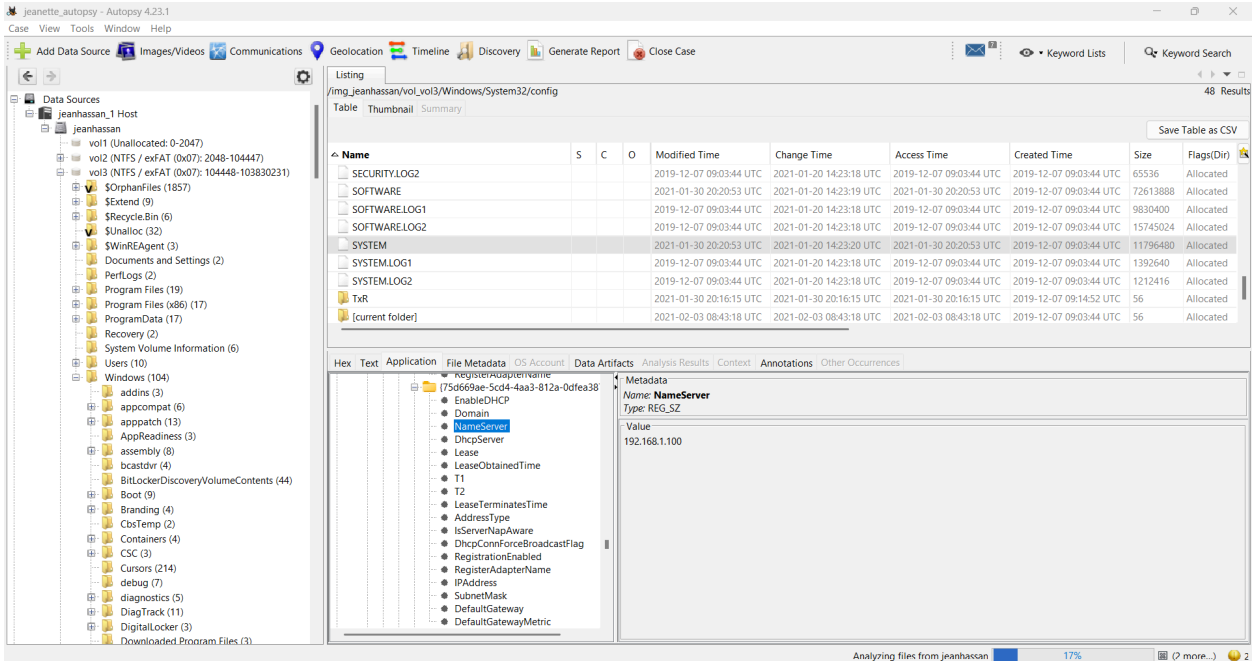
5.2.4.6 Jeanette_A_0006 Windows Security Files

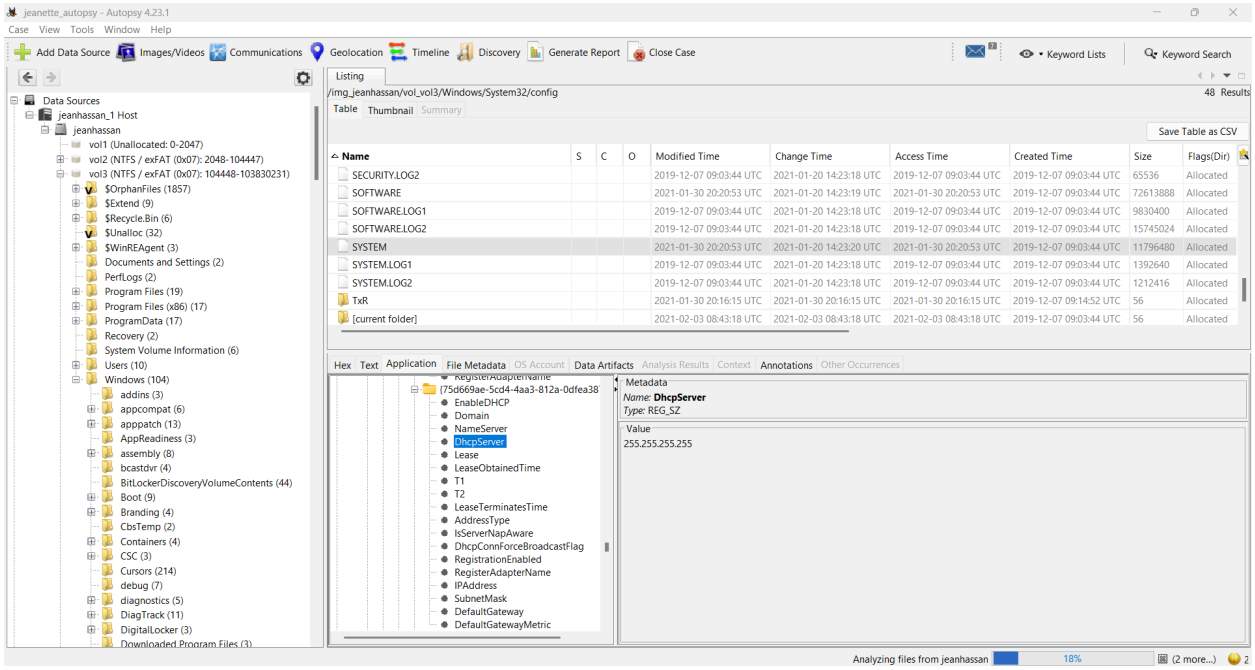


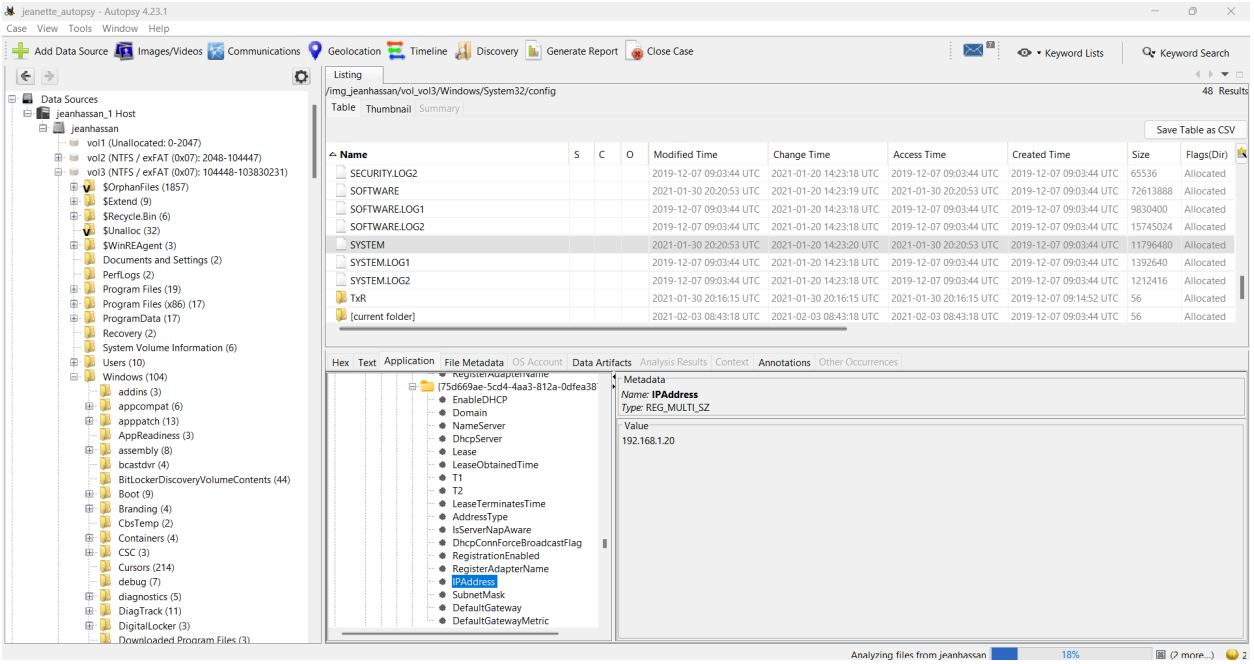
5.2.4.7 Jeanette_A_0007 Operating System

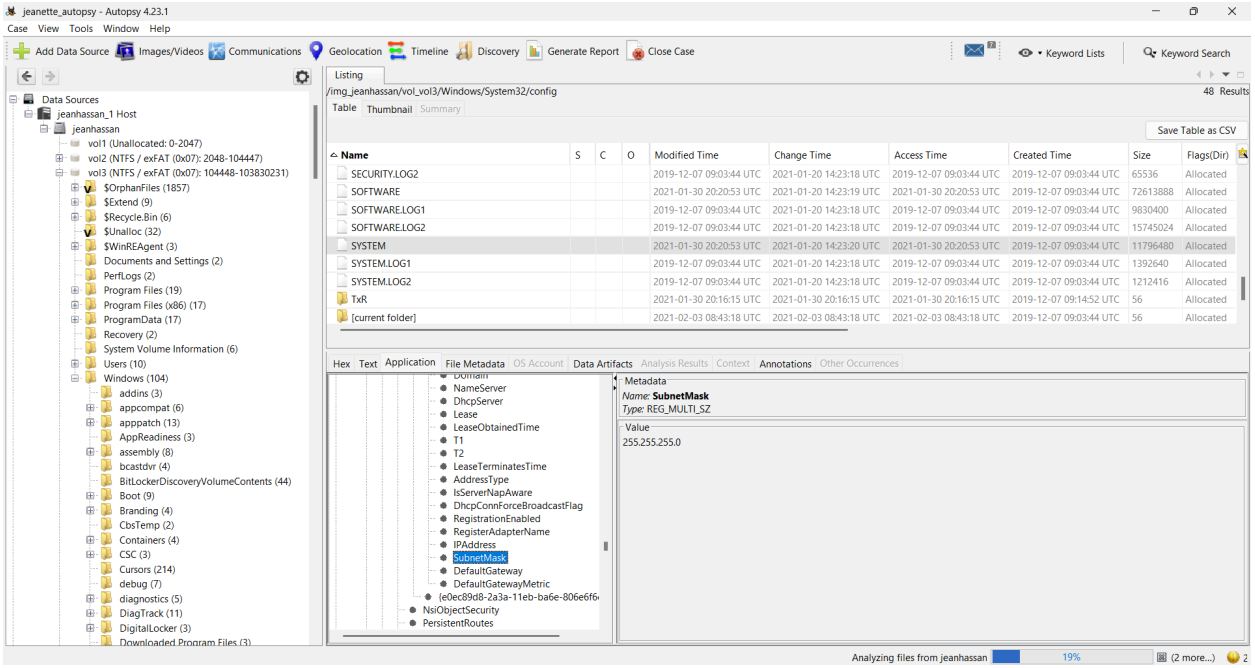


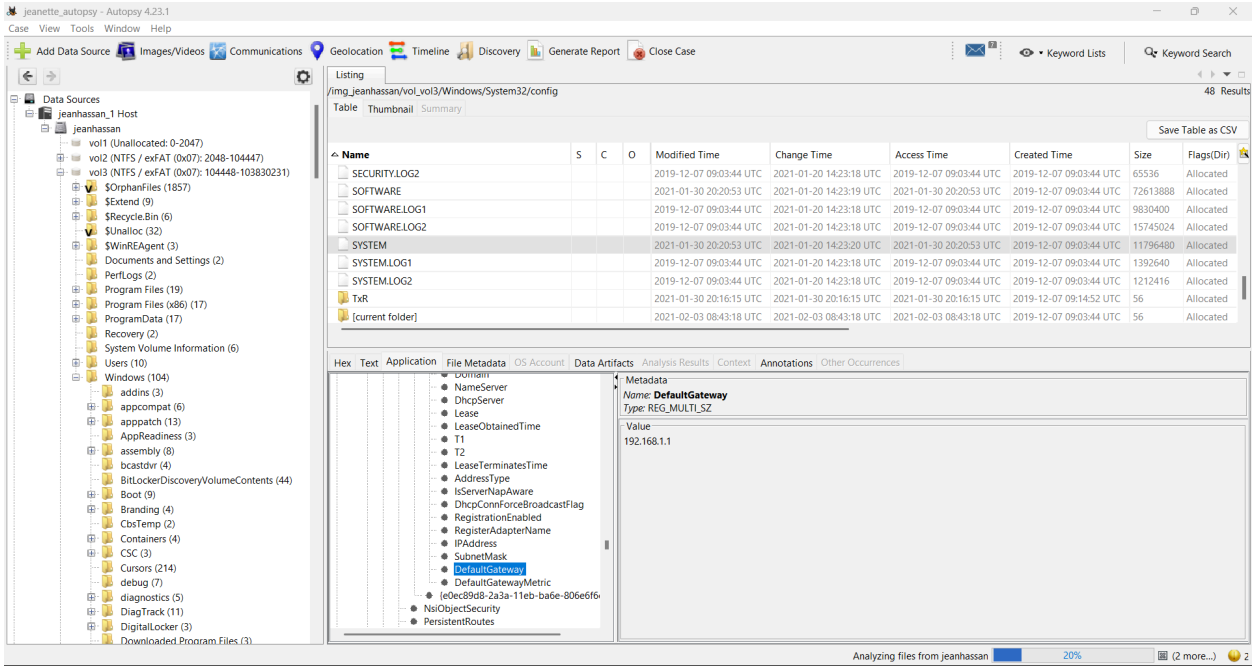
5.2.4.8 Jeanette_A_0008 IP Address



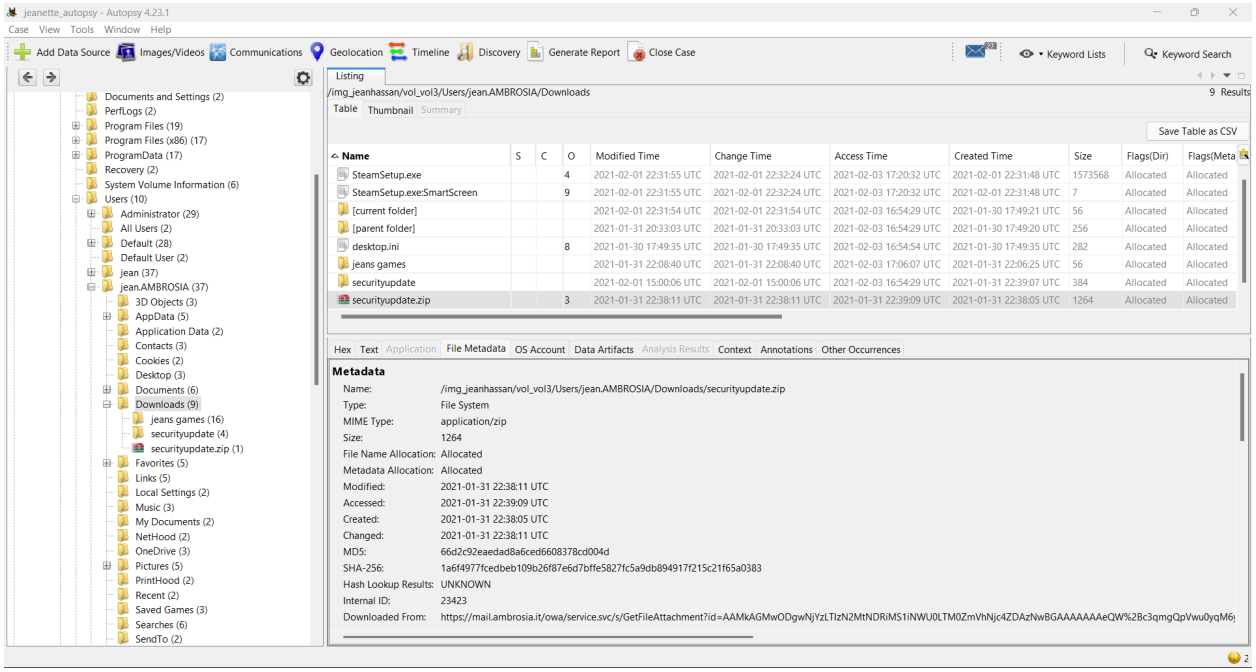


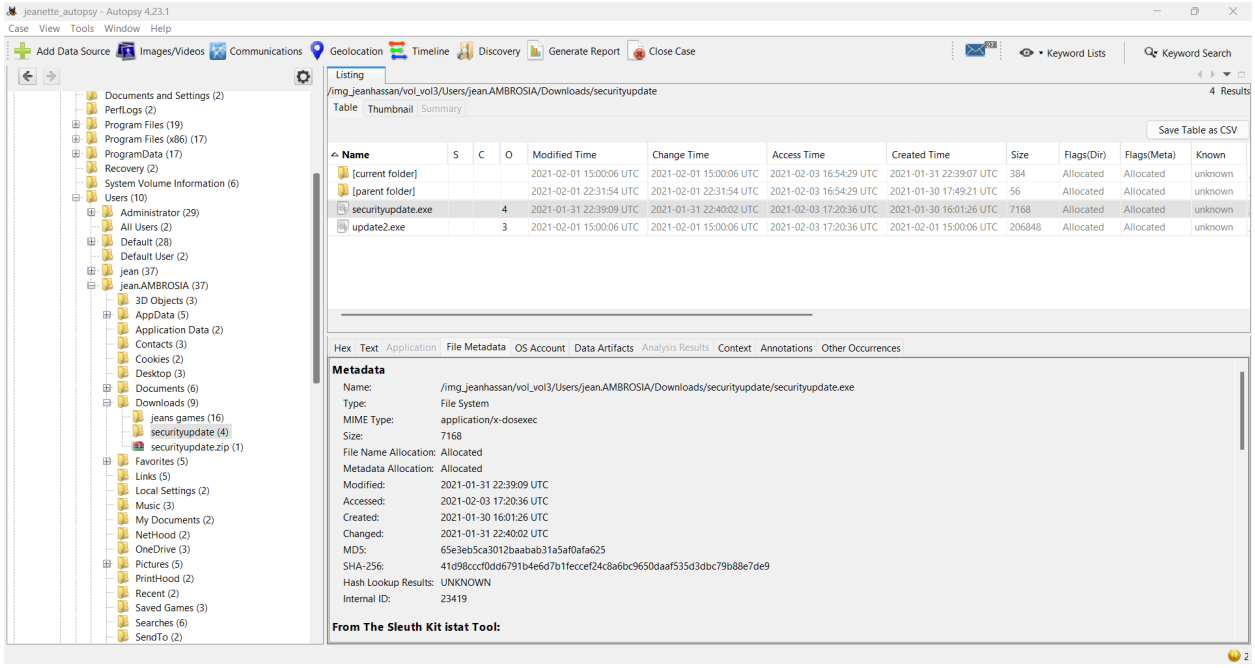

jeanette_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Data Sources
jeanhassan_1 Host
jeanhassan
vol1 (Unallocated: 0-2047)
vol2 (NTFS / exFAT (0x07): 2048-104447)
vol3 (NTFS / exFAT (0x07): 104448-103830231)
$OrphanFiles (1857)
$Extend (9)
$Recycle.Bin (6)
$Unalloc (32)
$WinREAgent (3)
Documents and Settings (2)
PerfLogs (2)
Program Files (19)
Program Files (x86) (17)
ProgramData (17)
Recovery (2)
System Volume Information (6)
Users (10)
Windows (104)
addins (3)
appcompat (6)
apppatch (13)
AppReadiness (3)
assembly (8)
bcastdvr (4)
BitLockerDiscoveryVolumeContents (44)
Boot (9)
Branding (4)
CbsTemp (2)
Containers (4)
CSC (3)
Cursors (214)
debug (7)
diagnostics (5)
DiagTrack (11)
DigitalLocker (3)
Downloaded Program Files (3)
Listing
/img_jeanhassan/vol_vol3/Windows/System32/config
48 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(Dir)
SECURITY.LOG2 2019-12-07 09:03:44 UTC 2021-01-20 14:23:18 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 65536 Allocated
SOFTWARE 2021-01-30 20:20:53 UTC 2021-01-20 14:23:19 UTC 2021-01-30 20:20:53 UTC 2019-12-07 09:03:44 UTC 72613888 Allocated
SOFTWARE.LOG1 2019-12-07 09:03:44 UTC 2021-01-20 14:23:18 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 9830400 Allocated
SOFTWARE.LOG2 2019-12-07 09:03:44 UTC 2021-01-20 14:23:18 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 15745024 Allocated
SYSTEM 2021-01-30 20:20:53 UTC 2021-01-20 14:23:20 UTC 2021-01-30 20:20:53 UTC 2019-12-07 09:03:44 UTC 11796480 Allocated
SYSTEM.LOG1 2019-12-07 09:03:44 UTC 2021-01-20 14:23:18 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 1392640 Allocated
SYSTEM.LOG2 2019-12-07 09:03:44 UTC 2021-01-20 14:23:18 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 1212416 Allocated
TxR 2021-01-30 20:16:15 UTC 2021-01-30 20:16:15 UTC 2021-01-30 20:16:15 UTC 2019-12-07 09:14:52 UTC 56 Allocated
[current folder] 2021-02-03 08:43:18 UTC 2021-02-03 08:43:18 UTC 2021-02-03 08:43:18 UTC 2019-12-07 09:03:44 UTC 56 Allocated
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
{75d669ae-5cd4-4aa3-812a-0dfea38
EnableDHCP
Domain
NameServer
DhcpServer
Lease
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
SubnetMask
DefaultGateway
DefaultGatewayMetric
Metadata
Name: IPAddress
Type: REG_MULTI_SZ
Value
192.168.1.20
Analyzing files from jeanhassan
18%
(2 more...)

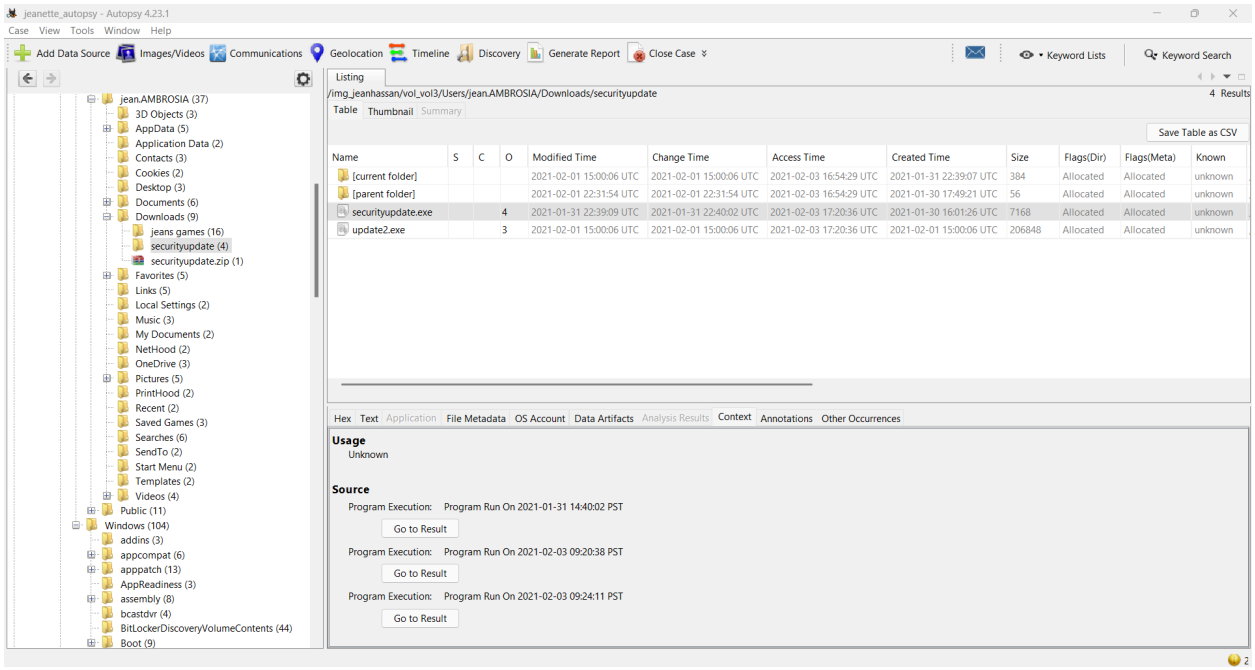

jeanette_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Data Sources
jeanhassan_1 Host
jeanhassan
vol1 (Unallocated: 0-2047)
vol2 (NTFS / exFAT (0x07): 2048-104447)
vol3 (NTFS / exFAT (0x07): 104448-103830231)
$OrphanFiles (1857)
$Extend (9)
$Recycle.Bin (6)
$Unalloc (32)
$WinREAgent (3)
Documents and Settings (2)
PerfLogs (2)
Program Files (19)
Program Files (x86) (17)
ProgramData (17)
Recovery (2)
System Volume Information (6)
Users (10)
Windows (104)
addins (3)
appcompat (6)
apppatch (13)
AppReadiness (3)
assembly (8)
bcastdvr (4)
BitLockerDiscoveryVolumeContents (44)
Boot (9)
Branding (4)
CbsTemp (2)
Containers (4)
CSC (3)
Cursors (214)
debug (7)
diagnostics (5)
DiagTrack (11)
DigitalLocker (3)
Downloaded Program Files (3)
Listing
/img_jeanhassan/vol_vol3/Windows/System32/config
48 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(Dir)
SECURITY.LOG2 2019-12-07 09:03:44 UTC 2021-01-20 14:23:18 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 65536 Allocated
SOFTWARE 2021-01-30 20:20:53 UTC 2021-01-20 14:23:19 UTC 2021-01-30 20:20:53 UTC 2019-12-07 09:03:44 UTC 72613888 Allocated
SOFTWARE.LOG1 2019-12-07 09:03:44 UTC 2021-01-20 14:23:18 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 9830400 Allocated
SOFTWARE.LOG2 2019-12-07 09:03:44 UTC 2021-01-20 14:23:18 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 15745024 Allocated
SYSTEM 2021-01-30 20:20:53 UTC 2021-01-20 14:23:20 UTC 2021-01-30 20:20:53 UTC 2019-12-07 09:03:44 UTC 11796480 Allocated
SYSTEM.LOG1 2019-12-07 09:03:44 UTC 2021-01-20 14:23:18 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 1392640 Allocated
SYSTEM.LOG2 2019-12-07 09:03:44 UTC 2021-01-20 14:23:18 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 1212416 Allocated
TxR 2021-01-30 20:16:15 UTC 2021-01-30 20:16:15 UTC 2021-01-30 20:16:15 UTC 2019-12-07 09:14:52 UTC 56 Allocated
[current folder] 2021-02-03 08:43:18 UTC 2021-02-03 08:43:18 UTC 2021-02-03 08:43:18 UTC 2019-12-07 09:03:44 UTC 56 Allocated
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
NameServer
DhcpServer
Lease
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
SubnetMask
DefaultGateway
DefaultGatewayMetric
{e0ec89d8-2a3a-11eb-ba6e-806e6f6
NsiObjectSecurity
PersistentRoutes
Metadata
Name: SubnetMask
Type: REG_MULTI_SZ
Value
255.255.255.0
Analyzing files from jeanhassan
19%
(2 more...)

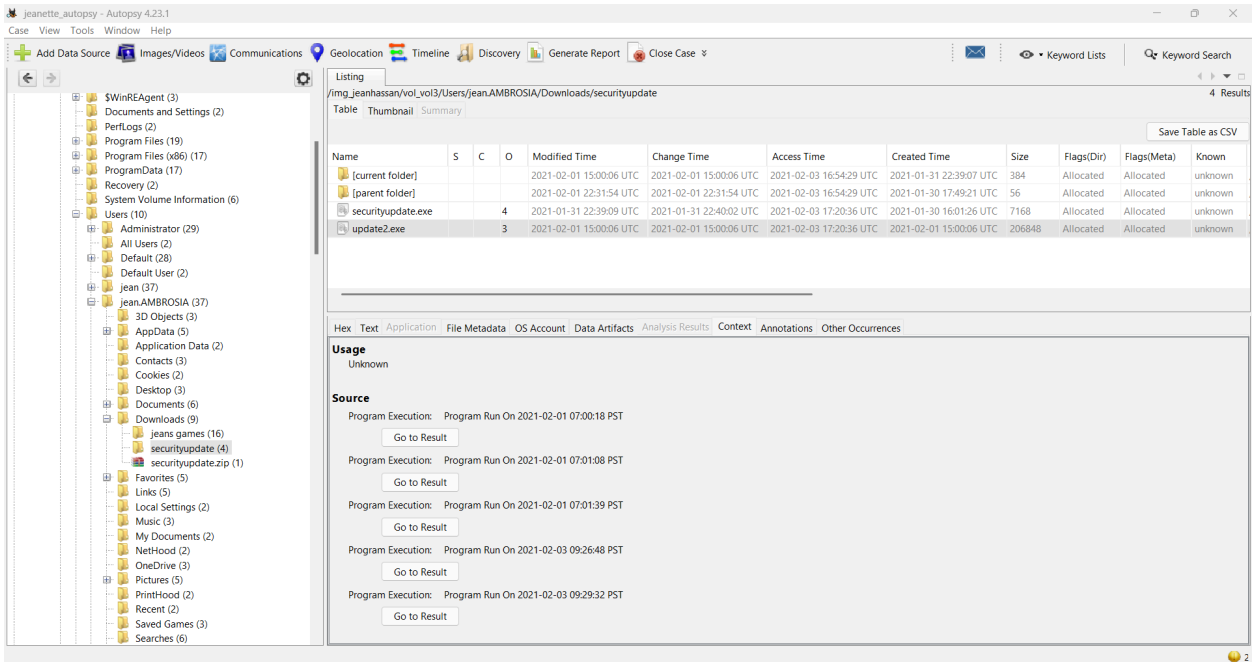


5.2.4.9 Jeanette_A_0009 Downloads

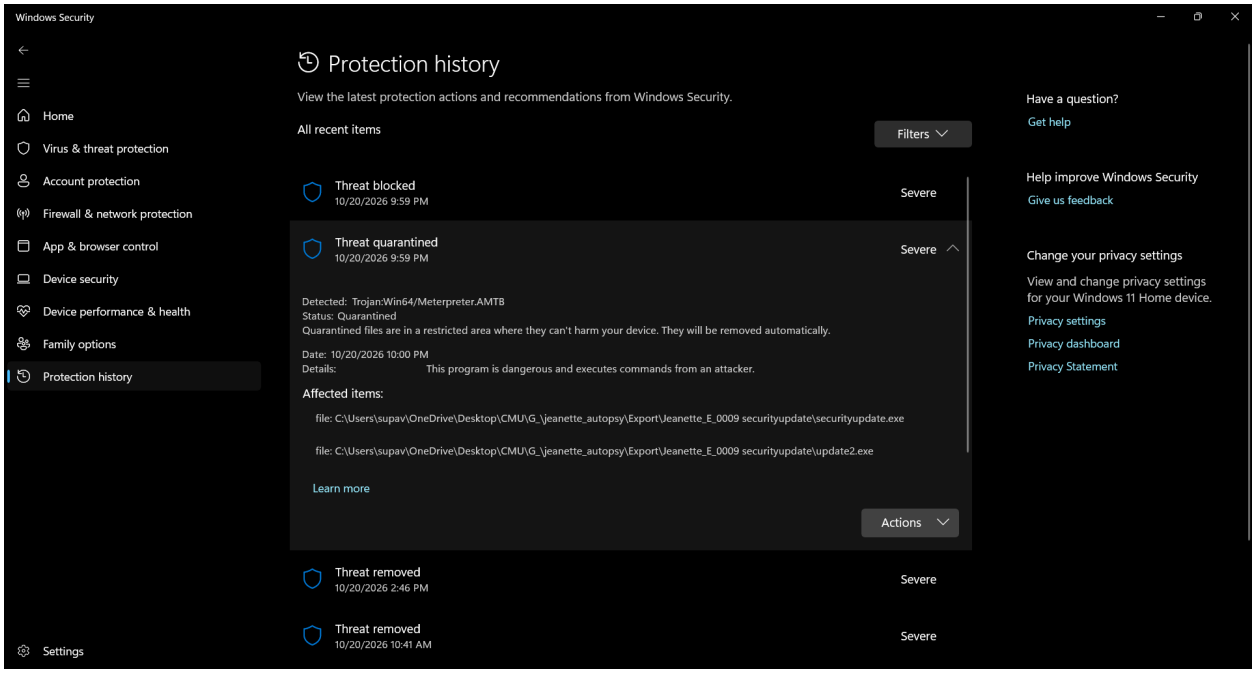



jeanette_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
Listing
/img_jeanhassan/vol_vol3/Users/jean.AMBROSIA/Downloads/securityupdate
4 Results
Table Thumbnail Summary
Save Table as CSV
Name S C O Modified Time Change Time Access Time Created Time Size Flags(Dir) Flags(Meta) Known
[current folder] 2021-02-01 15:00:06 UTC 2021-02-01 15:00:06 UTC 2021-02-03 16:54:29 UTC 2021-01-31 22:39:07 UTC 384 Allocated Allocated unknown
[parent folder] 2021-02-01 22:31:54 UTC 2021-02-01 22:31:54 UTC 2021-02-03 16:54:29 UTC 2021-01-30 17:49:21 UTC 56 Allocated Allocated unknown
securityupdate.exe 4 2021-01-31 22:39:09 UTC 2021-01-31 22:40:02 UTC 2021-02-03 17:20:36 UTC 2021-01-30 16:01:26 UTC 7168 Allocated Allocated unknown
update2.exe 3 2021-02-01 15:00:06 UTC 2021-02-01 15:00:06 UTC 2021-02-03 17:20:36 UTC 2021-02-01 15:00:06 UTC 206848 Allocated Allocated unknown
Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences
Metadata
Name: /img_jeanhassan/vol_vol3/Users/jean.AMBROSIA/Downloads/securityupdate/securityupdate.exe
Type: File System
MIME Type: application/x-dosexec
Size: 7168
File Name Allocation: Allocated
Metadata Allocation: Allocated
Modified: 2021-01-31 22:39:09 UTC
Accessed: 2021-02-03 17:20:36 UTC
Created: 2021-01-30 16:01:26 UTC
Changed: 2021-01-31 22:40:02 UTC
MD5: 65e3eb5ca3012baabab31a5af0afa625
SHA-256: 41d98cccf0dd6791b4e6d7b1feccef24c8a6bc9650daaf535d3dbc79b88e7de9
Hash Lookup Results: UNKNOWN
Internal ID: 23419
From The Sleuth Kit istat Tool:

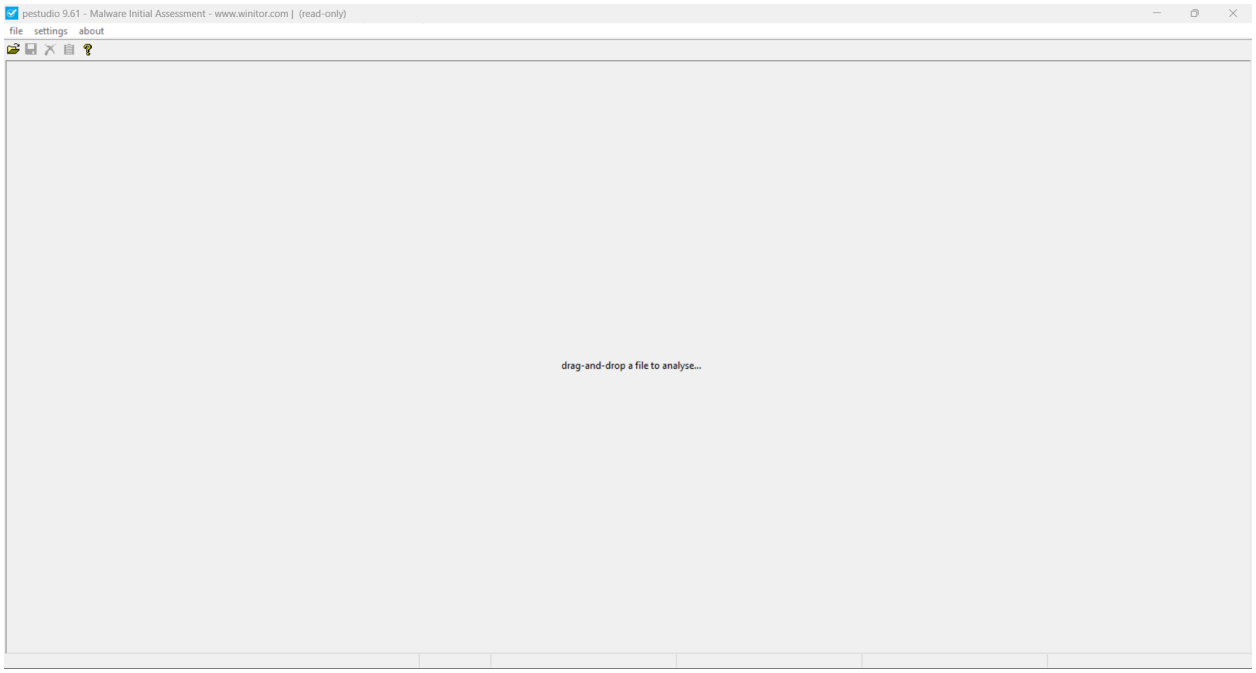

jeanette_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
Listing
/img_jeanhassan/vol_vol3/Users/jean.AMBROSIA/Downloads/securityupdate
4 Results
Table Thumbnail Summary
Save Table as CSV
Name S C O Modified Time Change Time Access Time Created Time Size Flags(Dir) Flags(Meta) Known
[current folder] 2021-02-01 15:00:06 UTC 2021-02-01 15:00:06 UTC 2021-02-03 16:54:29 UTC 2021-01-31 22:39:07 UTC 384 Allocated Allocated unknown
[parent folder] 2021-02-01 22:31:54 UTC 2021-02-01 22:31:54 UTC 2021-02-03 16:54:29 UTC 2021-01-30 17:49:21 UTC 56 Allocated Allocated unknown
securityupdate.exe 4 2021-01-31 22:39:09 UTC 2021-01-31 22:40:02 UTC 2021-02-03 17:20:36 UTC 2021-01-30 16:01:26 UTC 7168 Allocated Allocated unknown
update2.exe 3 2021-02-01 15:00:06 UTC 2021-02-01 15:00:06 UTC 2021-02-03 17:20:36 UTC 2021-02-01 15:00:06 UTC 206848 Allocated Allocated unknown
Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences
Usage
Unknown
Source
Program Execution: Program Run On 2021-01-31 14:40:02 PST
Go to Result
Program Execution: Program Run On 2021-02-03 09:20:38 PST
Go to Result
Program Execution: Program Run On 2021-02-03 09:24:11 PST
Go to Result

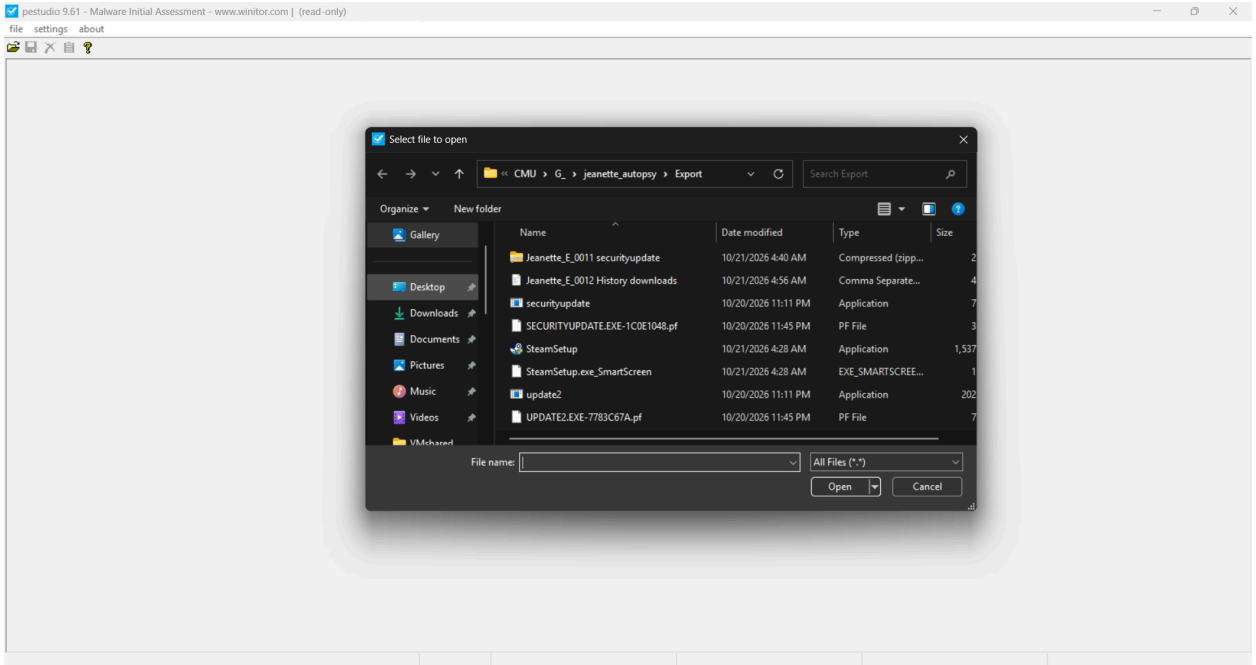


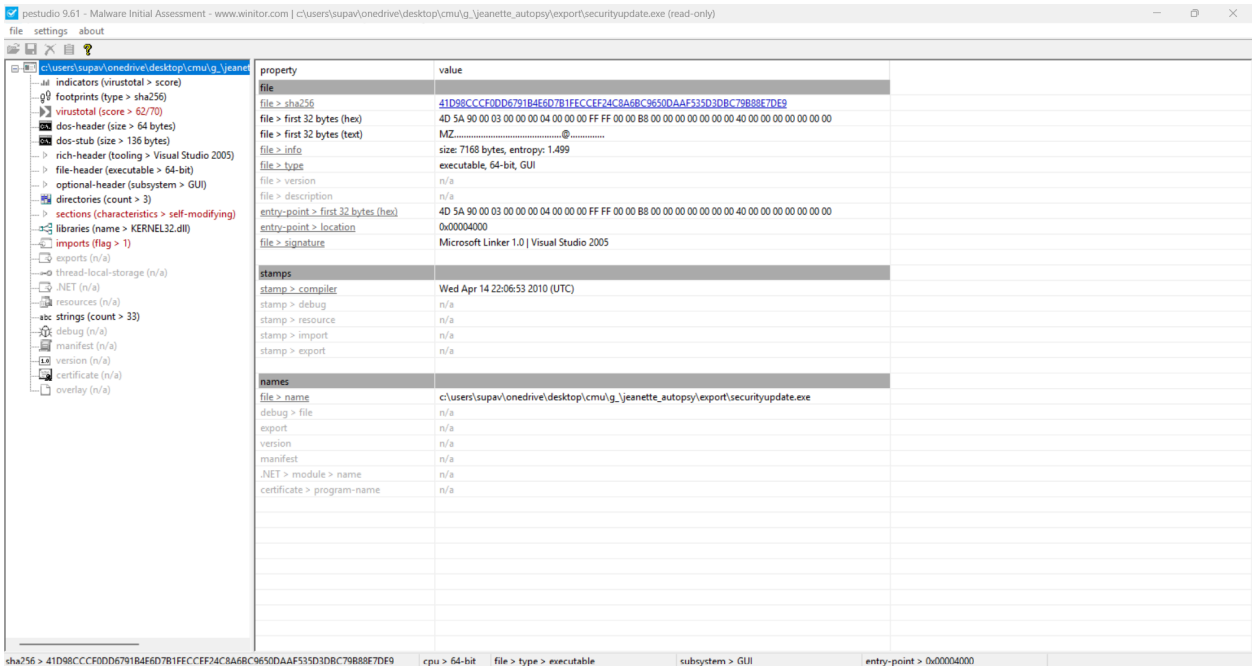
5.2.4.10 Jeanette_A_0010 Downloads Windows Security

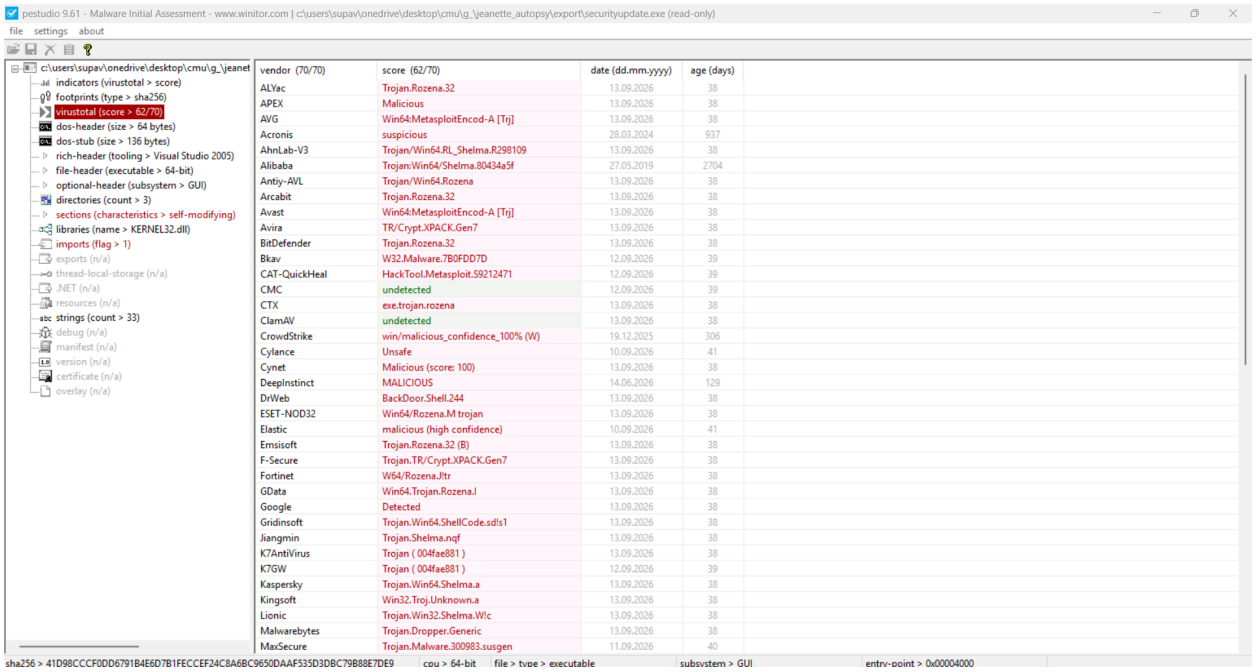


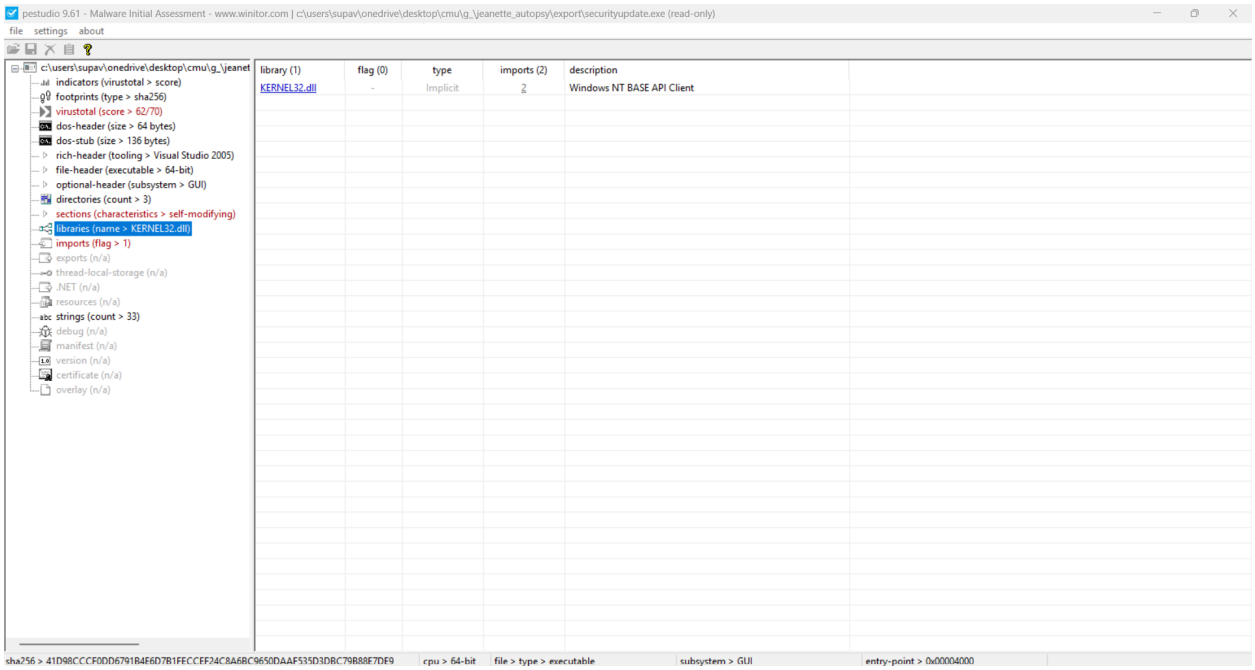
5.2.4.11 Jeanette_A_0011 securityupdate.exe PEstudio

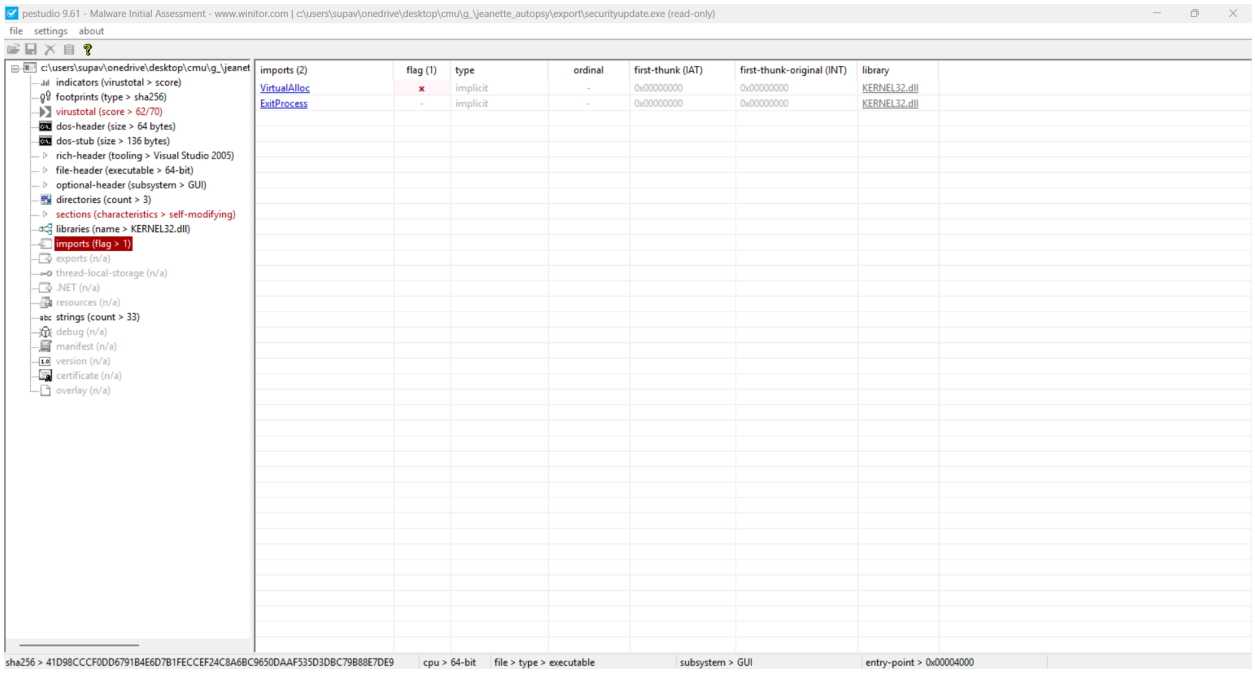


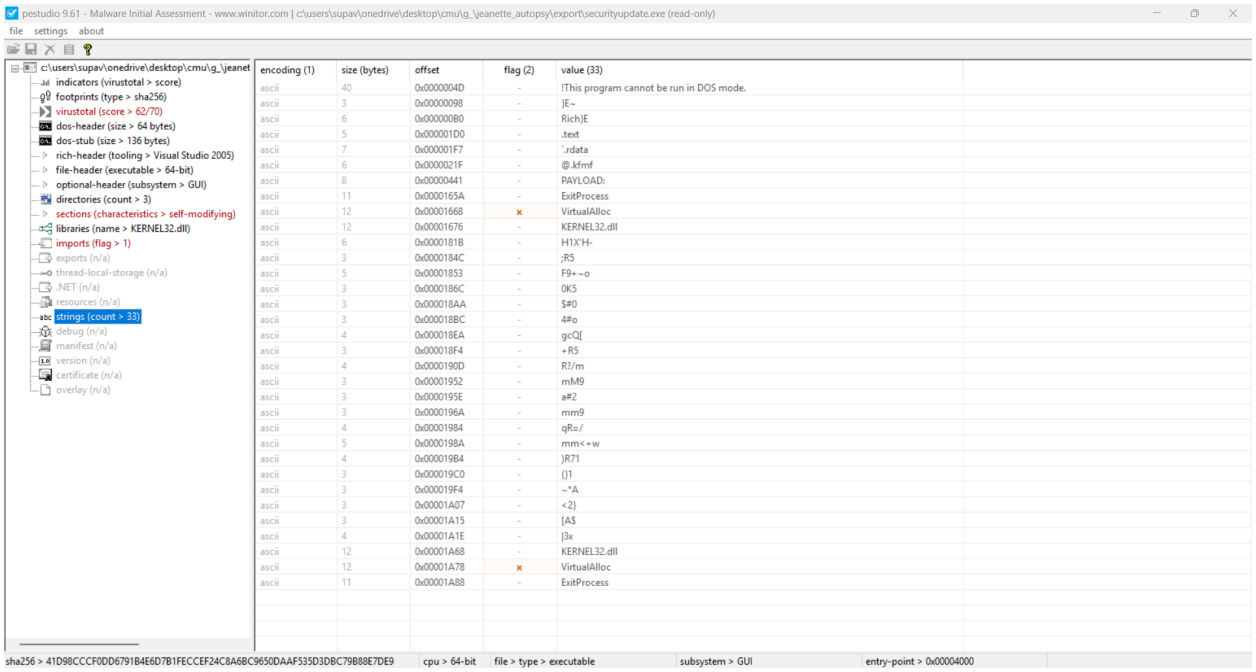



pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\supav\onedrive\desktop\cmu\g_\jeanette_autopsy\export\securityupdate.exe (read-only)
file settings about
c:\users\supav\onedrive\desktop\cmu\g_\jeanet
indicators (virustotal > score)
footprints (type > sha256)
virustotal (score > 62/70)
dos-header (size > 64 bytes)
dos-stub (size > 136 bytes)
rich-header (tooling > Visual Studio 2005)
file-header (executable > 64-bit)
optional-header (subsystem > GUI)
directories (count > 3)
sections (characteristics > self-modifying)
libraries (name > KERNEL32.dll)
imports (flag > 1)
exports (n/a)
thread-local-storage (n/a)
.NET (n/a)
resources (n/a)
strings (count > 33)
debug (n/a)
manifest (n/a)
version (n/a)
certificate (n/a)
overlay (n/a)
property value
file
file > sha256 41D98CCCF0DD6791B4E6D7B1FECCEF24C8A6BC9650DAAF535D3DBC79B88E7DE9
file > first 32 bytes (hex) 4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 B8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00
file > first 32 bytes (text) MZ.....@.....
file > info size: 7168 bytes, entropy: 1.499
file > type executable, 64-bit, GUI
file > version n/a
file > description n/a
entry-point > first 32 bytes (hex) 4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 B8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00
entry-point > location 0x00004000
file > signature Microsoft Linker 1.0 | Visual Studio 2005
stamps
stamp > compiler Wed Apr 14 22:06:53 2010 (UTC)
stamp > debug n/a
stamp > resource n/a
stamp > import n/a
stamp > export n/a
names
file > name c:\users\supav\onedrive\desktop\cmu\g_\jeanette_autopsy\export\securityupdate.exe
debug > file n/a
export n/a
version n/a
manifest n/a
.NET > module > name n/a
certificate > program-name n/a
sha256 > 41D98CCCF0DD6791B4E6D7B1FECCEF24C8A6BC9650DAAF535D3DBC79B88E7DE9
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x00004000

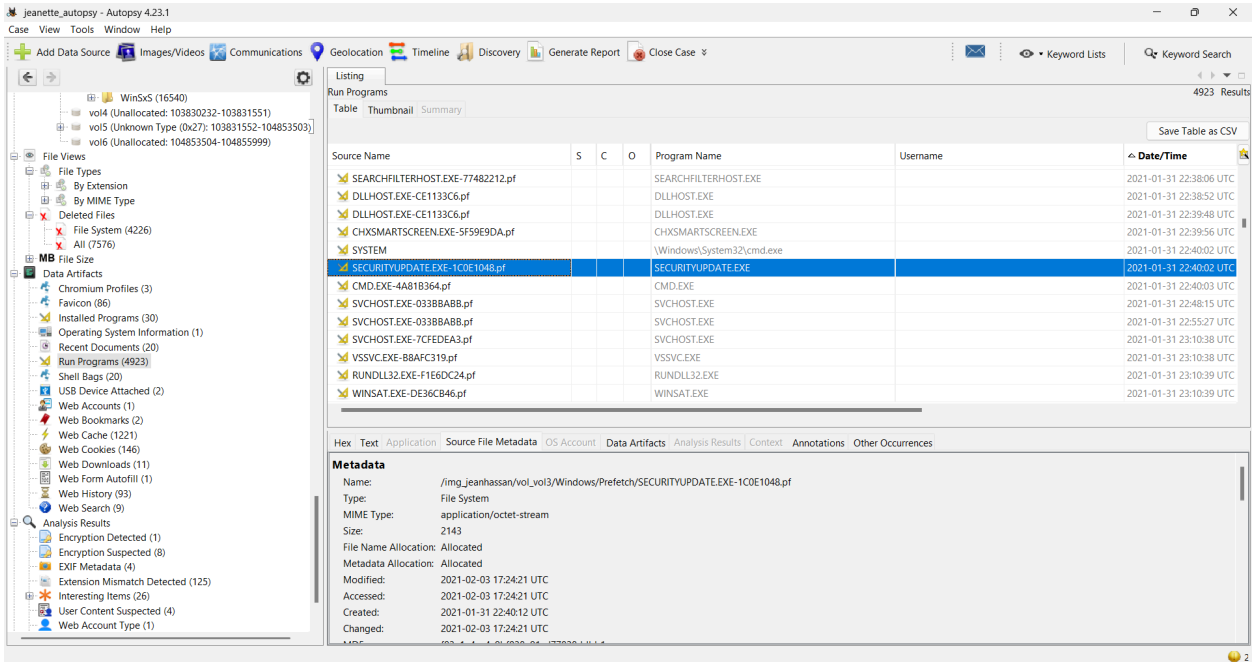

pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\supav\onedrive\desktop\cmu\g_\jeanette_autopsy\export\securityupdate.exe (read-only)
file settings about
vendor (70/70) score (62/70) date (dd.mm.yyyy) age (days)
ALYac Trojan.Rozena.32 13.09.2026 38
APEX Malicious 13.09.2026 38
AVG Win64:MetasploitEncod-A [Trj] 13.09.2026 38
Acronis suspicious 28.03.2024 937
AhnLab-V3 Trojan/Win64.RL_Shelma.R298109 13.09.2026 38
Alibaba Trojan:Win64/Shelma.80434a5f 27.05.2019 2704
Antiy-AVL Trojan/Win64.Rozena 13.09.2026 38
Arcabit Trojan.Rozena.32 13.09.2026 38
Avast Win64:MetasploitEncod-A [Trj] 13.09.2026 38
Avira TR/Crypt.XPACK.Gen7 13.09.2026 38
BitDefender Trojan.Rozena.32 13.09.2026 38
Bkav W32.Malware.7B0FDD7D 12.09.2026 39
CAT-QuickHeal HackTool.Metasploit.S9212471 12.09.2026 39
CMC undetected 12.09.2026 39
CTX exe.trojan.rozena 13.09.2026 38
ClamAV undetected 13.09.2026 38
CrowdStrike win/malicious_confidence_100% (W) 19.12.2025 306
Cylance Unsafe 10.09.2026 41
Cynet Malicious (score: 100) 13.09.2026 38
DeepInstinct MALICIOUS 14.06.2026 129
DrWeb BackDoor.Shell.244 13.09.2026 38
ESET-NOD32 Win64/Rozena.M trojan 13.09.2026 38
Elastic malicious (high confidence) 10.09.2026 41
Emsisoft Trojan.Rozena.32 (B) 13.09.2026 38
F-Secure Trojan.TR/Crypt.XPACK.Gen7 13.09.2026 38
Fortinet W64/Rozena.J!tr 13.09.2026 38
GData Win64.Trojan.Rozena.I 13.09.2026 38
Google Detected 13.09.2026 38
GridinSoft Trojan.Win64.ShellCode.sd!s1 13.09.2026 38
Jiangmin Trojan.Shelma.nqf 13.09.2026 38
K7AntiVirus Trojan (004fae881) 13.09.2026 38
K7GW Trojan (004fae881) 12.09.2026 39
Kaspersky Trojan.Win64.Shelma.a 13.09.2026 38
Kingsoft Win32.Troj.Unknown.a 13.09.2026 38
Lionic Trojan.Win32.Shelma.W!c 13.09.2026 38
Malwarebytes Trojan.Dropper.Generic 13.09.2026 38
MaxSecure Trojan.Malware.300983.susgen 11.09.2026 40


pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\supav\onedrive\desktop\cmu\g_\jeanette_autopsy\export\securityupdate.exe (read-only)
file settings about
c:\users\supav\onedrive\desktop\cmu\g_\jeanet
indicators (virustotal > score)
footprints (type > sha256)
virustotal (score > 62/70)
dos-header (size > 64 bytes)
dos-stub (size > 136 bytes)
rich-header (tooling > Visual Studio 2005)
file-header (executable > 64-bit)
optional-header (subsystem > GUI)
directories (count > 3)
sections (characteristics > self-modifying)
libraries (name > KERNEL32.dll)
imports (flag > 1)
exports (n/a)
thread-local-storage (n/a)
.NET (n/a)
resources (n/a)
strings (count > 33)
debug (n/a)
manifest (n/a)
version (n/a)
certificate (n/a)
overlay (n/a)
library (1) flag (0) type imports (2) description
KERNEL32.dll - Implicit 2 Windows NT BASE API Client
sha256 > 41D98CCCF0DD6791B4E6D7B1FECCEF24C8A6BC9650DAAF535D3DBC79B88E7DE9
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x00004000

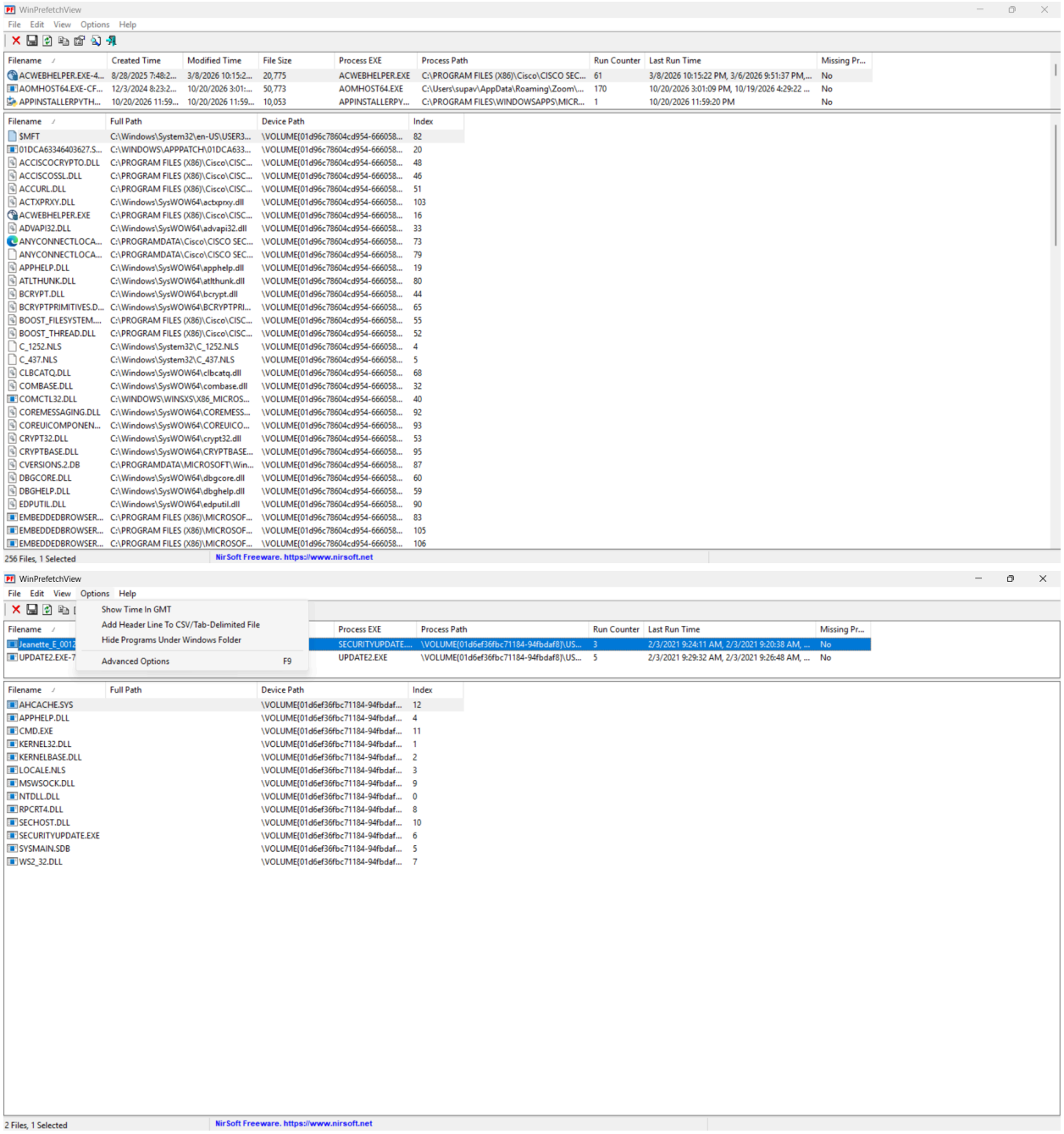

pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\supav\onedrive\desktop\cmu\g_\jeanette_autopsy\export\securityupdate.exe (read-only)
file settings about
c:\users\supav\onedrive\desktop\cmu\g_\jeanet
indicators (virustotal > score)
footprints (type > sha256)
virustotal (score > 62/70)
dos-header (size > 64 bytes)
dos-stub (size > 136 bytes)
rich-header (tooling > Visual Studio 2005)
file-header (executable > 64-bit)
optional-header (subsystem > GUI)
directories (count > 3)
sections (characteristics > self-modifying)
libraries (name > KERNEL32.dll)
imports (flag > 1)
exports (n/a)
thread-local-storage (n/a)
.NET (n/a)
resources (n/a)
strings (count > 33)
debug (n/a)
manifest (n/a)
version (n/a)
certificate (n/a)
overlay (n/a)
imports (2) flag (1) type ordinal first-thunk (IAT) first-thunk-original (INT) library
VirtualAlloc x implicit - 0x00000000 0x00000000 KERNEL32.dll
ExitProcess - implicit - 0x00000000 0x00000000 KERNEL32.dll
sha256 > 41D98CCCF0DD6791B4E6D7B1FECCEF24C8A6BC9650DAAF535D3DBC79B88E7DE9
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x00004000

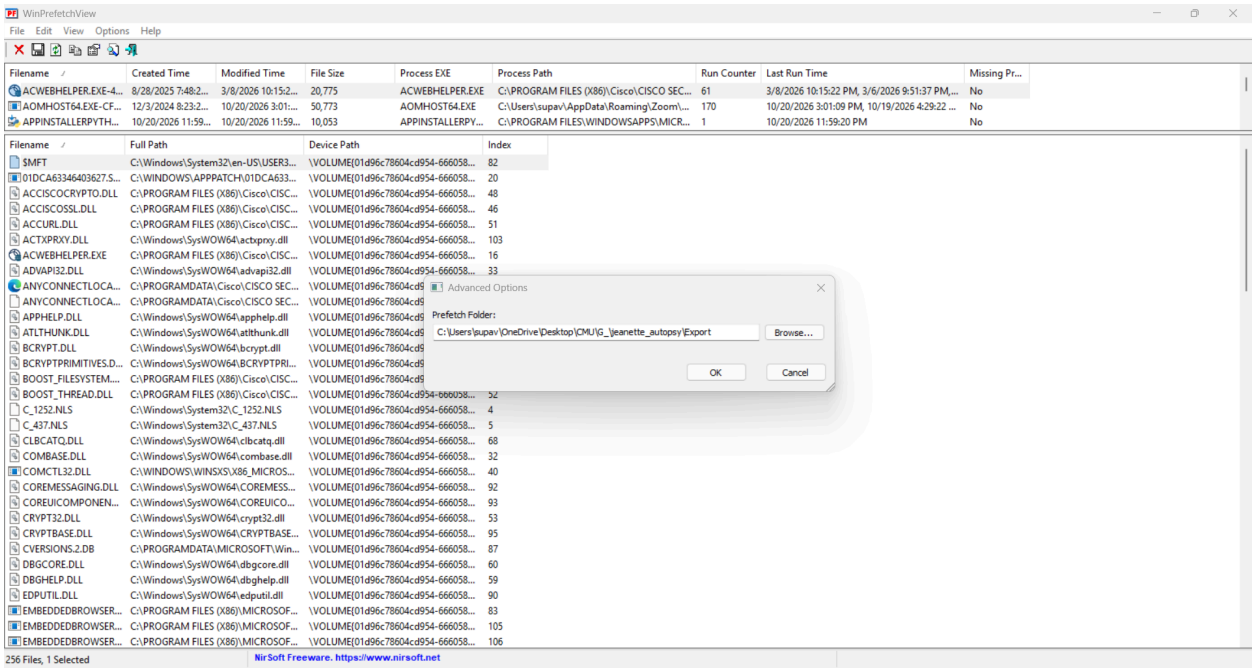


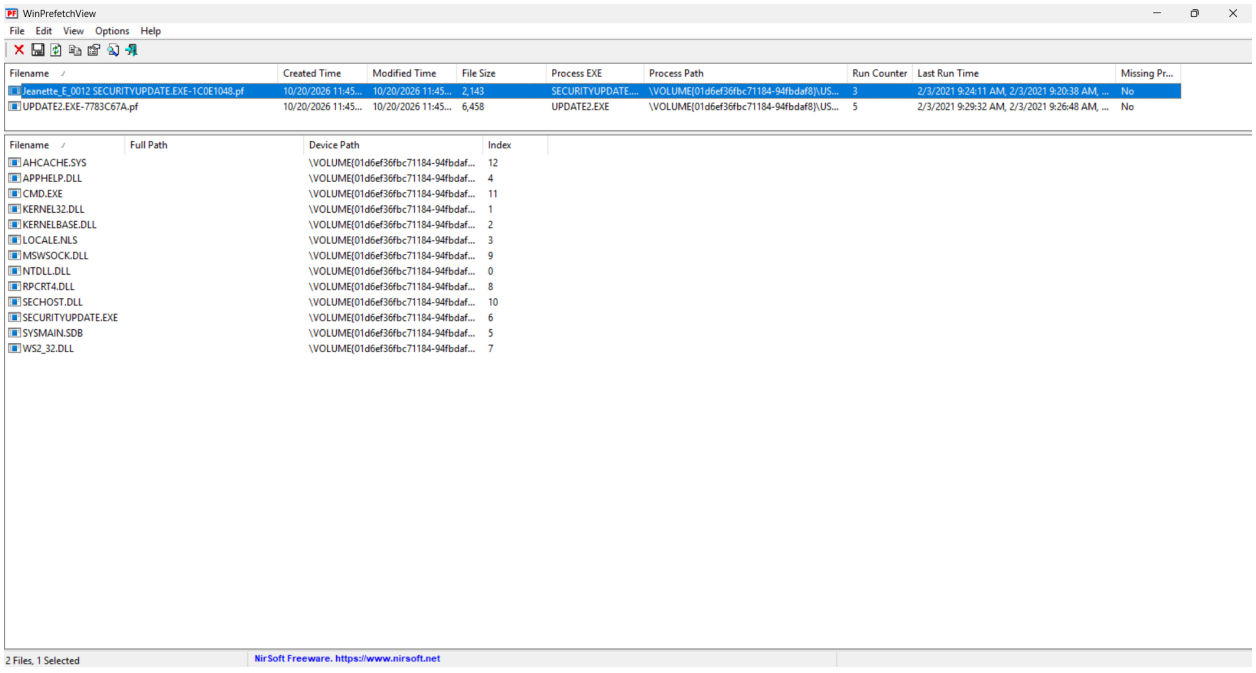
5.2.4.12 Jeanette_A_0012 securityupdate.exe Run Programs



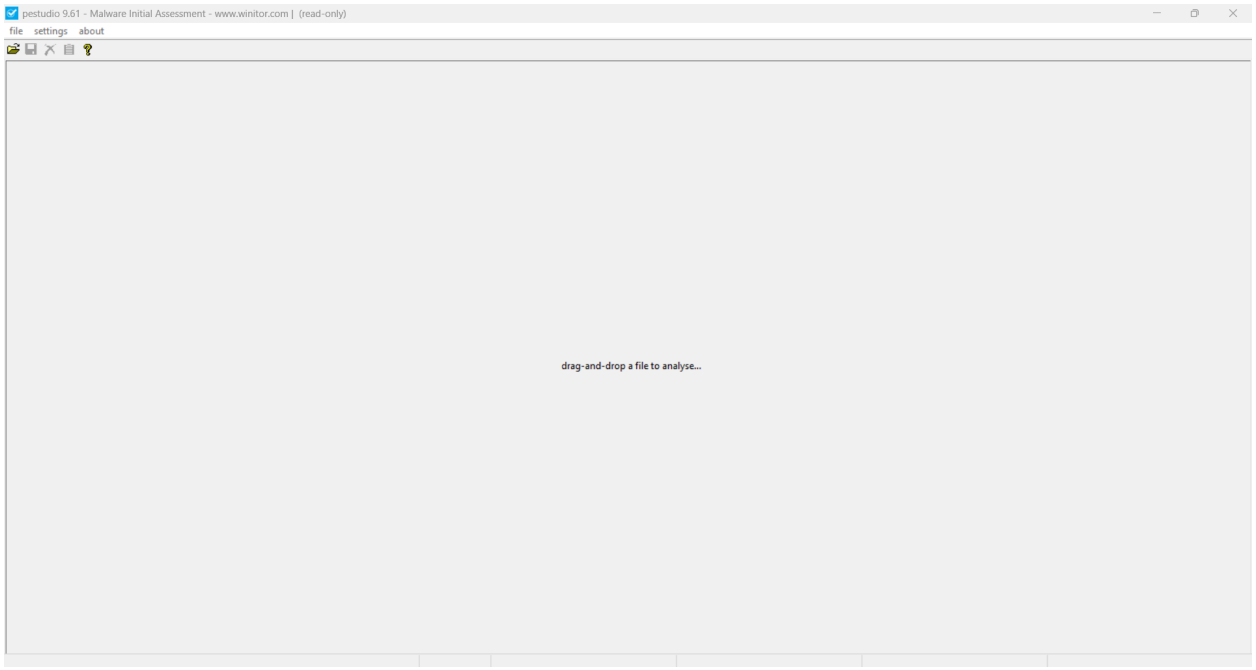
5.2.4.13 Jeanette_A_0013 securityupdate.exe WinPrefetchViewer

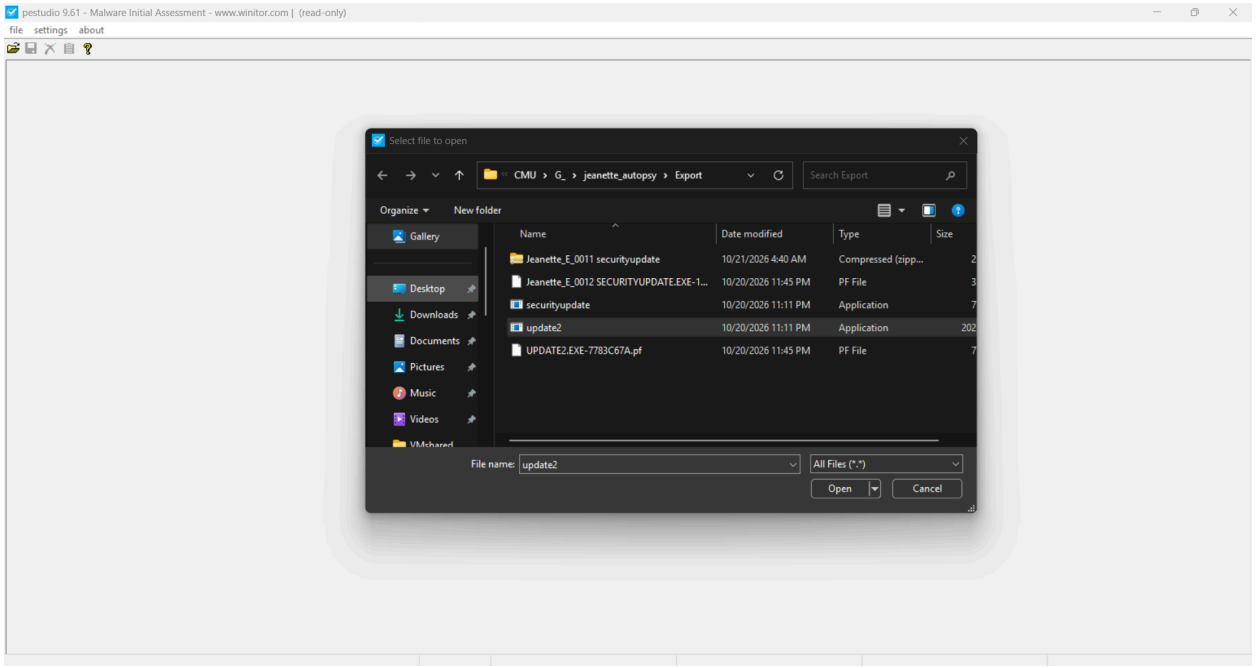


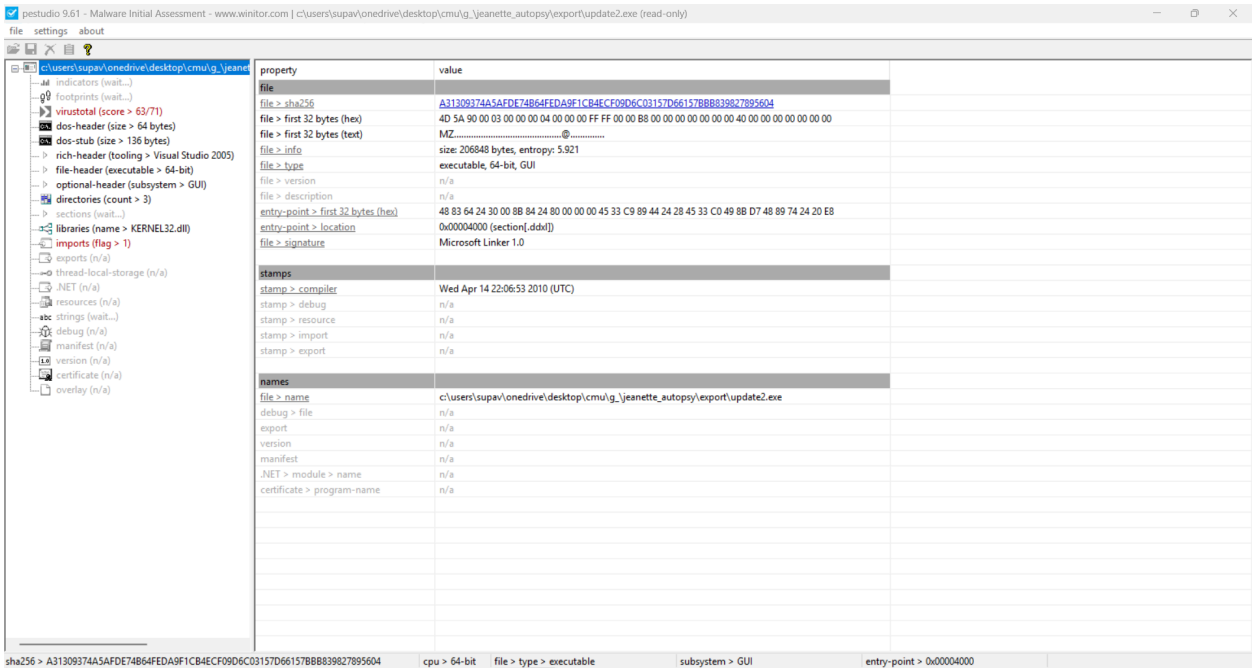

WinPrefetchView
File Edit View Options Help
Filename Created Time Modified Time File Size Process EXE Process Path Run Counter Last Run Time Missing Pr...
ACWEBHELPER.EXE-4... 8/28/2025 7:48:2... 3/8/2026 10:15:2... 20,775 ACWEBHELPER.EXE C:\PROGRAM FILES (X86)\Cisco\CISCO SEC... 61 3/8/2026 10:15:22 PM, 3/6/2026 9:51:37 PM,... No
AOMHOST64.EXE-CF... 12/3/2024 8:23:2... 10/20/2026 3:01:... 50,773 AOMHOST64.EXE C:\Users\supav\AppData\Roaming\Zoom\... 170 10/20/2026 3:01:09 PM, 10/19/2026 4:29:22 ... No
APPINSTALLERPYTH... 10/20/2026 11:59... 10/20/2026 11:59... 10,053 APPINSTALLERPY... C:\PROGRAM FILES\WINDOWSAPPS\MICR... 1 10/20/2026 11:59:20 PM No
Filename Full Path Device Path Index
$MFT C:\Windows\System32\en-US\USER3... \VOLUME{01d96c78604cd954-666058... 82
01DCA63346403627.S... C:\WINDOWS\APPPATCH\01DCA633... \VOLUME{01d96c78604cd954-666058... 20
ACCISCOCRYPTO.DLL C:\PROGRAM FILES (X86)\Cisco\CISC... \VOLUME{01d96c78604cd954-666058... 48
ACCISCOSSL.DLL C:\PROGRAM FILES (X86)\Cisco\CISC... \VOLUME{01d96c78604cd954-666058... 46
ACCURL.DLL C:\PROGRAM FILES (X86)\Cisco\CISC... \VOLUME{01d96c78604cd954-666058... 51
ACTXPRXY.DLL C:\Windows\SysWOW64\actxprxy.dll \VOLUME{01d96c78604cd954-666058... 103
ACWEBHELPER.EXE C:\PROGRAM FILES (X86)\Cisco\CISC... \VOLUME{01d96c78604cd954-666058... 16
ADVAPI32.DLL C:\Windows\SysWOW64\advapi32.dll \VOLUME{01d96c78604cd954-666058... 33
ANYCONNECTLOCA... C:\PROGRAMDATA\Cisco\CISCO SEC... \VOLUME{01d96c78604cd9
ANYCONNECTLOCA... C:\PROGRAMDATA\Cisco\CISCO SEC... \VOLUME{01d96c78604cd9
APPHELP.DLL C:\Windows\SysWOW64\apphelp.dll \VOLUME{01d96c78604cd9
ATLTHUNK.DLL C:\Windows\SysWOW64\atlthunk.dll \VOLUME{01d96c78604cd9
BCRYPT.DLL C:\Windows\SysWOW64\bcrypt.dll \VOLUME{01d96c78604cd9
BCRYPTPRIMITIVES.D... C:\Windows\SysWOW64\BCRYPTPRI... \VOLUME{01d96c78604cd9
BOOST_FILESYSTEM.... C:\PROGRAM FILES (X86)\Cisco\CISC... \VOLUME{01d96c78604cd9
BOOST_THREAD.DLL C:\PROGRAM FILES (X86)\Cisco\CISC... \VOLUME{01d96c78604cd954-666058... 52
C_1252.NLS C:\Windows\System32\C_1252.NLS \VOLUME{01d96c78604cd954-666058... 4
C_437.NLS C:\Windows\System32\C_437.NLS \VOLUME{01d96c78604cd954-666058... 5
CLBCATQ.DLL C:\Windows\SysWOW64\clbcatq.dll \VOLUME{01d96c78604cd954-666058... 68
COMBASE.DLL C:\Windows\SysWOW64\combase.dll \VOLUME{01d96c78604cd954-666058... 32
COMCTL32.DLL C:\WINDOWS\WINSXS\X86_MICROS... \VOLUME{01d96c78604cd954-666058... 40
COREMESSAGING.DLL C:\Windows\SysWOW64\COREMESS... \VOLUME{01d96c78604cd954-666058... 92
COREUICOMPONEN... C:\Windows\SysWOW64\COREUICO... \VOLUME{01d96c78604cd954-666058... 93
CRYPT32.DLL C:\Windows\SysWOW64\crypt32.dll \VOLUME{01d96c78604cd954-666058... 53
CRYPTBASE.DLL C:\Windows\SysWOW64\CRYPTBASE... \VOLUME{01d96c78604cd954-666058... 95
CVERSIONS.2.DB C:\PROGRAMDATA\MICROSOFT\Win... \VOLUME{01d96c78604cd954-666058... 87
DBGCORE.DLL C:\Windows\SysWOW64\dbgcore.dll \VOLUME{01d96c78604cd954-666058... 60
DBGHELP.DLL C:\Windows\SysWOW64\dbghelp.dll \VOLUME{01d96c78604cd954-666058... 59
EDPUTIL.DLL C:\Windows\SysWOW64\edputil.dll \VOLUME{01d96c78604cd954-666058... 90
EMBEDDEDBROWSER... C:\PROGRAM FILES (X86)\MICROSOF... \VOLUME{01d96c78604cd954-666058... 83
EMBEDDEDBROWSER... C:\PROGRAM FILES (X86)\MICROSOF... \VOLUME{01d96c78604cd954-666058... 105
EMBEDDEDBROWSER... C:\PROGRAM FILES (X86)\MICROSOF... \VOLUME{01d96c78604cd954-666058... 106
Advanced Options
Prefetch Folder:
C:\Users\supav\OneDrive\Desktop\CMU\G_\jeanette_autopsy\Export
Browse...
OK
Cancel
256 Files, 1 Selected
NirSoft Freeware. https://www.nirsoft.net

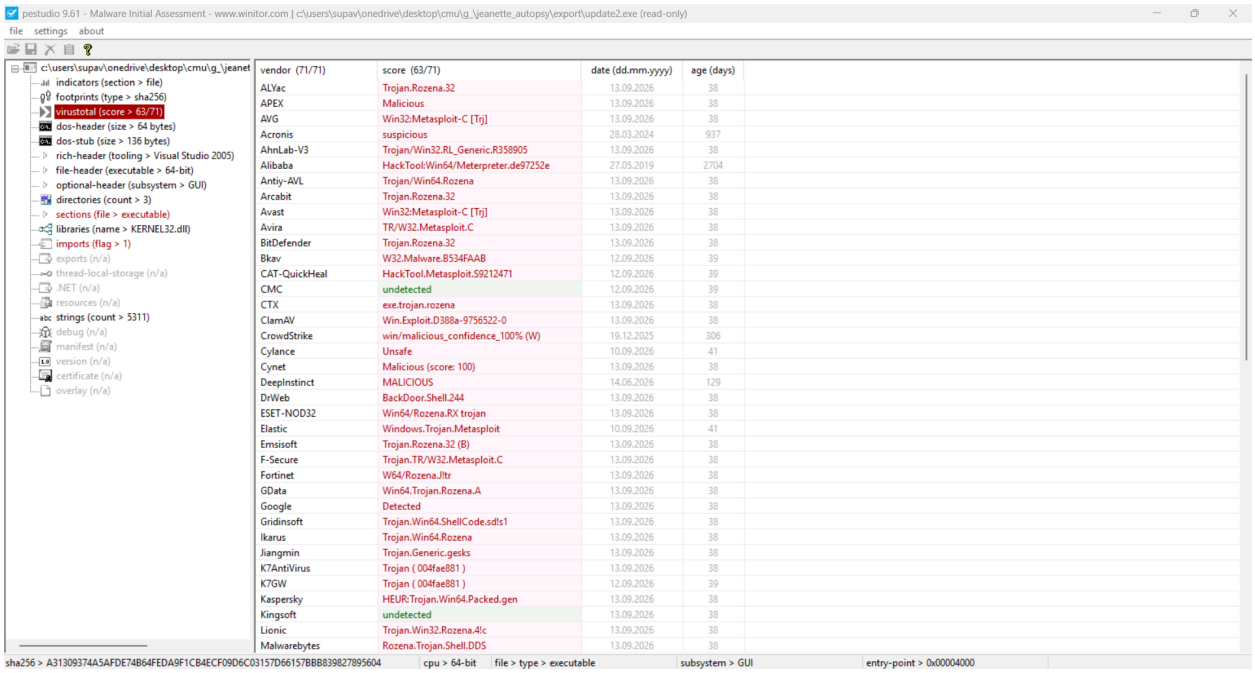

WinPrefetchView
File Edit View Options Help
Filename Created Time Modified Time File Size Process EXE Process Path Run Counter Last Run Time Missing Pr...
Jeanette_E_0012 SECURITYUPDATE.EXE-1C0E1048.pf 10/20/2026 11:45... 10/20/2026 11:45... 2,143 SECURITYUPDATE.... \VOLUME{01d6ef36fbc71184-94fbdaf8}\US... 3 2/3/2021 9:24:11 AM, 2/3/2021 9:20:38 AM, ... No
UPDATE2.EXE-7783C67A.pf 10/20/2026 11:45... 10/20/2026 11:45... 6,458 UPDATE2.EXE \VOLUME{01d6ef36fbc71184-94fbdaf8}\US... 5 2/3/2021 9:29:32 AM, 2/3/2021 9:26:48 AM, ... No
Filename Full Path Device Path Index
AHCACHE.SYS \VOLUME{01d6ef36fbc71184-94fbdaf... 12
APPHELP.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 4
CMD.EXE \VOLUME{01d6ef36fbc71184-94fbdaf... 11
KERNEL32.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 1
KERNELBASE.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 2
LOCALE.NLS \VOLUME{01d6ef36fbc71184-94fbdaf... 3
MSWSOCK.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 9
NTDLL.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 0
RPCRT4.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 8
SECHOST.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 10
SECURITYUPDATE.EXE \VOLUME{01d6ef36fbc71184-94fbdaf... 6
SYSMAIN.SDB \VOLUME{01d6ef36fbc71184-94fbdaf... 5
WS2_32.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 7
2 Files, 1 Selected
NirSoft Freeware. https://www.nirsoft.net

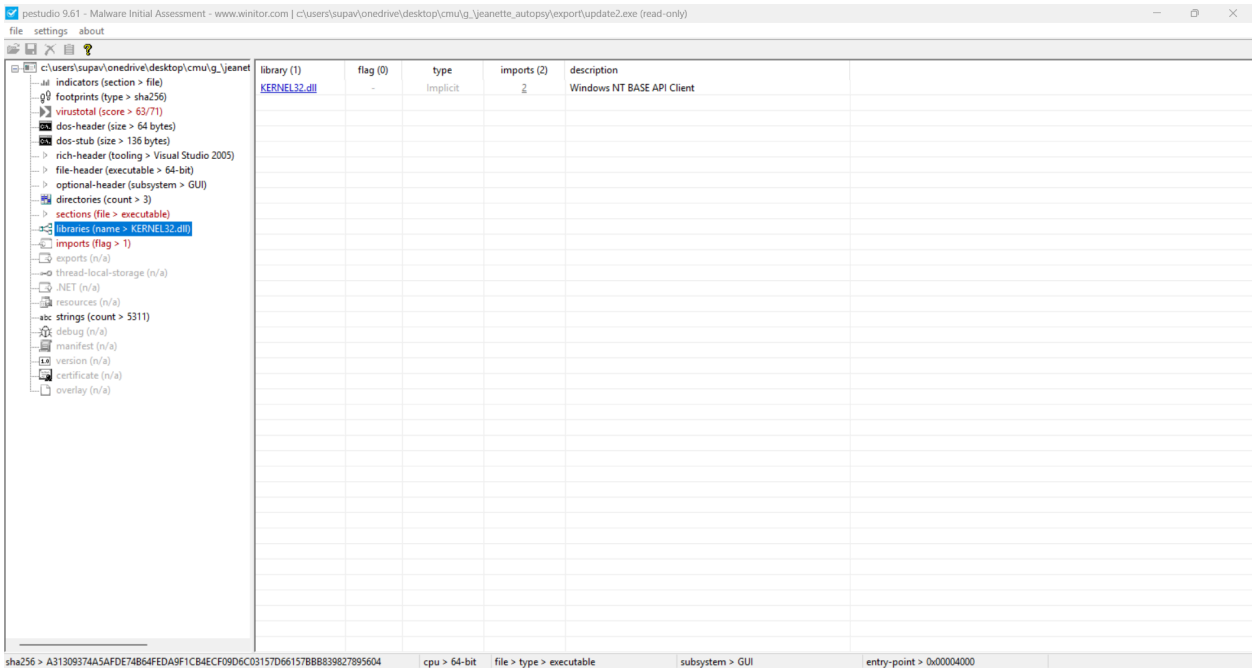
5.2.4.14 Jeanette_A_0014 update2.exe PEstudio

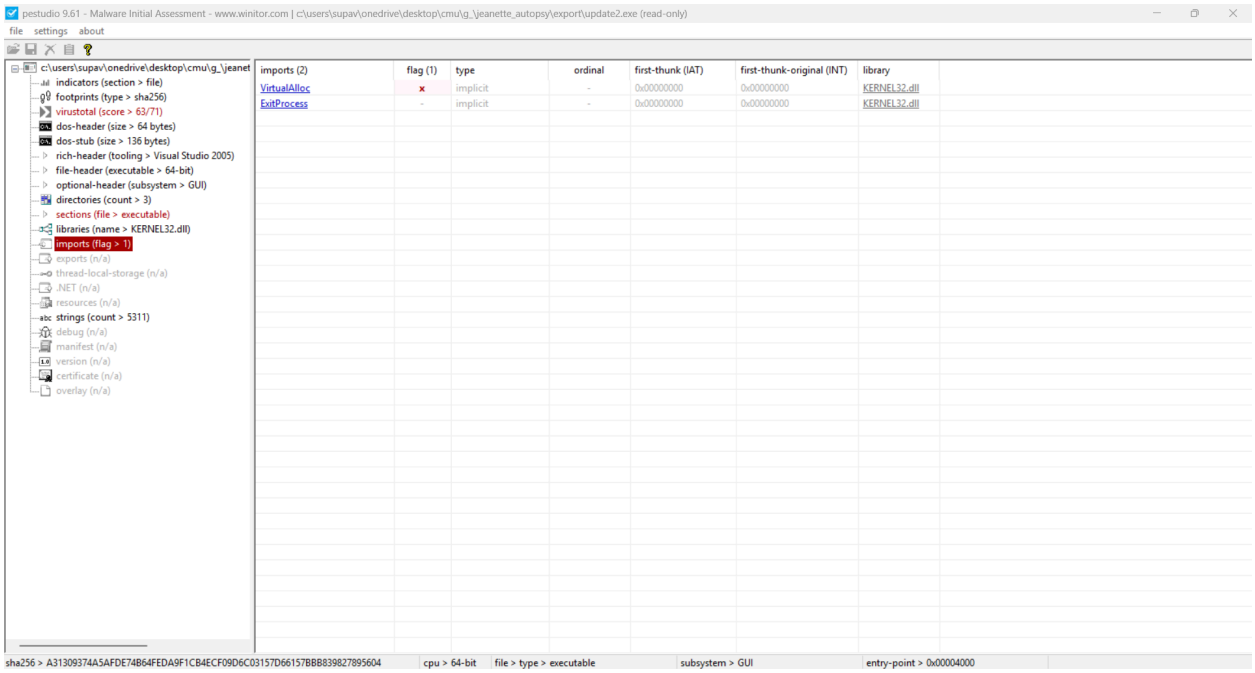


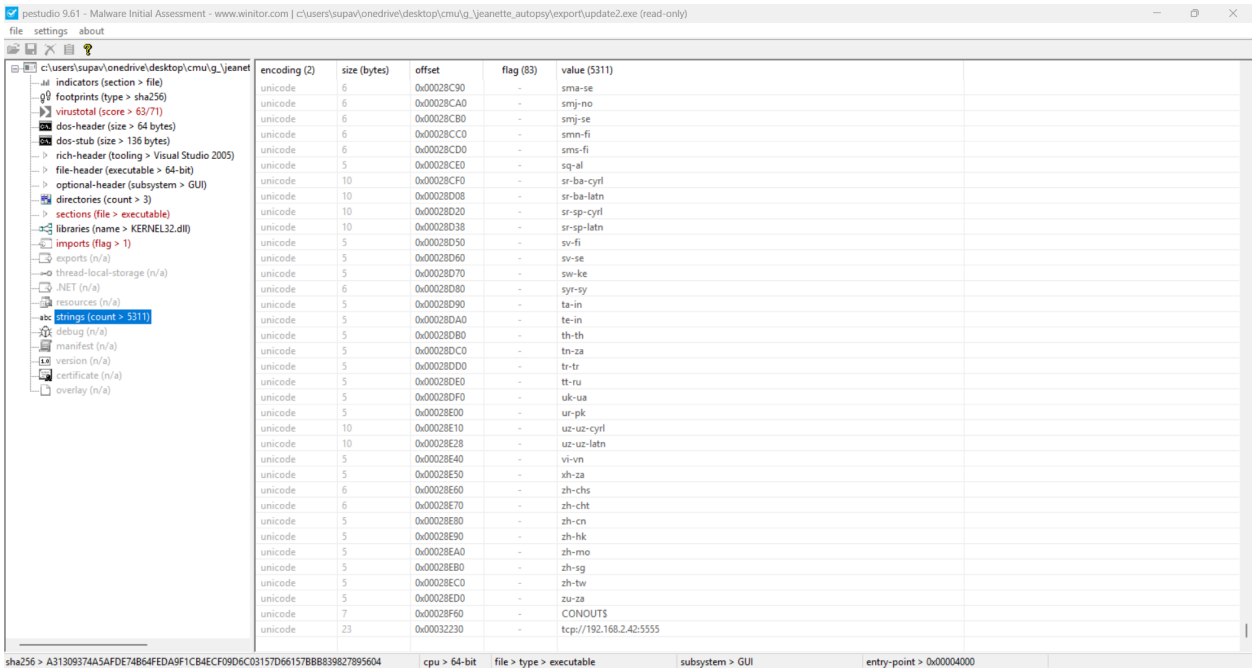


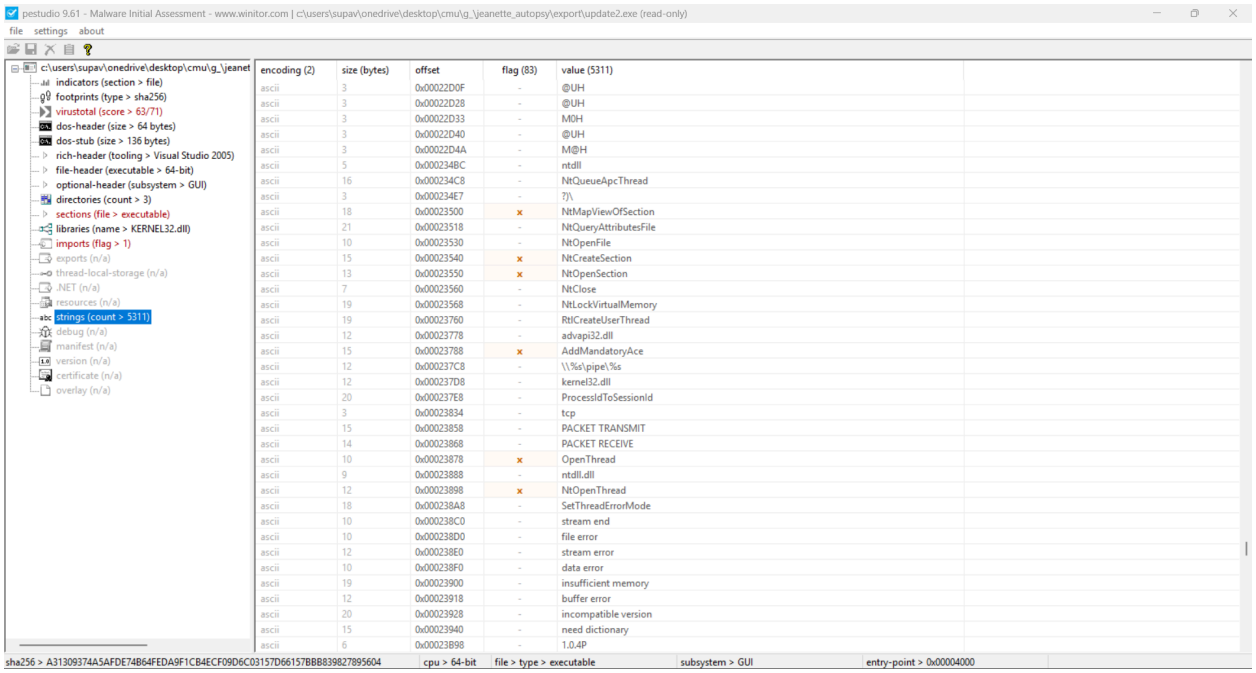

pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\supav\onedrive\desktop\cmu\g_\jeanette_autopsy\export\update2.exe (read-only)
file settings about
c:\users\supav\onedrive\desktop\cmu\g_\jeanet
indicators (wait...)
footprints (wait...)
virustotal (score > 63/71)
dos-header (size > 64 bytes)
dos-stub (size > 136 bytes)
rich-header (tooling > Visual Studio 2005)
file-header (executable > 64-bit)
optional-header (subsystem > GUI)
directories (count > 3)
sections (wait...)
libraries (name > KERNEL32.dll)
imports (flag > 1)
exports (n/a)
thread-local-storage (n/a)
.NET (n/a)
resources (n/a)
strings (wait...)
debug (n/a)
manifest (n/a)
version (n/a)
certificate (n/a)
overlay (n/a)
property value
file
file > sha256 A31309374A5AFDE74B64FEDA9F1CB4ECF09D6C03157D66157BBB839827895604
file > first 32 bytes (hex) 4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 B8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00
file > first 32 bytes (text) MZ.....@.....
file > info size: 206848 bytes, entropy: 5.921
file > type executable, 64-bit, GUI
file > version n/a
file > description n/a
entry-point > first 32 bytes (hex) 48 83 64 24 30 00 8B 84 24 80 00 00 00 45 33 C9 89 44 24 28 45 33 C0 49 8B D7 48 89 74 24 20 E8
entry-point > location 0x00004000 (section[.ddxl])
file > signature Microsoft Linker 1.0
stamps
stamp > compiler Wed Apr 14 22:06:53 2010 (UTC)
stamp > debug n/a
stamp > resource n/a
stamp > import n/a
stamp > export n/a
names
file > name c:\users\supav\onedrive\desktop\cmu\g_\jeanette_autopsy\export\update2.exe
debug > file n/a
export n/a
version n/a
manifest n/a
.NET > module > name n/a
certificate > program-name n/a
sha256 > A31309374A5AFDE74B64FEDA9F1CB4ECF09D6C03157D66157BBB839827895604
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x00004000

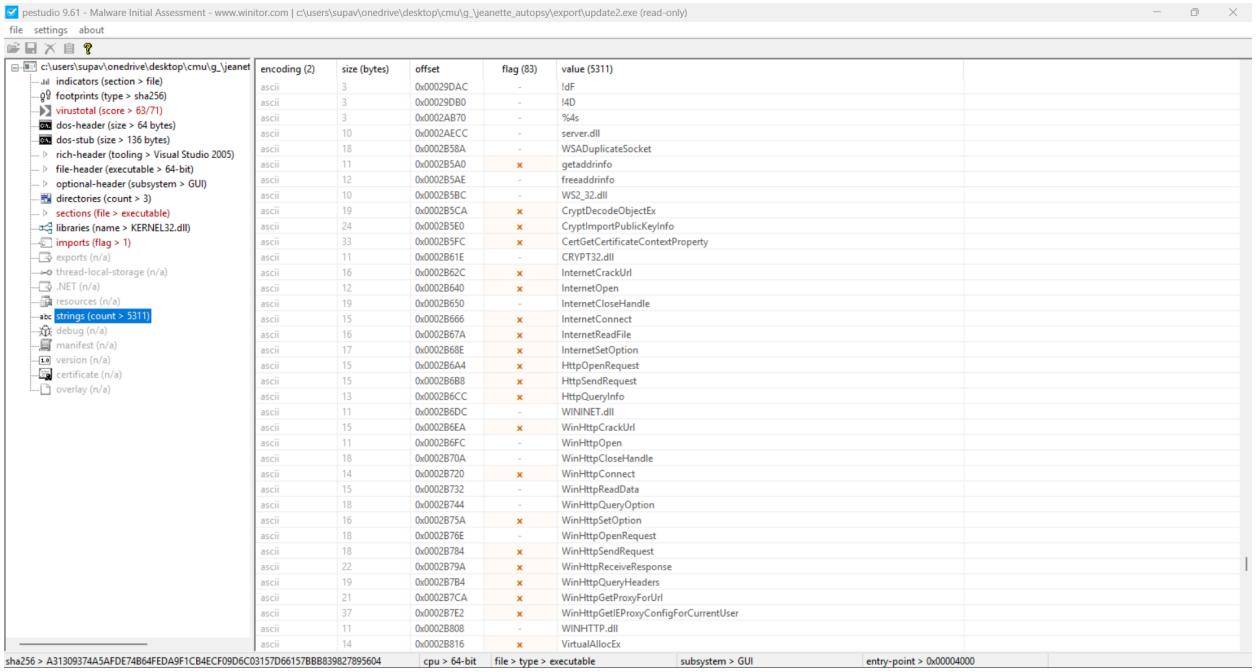

pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\supav\onedrive\desktop\cmu\g_\jeanette_autopsy\export\update2.exe (read-only)
file settings about
c:\users\supav\onedrive\desktop\cmu\g_\jeanet
indicators (section > file)
footprints (type > sha256)
virustotal (score > 63/71)
dos-header (size > 64 bytes)
dos-stub (size > 136 bytes)
rich-header (tooling > Visual Studio 2005)
file-header (executable > 64-bit)
optional-header (subsystem > GUI)
directories (count > 3)
sections (file > executable)
libraries (name > KERNEL32.dll)
imports (flag > 1)
exports (n/a)
thread-local-storage (n/a)
.NET (n/a)
resources (n/a)
strings (count > 5311)
debug (n/a)
manifest (n/a)
version (n/a)
certificate (n/a)
overlay (n/a)
vendor (71/71) score (63/71) date (dd.mm.yyyy) age (days)
ALYac Trojan.Rozena.32 13.09.2026 38
APEX Malicious 13.09.2026 38
AVG Win32:Metasploit-C [Trj] 13.09.2026 38
Acronis suspicious 28.03.2024 937
AhnLab-V3 Trojan/Win32.RL_Generic.R358905 13.09.2026 38
Alibaba HackTool:Win64/Meterpreter.de97252e 27.05.2019 2704
Antiy-AVL Trojan/Win64.Rozena 13.09.2026 38
Arcabit Trojan.Rozena.32 13.09.2026 38
Avast Win32:Metasploit-C [Trj] 13.09.2026 38
Avira TR/W32.Metasploit.C 13.09.2026 38
BitDefender Trojan.Rozena.32 13.09.2026 38
Bkav W32.Malware.B534FAAB 12.09.2026 39
CAT-QuickHeal HackTool.Metasploit.S9212471 12.09.2026 39
CMC undetected 12.09.2026 39
CTX exe.trojan.rozena 13.09.2026 38
ClamAV Win.Exploit.D388a-9756522-0 13.09.2026 38
CrowdStrike win/malicious_confidence_100% (W) 19.12.2025 306
Cylance Unsafe 10.09.2026 41
Cynet Malicious (score: 100) 13.09.2026 38
DeepInstinct MALICIOUS 14.06.2026 129
DrWeb BackDoor.Shell.244 13.09.2026 38
ESET-NOD32 Win64/Rozena.RX trojan 13.09.2026 38
Elastic Windows.Trojan.Metasploit 10.09.2026 41
Emsisoft Trojan.Rozena.32 (B) 13.09.2026 38
F-Secure Trojan.TR/W32.Metasploit.C 13.09.2026 38
Fortinet W64/Rozena.J!tr 13.09.2026 38
GData Win64.Trojan.Rozena.A 13.09.2026 38
Google Detected 13.09.2026 38
Gridinsoft Trojan.Win64.ShellCode.sd!s1 13.09.2026 38
Ikarus Trojan.Win64.Rozena 13.09.2026 38
Jiangmin Trojan.Generic.gesks 13.09.2026 38
K7AntiVirus Trojan (004fae881) 13.09.2026 38
K7GW Trojan (004fae881) 12.09.2026 39
Kaspersky HEUR:Trojan.Win64.Packed.gen 13.09.2026 38
Kingsoft undetected 13.09.2026 38
Lionic Trojan.Win32.Rozena.4!c 13.09.2026 38
Malwarebytes Rozena.Trojan.Shell.DDS 13.09.2026 38
sha256 > A31309374A5AFDE74B64FEDA9F1CB4ECF09D6C03157D66157BBB839827895604
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x00004000


pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\supav\onedrive\desktop\cmu\g_\jeanette_autopsy\export\update2.exe (read-only)
file settings about
c:\users\supav\onedrive\desktop\cmu\g_\jeanet
indicators (section > file)
footprints (type > sha256)
virustotal (score > 63/71)
dos-header (size > 64 bytes)
dos-stub (size > 136 bytes)
rich-header (tooling > Visual Studio 2005)
file-header (executable > 64-bit)
optional-header (subsystem > GUI)
directories (count > 3)
sections (file > executable)
libraries (name > KERNEL32.dll)
imports (flag > 1)
exports (n/a)
thread-local-storage (n/a)
.NET (n/a)
resources (n/a)
strings (count > 5311)
debug (n/a)
manifest (n/a)
version (n/a)
certificate (n/a)
overlay (n/a)
library (1) flag (0) type imports (2) description
KERNEL32.dll - Implicit 2 Windows NT BASE API Client
sha256 > A31309374A5AFDE74B64FEDA9F1CB4ECF09D6C03157D66157BBB839827895604
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x00004000

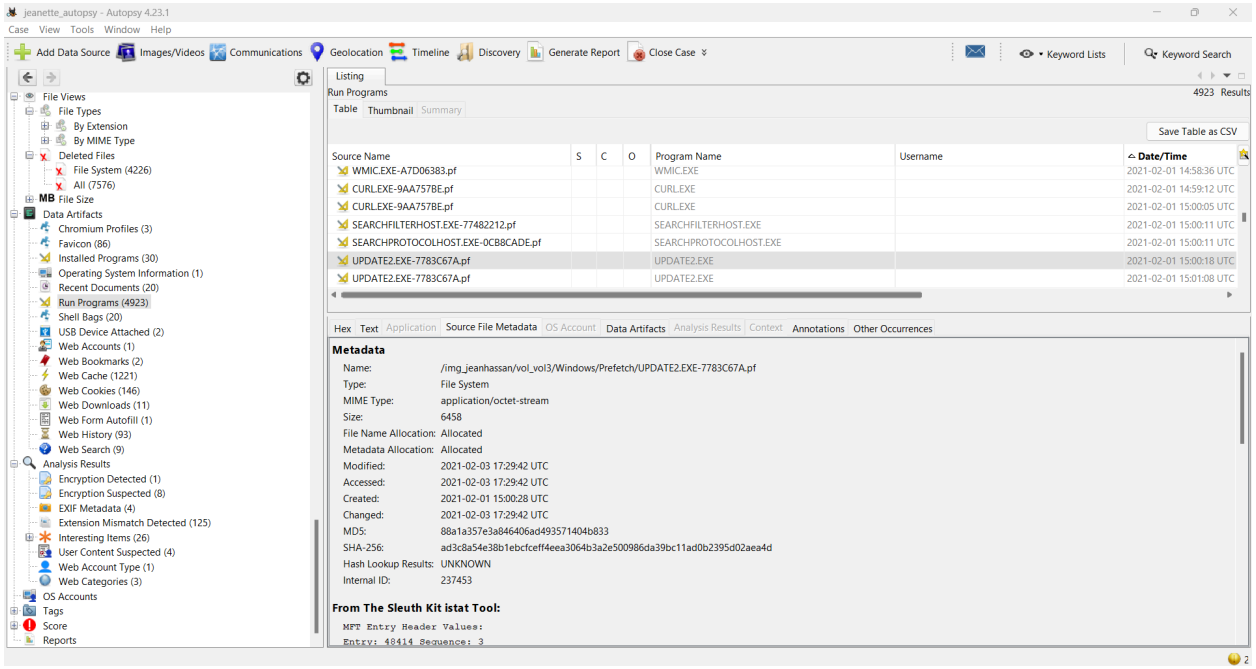

pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\supav\onedrive\desktop\cmu\g_\jeanette_autopsy\export\update2.exe (read-only)
file settings about
c:\users\supav\onedrive\desktop\cmu\g_\jeanet
indicators (section > file)
footprints (type > sha256)
virustotal (score > 63/71)
dos-header (size > 64 bytes)
dos-stub (size > 136 bytes)
rich-header (tooling > Visual Studio 2005)
file-header (executable > 64-bit)
optional-header (subsystem > GUI)
directories (count > 3)
sections (file > executable)
libraries (name > KERNEL32.dll)
imports (flag > 1)
exports (n/a)
thread-local-storage (n/a)
.NET (n/a)
resources (n/a)
strings (count > 5311)
debug (n/a)
manifest (n/a)
version (n/a)
certificate (n/a)
overlay (n/a)
imports (2) flag (1) type ordinal first-thunk (IAT) first-thunk-original (INT) library
VirtualAlloc x implicit - 0x00000000 0x00000000 KERNEL32.dll
ExitProcess - implicit - 0x00000000 0x00000000 KERNEL32.dll
sha256 > A31309374A5AFDE74B64FEDA9F1CB4ECF09D6C03157D66157BBB839827895604
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x00004000


pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\supav\onedrive\desktop\cmu\g_\jeanette_autopsy\export\update2.exe (read-only)
file settings about
c:\users\supav\onedrive\desktop\cmu\g_\jeanet
indicators (section > file)
footprints (type > sha256)
virustotal (score > 63/71)
dos-header (size > 64 bytes)
dos-stub (size > 136 bytes)
rich-header (tooling > Visual Studio 2005)
file-header (executable > 64-bit)
optional-header (subsystem > GUI)
directories (count > 3)
sections (file > executable)
libraries (name > KERNEL32.dll)
imports (flag > 1)
exports (n/a)
thread-local-storage (n/a)
.NET (n/a)
resources (n/a)
strings (count > 5311)
debug (n/a)
manifest (n/a)
version (n/a)
certificate (n/a)
overlay (n/a)
encoding (2) size (bytes) offset flag (83) value (5311)
unicode 6 0x00028C90 - sma-se
unicode 6 0x00028CA0 - smj-no
unicode 6 0x00028CB0 - smj-se
unicode 6 0x00028CC0 - smn-fi
unicode 6 0x00028CD0 - sms-fi
unicode 5 0x00028CE0 - sq-al
unicode 10 0x00028CF0 - sr-ba-cyrl
unicode 10 0x00028D08 - sr-ba-latn
unicode 10 0x00028D20 - sr-sp-cyrl
unicode 10 0x00028D38 - sr-sp-latn
unicode 5 0x00028D50 - sv-fi
unicode 5 0x00028D60 - sv-se
unicode 5 0x00028D70 - sw-ke
unicode 6 0x00028D80 - syr-sy
unicode 5 0x00028D90 - ta-in
unicode 5 0x00028DA0 - te-in
unicode 5 0x00028DB0 - th-th
unicode 5 0x00028DC0 - tn-za
unicode 5 0x00028DD0 - tr-tr
unicode 5 0x00028DE0 - tt-ru
unicode 5 0x00028DF0 - uk-ua
unicode 5 0x00028E00 - ur-pk
unicode 10 0x00028E10 - uz-uz-cyrl
unicode 10 0x00028E28 - uz-uz-latn
unicode 5 0x00028E40 - vi-vn
unicode 5 0x00028E50 - xh-za
unicode 6 0x00028E60 - zh-chs
unicode 6 0x00028E70 - zh-cht
unicode 5 0x00028E80 - zh-cn
unicode 5 0x00028E90 - zh-hk
unicode 5 0x00028EA0 - zh-mo
unicode 5 0x00028EB0 - zh-sg
unicode 5 0x00028EC0 - zh-tw
unicode 5 0x00028ED0 - zu-za
unicode 7 0x00028F60 - CONOUT$
unicode 23 0x00032230 - tcp://192.168.2.42:5555
sha256 > A31309374A5AFDE74B64FEDA9F1CB4ECF09D6C03157D66157BBB839827895604
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x00004000

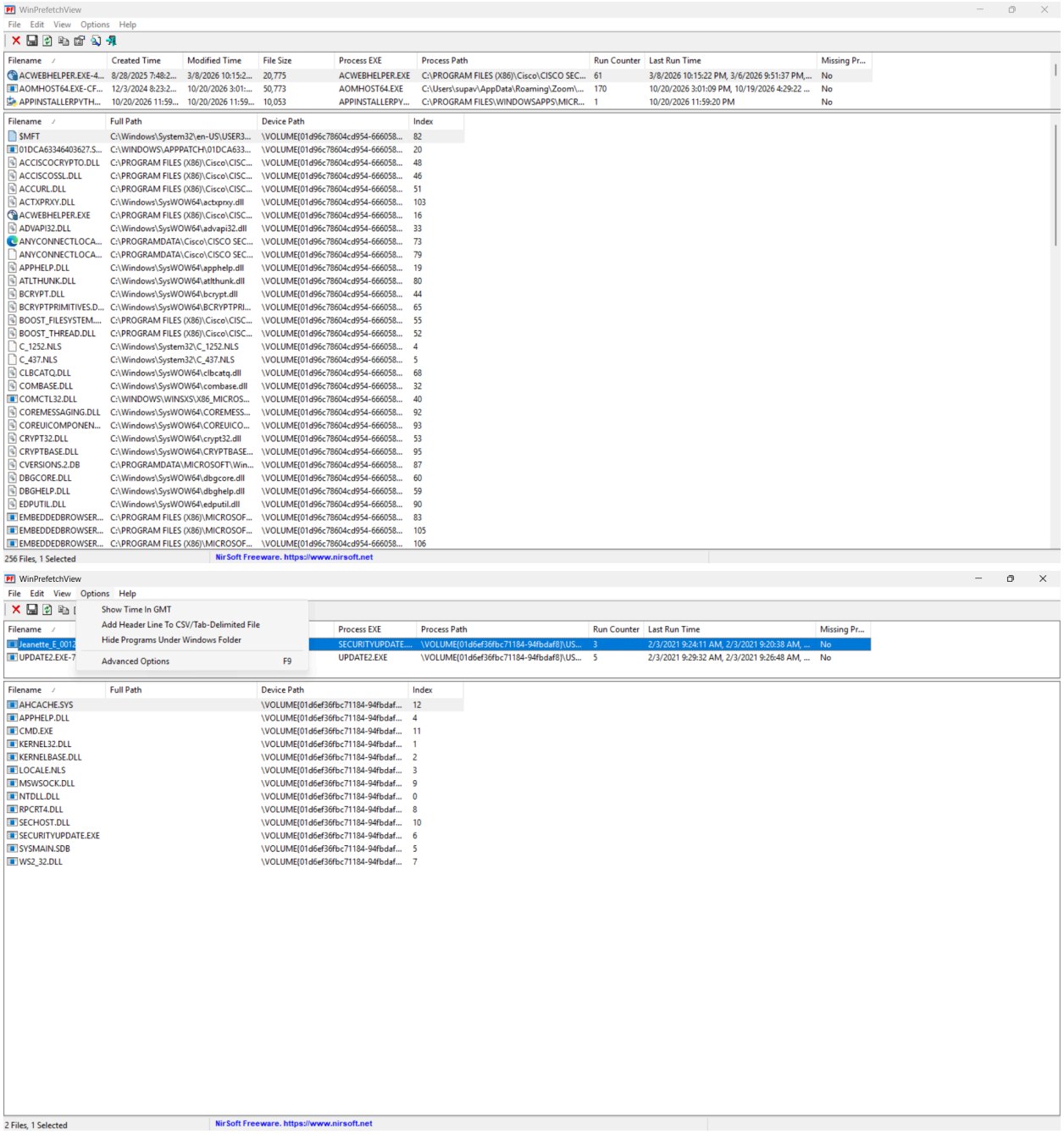

pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\supav\onedrive\desktop\cmu\g_\jeanette_autopsy\export\update2.exe (read-only)
file settings about
c:\users\supav\onedrive\desktop\cmu\g_\jeanet
indicators (section > file)
footprints (type > sha256)
virustotal (score > 63/71)
dos-header (size > 64 bytes)
dos-stub (size > 136 bytes)
rich-header (tooling > Visual Studio 2005)
file-header (executable > 64-bit)
optional-header (subsystem > GUI)
directories (count > 3)
sections (file > executable)
libraries (name > KERNEL32.dll)
imports (flag > 1)
exports (n/a)
thread-local-storage (n/a)
.NET (n/a)
resources (n/a)
strings (count > 5311)
debug (n/a)
manifest (n/a)
version (n/a)
certificate (n/a)
overlay (n/a)
encoding (2) size (bytes) offset flag (83) value (5311)
ascii 3 0x00022D0F - @UH
ascii 3 0x00022D28 - @UH
ascii 3 0x00022D33 - M0H
ascii 3 0x00022D40 - @UH
ascii 3 0x00022D4A - M@H
ascii 5 0x000234BC - ntdll
ascii 16 0x000234C8 - NtQueueApcThread
ascii 3 0x000234E7 - 7)\
ascii 18 0x00023500 x NtMapViewOfSection
ascii 21 0x00023518 - NtQueryAttributesFile
ascii 10 0x00023530 - NtOpenFile
ascii 15 0x00023540 x NtCreateSection
ascii 13 0x00023550 x NtOpenSection
ascii 7 0x00023560 - NtClose
ascii 19 0x00023568 - NtLockVirtualMemory
ascii 19 0x00023760 - RtlCreateUserThread
ascii 12 0x00023778 - advapi32.dll
ascii 15 0x00023788 x AddMandatoryAce
ascii 12 0x000237C8 - \\%s\pipe\%s
ascii 12 0x000237D8 - kernel32.dll
ascii 20 0x000237E8 - ProcessIdToSessionId
ascii 3 0x00023834 - tcp
ascii 15 0x00023858 - PACKET TRANSMIT
ascii 14 0x00023868 - PACKET RECEIVE
ascii 10 0x00023878 x OpenThread
ascii 9 0x00023888 - ntdll.dll
ascii 12 0x00023898 x NtOpenThread
ascii 18 0x000238A8 - SetThreadErrorMode
ascii 10 0x000238C0 - stream end
ascii 10 0x000238D0 - file error
ascii 12 0x000238E0 - stream error
ascii 10 0x000238F0 - data error
ascii 19 0x00023900 - insufficient memory
ascii 12 0x00023918 - buffer error
ascii 20 0x00023928 - incompatible version
ascii 15 0x00023940 - need dictionary
ascii 6 0x00023B98 - 1.0.4P
sha256 > A31309374A5AFDE74B64FEDA9F1CB4ECF09D6C03157D66157BBB839827895604
cpu > 64-bit
file > type > executable
subsystem > GUI
entry-point > 0x00004000

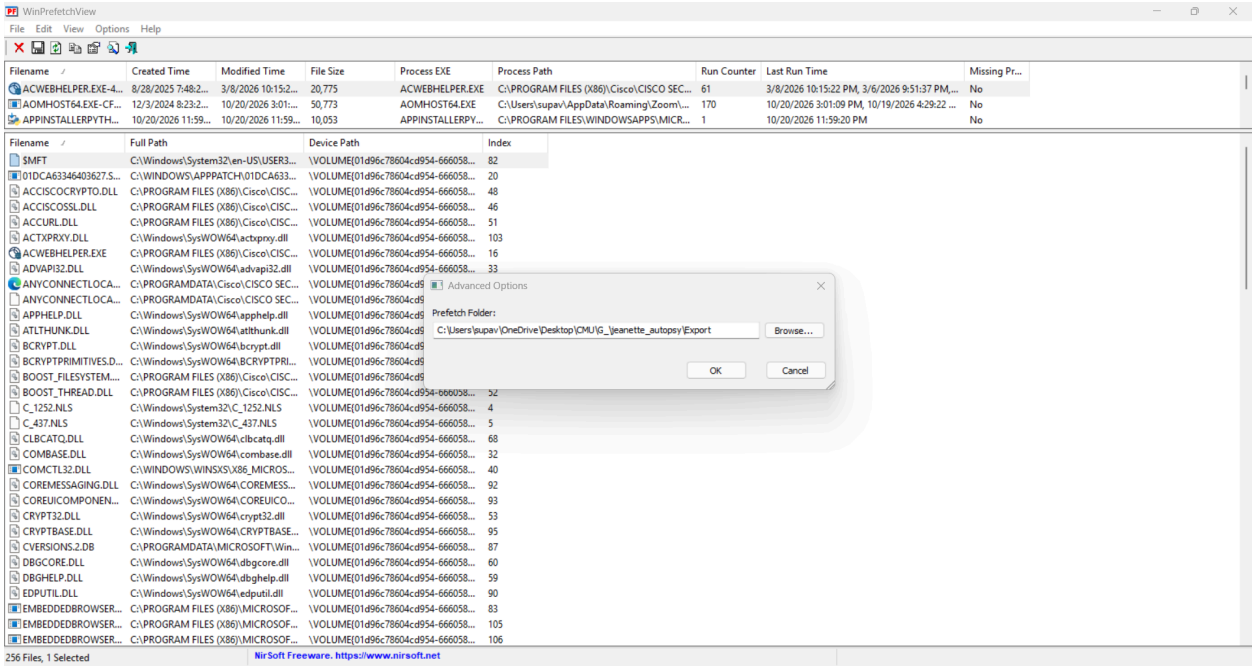


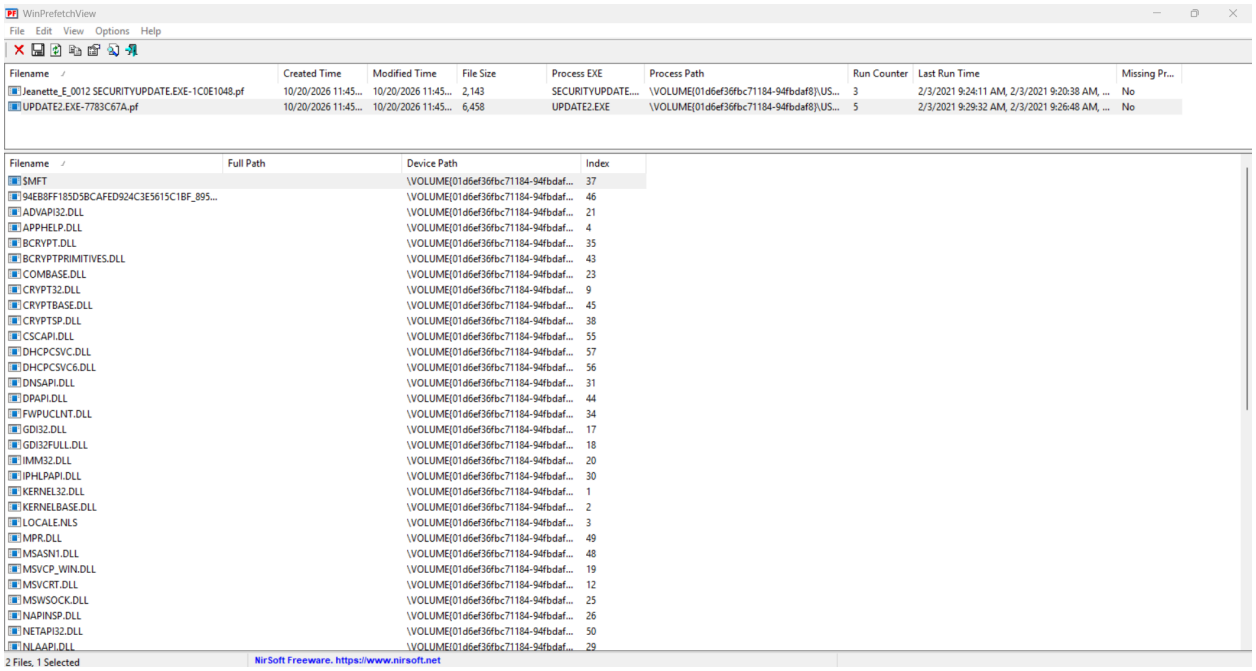
5.2.4.15 Jeanette_A_0015 update2.exe Run Programs



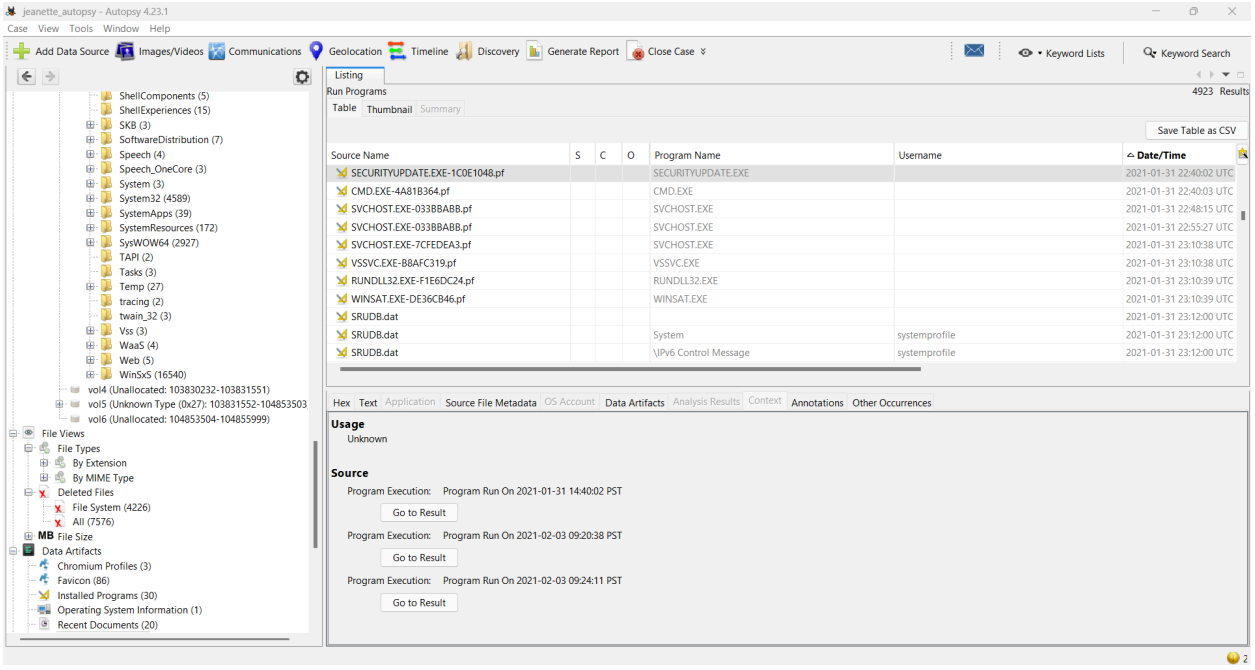
5.2.4.16 Jeanette_A_0016 update2.exe WinPrefetchViewer

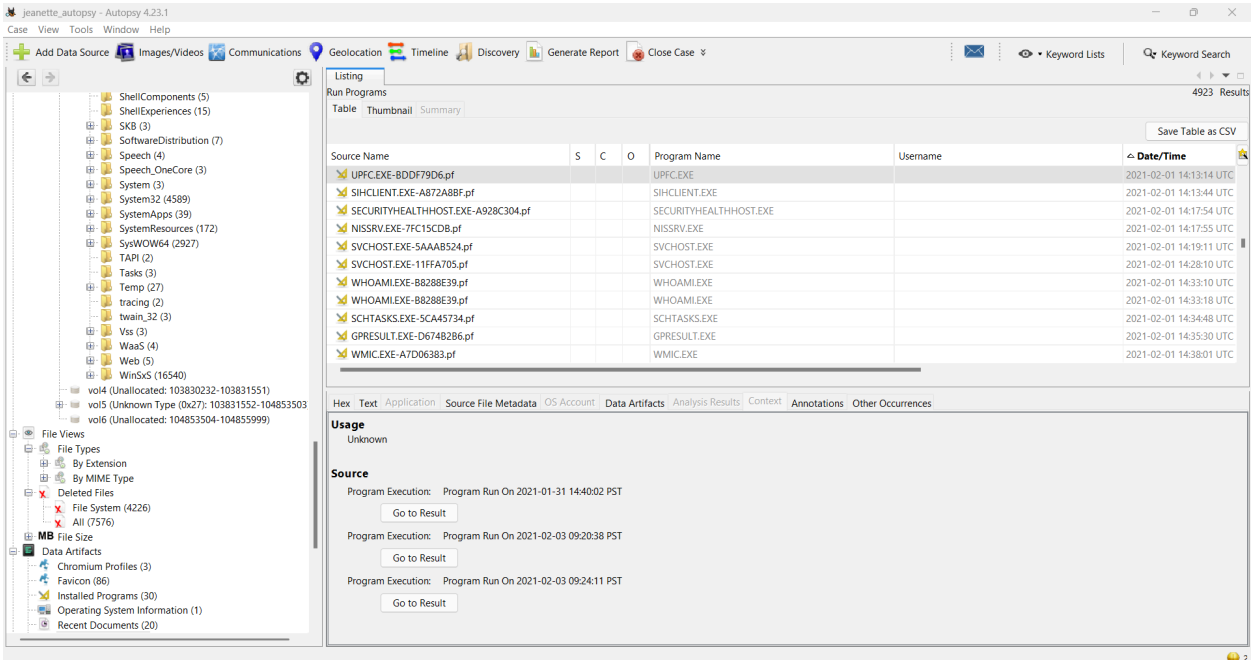


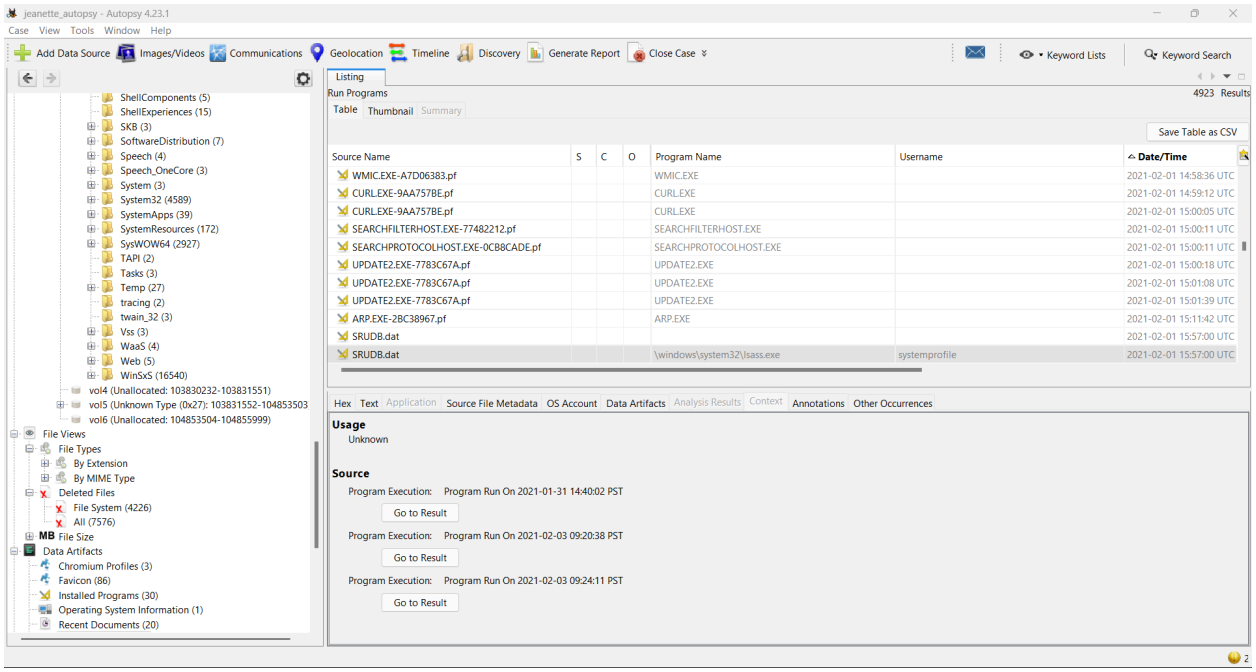

WinPrefetchView
File Edit View Options Help
Filename Created Time Modified Time File Size Process EXE Process Path Run Counter Last Run Time Missing Pr...
ACWEBHELPER.EXE-4... 8/28/2025 7:48:2... 3/8/2026 10:15:2... 20,775 ACWEBHELPER.EXE C:\PROGRAM FILES (X86)\Cisco\CISCO SEC... 61 3/8/2026 10:15:22 PM, 3/6/2026 9:51:37 PM,... No
AOMHOST64.EXE-CF... 12/3/2024 8:23:2... 10/20/2026 3:01:... 50,773 AOMHOST64.EXE C:\Users\supav\AppData\Roaming\Zoom\... 170 10/20/2026 3:01:09 PM, 10/19/2026 4:29:22 ... No
APPINSTALLERPYTH... 10/20/2026 11:59... 10/20/2026 11:59... 10,053 APPINSTALLERPY... C:\PROGRAM FILES\WINDOWSAPPS\MICR... 1 10/20/2026 11:59:20 PM No
Filename Full Path Device Path Index
$MFT C:\Windows\System32\en-US\USER3... \VOLUME{01d96c78604cd954-666058... 82
01DCA63346403627.S... C:\WINDOWS\APPPATCH\01DCA633... \VOLUME{01d96c78604cd954-666058... 20
ACCISCOCRYPTO.DLL C:\PROGRAM FILES (X86)\Cisco\CISC... \VOLUME{01d96c78604cd954-666058... 48
ACCISCOSSL.DLL C:\PROGRAM FILES (X86)\Cisco\CISC... \VOLUME{01d96c78604cd954-666058... 46
ACCURL.DLL C:\PROGRAM FILES (X86)\Cisco\CISC... \VOLUME{01d96c78604cd954-666058... 51
ACTXPRXY.DLL C:\Windows\SysWOW64\actxprxy.dll \VOLUME{01d96c78604cd954-666058... 103
ACWEBHELPER.EXE C:\PROGRAM FILES (X86)\Cisco\CISC... \VOLUME{01d96c78604cd954-666058... 16
ADVAPI32.DLL C:\Windows\SysWOW64\advapi32.dll \VOLUME{01d96c78604cd954-666058... 33
ANYCONNECTLOCA... C:\PROGRAMDATA\Cisco\CISCO SEC... \VOLUME{01d96c78604cd9
ANYCONNECTLOCA... C:\PROGRAMDATA\Cisco\CISCO SEC... \VOLUME{01d96c78604cd9
APPHELP.DLL C:\Windows\SysWOW64\apphelp.dll \VOLUME{01d96c78604cd9
ATLTHUNK.DLL C:\Windows\SysWOW64\atlthunk.dll \VOLUME{01d96c78604cd9
BCRYPT.DLL C:\Windows\SysWOW64\bcrypt.dll \VOLUME{01d96c78604cd9
BCRYPTPRIMITIVES.D... C:\Windows\SysWOW64\BCRYPTPRI... \VOLUME{01d96c78604cd9
BOOST_FILESYSTEM.... C:\PROGRAM FILES (X86)\Cisco\CISC... \VOLUME{01d96c78604cd9
BOOST_THREAD.DLL C:\PROGRAM FILES (X86)\Cisco\CISC... \VOLUME{01d96c78604cd954-666058... 52
C_1252.NLS C:\Windows\System32\C_1252.NLS \VOLUME{01d96c78604cd954-666058... 4
C_437.NLS C:\Windows\System32\C_437.NLS \VOLUME{01d96c78604cd954-666058... 5
CLBCATQ.DLL C:\Windows\SysWOW64\clbcatq.dll \VOLUME{01d96c78604cd954-666058... 68
COMBASE.DLL C:\Windows\SysWOW64\combase.dll \VOLUME{01d96c78604cd954-666058... 32
COMCTL32.DLL C:\WINDOWS\WINSXS\X86_MICROS... \VOLUME{01d96c78604cd954-666058... 40
COREMESSAGING.DLL C:\Windows\SysWOW64\COREMESS... \VOLUME{01d96c78604cd954-666058... 92
COREUICOMPONEN... C:\Windows\SysWOW64\COREUICO... \VOLUME{01d96c78604cd954-666058... 93
CRYPT32.DLL C:\Windows\SysWOW64\crypt32.dll \VOLUME{01d96c78604cd954-666058... 53
CRYPTBASE.DLL C:\Windows\SysWOW64\CRYPTBASE... \VOLUME{01d96c78604cd954-666058... 95
CVERSIONS.2.DB C:\PROGRAMDATA\MICROSOFT\Win... \VOLUME{01d96c78604cd954-666058... 87
DBGCORE.DLL C:\Windows\SysWOW64\dbgcore.dll \VOLUME{01d96c78604cd954-666058... 60
DBGHELP.DLL C:\Windows\SysWOW64\dbghelp.dll \VOLUME{01d96c78604cd954-666058... 59
EDPUTIL.DLL C:\Windows\SysWOW64\edputil.dll \VOLUME{01d96c78604cd954-666058... 90
EMBEDDEDBROWSER... C:\PROGRAM FILES (X86)\MICROSOF... \VOLUME{01d96c78604cd954-666058... 83
EMBEDDEDBROWSER... C:\PROGRAM FILES (X86)\MICROSOF... \VOLUME{01d96c78604cd954-666058... 105
EMBEDDEDBROWSER... C:\PROGRAM FILES (X86)\MICROSOF... \VOLUME{01d96c78604cd954-666058... 106
Advanced Options
Prefetch Folder:
C:\Users\supav\OneDrive\Desktop\CMU\G_\Jeanette_autopsy\Export
Browse...
OK
Cancel
256 Files, 1 Selected
NirSoft Freeware. https://www.nirsoft.net


WinPrefetchView
File Edit View Options Help
Filename Created Time Modified Time File Size Process EXE Process Path Run Counter Last Run Time Missing Pr...
Jeanette_E_0012 SECURITYUPDATE.EXE-1C0E1048.pf 10/20/2026 11:45... 10/20/2026 11:45... 2,143 SECURITYUPDATE.... \VOLUME{01d6ef36fbc71184-94fbdaf8}\US... 3 2/3/2021 9:24:11 AM, 2/3/2021 9:20:38 AM, ... No
UPDATE2.EXE-7783C67A.pf 10/20/2026 11:45... 10/20/2026 11:45... 6,458 UPDATE2.EXE \VOLUME{01d6ef36fbc71184-94fbdaf8}\US... 5 2/3/2021 9:29:32 AM, 2/3/2021 9:26:48 AM, ... No
Filename Full Path Device Path Index
$MFT \VOLUME{01d6ef36fbc71184-94fbdaf... 37
94EB8FF185D5BCAFED924C3E5615C1BF_895... \VOLUME{01d6ef36fbc71184-94fbdaf... 46
ADVAPI32.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 21
APPHELP.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 4
BCRYPT.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 35
BCRYPTPRIMITIVES.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 43
COMBASE.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 23
CRYPT32.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 9
CRYPTBASE.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 45
CRYPTSP.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 38
CSCAPI.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 55
DHCPCSVC.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 57
DHCPCSVC6.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 56
DNSAPI.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 31
DPAPI.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 44
FWPUCLNT.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 34
GDI32.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 17
GDI32FULL.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 18
IMM32.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 20
IPHLPAPI.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 30
KERNEL32.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 1
KERNELBASE.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 2
LOCALE.NLS \VOLUME{01d6ef36fbc71184-94fbdaf... 3
MPR.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 49
MSASN1.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 48
MSVCP_WIN.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 19
MSVCRT.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 12
MSWSOCK.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 25
NAPINSP.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 26
NETAPI32.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 50
NLAAPI.DLL \VOLUME{01d6ef36fbc71184-94fbdaf... 29
2 Files, 1 Selected
NirSoft Freeware. https://www.nirsoft.net

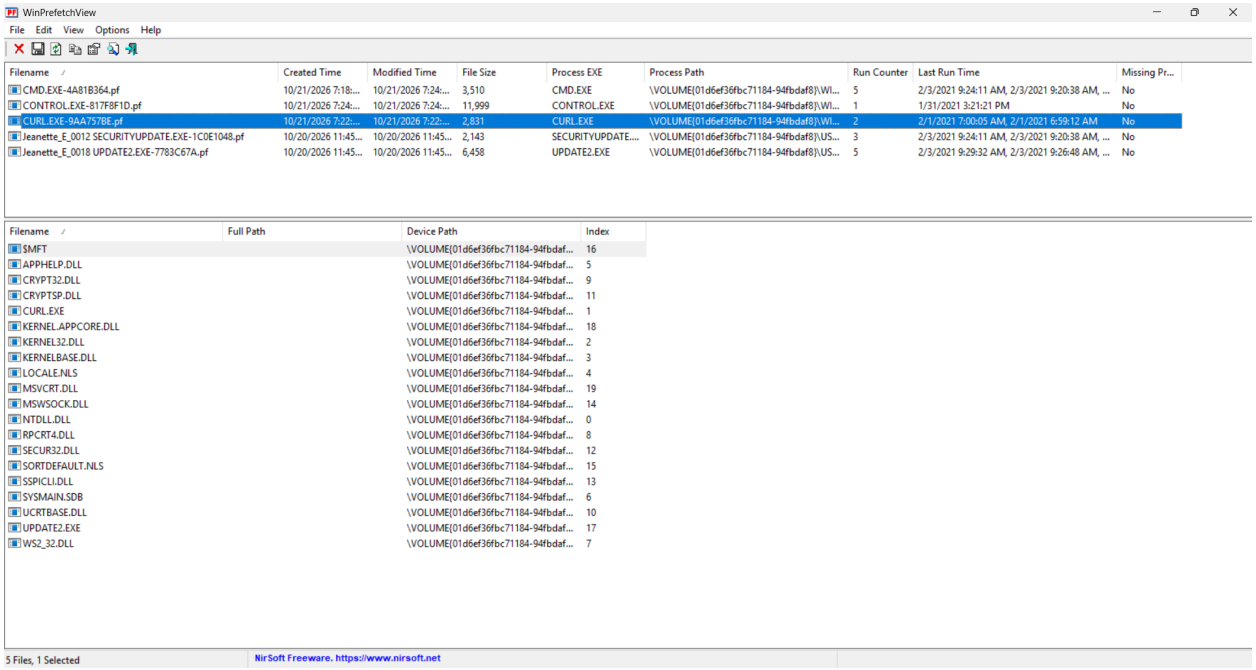
5.2.4.17 Jeanette_A_0017 Running Programs





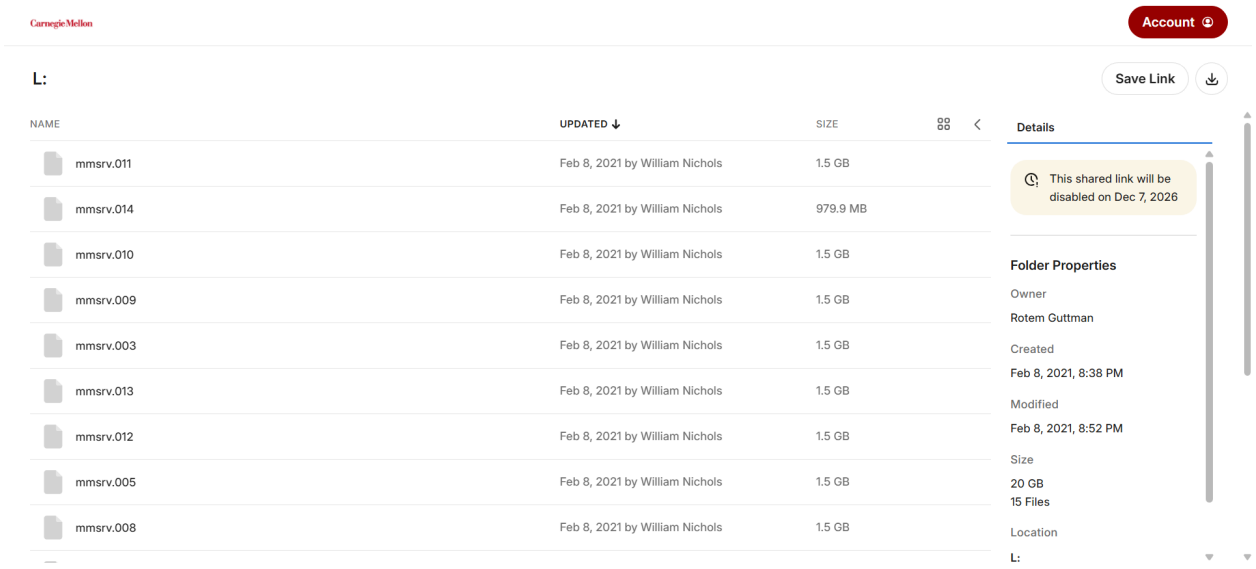


5.2.4.18 Jeanette_A_0018 curl.exe prefetch

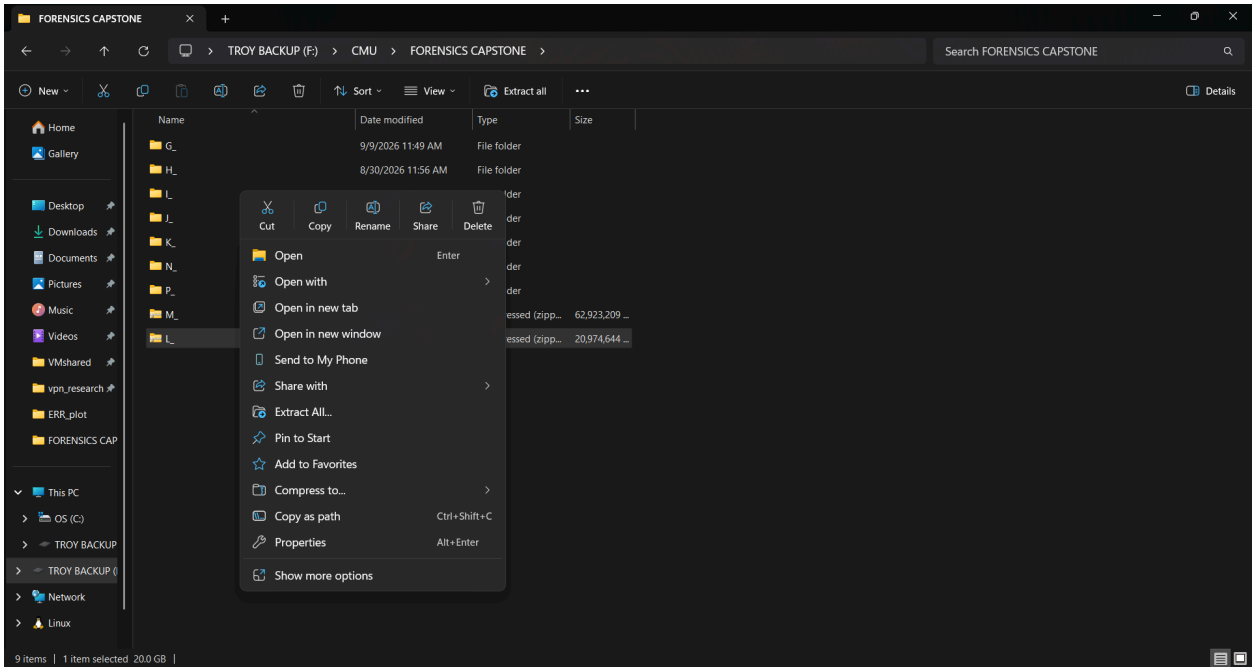


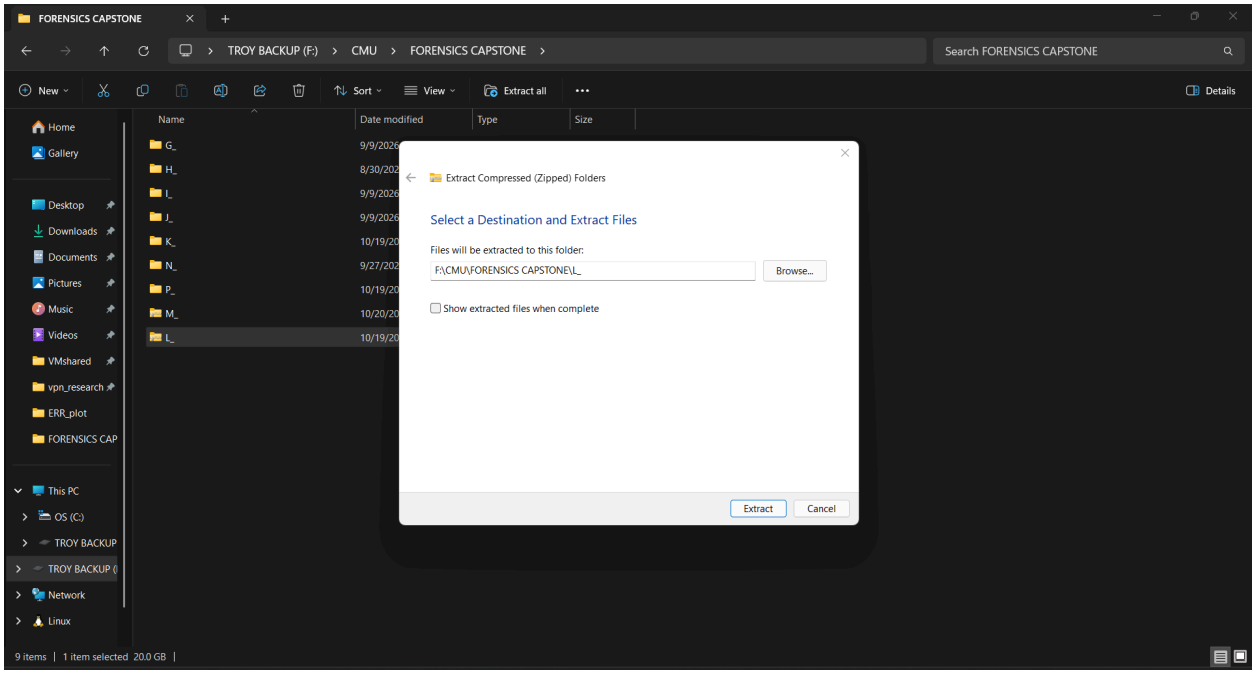
5.2.5 Mattermost Image (L)

5.2.5.1 Mattermost_A_0001 Image Download

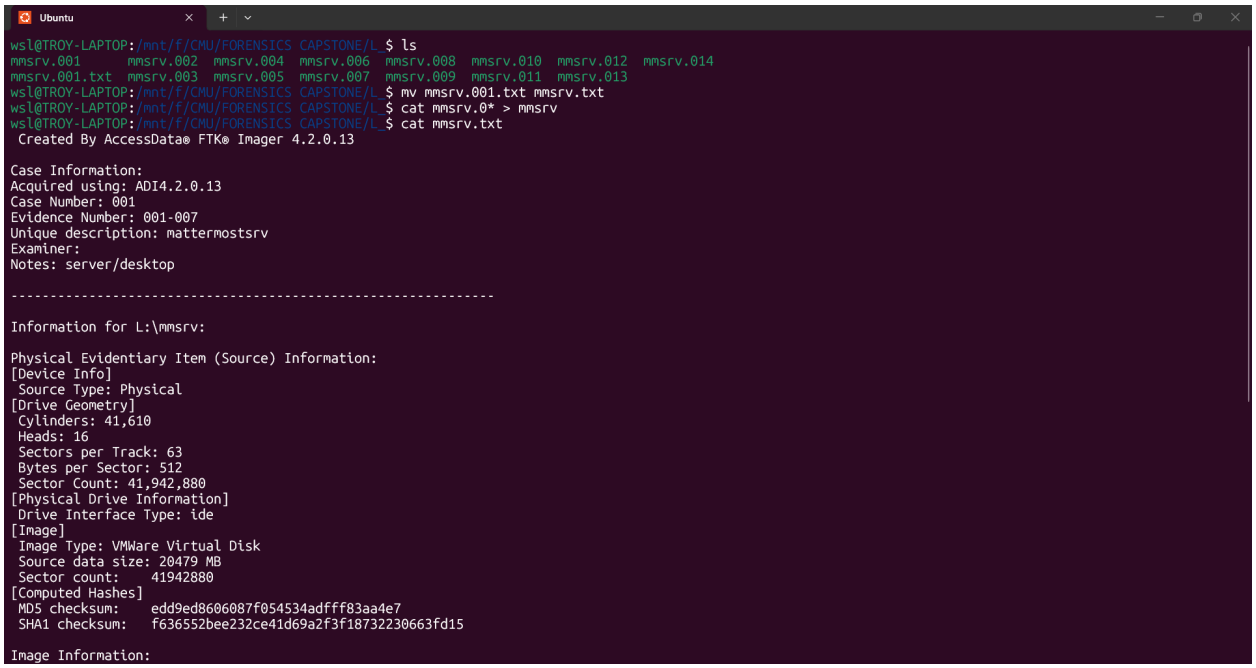


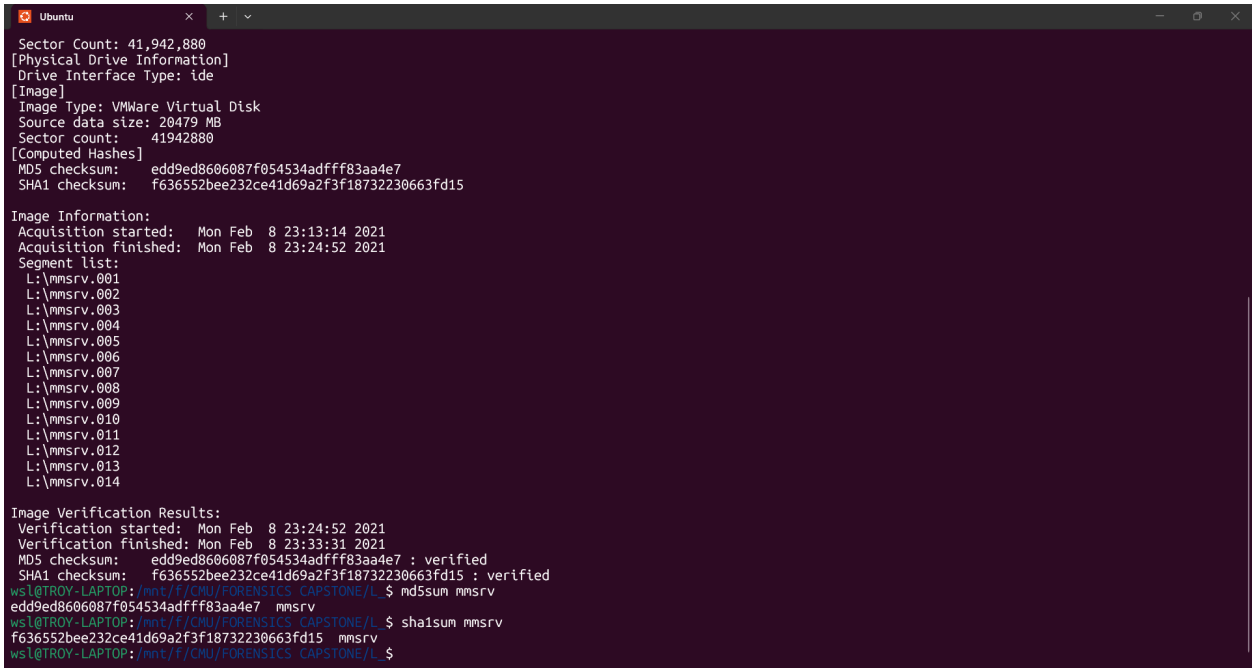
5.2.5.2 Mattermost_A_0002 Image Extraction



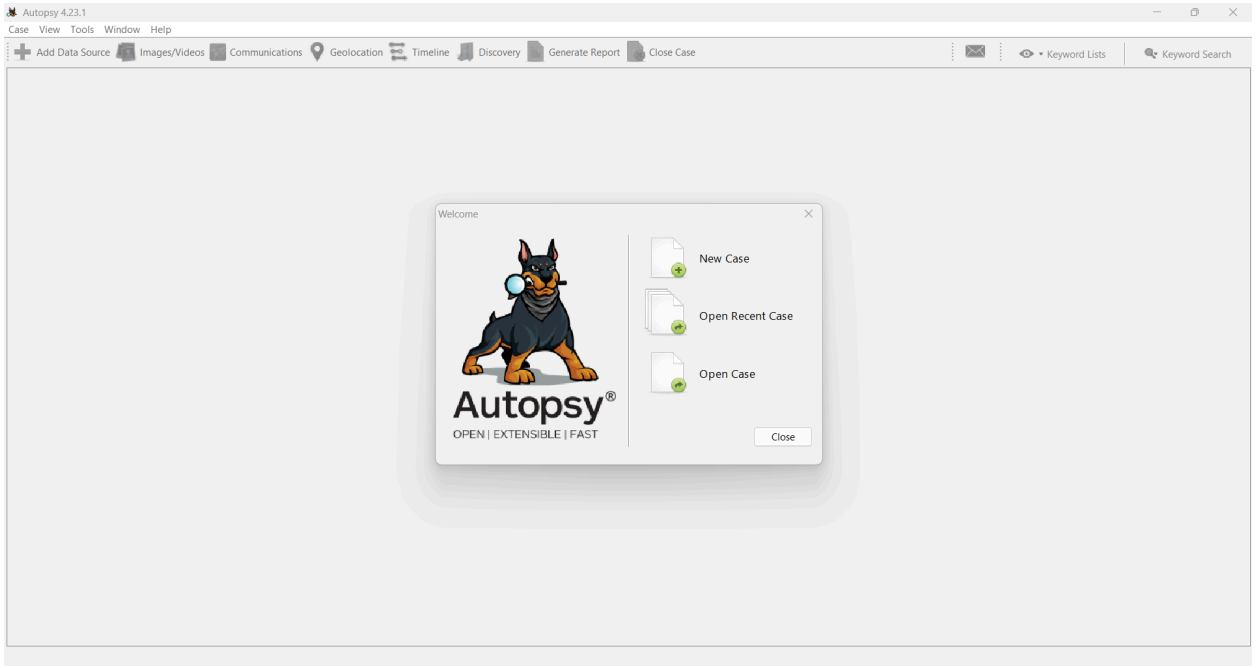


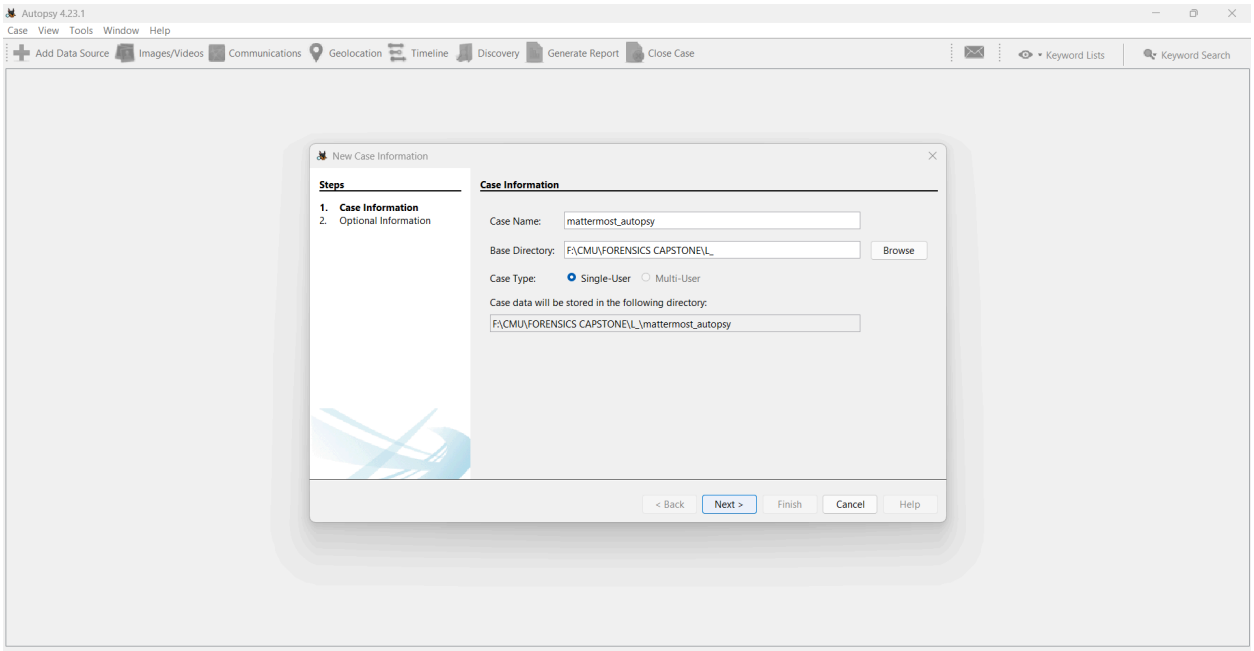
5.2.5.3 Mattermost_A_0003 Image Verification

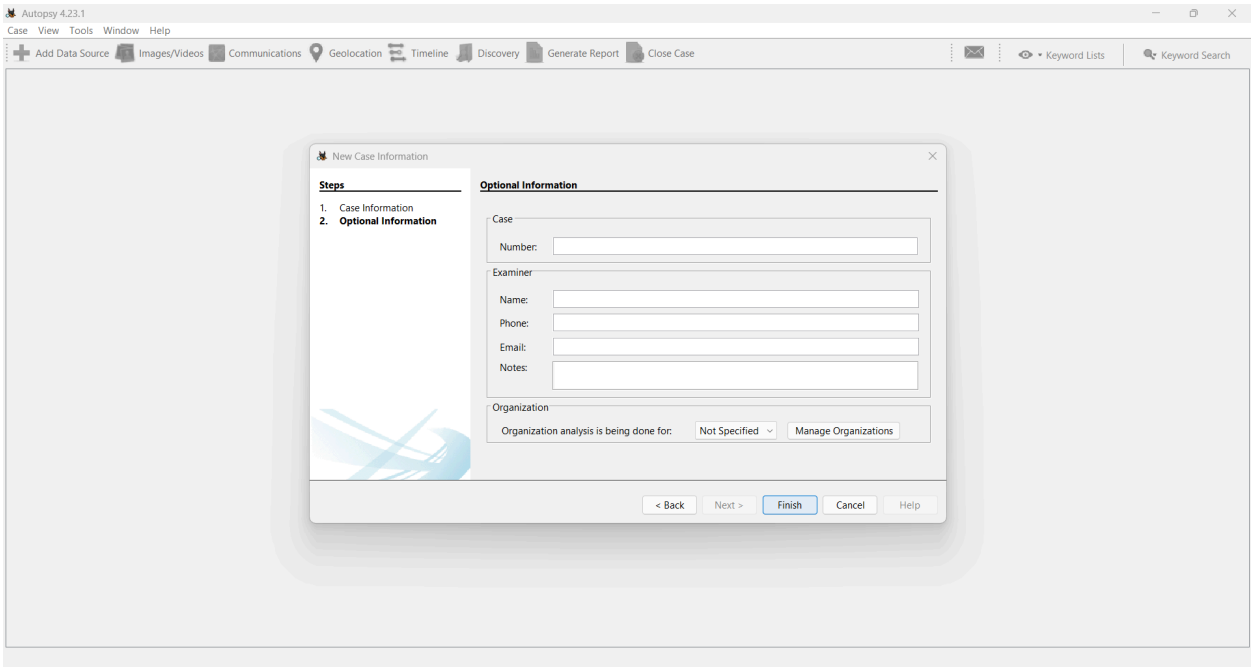


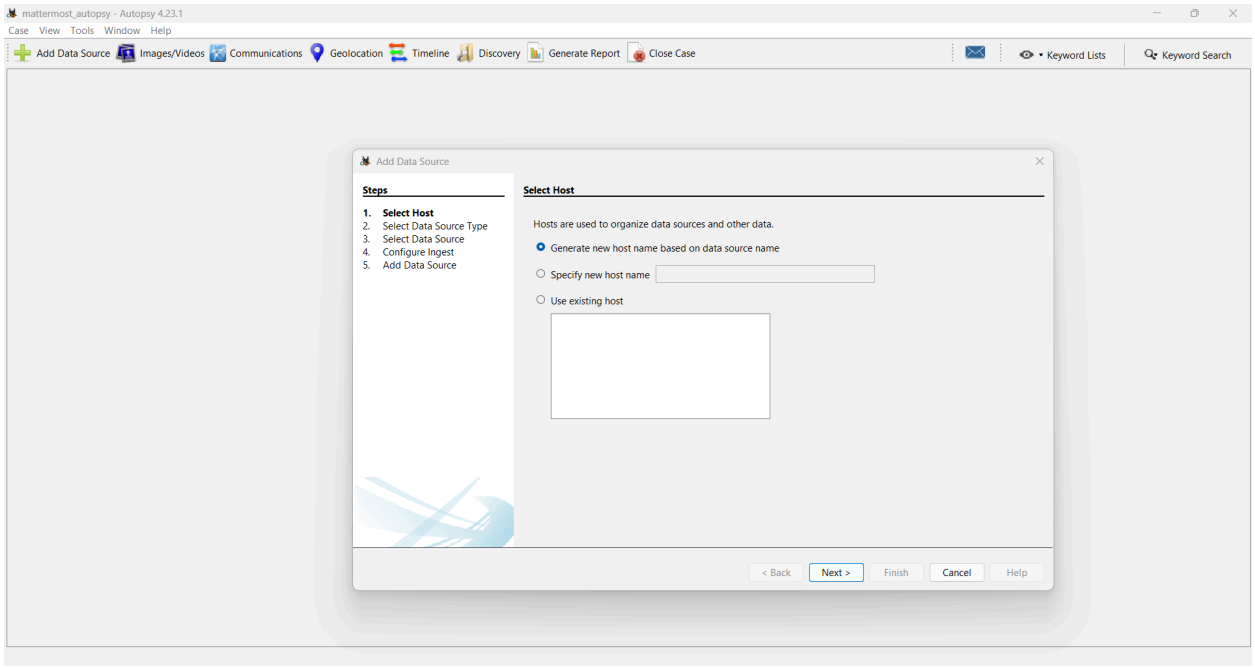


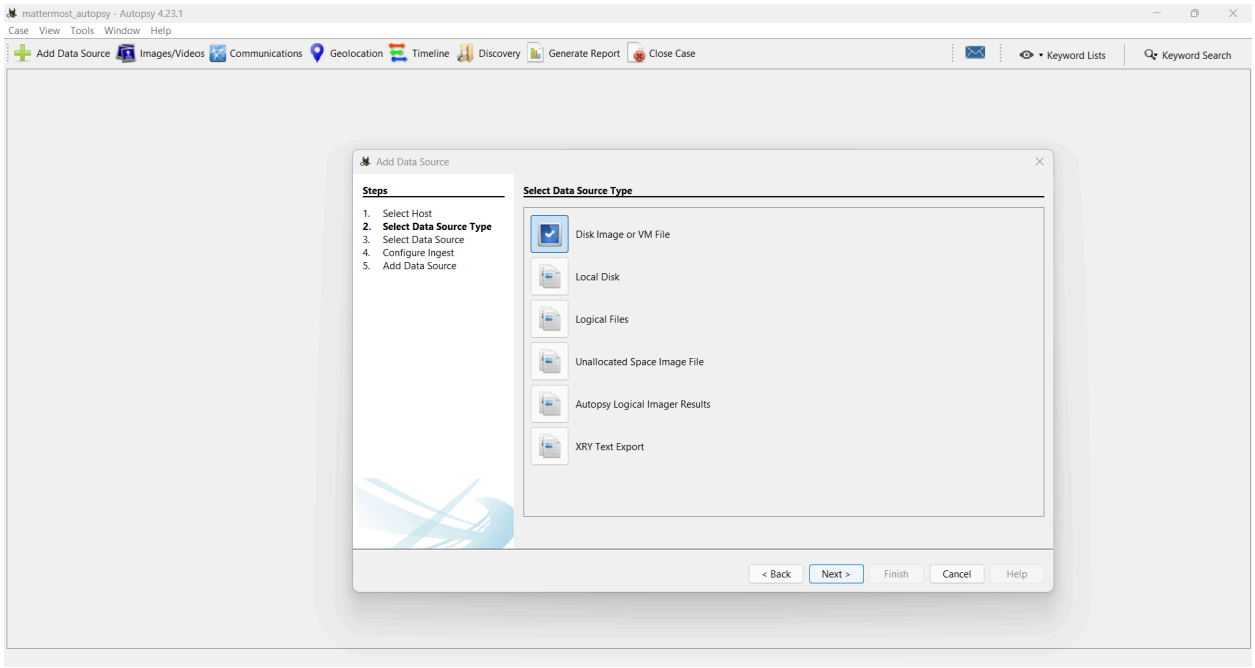
5.2.5.4 Mattermost_A_0004 Autopsy Case

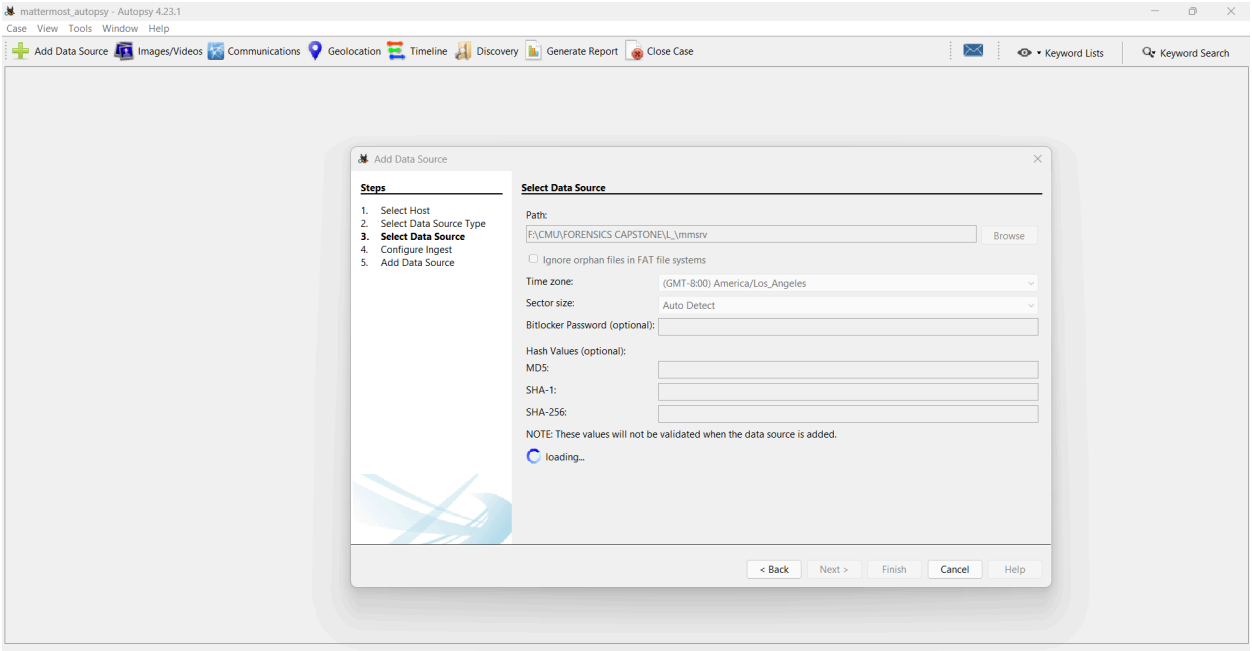


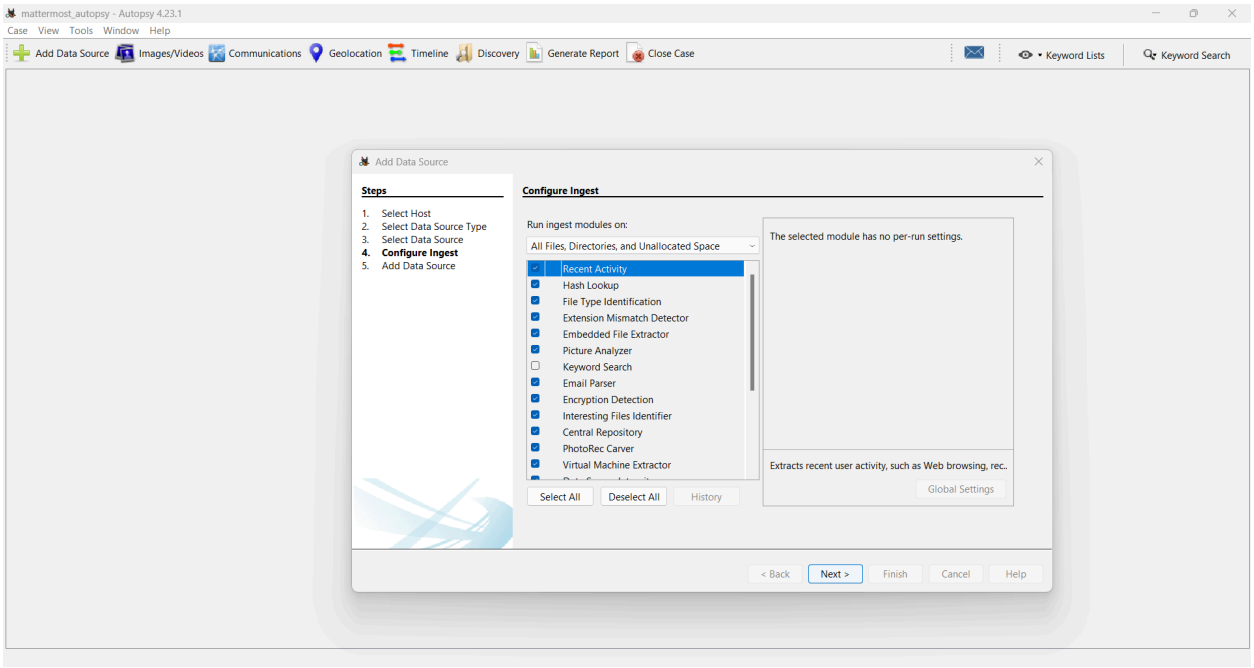

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Case Information
Case Name: mattermost_autopsy
Base Directory: F:\CMU\FORENSICS CAPSTONE\L_
Browse
Case Type: Single-User Multi-User
Case data will be stored in the following directory:
F:\CMU\FORENSICS CAPSTONE\L_\mattermost_autopsy
< Back Next > Finish Cancel Help

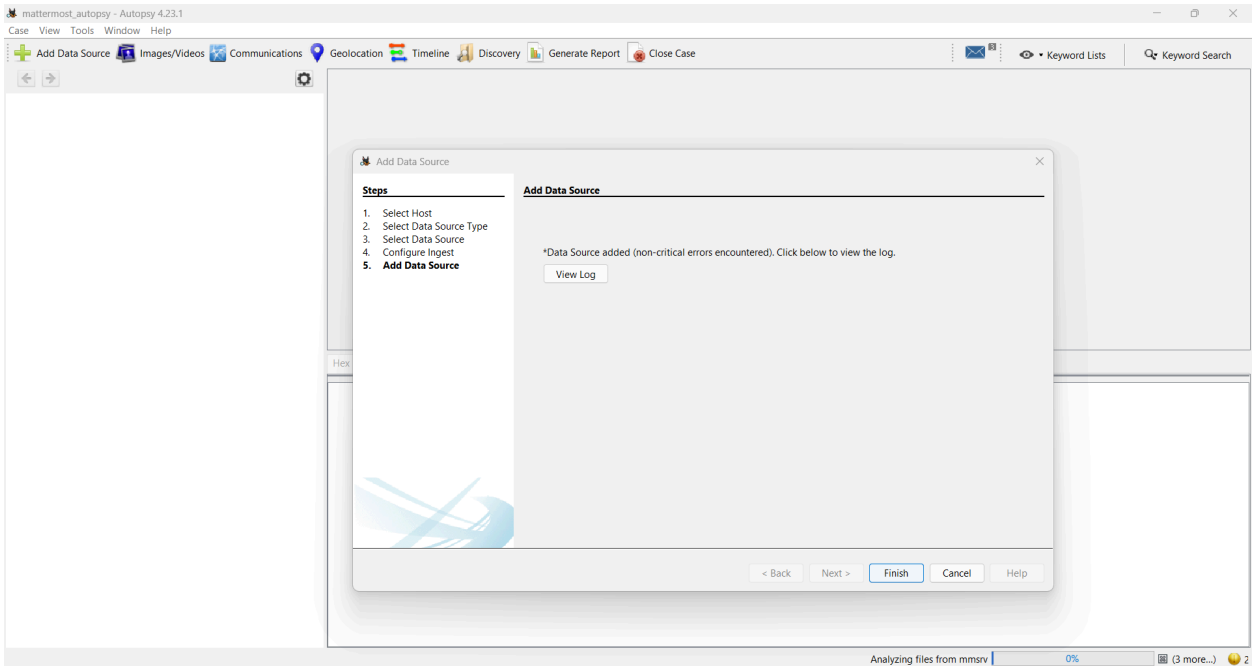

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Optional Information
Case
Number:
Examiner
Name:
Phone:
Email:
Notes:
Organization
Organization analysis is being done for: Not Specified Manage Organizations
< Back Next > Finish Cancel Help

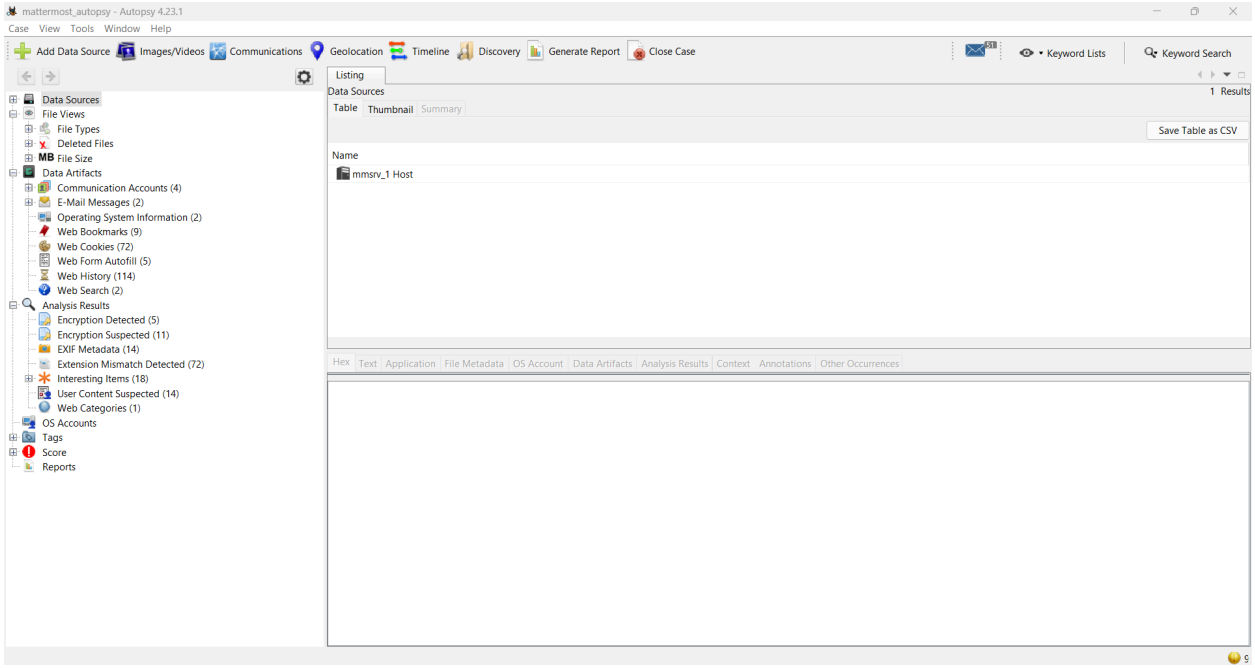

mattermost_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Host
Hosts are used to organize data sources and other data.
Generate new host name based on data source name
Specify new host name
Use existing host
< Back
Next >
Finish
Cancel
Help


mattermost_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source Type
Disk Image or VM File
Local Disk
Logical Files
Unallocated Space Image File
Autopsy Logical Imager Results
XRY Text Export
< Back
Next >
Finish
Cancel
Help

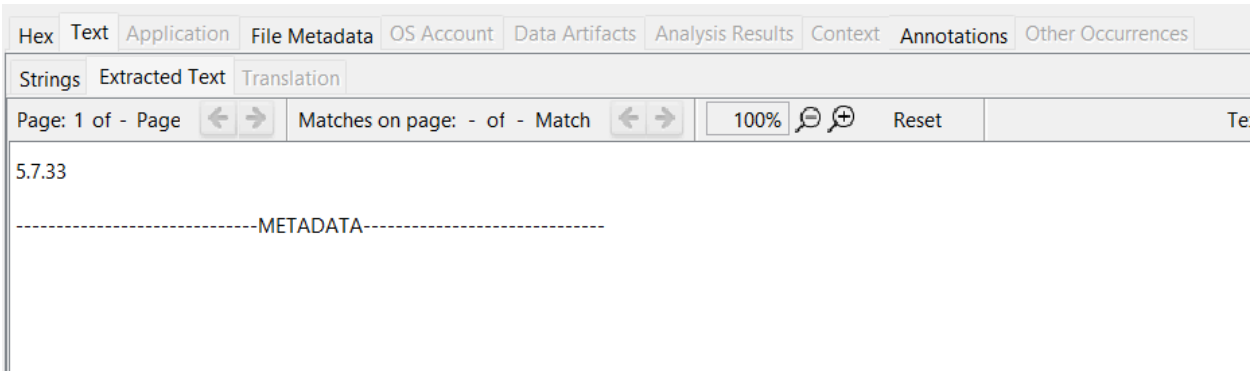

mattermost_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source
Path:
F:\CMU\FORENSICS CAPSTONE\L_\mmsrv
Browse
Ignore orphan files in FAT file systems
Time zone:
(GMT-8:00) America/Los_Angeles
Sector size:
Auto Detect
Bitlocker Password (optional):
Hash Values (optional):
MD5:
SHA-1:
SHA-256:
NOTE: These values will not be validated when the data source is added.
loading...
< Back
Next >
Finish
Cancel
Help


mattermost_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
Select All
Deselect All
History
The selected module has no per-run settings.
Extracts recent user activity, such as Web browsing, rec..
Global Settings
< Back
Next >
Finish
Cancel
Help


mattermost_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
*Data Source added (non-critical errors encountered). Click below to view the log.
View Log
< Back
Next >
Finish
Cancel
Help
Analyzing files from mmsrv
0%
(3 more...)


mattermost_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Communication Accounts (4)
E-Mail Messages (2)
Operating System Information (2)
Web Bookmarks (9)
Web Cookies (72)
Web Form Autofill (5)
Web History (114)
Web Search (2)
Analysis Results
Encryption Detected (5)
Encryption Suspected (11)
EXIF Metadata (14)
Extension Mismatch Detected (72)
Interesting Items (18)
User Content Suspected (14)
Web Categories (1)
OS Accounts
Tags
Score
Reports
Listing
Data Sources
1 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
mmsrv_1 Host
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences

5.2.5.5 MatterMost_A_0005 mysql_upgrade_info



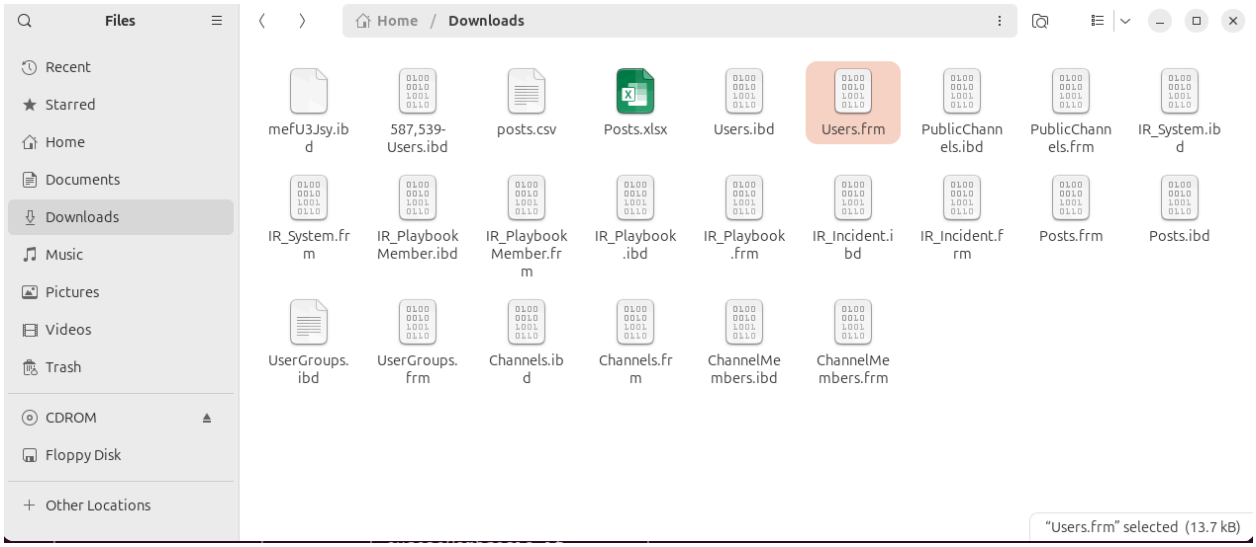
5.2.5.6 MatterMost_A_0006 Posts.idb, Posts.frm, Channels.ibd, Channels.frm, 587,539-Users.ibd, and Users.frm.

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flag
Sessions.frm				2021-01-18 21:41:39 PST	2021-02-02 22:08:30 PST	2021-02-02 22:08:43 PST	2021-01-18 21:41:39 PST	8936	Allocated	Allo
Schemes.ibd				2021-01-18 21:41:45 PST	2021-02-02 22:08:30 PST	2021-02-02 22:08:31 PST	2021-01-18 21:41:35 PST	163840	Allocated	Allo
Schemes.frm				2018-04-24 04:23:31 PDT	2021-02-02 22:11:12 PST	2021-02-02 22:11:12 PST	2021-02-02 22:11:12 PST	3759	Unallocated	Allo
Schemes.frm				2021-01-18 21:41:42 PST	2021-02-02 22:08:30 PST	2021-02-02 22:08:43 PST	2021-01-18 21:41:42 PST	13302	Allocated	Allo
Roles.ibd				2021-01-18 21:43:05 PST	2021-02-02 22:08:30 PST	2021-02-03 08:46:12 PST	2021-01-18 21:41:34 PST	114688	Allocated	Allo
Roles.frm				2021-01-18 21:41:34 PST	2021-02-02 22:08:30 PST	2021-02-02 22:08:43 PST	2021-01-18 21:41:34 PST	8916	Allocated	Allo
Reactions.ibd				2021-01-18 21:41:45 PST	2021-02-02 22:08:30 PST	2021-01-31 09:13:54 PST	2021-01-18 21:41:34 PST	98304	Allocated	Allo
Reactions.frm				2021-01-18 21:41:34 PST	2021-02-02 22:08:30 PST	2021-02-02 22:08:43 PST	2021-01-18 21:41:34 PST	8676	Allocated	Allo
PublicChannels.ibd				2021-01-24 23:22:29 PST	2021-02-02 22:08:30 PST	2021-02-02 16:35:36 PST	2021-01-18 21:41:36 PST	180224	Allocated	Allo
PublicChannels.frm				2021-01-18 21:41:36 PST	2021-02-02 22:08:30 PST	2021-02-02 22:08:43 PST	2021-01-18 21:41:36 PST	8772	Allocated	Allo
ProductNoticeViewState.ibd				2021-02-02 16:34:44 PST	2021-02-02 22:08:30 PST	2021-02-02 22:08:31 PST	2021-01-18 21:41:35 PST	163840	Allocated	Allo
ProductNoticeViewState.frm				2021-01-18 21:41:41 PST	2021-02-02 22:08:30 PST	2021-02-02 22:08:43 PST	2021-01-18 21:41:41 PST	8676	Allocated	Allo
Preferences.ibd				2021-02-03 09:13:18 PST	2021-02-03 09:13:18 PST	2021-02-02 22:08:31 PST	2021-01-18 21:41:34 PST	147456	Allocated	Allo
Preferences.frm				2021-01-18 21:41:41 PST	2021-02-02 22:08:30 PST	2021-02-02 22:08:43 PST	2021-01-18 21:41:41 PST	8850	Unallocated	Allo
Preferences.frm				2021-01-18 21:41:40 PST	2021-02-02 22:08:30 PST	2021-02-02 22:08:43 PST	2021-01-18 21:41:40 PST	8664	Allocated	Allo
Posts.ibd				2021-01-18 21:41:46 PST	2021-02-02 22:08:30 PST	2021-01-31 09:13:52 PST	2021-01-18 21:41:38 PST	98304	Unallocated	Allo
Posts.ibd				2021-02-03 08:49:19 PST	2021-02-03 08:49:19 PST	2021-02-02 22:08:31 PST	2021-01-18 21:41:37 PST	311296	Allocated	Allo

5.2.5.7 MatterMost_A_0007 MySQL 5.7.33 Docker Container Installation

```
vamp@vamp-VMware-Virtual-Platform:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED       STATUS
  PORTS                                                 NAMES
e4c2d3d8b8e9   mysql:5.7.33   "docker-entrypoint.s…"   3 weeks ago   Up 3 hours
  0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp   mysql57
vamp@vamp-VMware-Virtual-Platform:~$
```

5.2.5.8 MatterMost_A_0008 Downloaded Files



5.2.5.9 MatterMost_A_0009 Run Docker Container

```
vamp@vamp-VMware-Virtual-Platform:~$ sudo docker run --name mysql57 \
  -e MYSQL_ROOT_PASSWORD=YourPasswordHere \
  -p 3306:3306 \
  -d mysql:5.7.33
```

5.2.5.10 MatterMost_A_0010 Create Database

```
mysql> DROP DATABASE IF EXISTS recovery_test;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql> CREATE DATABASE recovery_test;
Query OK, 1 row affected (0.00 sec)
```

5.2.5.11 MatterMost_A_0011 Create Channels Table Structure

```
mysql> CREATE TABLE `Channels` (
    ->   `Id` varchar(26) COLLATE `utf8mb4_general_ci` NOT NULL,
    ->   `CreateAt` bigint(20) DEFAULT NULL,
    ->   `UpdateAt` bigint(20) DEFAULT NULL,
    ->   `DeleteAt` bigint(20) DEFAULT NULL,
    ->   `TeamId` varchar(26) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `Type` varchar(1) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `DisplayName` varchar(64) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `Name` varchar(64) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `Header` text COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `Purpose` varchar(250) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `LastPostAt` bigint(20) DEFAULT NULL,
    ->   `TotalMsgCount` bigint(20) DEFAULT NULL,
    ->   `ExtraUpdateAt` bigint(20) DEFAULT NULL,
    ->   `CreatorId` varchar(26) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `SchemeId` varchar(26) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `GroupConstrained` tinyint(1) DEFAULT NULL,
    ->
    ->   PRIMARY KEY (`Id`),
    ->   UNIQUE KEY `Name` (`Name`,`TeamId`),
    ->   KEY `idx_channels_team_id` (`TeamId`),
    ->   KEY `idx_channels_name` (`Name`),
    ->   KEY `idx_channels_update_at` (`UpdateAt`),
    ->   KEY `idx_channels_create_at` (`CreateAt`),
    ->   KEY `idx_channels_delete_at` (`DeleteAt`),
    ->   KEY `idx_channels_scheme_id` (`SchemeId`),
    ->
    ->   FULLTEXT KEY `idx_channel_search_txt`
    ->     (`Name`,`DisplayName`,`Purpose`)
    -> ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
Query OK, 0 rows affected (0.12 sec)
```

5.2.5.12 MatterMost_A_0012 Create Posts Table Structure

```
mysql> CREATE TABLE `Posts` (
    ->   `Id` varchar(26) COLLATE utf8mb4_general_ci NOT NULL,
    ->   `CreateAt` bigint(20) DEFAULT NULL,
    ->   `UpdateAt` bigint(20) DEFAULT NULL,
    ->   `EditAt` bigint(20) DEFAULT NULL,
    ->   `DeleteAt` bigint(20) DEFAULT NULL,
    ->   `IsPinned` tinyint(1) DEFAULT NULL,
    ->   `UserId` varchar(26) COLLATE utf8mb4_general_ci DEFAULT NULL,
    ->   `ChannelId` varchar(26) COLLATE utf8mb4_general_ci DEFAULT NULL,
    ->   `RootId` varchar(26) COLLATE utf8mb4_general_ci DEFAULT NULL,
    ->   `ParentId` varchar(26) COLLATE utf8mb4_general_ci DEFAULT NULL,
    ->   `OriginalId` varchar(26) COLLATE utf8mb4_general_ci DEFAULT NULL,
    ->   `Message` text COLLATE utf8mb4_general_ci,
    ->   `Type` varchar(26) COLLATE utf8mb4_general_ci DEFAULT NULL,
    ->   `Props` text COLLATE utf8mb4_general_ci,
    ->   `Hashtags` text COLLATE utf8mb4_general_ci,
    ->   `Filenames` text COLLATE utf8mb4_general_ci,
    ->   `FileIds` varchar(150) COLLATE utf8mb4_general_ci DEFAULT NULL,
    ->   `HasReactions` tinyint(1) DEFAULT NULL,
    ->   PRIMARY KEY (`Id`),
    ->   KEY `idx_posts_update_at` (`UpdateAt`),
    ->   KEY `idx_posts_create_at` (`CreateAt`),
    ->   KEY `idx_posts_delete_at` (`DeleteAt`),
    ->   KEY `idx_posts_channel_id` (`ChannelId`),
    ->   KEY `idx_posts_root_id` (`RootId`),
    ->   KEY `idx_posts_user_id` (`UserId`),
    ->   KEY `idx_posts_is_pinned` (`IsPinned`),
    ->   KEY `idx_posts_channel_id_update_at` (`ChannelId`,`UpdateAt`),
    ->   KEY `idx_posts_channel_id_delete_at_create_at`
    ->     (`ChannelId`,`DeleteAt`,`CreateAt`),
    ->   FULLTEXT KEY `idx_posts_message_txt` (`Message`),
    ->   FULLTEXT KEY `idx_posts_hashtags_txt` (`Hashtags`)
    -> ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
Query OK, 0 rows affected (0.08 sec)
```

5.2.5.13 MatterMost_A_0013 Create Users Table Structure

```
mysql> CREATE TABLE `Users` (
    ->   `Id` varchar(26) COLLATE `utf8mb4_general_ci` NOT NULL,
    ->   `CreateAt` bigint(20) DEFAULT NULL,
    ->   `UpdateAt` bigint(20) DEFAULT NULL,
    ->   `DeleteAt` bigint(20) DEFAULT NULL,
    ->   `Username` varchar(64) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `Password` varchar(128) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `AuthData` varchar(128) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `AuthService` varchar(32) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `Email` varchar(128) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `EmailVerified` tinyint(1) DEFAULT NULL,
    ->   `Nickname` varchar(64) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `FirstName` varchar(64) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `LastName` varchar(64) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `Position` varchar(128) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `Roles` text COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `AllowMarketing` tinyint(1) DEFAULT NULL,
    ->   `Props` text COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `NotifyProps` text COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `LastPasswordUpdate` bigint(20) DEFAULT NULL,
    ->   `LastPictureUpdate` bigint(20) DEFAULT NULL,
    ->   `FailedAttempts` int(11) DEFAULT NULL,
    ->   `Locale` varchar(5) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `Timezone` text COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->   `MfaActive` tinyint(1) DEFAULT NULL,
    ->   `MfaSecret` varchar(128) COLLATE `utf8mb4_general_ci` DEFAULT NULL,
    ->
    ->   PRIMARY KEY (`Id`),
    ->   UNIQUE KEY `Username` (`Username`),
    ->   UNIQUE KEY `AuthData` (`AuthData`),
    ->   UNIQUE KEY `Email` (`Email`),
    ->   KEY `idx_users_email` (`Email`),
    ->   KEY `idx_users_update_at` (`UpdateAt`),
    ->   KEY `idx_users_create_at` (`CreateAt`),
    ->   KEY `idx_users_delete_at` (`DeleteAt`),
    ->
    ->   FULLTEXT KEY `idx_users_all_txt`
    ->     (`Username`,`FirstName`,`LastName`,`Nickname`,`Email`),
    ->
    ->   FULLTEXT KEY `idx_users_all_no_full_name_txt`
    ->     (`Username`,`Nickname`,`Email`),
    ->
    ->   FULLTEXT KEY `idx_users_names_txt`
    ->     (`Username`,`FirstName`,`LastName`,`Nickname`),
    ->
    ->   FULLTEXT KEY `idx_users_names_no_full_name_txt`
    ->     (`Username`,`Nickname`)
    -> ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
Query OK, 0 rows affected (0.19 sec)
```

5.2.5.14 MatterMost_A_0014 Discard Empty Tablespace

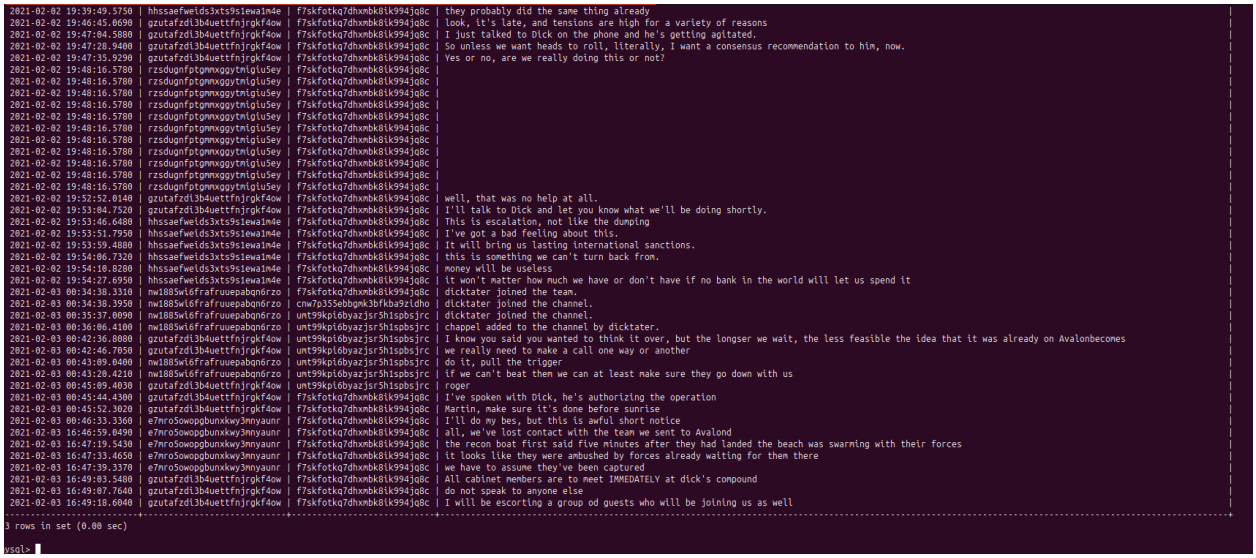
```
mysql> ALTER TABLE Channels DISCARD TABLESPACE;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> ALTER TABLE Posts DISCARD TABLESPACE;
Query OK, 0 rows affected (0.02 sec)

mysql> EXIT;
Bye
```

```
mysql> ALTER TABLE Users DISCARD TABLESPACE;
Query OK, 0 rows affected (0.06 sec)
```

5.2.5.15 MatterMost_A_0015 Posts Table

CreateAt	UserId	ChannelId	Message
2021-01-25 07:22:26.6240	khf8znhnbbyjmdp88brqfy63na	f7skfotkq7dhxnbk8lk994jq8c	adminlstrator joined the team.
2021-01-25 07:22:26.9030	khf8znhnbbyjmdp88brqfy63na	cnw7p35ebbgmk3bfkba9zidho	administrator joined the channel.
2021-01-31 20:10:40.6720	gzutafzdl3b4uettfnjrgkf4ow	f7skfotkq7dhxnbk8lk994jq8c	chappel joined the team.
2021-01-31 20:10:44.2760	gzutafzdl3b4uettfnjrgkf4ow	cnw7p35ebbgmk3bfkba9zidho	chappel joined the channel.
2021-01-31 20:29:50.8590	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	eugene joined the team.
2021-01-31 20:29:52.1420	hhssaefweids3xts9s1ewa1m4e	cnw7p35ebbgmk3bfkba9zidho	eugene joined the channel.
2021-01-31 20:30:05.3330	qbdy4hrz63bhmkrdc13m7465sw	f7skfotkq7dhxnbk8lk994jq8c	julia joined the team.
2021-01-31 20:30:06.4290	qbdy4hrz63bhmkrdc13m7465sw	cnw7p35ebbgmk3bfkba9zidho	julia joined the channel.
2021-01-31 20:37:20.9840	1os4fw9m3brtmepb7x8pnwarda	f7skfotkq7dhxnbk8lk994jq8c	jean joined the team.
2021-01-31 20:37:21.5870	1os4fw9m3brtmepb7x8pnwarda	cnw7p35ebbgmk3bfkba9zidho	jean joined the channel.
2021-01-31 20:39:20.0340	1os4fw9m3brtmepb7x8pnwarda	f7skfotkq7dhxnbk8lk994jq8c	hi all
2021-01-31 20:42:48.8030	bzhp1dfy7fr9mfyj8md5pn8swc	f7skfotkq7dhxnbk8lk994jq8c	richard joined the team.
2021-01-31 20:42:49.8210	bzhp1dfy7fr9mfyj8md5pn8swc	cnw7p35ebbgmk3bfkba9zidho	richard joined the channel.
2021-01-31 20:50:48.1580	e7mroSowopgbunxkwy3mnyaunr	f7skfotkq7dhxnbk8lk994jq8c	martin joined the team.
2021-01-31 20:50:49.1490	e7mroSowopgbunxkwy3mnyaunr	cnw7p35ebbgmk3bfkba9zidho	martin joined the channel.
2021-01-31 20:51:00.8770	trocmpjyhtga8b4mer3wn6dc9o	f7skfotkq7dhxnbk8lk994jq8c	eleanor joined the team.
2021-01-31 20:51:01.4010	trocmpjyhtga8b4mer3wn6dc9o	cnw7p35ebbgmk3bfkba9zidho	eleanor joined the channel.
2021-01-31 22:59:59.0130	rzsdugnfptgmnxggytmlgiu5ey	f7skfotkq7dhxnbk8lk994jq8c	
2021-01-31 22:59:59.0130	rzsdugnfptgmnxggytmlgiu5ey	f7skfotkq7dhxnbk8lk994jq8c	
2021-01-31 22:59:59.0130	rzsdugnfptgmnxggytmlgiu5ey	f7skfotkq7dhxnbk8lk994jq8c	
2021-01-31 22:59:59.0130	rzsdugnfptgmnxggytmlgiu5ey	f7skfotkq7dhxnbk8lk994jq8c	
2021-01-31 22:59:59.0130	rzsdugnfptgmnxggytmlgiu5ey	f7skfotkq7dhxnbk8lk994jq8c	
2021-01-31 22:59:59.0130	rzsdugnfptgmnxggytmlgiu5ey	f7skfotkq7dhxnbk8lk994jq8c	
2021-01-31 22:59:59.0130	rzsdugnfptgmnxggytmlgiu5ey	f7skfotkq7dhxnbk8lk994jq8c	
2021-01-31 22:59:59.0130	rzsdugnfptgmnxggytmlgiu5ey	f7skfotkq7dhxnbk8lk994jq8c	
2021-01-31 22:59:59.0130	rzsdugnfptgmnxggytmlgiu5ey	f7skfotkq7dhxnbk8lk994jq8c	
2021-01-31 23:19:36.6150	gzutafzdl3b4uettfnjrgkf4ow	f7skfotkq7dhxnbk8lk994jq8c	Let's get going ladies and gentlemen
2021-01-31 23:19:54.1310	e7mroSowopgbunxkwy3mnyaunr	f7skfotkq7dhxnbk8lk994jq8c	Is the poll done?
2021-02-02 19:33:24.3600	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	Okay, the rumors are flying all over the place.
2021-02-02 19:33:58.3470	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	Avalon's banana futures are through the roof as they just announced mandatory screenings for EVERYTHING coming onto the island.
2021-02-02 19:34:21.0250	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	Their press release even said they're the 'only producer in the reigon unaffected by the blight'
2021-02-02 19:34:38.9170	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	Not only will this entirely ruin our plans, they could start taking our share of the market!
2021-02-02 19:34:45.5670	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	we have to do something here
2021-02-02 19:34:48.3160	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	something extreme
2021-02-02 19:34:55.6830	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	and we have to do it RIGHT NOW
2021-02-02 19:35:27.7210	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	the only way we keep them from coming out ahead of us is to make sure they're infected too.
2021-02-02 19:35:54.1100	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	I mean, all we'd need to do is take some of those black bananas from the infected farm and make sure they end up scattered around a few farms on Avalon.
2021-02-02 19:36:20.6130	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	If we do it right away, they'll just assume the blight had already spread there before the put in these draconian measures.
2021-02-02 19:37:13.2890	qbdy4hrz63bhmkrdc13m7465sw	f7skfotkq7dhxnbk8lk994jq8c	Normally, i'd just ignore your after hours rambling eugene, but that is beyond nuts.
2021-02-02 19:37:23.8450	qbdy4hrz63bhmkrdc13m7465sw	f7skfotkq7dhxnbk8lk994jq8c	what you're talking about is industrial sabotage
2021-02-02 19:37:29.1860	qbdy4hrz63bhmkrdc13m7465sw	f7skfotkq7dhxnbk8lk994jq8c	probably even bio terrorism
2021-02-02 19:37:45.1610	qbdy4hrz63bhmkrdc13m7465sw	f7skfotkq7dhxnbk8lk994jq8c	Do you have any idea how hard the world community would come down on us if they found out?
2021-02-02 19:37:55.9270	qbdy4hrz63bhmkrdc13m7465sw	f7skfotkq7dhxnbk8lk994jq8c	We're not a super-power that people will be afraid to sanction
2021-02-02 19:38:51.1000	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	I know what i'm suggesting, but it would be impossible to track back to us
2021-02-02 19:38:58.8040	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	certainly not to prove it was done with malice
2021-02-02 19:39:10.1740	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	unless we were litterally caught red handed
2021-02-02 19:39:16.9850	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	we'd jsut say it was an accident
2021-02-02 19:39:30.0740	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	some tourist had some fruit in their bag, hardly something you can prove
2021-02-02 19:39:43.5670	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	heck, for all we know, they're the ones who made sure we got infected
2021-02-02 19:39:49.5750	hhssaefweids3xts9s1ewa1m4e	f7skfotkq7dhxnbk8lk994jq8c	they probably did the same thing already
2021-02-02 19:46:45.0690	gzutafzdl3b4uettfnjrgkf4ow	f7skfotkq7dhxnbk8lk994jq8c	look, it's late, and tensions are high for a variety of reasons
2021-02-02 19:47:04.5880	gzutafzdl3b4uettfnjrgkf4ow	f7skfotkq7dhxnbk8lk994jq8c	I just talked to Dick on the phone and he's getting agitated.
2021-02-02 19:47:28.0400	gzutafzdl3b4uettfnjrgkf4ow	f7skfotkq7dhxnbk8lk994jq8c	So unless we want heads to roll, literally, I want a consensus recommendation to him, now.
2021-02-02 19:47:35.9290	gzutafzdl3b4uettfnjrgkf4ow	f7skfotkq7dhxnbk8lk994jq8c	Yes or no, are we really doing this or not?



```
2021-02-02 19:39:49.5750 | hhssaefweids3xts9s1ewa1m4e | f7skfotkq7dhxmbk8ik994jq8c | they probably did the same thing already
2021-02-02 19:46:45.0690 | gzutafzdi3b4uettfnjrgkf4ow | f7skfotkq7dhxmbk8ik994jq8c | look, it's late, and tensions are high for a variety of reasons
2021-02-02 19:47:04.5800 | gzutafzdi3b4uettfnjrgkf4ow | f7skfotkq7dhxmbk8ik994jq8c | I just talked to Dick on the phone and he's getting agitated.
2021-02-02 19:47:28.9400 | gzutafzdi3b4uettfnjrgkf4ow | f7skfotkq7dhxmbk8ik994jq8c | So unless we want heads to roll, literally, I want a consensus recommendation to him, now.
2021-02-02 19:47:35.9290 | gzutafzdi3b4uettfnjrgkf4ow | f7skfotkq7dhxmbk8ik994jq8c | Yes or no, are we really doing this or not?
2021-02-02 19:48:16.5780 | rzsdugnfptgmmxggytmigiu5ey | f7skfotkq7dhxmbk8ik994jq8c |
2021-02-02 19:48:16.5780 | rzsdugnfptgmmxggytmigiu5ey | f7skfotkq7dhxmbk8ik994jq8c |
2021-02-02 19:48:16.5780 | rzsdugnfptgmmxggytmigiu5ey | f7skfotkq7dhxmbk8ik994jq8c |
2021-02-02 19:48:16.5780 | rzsdugnfptgmmxggytmigiu5ey | f7skfotkq7dhxmbk8ik994jq8c |
2021-02-02 19:48:16.5780 | rzsdugnfptgmmxggytmigiu5ey | f7skfotkq7dhxmbk8ik994jq8c |
2021-02-02 19:48:16.5780 | rzsdugnfptgmmxggytmigiu5ey | f7skfotkq7dhxmbk8ik994jq8c |
2021-02-02 19:48:16.5780 | rzsdugnfptgmmxggytmigiu5ey | f7skfotkq7dhxmbk8ik994jq8c |
2021-02-02 19:48:16.5780 | rzsdugnfptgmmxggytmigiu5ey | f7skfotkq7dhxmbk8ik994jq8c |
2021-02-02 19:48:16.5780 | rzsdugnfptgmmxggytmigiu5ey | f7skfotkq7dhxmbk8ik994jq8c |
2021-02-02 19:52:52.0140 | gzutafzdi3b4uettfnjrgkf4ow | f7skfotkq7dhxmbk8ik994jq8c | well, that was no help at all.
2021-02-02 19:53:04.7520 | gzutafzdi3b4uettfnjrgkf4ow | f7skfotkq7dhxmbk8ik994jq8c | I'll talk to Dick and let you know what we'll be doing shortly.
2021-02-02 19:53:46.6480 | hhssaefweids3xts9s1ewa1m4e | f7skfotkq7dhxmbk8ik994jq8c | This is escalation, not like the dumping
2021-02-02 19:53:51.7950 | hhssaefweids3xts9s1ewa1m4e | f7skfotkq7dhxmbk8ik994jq8c | I've got a bad feeling about this.
2021-02-02 19:53:59.4880 | hhssaefweids3xts9s1ewa1m4e | f7skfotkq7dhxmbk8ik994jq8c | It will bring us lasting international sanctions.
2021-02-02 19:54:06.7320 | hhssaefweids3xts9s1ewa1m4e | f7skfotkq7dhxmbk8ik994jq8c | this is something we can't turn back from.
2021-02-02 19:54:10.8280 | hhssaefweids3xts9s1ewa1m4e | f7skfotkq7dhxmbk8ik994jq8c | money will be useless
2021-02-02 19:54:27.6950 | hhssaefweids3xts9s1ewa1m4e | f7skfotkq7dhxmbk8ik994jq8c | it won't matter how much we have or don't have if no bank in the world will let us spend it
2021-02-03 00:34:38.3310 | nw1885wi6frafruuepabqn6rzo | f7skfotkq7dhxmbk8ik994jq8c | dicktater joined the team.
2021-02-03 00:34:38.3950 | nw1885wi6frafruuepabqn6rzo | cnw7p355ebbgmk3bfkba9zidho | dicktater joined the channel.
2021-02-03 00:35:37.0090 | nw1885wi6frafruuepabqn6rzo | umt99kpi6byazjsr5h1spbsjrc | dicktater joined the channel.
2021-02-03 00:36:06.4100 | nw1885wi6frafruuepabqn6rzo | umt99kpi6byazjsr5h1spbsjrc | chappel added to the channel by dicktater.
2021-02-03 00:42:36.8080 | gzutafzdi3b4uettfnjrgkf4ow | umt99kpi6byazjsr5h1spbsjrc | I know you said you wanted to think it over, but the longser we wait, the less feasible the idea that it was already on Avalonbecomes
2021-02-03 00:42:46.7050 | gzutafzdi3b4uettfnjrgkf4ow | umt99kpi6byazjsr5h1spbsjrc | we really need to make a call one way or another
2021-02-03 00:43:09.0400 | nw1885wi6frafruuepabqn6rzo | umt99kpi6byazjsr5h1spbsjrc | do it, pull the trigger
2021-02-03 00:43:20.4210 | nw1885wi6frafruuepabqn6rzo | umt99kpi6byazjsr5h1spbsjrc | if we can't beat them we can at least make sure they go down with us
2021-02-03 00:45:09.4030 | gzutafzdi3b4uettfnjrgkf4ow | umt99kpi6byazjsr5h1spbsjrc | roger
2021-02-03 00:45:44.4300 | gzutafzdi3b4uettfnjrgkf4ow | f7skfotkq7dhxmbk8ik994jq8c | I've spoken with Dick, he's authorizing the operation
2021-02-03 00:45:52.3020 | gzutafzdi3b4uettfnjrgkf4ow | f7skfotkq7dhxmbk8ik994jq8c | Martin, make sure it's done before sunrise
2021-02-03 00:46:33.3360 | e7mro5owopgbunxkwy3mnyaunr | f7skfotkq7dhxmbk8ik994jq8c | I'll do my bes, but this is awful short notice
2021-02-03 16:46:59.0490 | e7mro5owopgbunxkwy3mnyaunr | f7skfotkq7dhxmbk8ik994jq8c | all, we've lost contact with the team we sent to Avalond
2021-02-03 16:47:19.5430 | e7mro5owopgbunxkwy3mnyaunr | f7skfotkq7dhxmbk8ik994jq8c | the recon boat first said five minutes after they had landed the beach was swarming with their forces
2021-02-03 16:47:33.4650 | e7mro5owopgbunxkwy3mnyaunr | f7skfotkq7dhxmbk8ik994jq8c | it looks like they were ambushed by forces already waiting for them there
2021-02-03 16:47:39.3370 | e7mro5owopgbunxkwy3mnyaunr | f7skfotkq7dhxmbk8ik994jq8c | we have to assume they've been captured
2021-02-03 16:49:03.5400 | gzutafzdi3b4uettfnjrgkf4ow | f7skfotkq7dhxmbk8ik994jq8c | All cabinet members are to meet IMMEDATELY at dick's compound
2021-02-03 16:49:07.7640 | gzutafzdi3b4uettfnjrgkf4ow | f7skfotkq7dhxmbk8ik994jq8c | do not speak to anyone else
2021-02-03 16:49:18.6040 | gzutafzdi3b4uettfnjrgkf4ow | f7skfotkq7dhxmbk8ik994jq8c | I will be escorting a group od guests who will be joining us as well
3 rows in set (0.00 sec)

ysql>
```

5.2.5.16 MatterMost_A_0016 Channels Table

```
mysql> SELECT * FROM Channels LIMIT 10;
+----------------------------+---------------+---------------+----------+----------------------------+------+-------------+-------------+
---+---------------+----------------------------+----------+-----------------+
| Id                         | CreateAt      | UpdateAt      | DeleteAt | TeamId                     | Type | DisplayName | Name        |
unt | ExtraUpdateAt | CreatorId                  | SchemeId | GroupConstrained |
+----------------------------+---------------+---------------+----------+----------------------------+------+-------------+-------------+
---+---------------+----------------------------+----------+-----------------+
| cnw7p355ebbgmk3bfkba9zidho | 1611559346113 | 1611559346113 |        0 | qwzgif3spjb9pb3kc795ponnnc | O    | Off-Topic   | off-topic   |
  0 |             0 |                            |          |                 |
| f7skfotkq7dhxmbk8ik994jq8c | 1611559345842 | 1611559345842 |        0 | qwzgif3spjb9pb3kc795ponnnc | O    | Town Square | town-square |
 49 |             0 |                            |          |                 |
| umt99kpi6byazjsr5h1spbsjrc | 1612312536921 | 1612312536921 |        0 | qwzgif3spjb9pb3kc795ponnnc | P    | BossmanChat | bossmanchat |
  5 |             0 | nw1885wi6frafruuepabqn6rzo |          |               0 |
+----------------------------+---------------+---------------+----------+----------------------------+------+-------------+-------------+
```

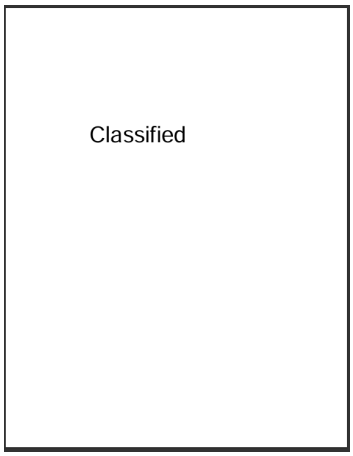
5.2.5.17 MatterMost_A_0017 Users Table

```
mysql> SELECT Id, Username, FirstName, LastName, Email
    -> FROM Users;
+----------------------------+----------------+---------------------+----------+----------------------------+
| Id                         | Username       | FirstName           | LastName | Email                      |
+----------------------------+----------------+---------------------+----------+----------------------------+
| 17fg3iexi7rruj15reygya4ojh | channelexport  | Channel Export Bot  |          | channelexport@localhost    |
| 1os4fw9m3brtmepb7x8pmwarda | jean           |                     |          | jean@ambrosia.it           |
| 41dz4fqso7fdjfsak7n8am5toy | surveybot      | Surveybot           |          | surveybot@localhost        |
| 7a7bmebjmbyu5cr5c9cqhf7u6h | incident       | Incident Bot        |          | incident@localhost         |
| bzhp1dfy7fr9mfyj8md5pn8swc | richard        |                     |          | richard@ambrosia.it        |
| e7mro5owopgbunxkwy3mnyaunr | martin         |                     |          | martin@ambrosia.it         |
| gzutafzdi3b4uettfnjrgkf4ow | chappel        |                     |          | chappel@ambrosia.it        |
| hhssaefweids3xts9s1ewa1m4e | eugene         |                     |          | eugene@ambrosia.it         |
| khf8znhnbbyjmdp88brqfy63na | administrator  |                     |          | administrator@ambrosia.it  |
| nw1885wi6frafruuepabqn6rzo | dicktater      |                     |          | ater@ambrosia.it           |
| qbdy4hrz63bhmkrdc13m7465sw | julia          |                     |          | julia@ambrosia.it          |
| rzsdugnfptgmmxggytmigiu5ey | matterpoll     | Matterpoll          |          | matterpoll@localhost       |
| trocmpjyhtga8b4mer3wn6dc9o | eleanor        |                     |          | eleanor@ambrosia.it        |
+----------------------------+----------------+---------------------+----------+----------------------------+
13 rows in set (0.00 sec)
```

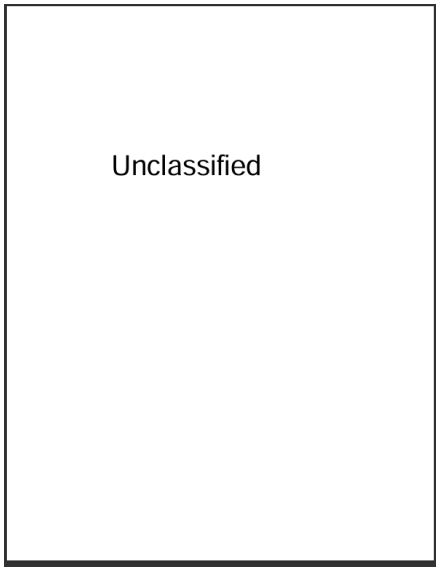
5.2.5.18 MatterMost_A_0018 Command to view Recovered Posts Messages as a readable CSV

```
mysql> SELECT
    ->     Id,
    ->     FROM_UNIXTIME(CreateAt / 1000) AS CreateAt,
    ->     FROM_UNIXTIME(UpdateAt / 1000) AS UpdateAt,
    ->     FROM_UNIXTIME(EditAt / 1000) AS EditAt,
    ->     FROM_UNIXTIME(DeleteAt / 1000) AS DeleteAt,
    ->     IsPinned,
    ->     UserId,
    ->     ChannelId,
    ->     RootId,
    ->     ParentId,
    ->     OriginalId,
    ->     Message,
    ->     Type,
    ->     Props,
    ->     Hashtags,
    ->     Filenames,
    ->     FileIds,
    ->     HasReactions
    -> FROM Posts
    -> ORDER BY CreateAt
    -> INTO OUTFILE '/var/lib/mysql-files/posts_all_columns.csv'
    -> FIELDS TERMINATED BY ','
    -> OPTIONALLY ENCLOSED BY '"'
    -> LINES TERMINATED BY '\n';
Query OK, 93 rows affected (0.01 sec)
```

5.2.5.19 MatterMost_A_0019 Classified.pdf

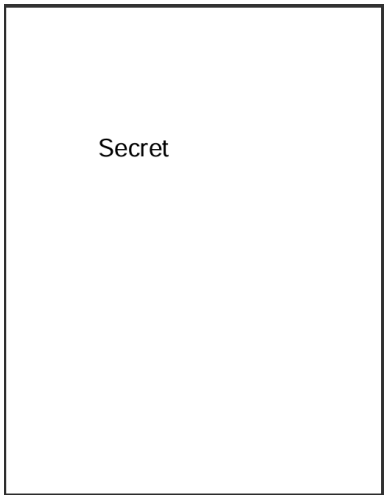


5.2.5.20 MatterMost_A_0020 Unclassified.pdf

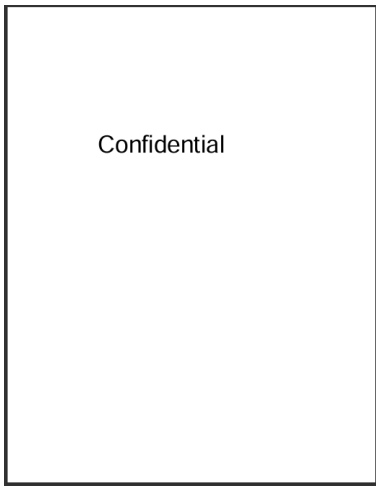


5.2.5.21 MatterMost_A_0021 Top secret.pdf

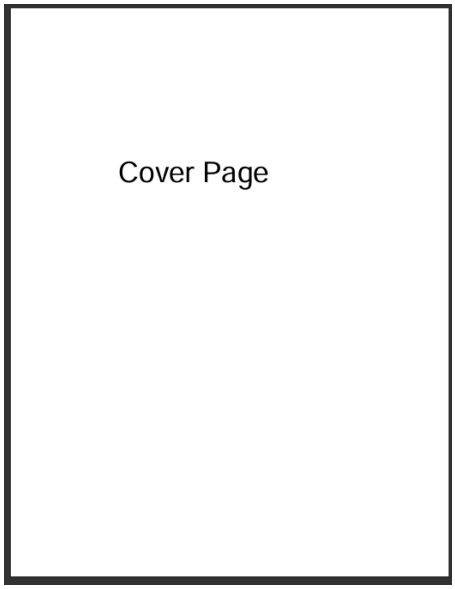
5.2.5.22 MatterMost_A_0022 Secret.pdf



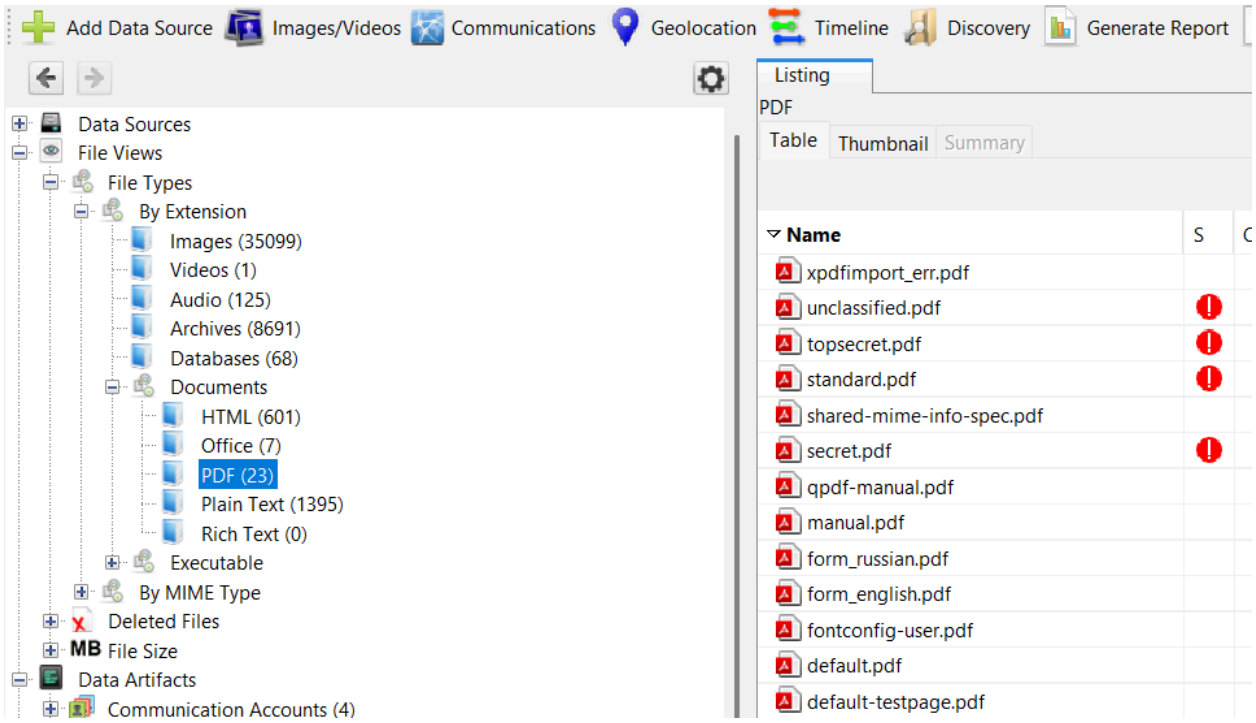
5.2.5.23 MatterMost_A_0023 Confidential.pdf



5.2.5.24 MatterMost_A_0024 Cover Page

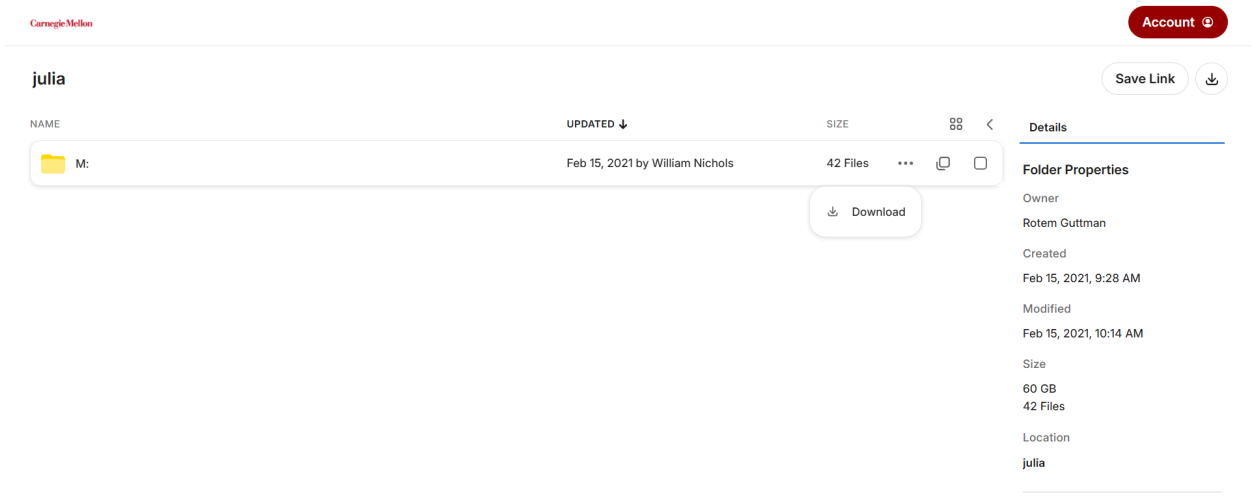


5.2.25 MatterMost_A_0025 Files

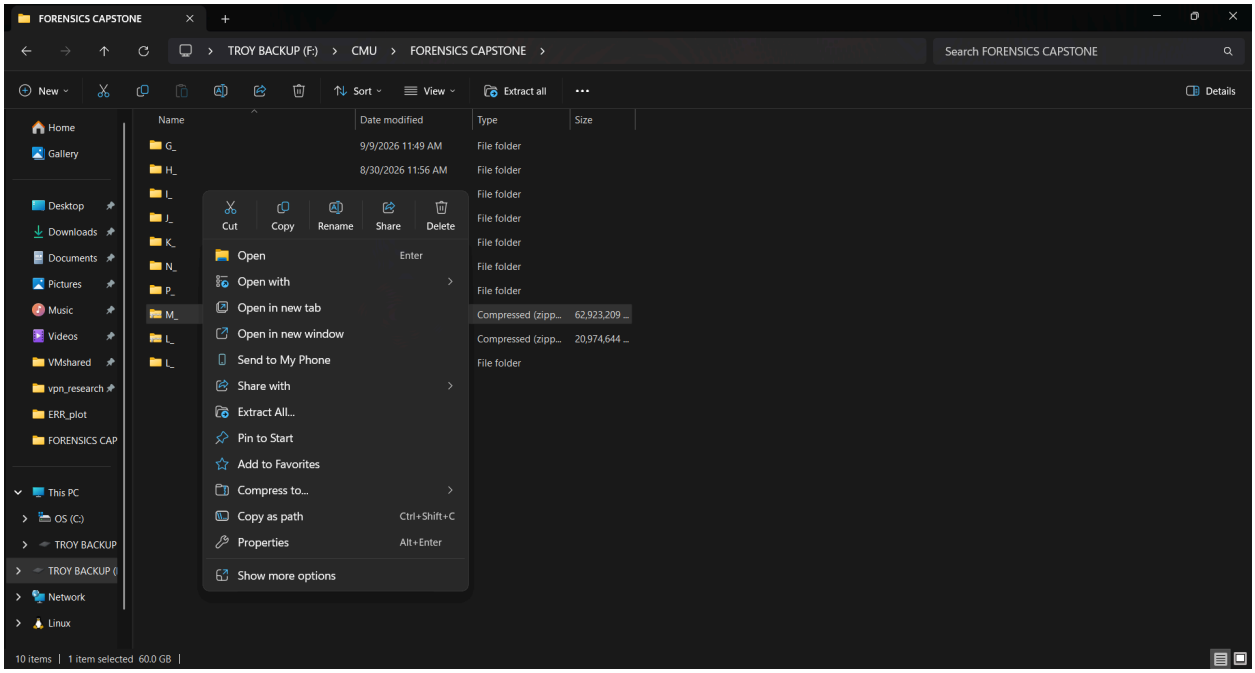


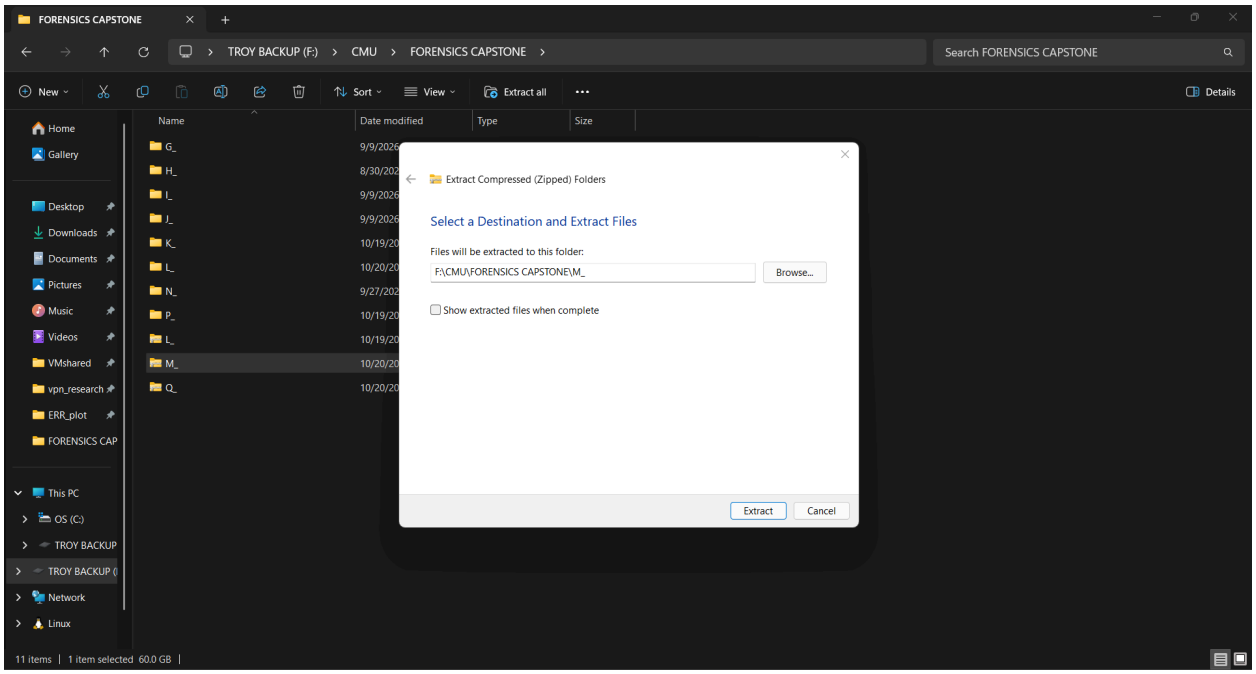
5.2.6 Julia Image (M)

5.2.6.1 Julia_A_0001 Image Download

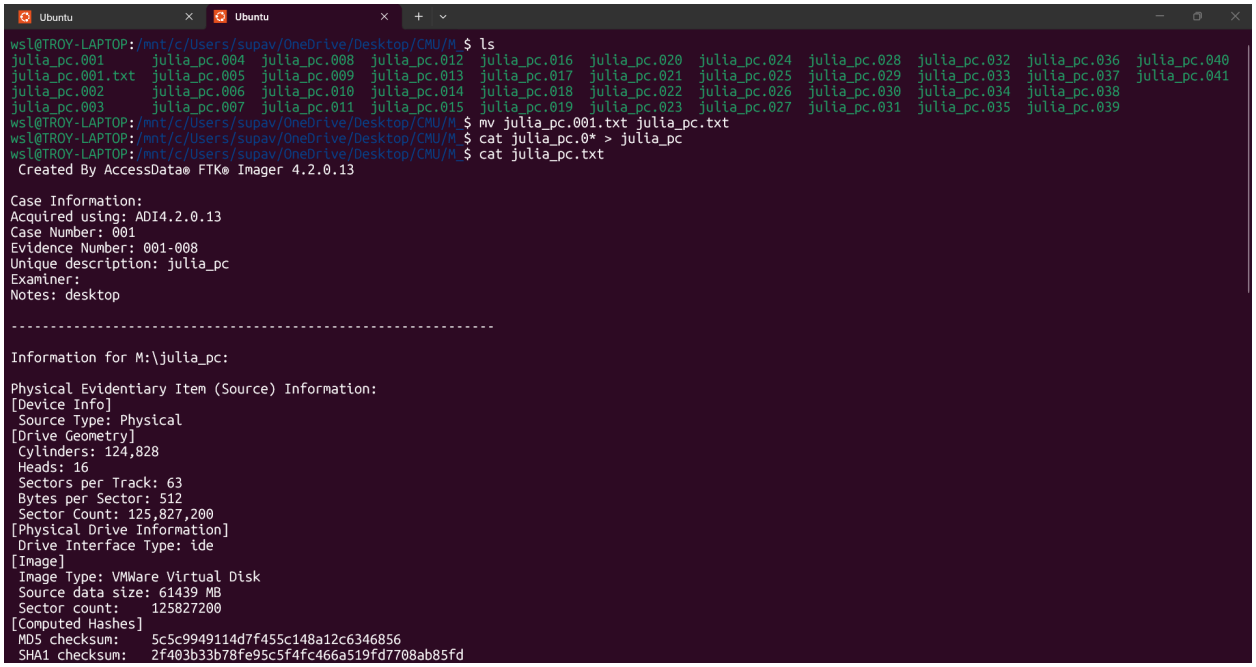


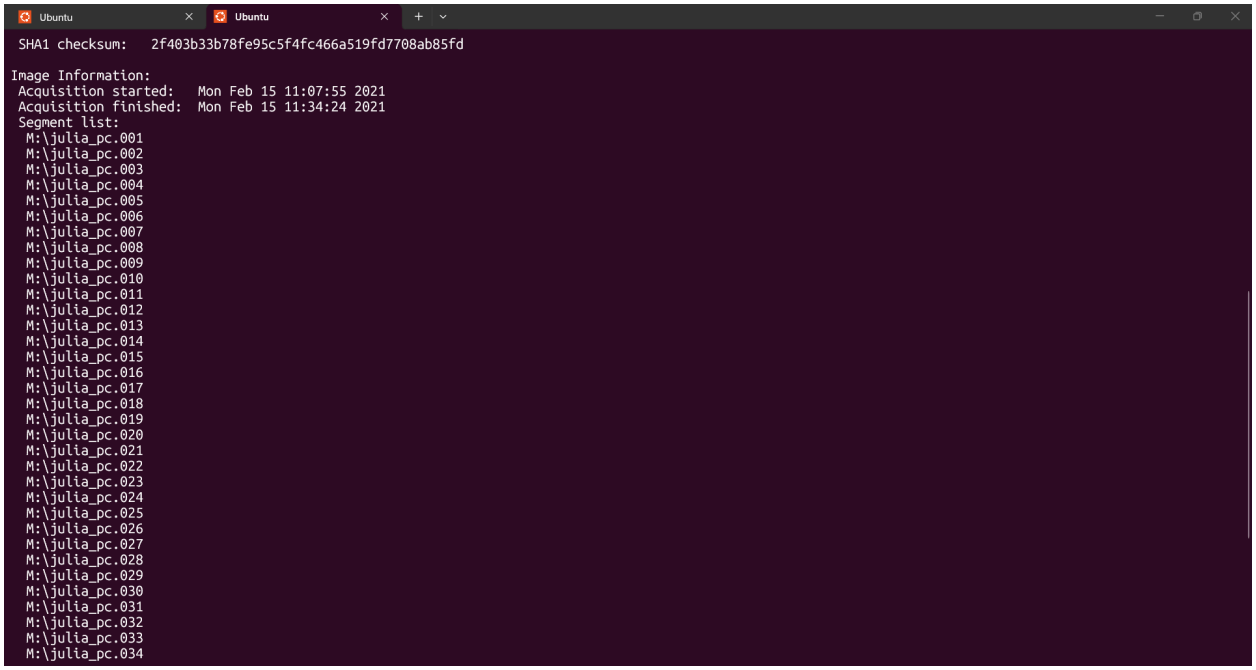
5.2.6.2 Julia_A_0002 Image Extraction

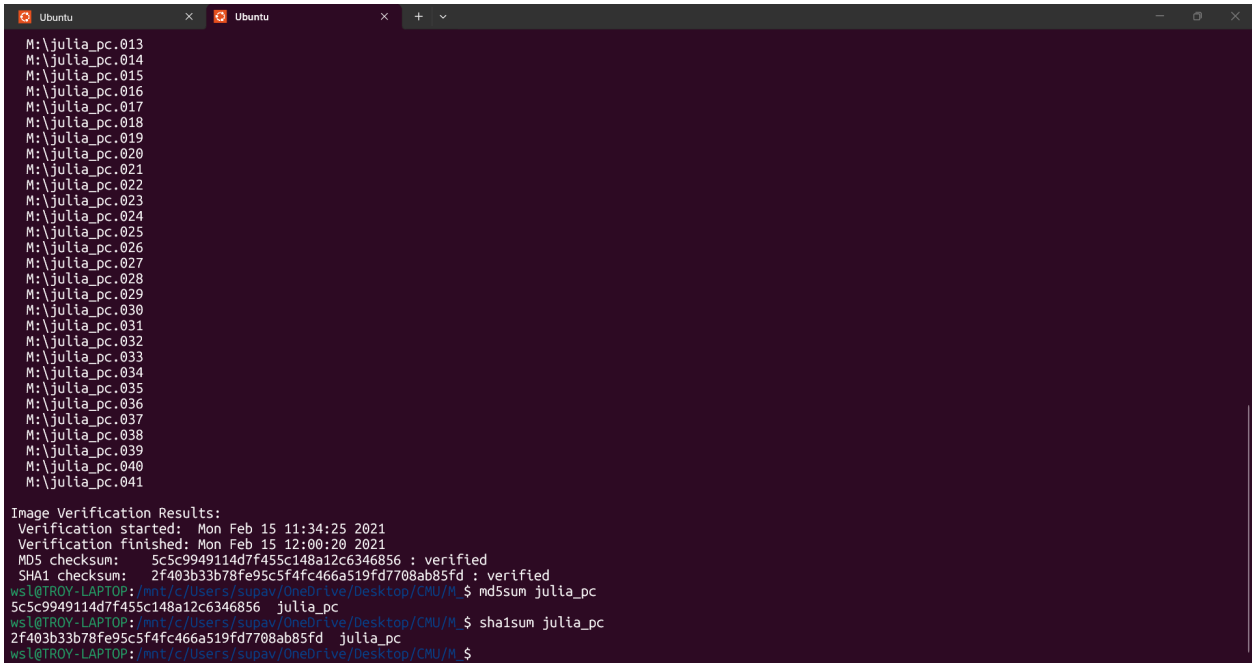




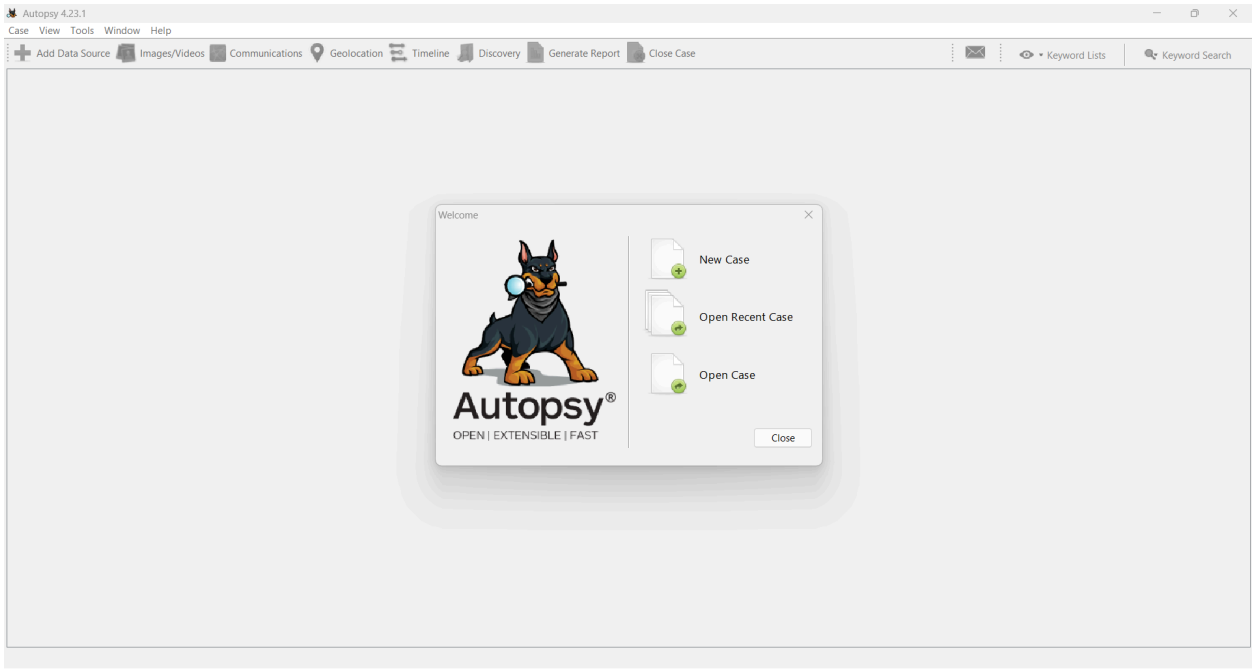
5.2.6.3 Julia_A_0003 Image Verification

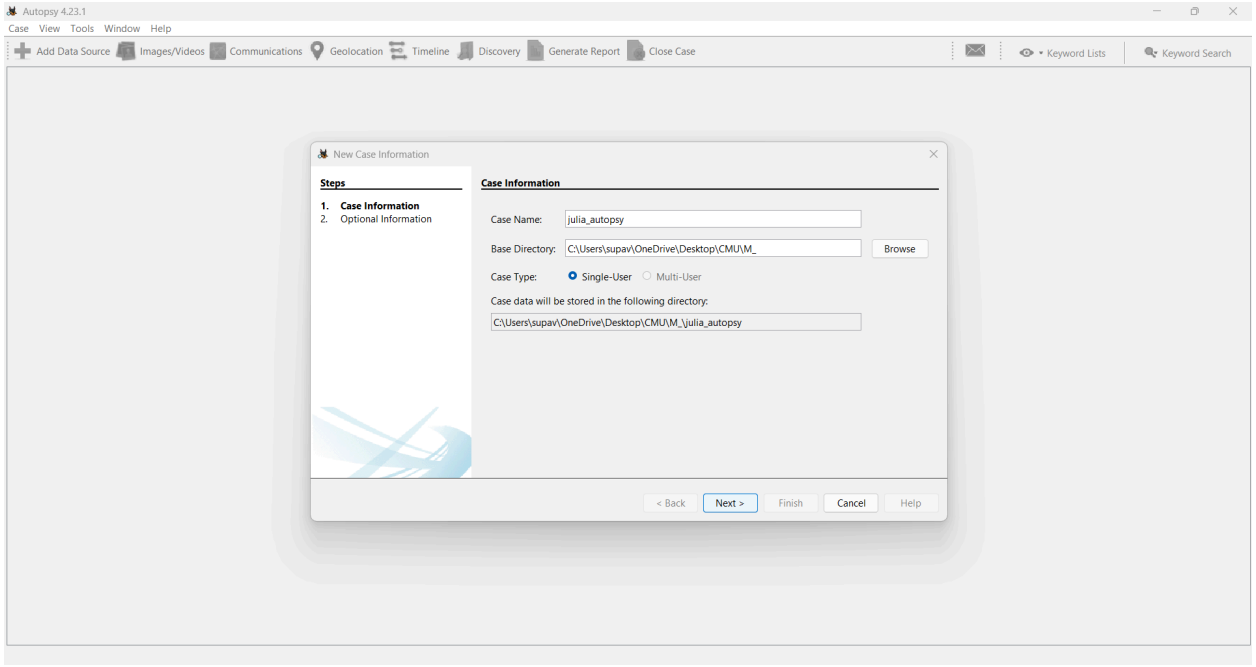


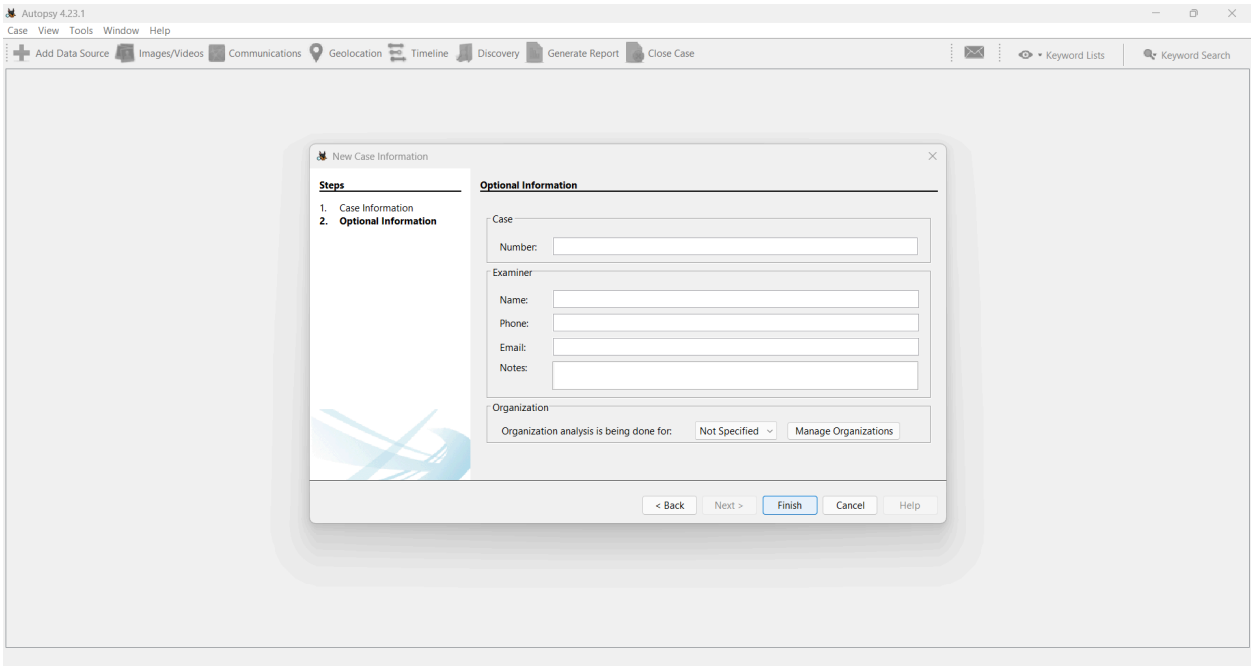

Ubuntu
Ubuntu
SHA1 checksum: 2f403b33b78fe95c5f4fc466a519fd7708ab85fd
Image Information:
Acquisition started: Mon Feb 15 11:07:55 2021
Acquisition finished: Mon Feb 15 11:34:24 2021
Segment list:
M:\julia_pc.001
M:\julia_pc.002
M:\julia_pc.003
M:\julia_pc.004
M:\julia_pc.005
M:\julia_pc.006
M:\julia_pc.007
M:\julia_pc.008
M:\julia_pc.009
M:\julia_pc.010
M:\julia_pc.011
M:\julia_pc.012
M:\julia_pc.013
M:\julia_pc.014
M:\julia_pc.015
M:\julia_pc.016
M:\julia_pc.017
M:\julia_pc.018
M:\julia_pc.019
M:\julia_pc.020
M:\julia_pc.021
M:\julia_pc.022
M:\julia_pc.023
M:\julia_pc.024
M:\julia_pc.025
M:\julia_pc.026
M:\julia_pc.027
M:\julia_pc.028
M:\julia_pc.029
M:\julia_pc.030
M:\julia_pc.031
M:\julia_pc.032
M:\julia_pc.033
M:\julia_pc.034

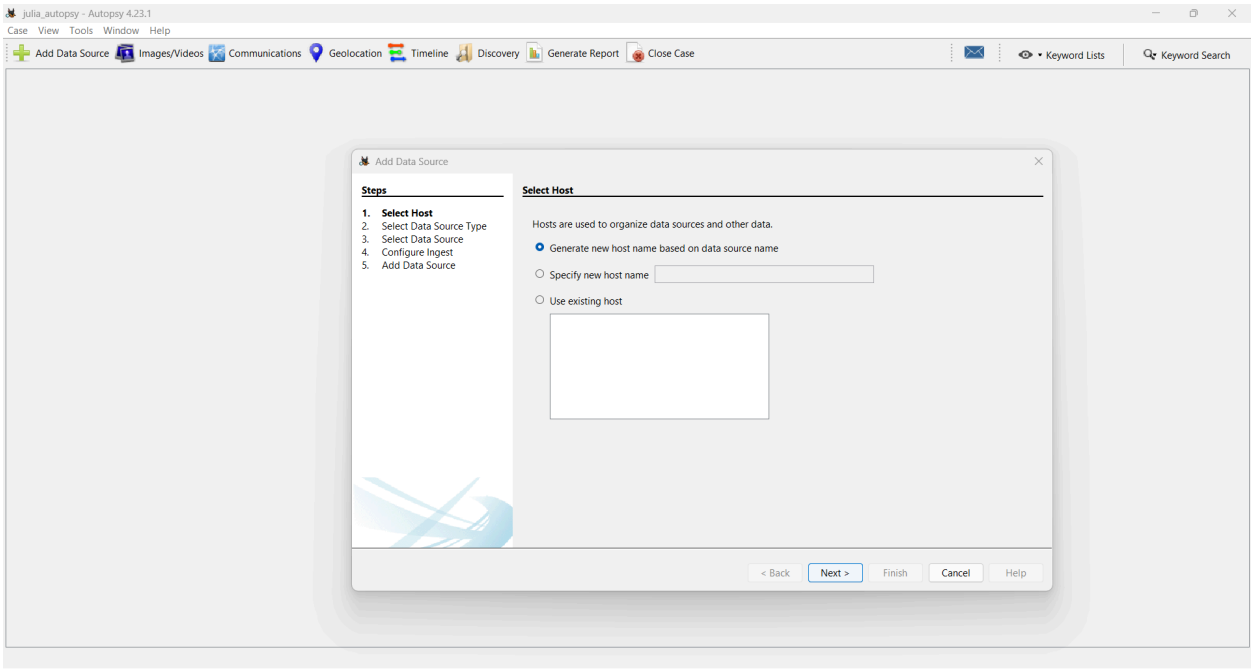

Ubuntu
Ubuntu
M:\julia_pc.013
M:\julia_pc.014
M:\julia_pc.015
M:\julia_pc.016
M:\julia_pc.017
M:\julia_pc.018
M:\julia_pc.019
M:\julia_pc.020
M:\julia_pc.021
M:\julia_pc.022
M:\julia_pc.023
M:\julia_pc.024
M:\julia_pc.025
M:\julia_pc.026
M:\julia_pc.027
M:\julia_pc.028
M:\julia_pc.029
M:\julia_pc.030
M:\julia_pc.031
M:\julia_pc.032
M:\julia_pc.033
M:\julia_pc.034
M:\julia_pc.035
M:\julia_pc.036
M:\julia_pc.037
M:\julia_pc.038
M:\julia_pc.039
M:\julia_pc.040
M:\julia_pc.041
Image Verification Results:
Verification started: Mon Feb 15 11:34:25 2021
Verification finished: Mon Feb 15 12:00:20 2021
MD5 checksum: 5c5c9949114d7f455c148a12c6346856 : verified
SHA1 checksum: 2f403b33b78fe95c5f4fc466a519fd7708ab85fd : verified
wsl@TROY-LAPTOP:/mnt/c/Users/supav/OneDrive/Desktop/CMU/M_$ md5sum julia_pc
5c5c9949114d7f455c148a12c6346856 julia_pc
wsl@TROY-LAPTOP:/mnt/c/Users/supav/OneDrive/Desktop/CMU/M_$ sha1sum julia_pc
2f403b33b78fe95c5f4fc466a519fd7708ab85fd julia_pc
wsl@TROY-LAPTOP:/mnt/c/Users/supav/OneDrive/Desktop/CMU/M_$

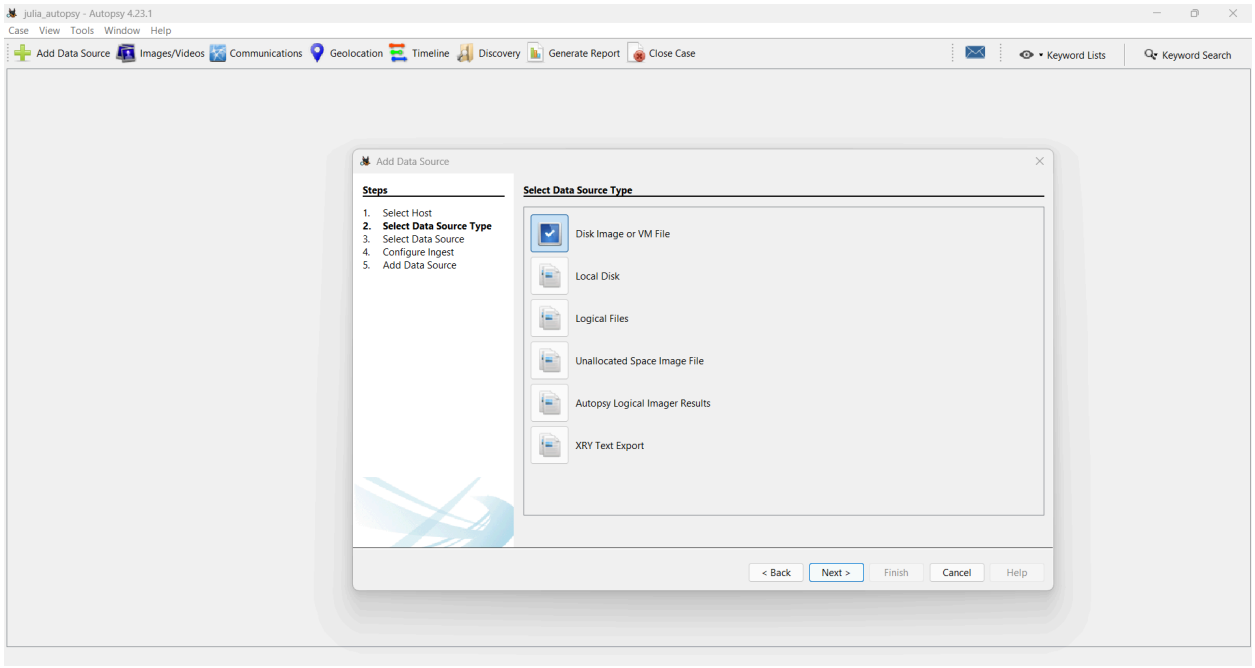
5.2.6.4 Julia_A_0004 Autopsy Case

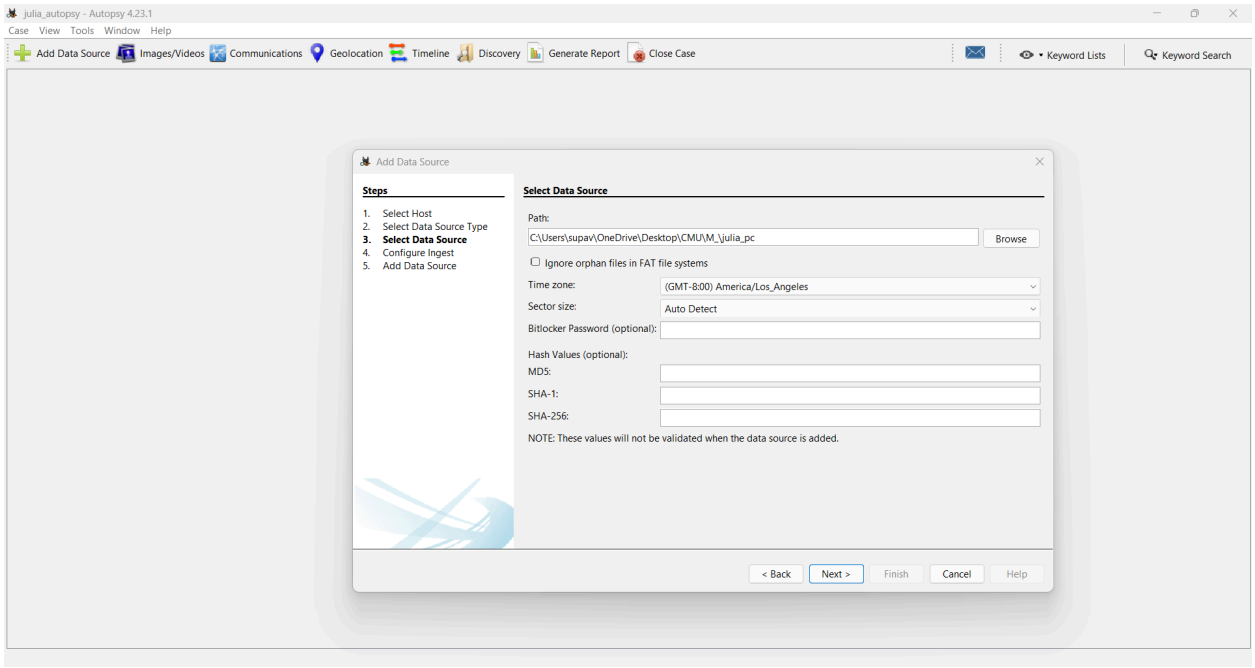


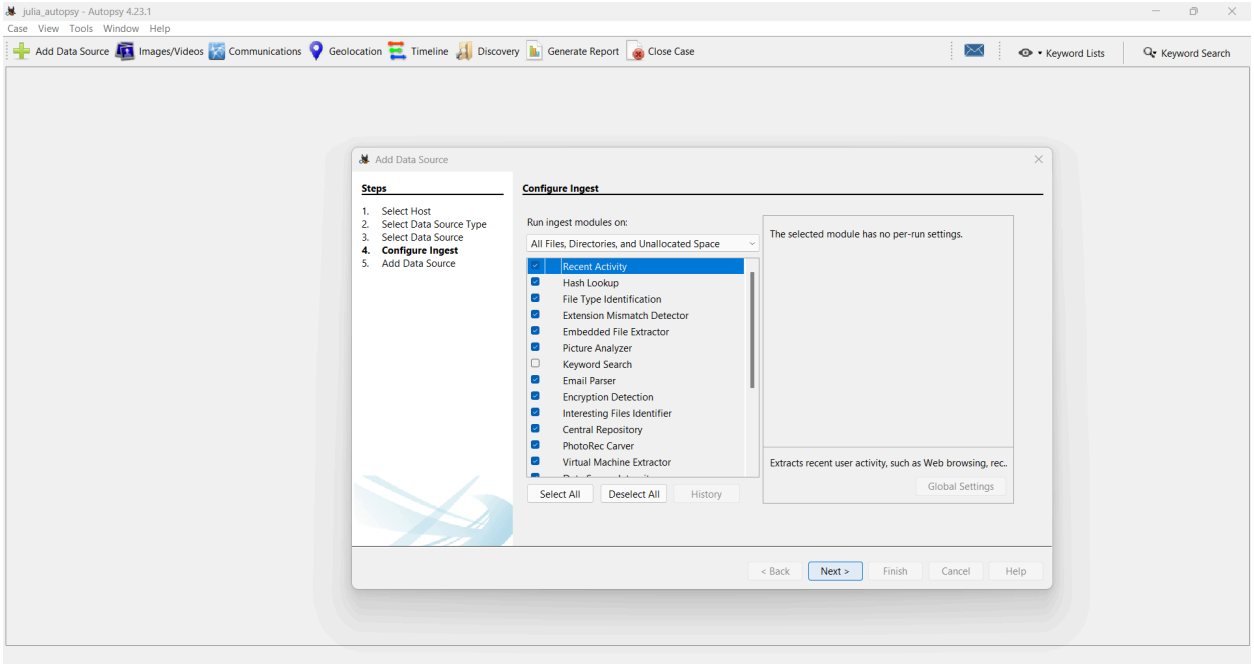


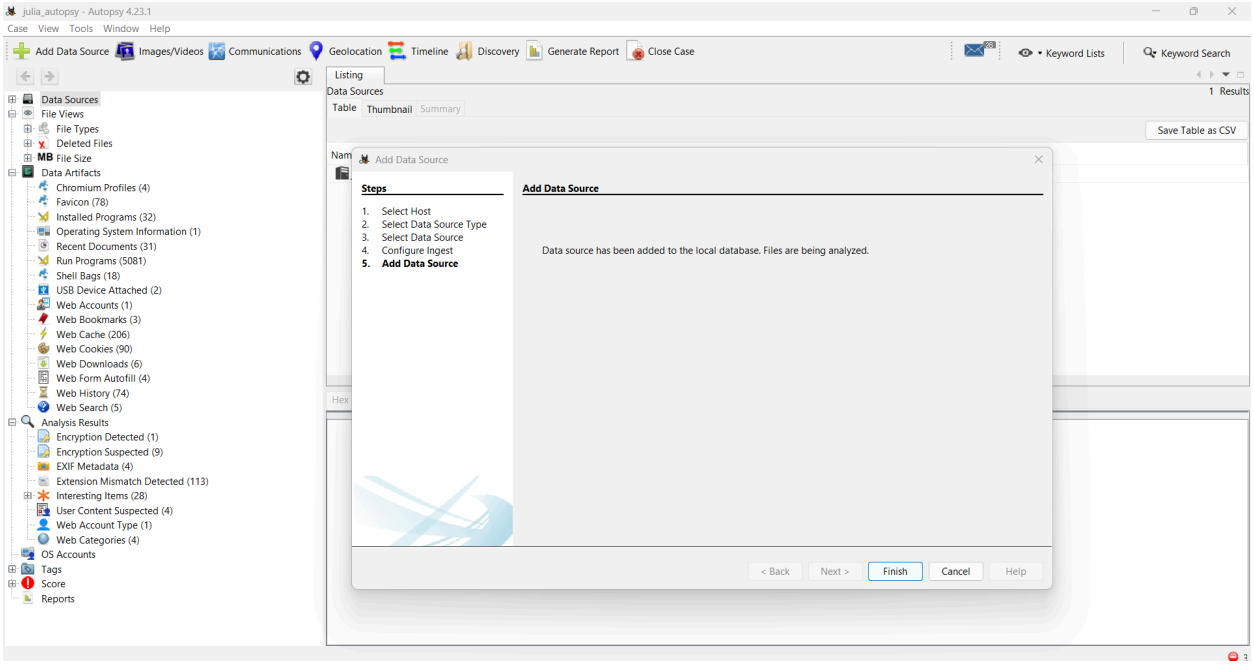

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Optional Information
Case
Number:
Examiner
Name:
Phone:
Email:
Notes:
Organization
Organization analysis is being done for:
Not Specified
Manage Organizations
< Back Next > Finish Cancel Help

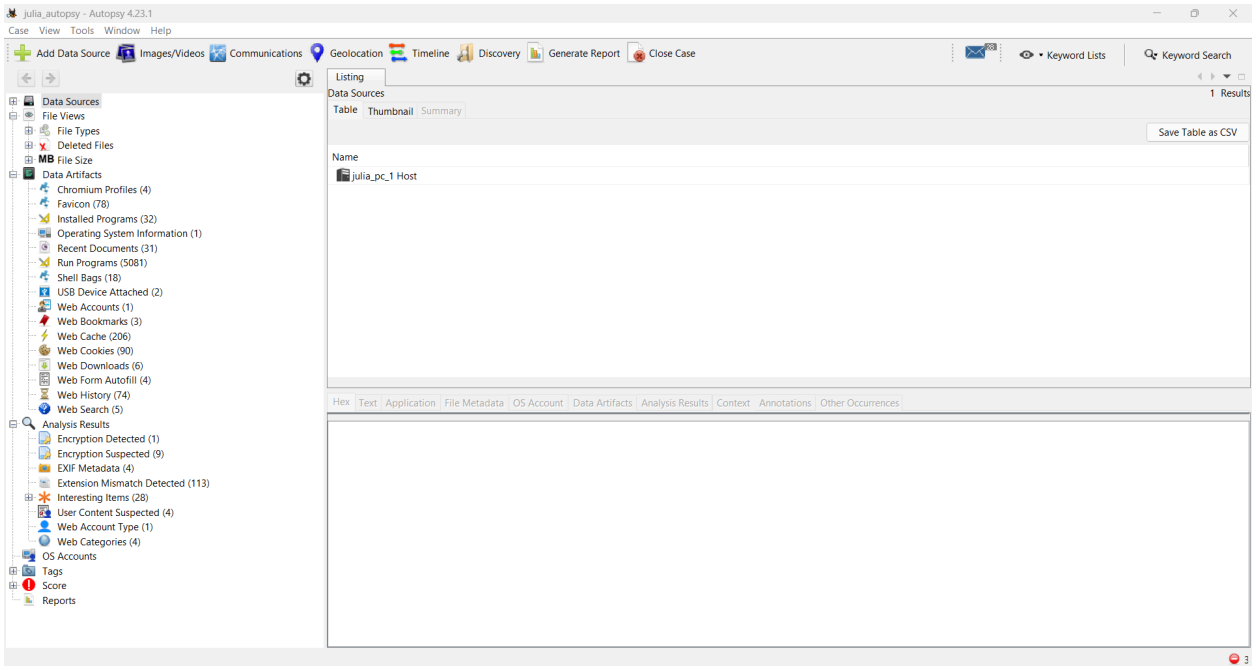

julia_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Host
Hosts are used to organize data sources and other data.
Generate new host name based on data source name
Specify new host name
Use existing host
< Back Next > Finish Cancel Help


julia_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source Type
Disk Image or VM File
Local Disk
Logical Files
Unallocated Space Image File
Autopsy Logical Imager Results
XRY Text Export
< Back
Next >
Finish
Cancel
Help

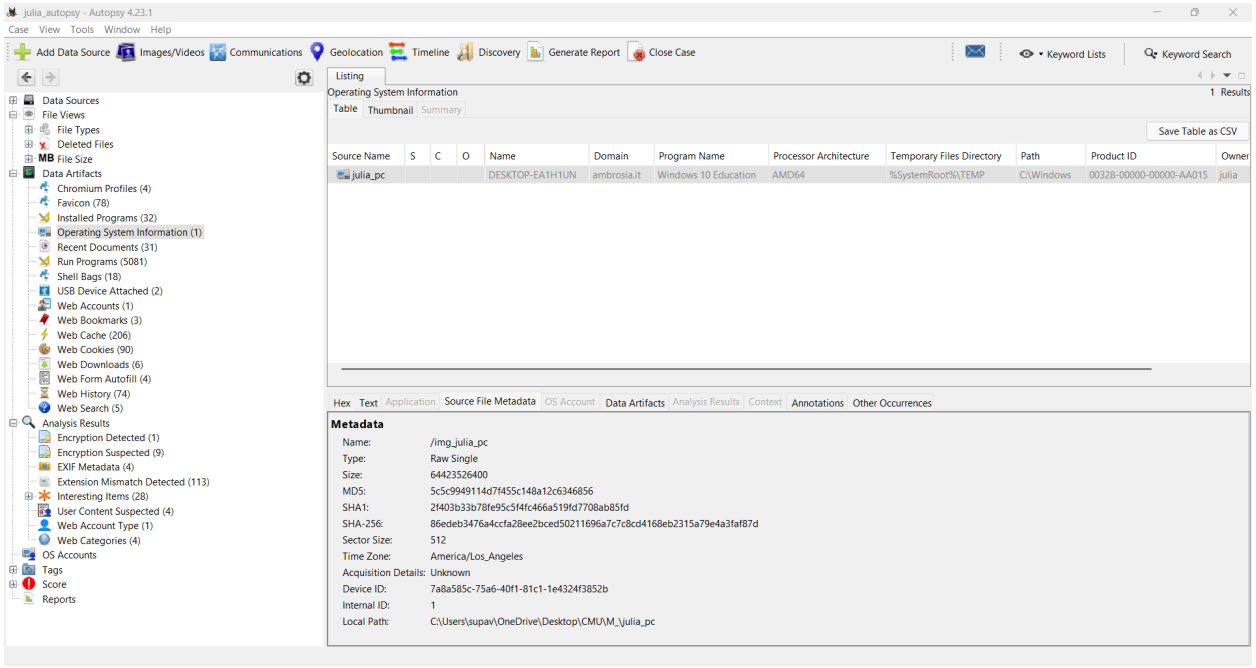

julia_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source
Path:
C:\Users\supav\OneDrive\Desktop\CMU\M_julia_pc
Browse
Ignore orphan files in FAT file systems
Time zone:
(GMT-8:00) America/Los_Angeles
Sector size:
Auto Detect
Bitlocker Password (optional):
Hash Values (optional):
MD5:
SHA-1:
SHA-256:
NOTE: These values will not be validated when the data source is added.
< Back
Next >
Finish
Cancel
Help


julia_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
The selected module has no per-run settings.
Extracts recent user activity, such as Web browsing, rec..
Global Settings
Select All
Deselect All
History
< Back
Next >
Finish
Cancel
Help

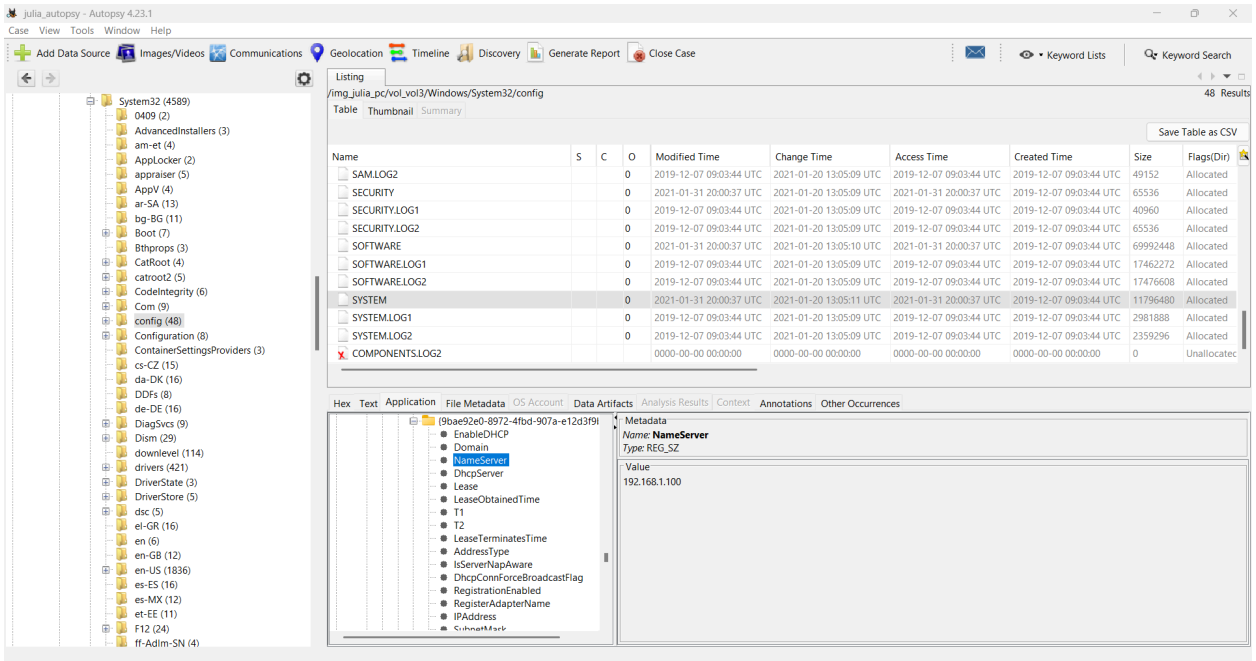

julia_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Listing
Data Sources
1 Results
Table
Thumbnail
Summary
Save Table as CSV
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Chromium Profiles (4)
Favicon (78)
Installed Programs (32)
Operating System Information (1)
Recent Documents (31)
Run Programs (5081)
Shell Bags (18)
USB Device Attached (2)
Web Accounts (1)
Web Bookmarks (3)
Web Cache (206)
Web Cookies (90)
Web Downloads (6)
Web Form Autofill (4)
Web History (74)
Web Search (5)
Analysis Results
Encryption Detected (1)
Encryption Suspected (9)
EXIF Metadata (4)
Extension Mismatch Detected (113)
Interesting Items (28)
User Content Suspected (4)
Web Account Type (1)
Web Categories (4)
OS Accounts
Tags
Score
Reports
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
Data source has been added to the local database. Files are being analyzed.
< Back
Next >
Finish
Cancel
Help

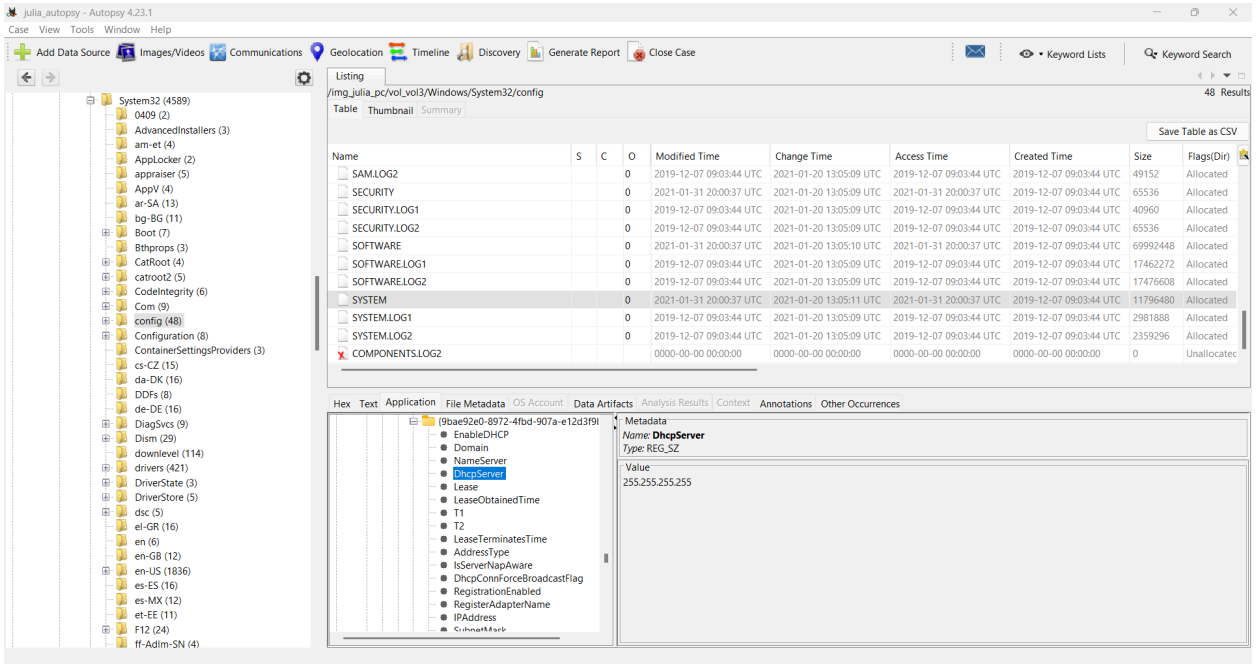


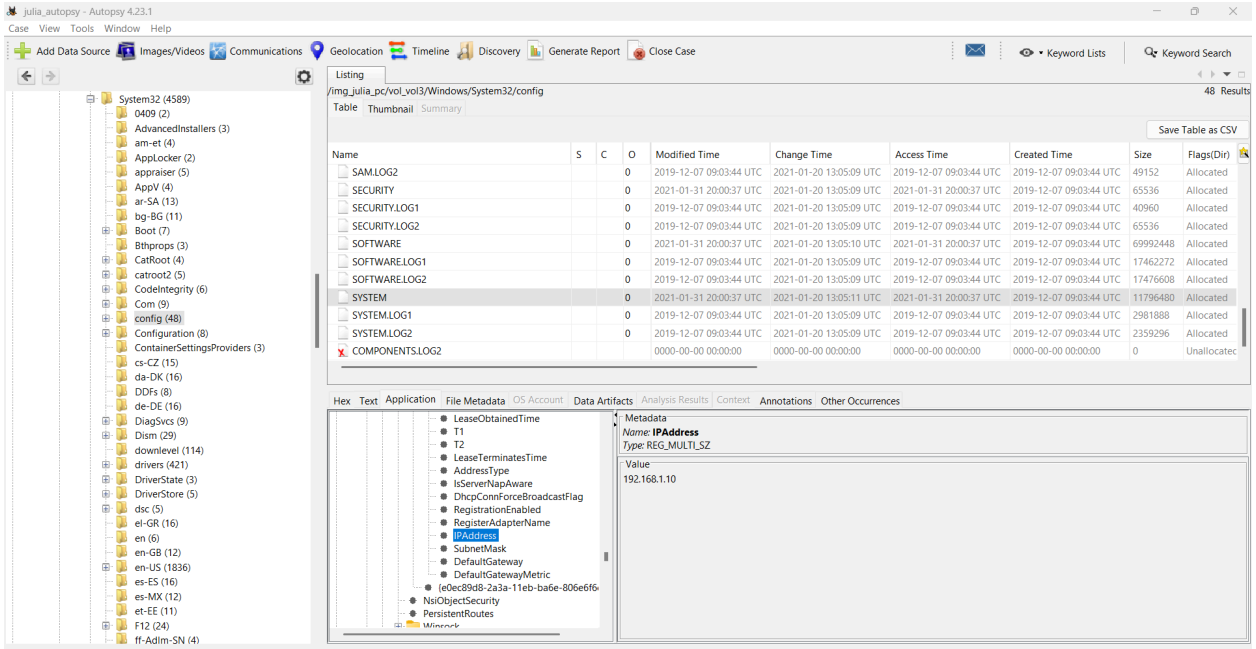
5.2.6.5 Julia_A_0005 Operating System

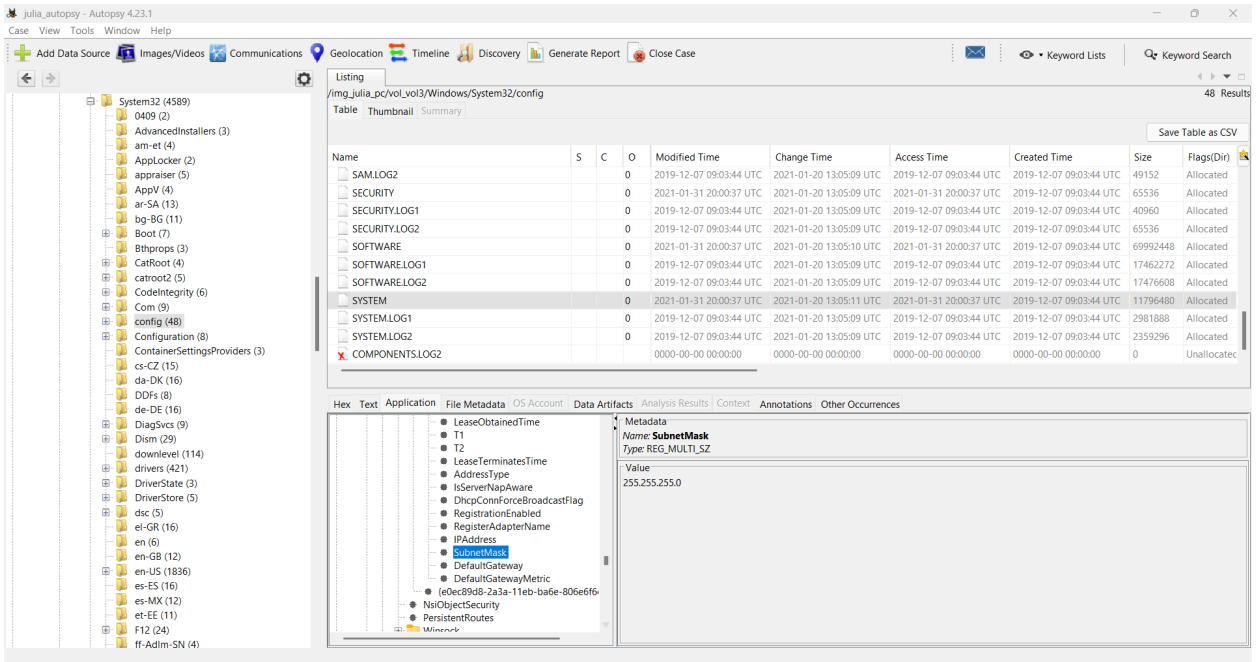


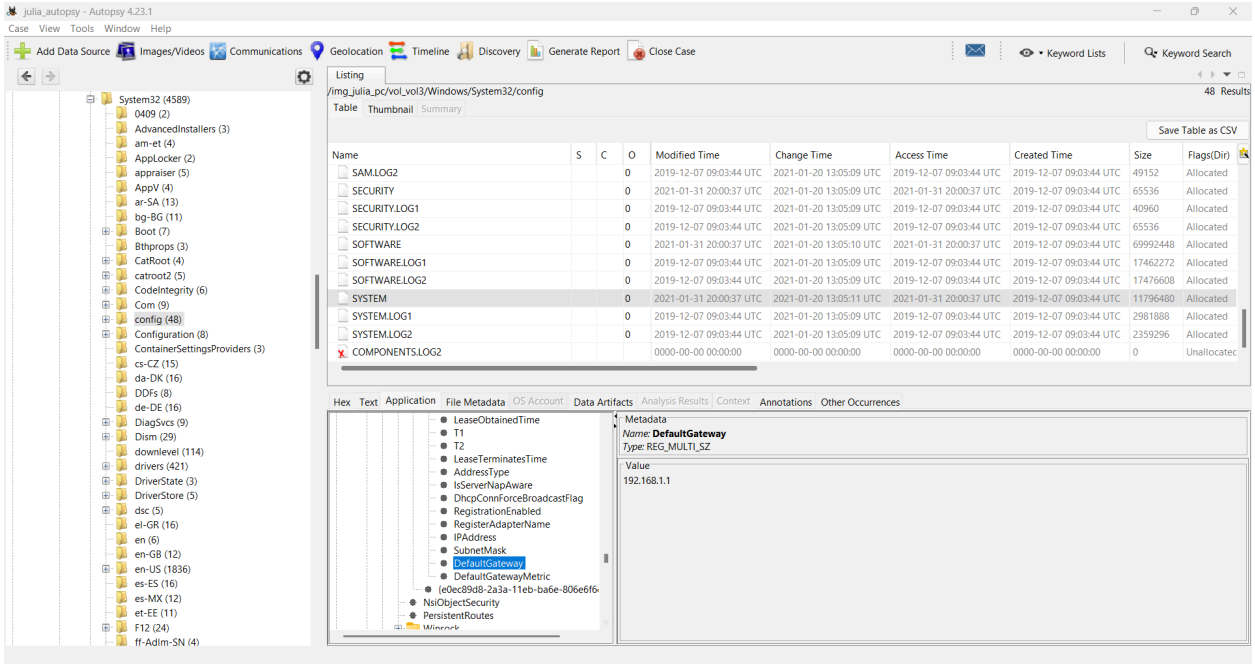
5.2.6.6 Julia_A_0006 IP Address



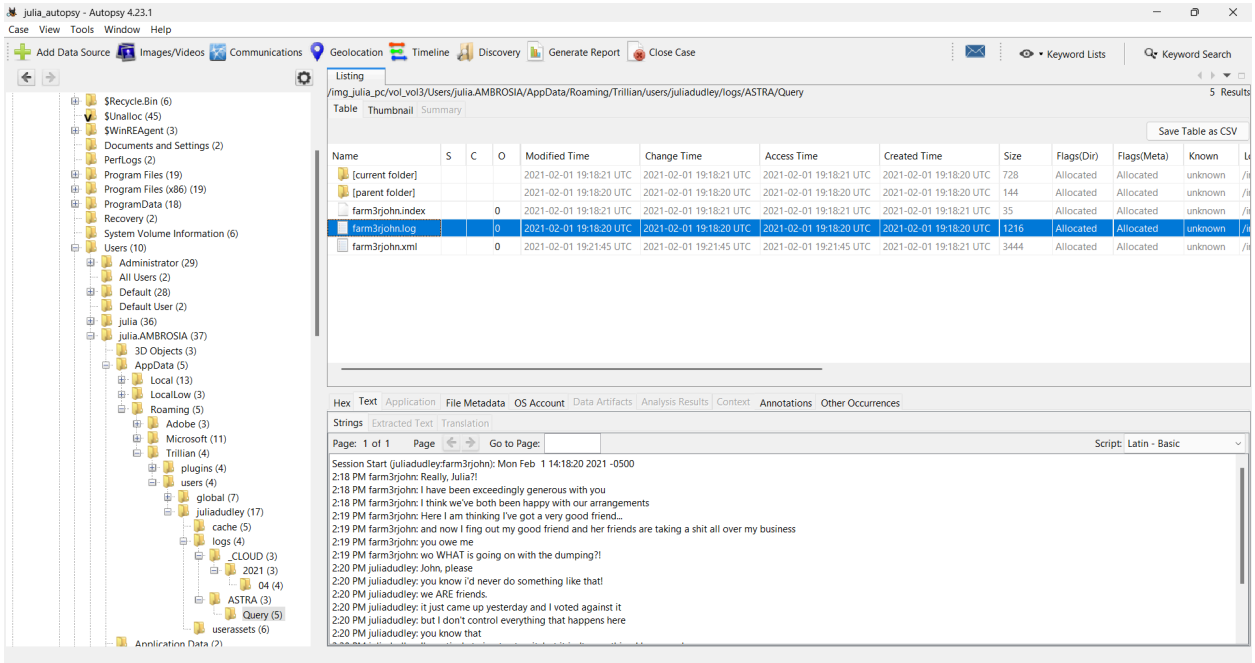



julia_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
Listing
/img_julia_pc/vol_vol3/Windows/System32/config
48 Results
Table Thumbnail Summary
Save Table as CSV
Name S C O Modified Time Change Time Access Time Created Time Size Flags(Dir)
SAM.LOG2 0 2019-12-07 09:03:44 UTC 2021-01-20 13:05:09 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 49152 Allocated
SECURITY 0 2021-01-31 20:00:37 UTC 2021-01-20 13:05:09 UTC 2021-01-31 20:00:37 UTC 2019-12-07 09:03:44 UTC 65536 Allocated
SECURITY.LOG1 0 2019-12-07 09:03:44 UTC 2021-01-20 13:05:09 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 40960 Allocated
SECURITY.LOG2 0 2019-12-07 09:03:44 UTC 2021-01-20 13:05:09 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 65536 Allocated
SOFTWARE 0 2021-01-31 20:00:37 UTC 2021-01-20 13:05:10 UTC 2021-01-31 20:00:37 UTC 2019-12-07 09:03:44 UTC 69992448 Allocated
SOFTWARE.LOG1 0 2019-12-07 09:03:44 UTC 2021-01-20 13:05:09 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 17462272 Allocated
SOFTWARE.LOG2 0 2019-12-07 09:03:44 UTC 2021-01-20 13:05:09 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 17476608 Allocated
SYSTEM 0 2021-01-31 20:00:37 UTC 2021-01-20 13:05:11 UTC 2021-01-31 20:00:37 UTC 2019-12-07 09:03:44 UTC 11796480 Allocated
SYSTEM.LOG1 0 2019-12-07 09:03:44 UTC 2021-01-20 13:05:09 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 2981888 Allocated
SYSTEM.LOG2 0 2019-12-07 09:03:44 UTC 2021-01-20 13:05:09 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 2359296 Allocated
COMPONENTS.LOG2 0000-00-00 00:00:00 0000-00-00 00:00:00 0000-00-00 00:00:00 0000-00-00 00:00:00 0 Unallocatec
Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
SubnetMask
DefaultGateway
DefaultGatewayMetric
{e0ec89d8-2a3a-11eb-ba6e-806e6f6
NsiObjectSecurity
PersistentRoutes
Winsock
Metadata
Name: IPAddress
Type: REG_MULTI_SZ
Value
192.168.1.10
System32 (4589)
0409 (2)
AdvancedInstallers (3)
am-et (4)
AppLocker (2)
appraiser (5)
AppV (4)
ar-SA (13)
bg-BG (11)
Boot (7)
Bthprops (3)
CatRoot (4)
catroot2 (5)
CodeIntegrity (6)
Com (9)
config (48)
Configuration (8)
ContainerSettingsProviders (3)
cs-CZ (15)
da-DK (16)
DDFs (8)
de-DE (16)
DiagSvcs (9)
Dism (29)
downlevel (114)
drivers (421)
DriverState (3)
DriverStore (5)
dsc (5)
el-GR (16)
en (6)
en-GB (12)
en-US (1836)
es-ES (16)
es-MX (12)
et-EE (11)
F12 (24)
ff-Adlm-SN (4)

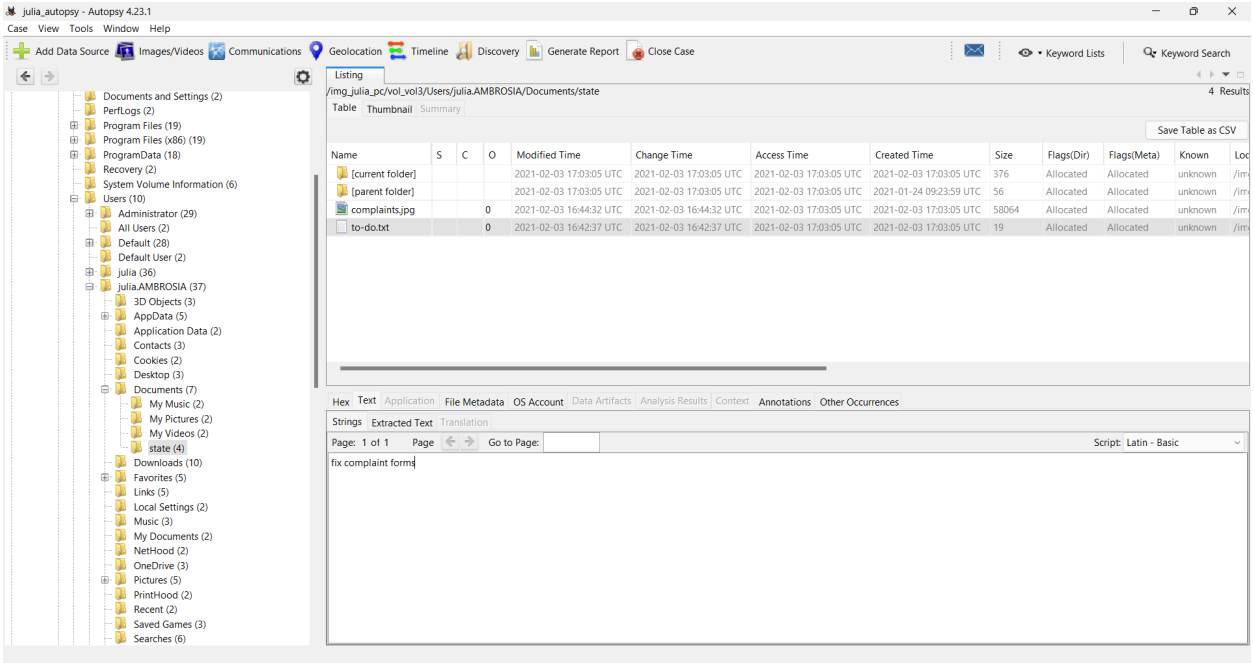

julia_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
Listing
/img_julia_pc/vol_vol3/Windows/System32/config
48 Results
Table Thumbnail Summary
Save Table as CSV
Name S C O Modified Time Change Time Access Time Created Time Size Flags(Dir)
SAM.LOG2 0 2019-12-07 09:03:44 UTC 2021-01-20 13:05:09 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 49152 Allocated
SECURITY 0 2021-01-31 20:00:37 UTC 2021-01-20 13:05:09 UTC 2021-01-31 20:00:37 UTC 2019-12-07 09:03:44 UTC 65536 Allocated
SECURITY.LOG1 0 2019-12-07 09:03:44 UTC 2021-01-20 13:05:09 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 40960 Allocated
SECURITY.LOG2 0 2019-12-07 09:03:44 UTC 2021-01-20 13:05:09 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 65536 Allocated
SOFTWARE 0 2021-01-31 20:00:37 UTC 2021-01-20 13:05:10 UTC 2021-01-31 20:00:37 UTC 2019-12-07 09:03:44 UTC 69992448 Allocated
SOFTWARE.LOG1 0 2019-12-07 09:03:44 UTC 2021-01-20 13:05:09 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 17462272 Allocated
SOFTWARE.LOG2 0 2019-12-07 09:03:44 UTC 2021-01-20 13:05:09 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 17476608 Allocated
SYSTEM 0 2021-01-31 20:00:37 UTC 2021-01-20 13:05:11 UTC 2021-01-31 20:00:37 UTC 2019-12-07 09:03:44 UTC 11796480 Allocated
SYSTEM.LOG1 0 2019-12-07 09:03:44 UTC 2021-01-20 13:05:09 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 2981888 Allocated
SYSTEM.LOG2 0 2019-12-07 09:03:44 UTC 2021-01-20 13:05:09 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 2359296 Allocated
COMPONENTS.LOG2 0000-00-00 00:00:00 0000-00-00 00:00:00 0000-00-00 00:00:00 0000-00-00 00:00:00 0 Unallocatec
Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
SubnetMask
DefaultGateway
DefaultGatewayMetric
{e0ec89d8-2a3a-11eb-ba6e-806e6f6
NsiObjectSecurity
PersistentRoutes
Winsock
Metadata
Name: SubnetMask
Type: REG_MULTI_SZ
Value
255.255.255.0
System32 (4589)
0409 (2)
AdvancedInstallers (3)
am-et (4)
AppLocker (2)
appraiser (5)
AppV (4)
ar-SA (13)
bg-BG (11)
Boot (7)
Bthprops (3)
CatRoot (4)
catroot2 (5)
CodeIntegrity (6)
Com (9)
config (48)
Configuration (8)
ContainerSettingsProviders (3)
cs-CZ (15)
da-DK (16)
DDFs (8)
de-DE (16)
DiagSvcs (9)
Dism (29)
downlevel (114)
drivers (421)
DriverState (3)
DriverStore (5)
dsc (5)
el-GR (16)
en (6)
en-GB (12)
en-US (1836)
es-ES (16)
es-MX (12)
et-EE (11)
F12 (24)
ff-Adlm-SN (4)



5.2.6.7 Julia_A_0007 Trillian Log

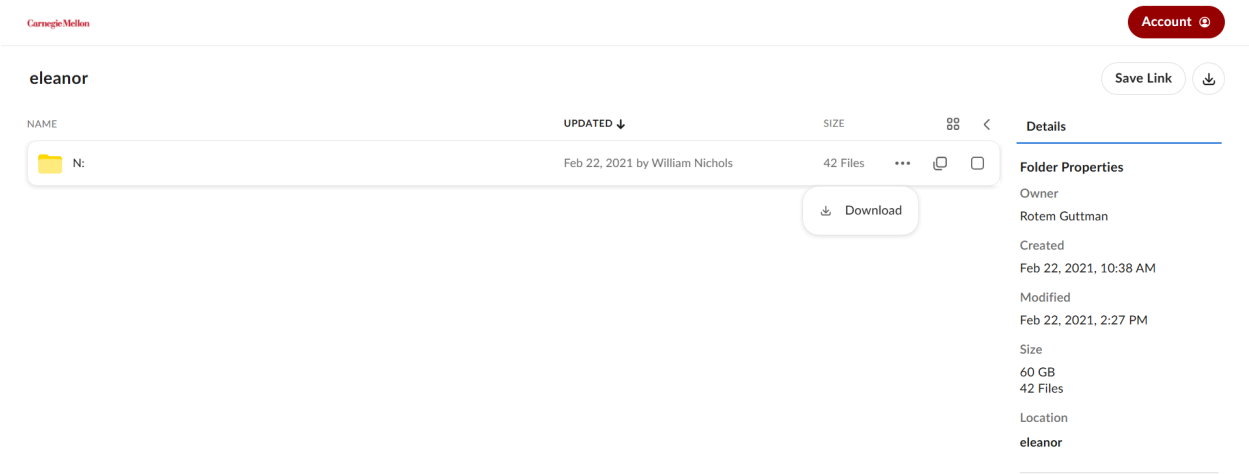


5.2.6.8 Julia_A_0008 To-do List

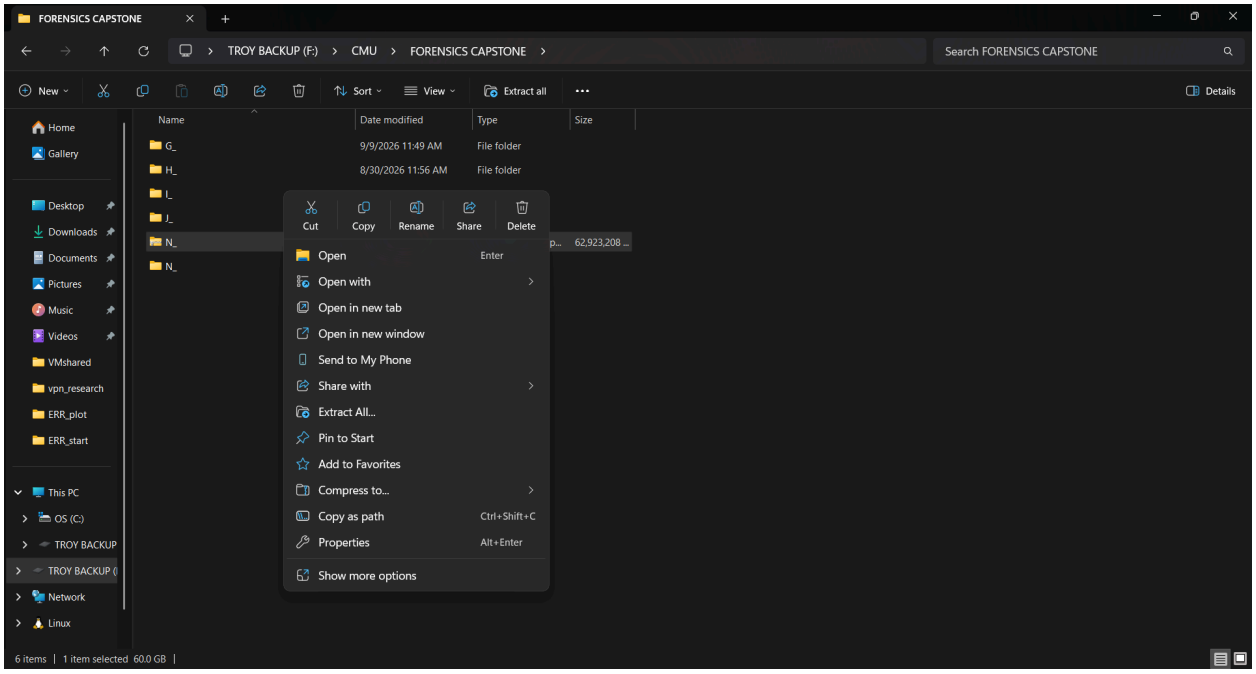


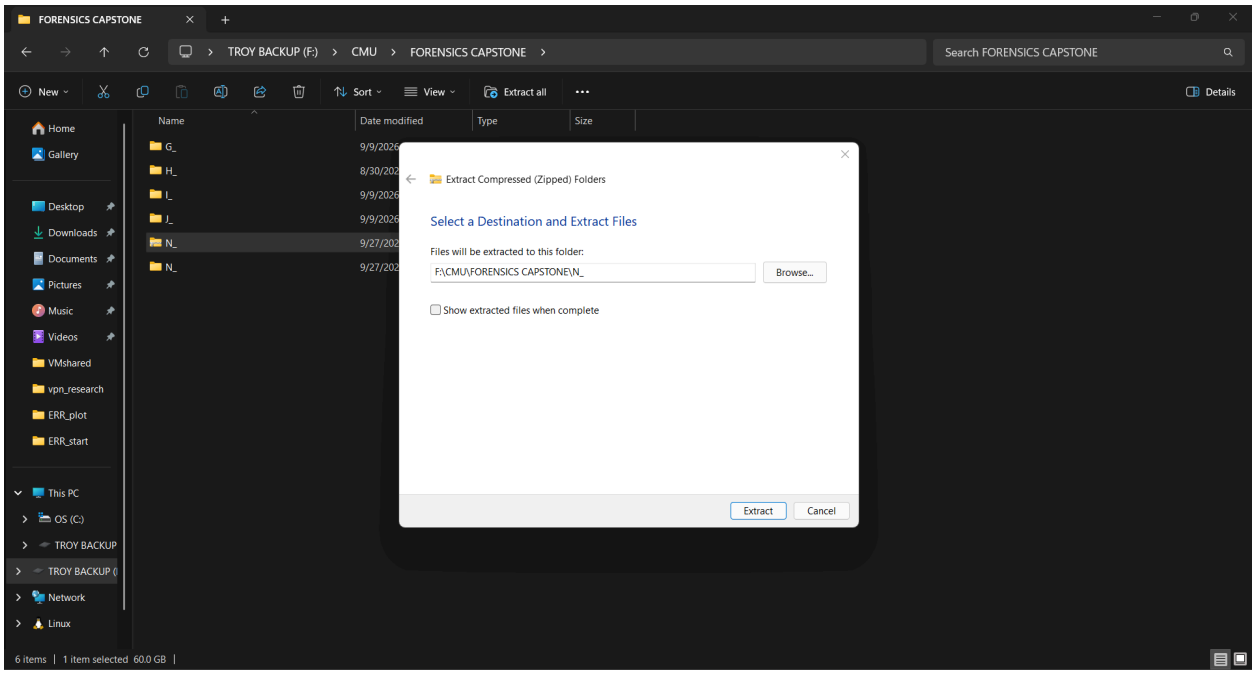
5.2.7 Eleanor Image (N)

5.2.7.1 Eleanor_A_0001 Image Download

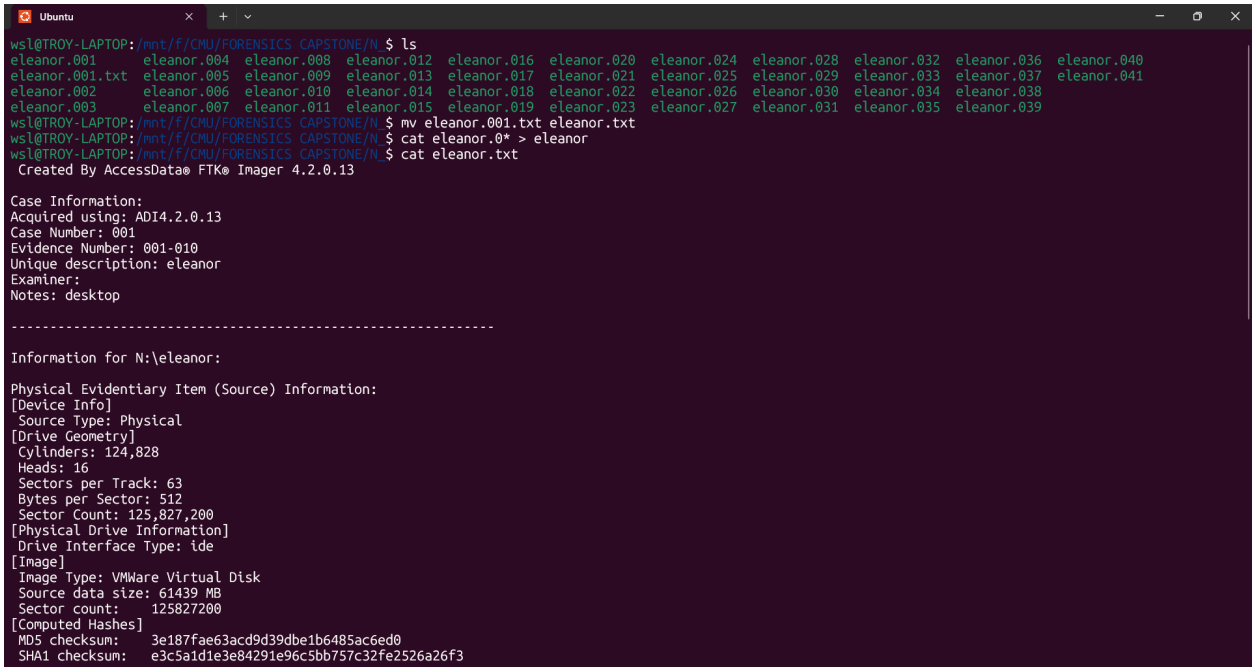


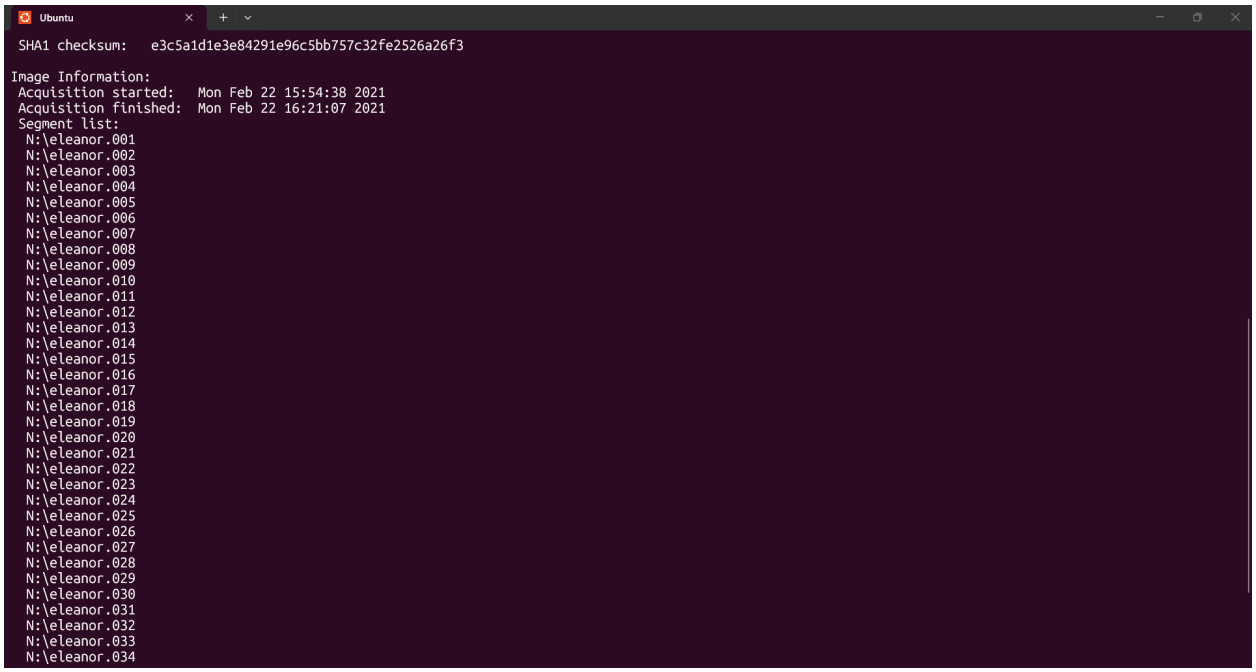
5.2.7.2 Eleanor_A_0002 Image Extract

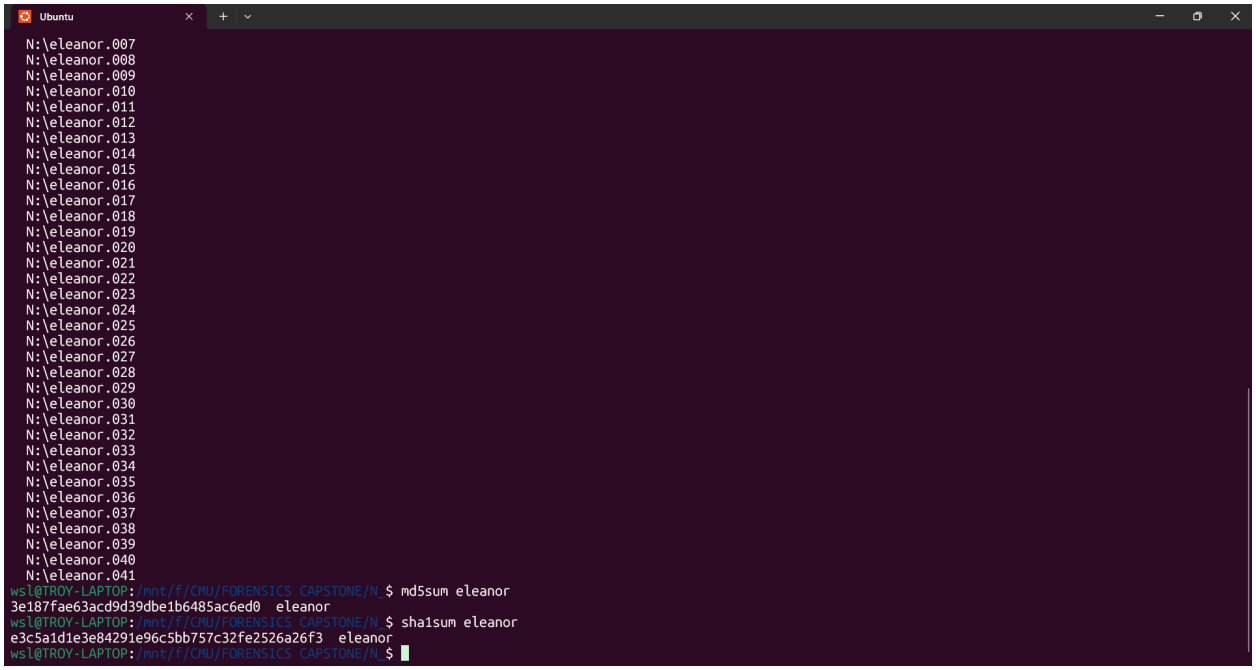




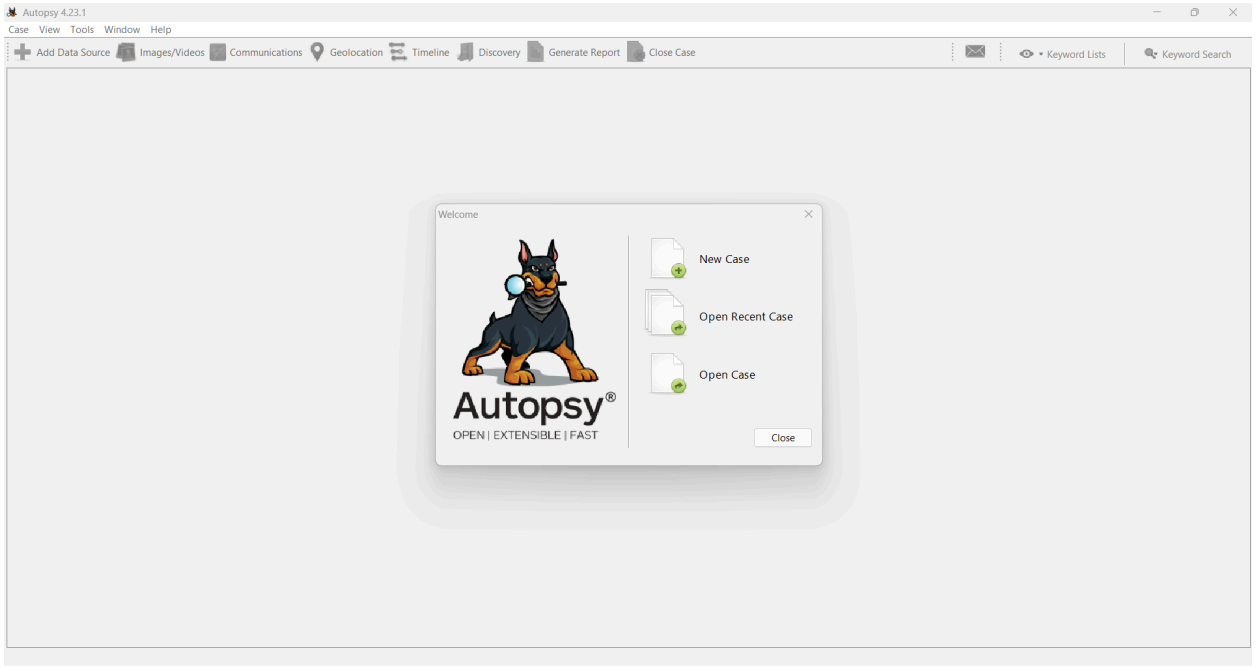
5.2.7.3 Eleanor_A_0003 Image Verification

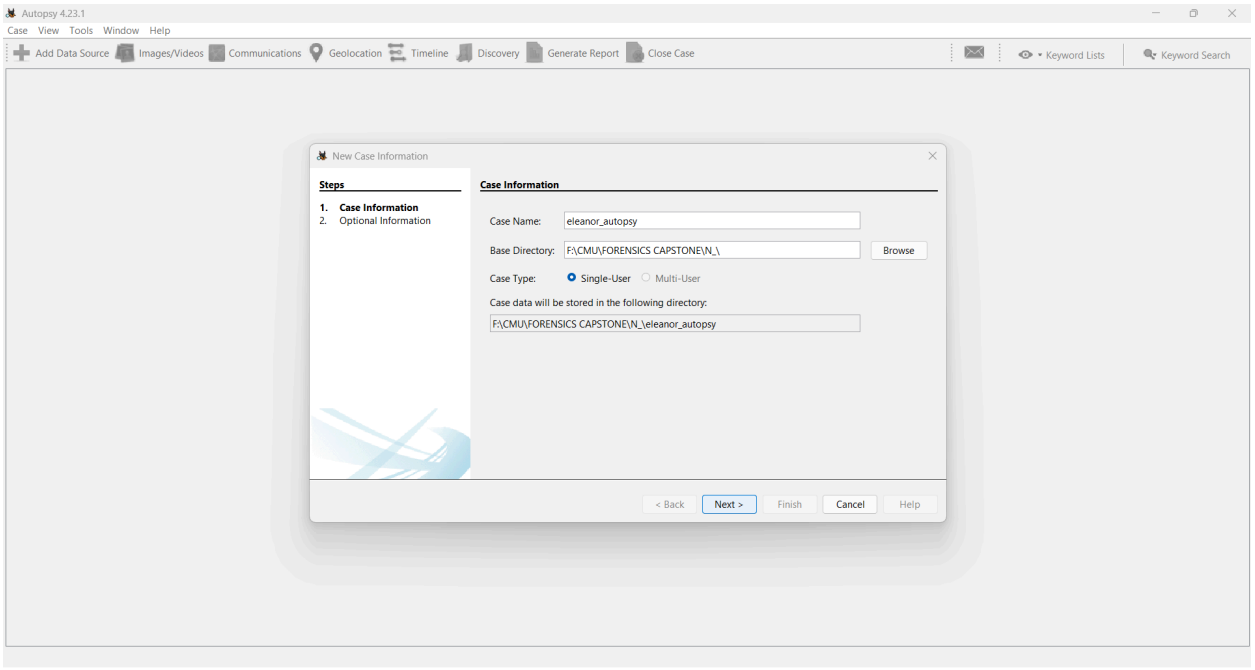


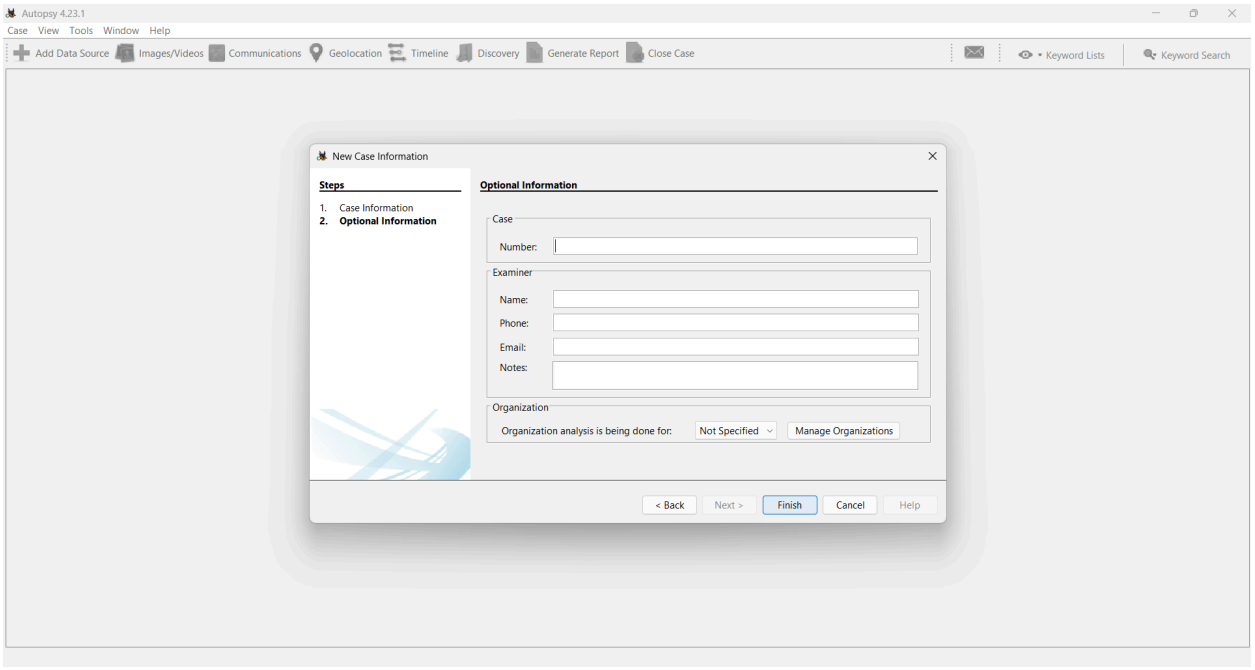


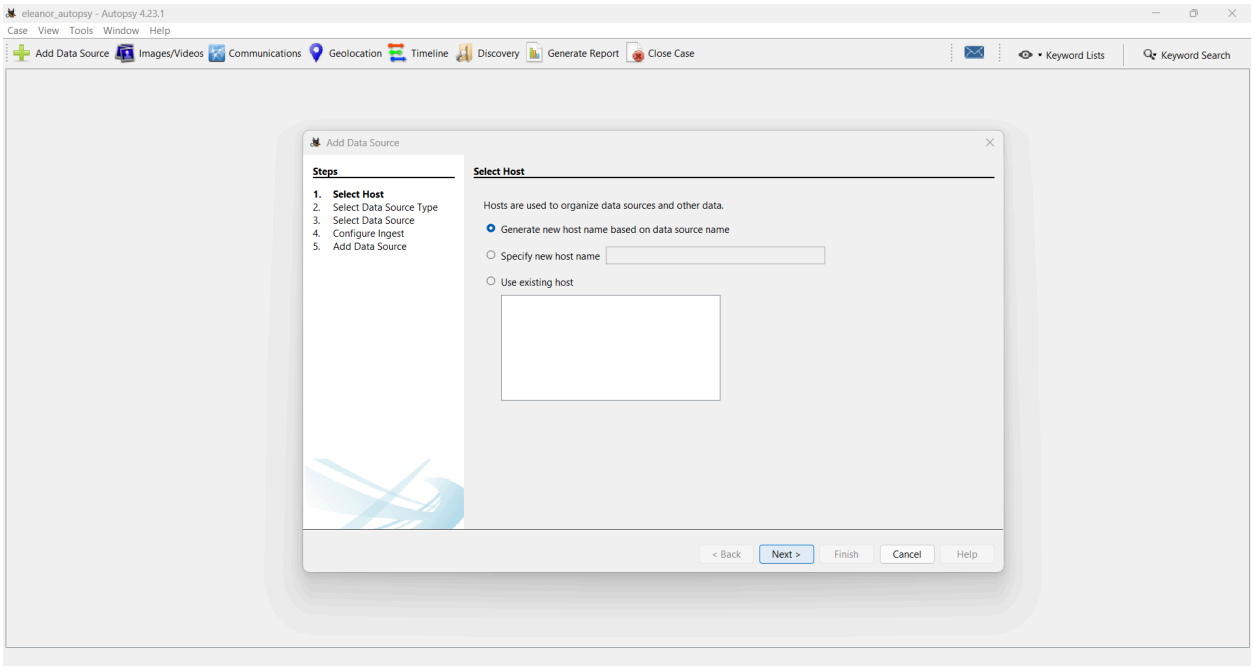


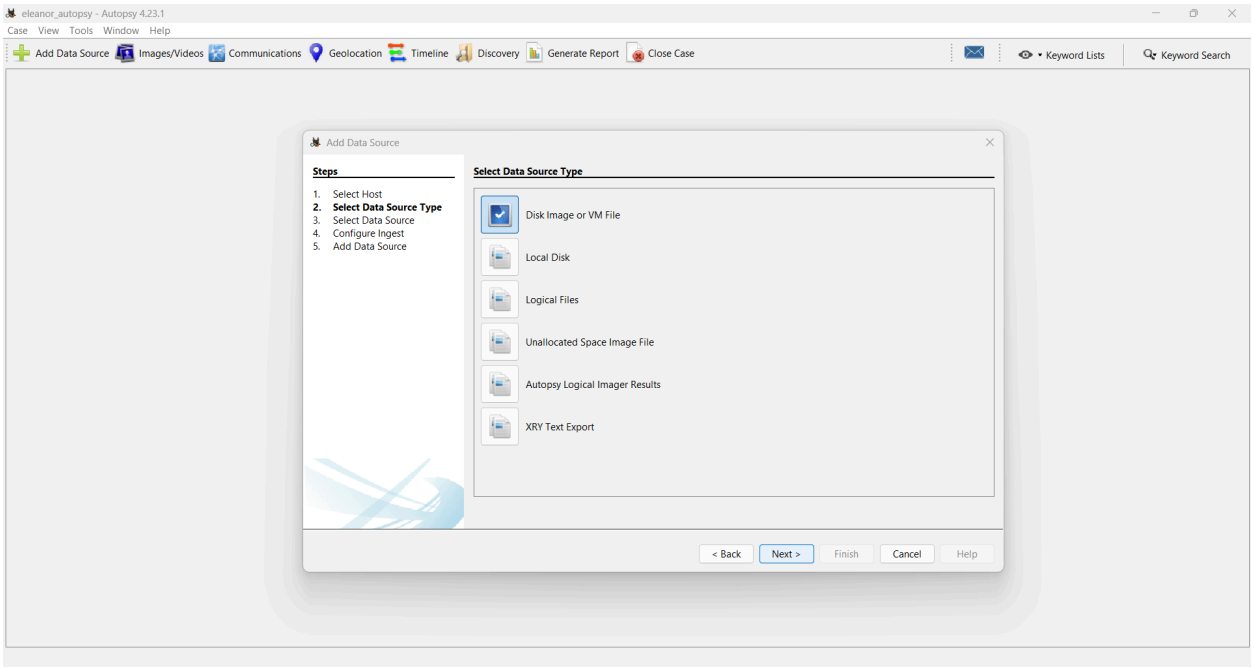
5.2.7.4 Eleanor_A_0004 Autopsy Case

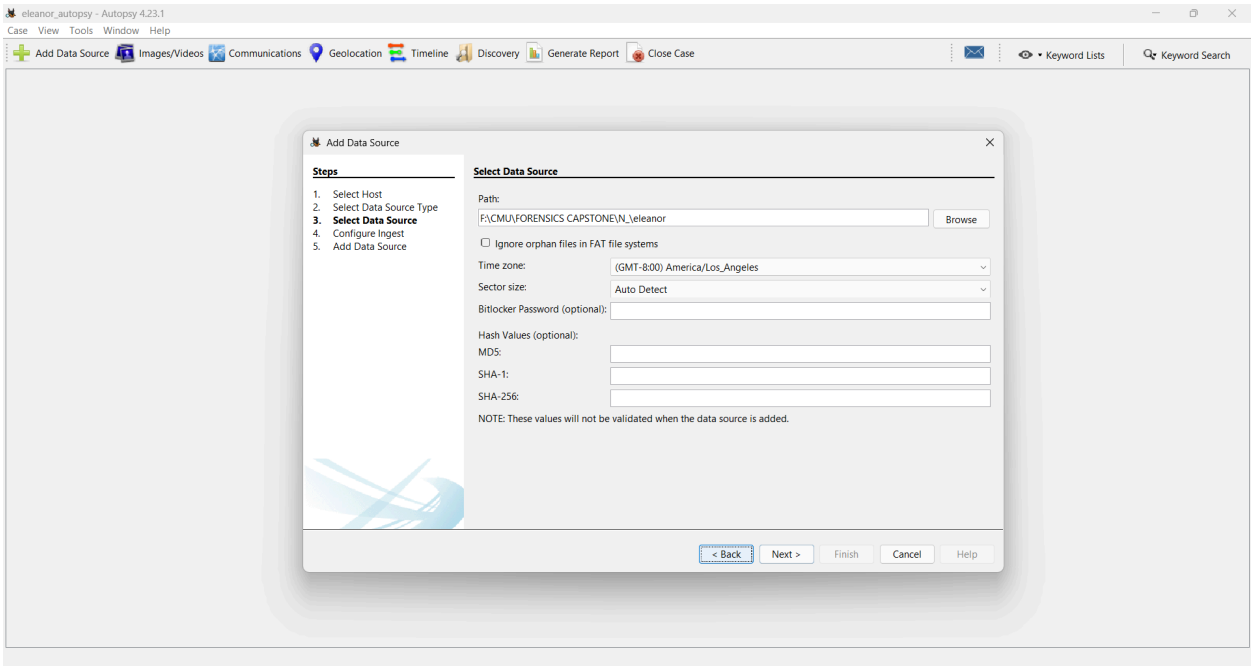


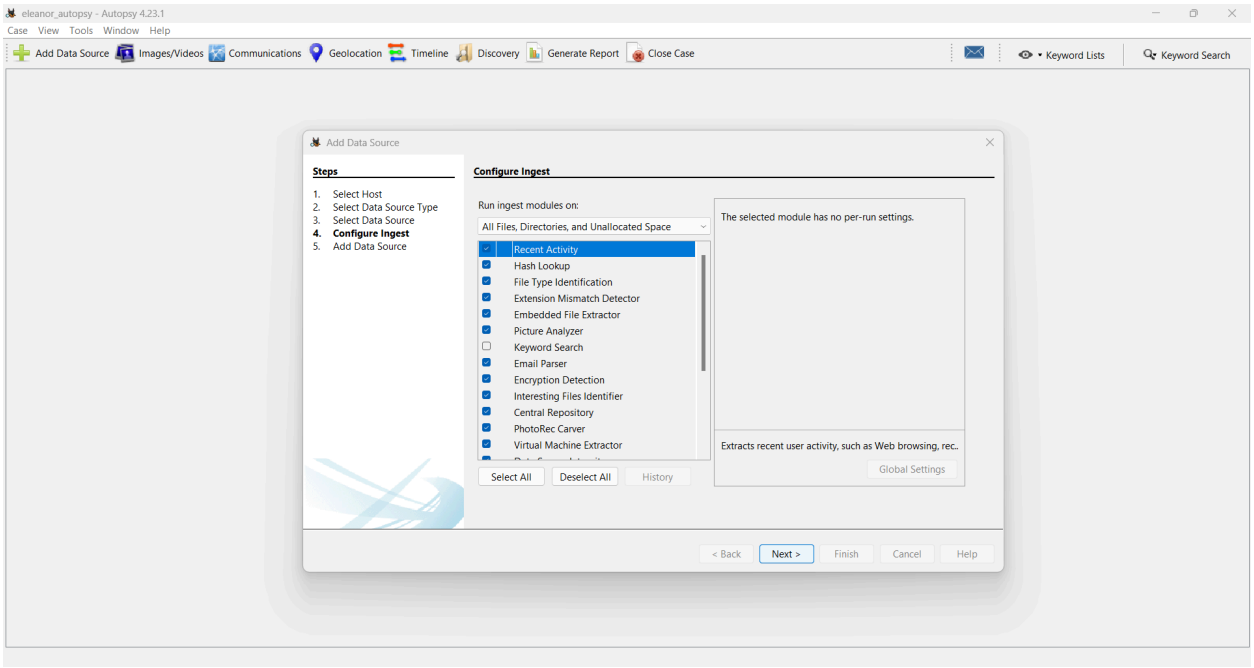

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Case Information
Case Name: eleanor_autopsy
Base Directory: F:\CMU\FORENSICS CAPSTONE\N_\
Browse
Case Type: Single-User Multi-User
Case data will be stored in the following directory:
F:\CMU\FORENSICS CAPSTONE\N_\eleanor_autopsy
< Back Next > Finish Cancel Help

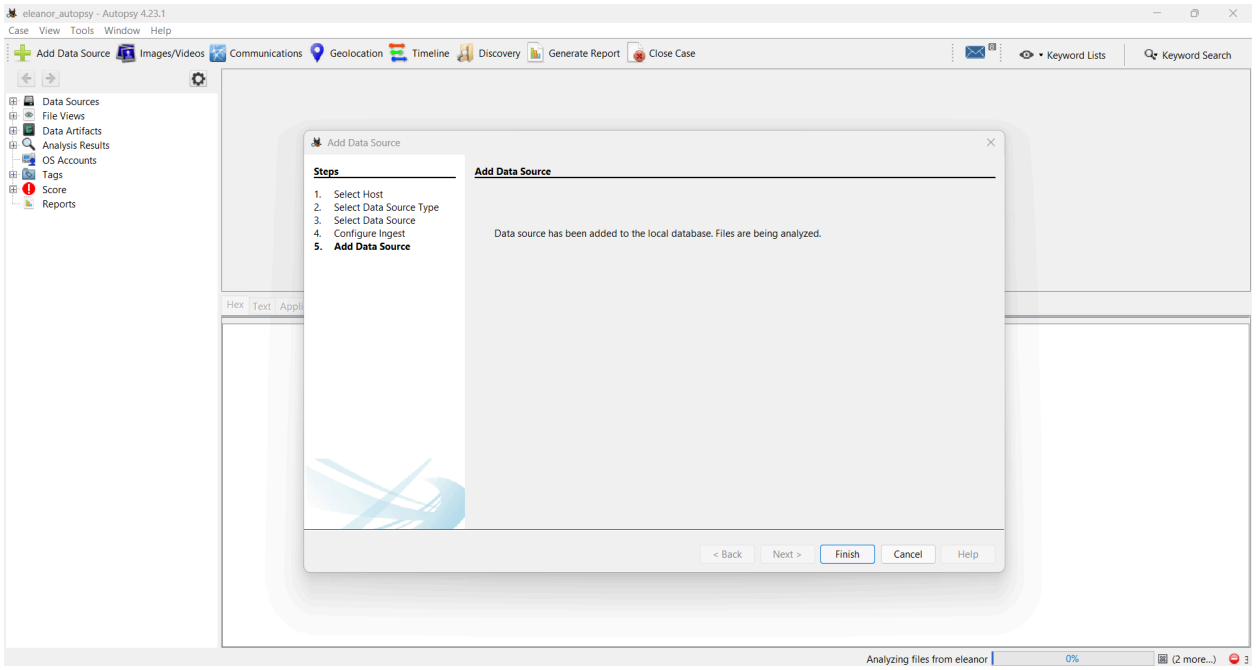

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Optional Information
Case
Number:
Examiner
Name:
Phone:
Email:
Notes:
Organization
Organization analysis is being done for: Not Specified
Manage Organizations
< Back Next > Finish Cancel Help

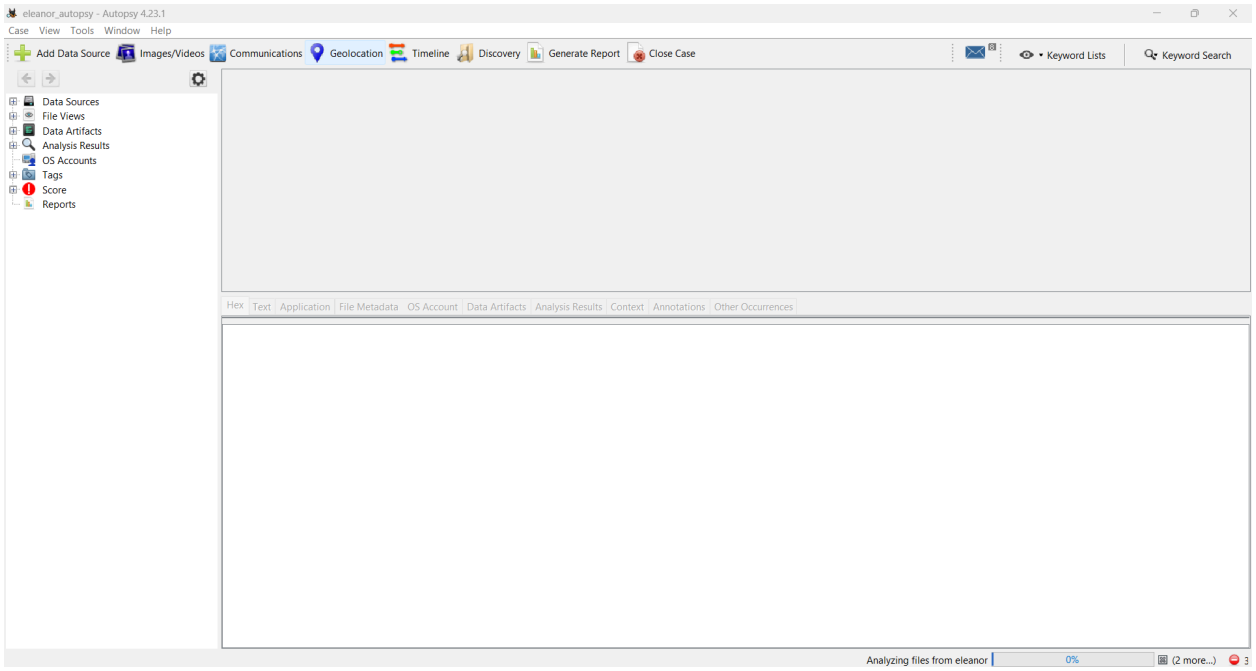

eleanor_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Host
Hosts are used to organize data sources and other data.
Generate new host name based on data source name
Specify new host name
Use existing host
< Back
Next >
Finish
Cancel
Help


eleanor_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source Type
Disk Image or VM File
Local Disk
Logical Files
Unallocated Space Image File
Autopsy Logical Imager Results
XRY Text Export
< Back
Next >
Finish
Cancel
Help

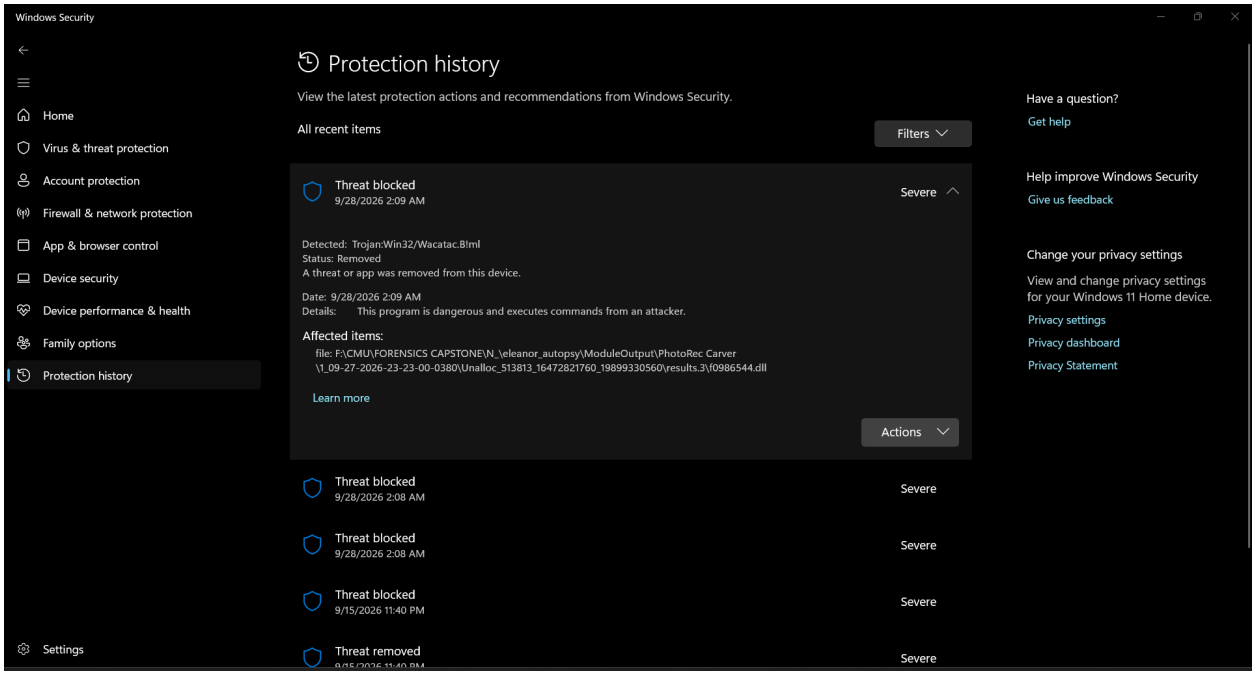

eleanor_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source
Path:
F:\CMU\FORENSICS CAPSTONE\N_\eleanor
Browse
Ignore orphan files in FAT file systems
Time zone:
(GMT-8:00) America/Los_Angeles
Sector size:
Auto Detect
Bitlocker Password (optional):
Hash Values (optional):
MD5:
SHA-1:
SHA-256:
NOTE: These values will not be validated when the data source is added.
< Back
Next >
Finish
Cancel
Help

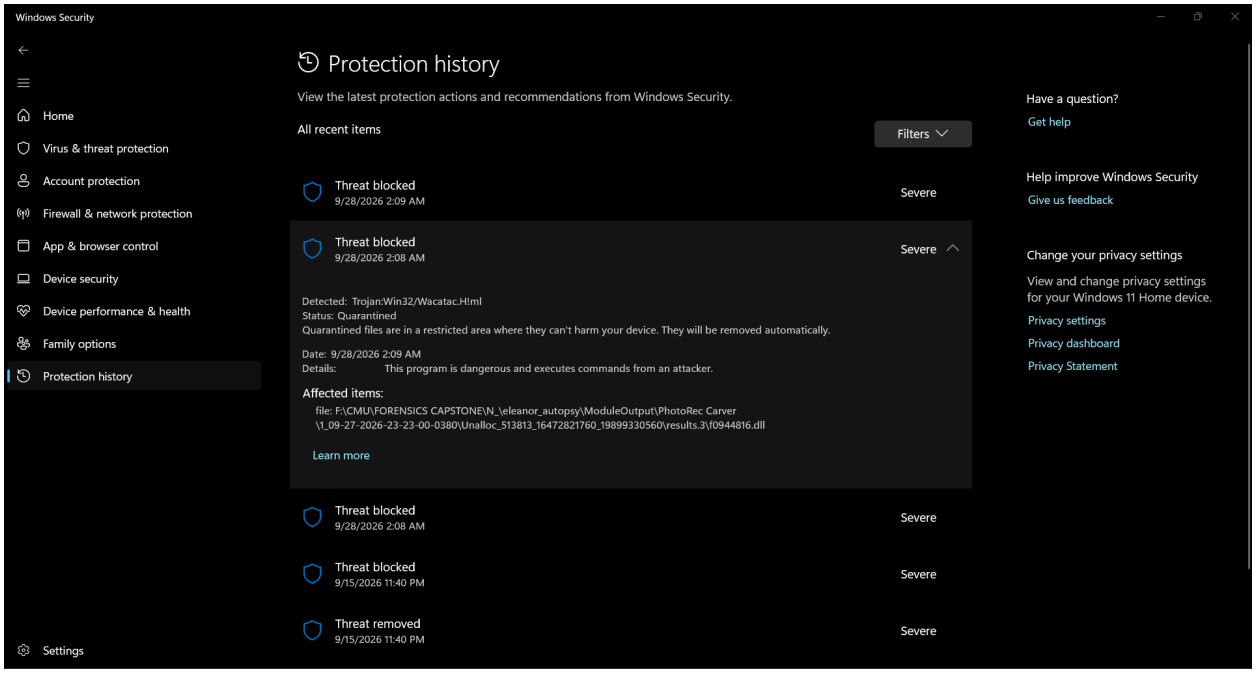

eleanor_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
The selected module has no per-run settings.
Extracts recent user activity, such as Web browsing, rec..
Global Settings
Select All
Deselect All
History
< Back
Next >
Finish
Cancel
Help

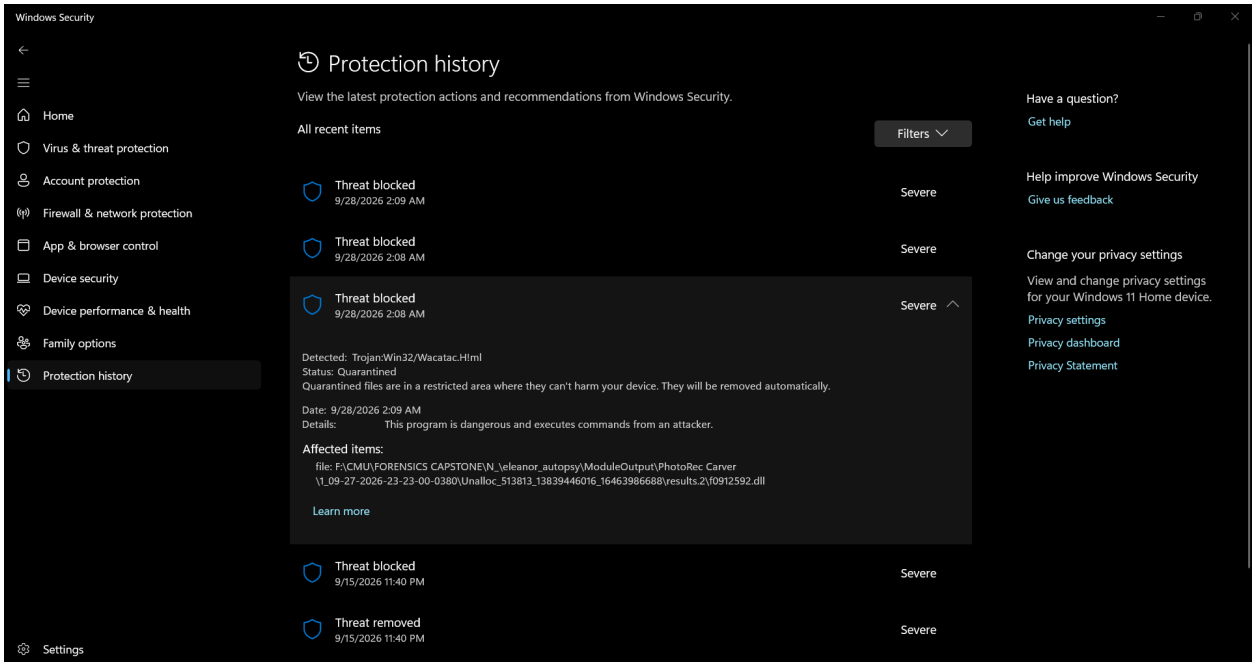

eleanor_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Data Sources
File Views
Data Artifacts
Analysis Results
OS Accounts
Tags
Score
Reports
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
Data source has been added to the local database. Files are being analyzed.
< Back
Next >
Finish
Cancel
Help
Hex
Text
Analyzing files from eleanor
0%
(2 more...)


eleanor_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Data Sources
File Views
Data Artifacts
Analysis Results
OS Accounts
Tags
Score
Reports
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Analyzing files from eleanor
0%
(2 more...)

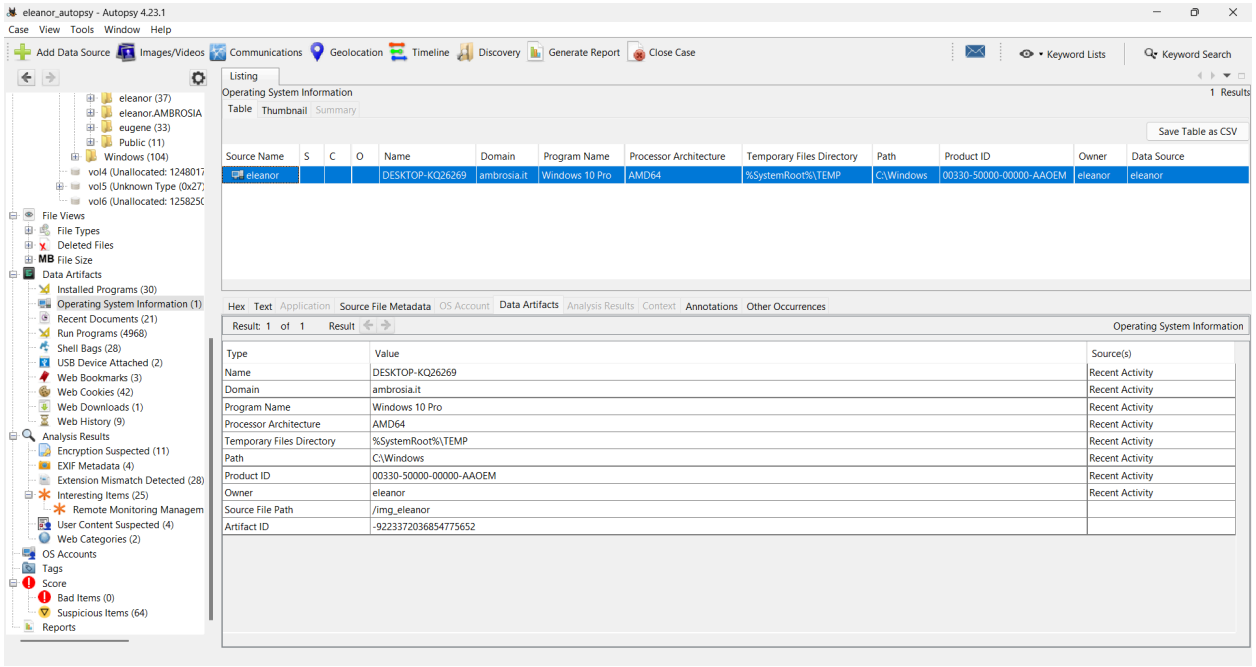
5.2.7.5 Eleanor_A_0005 Windows Security



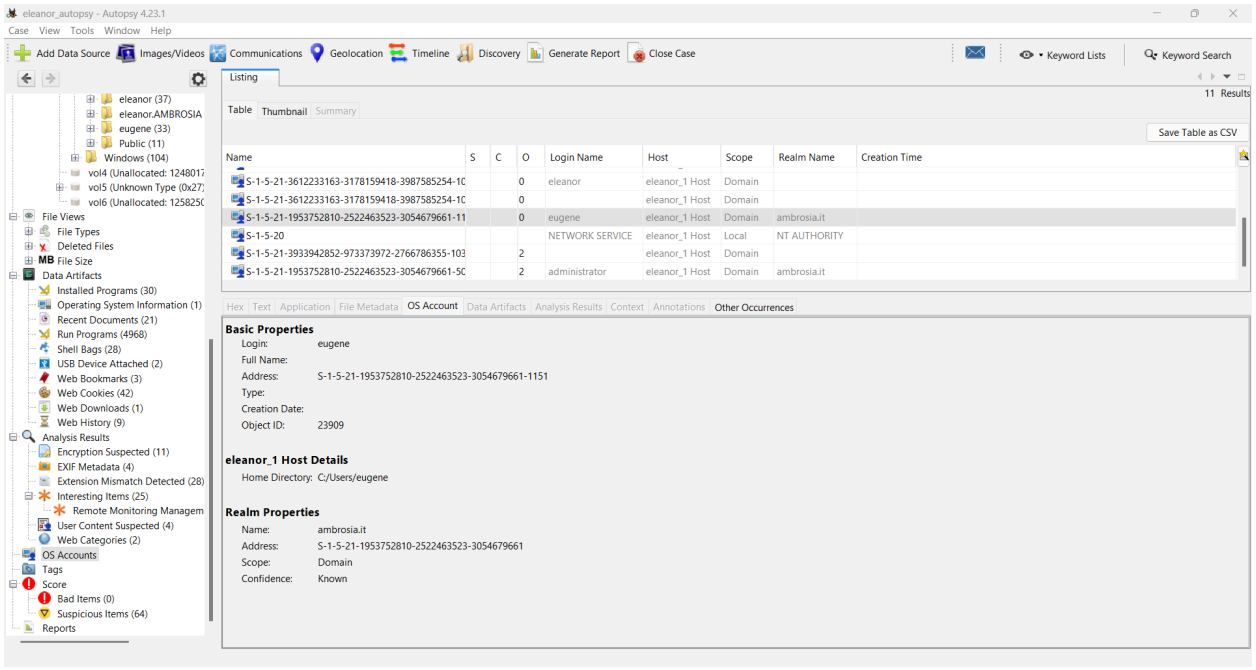




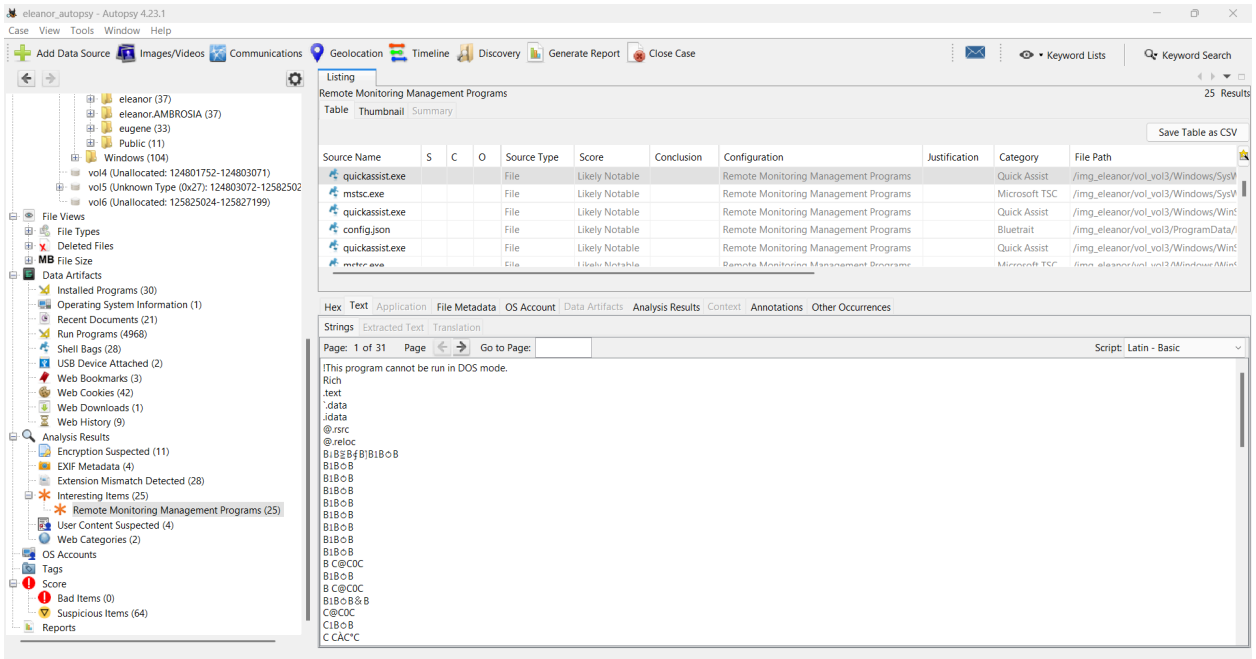
5.2.7.6 Eleanor_A_0006 Device Information



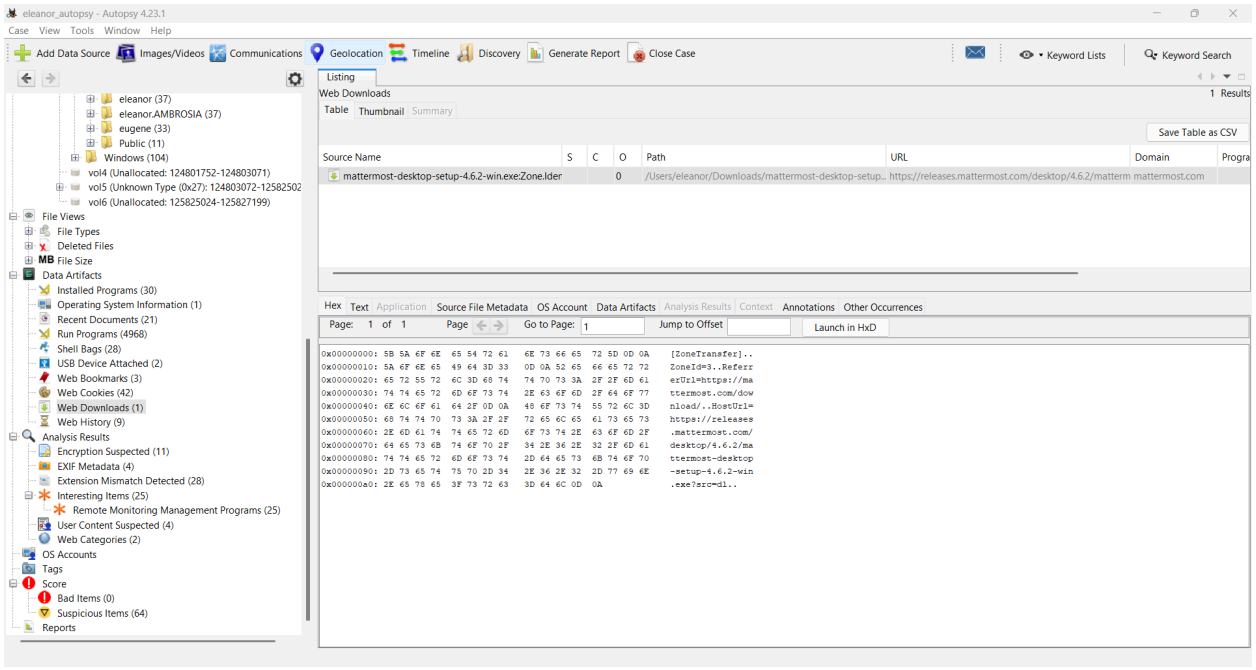
5.2.7.7 Eleanor_A_0007 Eugene Account



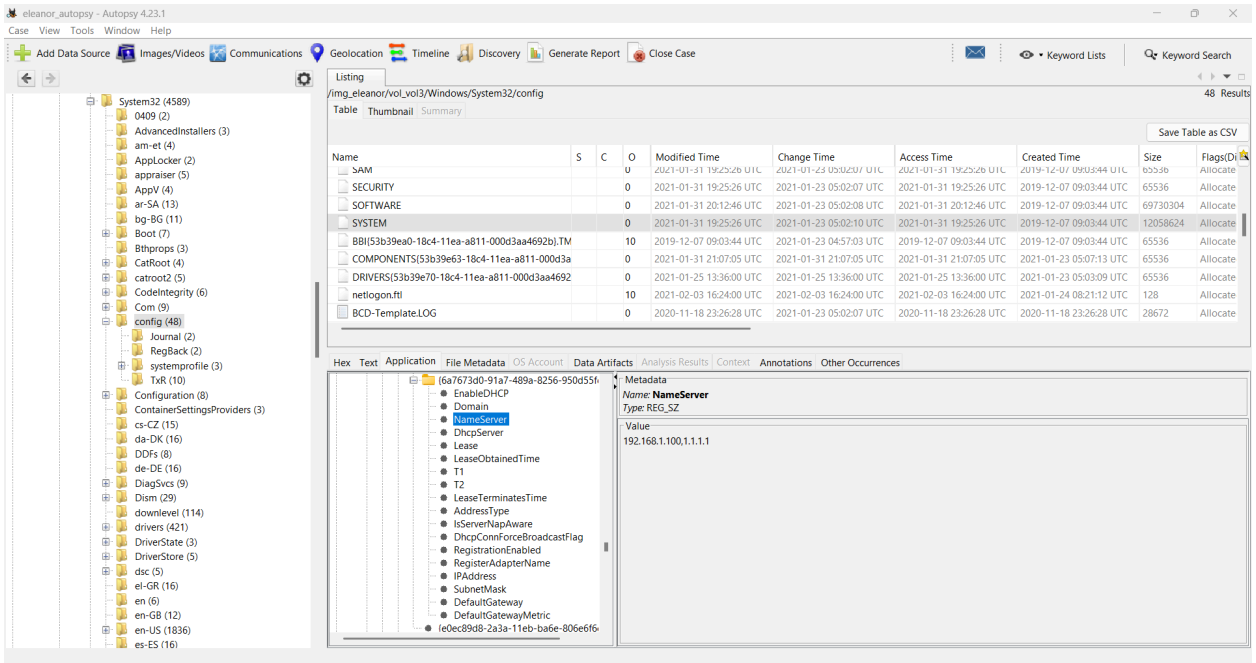
5.2.7.8 Eleanor_A_0008 Remote Monitoring Management Program

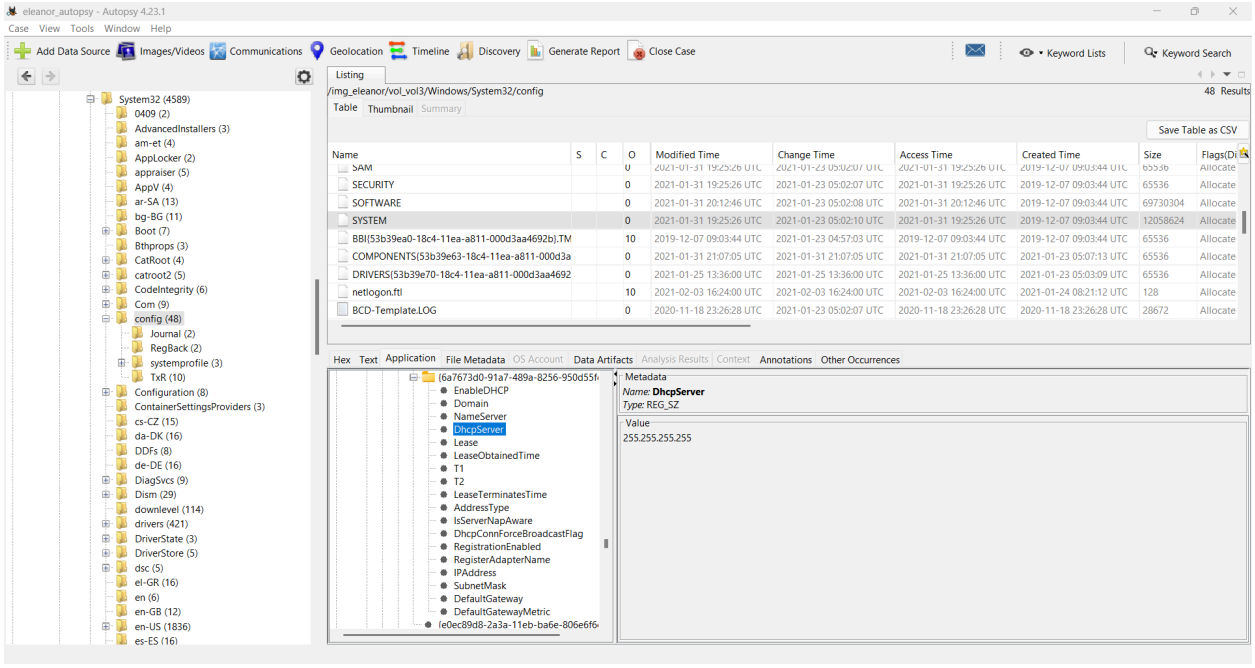


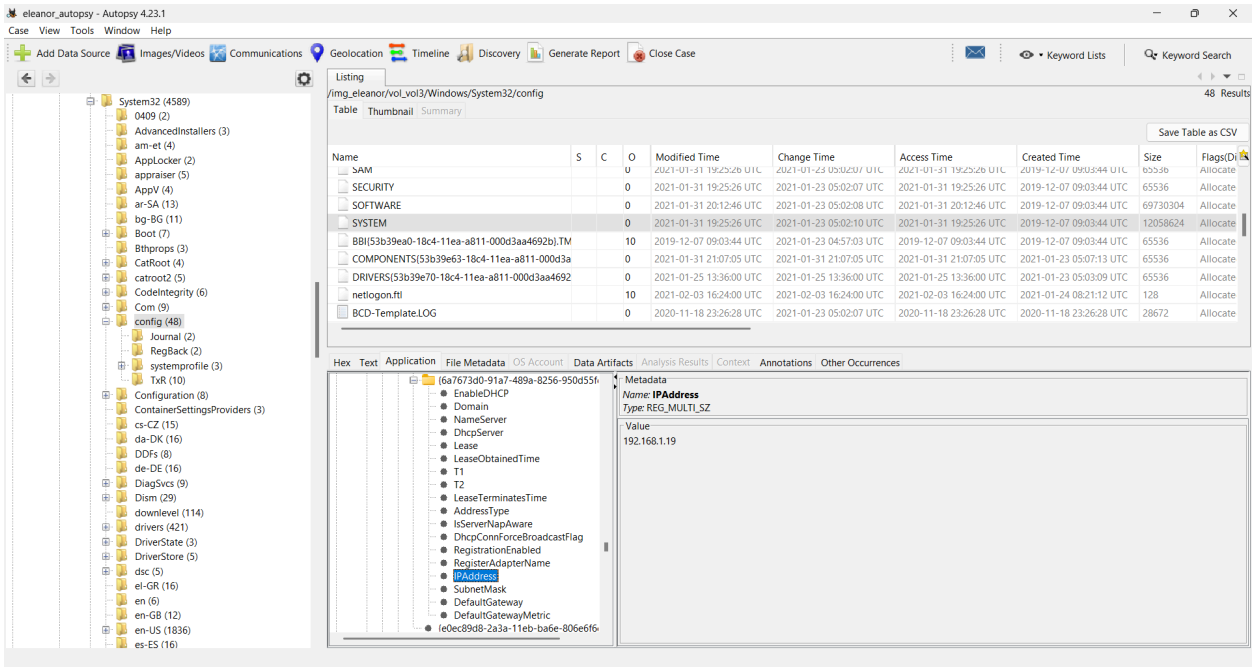
5.2.7.9 Eleanor_A_0009 Mattermost Download

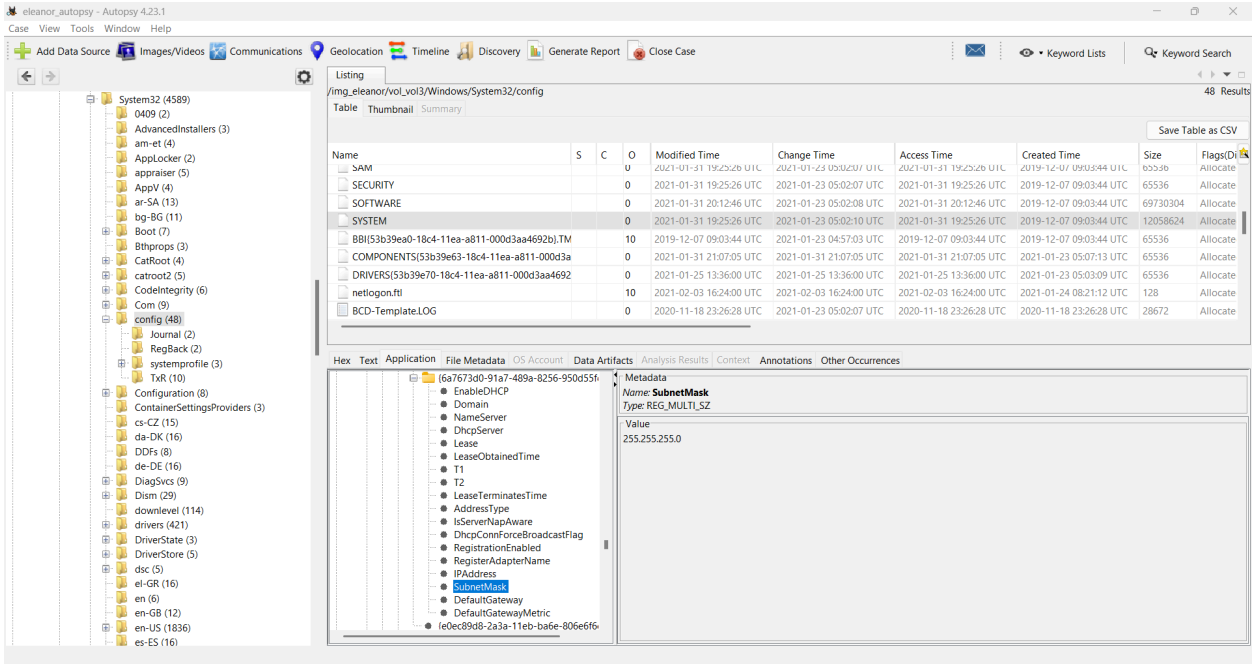


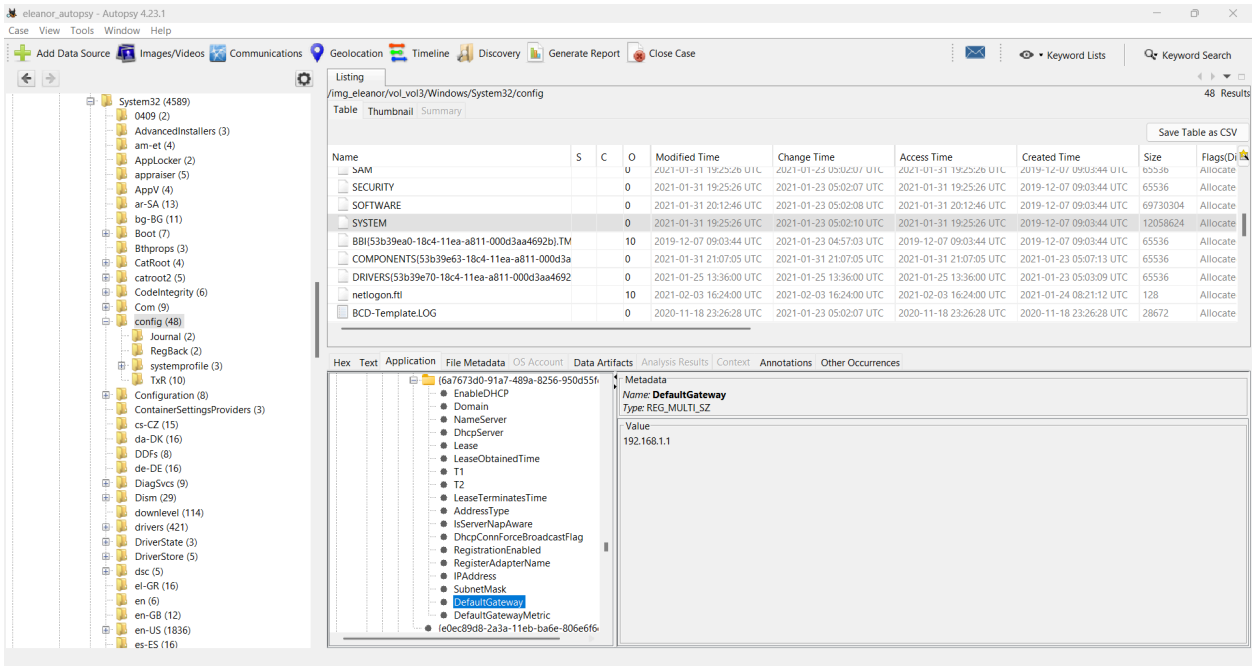
5.2.7.10 Eleanor_A_0010 IP Address



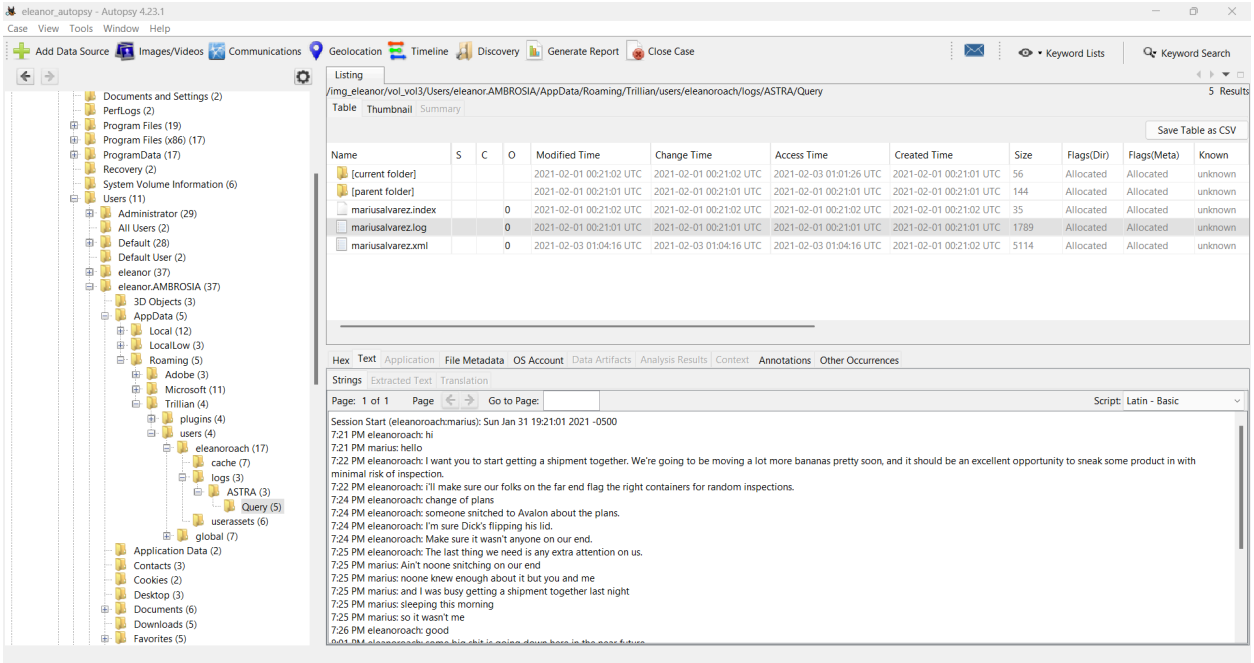

eleanor_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Listing
/img_eleanor/vol_vol3/Windows/System32/config
48 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(Di
SECURITY 0 2021-01-31 19:25:26 UTC 2021-01-23 05:02:07 UTC 2021-01-31 19:25:26 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
SOFTWARE 0 2021-01-31 20:12:46 UTC 2021-01-23 05:02:08 UTC 2021-01-31 20:12:46 UTC 2019-12-07 09:03:44 UTC 69730304 Allocate
SYSTEM 0 2021-01-31 19:25:26 UTC 2021-01-23 05:02:10 UTC 2021-01-31 19:25:26 UTC 2019-12-07 09:03:44 UTC 12058624 Allocate
BBI{53b39ea0-18c4-11ea-a811-000d3aa4692b}.TM 10 2019-12-07 09:03:44 UTC 2021-01-23 04:57:03 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
COMPONENTS{53b39e63-18c4-11ea-a811-000d3a 0 2021-01-31 21:07:05 UTC 2021-01-31 21:07:05 UTC 2021-01-31 21:07:05 UTC 2021-01-23 05:07:13 UTC 65536 Allocate
DRIVERS{53b39e70-18c4-11ea-a811-000d3aa4692 0 2021-01-25 13:36:00 UTC 2021-01-25 13:36:00 UTC 2021-01-25 13:36:00 UTC 2021-01-23 05:03:09 UTC 65536 Allocate
netlogon.ftl 10 2021-02-03 16:24:00 UTC 2021-02-03 16:24:00 UTC 2021-02-03 16:24:00 UTC 2021-01-24 08:21:12 UTC 128 Allocate
BCD-Template.LOG 0 2020-11-18 23:26:28 UTC 2021-01-23 05:02:07 UTC 2020-11-18 23:26:28 UTC 2020-11-18 23:26:28 UTC 28672 Allocate
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
{6a7673d0-91a7-489a-8256-950d55f
EnableDHCP
Domain
NameServer
DhcpServer
Lease
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
SubnetMask
DefaultGateway
DefaultGatewayMetric
{e0ec89d8-2a3a-11eb-ba6e-806e6f6
Metadata
Name: DhcpServer
Type: REG_SZ
Value
255.255.255.255
System32 (4589)
0409 (2)
AdvancedInstallers (3)
am-et (4)
AppLocker (2)
appraiser (5)
AppV (4)
ar-SA (13)
bg-BG (11)
Boot (7)
Bthprops (3)
CatRoot (4)
catroot2 (5)
CodeIntegrity (6)
Com (9)
config (48)
Journal (2)
RegBack (2)
systemprofile (3)
TxR (10)
Configuration (8)
ContainerSettingsProviders (3)
cs-CZ (15)
da-DK (16)
DDFs (8)
de-DE (16)
DiagSvcs (9)
Dism (29)
downlevel (114)
drivers (421)
DriverState (3)
DriverStore (5)
dsc (5)
el-GR (16)
en (6)
en-GB (12)
en-US (1836)
es-ES (16)


eleanor_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Listing
/img_eleanor/vol_vol3/Windows/System32/config
48 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(Di
SECURITY 0 2021-01-31 19:25:26 UTC 2021-01-23 05:02:07 UTC 2021-01-31 19:25:26 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
SOFTWARE 0 2021-01-31 20:12:46 UTC 2021-01-23 05:02:08 UTC 2021-01-31 20:12:46 UTC 2019-12-07 09:03:44 UTC 69730304 Allocate
SYSTEM 0 2021-01-31 19:25:26 UTC 2021-01-23 05:02:10 UTC 2021-01-31 19:25:26 UTC 2019-12-07 09:03:44 UTC 12058624 Allocate
BBI{53b39ea0-18c4-11ea-a811-000d3aa4692b}.TM 10 2019-12-07 09:03:44 UTC 2021-01-23 04:57:03 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
COMPONENTS{53b39e63-18c4-11ea-a811-000d3a 0 2021-01-31 21:07:05 UTC 2021-01-31 21:07:05 UTC 2021-01-31 21:07:05 UTC 2021-01-23 05:07:13 UTC 65536 Allocate
DRIVERS{53b39e70-18c4-11ea-a811-000d3aa4692 0 2021-01-25 13:36:00 UTC 2021-01-25 13:36:00 UTC 2021-01-25 13:36:00 UTC 2021-01-23 05:03:09 UTC 65536 Allocate
netlogon.ftl 10 2021-02-03 16:24:00 UTC 2021-02-03 16:24:00 UTC 2021-02-03 16:24:00 UTC 2021-01-24 08:21:12 UTC 128 Allocate
BCD-Template.LOG 0 2020-11-18 23:26:28 UTC 2021-01-23 05:02:07 UTC 2020-11-18 23:26:28 UTC 2020-11-18 23:26:28 UTC 28672 Allocate
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
{6a7673d0-91a7-489a-8256-950d55f
EnableDHCP
Domain
NameServer
DhcpServer
Lease
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
SubnetMask
DefaultGateway
DefaultGatewayMetric
{e0ec89d8-2a3a-11eb-ba6e-806e6f6
Metadata
Name: IPAddress
Type: REG_MULTI_SZ
Value
192.168.1.19
System32 (4589)
0409 (2)
AdvancedInstallers (3)
am-et (4)
AppLocker (2)
appraiser (5)
AppV (4)
ar-SA (13)
bg-BG (11)
Boot (7)
Bthprops (3)
CatRoot (4)
catroot2 (5)
CodeIntegrity (6)
Com (9)
config (48)
Journal (2)
RegBack (2)
systemprofile (3)
TxR (10)
Configuration (8)
ContainerSettingsProviders (3)
cs-CZ (15)
da-DK (16)
DDFs (8)
de-DE (16)
DiagSvcs (9)
Dism (29)
downlevel (114)
drivers (421)
DriverState (3)
DriverStore (5)
dsc (5)
el-GR (16)
en (6)
en-GB (12)
en-US (1836)
es-ES (16)


eleanor_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Listing
/img_eleanor/vol_vol3/Windows/System32/config
48 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(Di
SECURITY 0 2021-01-31 19:25:26 UTC 2021-01-23 05:02:07 UTC 2021-01-31 19:25:26 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
SOFTWARE 0 2021-01-31 20:12:46 UTC 2021-01-23 05:02:08 UTC 2021-01-31 20:12:46 UTC 2019-12-07 09:03:44 UTC 69730304 Allocate
SYSTEM 0 2021-01-31 19:25:26 UTC 2021-01-23 05:02:10 UTC 2021-01-31 19:25:26 UTC 2019-12-07 09:03:44 UTC 12058624 Allocate
BBI{53b39ea0-18c4-11ea-a811-000d3aa4692b}.TM 10 2019-12-07 09:03:44 UTC 2021-01-23 04:57:03 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
COMPONENTS{53b39e63-18c4-11ea-a811-000d3a 0 2021-01-31 21:07:05 UTC 2021-01-31 21:07:05 UTC 2021-01-31 21:07:05 UTC 2021-01-23 05:07:13 UTC 65536 Allocate
DRIVERS{53b39e70-18c4-11ea-a811-000d3aa4692 0 2021-01-25 13:36:00 UTC 2021-01-25 13:36:00 UTC 2021-01-25 13:36:00 UTC 2021-01-23 05:03:09 UTC 65536 Allocate
netlogon.ftl 10 2021-02-03 16:24:00 UTC 2021-02-03 16:24:00 UTC 2021-02-03 16:24:00 UTC 2021-01-24 08:21:12 UTC 128 Allocate
BCD-Template.LOG 0 2020-11-18 23:26:28 UTC 2021-01-23 05:02:07 UTC 2020-11-18 23:26:28 UTC 2020-11-18 23:26:28 UTC 28672 Allocate
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
EnableDHCP
Domain
NameServer
DhcpServer
Lease
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
SubnetMask
DefaultGateway
DefaultGatewayMetric
Metadata
Name: SubnetMask
Type: REG_MULTI_SZ
Value
255.255.255.0
System32 (4589)
0409 (2)
AdvancedInstallers (3)
am-et (4)
AppLocker (2)
appraiser (5)
AppV (4)
ar-SA (13)
bg-BG (11)
Boot (7)
Bthprops (3)
CatRoot (4)
catroot2 (5)
CodeIntegrity (6)
Com (9)
config (48)
Journal (2)
RegBack (2)
systemprofile (3)
TxR (10)
Configuration (8)
ContainerSettingsProviders (3)
cs-CZ (15)
da-DK (16)
DDFs (8)
de-DE (16)
DiagSvcs (9)
Dism (29)
downlevel (114)
drivers (421)
DriverState (3)
DriverStore (5)
dsc (5)
el-GR (16)
en (6)
en-GB (12)
en-US (1836)
es-ES (16)

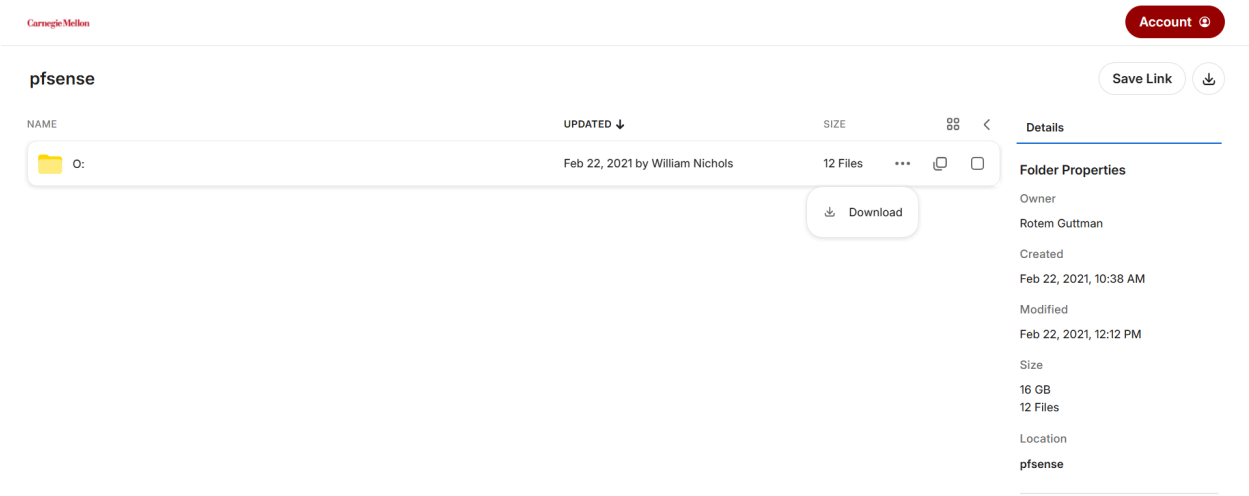

eleanor_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Listing
/img_eleanor/vol_vol3/Windows/System32/config
48 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(Di
SECURITY 0 2021-01-31 19:25:26 UTC 2021-01-23 05:02:07 UTC 2021-01-31 19:25:26 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
SOFTWARE 0 2021-01-31 20:12:46 UTC 2021-01-23 05:02:08 UTC 2021-01-31 20:12:46 UTC 2019-12-07 09:03:44 UTC 69730304 Allocate
SYSTEM 0 2021-01-31 19:25:26 UTC 2021-01-23 05:02:10 UTC 2021-01-31 19:25:26 UTC 2019-12-07 09:03:44 UTC 12058624 Allocate
BBI{53b39ea0-18c4-11ea-a811-000d3aa4692b}.TM 10 2019-12-07 09:03:44 UTC 2021-01-23 04:57:03 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
COMPONENTS{53b39e63-18c4-11ea-a811-000d3a 0 2021-01-31 21:07:05 UTC 2021-01-31 21:07:05 UTC 2021-01-31 21:07:05 UTC 2021-01-23 05:07:13 UTC 65536 Allocate
DRIVERS{53b39e70-18c4-11ea-a811-000d3aa4692 0 2021-01-25 13:36:00 UTC 2021-01-25 13:36:00 UTC 2021-01-25 13:36:00 UTC 2021-01-23 05:03:09 UTC 65536 Allocate
netlogon.ftl 10 2021-02-03 16:24:00 UTC 2021-02-03 16:24:00 UTC 2021-02-03 16:24:00 UTC 2021-01-24 08:21:12 UTC 128 Allocate
BCD-Template.LOG 0 2020-11-18 23:26:28 UTC 2021-01-23 05:02:07 UTC 2020-11-18 23:26:28 UTC 2020-11-18 23:26:28 UTC 28672 Allocate
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
EnableDHCP
Domain
NameServer
DhcpServer
Lease
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
SubnetMask
DefaultGateway
DefaultGatewayMetric
Metadata
Name: DefaultGateway
Type: REG_MULTI_SZ
Value
192.168.1.1
System32 (4589)
0409 (2)
AdvancedInstallers (3)
am-et (4)
AppLocker (2)
appraiser (5)
AppV (4)
ar-SA (13)
bg-BG (11)
Boot (7)
Bthprops (3)
CatRoot (4)
catroot2 (5)
CodeIntegrity (6)
Com (9)
config (48)
Journal (2)
RegBack (2)
systemprofile (3)
TxR (10)
Configuration (8)
ContainerSettingsProviders (3)
cs-CZ (15)
da-DK (16)
DDFs (8)
de-DE (16)
DiagSvcs (9)
Dism (29)
downlevel (114)
drivers (421)
DriverState (3)
DriverStore (5)
dsc (5)
el-GR (16)
en (6)
en-GB (12)
en-US (1836)
es-ES (16)

5.2.7.11 Eleanor_A_0011 Trillian Log

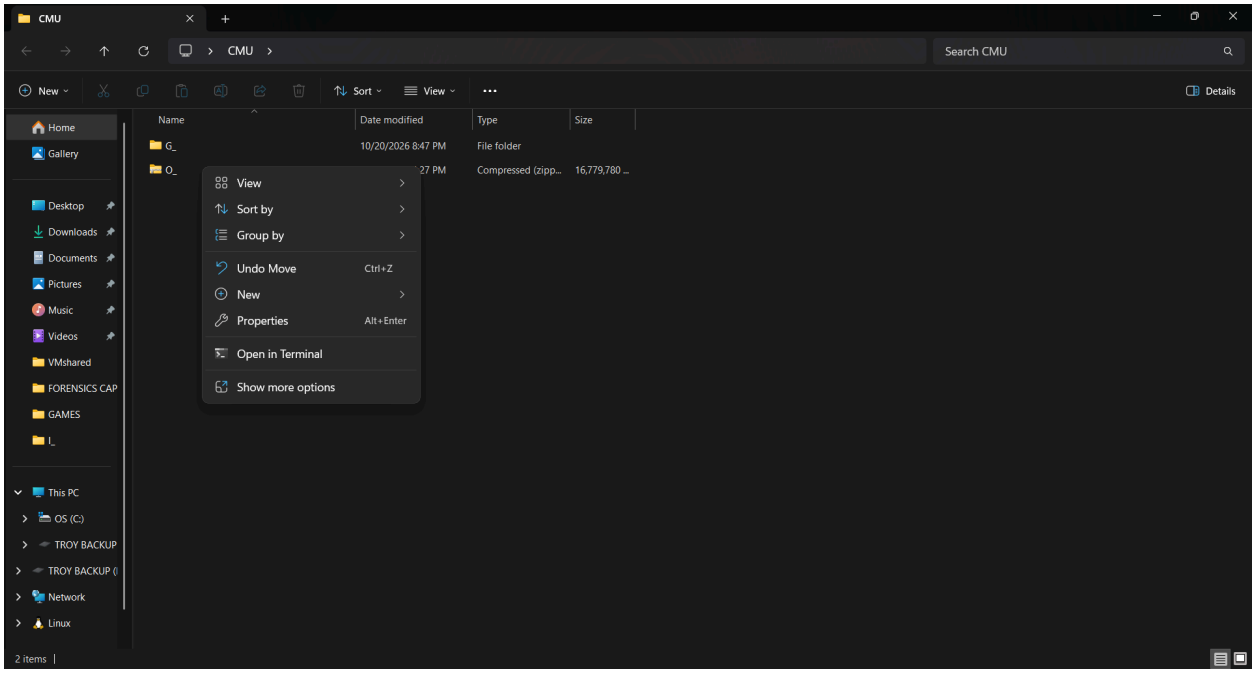


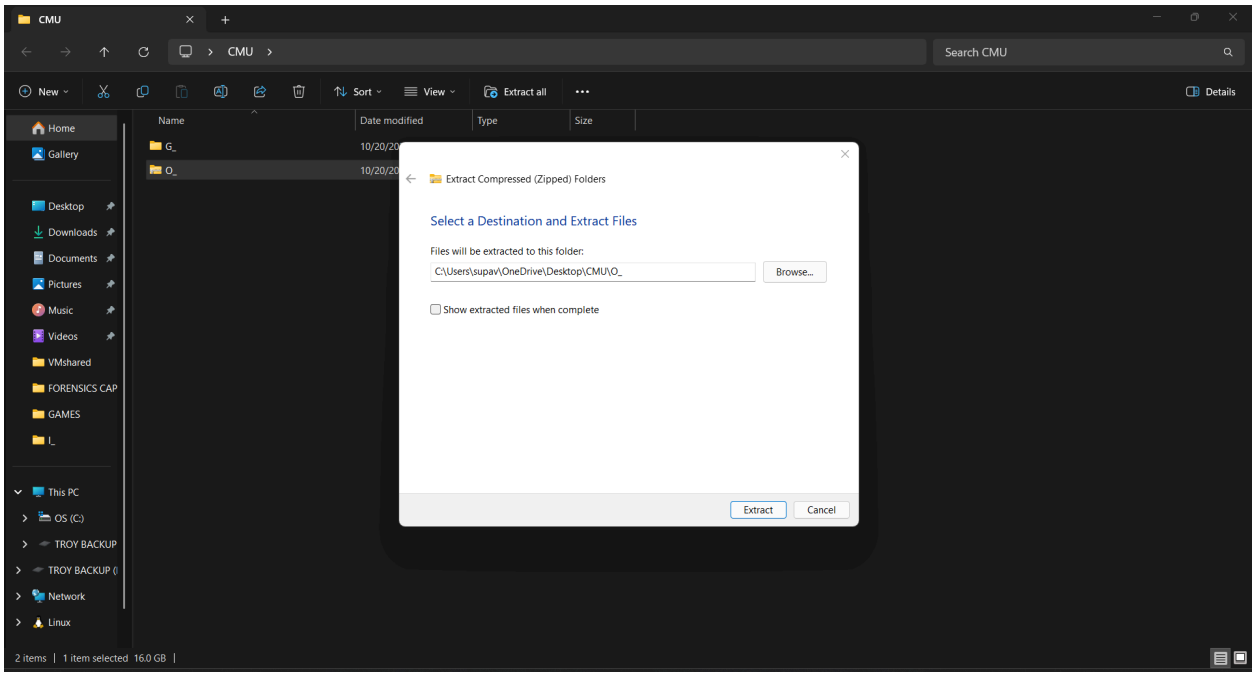
5.2.8 PFsense Image (O)

5.2.8.1 Pfsense_A_0001 Image Download

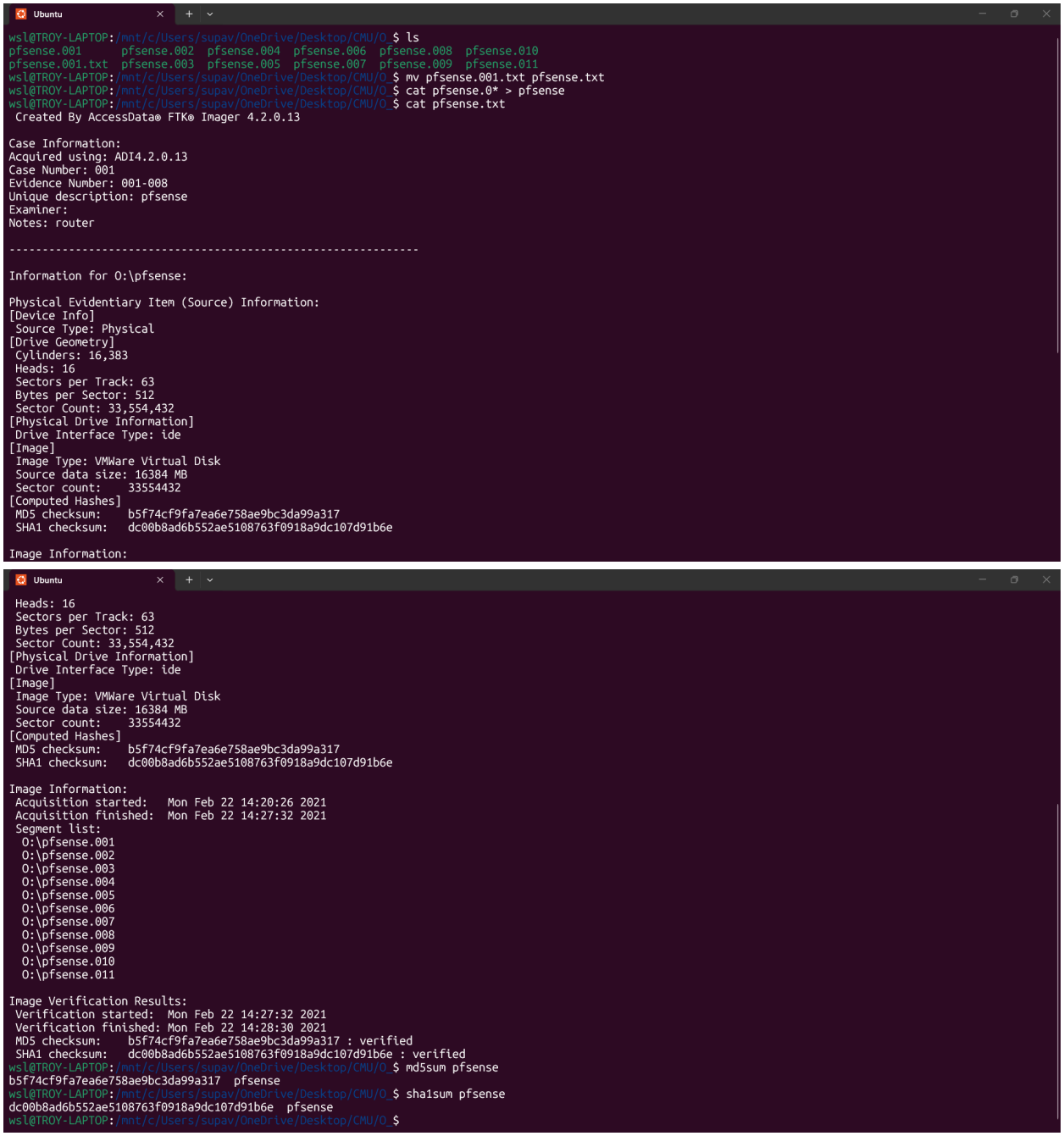


5.2.8.2 Pfsense_A_0002 Image Extraction

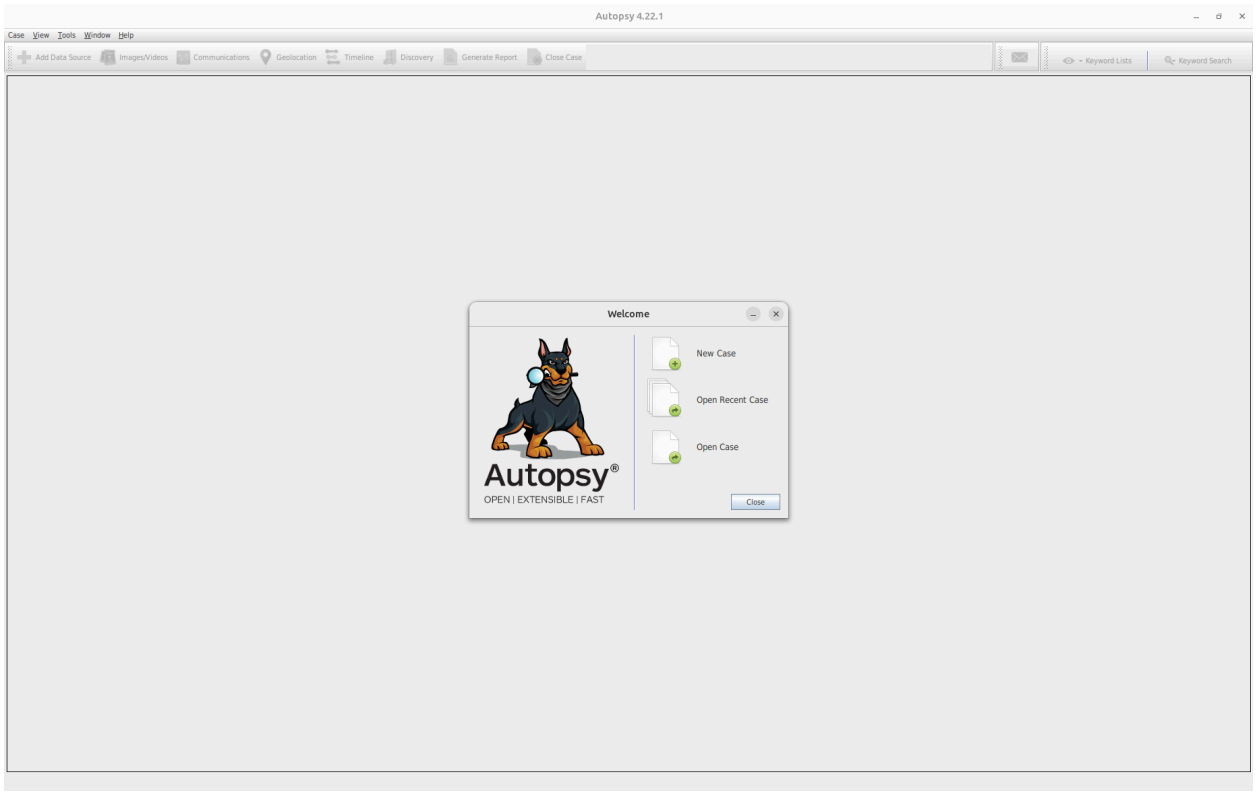


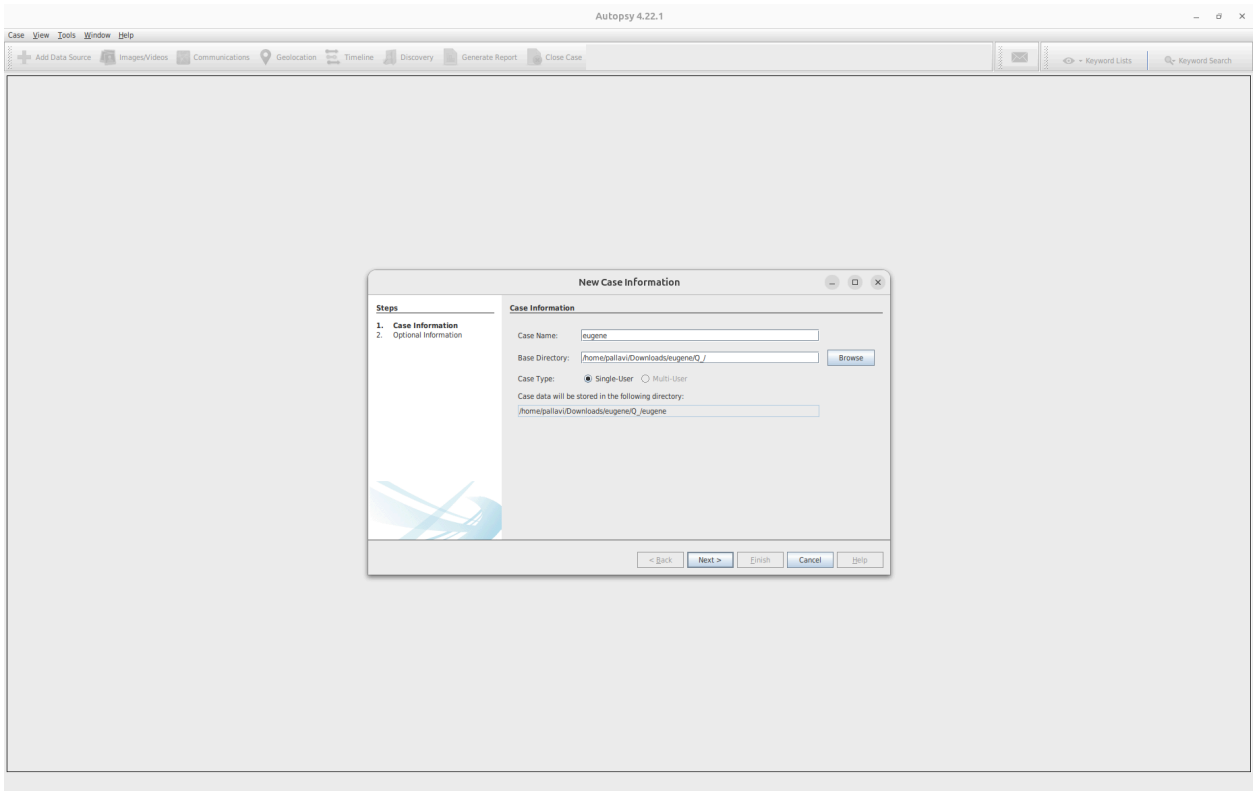


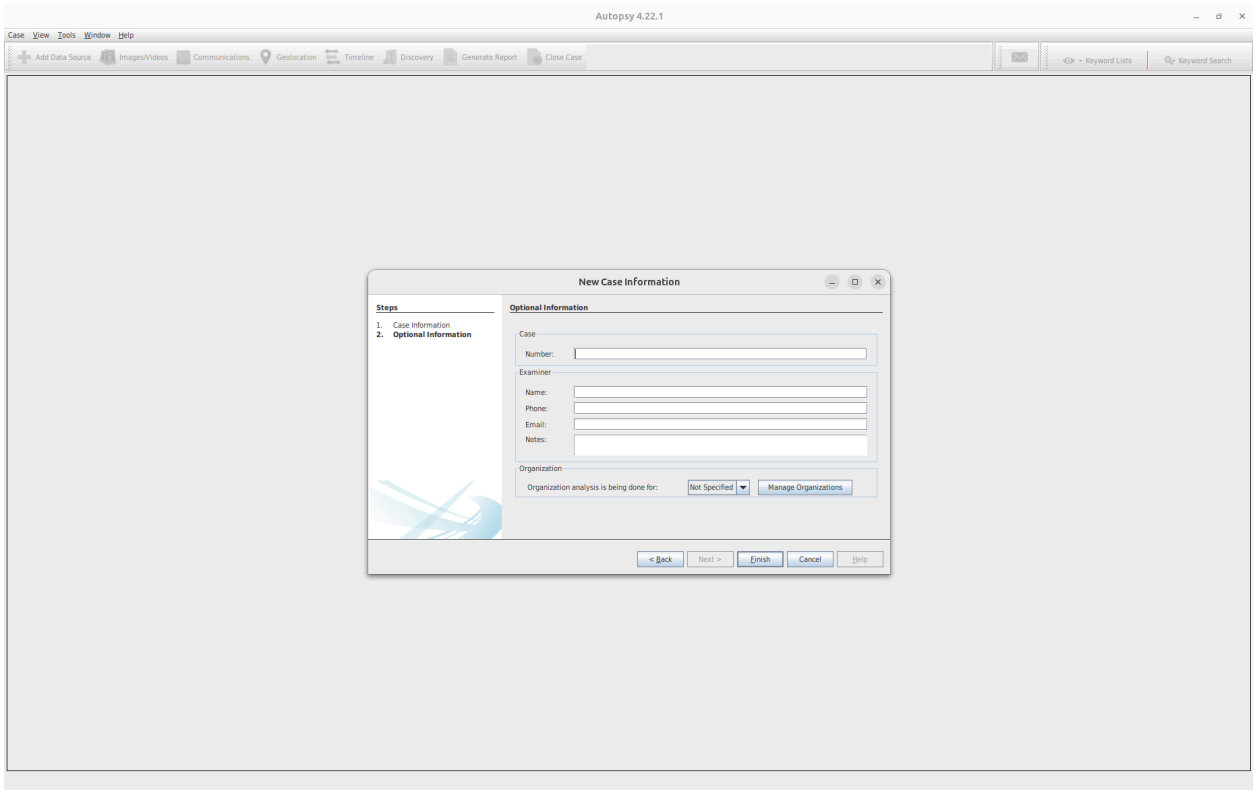
5.2.8.3 Pfsense_A_0003 Image Verification

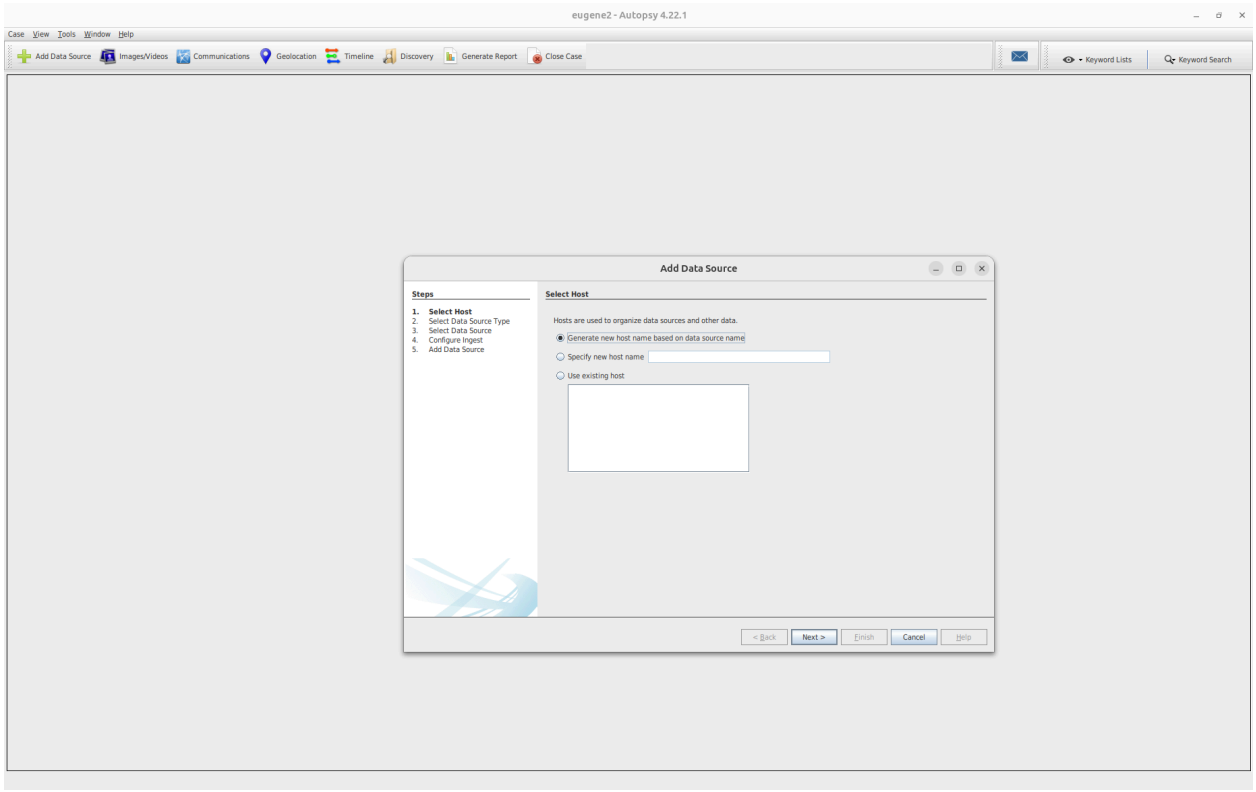


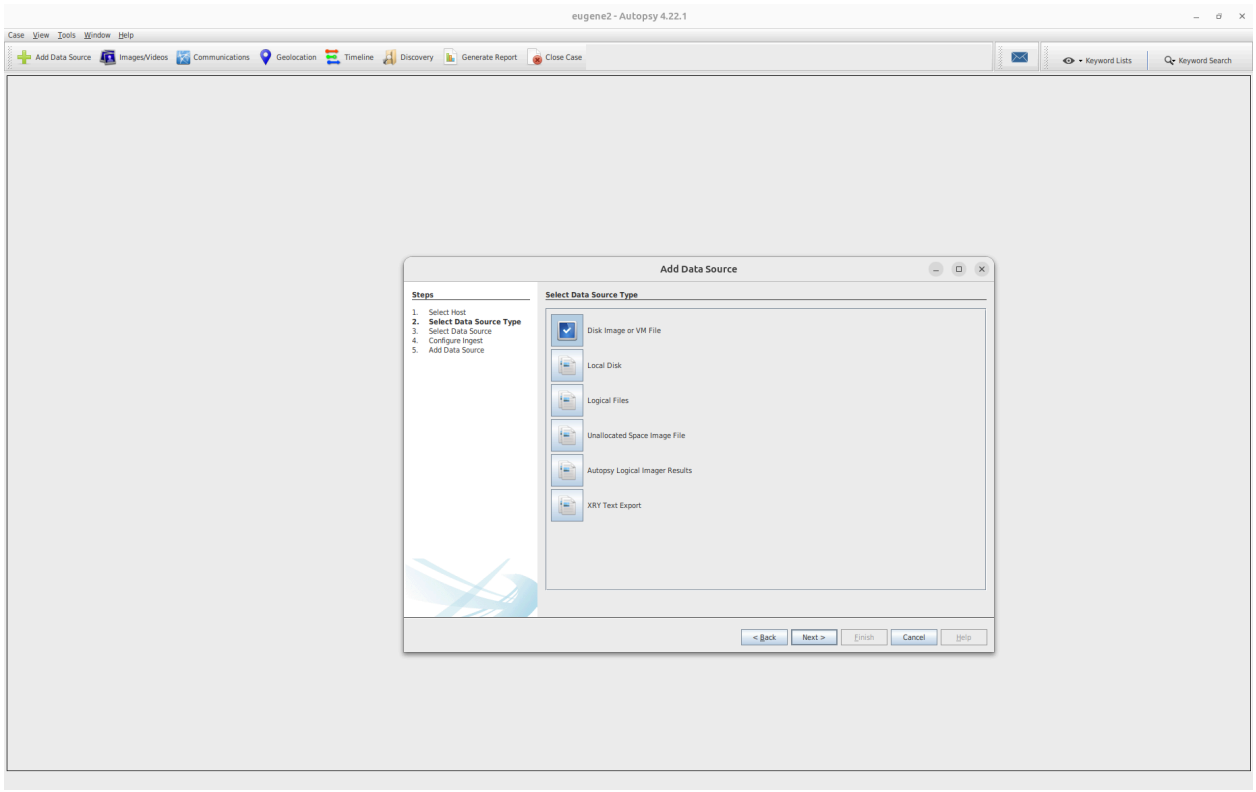
5.2.8.4 PFsense_A_0004 Autopsy Case

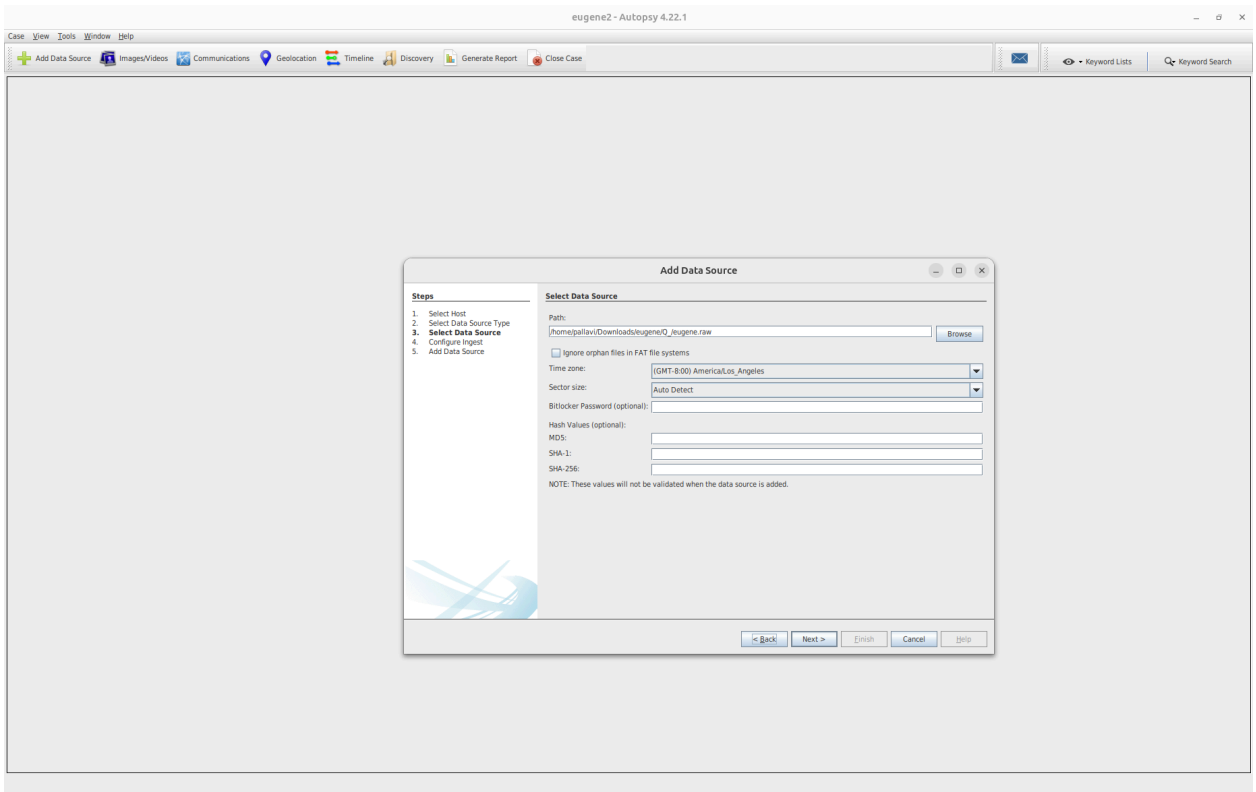


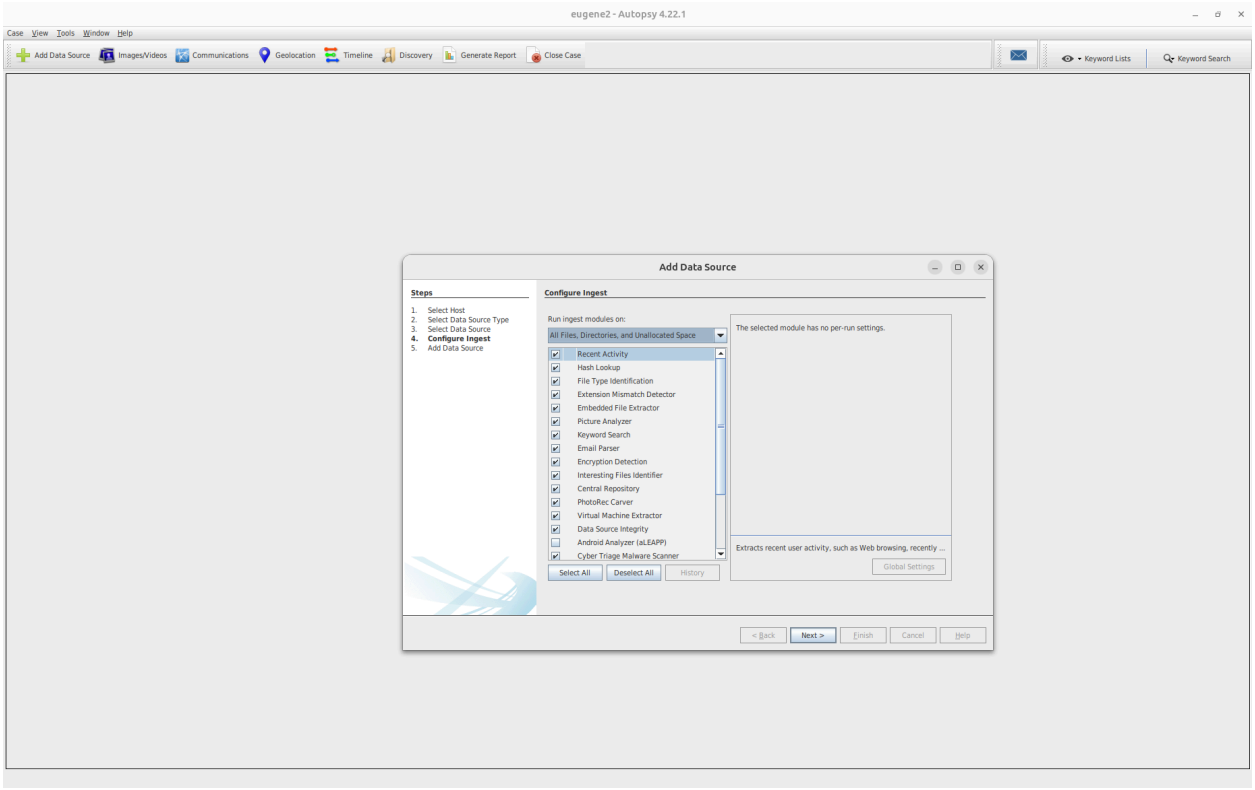


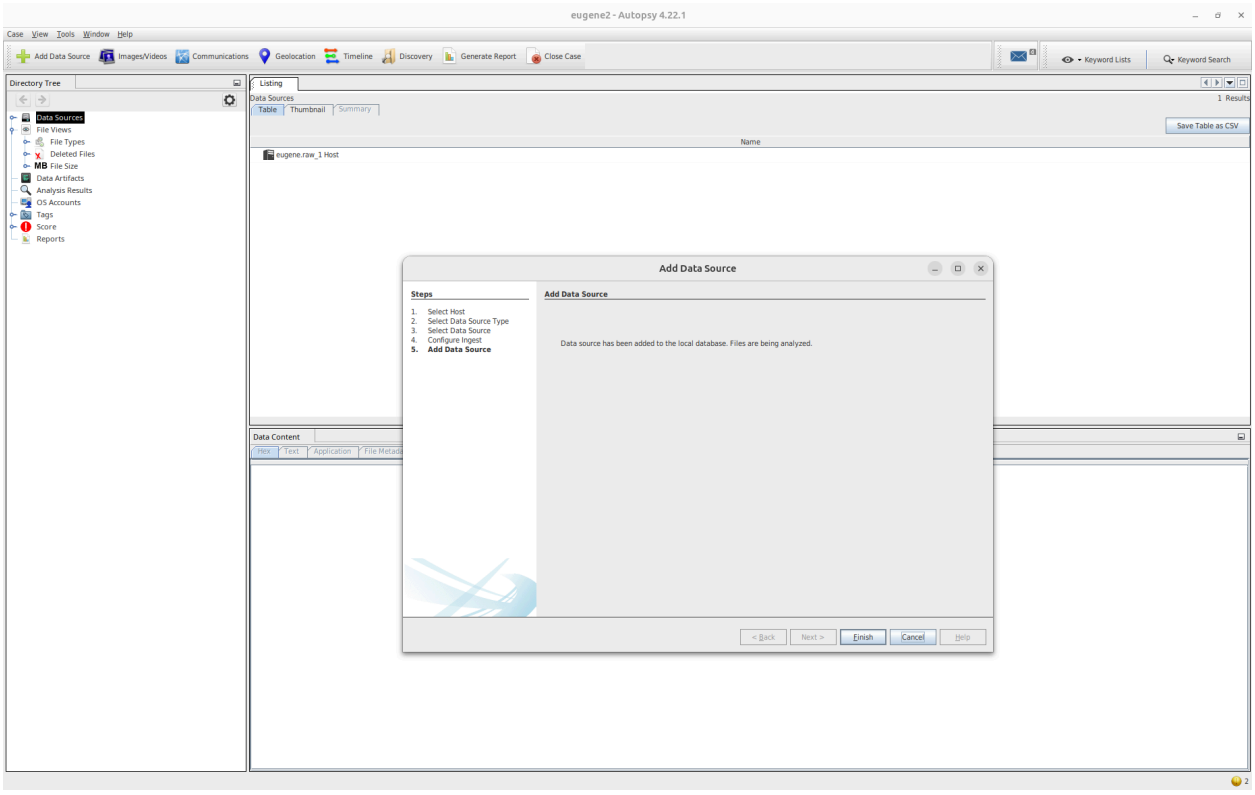

Autopsy 4.22.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Optional Information
Case
Number:
Examiner
Name:
Phone:
Email:
Notes:
Organization
Organization analysis is being done for:
Not Specified
Manage Organizations
< Back
Next >
Finish
Cancel
Help

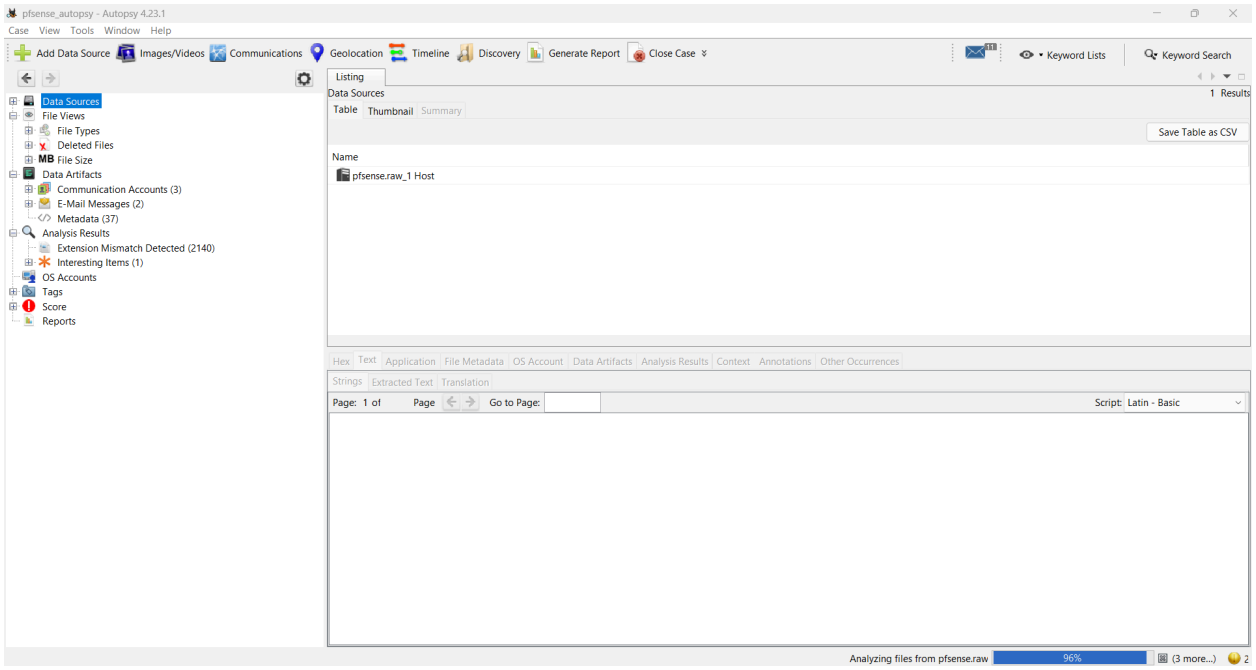

eugene2 - Autopsy 4.22.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Host
Hosts are used to organize data sources and other data.
Generate new host name based on data source name
Specify new host name
Use existing host
< Back
Next >
Finish
Cancel
Help


eugene2 - Autopsy 4.22.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source Type
Disk Image or VM File
Local Disk
Logical Files
Unallocated Space Image File
Autopsy Logical Imager Results
XRY Text Export
< Back
Next >
Finish
Cancel
Help


eugene2 - Autopsy 4.22.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source
Path:
/home/pallavi/Downloads/eugene/0_Jeugene.raw
Browse
Ignore orphan files in FAT file systems
Time zone:
(GMT-8:00) America/Los_Angeles
Sector size:
Auto Detect
Bitlocker Password (optional):
Hash Values (optional):
MD5:
SHA-1:
SHA-256:
NOTE: These values will not be validated when the data source is added.
< Back
Next >
Finish
Cancel
Help


eugene2 - Autopsy 4.22.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
The selected module has no per-run settings.
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
Data Source Integrity
Android Analyzer (aLEAPP)
Cyber Triage Malware Scanner
Extracts recent user activity, such as Web browsing, recently ...
Global Settings
Select All
Deselect All
History
< Back
Next >
Finish
Cancel
Help

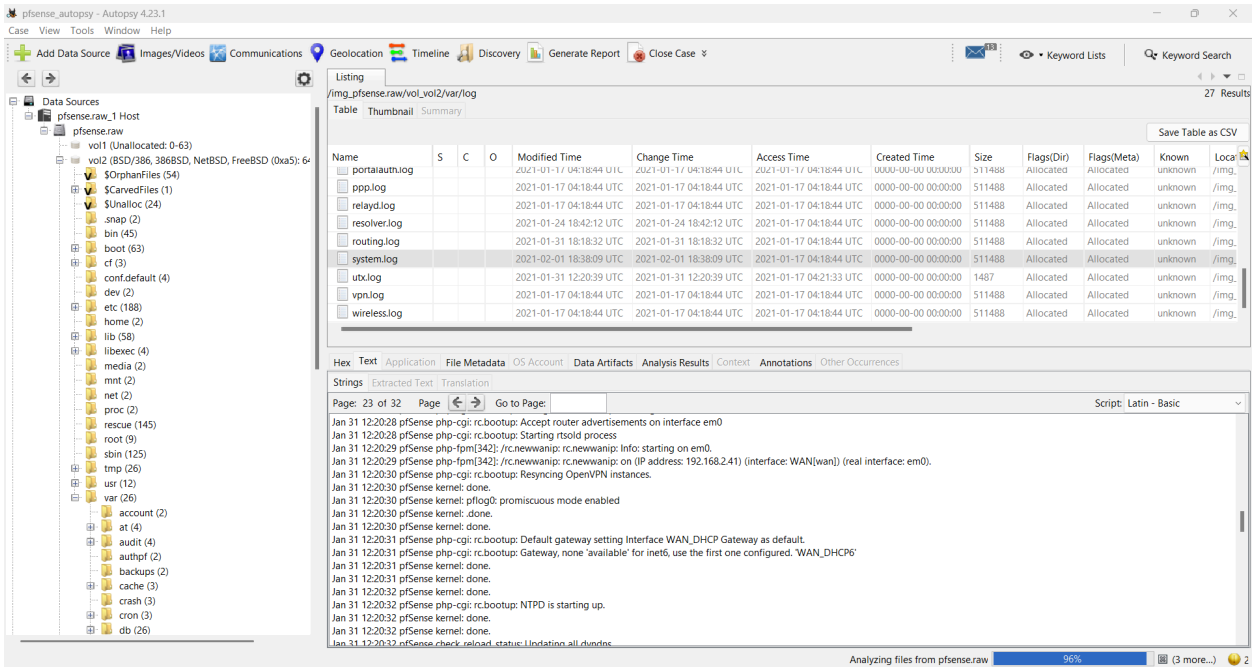

eugene2 - Autopsy 4.22.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Directory Tree
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Analysis Results
OS Accounts
Tags
Score
Reports
Listing
Data Sources
Table
Thumbnail
Summary
1 Results
Save Table as CSV
Name
eugene.raw_1 Host
Data Content
Hex
Text
Application
File Metada
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
Data source has been added to the local database. Files are being analyzed.
< Back
Next >
Finish
Cancel
Help



5.2.8.5 Pfsense_A_0005 Potential Data Exfiltration

```
NOTABLE -- POTENTIAL DATA EXFILTRATION
Feb  3 16:58:22 pfSense filterlog: 5,,,1000000103,em0,match,block,in,4,0x0,,64,48803,0,DF,6,tcp,51,192.168.2.42,192.168.2.41,4444,45209,11,PA,3811215915:3811215926,2355336458,501,,
Feb  3 16:58:22 pfSense filterlog: 5,,,1000000103,em0,match,block,in,4,0x0,,64,48804,0,DF,6,tcp,51,192.168.2.42,192.168.2.41,4444,45209,11,PA,3811215915:3811215926,2355336458,501,,
Feb  3 16:58:22 pfSense filterlog: 5,,,1000000103,em0,match,block,in,4,0x0,,64,48805,0,DF,6,tcp,51,192.168.2.42,192.168.2.41,4444,45209,11,PA,3811215915:3811215926,2355336458,501,,
Feb  3 16:58:23 pfSense filterlog: 5,,,1000000103,em0,match,block,in,4,0x0,,64,48806,0,DF,6,tcp,51,192.168.2.42,192.168.2.41,4444,45209,11,PA,3811215915:3811215926,2355336458,501,,
Feb  3 16:58:24 pfSense filterlog: 5,,,1000000103,em0,match,block,in,4,0x0,,64,48807,0,DF,6,tcp,51,192.168.2.42,192.168.2.41,4444,45209,11,PA,3811215915:3811215926,2355336458,501,,
Feb  3 16:58:25 pfSense filterlog: 5,,,1000000103,em0,match,block,in,4,0x0,,64,48808,0,DF,6,tcp,51,192.168.2.42,192.168.2.41,4444,45209,11,PA,3811215915:3811215926,2355336458,501,,
Feb  3 16:58:29 pfSense filterlog: 5,,,1000000103,em0,match,block,in,4,0x0,,64,48809,0,DF,6,tcp,51,192.168.2.42,192.168.2.41,4444,45209,11,PA,3811215915:3811215926,2355336458,501,,
Feb  3 16:58:35 pfSense filterlog: 5,,,1000000103,em0,match,block,in,4,0x0,,64,48810,0,DF,6,tcp,51,192.168.2.42,192.168.2.41,4444,45209,11,PA,3811215915:3811215926,2355336458,501,,
Feb  3 16:58:48 pfSense filterlog: 5,,,1000000103,em0,match,block,in,4,0x0,,64,48811,0,DF,6,tcp,51,192.168.2.42,192.168.2.41,4444,45209,11,PA,3811215915:3811215926,2355336458,501,,
Feb  3 16:59:14 pfSense filterlog: 5,,,1000000103,em0,match,block,in,4,0x0,,64,48812,0,DF,6,tcp,57,192.168.2.42,192.168.2.41,4444,45209,17,FPA,3811215926:3811215943,2355336458,501,,
Feb  3 16:59:15 pfSense filterlog: 5,,,1000000103,em0,match,block,in,4,0x0,,64,48813,0,DF,6,tcp,68,192.168.2.42,192.168.2.41,4444,45209,28,FPA,3811215915:3811215943,2355336458,501,,
END
```

5.2.8.6 PFSense_A_0006 system.log WAN IP Assignment



5.2.8.7 PFSense_A_0007 f0000184.txt

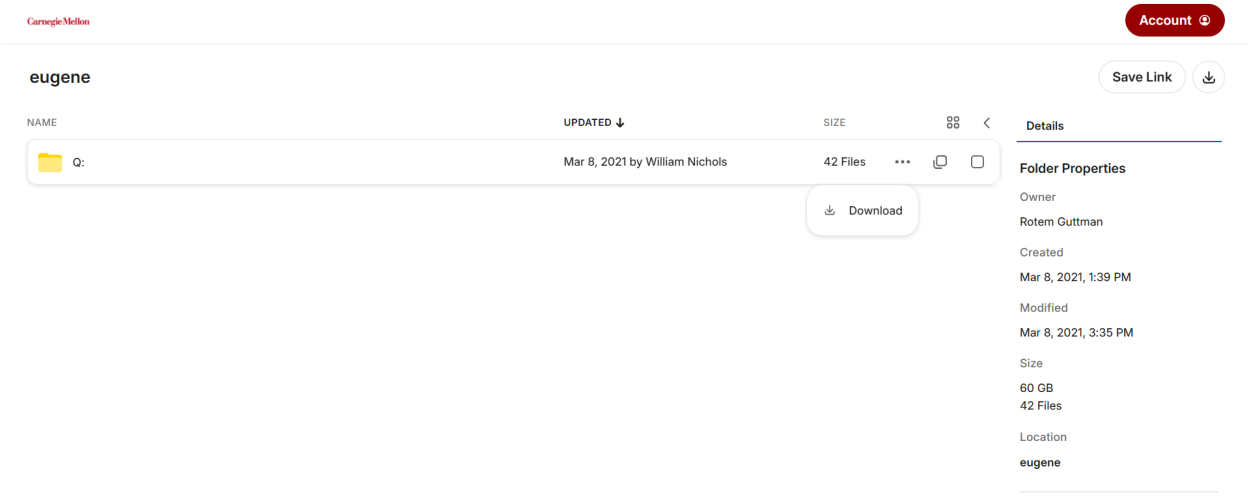
em1 tcp 192.168.2.42:4444 <- 192.168.1.20:63119 TIME_WAIT:TIME_WAIT
em0 tcp 192.168.2.41:5657 (192.168.1.20:63119) -> 192.168.2.42:4444 TIME_WAIT:TIME_WAIT
em1 tcp 104.82.52.231:80 <- 192.168.1.19:57957 FIN_WAIT_2:FIN_WAIT_2
em0 tcp 192.168.2.41:64290 (192.168.1.19:57957) -> 104.82.52.231:80 FIN_WAIT_2:FIN_WAIT_2
em1 tcp 99.86.189.23:443 <- 192.168.1.100:42838 ESTABLISHED:ESTABLISHED
em0 tcp 192.168.2.41:11463 (192.168.1.100:42838) -> 99.86.189.23:443 ESTABLISHED:ESTABLISHED
em1 tcp 99.86.189.32:443 <- 192.168.1.19:57964 ESTABLISHED:ESTABLISHED
em0 tcp 192.168.2.41:1838 (192.168.1.19:57964) -> 99.86.189.32:443 ESTABLISHED:ESTABLISHED
em1 tcp 20.54.24.148:443 <- 192.168.1.19:57965 ESTABLISHED:ESTABLISHED
em0 tcp 192.168.2.41:58918 (192.168.1.19:57965) -> 20.54.24.148:443 ESTABLISHED:ESTABLISHED
em1 tcp 99.86.189.62:443 <- 192.168.1.12:61997 ESTABLISHED:ESTABLISHED
em0 tcp 192.168.2.41:51582 (192.168.1.12:61997) -> 99.86.189.62:443 ESTABLISHED:ESTABLISHED
em1 tcp 192.168.2.30:7680 <- 192.168.1.20:63144 FIN_WAIT_2:FIN_WAIT_2
em0 tcp 192.168.2.41:46630 (192.168.1.20:63144) -> 192.168.2.30:7680 FIN_WAIT_2:FIN_WAIT_2
em1 tcp 192.168.2.42:4444 <- 192.168.1.20:63146 ESTABLISHED:ESTABLISHED
em0 tcp 192.168.2.41:22671 (192.168.1.20:63146) -> 192.168.2.42:4444 ESTABLISHED:ESTABLISHED

5.2.8.8 PFSense_A_0008 suricata.yaml

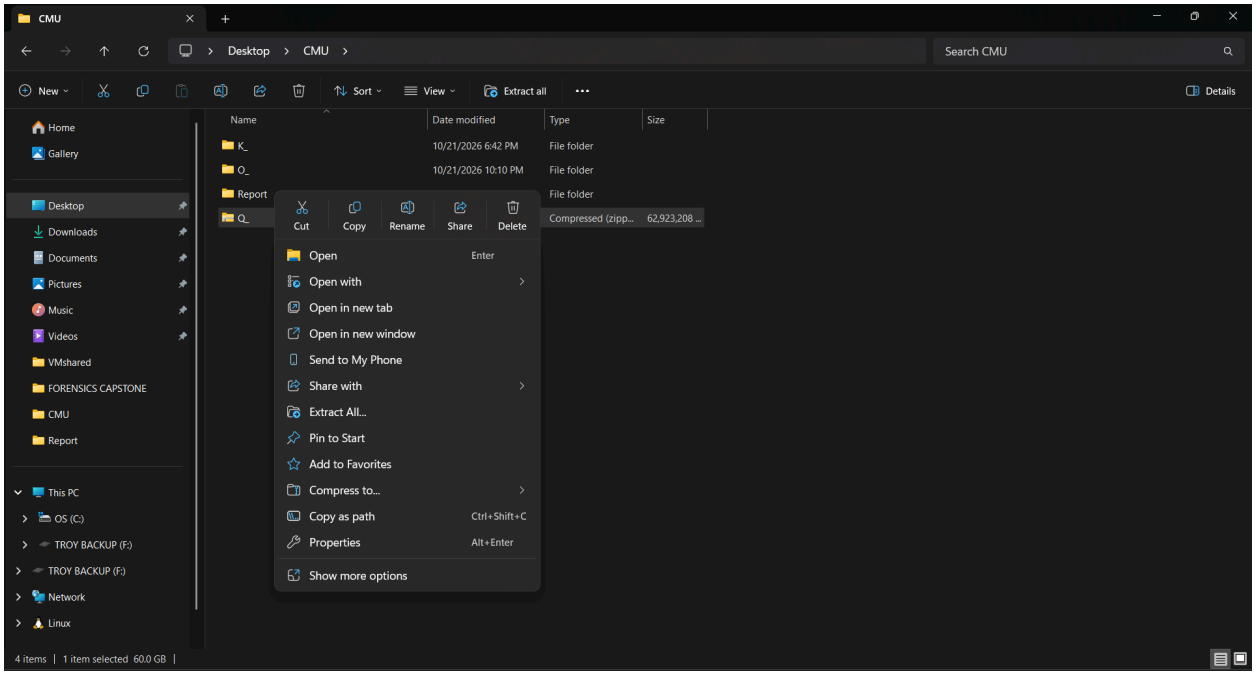
```
vars:
  # Holds the address group vars that would be passed in a Signature.
  address-groups:
    HOME_NET: "[1.1.1.1/32, 127.0.0.1/32, 192.168.1.0/24, 192.168.1.100/32, 192.168.2.1/32, 192.168.2.29/32, 192.168.2.41/32, ::1/128, fe80::1:1/128, fe80::a00:27ff:fe5d:149b/128]"
    EXTERNAL_NET: "[!1.1.1.1/32, !127.0.0.1/32, !192.168.1.0/24, !192.168.1.100/32, !192.168.2.1/32, !192.168.2.29/32, !192.168.2.41/32, !::1/128, !fe80::1:1/128, !fe80::a00:27ff:fe5d:149b/128]"
    DNS_SERVERS: "$HOME_NET"
    SMTP_SERVERS: "$HOME_NET"
    HTTP_SERVERS: "$HOME_NET"
    SQL_SERVERS: "$HOME_NET"
```

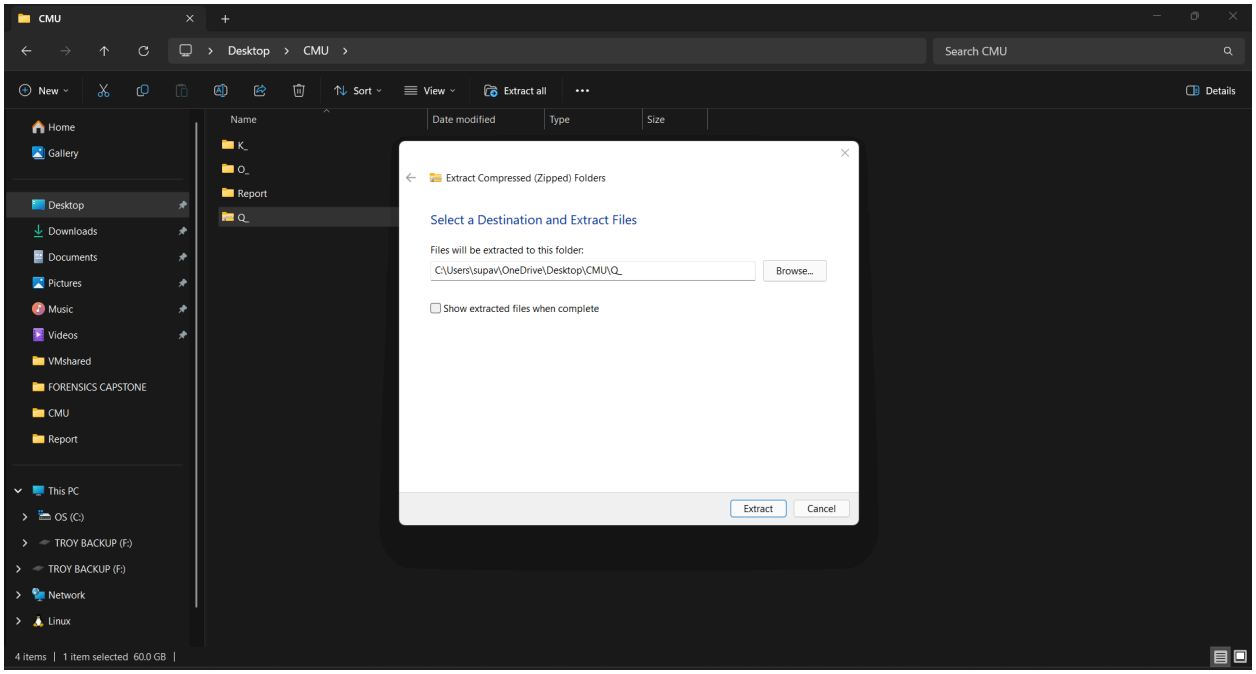
5.2.9 Eugene Image (Q)

5.2.9.1 Eugene_A_0001 Image Download



5.2.9.2 Eugene_A_0002 Image Extraction



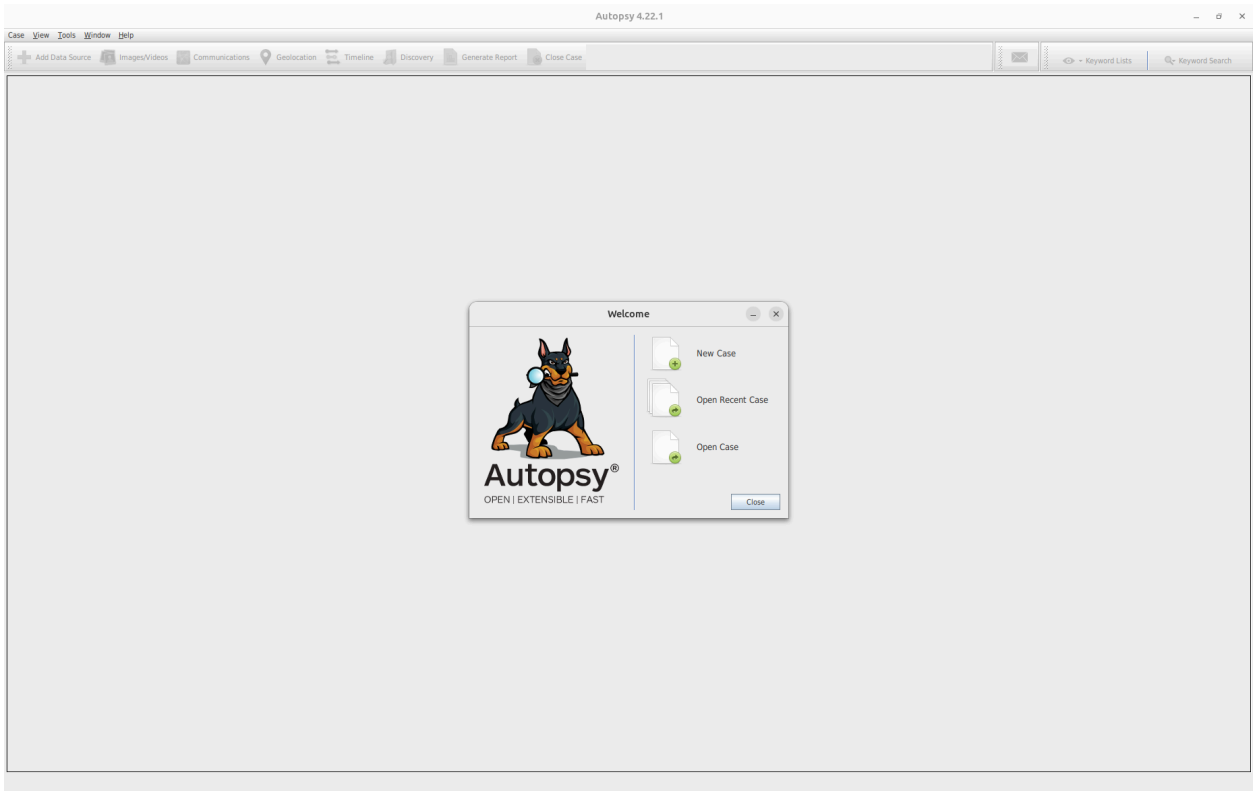

CMU
Desktop
CMU
Search CMU
New
Sort
View
Extract all
Details
Home
Gallery
Desktop
Downloads
Documents
Pictures
Music
Videos
VMshared
FORENSICS CAPSTONE
CMU
Report
This PC
OS (C:)
TROY BACKUP (F:)
TROY BACKUP (F:)
Network
Linux
Name
Date modified
Type
Size
K_
O_
Report
Q_
Extract Compressed (Zipped) Folders
Select a Destination and Extract Files
Files will be extracted to this folder:
C:\Users\supav\OneDrive\Desktop\CMU\Q_
Browse...
Show extracted files when complete
Extract
Cancel
4 items | 1 item selected 60.0 GB

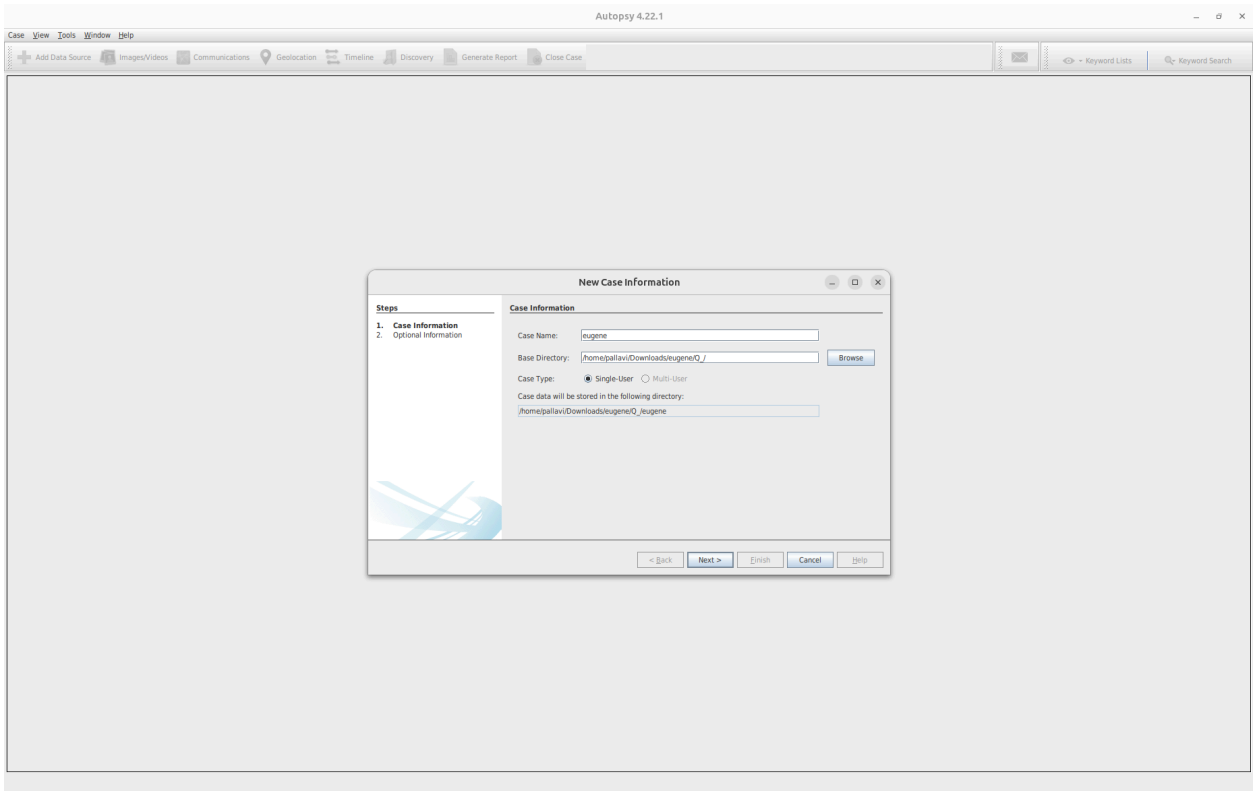
5.2.9.3 Eugene_A_0003 Image Verification

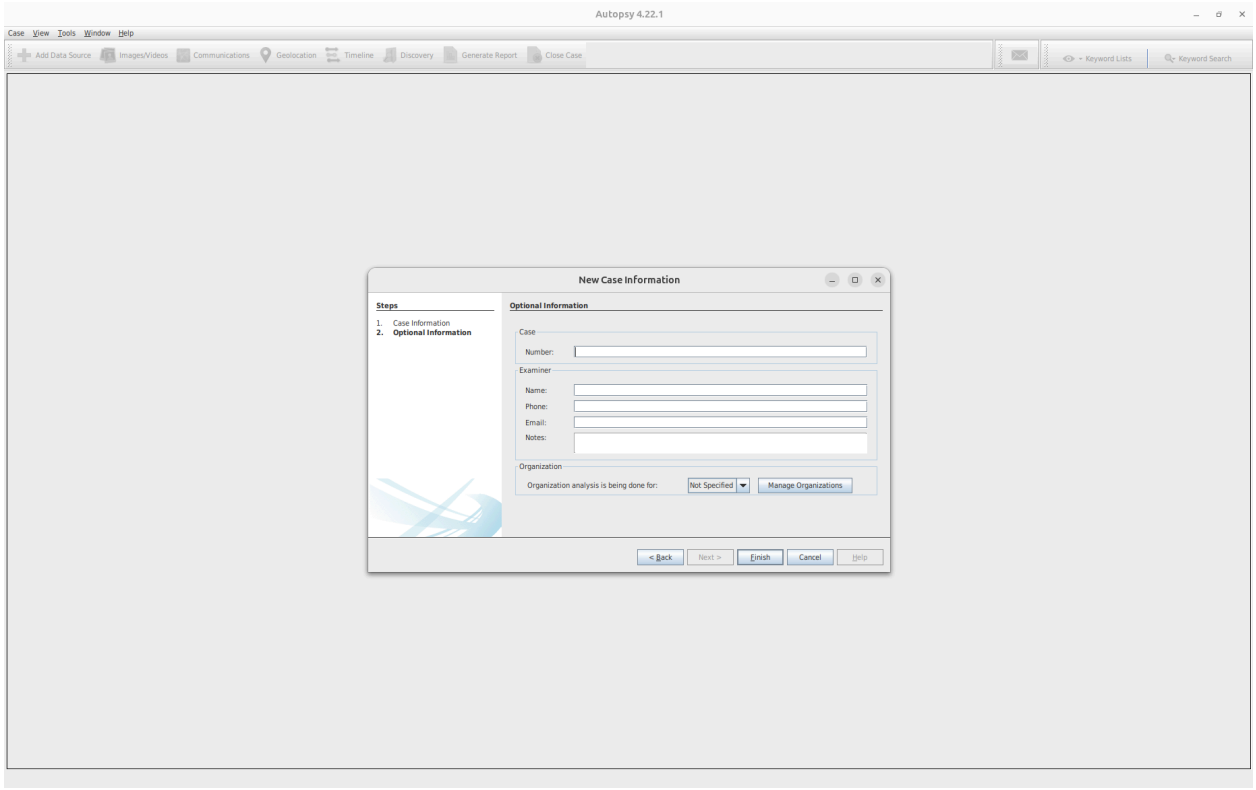
```
pallavi@pallavi-ASUS-Zenbook-14-UX3405CA-UX3405CA:~/Downloads/eugene/Q_$ cat eugenepc.001.txt
Created By AccessData® FTK® Imager 4.2.0.13

Case Information:
Acquired using: ADI4.2.0.13
Case Number: 001
Evidence Number: 001-010
Unique description: eugene
Examiner:
Notes: desktop

------------------------------------------------------------

Information for Q:\eugenepc:

Physical Evidentiary Item (Source) Information:
[Device Info]
 Source Type: Physical
[Drive Geometry]
 Cylinders: 124,828
 Heads: 16
 Sectors per Track: 63
 Bytes per Sector: 512
 Sector Count: 125,827,200
[Physical Drive Information]
 Drive Interface Type: ide
[Image]
 Image Type: VMWare Virtual Disk
 Source data size: 61439 MB
 Sector count:    125827200
[Computed Hashes]
 MD5 checksum:    54f8043c8d73b577846db6713c02210f
 SHA1 checksum:   b57df8fc537112de075373922c7759a0755804fe

Image Information:
 Acquisition started:   Mon Mar  8 16:50:49 2021
 Acquisition finished:  Mon Mar  8 17:17:51 2021
 Segment list:
  Q:\eugenepc.001
  Q:\eugenepc.002
  Q:\eugenepc.003
  Q:\eugenepc.004
  Q:\eugenepc.005
  Q:\eugenepc.006
  Q:\eugenepc.007
  Q:\eugenepc.008
  Q:\eugenepc.009
  Q:\eugenepc.010
  Q:\eugenepc.011
  Q:\eugenepc.012
  Q:\eugenepc.013
  Q:\eugenepc.014
  Q:\eugenepc.015
  Q:\eugenepc.016
  Q:\eugenepc.017
  Q:\eugenepc.018
  Q:\eugenepc.019
  Q:\eugenepc.020
  Q:\eugenepc.021
  Q:\eugenepc.022
  Q:\eugenepc.023
  Q:\eugenepc.024
  Q:\eugenepc.025
  Q:\eugenepc.026
  Q:\eugenepc.027
  Q:\eugenepc.028
```

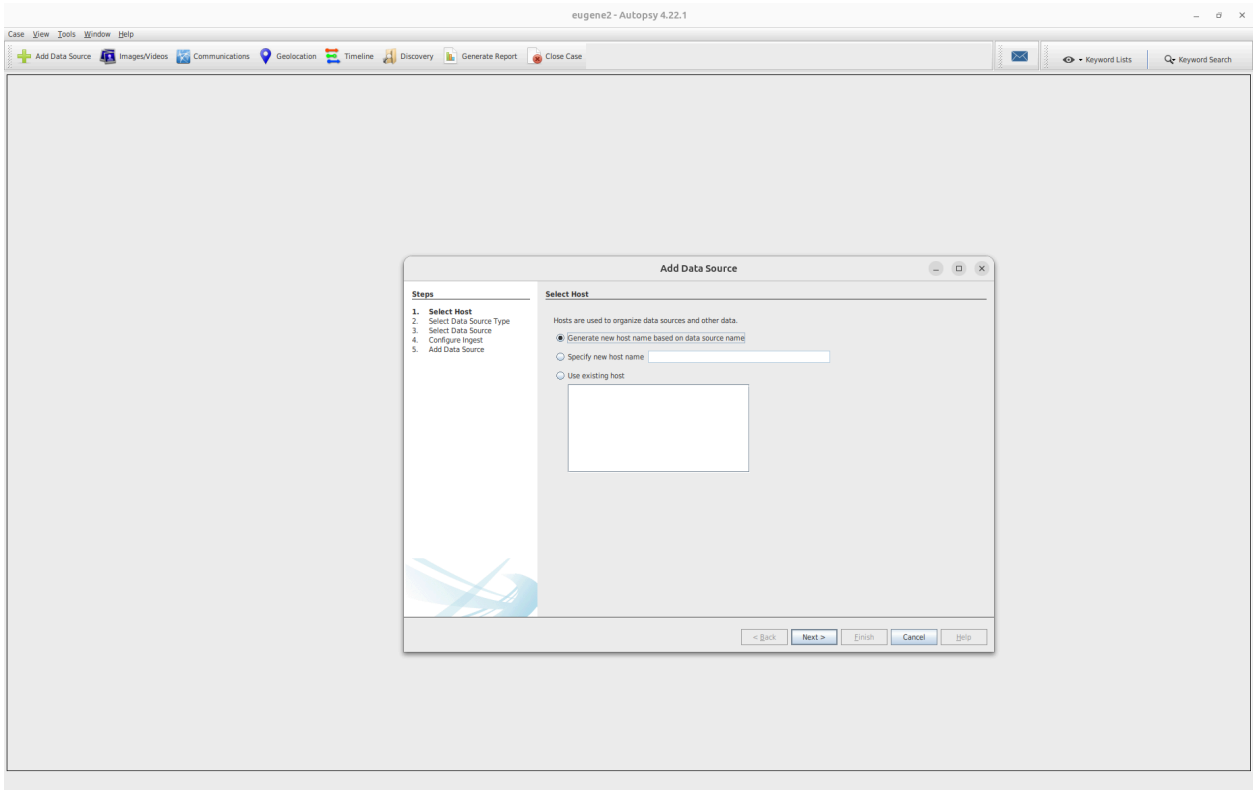
```
pallavi@pallavi-ASUS-Zenbook-14-UX3405CA-UX3405CA:~/Downloads/eugene/Q_$ md5sum eugene.raw
54f8043c8d73b577846db6713c02210f  eugene.raw
pallavi@pallavi-ASUS-Zenbook-14-UX3405CA-UX3405CA:~/Downloads/eugene/Q_$ sha1sum eugene.raw
b57df8fc537112de075373922c7759a0755804fe  eugene.raw
pallavi@pallavi-ASUS-Zenbook-14-UX3405CA-UX3405CA:~/Downloads/eugene/Q_$
```

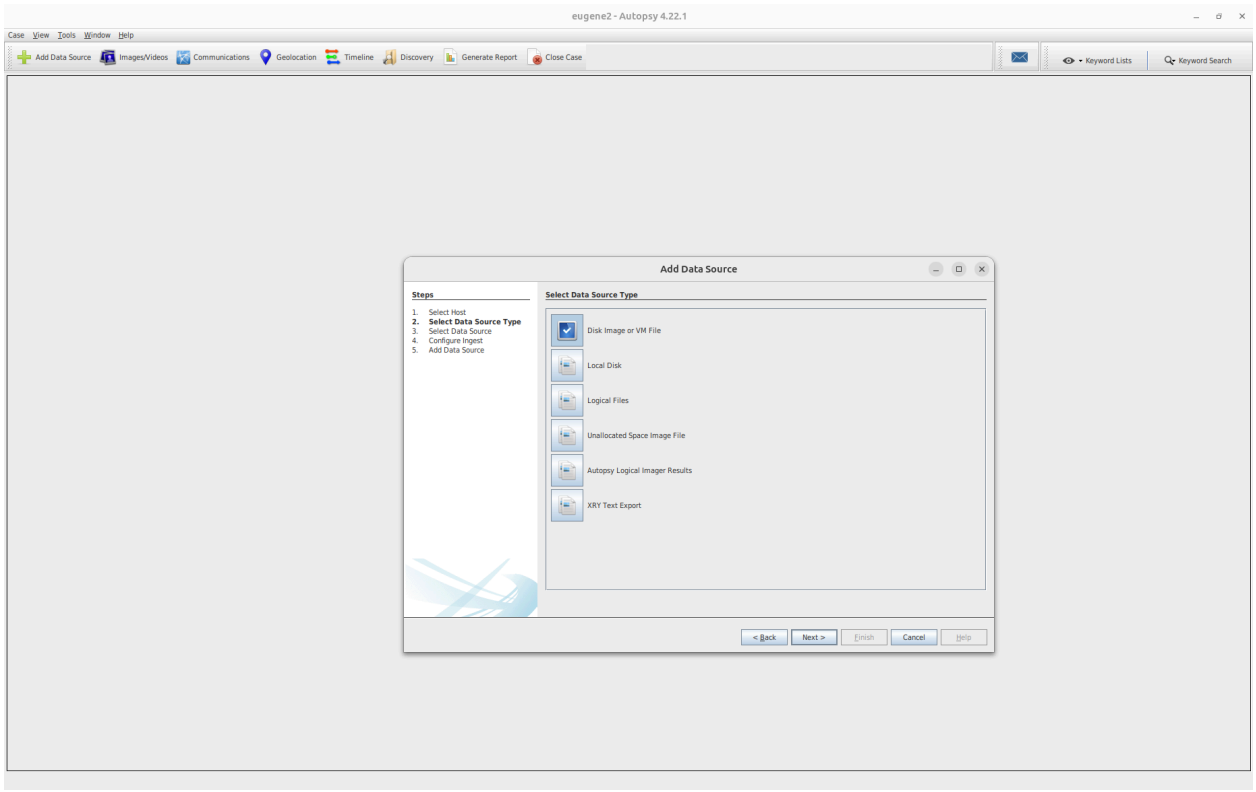
5.2.9.4 Eugene_A_0004 Autopsy Case

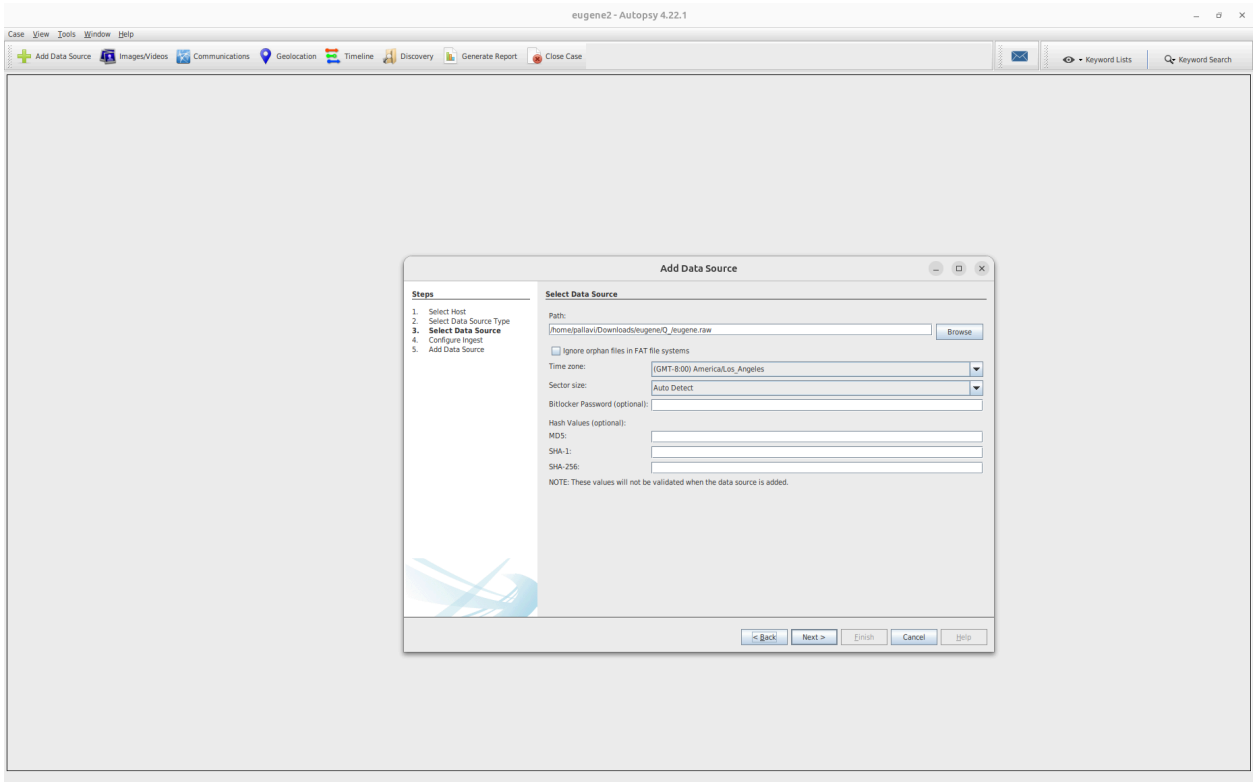


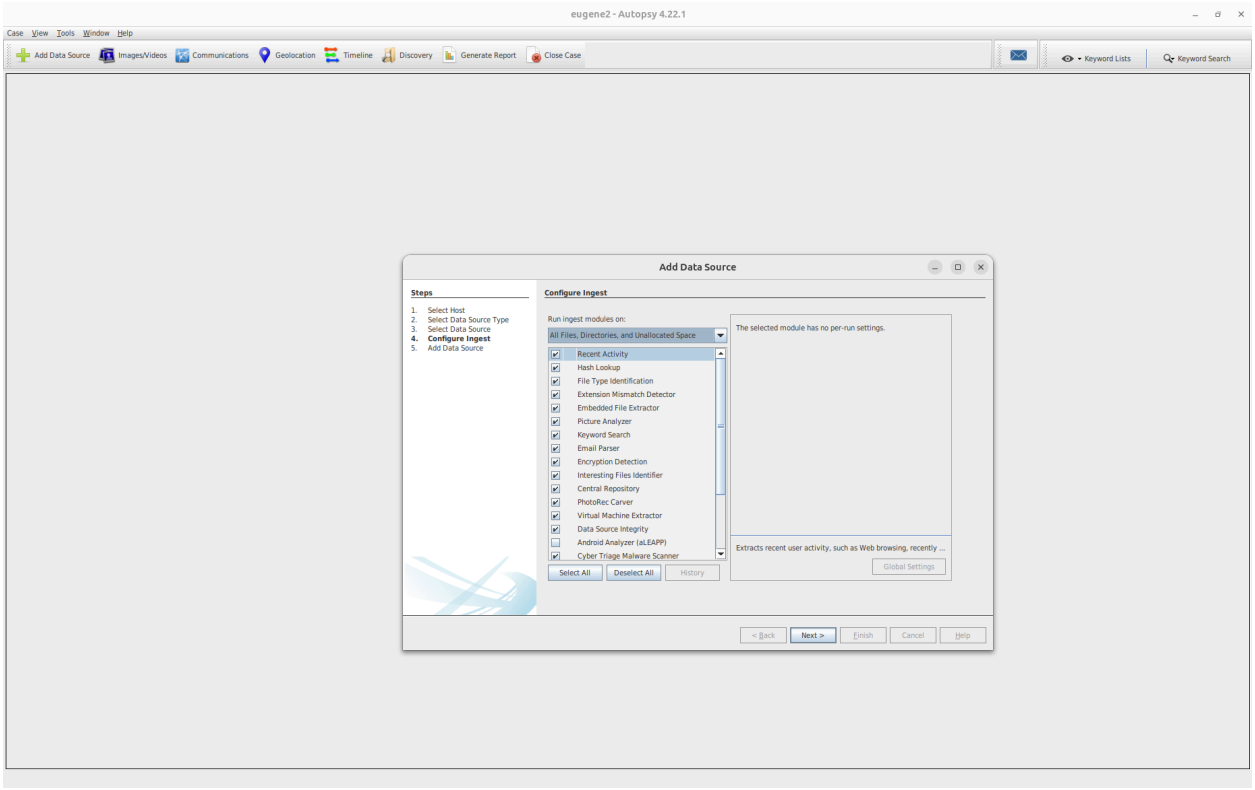


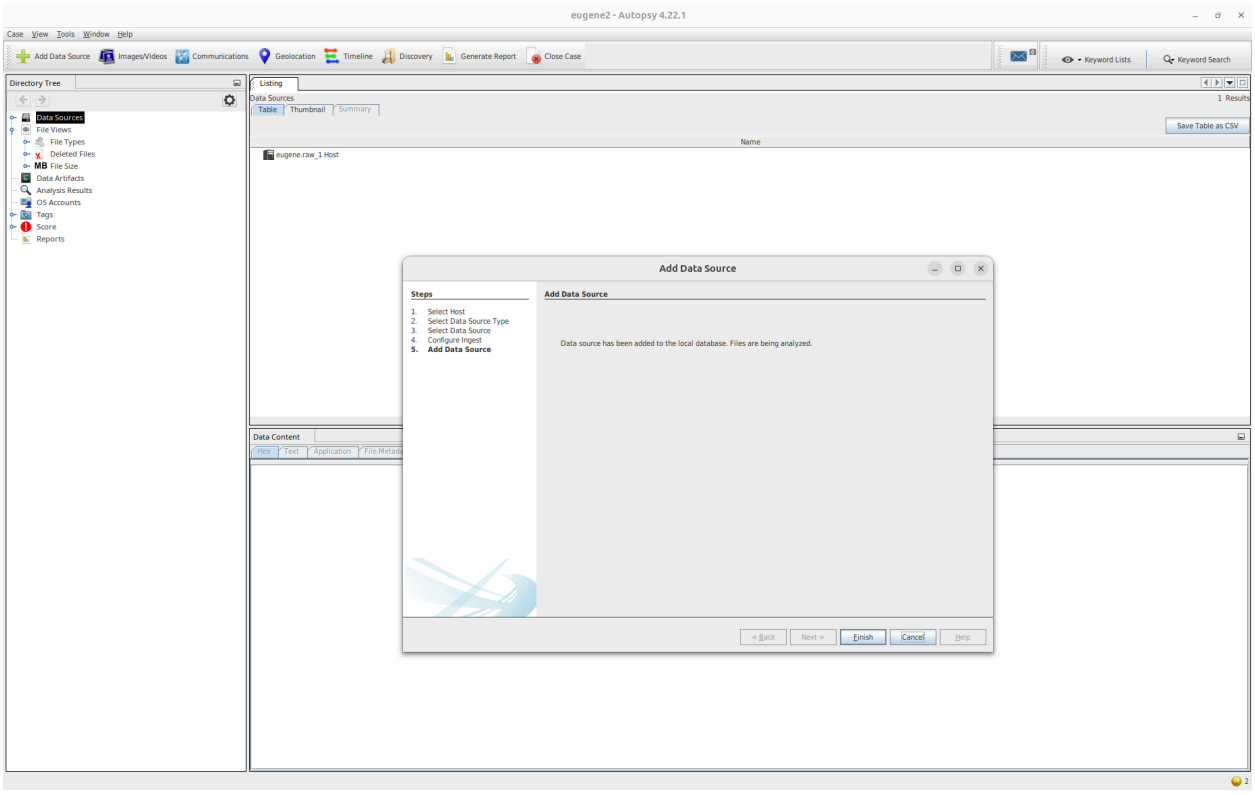

Autopsy 4.22.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Optional Information
Case
Number:
Examiner
Name:
Phone:
Email:
Notes:
Organization
Organization analysis is being done for:
Not Specified
Manage Organizations
< Back
Next >
Finish
Cancel
Help


eugene2 - Autopsy 4.22.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Host
Hosts are used to organize data sources and other data.
Generate new host name based on data source name
Specify new host name
Use existing host
< Back
Next >
Finish
Cancel
Help

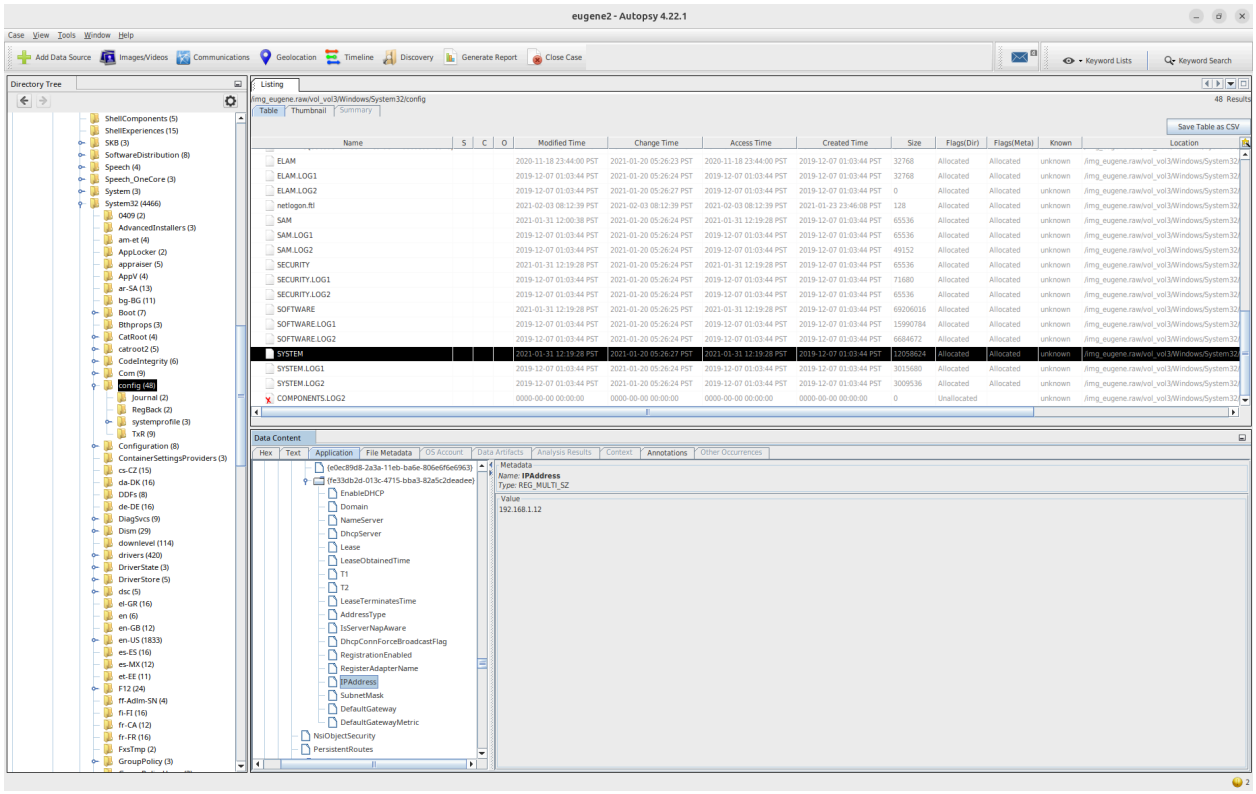

eugene2 - Autopsy 4.22.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source Type
Disk Image or VM File
Local Disk
Logical Files
Unallocated Space Image File
Autopsy Logical Imager Results
XRY Text Export
< Back
Next >
Finish
Cancel
Help

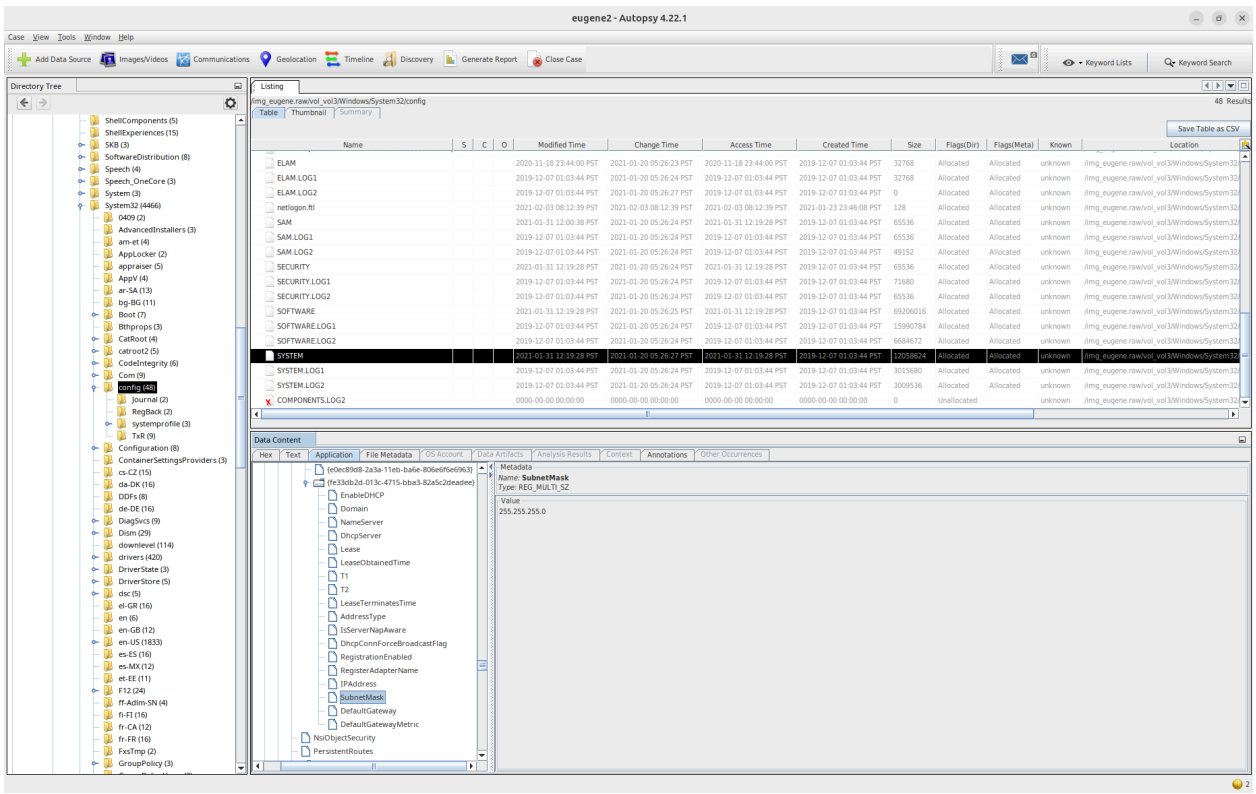

eugene2 - Autopsy 4.22.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source
Path:
/home/pallavi/Downloads/eugene/0_Jeugene.raw
Browse
Ignore orphan files in FAT file systems
Time zone:
(GMT-8:00) America/Los_Angeles
Sector size:
Auto Detect
Bitlocker Password (optional):
Hash Values (optional):
MD5:
SHA-1:
SHA-256:
NOTE: These values will not be validated when the data source is added.
< Back
Next >
Finish
Cancel
Help

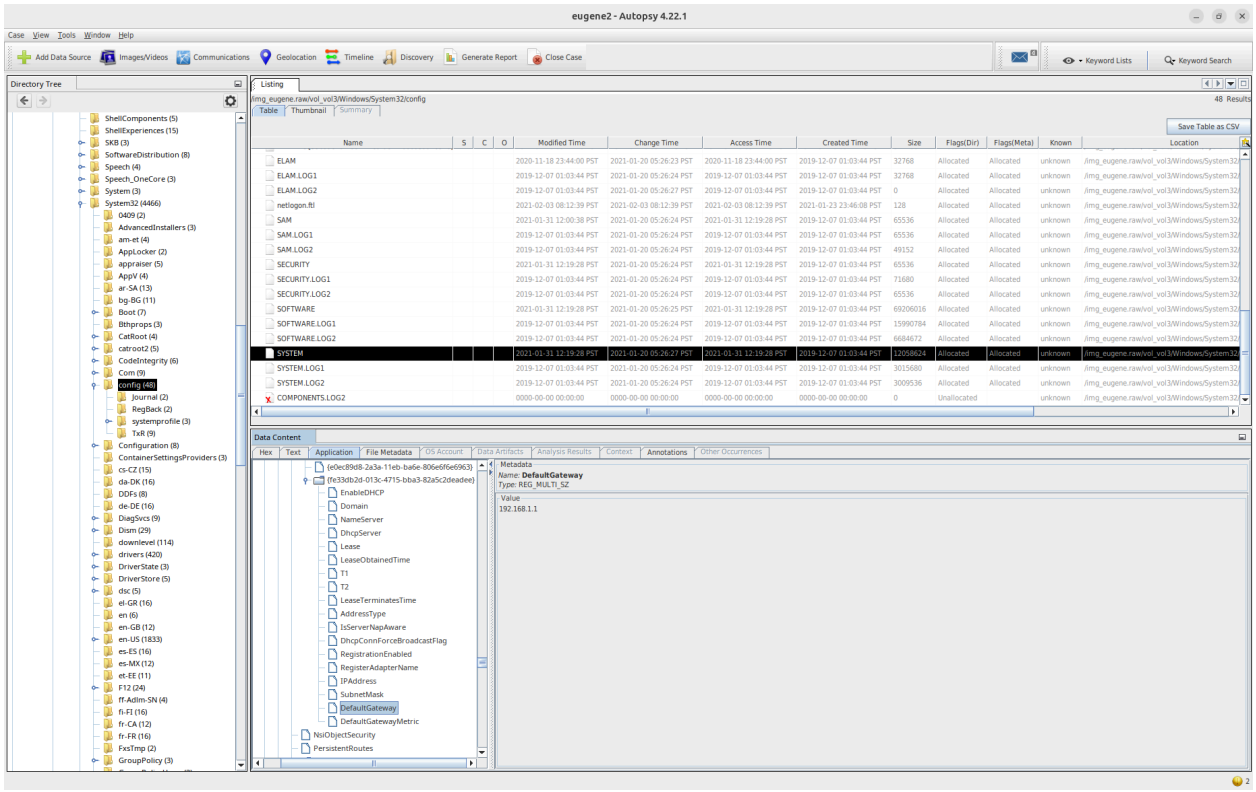

eugene2 - Autopsy 4.22.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
The selected module has no per-run settings.
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
Data Source Integrity
Android Analyzer (aLEAPP)
Cyber Triage Malware Scanner
Extracts recent user activity, such as Web browsing, recently ...
Global Settings
Select All
Deselect All
History
< Back
Next >
Finish
Cancel
Help

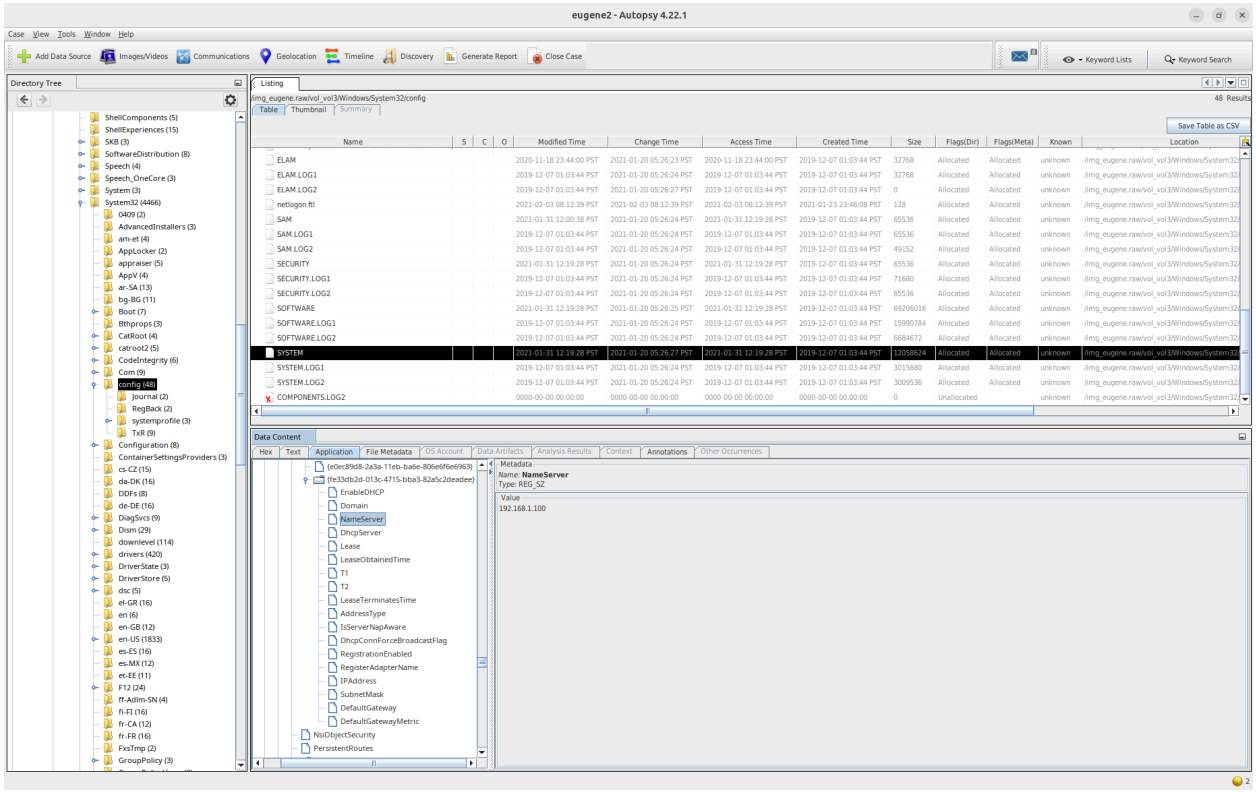

eugene2 - Autopsy 4.22.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Directory Tree
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Analysis Results
OS Accounts
Tags
Score
Reports
Listing
Data Sources
Table
Thumbnail
Summary
1 Results
Save Table as CSV
Name
eugene.raw_1 Host
Data Content
Hex
Text
Application
File Metada
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
Data source has been added to the local database. Files are being analyzed.
< Back
Next >
Finish
Cancel
Help

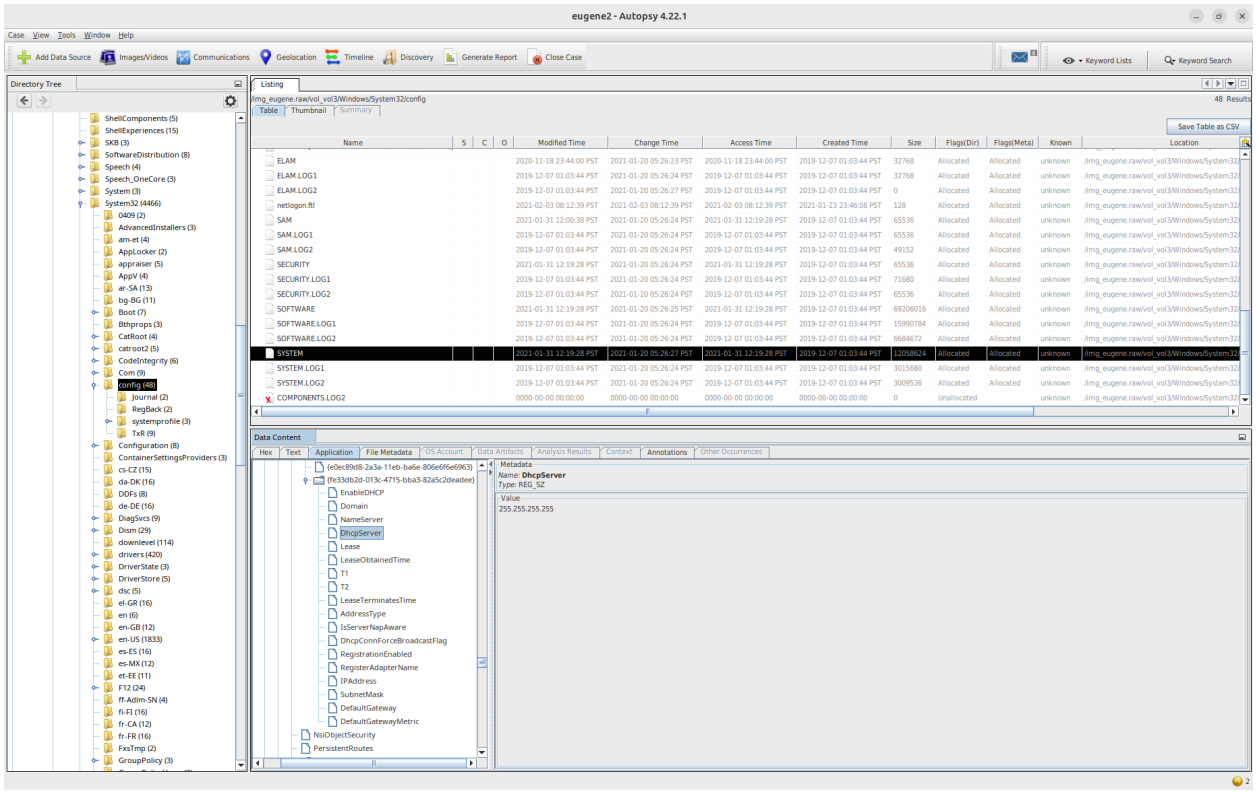
5.2.9.5 Eugene_A_0005 IP Addresses



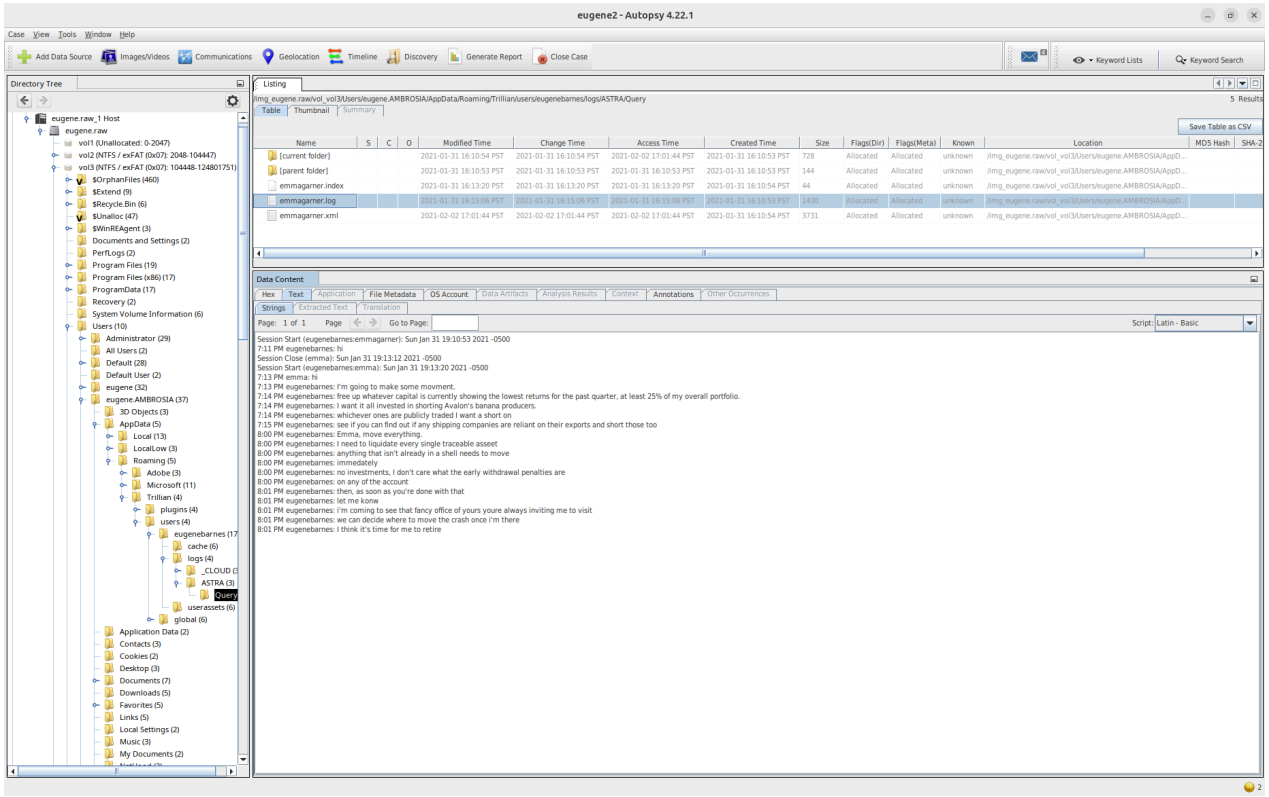



eugene2 - Autopsy 4.22.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Directory Tree
Listing
/img_eugene.raw/vol_vol3/Windows/System32/config
48 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(Dir)
Flags(Meta)
Known
Location
ELAM
ELAM.LOG1
ELAM.LOG2
netlogon.ftl
SAM
SAM.LOG1
SAM.LOG2
SECURITY
SECURITY.LOG1
SECURITY.LOG2
SOFTWARE
SOFTWARE.LOG1
SOFTWARE.LOG2
SYSTEM
SYSTEM.LOG1
SYSTEM.LOG2
COMPONENTS.LOG2
Data Content
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Metadata
Name: DefaultGateway
Type: REG_MULTI_SZ
Value
192.168.1.1
EnableDHCP
Domain
NameServer
DhcpServer
Lease
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
SubnetMask
DefaultGateway
DefaultGatewayMetric
NsiObjectSecurity
PersistentRoutes

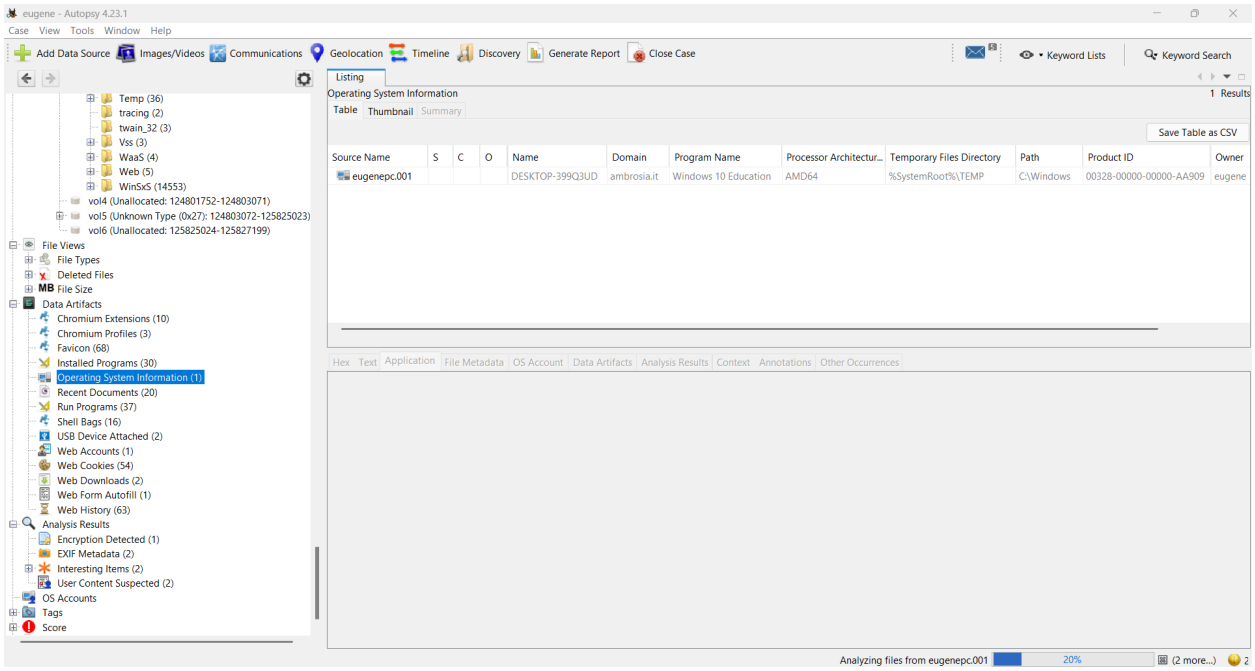

eugene2 - Autopsy 4.22.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Directory Tree
Listing
/img_eugene.raw/vol_vol3/Windows/System32/config
48 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(Dir)
Flags(Meta)
Known
Location
ELAM
ELAM.LOG1
ELAM.LOG2
netlogon.ftl
SAM
SAM.LOG1
SAM.LOG2
SECURITY
SECURITY.LOG1
SECURITY.LOG2
SOFTWARE
SOFTWARE.LOG1
SOFTWARE.LOG2
SYSTEM
SYSTEM.LOG1
SYSTEM.LOG2
COMPONENTS.LOG2
Data Content
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Metadata
Name: NameServer
Type: REG_SZ
Value
192.168.1.100
EnableDHCP
Domain
NameServer
DhcpServer
Lease
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
SubnetMask
DefaultGateway
DefaultGatewayMetric
NsiObjectSecurity
PersistentRoutes



5.2.9.6 Eugene_A_0006 Trillian Logs

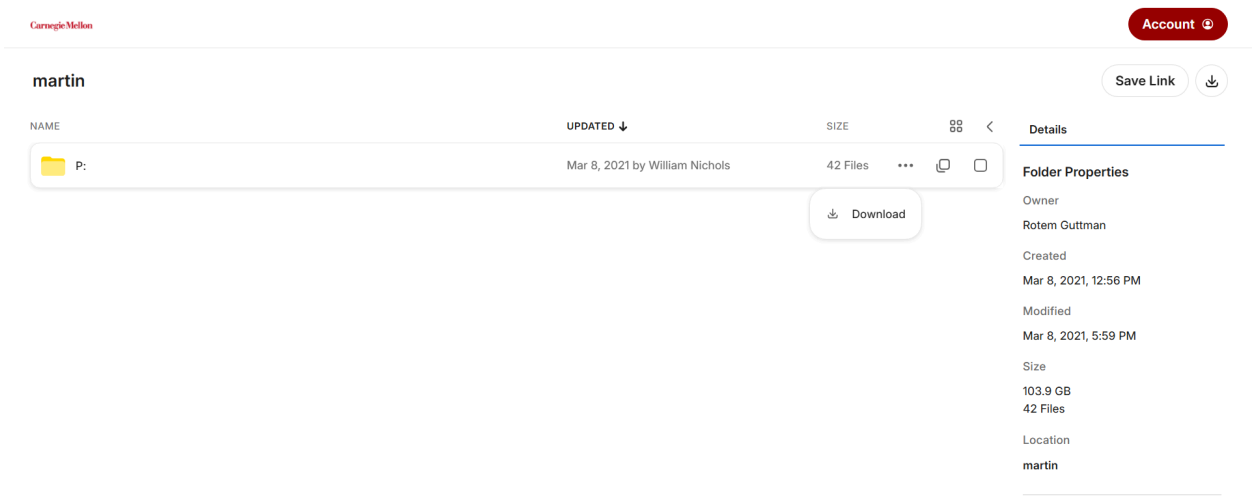


5.2.9.7 Eugene_A_0007 Operating System

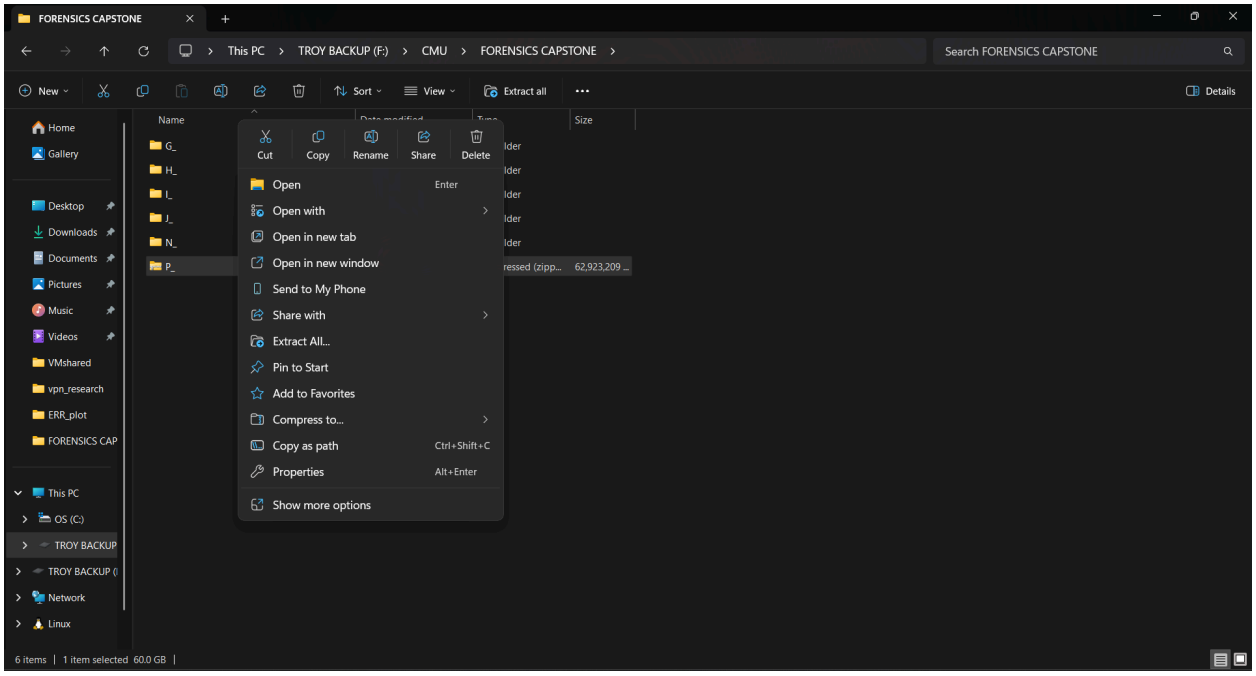


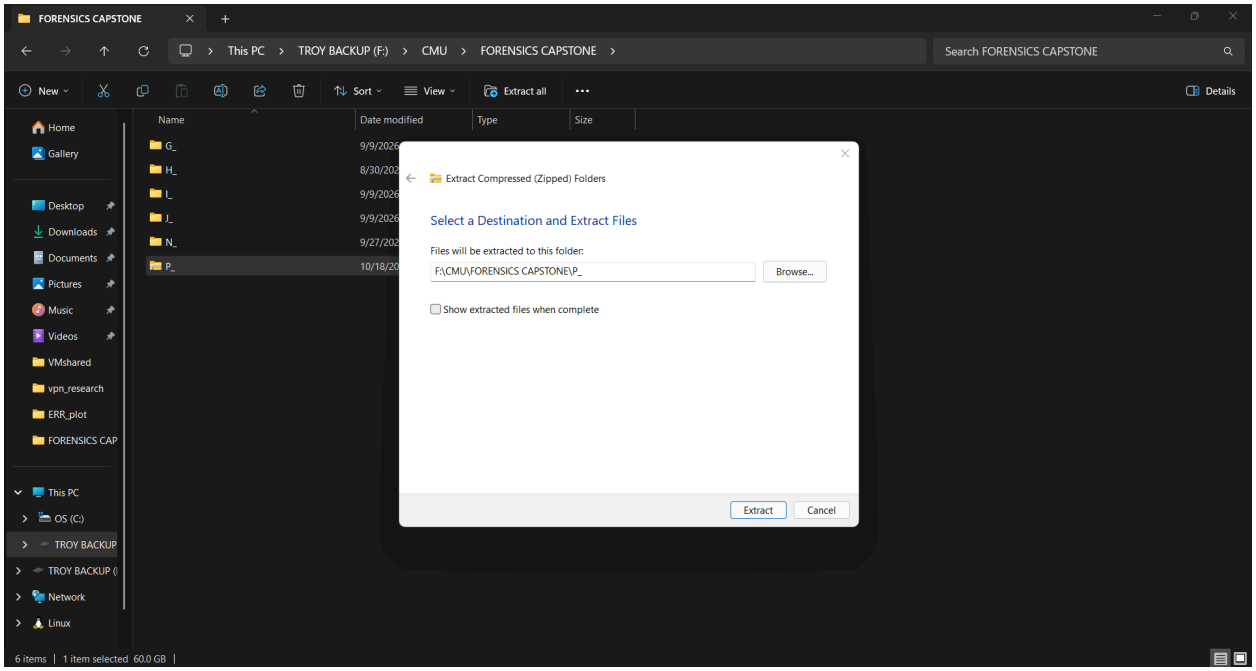
5.2.10 Martin Image (P)

5.2.10.1 Martin_A_0001 Image Download

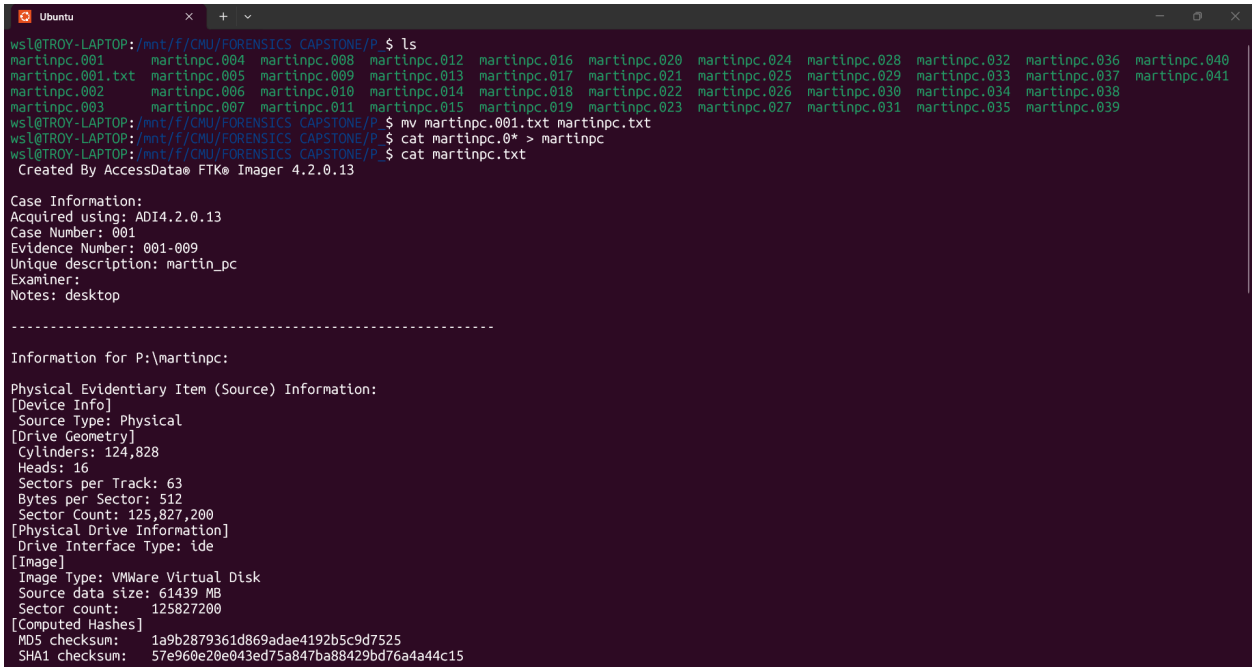


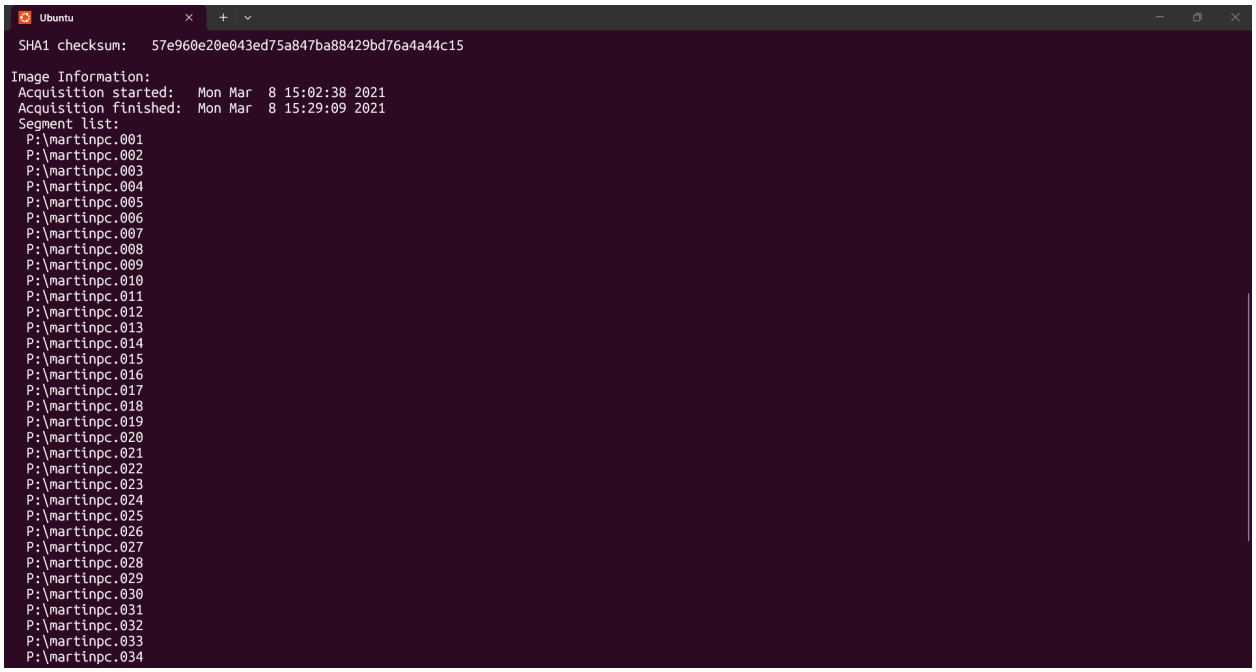
5.2.10.2 Martin_A_0002 Image Extraction

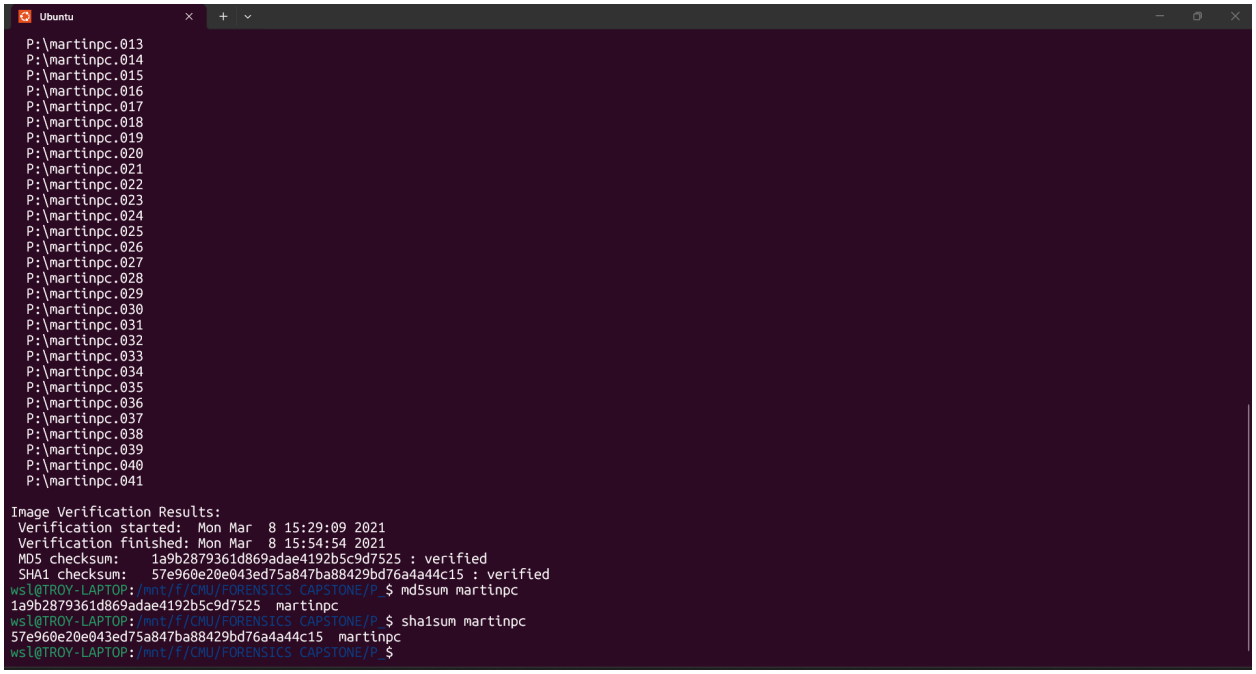




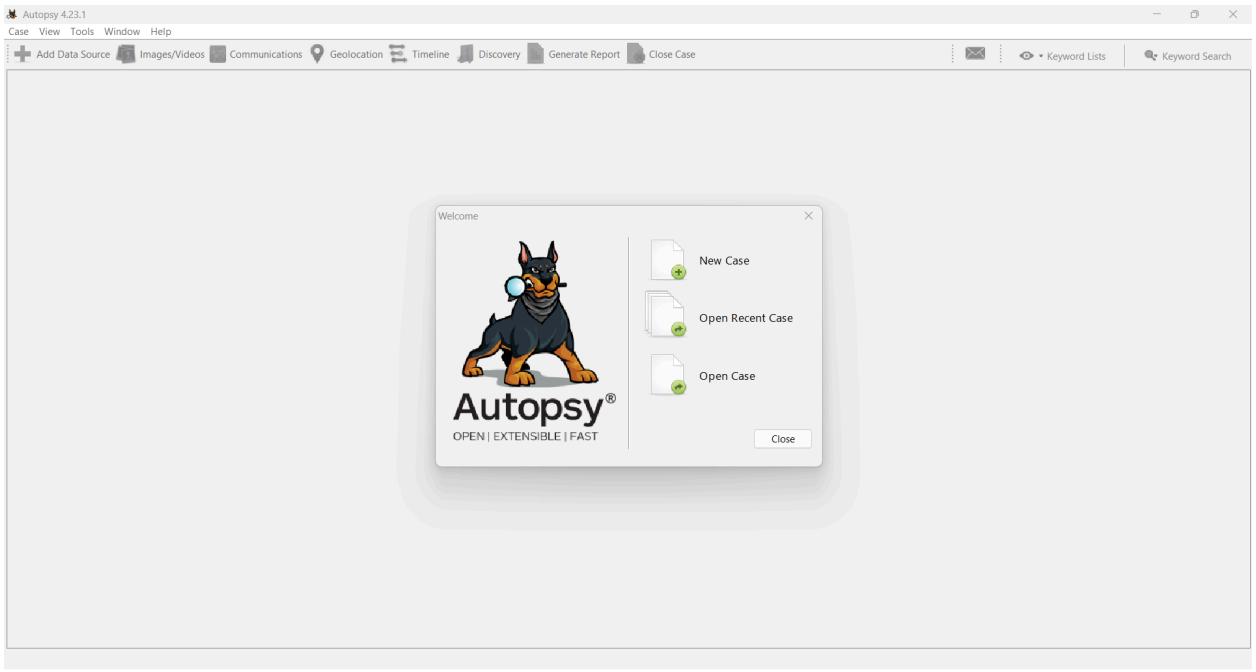
5.2.10.3 Martin_A_0003 Image Verification

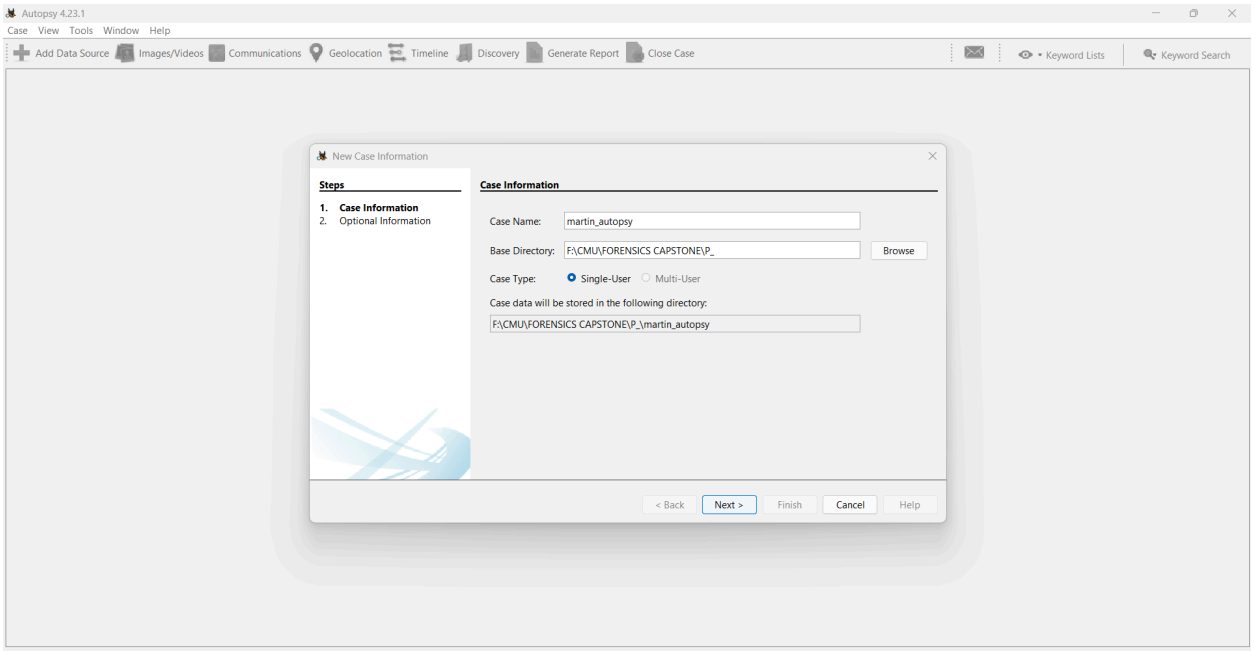


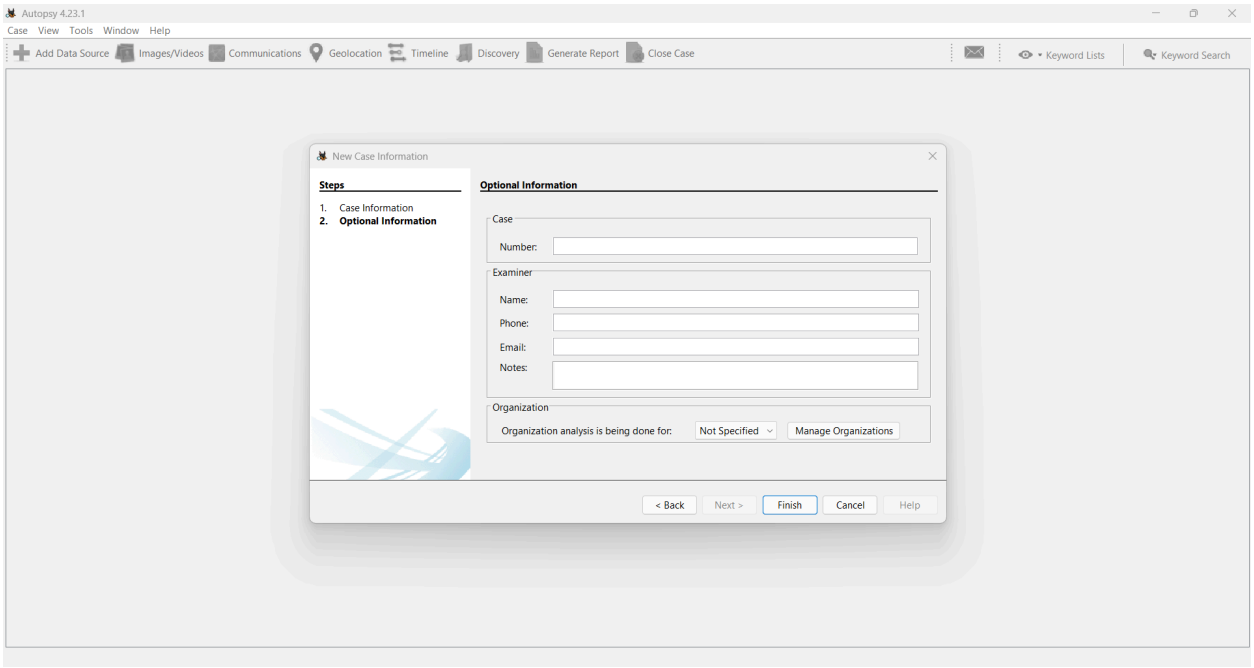


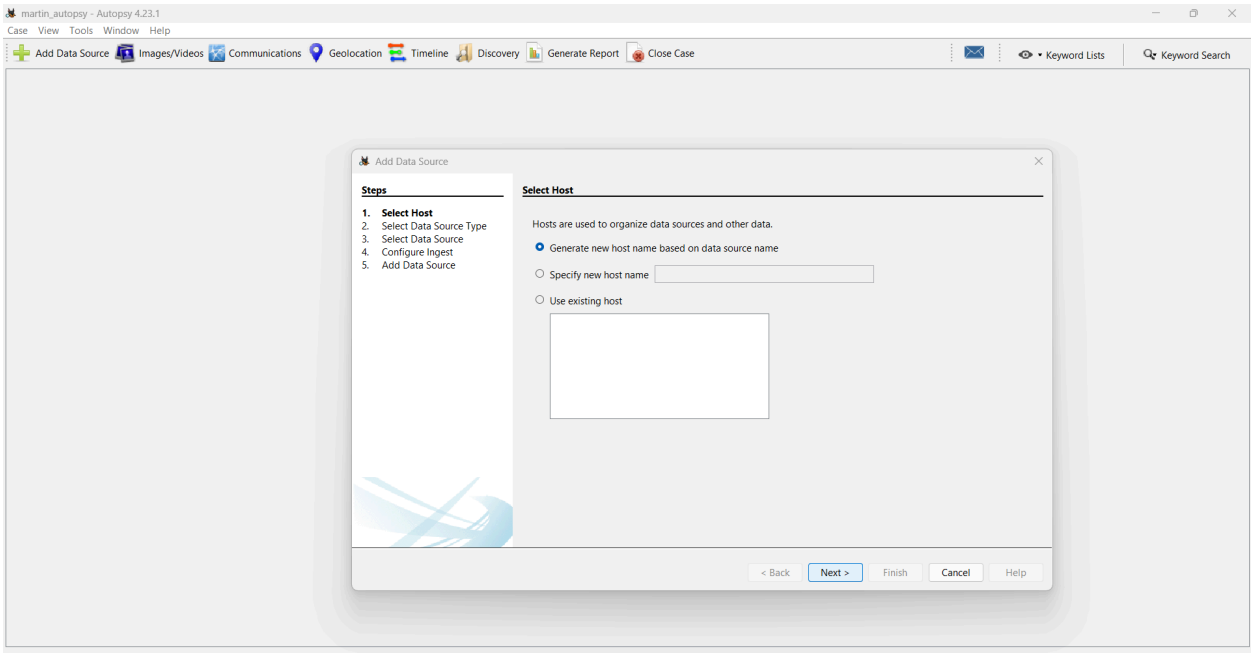


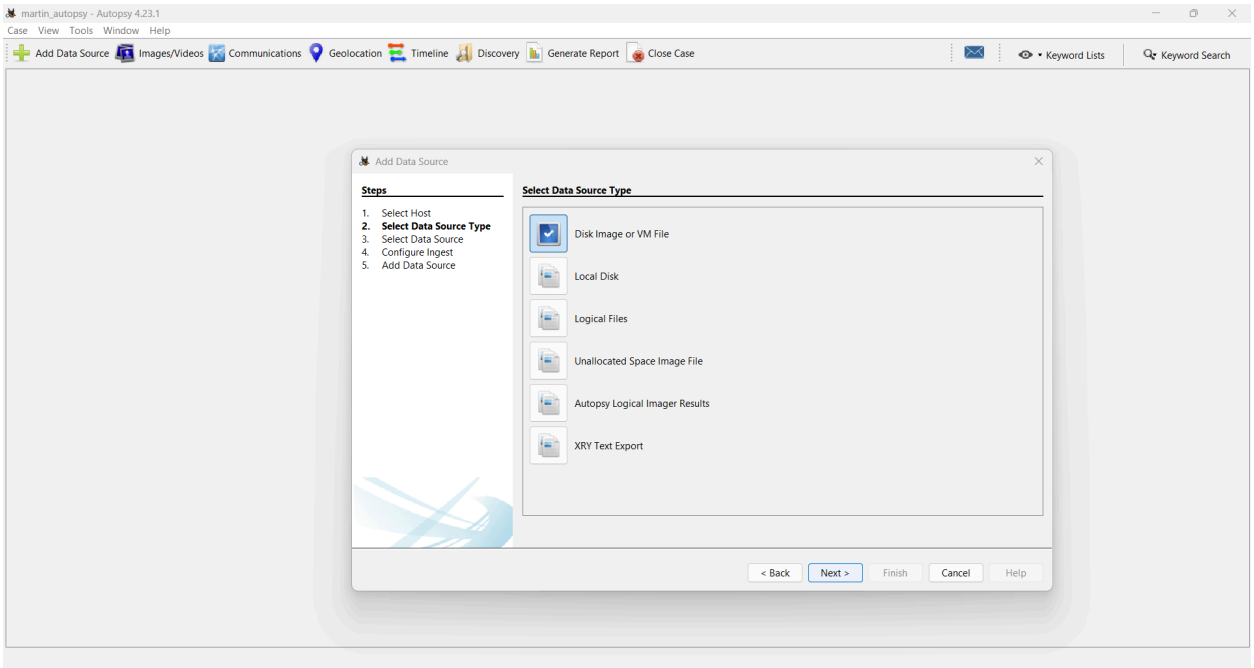
5.2.10.4 Martin_A_0004 Autopsy Case

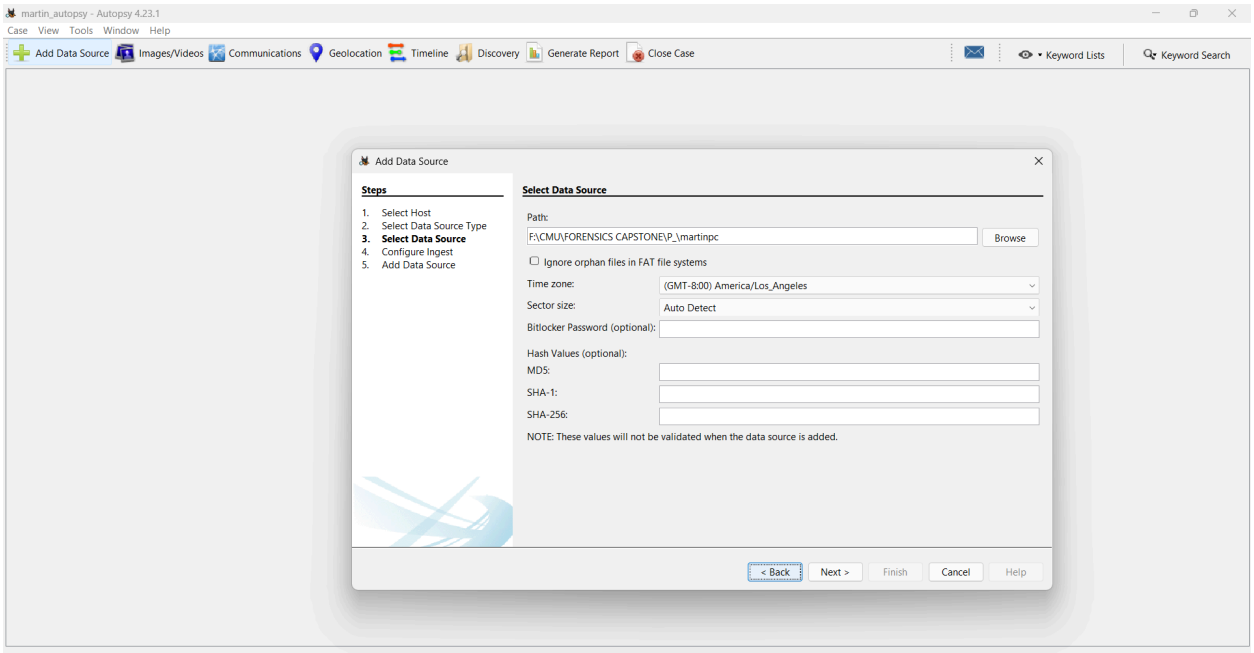


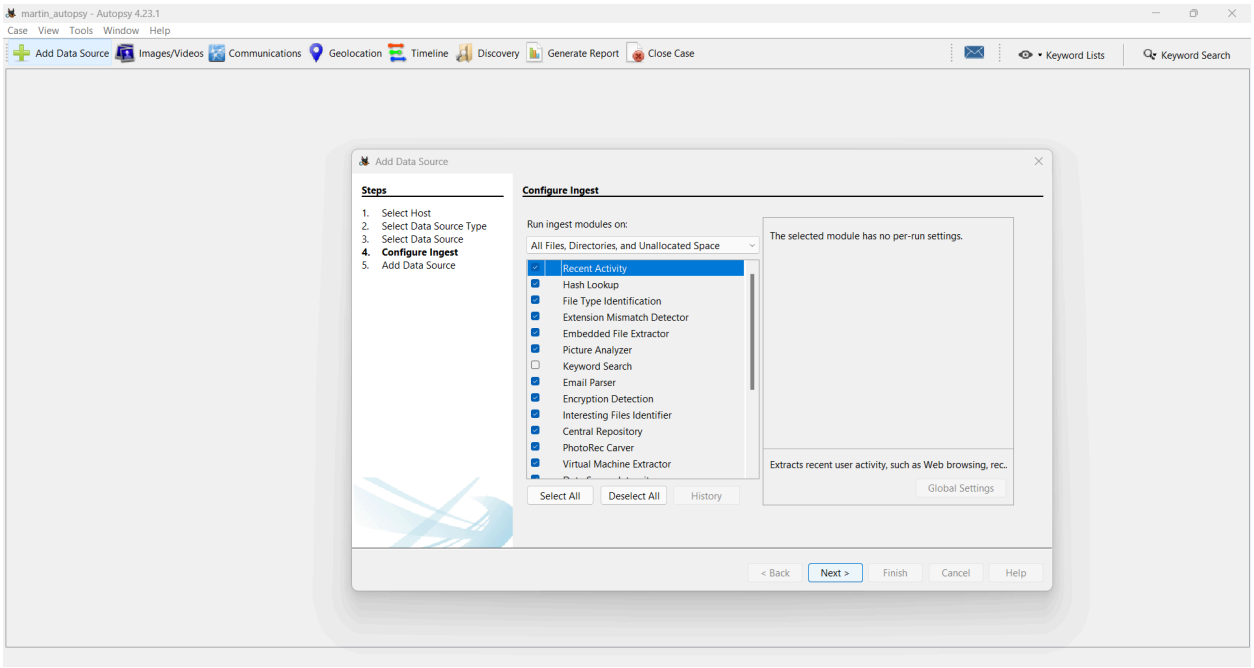

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Case Information
Case Name: martin_autopsy
Base Directory: F:\CMU\FORENSICS CAPSTONE\P_
Browse
Case Type: Single-User Multi-User
Case data will be stored in the following directory:
F:\CMU\FORENSICS CAPSTONE\P_\martin_autopsy
< Back Next > Finish Cancel Help

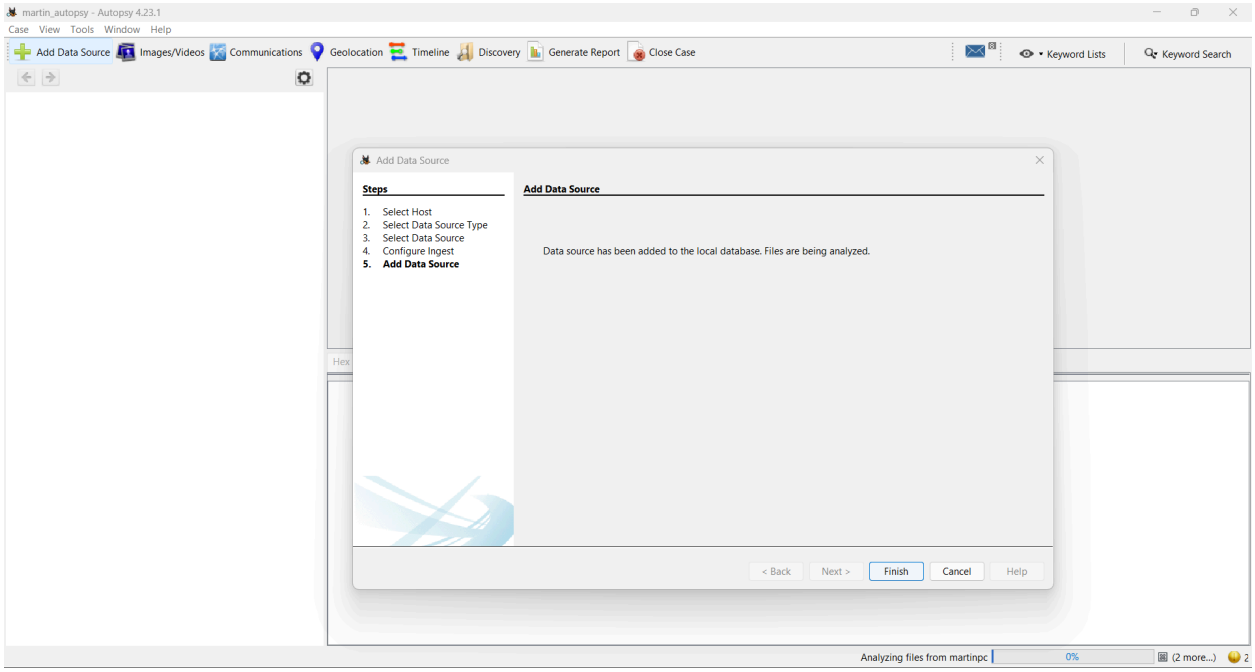

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Optional Information
Case
Number:
Examiner
Name:
Phone:
Email:
Notes:
Organization
Organization analysis is being done for: Not Specified Manage Organizations
< Back Next > Finish Cancel Help

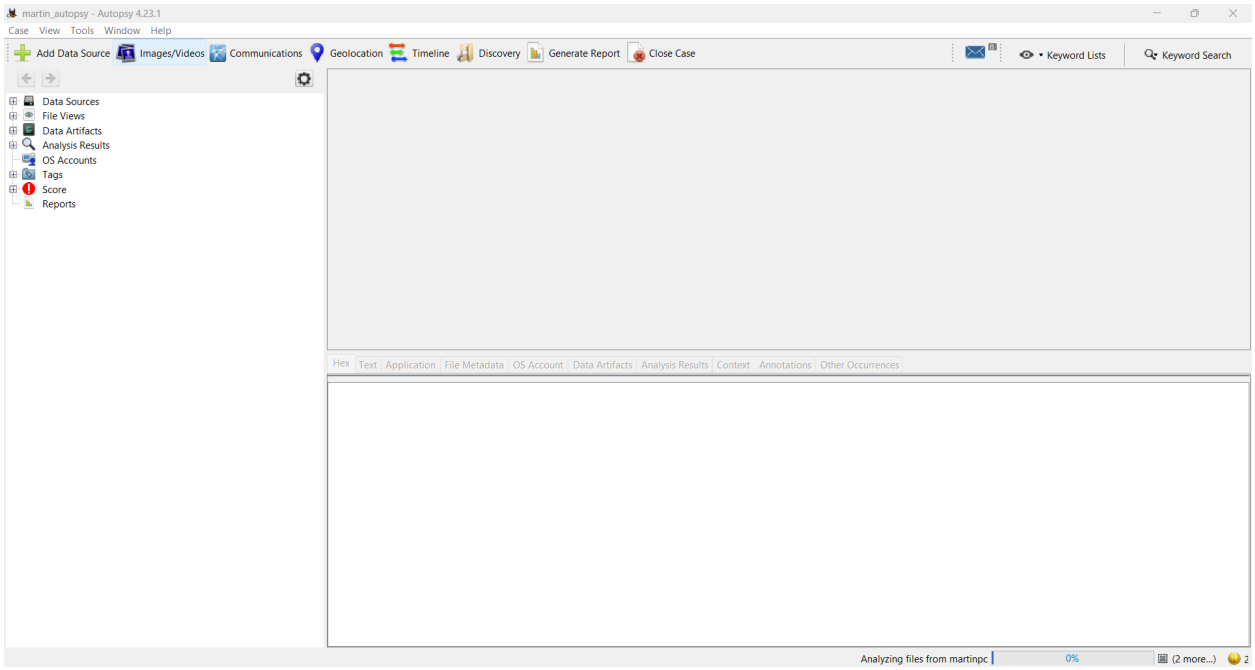

martin_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Host
Hosts are used to organize data sources and other data.
Generate new host name based on data source name
Specify new host name
Use existing host
< Back
Next >
Finish
Cancel
Help


martin_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source Type
Disk Image or VM File
Local Disk
Logical Files
Unallocated Space Image File
Autopsy Logical Imager Results
XRY Text Export
< Back
Next >
Finish
Cancel
Help

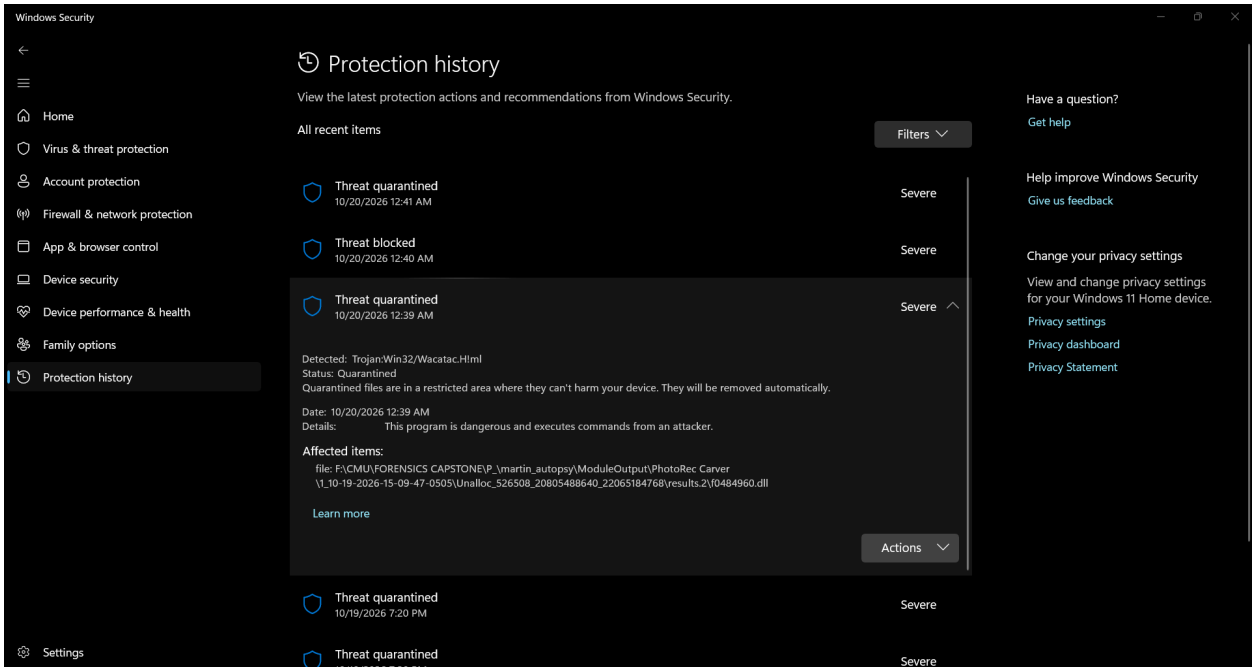

martin_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source
Path:
F:\CMU\FORENSICS CAPSTONE\P_\martinpc
Browse
Ignore orphan files in FAT file systems
Time zone:
(GMT-8:00) America/Los_Angeles
Sector size:
Auto Detect
Bitlocker Password (optional):
Hash Values (optional):
MD5:
SHA-1:
SHA-256:
NOTE: These values will not be validated when the data source is added.
< Back
Next >
Finish
Cancel
Help

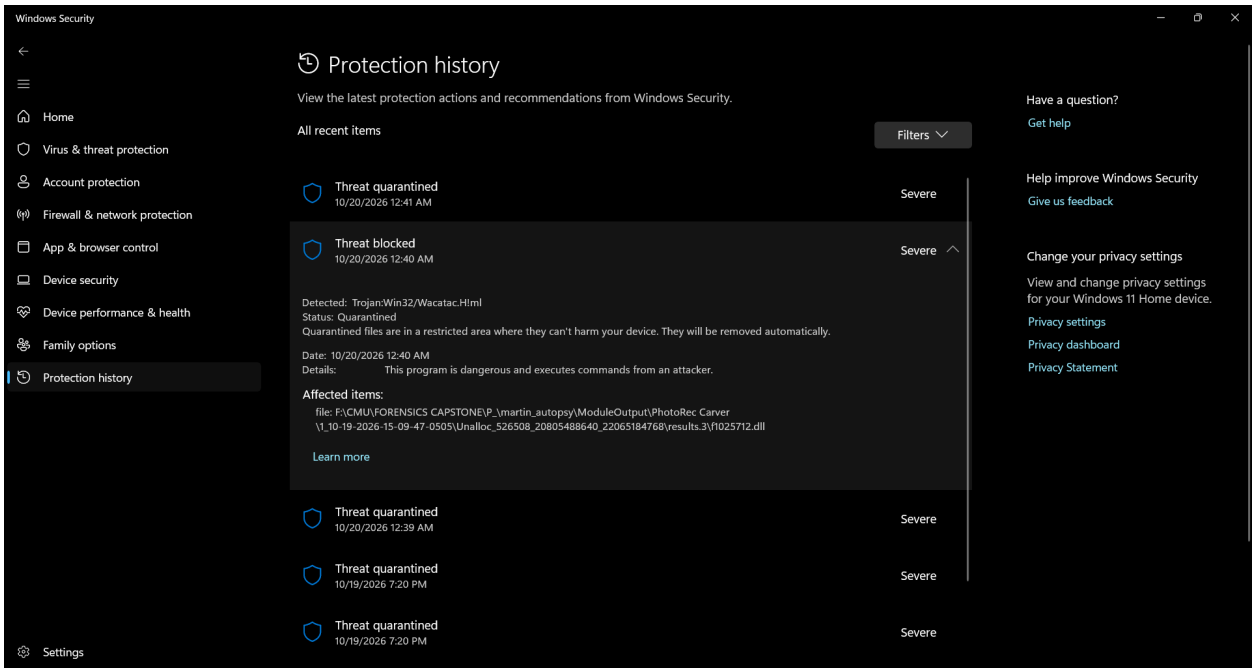

martin_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
Select All
Deselect All
History
The selected module has no per-run settings.
Extracts recent user activity, such as Web browsing, rec..
Global Settings
< Back
Next >
Finish
Cancel
Help

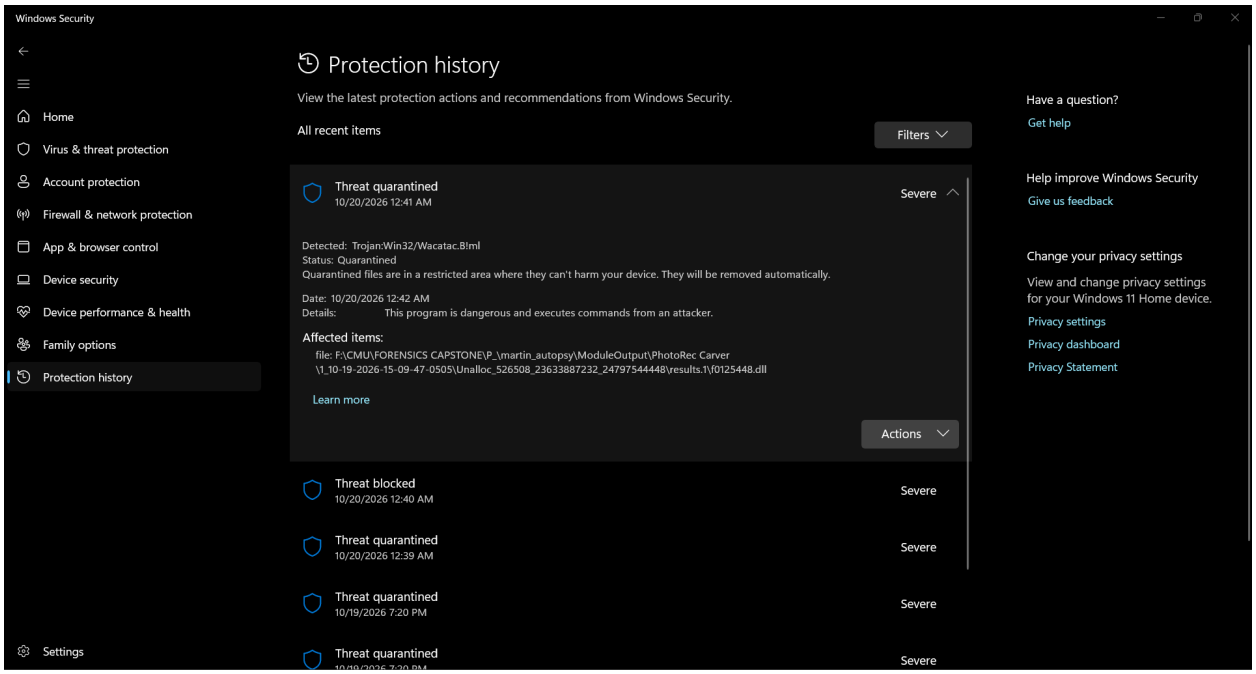

martin_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
Data source has been added to the local database. Files are being analyzed.
< Back
Next >
Finish
Cancel
Help
Hex
Analyzing files from martinpc
0%
(2 more...)


martin_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Data Sources
File Views
Data Artifacts
Analysis Results
OS Accounts
Tags
Score
Reports
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Analyzing files from martinpc
0%
(2 more...)

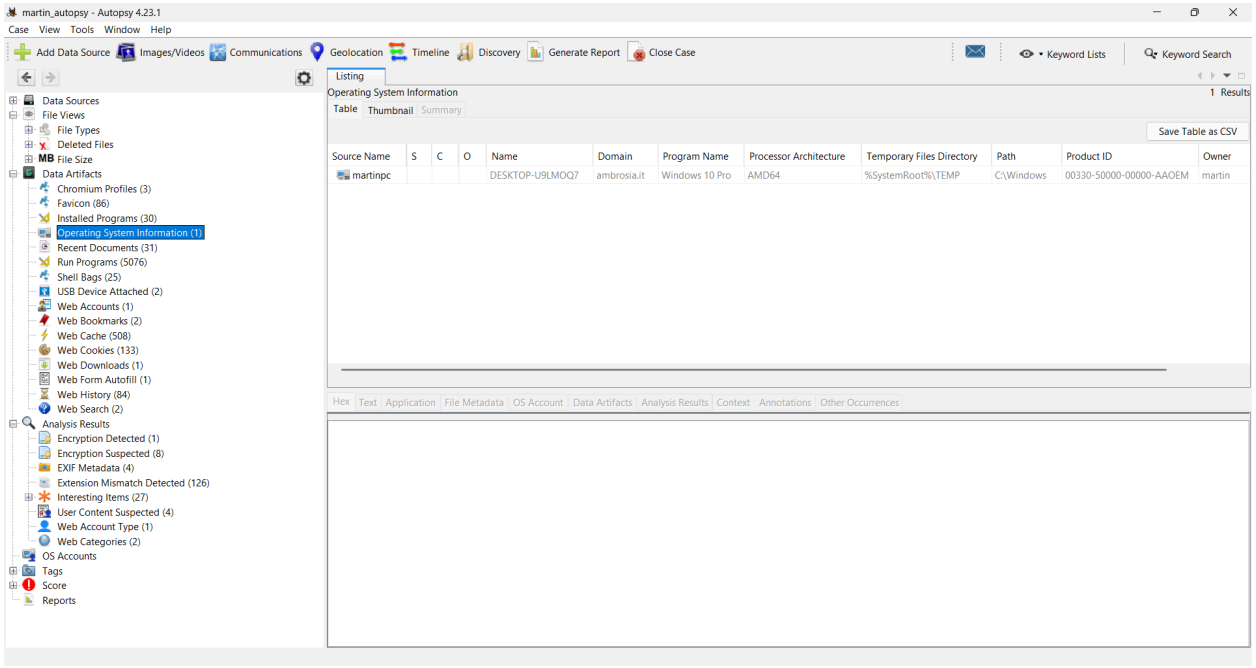
5.2.10.5 Martin_A_0005 Windows Security



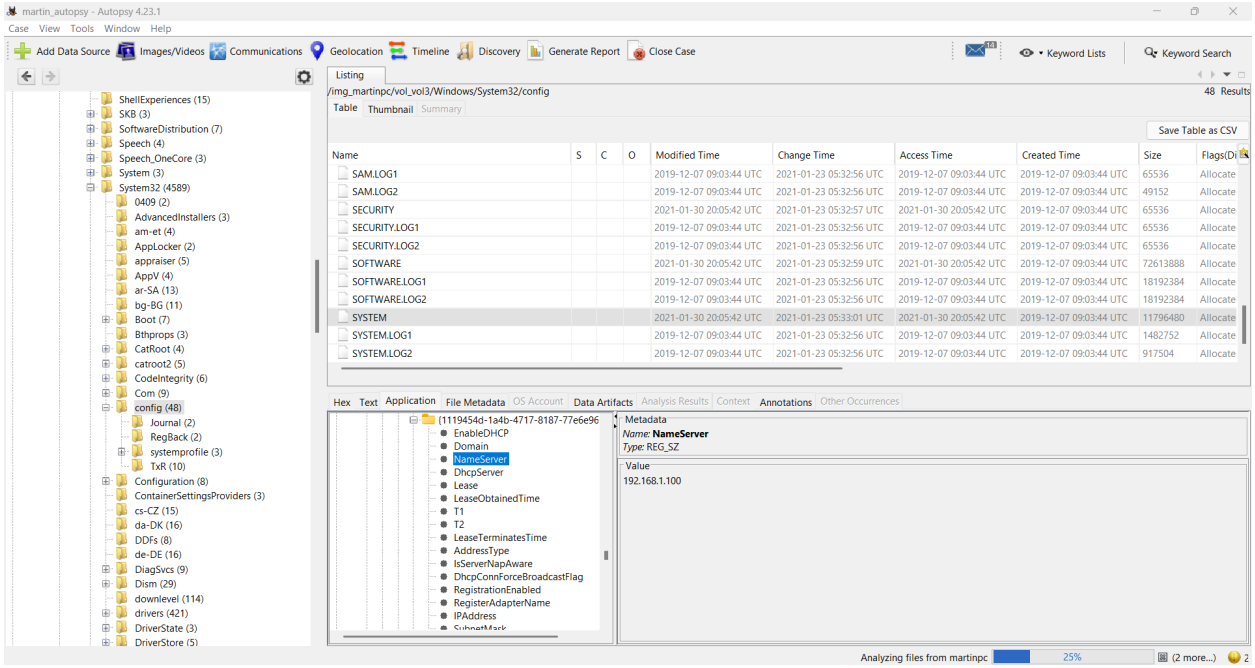


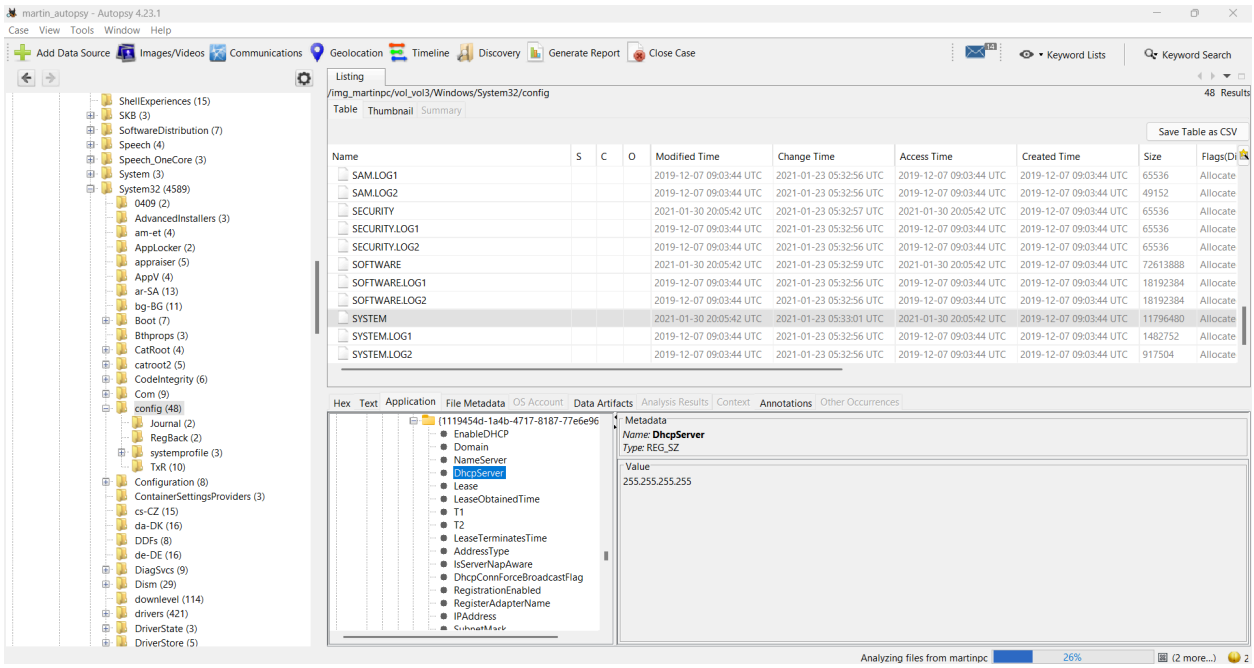


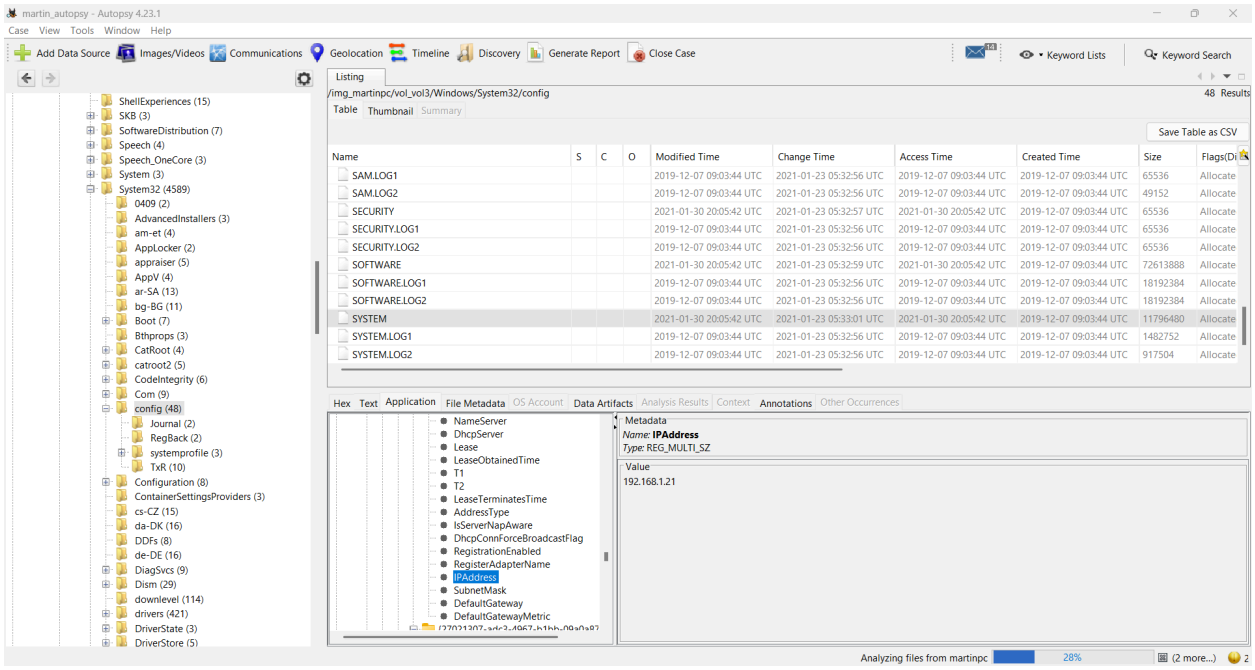
5.2.10.6 Martin_A_0006 Operating System

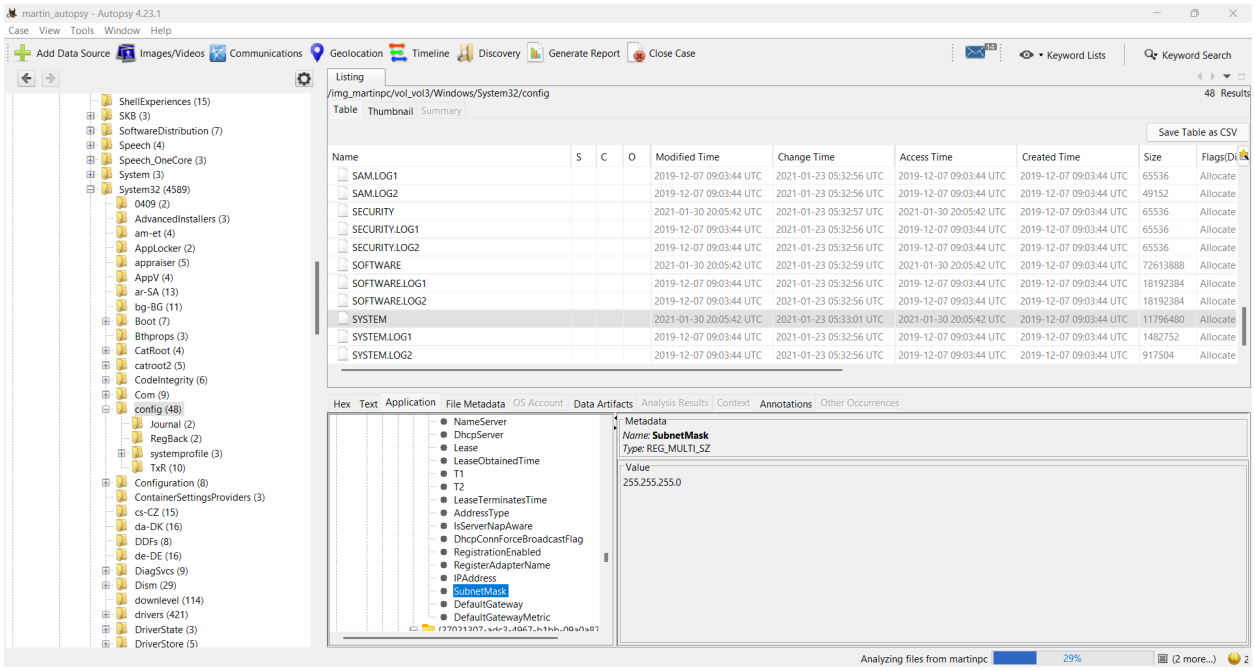


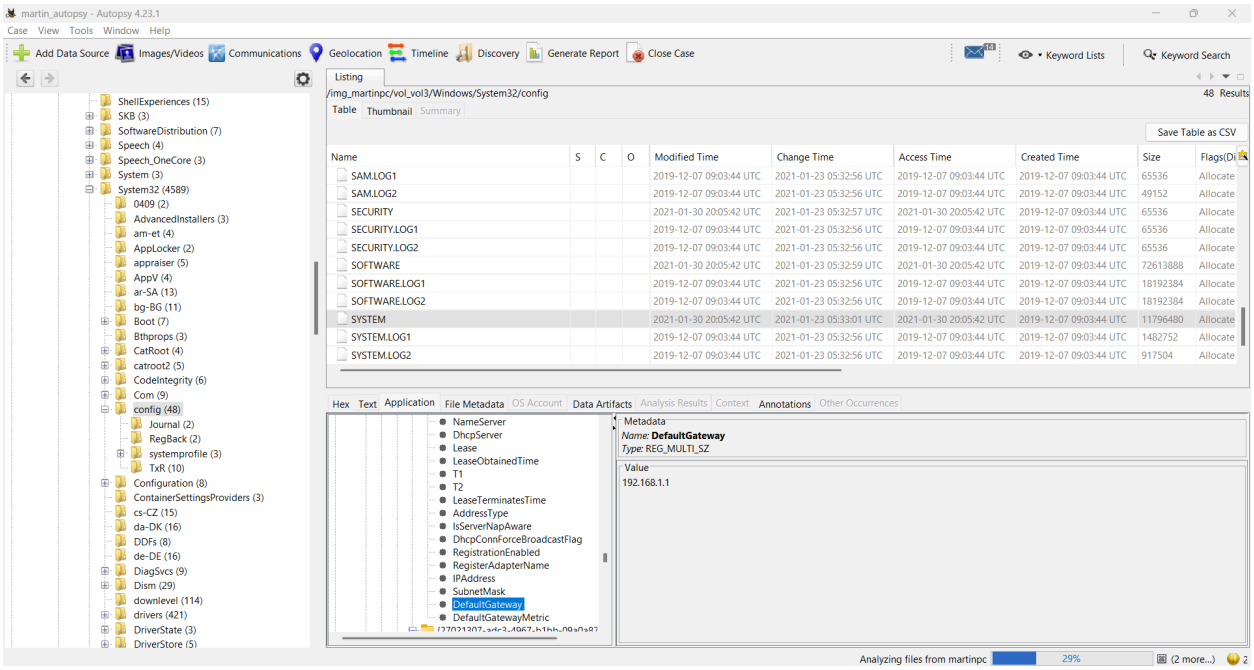
5.2.10.7 Martin_A_0007 IP Addresses





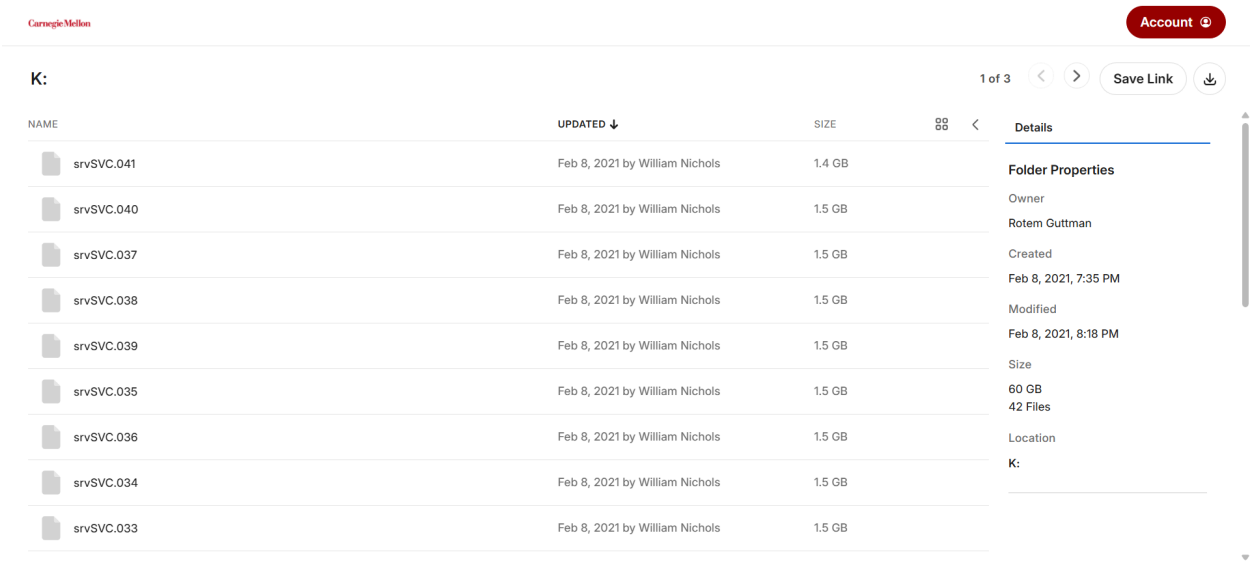

martin_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Listing
/img_martinpc/vol_vol3/Windows/System32/config
48 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(D
SAM.LOG1 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
SAM.LOG2 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 49152 Allocate
SECURITY 2021-01-30 20:05:42 UTC 2021-01-23 05:32:57 UTC 2021-01-30 20:05:42 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
SECURITY.LOG1 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
SECURITY.LOG2 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
SOFTWARE 2021-01-30 20:05:42 UTC 2021-01-23 05:32:59 UTC 2021-01-30 20:05:42 UTC 2019-12-07 09:03:44 UTC 72613888 Allocate
SOFTWARE.LOG1 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 18192384 Allocate
SOFTWARE.LOG2 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 18192384 Allocate
SYSTEM 2021-01-30 20:05:42 UTC 2021-01-23 05:33:01 UTC 2021-01-30 20:05:42 UTC 2019-12-07 09:03:44 UTC 11796480 Allocate
SYSTEM.LOG1 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 1482752 Allocate
SYSTEM.LOG2 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 917504 Allocate
Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences
NameServer
DhcpServer
Lease
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
SubnetMask
DefaultGateway
DefaultGatewayMetric
Metadata
Name: IPAddress
Type: REG_MULTI_SZ
Value
192.168.1.21
Analyzing files from martinpc
28%
(2 more...)


martin_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Listing
/img_martinpc/vol_vol3/Windows/System32/config
48 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
S
C
O
Modified Time
Change Time
Access Time
Created Time
Size
Flags(D
SAM.LOG1 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
SAM.LOG2 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 49152 Allocate
SECURITY 2021-01-30 20:05:42 UTC 2021-01-23 05:32:57 UTC 2021-01-30 20:05:42 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
SECURITY.LOG1 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
SECURITY.LOG2 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 65536 Allocate
SOFTWARE 2021-01-30 20:05:42 UTC 2021-01-23 05:32:59 UTC 2021-01-30 20:05:42 UTC 2019-12-07 09:03:44 UTC 72613888 Allocate
SOFTWARE.LOG1 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 18192384 Allocate
SOFTWARE.LOG2 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 18192384 Allocate
SYSTEM 2021-01-30 20:05:42 UTC 2021-01-23 05:33:01 UTC 2021-01-30 20:05:42 UTC 2019-12-07 09:03:44 UTC 11796480 Allocate
SYSTEM.LOG1 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 1482752 Allocate
SYSTEM.LOG2 2019-12-07 09:03:44 UTC 2021-01-23 05:32:56 UTC 2019-12-07 09:03:44 UTC 2019-12-07 09:03:44 UTC 917504 Allocate
Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences
NameServer
DhcpServer
Lease
LeaseObtainedTime
T1
T2
LeaseTerminatesTime
AddressType
IsServerNapAware
DhcpConnForceBroadcastFlag
RegistrationEnabled
RegisterAdapterName
IPAddress
SubnetMask
DefaultGateway
DefaultGatewayMetric
Metadata
Name: SubnetMask
Type: REG_MULTI_SZ
Value
255.255.255.0
Analyzing files from martinpc
29%
(2 more...)

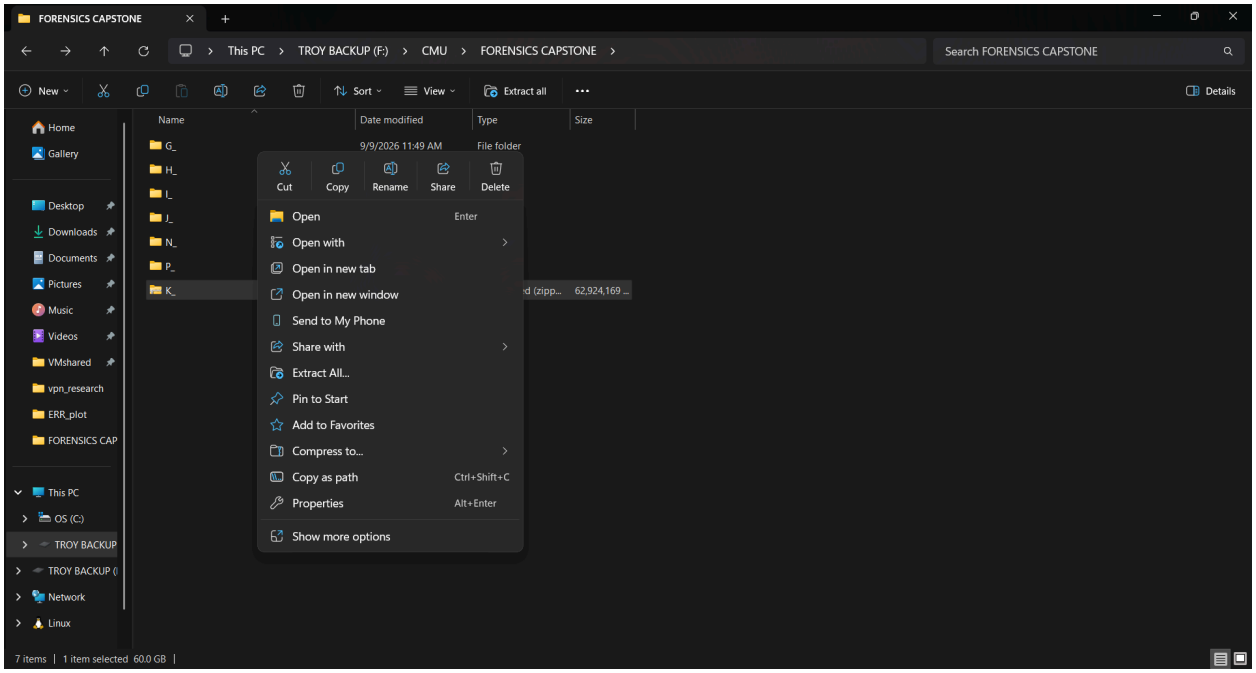


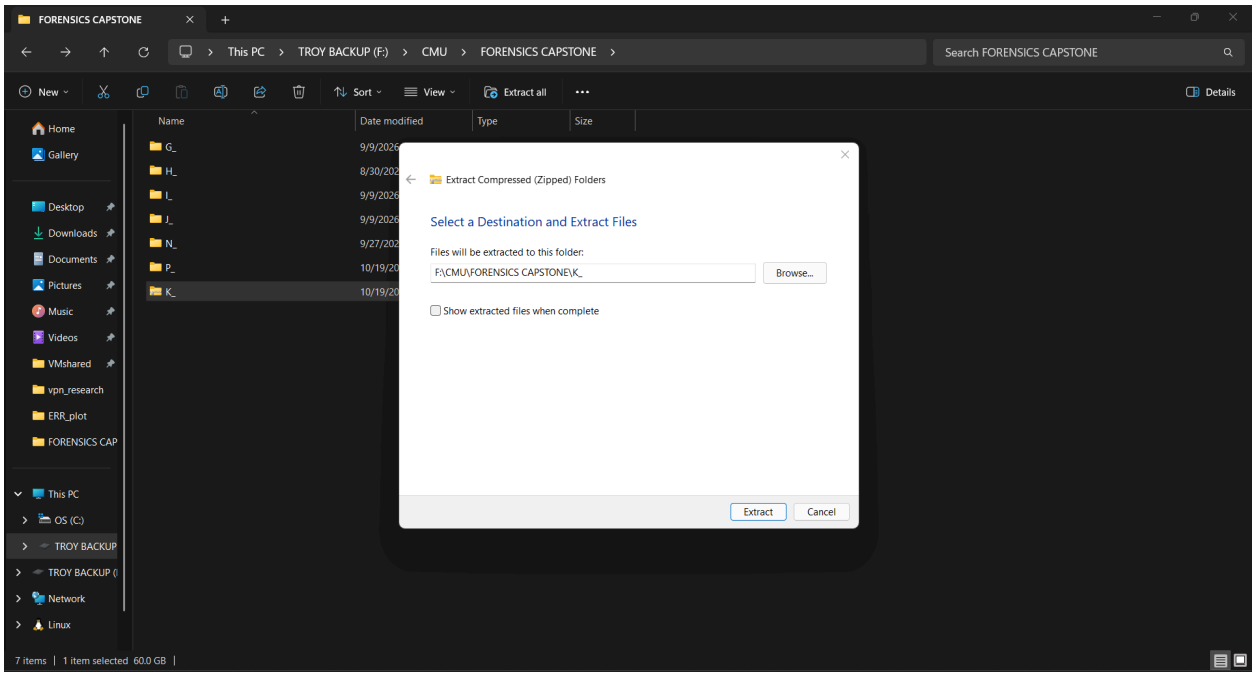
5.2.11 Services Image (K)

5.2.11.1 Services_A_0001 Image Download

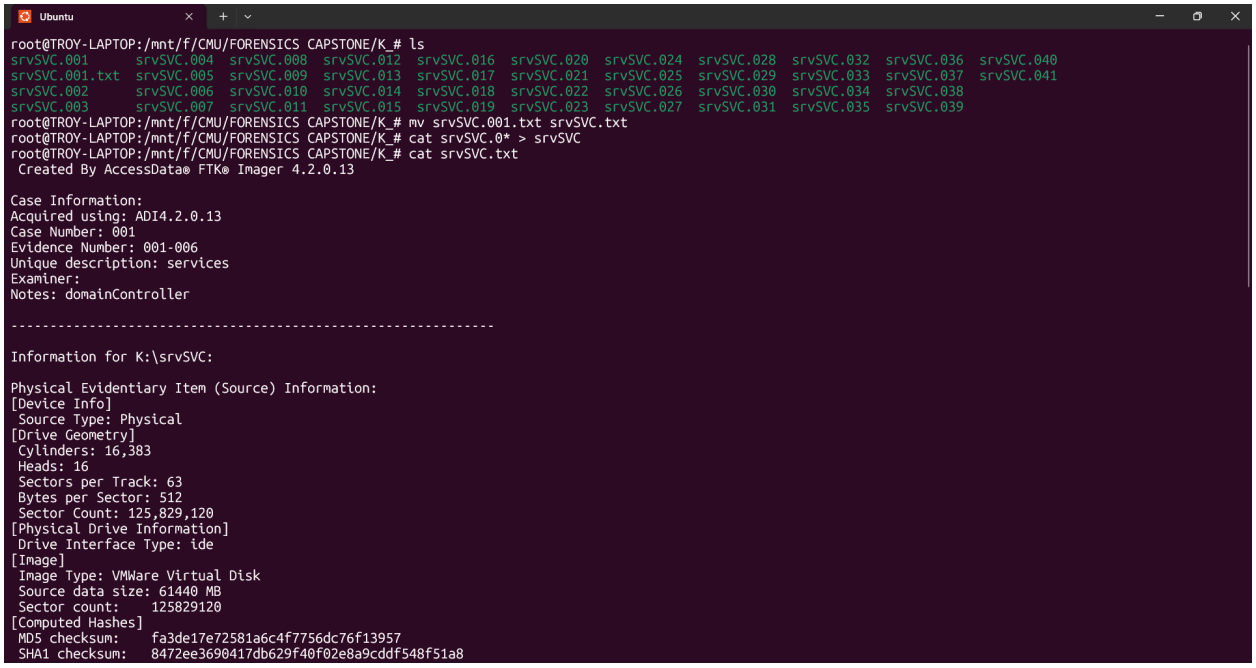


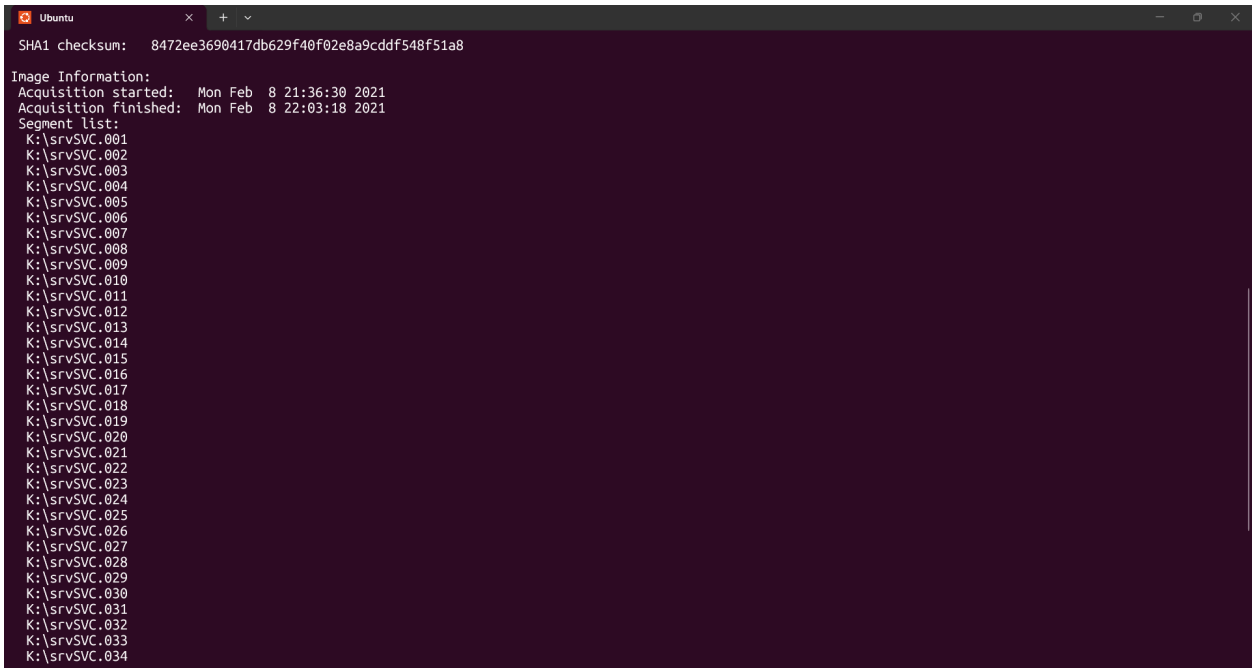
5.2.11.2 Services_A_0002 Image Extraction

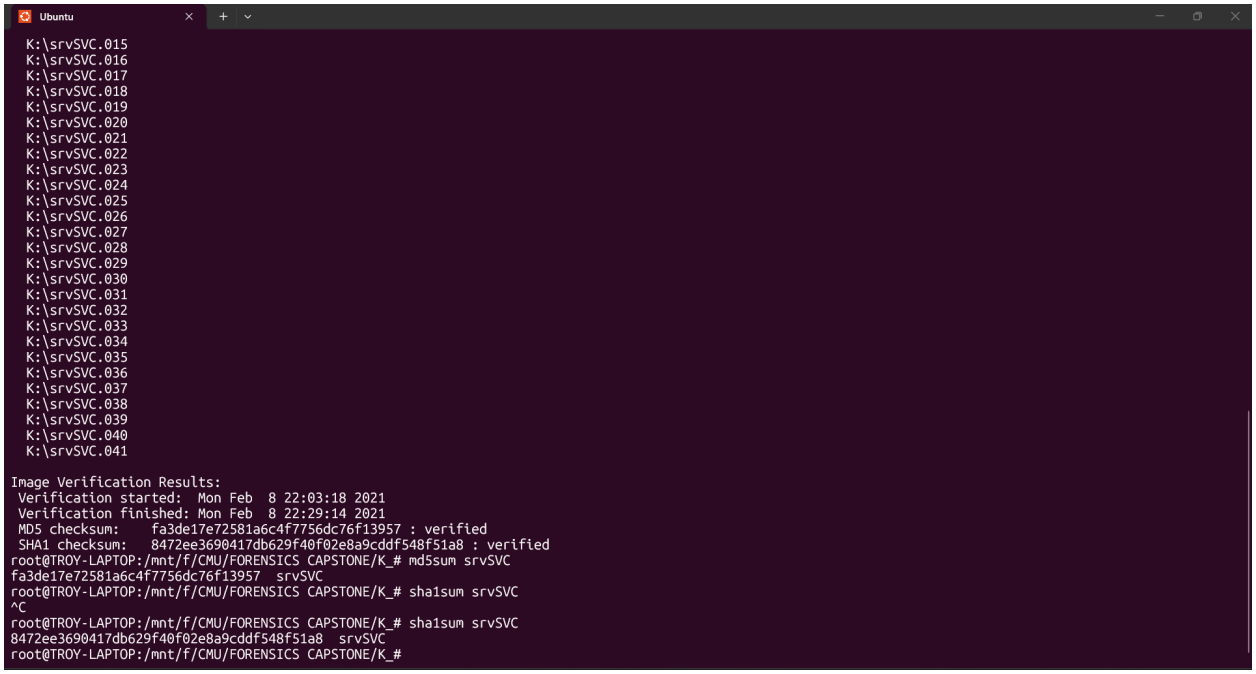




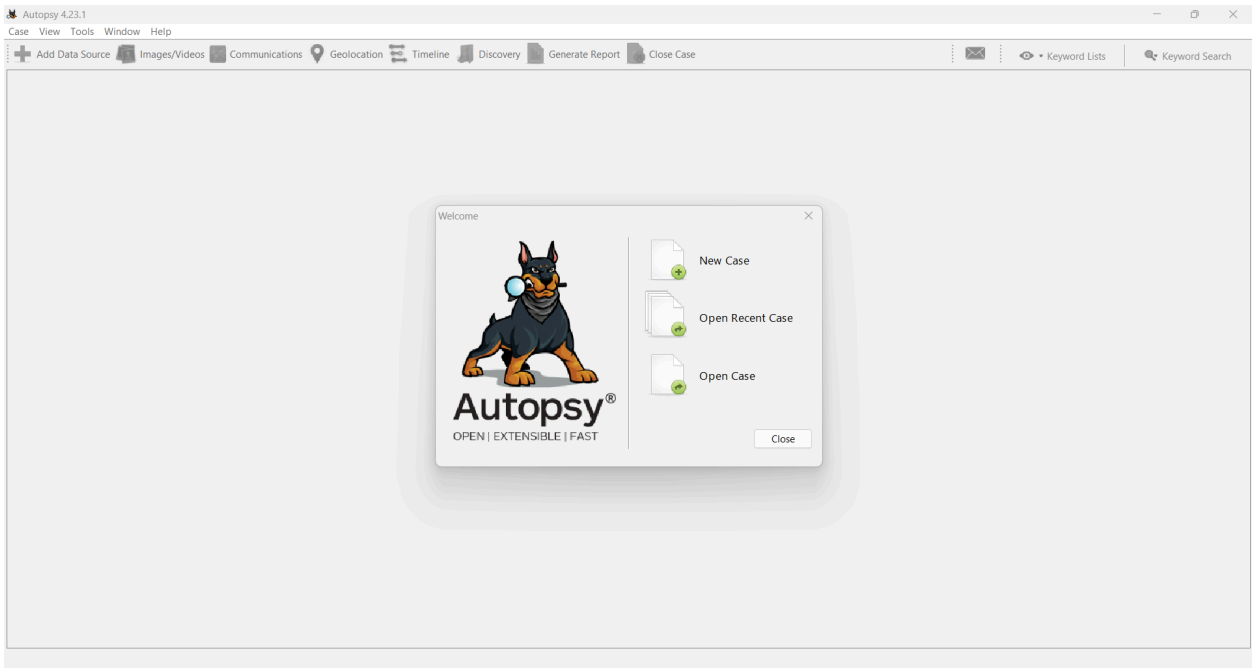
5.2.11.3 Services_A_0003 Image Verification

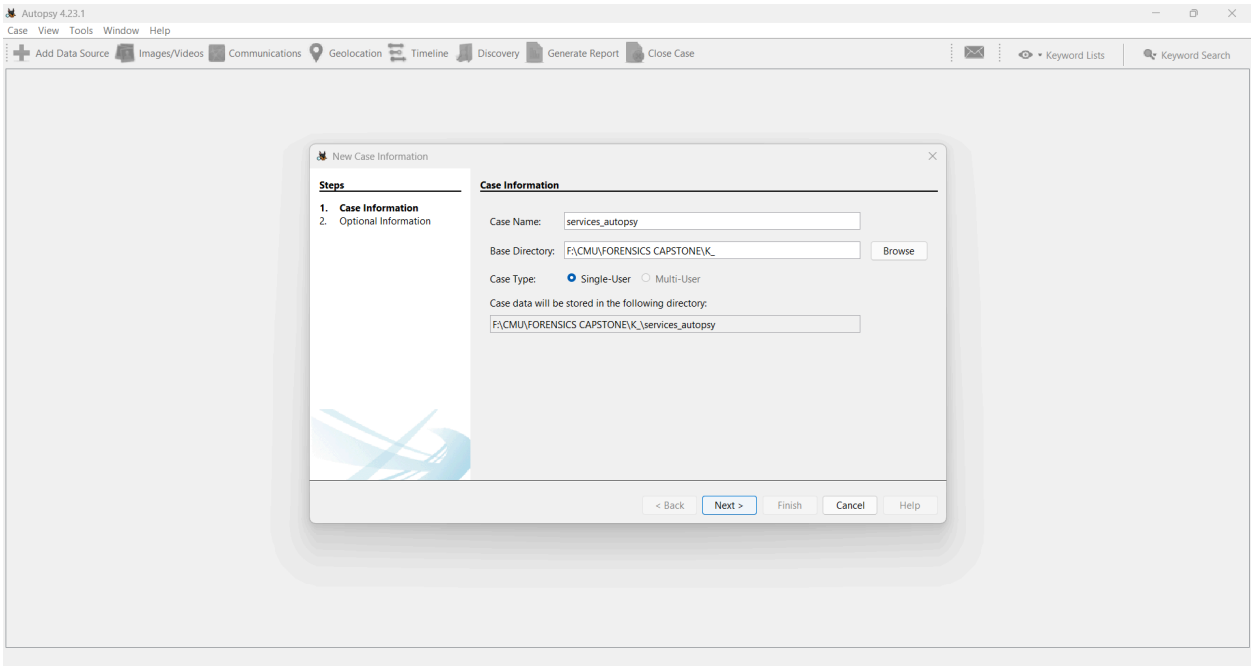


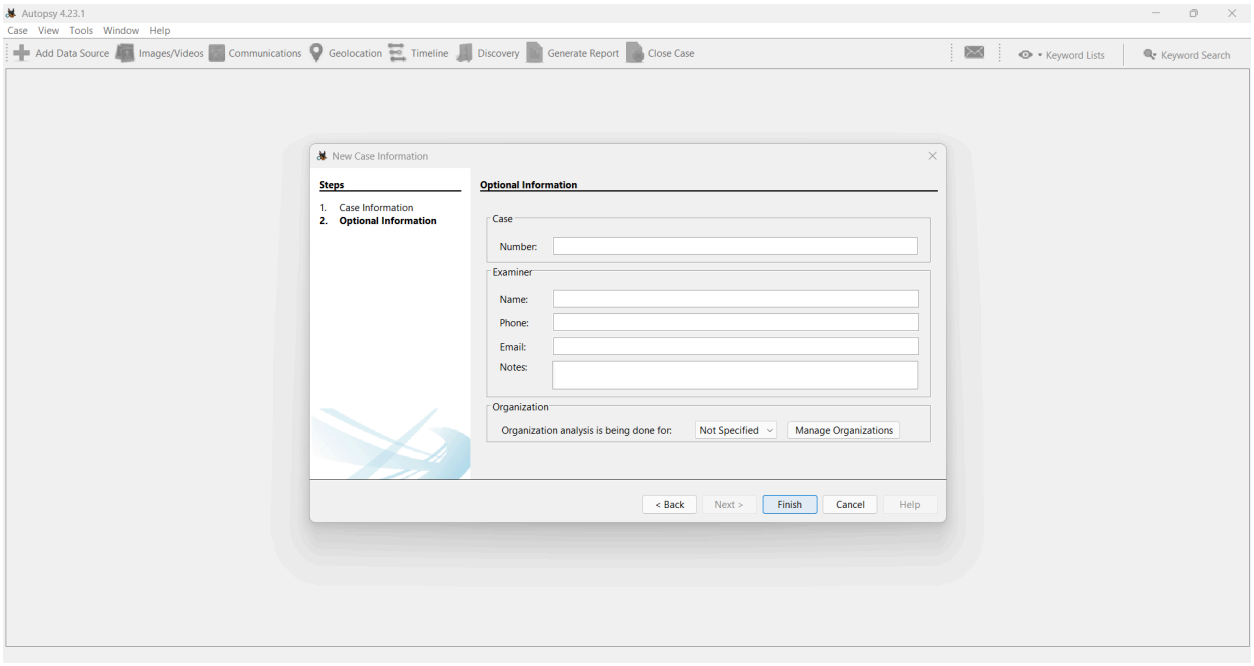


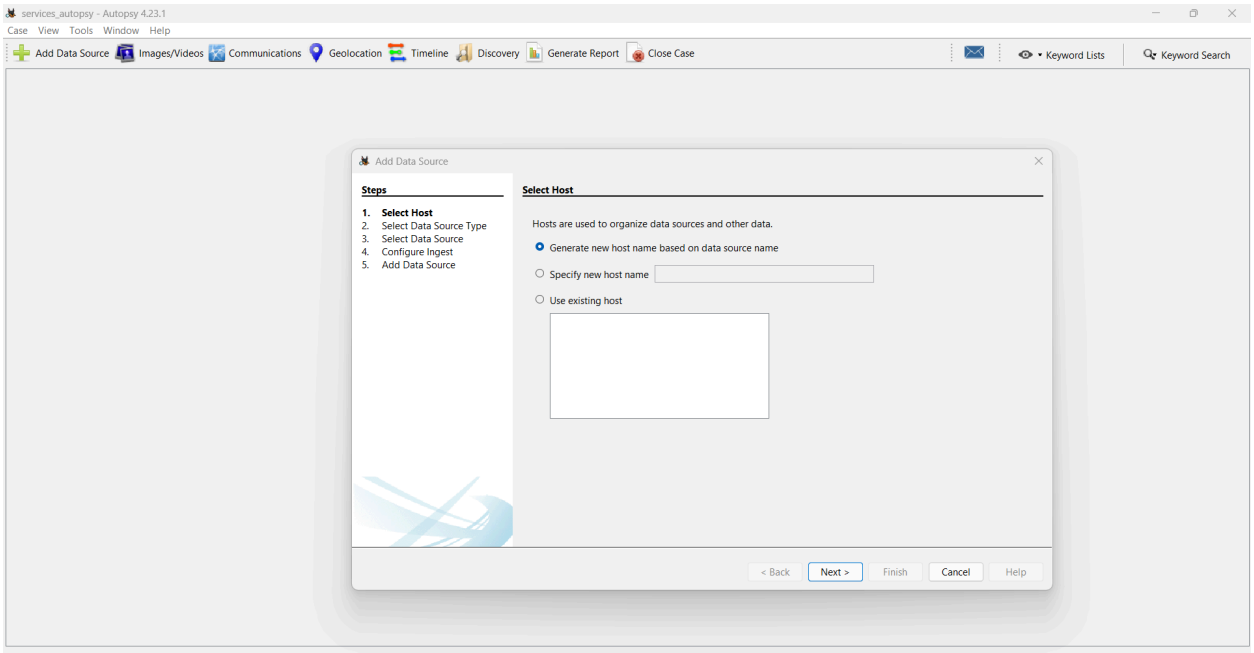


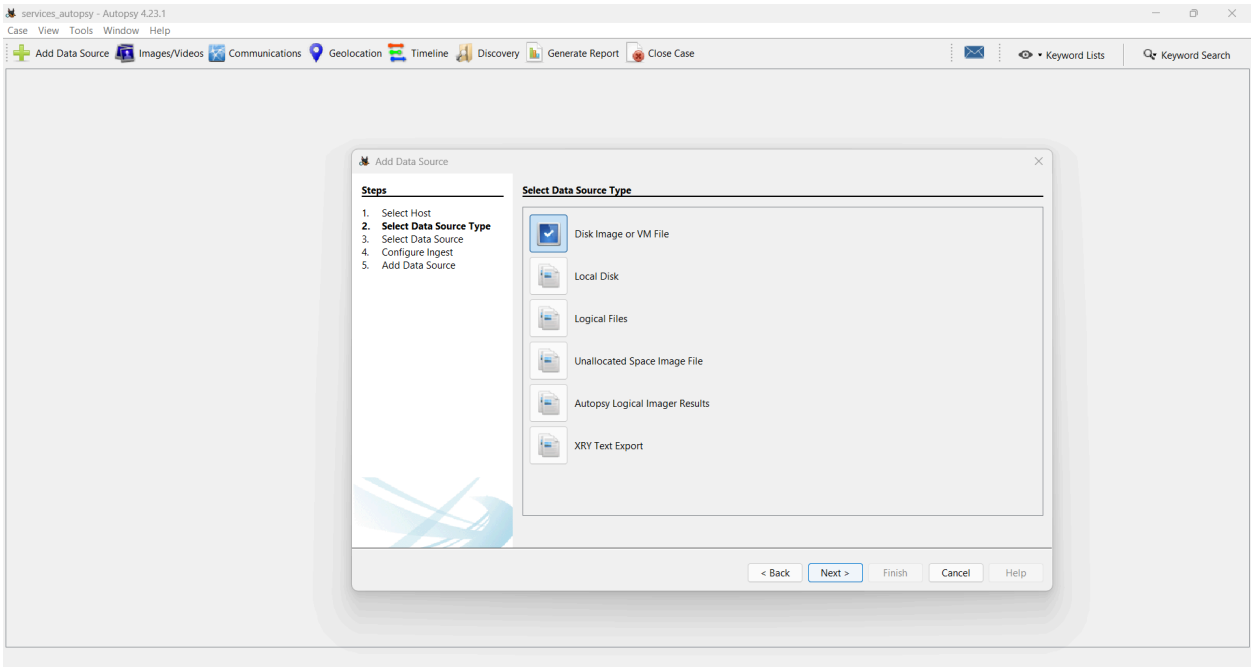
5.2.11.4 Services_A_0004 Autopsy Case

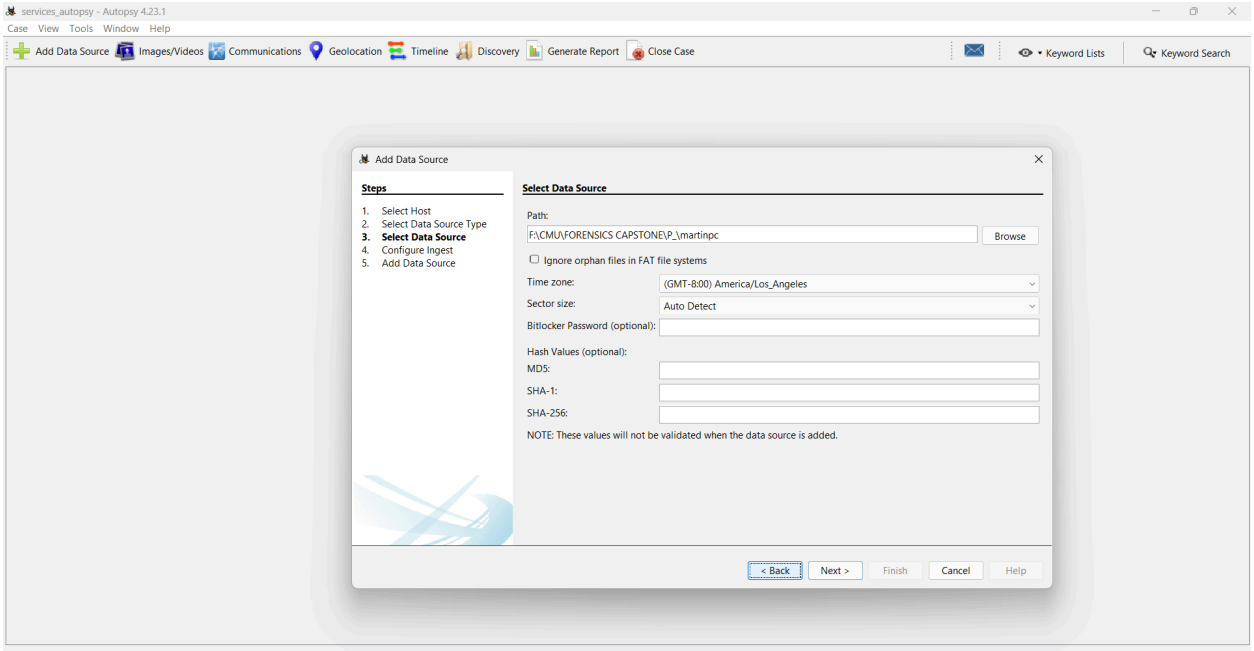


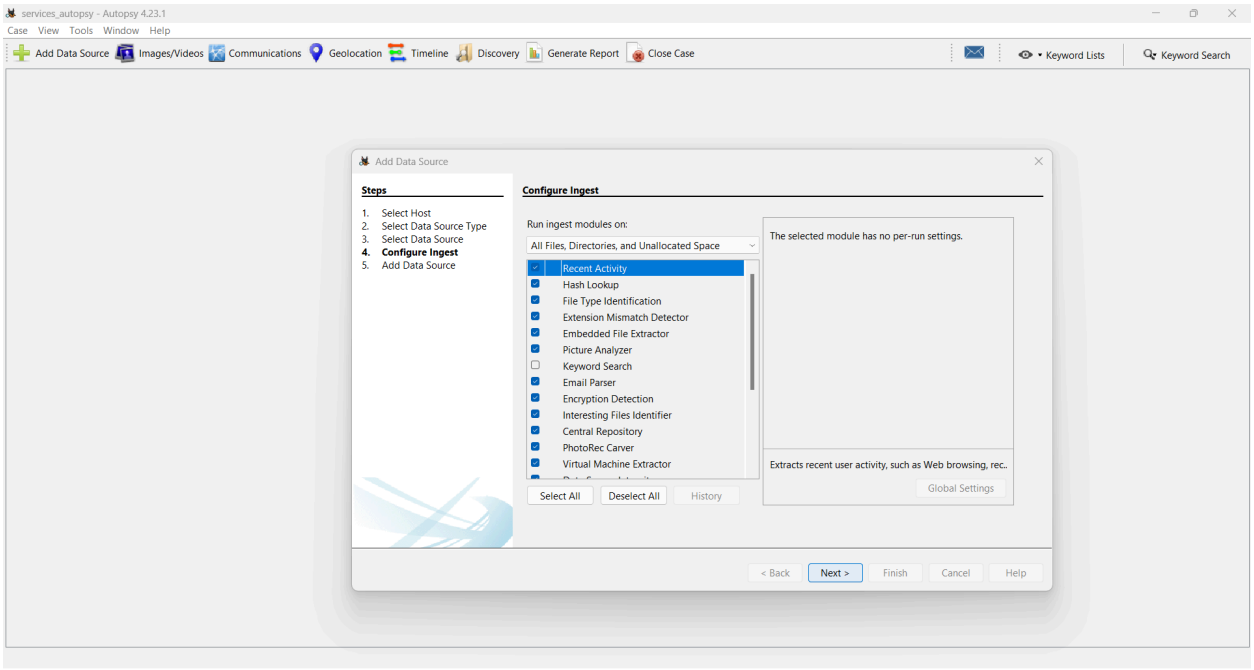

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Case Information
Case Name: services_autopsy
Base Directory: F:\CMU\FORENSICS CAPSTONE\K_
Browse
Case Type: Single-User Multi-User
Case data will be stored in the following directory:
F:\CMU\FORENSICS CAPSTONE\K_\services_autopsy
< Back Next > Finish Cancel Help

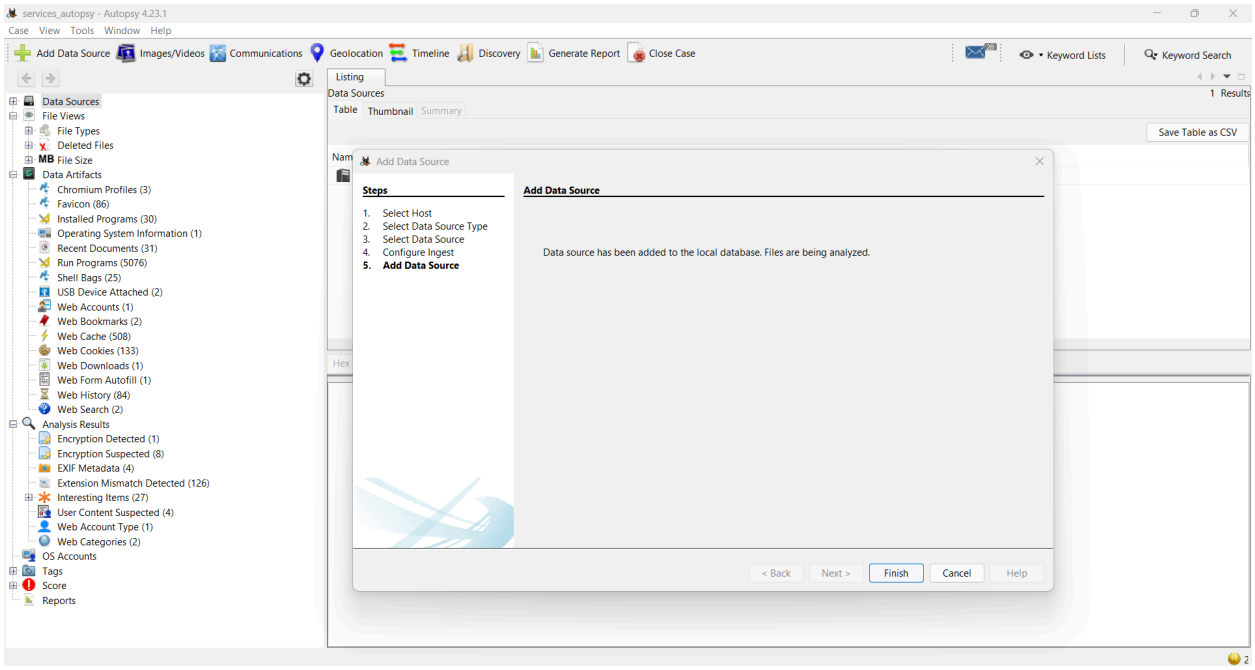

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Optional Information
Case
Number:
Examiner
Name:
Phone:
Email:
Notes:
Organization
Organization analysis is being done for: Not Specified Manage Organizations
< Back Next > Finish Cancel Help

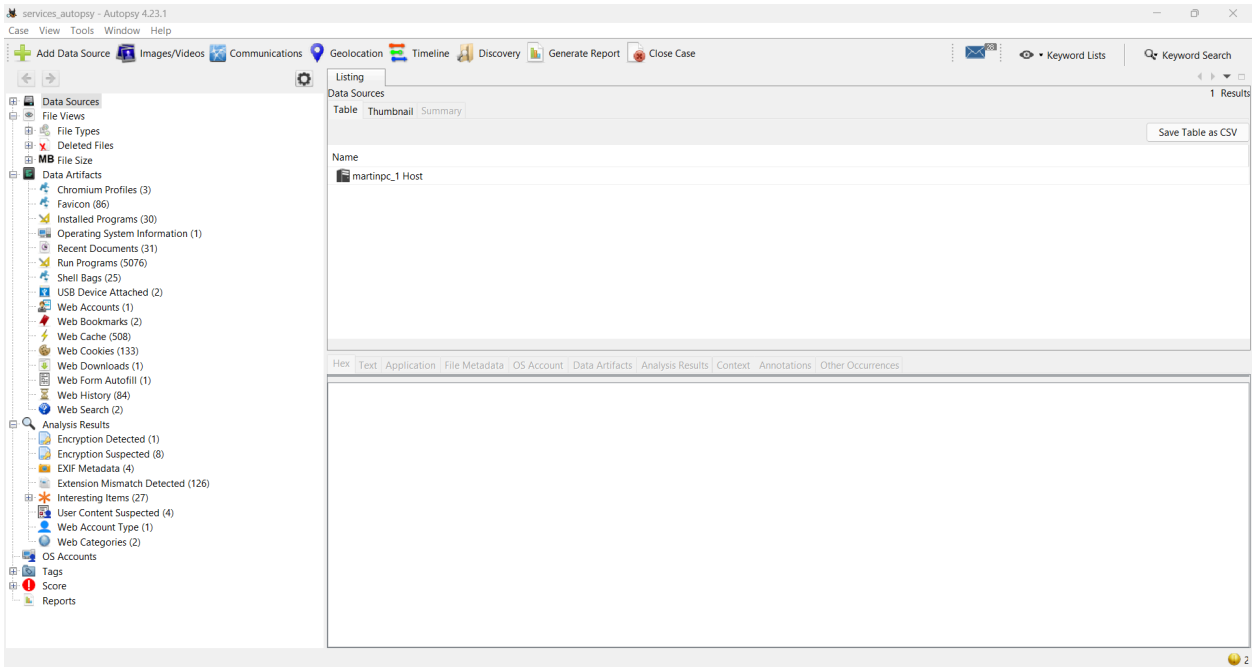

services_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Host
Hosts are used to organize data sources and other data.
Generate new host name based on data source name
Specify new host name
Use existing host
< Back
Next >
Finish
Cancel
Help


services_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source Type
Disk Image or VM File
Local Disk
Logical Files
Unallocated Space Image File
Autopsy Logical Imager Results
XRY Text Export
< Back
Next >
Finish
Cancel
Help

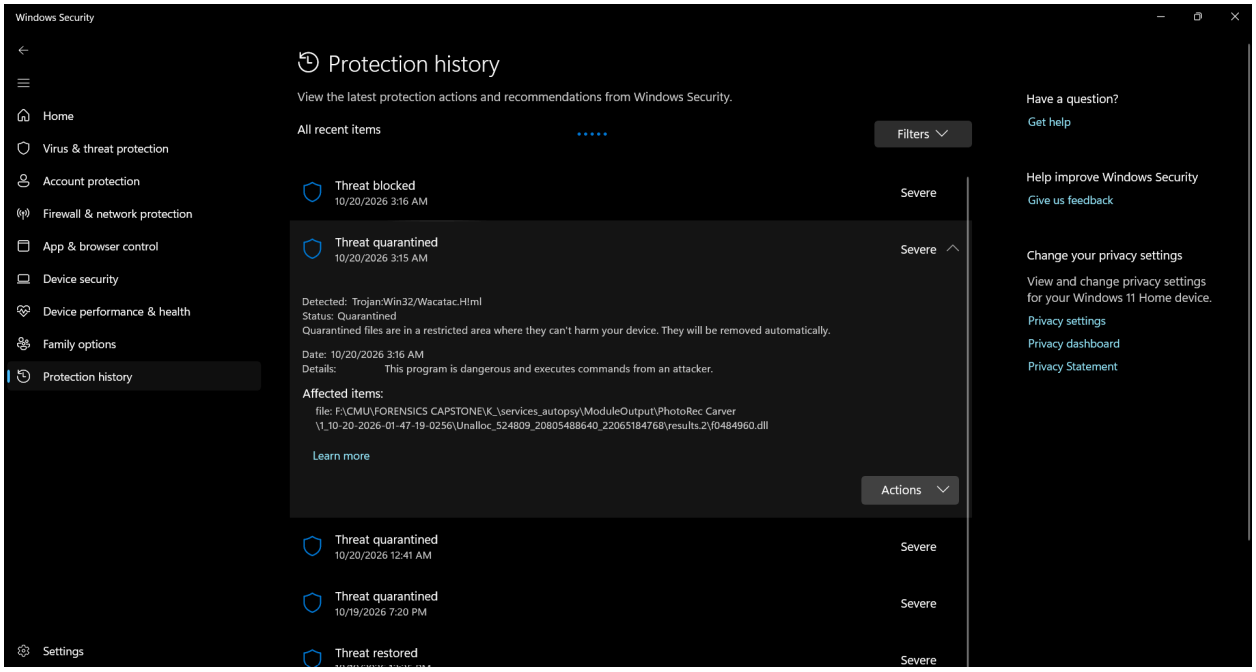

services_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source
Path:
F:\CMU\FORENSICS CAPSTONE\P_\martinpc
Browse
Ignore orphan files in FAT file systems
Time zone:
(GMT-8:00) America/Los_Angeles
Sector size:
Auto Detect
Bitlocker Password (optional):
Hash Values (optional):
MD5:
SHA-1:
SHA-256:
NOTE: These values will not be validated when the data source is added.
< Back
Next >
Finish
Cancel
Help

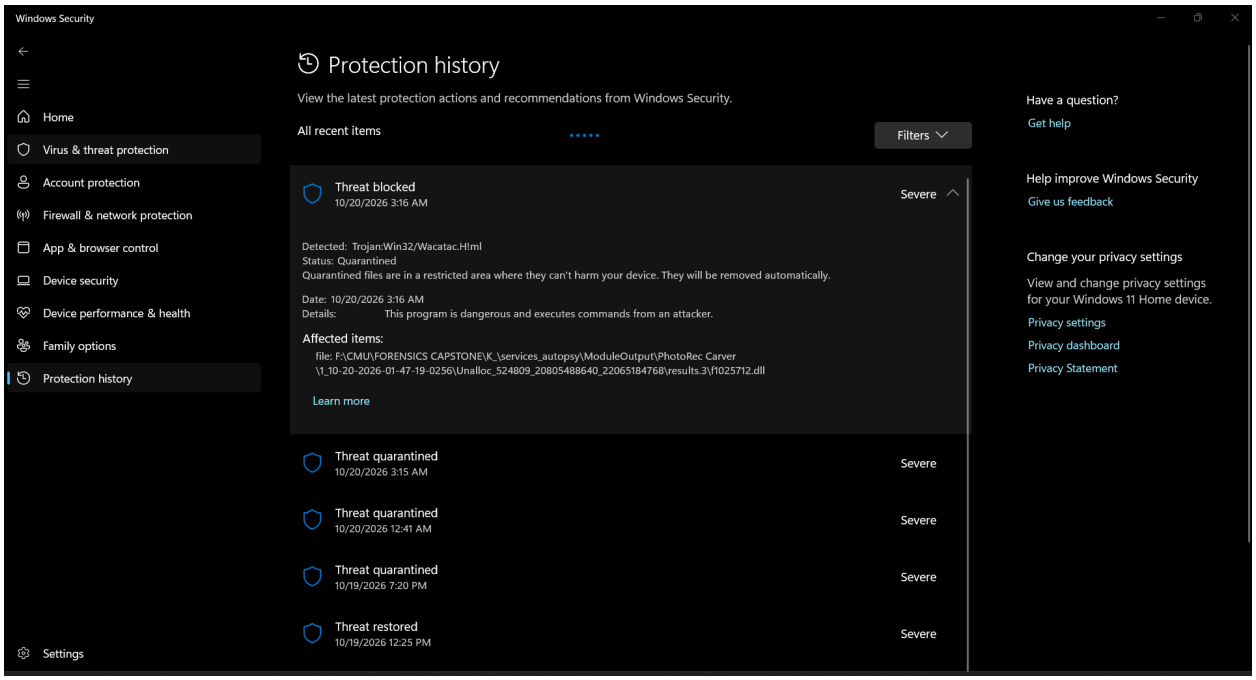

services_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
Select All
Deselect All
History
The selected module has no per-run settings.
Extracts recent user activity, such as Web browsing, rec..
Global Settings
< Back
Next >
Finish
Cancel
Help


services_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Chromium Profiles (3)
Favicon (86)
Installed Programs (30)
Operating System Information (1)
Recent Documents (31)
Run Programs (5076)
Shell Bags (25)
USB Device Attached (2)
Web Accounts (1)
Web Bookmarks (2)
Web Cache (508)
Web Cookies (133)
Web Downloads (1)
Web Form Autofill (1)
Web History (84)
Web Search (2)
Analysis Results
Encryption Detected (1)
Encryption Suspected (8)
EXIF Metadata (4)
Extension Mismatch Detected (126)
Interesting Items (27)
User Content Suspected (4)
Web Account Type (1)
Web Categories (2)
OS Accounts
Tags
Score
Reports
Listing
Data Sources
1 Results
Table
Thumbnail
Summary
Save Table as CSV
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
Data source has been added to the local database. Files are being analyzed.
< Back
Next >
Finish
Cancel
Help

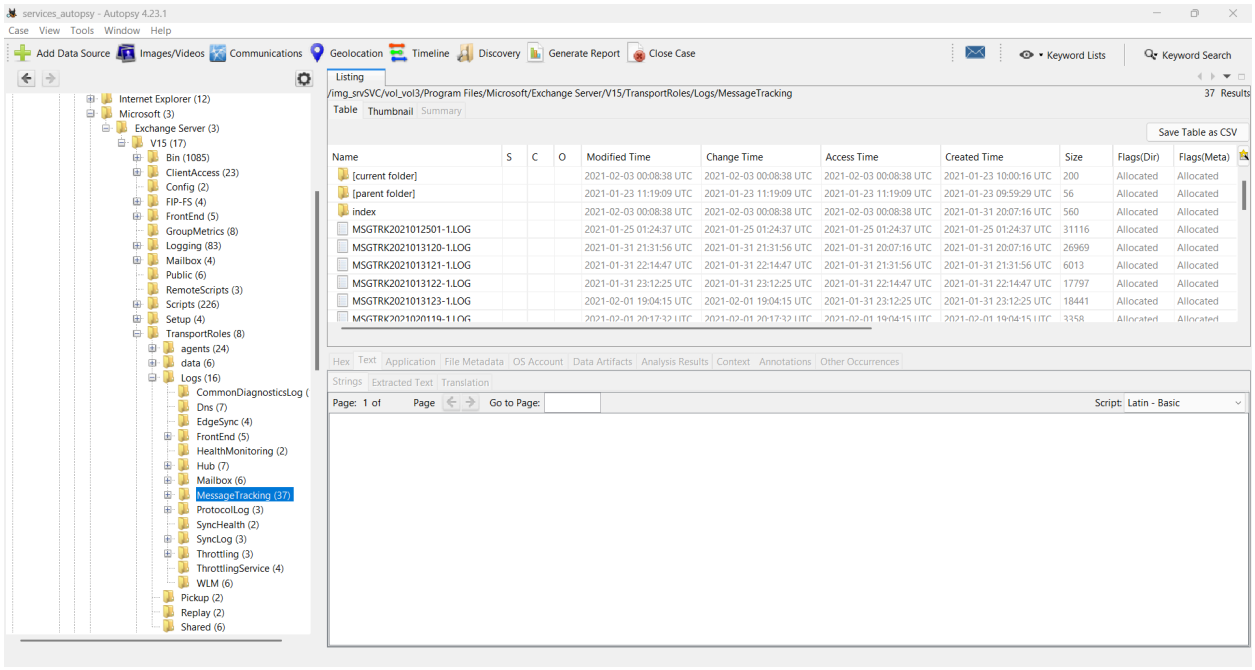

services_autopsy - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Close Case
Keyword Lists
Keyword Search
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Chromium Profiles (3)
Favicon (86)
Installed Programs (30)
Operating System Information (1)
Recent Documents (31)
Run Programs (5076)
Shell Bags (25)
USB Device Attached (2)
Web Accounts (1)
Web Bookmarks (2)
Web Cache (508)
Web Cookies (133)
Web Downloads (1)
Web Form Autofill (1)
Web History (84)
Web Search (2)
Analysis Results
Encryption Detected (1)
Encryption Suspected (8)
EXIF Metadata (4)
Extension Mismatch Detected (126)
Interesting Items (27)
User Content Suspected (4)
Web Account Type (1)
Web Categories (2)
OS Accounts
Tags
Score
Reports
Listing
Data Sources
1 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
martinpc_1 Host
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences

5.2.11.5 Services_A_0005 Windows Security

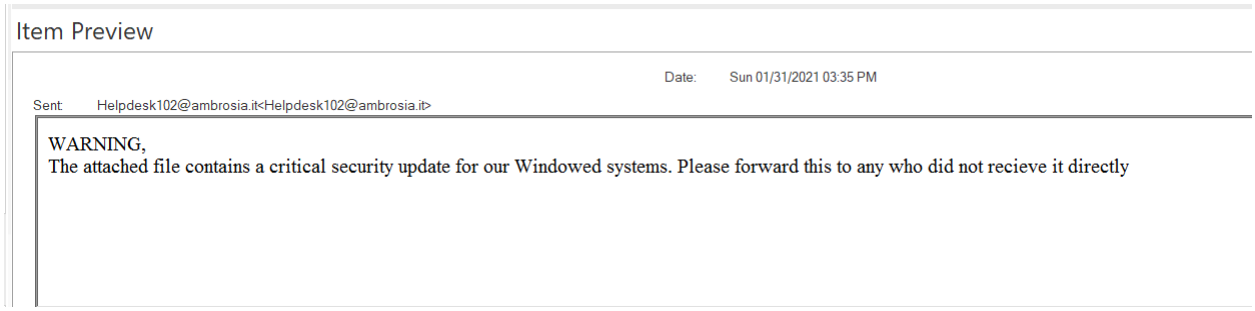




5.2.11.6 Services_A_0006 Email log files

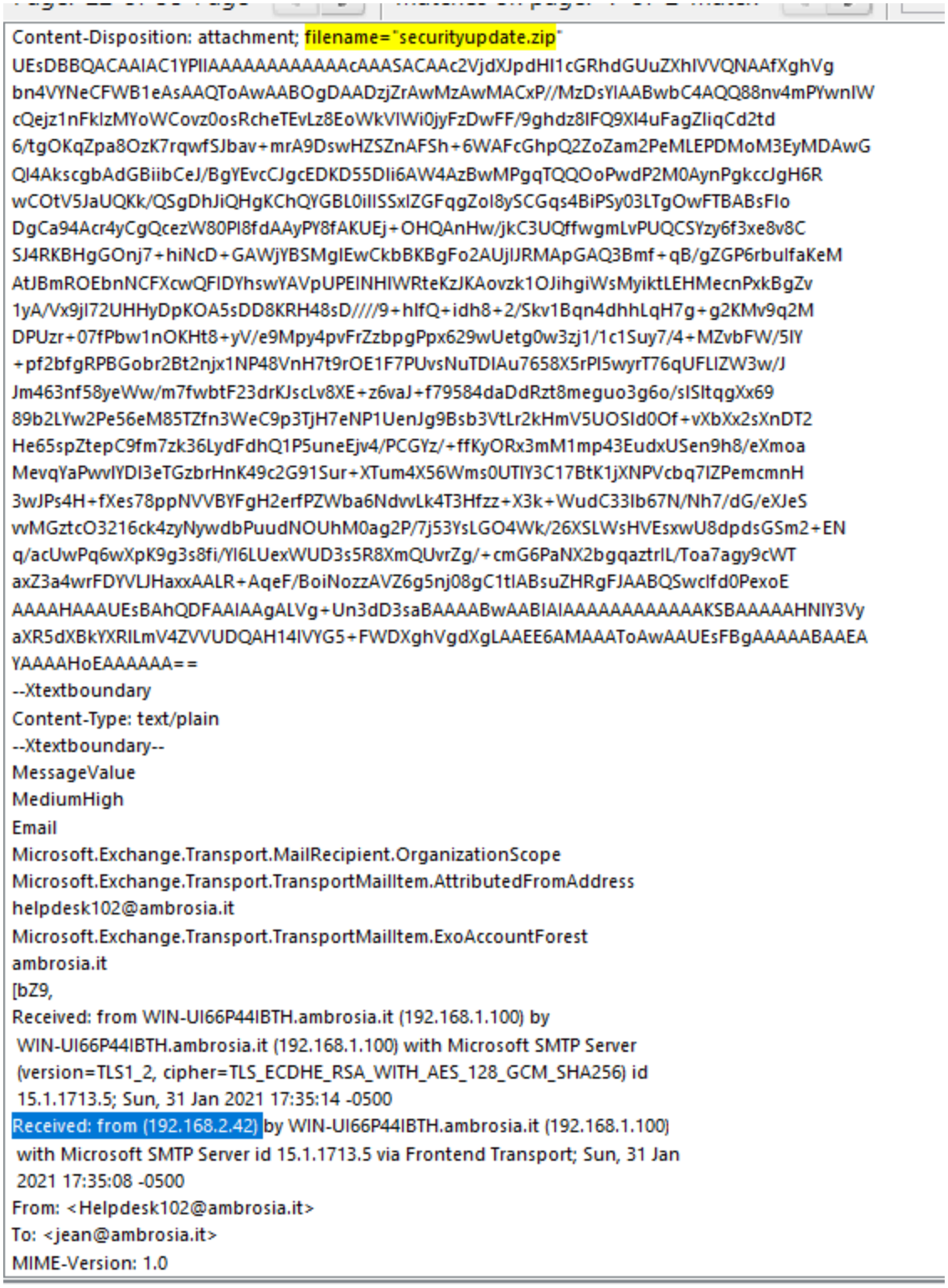


5.2.11.7 Services_A_0007 Email HelpDesk102@ambrosia.it



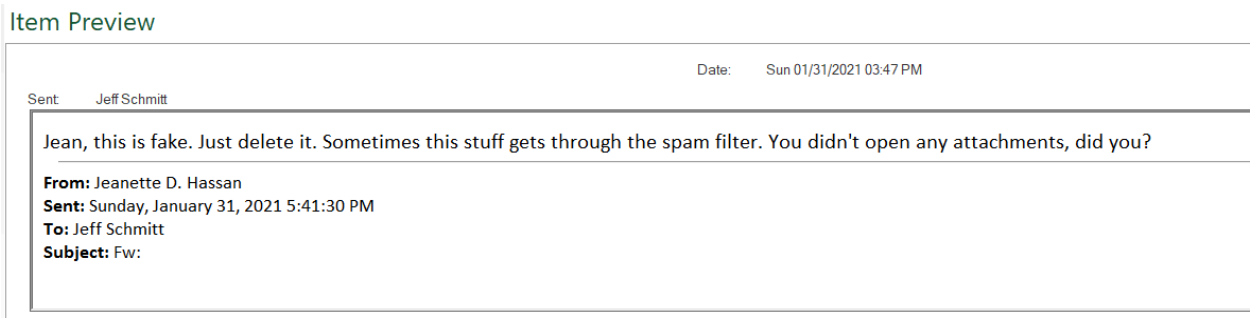
5.2.11.8 Services_A_0008 192.168.2.42 from trn000000001.log

```
Date: Sun, 31 Jan 2021 17:35:08 -0500
X-MS-Exchange-Organization-OriginalArrivalTime: 31 Jan 2021 22:35:08.0885
 (UTC)
X-MS-Exchange-Organization-Network-Message-Id: b686c20f-3838-4298-a01f-08d8c6387a5d
X-MS-Exchange-Organization-OriginalClientIPAddress: 192.168.2.42
X-MS-Exchange-Organization-OriginalServerIPAddress: 192.168.1.100
X-MS-Exchange-Organization-Cross-Premises-Headers-Processed: WIN-UI66P44IBTH.ambrosia.it
X-MS-Exchange-Organization-OrderedPrecisionLatencyInProgress: LSRV=WIN-UI66P44IBTH.ambrosia.it:TOTAL-FE=6.566|SMR=6.565(SMRPI=0.018(SMRPI
T22:35:14.654Z
X-MS-Exchange-Forest-ArrivalHubServer: WIN-UI66P44IBTH.ambrosia.it
X-MS-Exchange-Organization-AuthSource: WIN-UI66P44IBTH.ambrosia.it
X-MS-Exchange-Organization-AuthAs: Anonymous
X-MS-Exchange-Organization-FromEntityHeader: Internet
X-MS-Exchange-Organization-OriginalSize: 3256
--Xtextboundary
Content-Type: text/plain
WARNING,
The attached file contains a critical security update for our Windowed systems. Please forward this to any who did not recieve it directly
--Xtextboundary
Content-Type: application/zip; name="securityupdate.zip"
Content-Transfer-Encoding: base64
Content-Disposition: attachment; filename="securityupdate.zip"
```

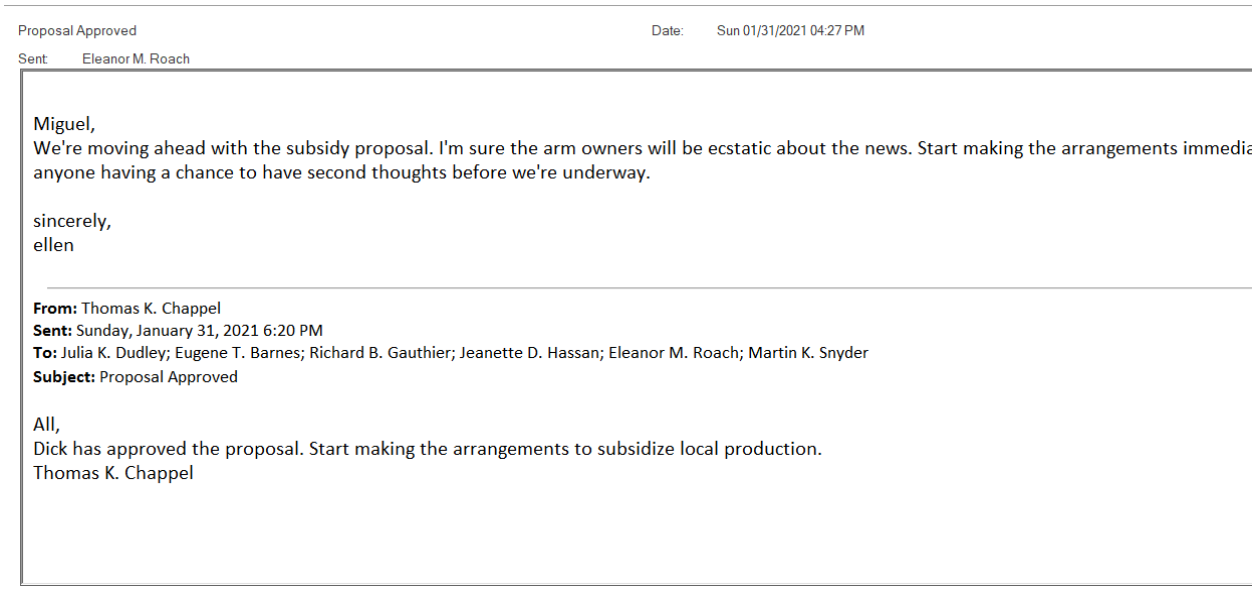


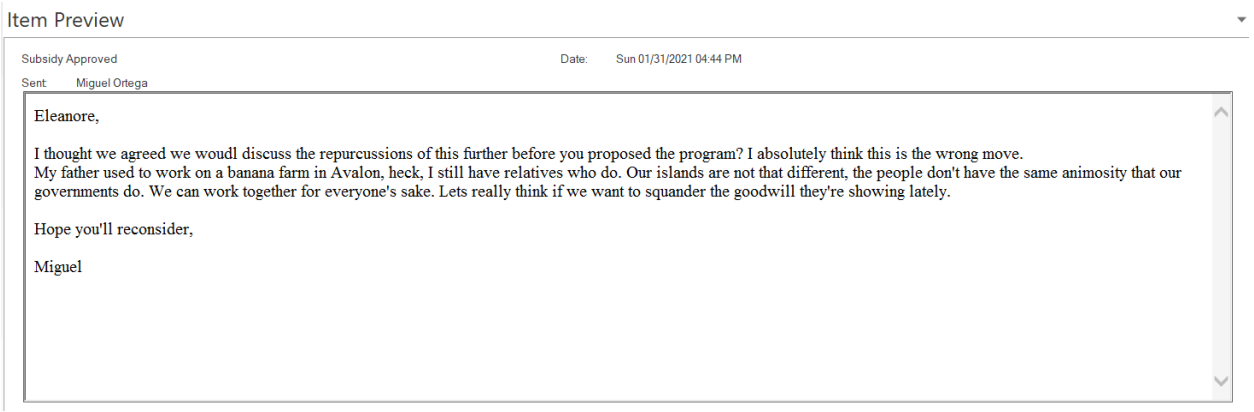
```
Content-Disposition: attachment; filename="securityupdate.zip"
UEsDBBQACAAIAC1YPIIAAAAAAAAAAAAcAAASACAAc2VjdXJpdHI1cGRhdGUuZXhlIVVQNAAfXghVg
bn4VYNeCFWB1eAsAAQToAwAABOgDAADzjZrAwMzAwMACxP//MzDsYIAABwbC4AQQ88nv4mPYwnIW
cQejz1nFkIzMYoWCovz0osRcheTEvLz8EoWkVIWi0jyFzDwFF/9ghdz8IFQ9XI4uFagZIiqCd2td
6/tgOKqZpa8OzK7rqwfSJbav+mrA9DswHZSZnAFSh+6WAFcGhpQ2ZoZam2PeMLEPDMoM3EyMDAwG
Ql4AkscgbAdGBiibCeJ/BgYEvcCJgcEDKD55Dli6AW4AzBwMPgqTQQOoPwdP2M0AynPgkccJgH6R
wCOtV5JaUQKk/QSgDhJiQHgKChQYGBL0iIlSSxIZGFqgZoI8ySCGqs4BiPSy03LTgOwFTBABsFIo
DgCa94Acr4yCgQcezW80PI8fdAAyPY8fAKUEj+OHQAnHw/jkC3UQffwgmLvPUQCSYzy6f3xe8v8C
SJ4RKBHgGOnj7+hiNcD+GAWjYBSMgIEwCkbBKBgFo2AUjIJRMApGAQ3Bmf+qB/gZGP6rbulfaKeM
AtJBmROEbnNCFXcwQFIDYhswYAVpUPEINHIWRteKzJKAovzk1OJihgiWsMyiktLEHMecnPxkBgZv
1yA/Vx9jI72UHHyDpKOA5sDD8KRH48sD////9+hIfQ+idh8+2/Skv1Bqn4dhhLqH7g+g2KMv9q2M
DPUzr+07fPbw1nOKHt8+yV/e9Mpy4pvFrZzbpgPpx629wUetg0w3zj1/1c1Suy7/4+MZvbFW/5IY
+pf2bfgRPBGobr2Bt2njx1NP48VnH7t9rOE1F7PUvsNuTDIAu7658X5rPI5wyrT76qUFLIZW3w/J
Jm463nf58yeWw/m7fwbtF23drKJscLv8XE+z6vaJ+f79584daDdRzt8meguo3g6o/sISltqgXx69
89b2LYw2Pe56eM85TZfn3WeC9p3TjH7eNP1UenJg9Bsb3VtLr2kHmV5UOSId0Of+vXbXx2sXnDT2
He65spZtepC9fm7zk36LydFdhQ1P5uneEjv4/PCGYz/+ffKyORx3mM1mp43EudxUSen9h8/eXmoa
MevqYaPwvIYDI3eTGzbrHnK49c2G91Sur+XTum4X56Wms0UTIY3C17BtK1jXNPVcbq7IZPemcmnH
3wJPs4H+fXes78ppNVVBYFgH2erfPZWba6NdvvLk4T3Hfzz+X3k+WudC33Ib67N/Nh7/dG/eXJeS
vvMGztcO3216ck4zyNywdbPuudNOUhM0ag2P/7j53YsLGO4Wk/26XSLWsHVEsxwU8dpdsGSm2+EN
q/acUwPq6wXpK9g3s8fi/YI6LUexWUD3s5R8XmQUvrZg/+cmG6PaNX2bgqaztrIL/Toa7agy9cWT
axZ3a4wrFDYVLJHaxxAALR+AqeF/BoiNozzAVZ6g5nj08gC1tIABsuZHRgFJAABQSwcIfd0PexoE
AAAAHAAAUEsBAhQDFAAIAAgALVg+Un3dD3saBAAAABwAABIAIAAAAAAAAAAAAKSBAAAAAHNIY3Vy
aXR5dXBkYXRlLmV4ZVVUDQAH14IVYG5+FWDXghVgdXgLAAEE6AMAAAToAwAAUEsFBgAAAAABAAEA
YAAAAHoEAAAAAA==
--Xtextboundary
Content-Type: text/plain
--Xtextboundary--
MessageValue
MediumHigh
Email
Microsoft.Exchange.Transport.MailRecipient.OrganizationScope
Microsoft.Exchange.Transport.TransportMailItem.AttributedFromAddress
helpdesk102@ambrosia.it
Microsoft.Exchange.Transport.TransportMailItem.ExoAccountForest
ambrosia.it
[bZ9,
Received: from WIN-UI66P44IBTH.ambrosia.it (192.168.1.100) by
 WIN-UI66P44IBTH.ambrosia.it (192.168.1.100) with Microsoft SMTP Server
 (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256) id
 15.1.1713.5; Sun, 31 Jan 2021 17:35:14 -0500
Received: from (192.168.2.42) by WIN-UI66P44IBTH.ambrosia.it (192.168.1.100)
 with Microsoft SMTP Server id 15.1.1713.5 via Frontend Transport; Sun, 31 Jan
 2021 17:35:08 -0500
From: <Helpdesk102@ambrosia.it>
To: <jean@ambrosia.it>
MIME-Version: 1.0
```

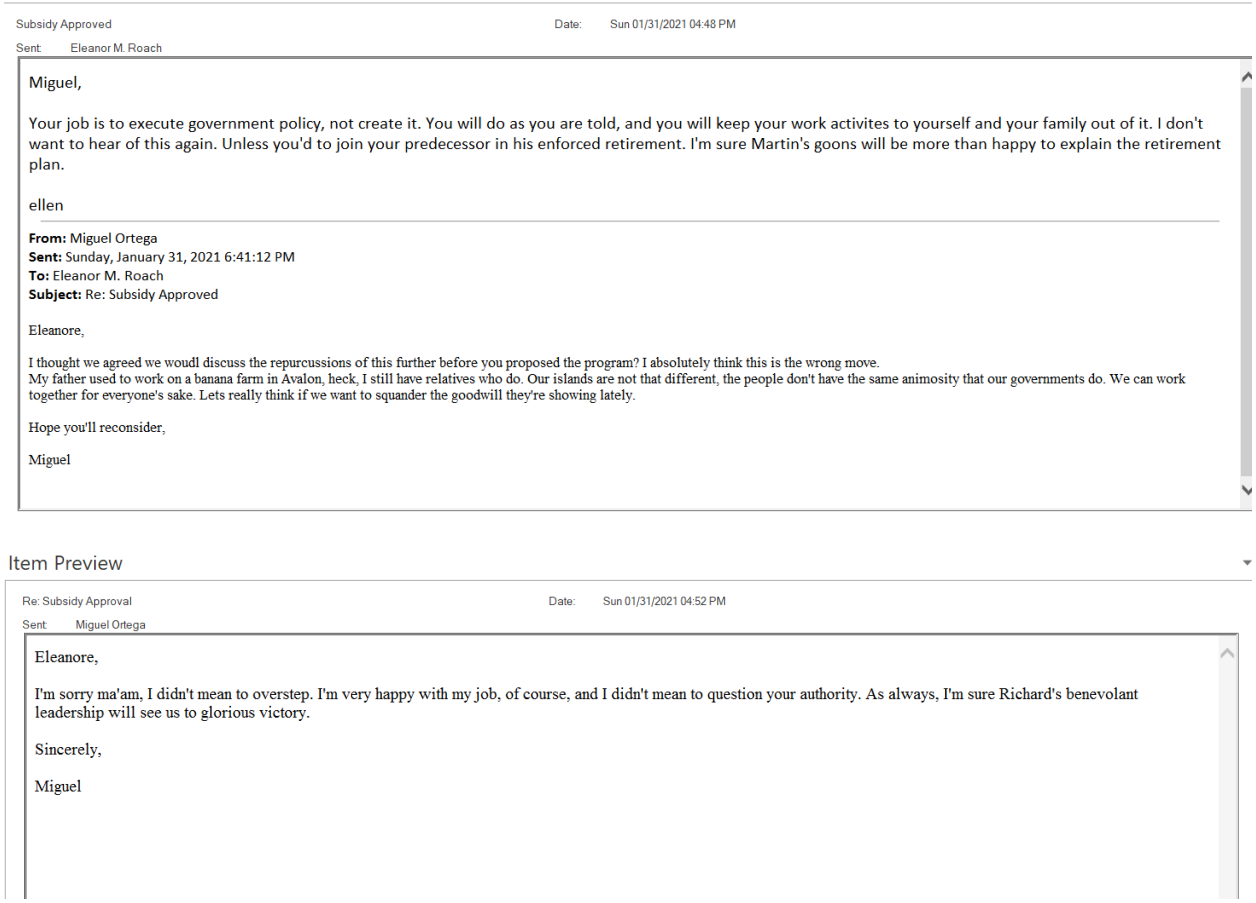
5.2.11.9 Services_A_0009 Jeannette Forward Email to Jeff Schmitt



5.2.11.10 Services_A_00010 Subject: Subsidy Proposal between Eleanor and Miguel







Subsidy Approved

Date: Sun 01/31/2021 04:48 PM

Sent: Eleanor M. Roach

Miguel,

Your job is to execute government policy, not create it. You will do as you are told, and you will keep your work activites to yourself and your family out of it. I don't want to hear of this again. Unless you'd to join your predecessor in his enforced retirement. I'm sure Martin's goons will be more than happy to explain the retirement plan.

ellen

From: Miguel Ortega
Sent: Sunday, January 31, 2021 6:41:12 PM
To: Eleanor M. Roach
Subject: Re: Subsidy Approved

Eleanore,

I thought we agreed we woudl discuss the repurcussions of this further before you proposed the program? I absolutely think this is the wrong move.
My father used to work on a banana farm in Avalon, heck, I still have relatives who do. Our islands are not that different, the people don't have the same animosity that our governments do. We can work together for everyone's sake. Lets really think if we want to squander the goodwill they're showing lately.

Hope you'll reconsider,

Miguel

Item Preview

Re: Subsidy Approval

Date: Sun 01/31/2021 04:52 PM

Sent: Miguel Ortega

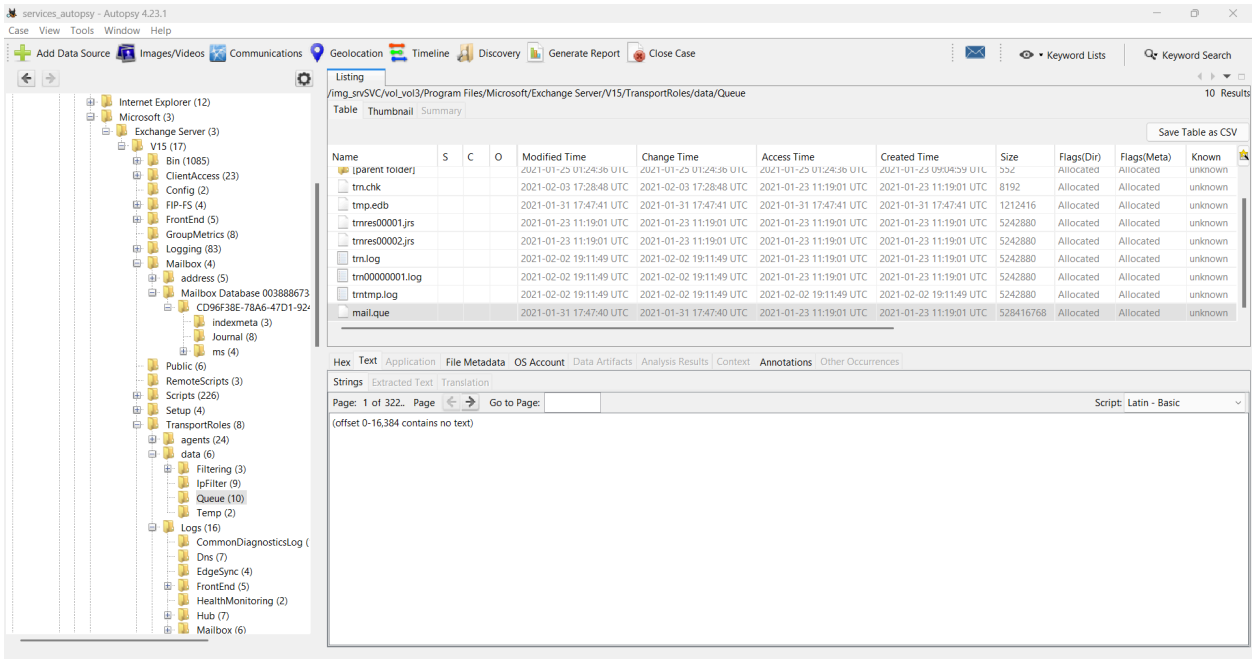
Eleanore,

I'm sorry ma'am, I didn't mean to overstep. I'm very happy with my job, of course, and I didn't mean to question your authority. As always, I'm sure Richard's benevolant leadership will see us to glorious victory.

Sincerely,

Miguel

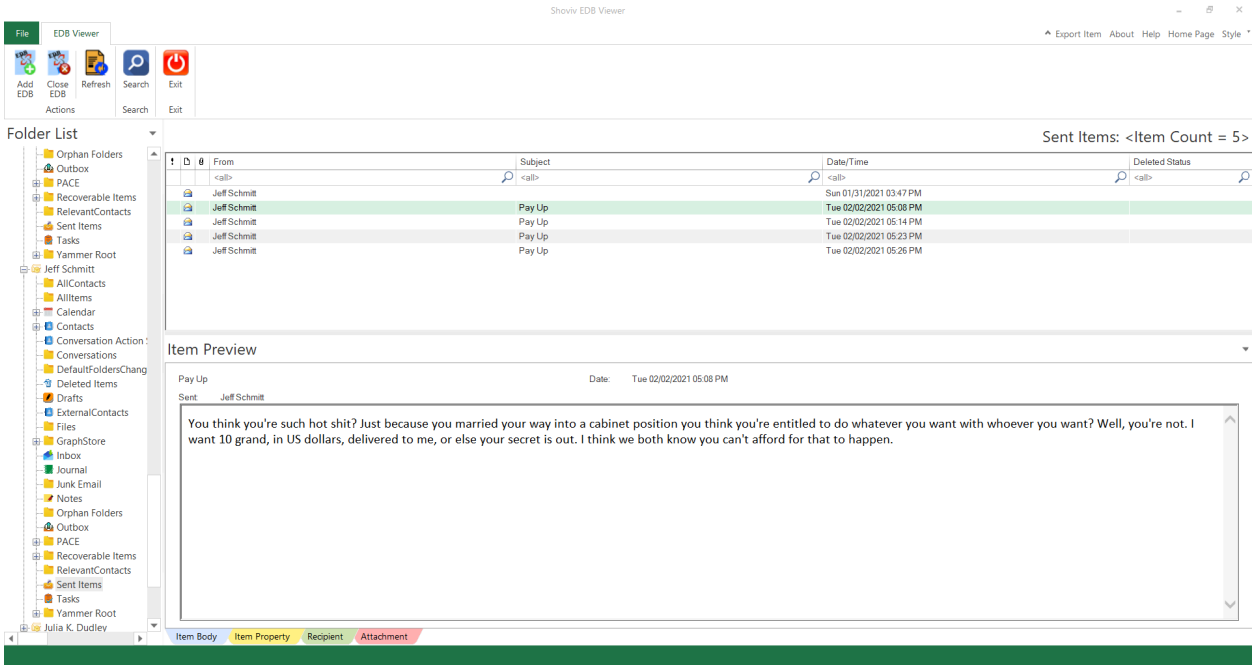
5.2.11.11 Services_A_0011 Email Metadata

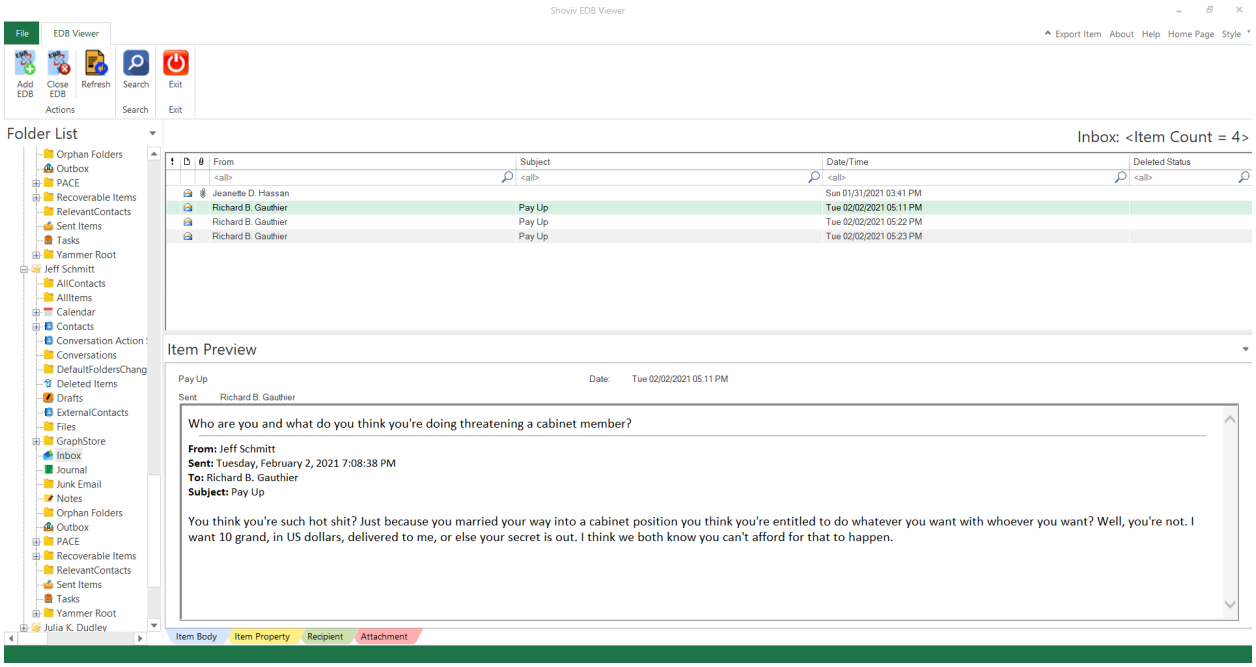


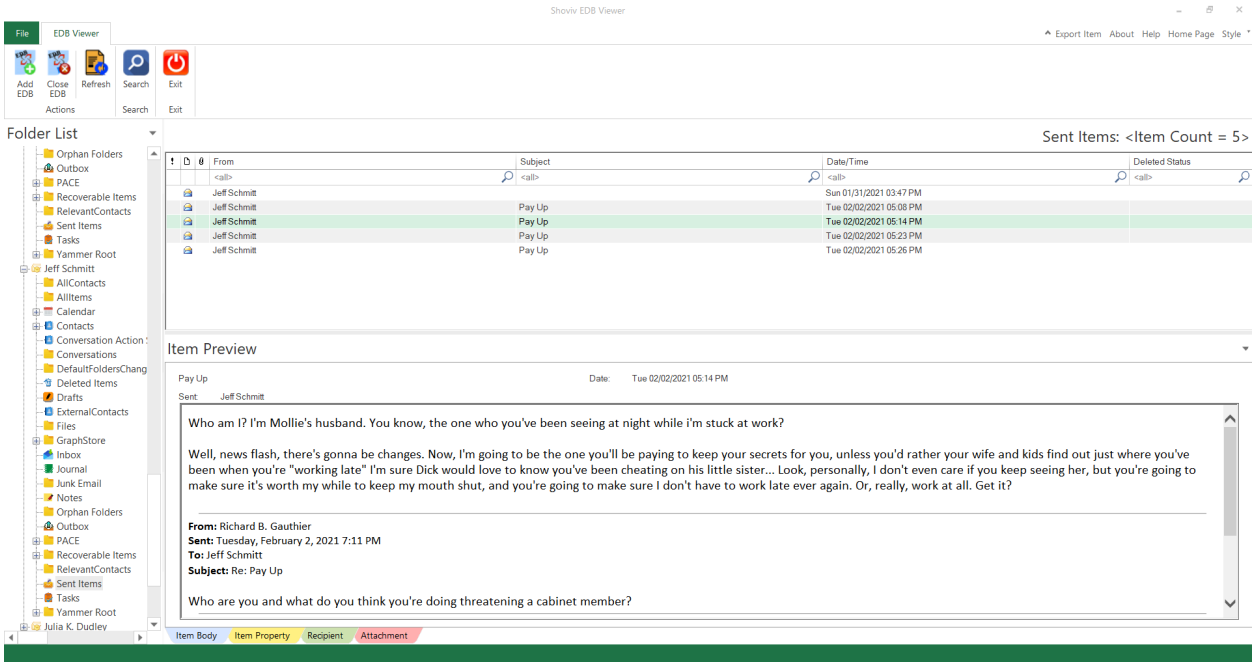
Received: from (192.168.2.42) by WIN-UI66P44IBTH.ambrosia.it (192.168.1.100)
 with Microsoft SMTP Server id 15.1.1713.5 via Frontend Transport; Sun, 31 Jan
 2021 18:36:12 -0500
Subject: Re: Subsidy Approval
From: Miguel Ortega
To: <Eleanore@Ambrosia.it>
Message-ID: <1ef5ce76-5efa-4ecc-a90a-44146900a851@WIN-UI66P44IBTH.ambrosia.it>
Return-Path: XxmiguelortegaxX3456@yahoo.com
Date: Sun, 31 Jan 2021 18:36:12 -0500
X-MS-Exchange-Organization-OriginalArrivalTime: 31 Jan 2021 23:36:12.0239
 (UTC)
X-MS-Exchange-Organization-Network-Message-Id: 3d6454e1-4d80-4030-2986-08d8c6413234
X-MS-Exchange-Organization-OriginalClientIPAddress: 192.168.2.42
X-MS-Exchange-Organization-OriginalServerIPAddress: 192.168.1.100

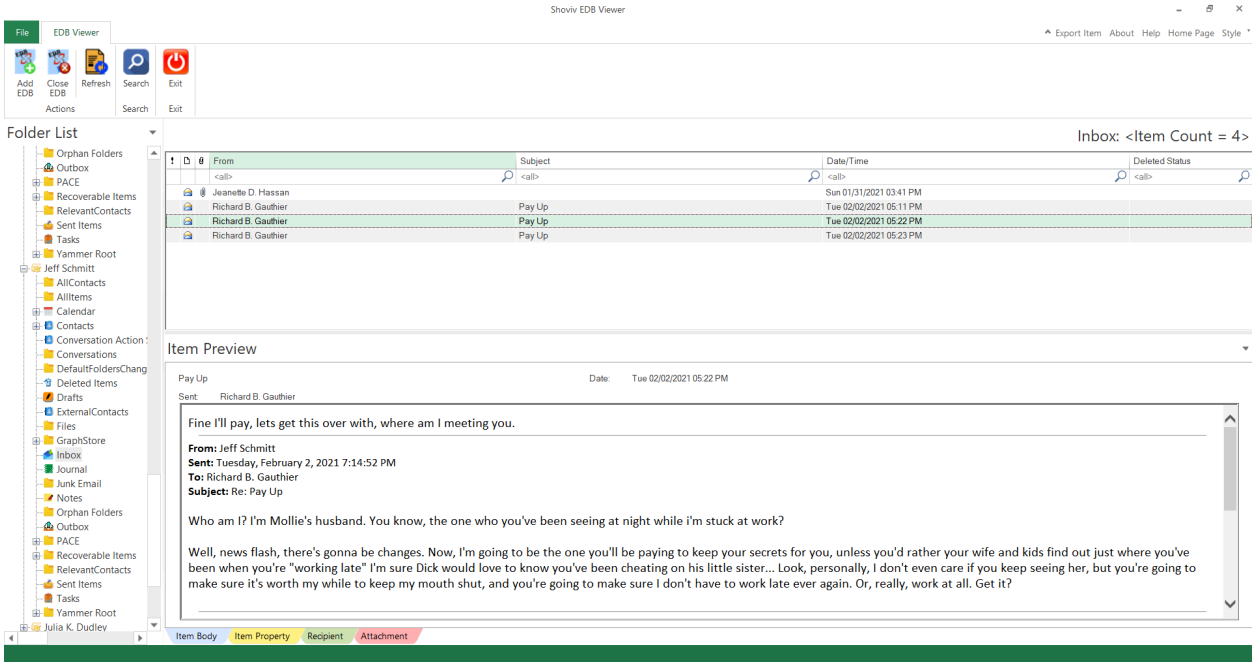
Received: from WIN-UI66P44IBTH.ambrosia.it (192.168.1.100) by
WIN-UI66P44IBTH.ambrosia.it (192.168.1.100) with Microsoft SMTP Server
(version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256) id
15.1.1713.5; Sun, 31 Jan 2021 17:35:14 -0500
Received: from (192.168.2.42) by WIN-UI66P44IBTH.ambrosia.it (192.168.1?
100)
with Microsoft SMTP Server id 15.1.1713.5 via Frontend Transport; Sun, 31 Jan
2021 17:35:08 -0500
From: <Helpdesk102@ambrosia.it>
To: jean@ambrosia.it
MIME-Version: 1.0
Content-Type: multipart/mixed; boundary="Xtextboundary"
Message-ID: <7f35d38a-0c02-447a-96fc-5af97e2f80af@WIN-UI66P44IBTH.ambrosia.it>
Return-Path: helpdesk102@ambrosia.it
Date: Sun, 31 Jan 2021 17:35:08 -0500
X-MS-Exchange-Organization-OriginalArrivalTime: 31 Jan 2021 22:35:08.0885
(UTC)

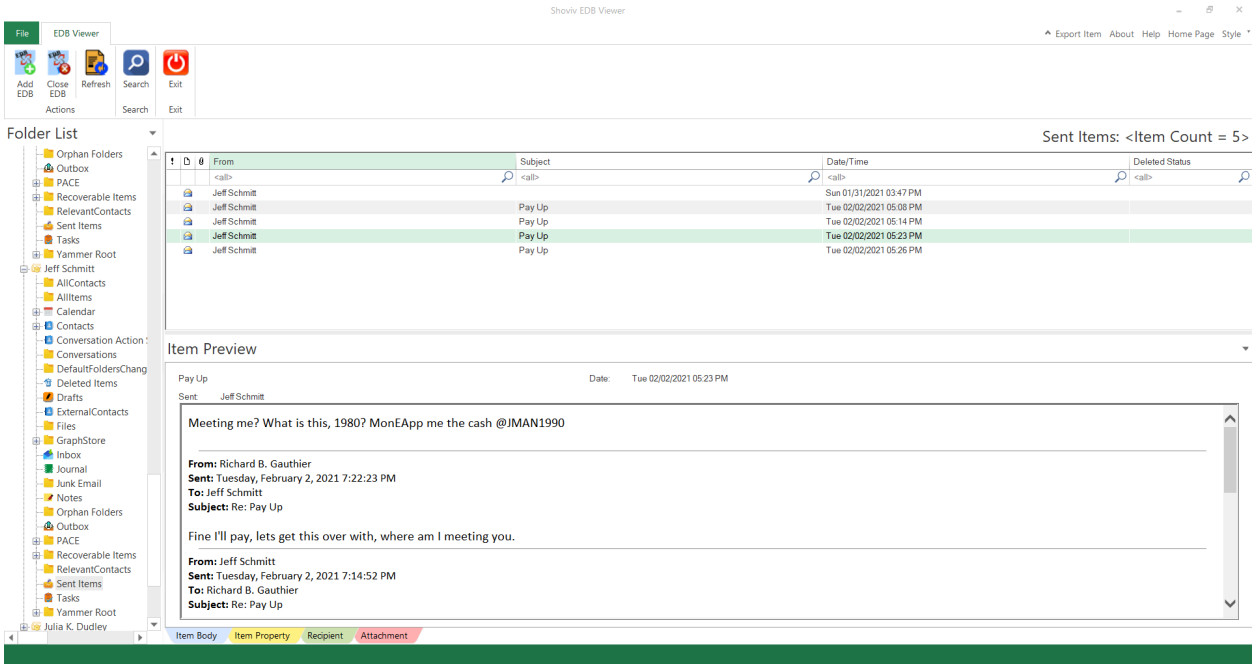
5.2.11.12 Services_A_0012 Jeff Schmitt x Richard Gauthier - Pay Up

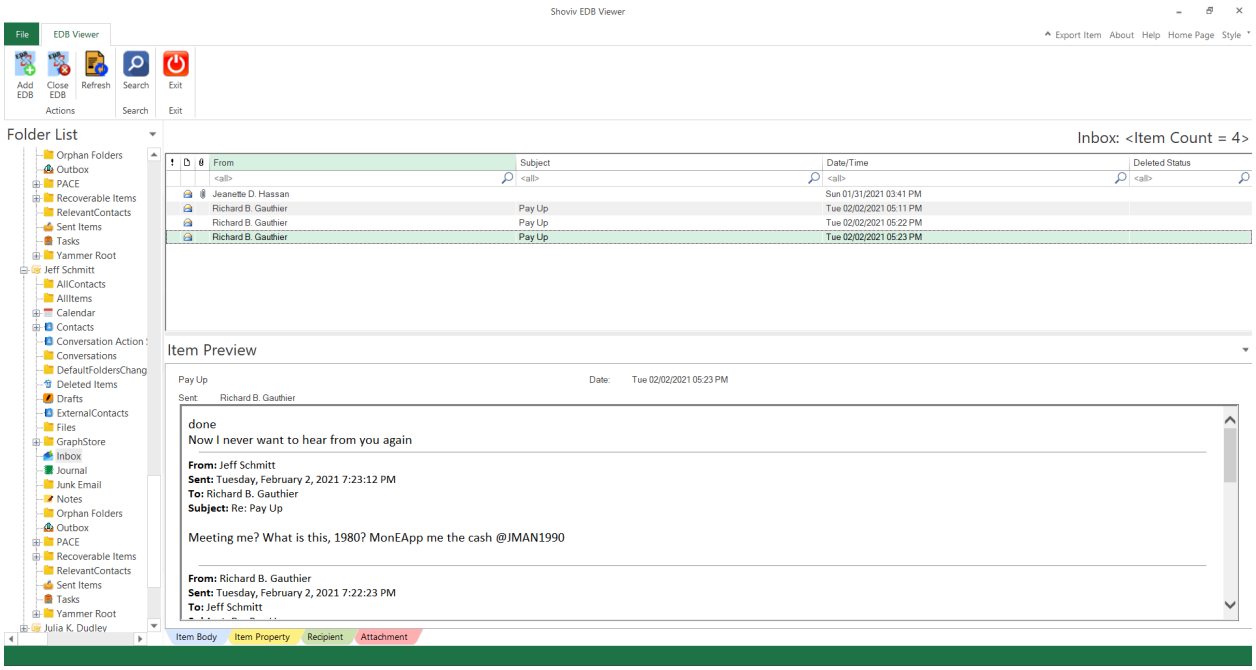


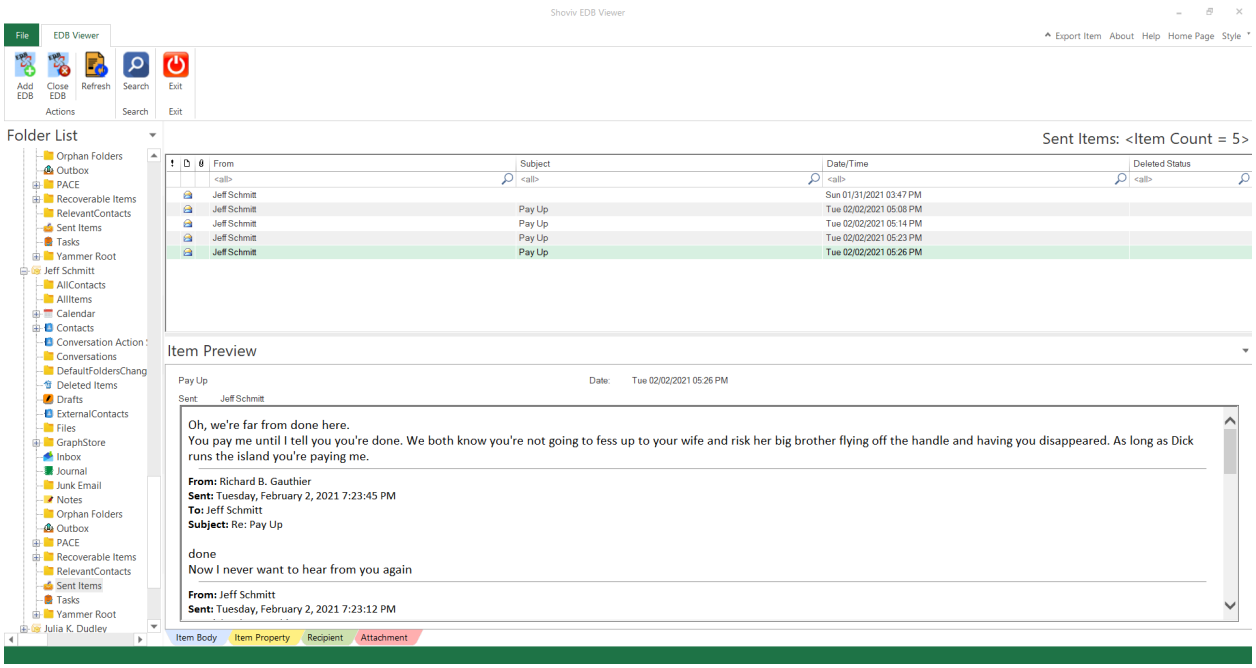



Shoviv EDB Viewer
File
EDB Viewer
Export Item
About
Help
Home Page
Style
Add EDB
Close EDB
Refresh
Search
Exit
Actions
Search
Exit
Folder List
Sent Items: <Item Count = 5>
From
Subject
Date/Time
Deleted Status
Jeff Schmitt
Sun 01/31/2021 03:47 PM
Jeff Schmitt
Pay Up
Tue 02/02/2021 05:08 PM
Jeff Schmitt
Pay Up
Tue 02/02/2021 05:14 PM
Jeff Schmitt
Pay Up
Tue 02/02/2021 05:23 PM
Jeff Schmitt
Pay Up
Tue 02/02/2021 05:26 PM
Item Preview
Pay Up
Date: Tue 02/02/2021 05:14 PM
Sent: Jeff Schmitt
Who am I? I'm Mollie's husband. You know, the one who you've been seeing at night while i'm stuck at work?
Well, news flash, there's gonna be changes. Now, I'm going to be the one you'll be paying to keep your secrets for you, unless you'd rather your wife and kids find out just where you've been when you're "working late" I'm sure Dick would love to know you've been cheating on his little sister... Look, personally, I don't even care if you keep seeing her, but you're going to make sure it's worth my while to keep my mouth shut, and you're going to make sure I don't have to work late ever again. Or, really, work at all. Get it?
From: Richard B. Gauthier
Sent: Tuesday, February 2, 2021 7:11 PM
To: Jeff Schmitt
Subject: Re: Pay Up
Who are you and what do you think you're doing threatening a cabinet member?
Item Body
Item Property
Recipient
Attachment


Shoviv EDB Viewer
File
EDB Viewer
Export Item
About
Help
Home Page
Style
Add EDB
Close EDB
Refresh
Search
Exit
Actions
Search
Exit
Folder List
Inbox: <Item Count = 4>
From
Subject
Date/Time
Deleted Status
Jeanette D. Hassan
Sun 01/31/2021 03:41 PM
Richard B. Gauthier
Pay Up
Tue 02/02/2021 05:11 PM
Richard B. Gauthier
Pay Up
Tue 02/02/2021 05:22 PM
Richard B. Gauthier
Pay Up
Tue 02/02/2021 05:23 PM
Item Preview
Pay Up
Date: Tue 02/02/2021 05:22 PM
Sent: Richard B. Gauthier
Fine I'll pay, lets get this over with, where am I meeting you.
From: Jeff Schmitt
Sent: Tuesday, February 2, 2021 7:14:52 PM
To: Richard B. Gauthier
Subject: Re: Pay Up
Who am I? I'm Mollie's husband. You know, the one who you've been seeing at night while i'm stuck at work?
Well, news flash, there's gonna be changes. Now, I'm going to be the one you'll be paying to keep your secrets for you, unless you'd rather your wife and kids find out just where you've been when you're "working late" I'm sure Dick would love to know you've been cheating on his little sister... Look, personally, I don't even care if you keep seeing her, but you're going to make sure it's worth my while to keep my mouth shut, and you're going to make sure I don't have to work late ever again. Or, really, work at all. Get it?
Item Body
Item Property
Recipient
Attachment


Shoviv EDB Viewer
File
EDB Viewer
Export Item
About
Help
Home Page
Style
Add EDB
Close EDB
Refresh
Search
Exit
Actions
Search
Exit
Folder List
Sent Items: <Item Count = 5>
From
Subject
Date/Time
Deleted Status
Jeff Schmitt
Sun 01/31/2021 03:47 PM
Pay Up
Tue 02/02/2021 05:08 PM
Tue 02/02/2021 05:14 PM
Tue 02/02/2021 05:23 PM
Tue 02/02/2021 05:26 PM
Item Preview
Date: Tue 02/02/2021 05:23 PM
Sent: Jeff Schmitt
Meeting me? What is this, 1980? MonEApp me the cash @JMAN1990
From: Richard B. Gauthier
Sent: Tuesday, February 2, 2021 7:22:23 PM
To: Jeff Schmitt
Subject: Re: Pay Up
Fine I'll pay, lets get this over with, where am I meeting you.
From: Jeff Schmitt
Sent: Tuesday, February 2, 2021 7:14:52 PM
To: Richard B. Gauthier
Subject: Re: Pay Up
Item Body
Item Property
Recipient
Attachment


Shoviv EDB Viewer
File
EDB Viewer
Export Item
About
Help
Home Page
Style
Add EDB
Close EDB
Refresh
Search
Exit
Actions
Search
Exit
Folder List
Inbox: <Item Count = 4>
From
Subject
Date/Time
Deleted Status
Jeanette D. Hassan
Sun 01/31/2021 03:41 PM
Richard B. Gauthier
Tue 02/02/2021 05:11 PM
Tue 02/02/2021 05:22 PM
Tue 02/02/2021 05:23 PM
Item Preview
Date: Tue 02/02/2021 05:23 PM
Sent: Richard B. Gauthier
done
Now I never want to hear from you again
From: Jeff Schmitt
Sent: Tuesday, February 2, 2021 7:23:12 PM
To: Richard B. Gauthier
Subject: Re: Pay Up
Meeting me? What is this, 1980? MonEApp me the cash @JMAN1990
From: Richard B. Gauthier
Sent: Tuesday, February 2, 2021 7:22:23 PM
To: Jeff Schmitt
Item Body
Item Property
Recipient
Attachment


Shoviv EDB Viewer
File
EDB Viewer
Export Item
About
Help
Home Page
Style
Add EDB
Close EDB
Refresh
Search
Exit
Actions
Search
Exit
Folder List
Orphan Folders
Outbox
PACE
Recoverable Items
RelevantContacts
Sent Items
Tasks
Yammer Root
Jeff Schmitt
AllContacts
AllItems
Calendar
Contacts
Conversation Action
Conversations
DefaultFoldersChang
Deleted Items
Drafts
ExternalContacts
Files
GraphStore
Inbox
Journal
Junk Email
Notes
Orphan Folders
Outbox
PACE
Recoverable Items
RelevantContacts
Sent Items
Tasks
Yammer Root
Julia K. Dudley
Sent Items: <Item Count = 5>
From
Subject
Date/Time
Deleted Status
Jeff Schmitt
Sun 01/31/2021 03:47 PM
Jeff Schmitt
Pay Up
Tue 02/02/2021 05:08 PM
Jeff Schmitt
Pay Up
Tue 02/02/2021 05:14 PM
Jeff Schmitt
Pay Up
Tue 02/02/2021 05:23 PM
Jeff Schmitt
Pay Up
Tue 02/02/2021 05:26 PM
Item Preview
Pay Up
Date: Tue 02/02/2021 05:26 PM
Sent: Jeff Schmitt
Oh, we're far from done here.
You pay me until I tell you you're done. We both know you're not going to fess up to your wife and risk her big brother flying off the handle and having you disappeared. As long as Dick runs the island you're paying me.
From: Richard B. Gauthier
Sent: Tuesday, February 2, 2021 7:23:45 PM
To: Jeff Schmitt
Subject: Re: Pay Up
done
Now I never want to hear from you again
From: Jeff Schmitt
Sent: Tuesday, February 2, 2021 7:23:12 PM
Item Body
Item Property
Recipient
Attachment